Eclipses by Appointment - Supplement

Volume 2 of 4

Fixed magnitude comparison and R217-R409

Original continuous pages 1187-1947 of 3269

Navigation: each volume adds this one-page guide.

A page in this volume = original page minus 1186 plus 1.

The public supplement-review-pages.csv maps every R key to volume

and exact page. The scientific-review-register.csv maps status to report.

Volume 1: Original A catalogue and R001-R216 (original pages 1-1186).

Volume 2: Fixed magnitude comparison and R217-R409 (original pages 1187-1947).

Volume 3: R410-R539 (original pages 1948-2450).

Volume 4: R540-R675 and remaining reviewed evidence (original pages 2451-3269).

Original illustrated A entries and all R full reports remain in the

continuous source HTML and the public reports/evidence archive.

Cross-volume references use the page map above.

R217. TIC 428691463

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 428691463: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 15 distinct eclipse events in selected sectors[59, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 15 same-kind midtimes give P=2.2025512596 ±0.0000032099d, reference eclipse epoch BTJD2912.7110288, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S59 deficit peak[7, 7] vs target pixel[8, 7]; deficit-target separation3.708arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.815. Capable neighbours: Gaia 251403200552910592: 6.95 arcsec, G12.261, required fractional eclipse0.023, pixel fractional loss0.005518074279438695; Gaia 251403200553995008: 13.52 arcsec, G16.015, required fractional eclipse0.465, pixel fractional loss0.005042044222833572; Gaia 251403234912639616: 39.32 arcsec, G15.736, required fractional eclipse0.361, pixel fractional loss0.0023197334242561615
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths5.316/4.931/4.325ppt, with 10 stacked events. QLP selected-kind fitted median10.135ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3251403200552912128, P=2.2025752205±0.0000738127d, e=0.023863, omega=315.1974deg. Closest edge-on conjunction offset-0.031275d (-0.014199phase), full-SB1-covariance linear propagated Gaia uncertainty0.077205d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.003816d (photometric period projection allowance0.003679d). The approximate central duration3.88h vs observed median1.34h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unresolved attribution: capable neighbor Gaia251403200552910592 at6.95arcsec needs only2.3% fractional eclipse. Deficit centroid lies between target and neighbor; PSF residual0.815 and uncalibrated centroid cannot discriminate. Real repeated eclipses remain probable for this target.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

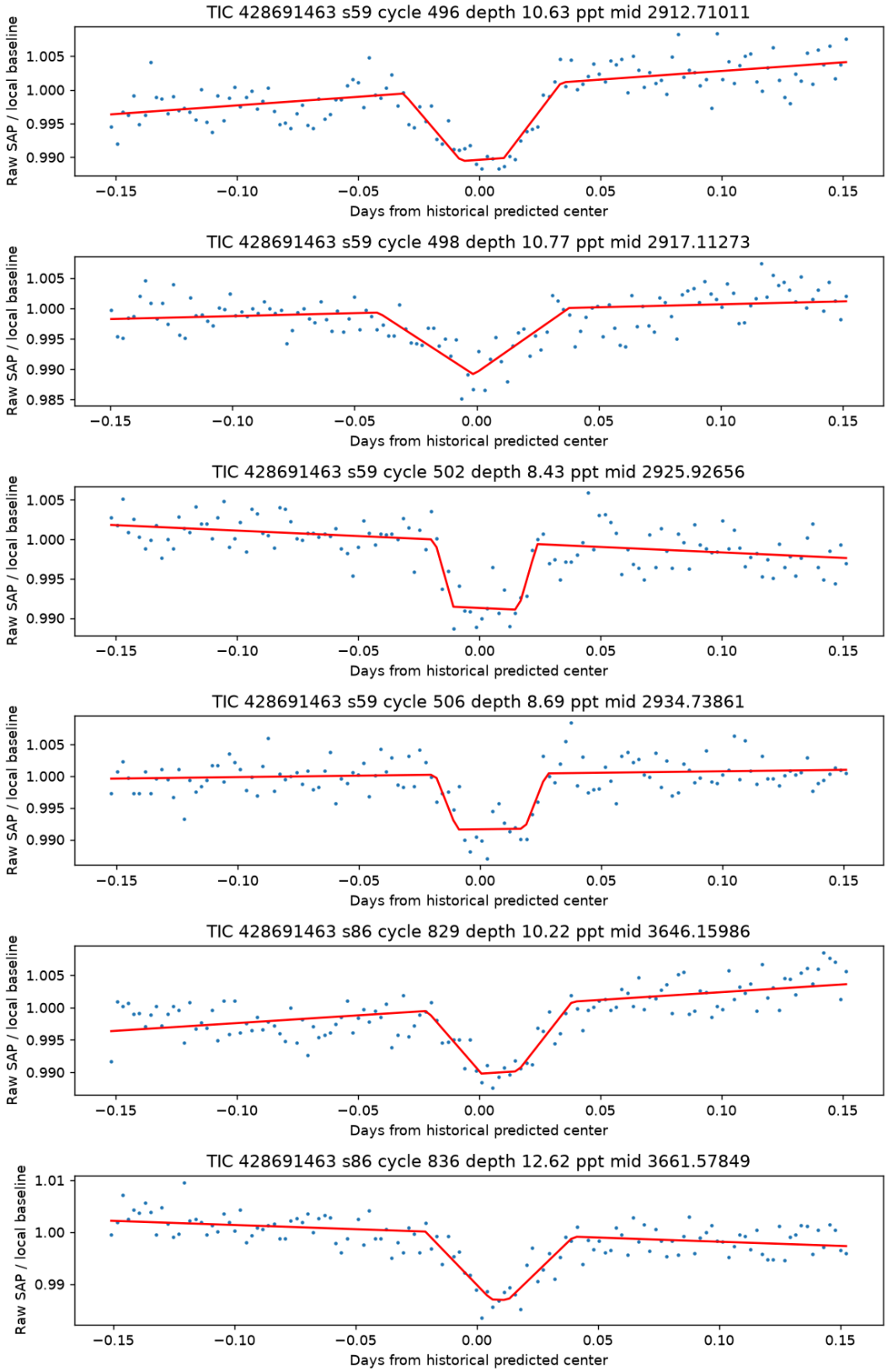
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

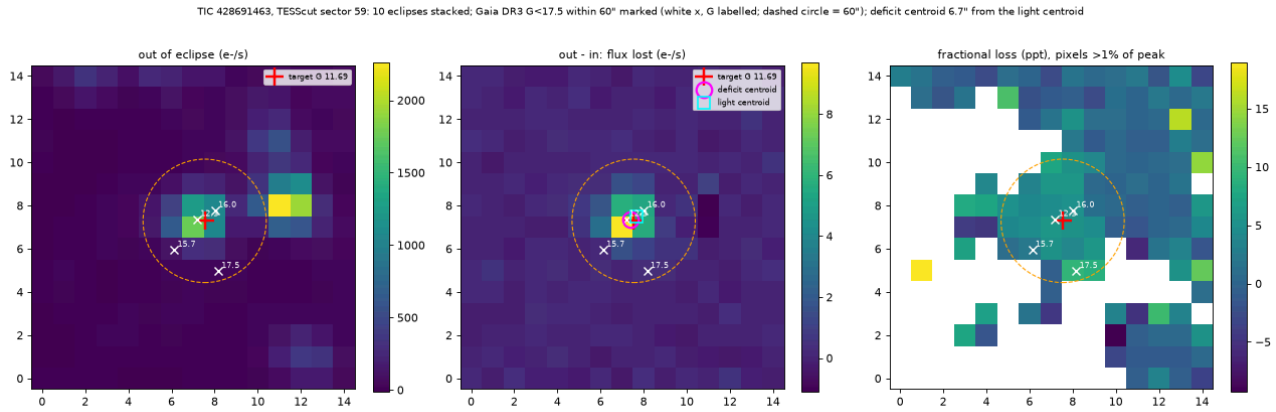
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
59	496	2912.7101110	0.002139	10.628	usable; same-kind selection applies
59	497	2914.9109222	0.001157	8.703	usable; same-kind selection applies
59	498	2917.1127333	0.001780	10.766	usable; same-kind selection applies
59	499	2919.3221369	0.001960	11.576	usable; same-kind selection applies
59	500	2921.5194762	0.002219	8.957	usable; same-kind selection applies
59	502	2925.9265644	0.001157	8.434	usable; same-kind selection applies
59	503	2928.1295658	0.001157	10.135	usable; same-kind selection applies
59	505	2932.5369579	0.002308	12.706	usable; same-kind selection applies
59	506	2934.7386079	0.001157	8.687	usable; same-kind selection applies
86	826	3639.5553104	0.002137	7.859	usable; same-kind selection applies
86	828	3643.9576304	0.001347	10.204	usable; same-kind selection applies
86	829	3646.1598615	0.001841	10.219	usable; same-kind selection applies
86	830	3648.3627624	0.002785	8.065	usable; same-kind selection applies
86	835	3659.3751737	0.002683	8.426	usable; same-kind selection applies
86	836	3661.5784863	0.001237	12.621	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s59.json, historical/diff_s59.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R217. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R217; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R217; diff_s59.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R218. TIC 167105778

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 167105778: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[15, 75, 83]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=8.5865668060 ±0.0000273371d, reference eclipse epoch BTJD1718.8649126, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S56 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.804arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.490. Capable neighbours: Gaia 1968960312990727296: 10.12 arcsec, G14.833, required fractional eclipse0.441, pixel fractional loss0.02316814965491679; Gaia 1968960278623106176: 55.49 arcsec, G14.881, required fractional eclipse0.460, pixel fractional loss-0.001106644215653597
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths23.168/22.598/20.538ppt, with 4 stacked events. QLP selected-kind fitted median24.975ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3196896031299072588 8, P=8.5870099785±0.0010707120d, e=0.017110, omega=153.5675deg. Closest edge-on conjunction offset-0.155334d (-0.018089phase), full-SB1-covariance linear propagated Gaia uncertainty0.199645d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.086642d (photometric period projection allowance0.004237d). The approximate central

duration9.82h vs observed median6.95h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Capable G14.83 neighbor10.1arcsec away requires44% eclipse. Deficit is0.18pixel from target and0.47pixel from neighbor; image favors target, with uncalibrated centroid resolution retained as limitation.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

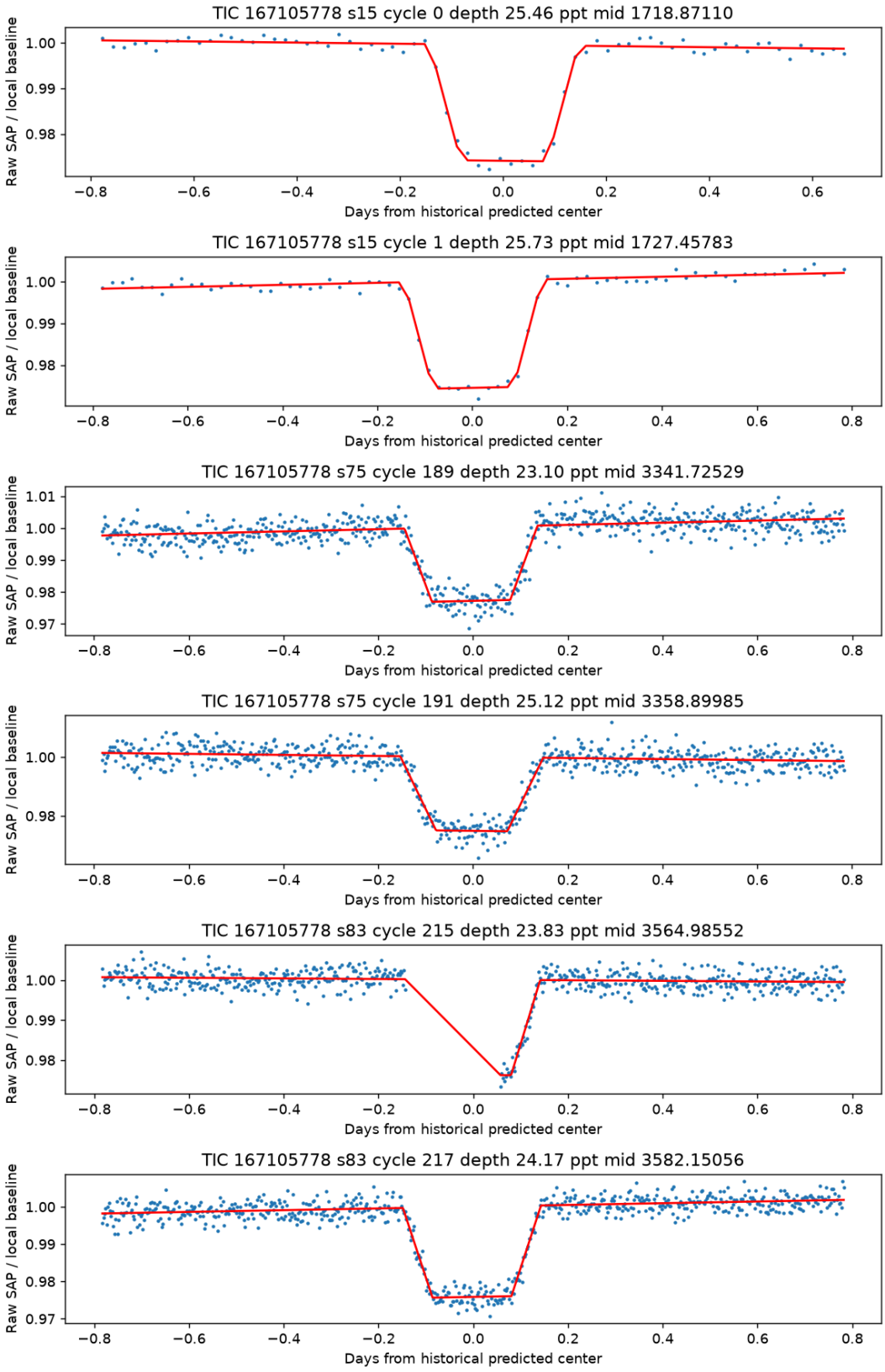
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

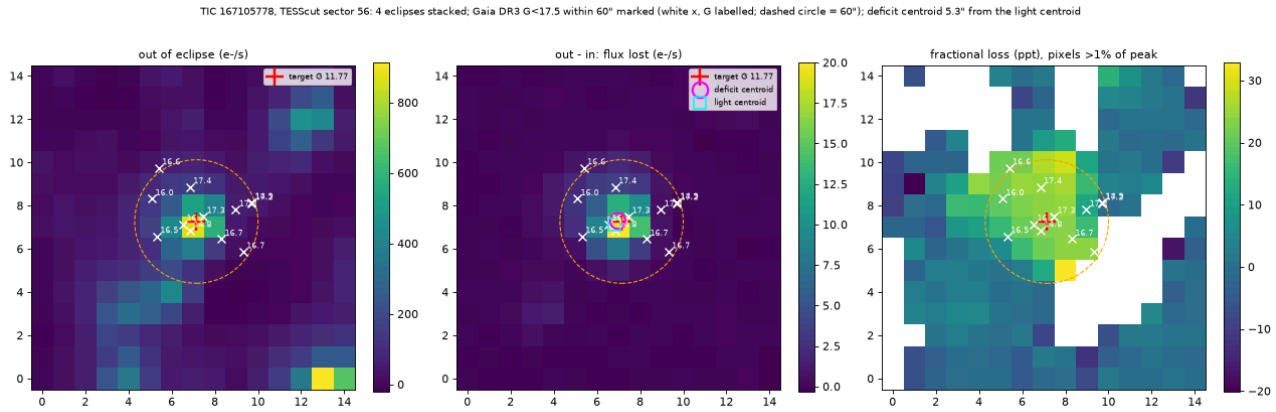
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1718.8711042	0.010417	25.465	usable; same-kind selection applies
15	1	1727.4578265	0.010417	25.730	usable; same-kind selection applies
75	189	3341.7252905	0.001282	23.103	usable; same-kind selection applies
75	190	3350.3114455	0.001157	24.975	usable; same-kind selection applies
75	191	3358.8998480	0.001274	25.117	usable; same-kind selection applies
83	215	3564.9855222	74778.289228	23.830	excluded singular timing fit
83	216	3573.5637494	0.001157	24.259	usable; same-kind selection applies
83	217	3582.1505608	0.001157	24.167	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s56.json, historical/diff_s56.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R218. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R218; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R218; diff_s56.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R219. TIC 48212656

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 48212656: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 15 distinct eclipse events in selected sectors[36, 62, 89]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=9.0832698961 ±0.0000108660d, reference eclipse epoch BTJD2290.1567712, kind: Deeper primary (odd half-period cycles). Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S62 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation0.773arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.179. Capable neighbours: Gaia 5463218573976102400: 45.22 arcsec, G13.926, required fractional eclipse0.509, pixel fractional loss0.010378443232646176
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths48.360/47.433/42.752ppt, with 5 stacked events. QLP selected-kind fitted median73.015ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35463218604039665408, P=9.0837171230±0.0011010123d, e=0.012056, omega=249.6383deg. Closest edge-on conjunction offset-0.076515d (-0.008423phase), full-SB1-covariance linear propagated Gaia uncertainty0.253510d at the measured epoch. Near-Gaia-reference projected eclipse offset0.016955d (photometric period projection allowance0.002271d). The approximate central duration8.20h vs observed median5.58h permits noncentral geometry; this radius/minimum-mass estimate is not a

detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses approximately40/73ppt establish doubled orbital period. Odd historical half-period cycles are deeper primaries and alone define ephemeris. Historical cutout mixes eclipse kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

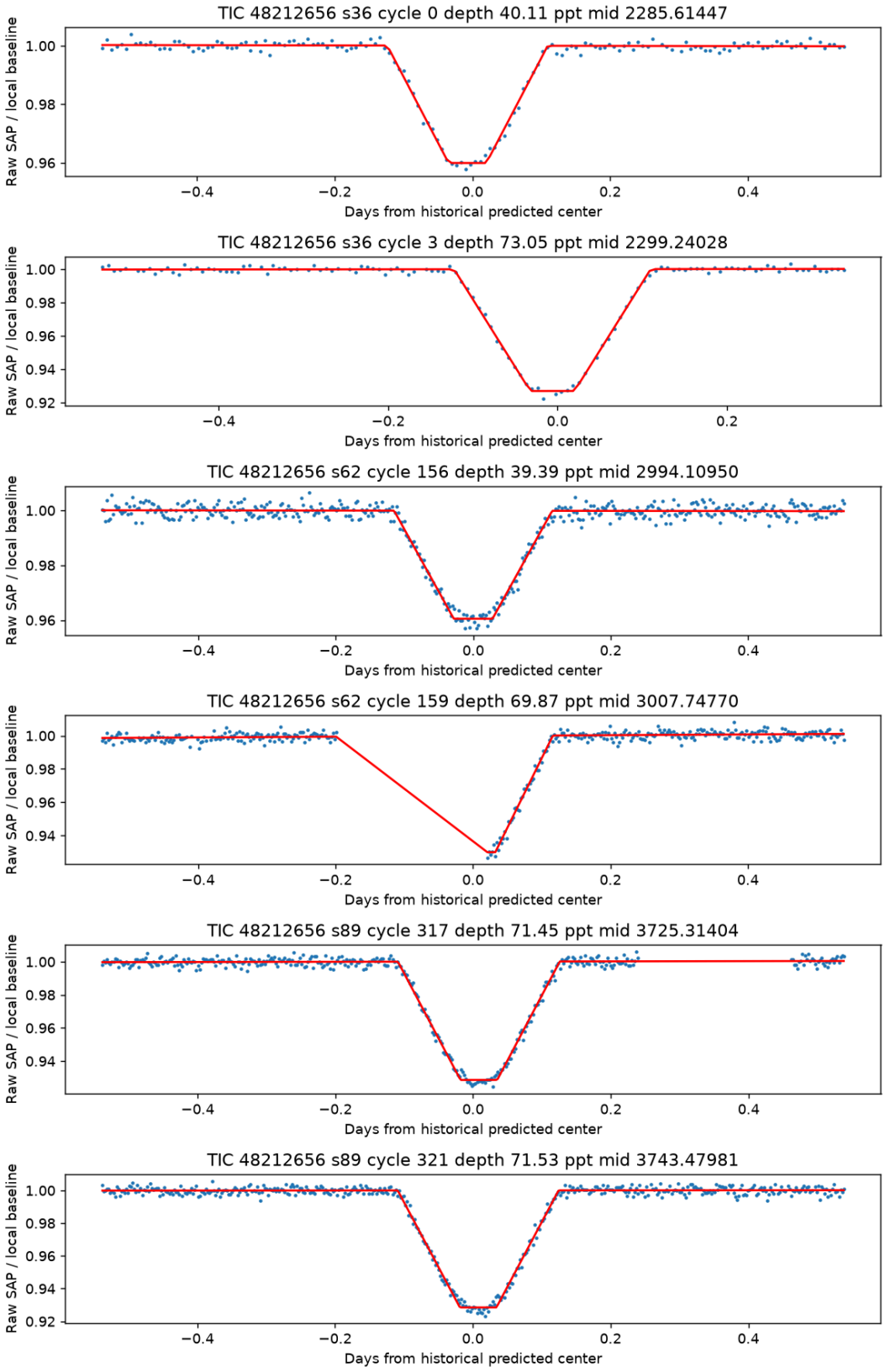
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

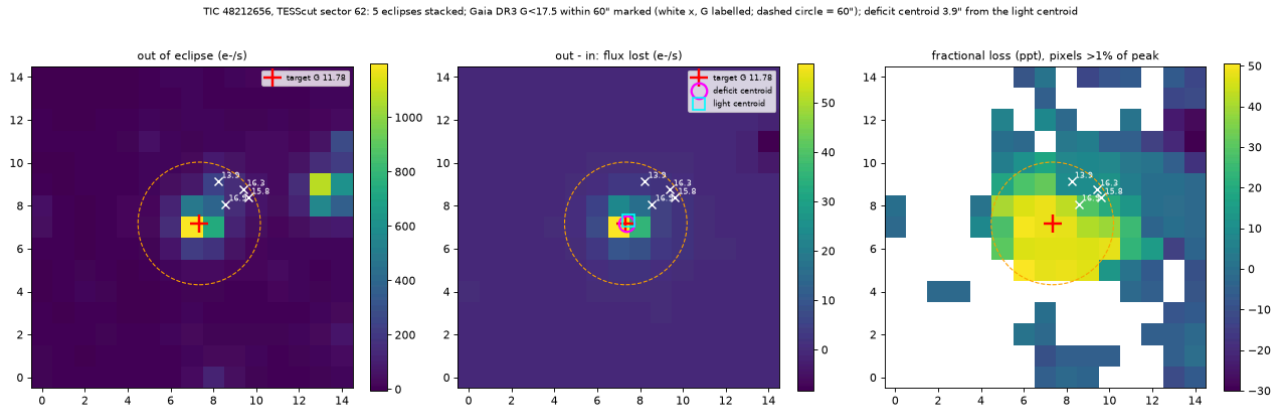
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
36	0	2285.6144732	0.003472	40.112	usable; same-kind selection applies
36	1	2290.1571543	0.003472	73.015	usable; same-kind selection applies
36	3	2299.2402837	0.003472	73.048	usable; same-kind selection applies
36	4	2303.7814680	0.003472	40.339	usable; same-kind selection applies
62	155	2989.5682938	0.001157	73.938	usable; same-kind selection applies
62	156	2994.1094979	0.001157	39.389	usable; same-kind selection applies
62	157	2998.6519402	0.001157	73.423	usable; same-kind selection applies
62	158	3003.1922367	0.001157	40.036	usable; same-kind selection applies
62	159	3007.7476989	60271.664713	69.865	excluded singular timing fit
62	160	3012.2751753	0.001157	38.834	usable; same-kind selection applies
89	316	3720.7710087	0.001157	39.192	usable; same-kind selection applies
89	317	3725.3140402	0.001157	71.449	usable; same-kind selection applies
89	318	3729.8537742	0.001157	39.552	usable; same-kind selection applies
89	319	3734.3962781	0.001157	72.160	usable; same-kind selection applies
89	321	3743.4798087	0.001157	71.532	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R219. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R219; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R219; diff_s62.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R220. TIC 442662494

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 442662494: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[9, 37, 90]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=15.8202440581 ±0.0000267968d, reference eclipse epoch BTJD1555.0654980, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S36 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.803arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.968. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths279.938/261.135/234.930ppt, with 2 stacked events. QLP selected-kind fitted median317.962ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3525708361458224128, P=15.8200347806±0.0026074222d, e=0.103885, omega=111.0794deg. Closest edge-on conjunction offset0.003691d (0.000233phase), full-SB1-covariance linear propagated Gaia uncertainty0.256392d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.011796d (photometric period projection allowance0.001983d). The approximate central duration7.35h vs observed median5.41h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Historical PSF fit residual0.968 is poor, though target-centered deficit and no capable catalogued neighbor support attribution. Aperture/sector-dependent depths approximately277-331ppt; no intrinsic undiluted depth is inferred.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

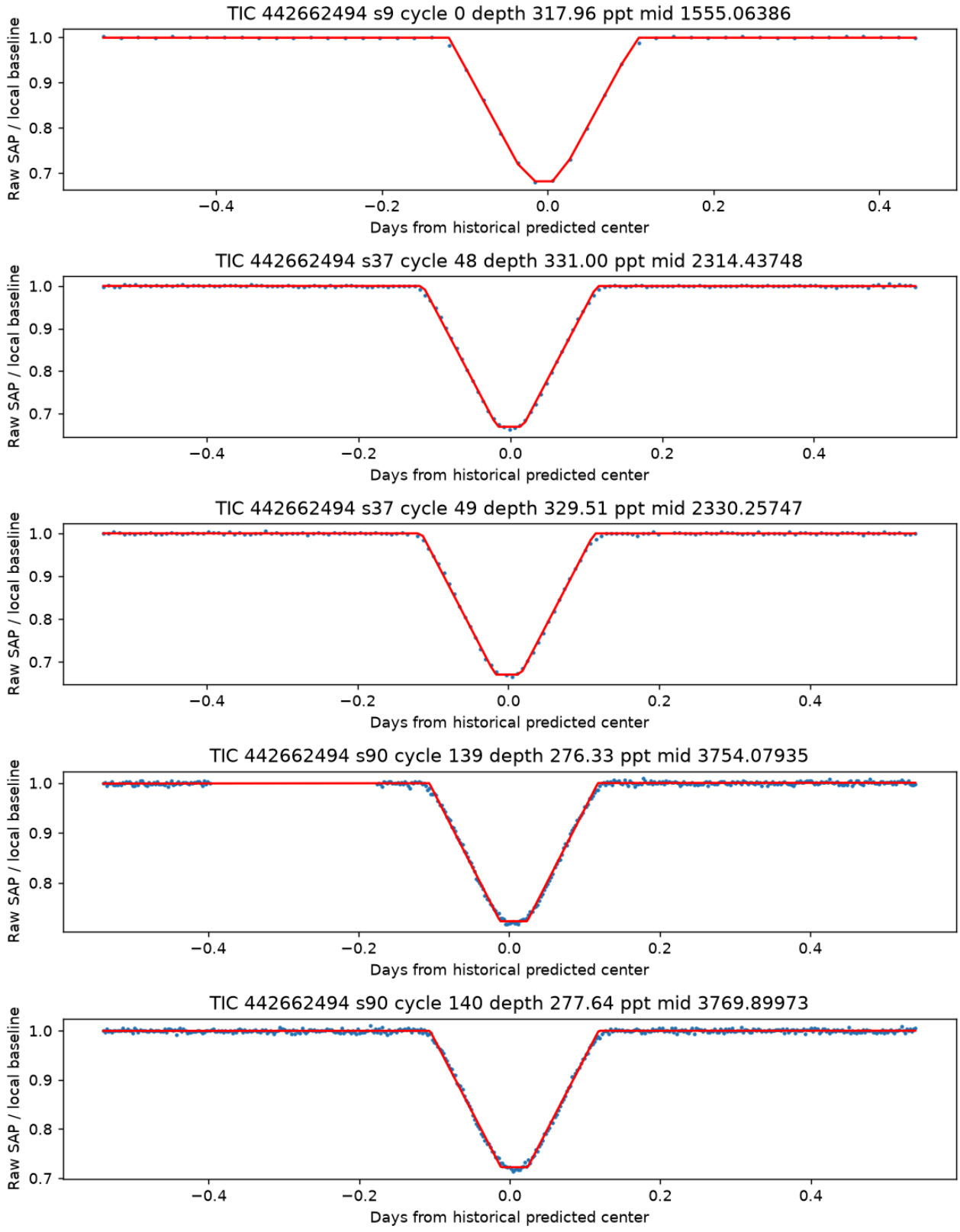
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

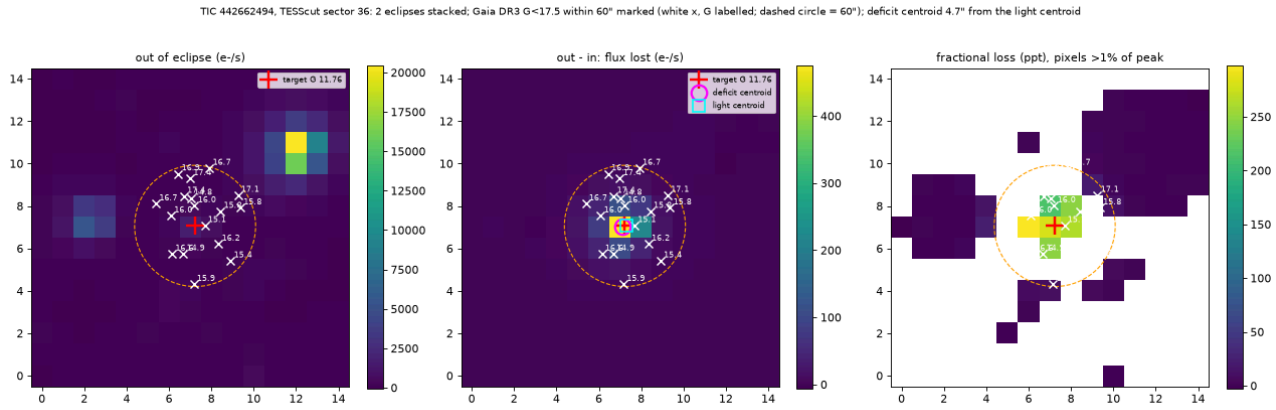
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	0	1555.0638588	0.010417	317.962	usable; same-kind selection applies
37	48	2314.4374794	0.003472	331.002	usable; same-kind selection applies
37	49	2330.2574750	0.003472	329.508	usable; same-kind selection applies
90	139	3754.0793461	0.001157	276.333	usable; same-kind selection applies
90	140	3769.8997306	0.001157	277.645	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s36.json, historical/diff_s36.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R220. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R220; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R220; diff_s36.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R221. TIC 280881615

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 280881615: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[59, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=5.2631068082 ±0.0000072351d, reference eclipse epoch BTJD2912.9426599, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S53 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.424arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.210. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths133.539/131.270/126.773ppt, with 5 stacked events. QLP selected-kind fitted median139.117ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3500313662204983936, P=5.2629890551±0.0002254657d, e=0.010989, omega=299.7018deg. Closest edge-on conjunction offset0.087594d (0.016643phase), full-SB1-covariance linear propagated Gaia uncertainty0.099605d at the measured epoch. Near-Gaia-reference projected eclipse offset0.031073d (photometric period projection allowance0.003473d). The approximate central duration5.07h vs observed median3.46h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Triangular ingress and egress repeat cleanly. No capable neighbor under the approximate G-band flux screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

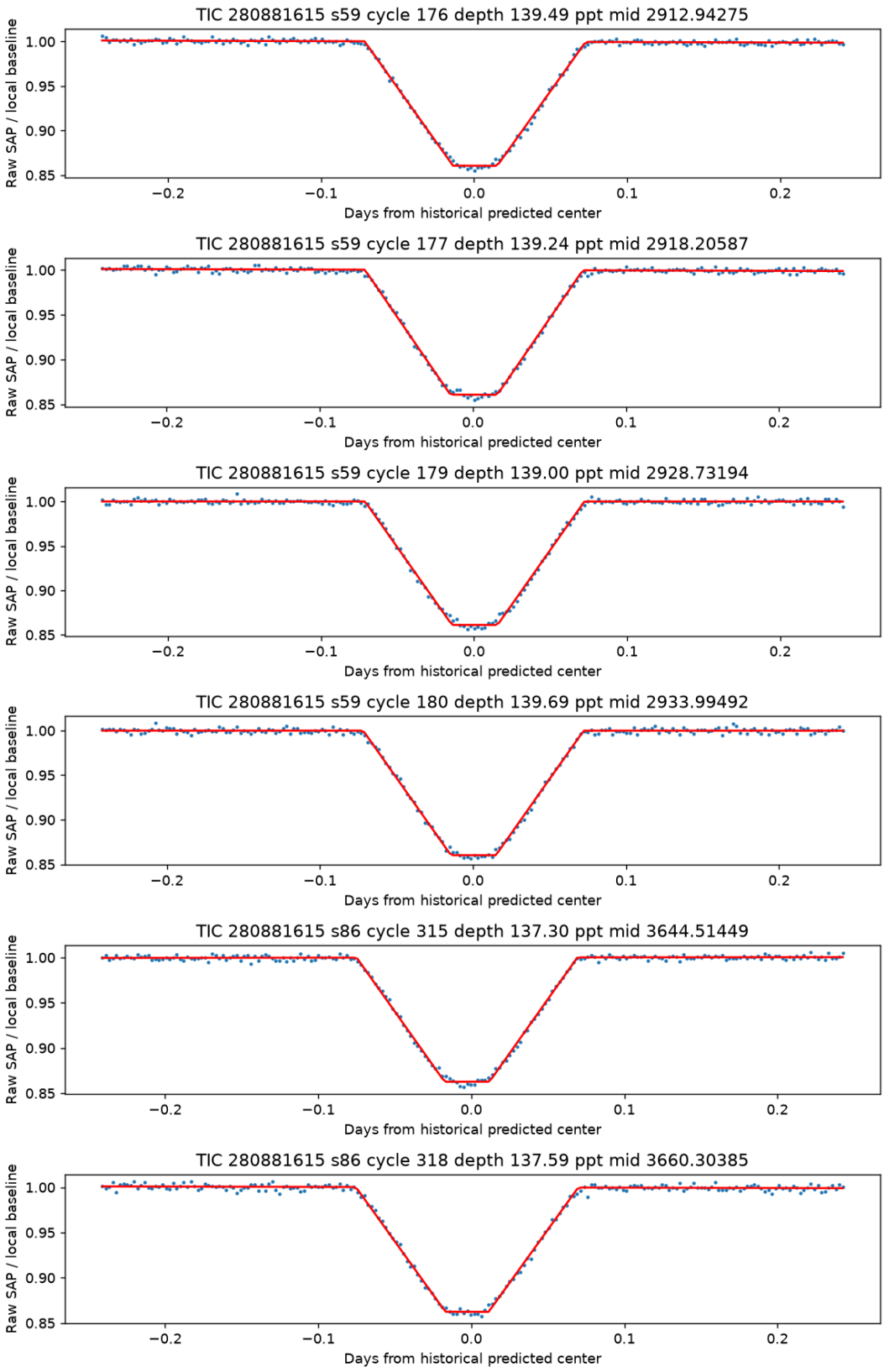
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

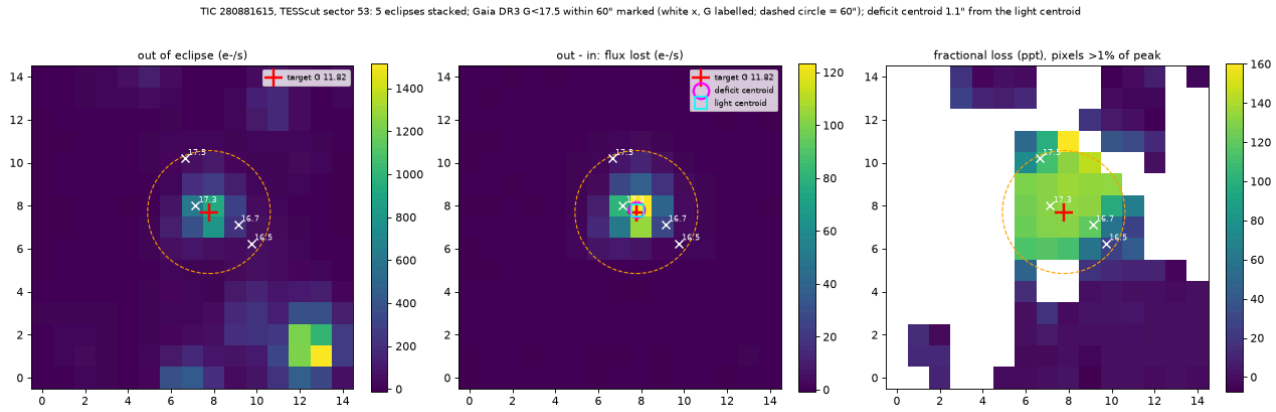
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
59	176	2912.9427546	0.001157	139.489	usable; same-kind selection applies
59	177	2918.2058723	0.001157	139.237	usable; same-kind selection applies
59	179	2928.7319403	0.001157	138.997	usable; same-kind selection applies
59	180	2933.9949223	0.001157	139.691	usable; same-kind selection applies
86	315	3644.5144895	0.001157	137.296	usable; same-kind selection applies
86	318	3660.3038478	0.001157	137.591	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s53.json, historical/diff_s53.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R221. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R221; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R221; diff_s53.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R222. TIC 238857132

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 238857132: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 21 distinct eclipse events in selected sectors[19, 59, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=4.1681777841 ±0.0000043833d, reference eclipse epoch BTJD1819.0960960, kind: Deeper primary (even half-period cycles). Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S59 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.364arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.500. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths102.438/99.296/95.399ppt, with 12 stacked events. QLP selected-kind fitted median145.901ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3179205929877639680, P=4.1679867378±0.0002603474d, e=0.021003, omega=258.5500deg. Closest edge-on conjunction offset0.077346d (0.018557phase), full-SB1-covariance linear propagated Gaia uncertainty0.096978d at the measured epoch. Near-Gaia-reference projected eclipse offset0.011818d (photometric period projection allowance0.001503d). The approximate central duration7.74h vs observed median5.00h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses approximately75/147ppt establish doubled orbital period. Even half-period cycles are deeper primaries and alone define ephemeris. Historical cutout mixes eclipse kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

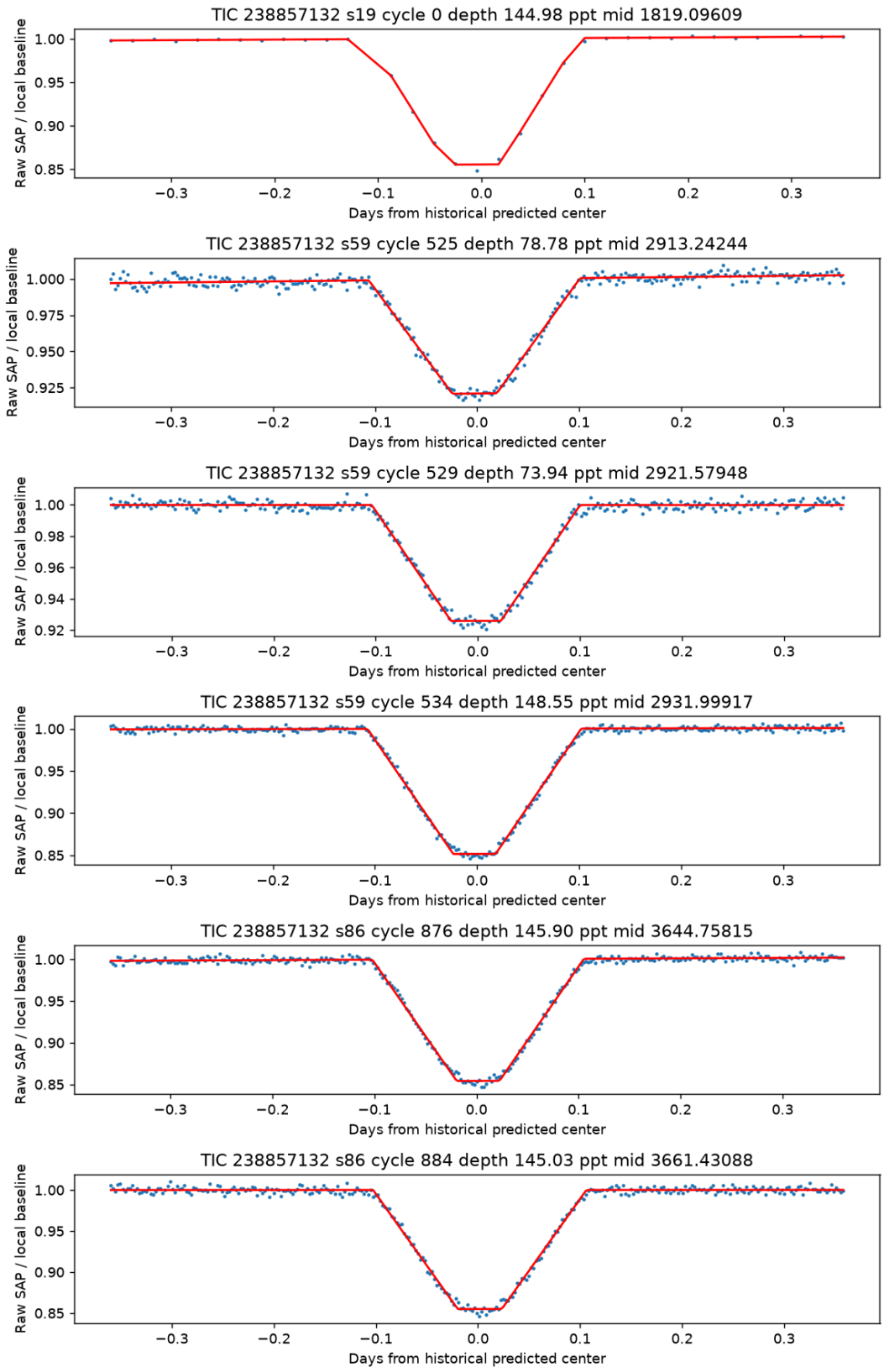
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

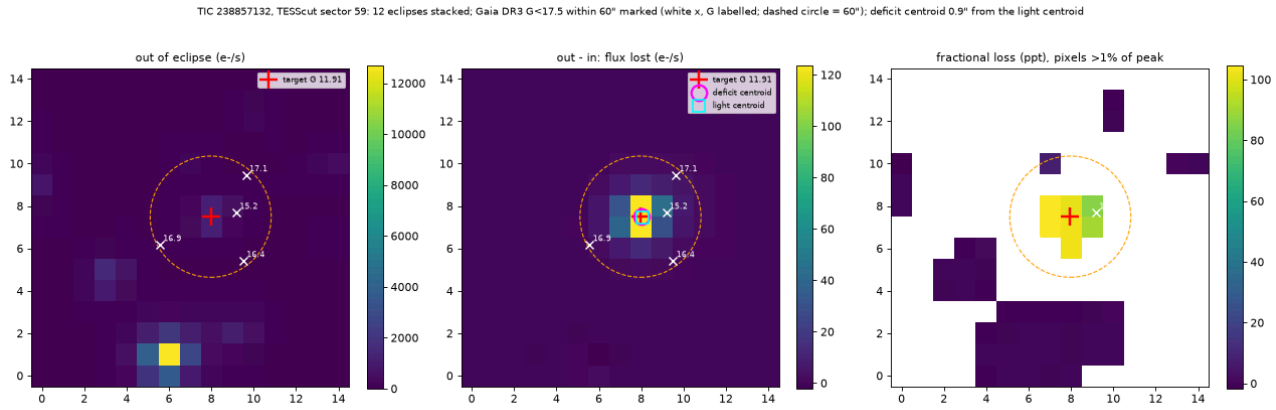
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
19	0	1819.0960855	0.010417	144.979	usable; same-kind selection applies
19	3	1825.3565265	0.010417	66.421	usable; same-kind selection applies
19	4	1827.4361486	0.010417	130.797	usable; same-kind selection applies
19	9	1837.8573728	0.010417	62.025	usable; same-kind selection applies
59	525	2913.2424422	0.001157	78.780	usable; same-kind selection applies
59	526	2915.3273027	0.001157	145.590	usable; same-kind selection applies
59	527	2917.4112655	0.001157	74.574	usable; same-kind selection applies
59	528	2919.4950535	0.001157	146.996	usable; same-kind selection applies
59	529	2921.5794780	0.001157	73.944	usable; same-kind selection applies
59	531	2925.7483054	0.001157	76.121	usable; same-kind selection applies
59	532	2927.8315850	0.001157	148.491	usable; same-kind selection applies
59	533	2929.9158391	0.001157	75.967	usable; same-kind selection applies
59	534	2931.9991674	0.001157	148.551	usable; same-kind selection applies
59	535	2934.0837257	0.001157	75.886	usable; same-kind selection applies
59	536	2936.1673412	0.001157	147.335	usable; same-kind selection applies
86	874	3640.5897800	0.001157	146.991	usable; same-kind selection applies
86	876	3644.7581530	0.001157	145.901	usable; same-kind selection applies
86	877	3646.8426710	0.001157	75.747	usable; same-kind selection applies
86	878	3648.9258390	0.001157	144.066	usable; same-kind selection applies
86	883	3659.3478677	0.001157	73.359	usable; same-kind selection applies
86	884	3661.4308826	0.001157	145.028	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s59.json, historical/diff_s59.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R222. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R222; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R222; diff_s59.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R223. TIC 60249512

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 60249512: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[7, 61, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=15.3007645492 ±0.0000224480d, reference eclipse epoch BTJD1497.6542271, kind: Selected conjunction, even half-period cycles; primary/secondary depth identity ambiguous. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S88 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.783arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.113. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths273.940/271.050/261.562ppt, with 4 stacked events. QLP selected-kind fitted median309.258ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3551236948821908774 4, P=15.2997416314±0.0023368048d, e=0.023463, omega=147.8744deg. Closest edge-on conjunction offset0.070489d (0.004607phase), full-SB1-covariance linear propagated Gaia uncertainty0.159401d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.003161d (photometric period projection allowance0.001616d). The approximate central duration10.68h vs observed median8.42h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Similar eclipse depths do not establish half orbital period. Consecutive sector88 spacings7.682814/7.617839/7.682757d alternate; same-kind spacings15.300653/15.300596d support full orbital period. Even half-period cycles define selected-conjunction epoch; deeper-primary identity is not claimed. Historical cutout mixes eclipse kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

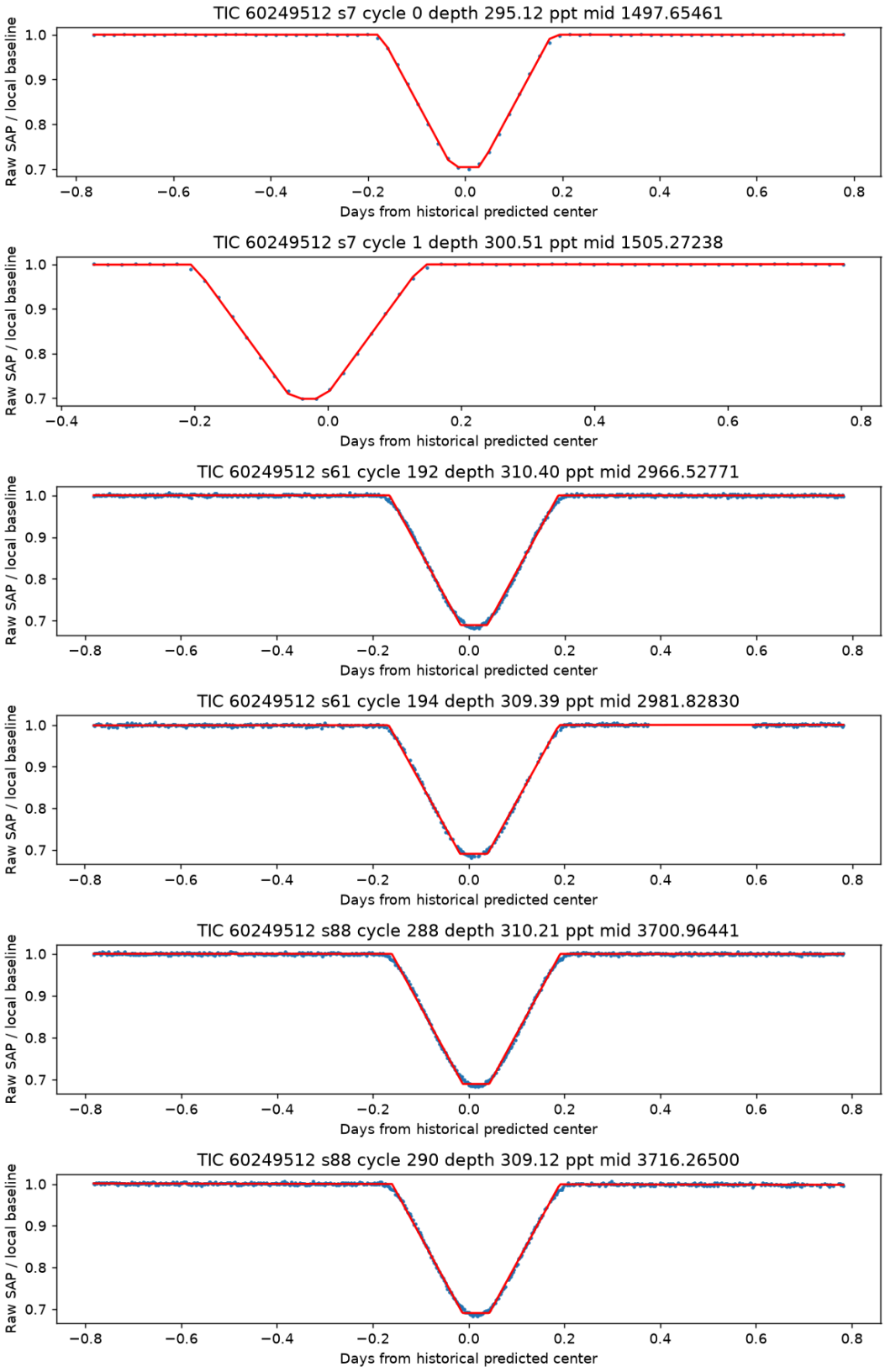
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

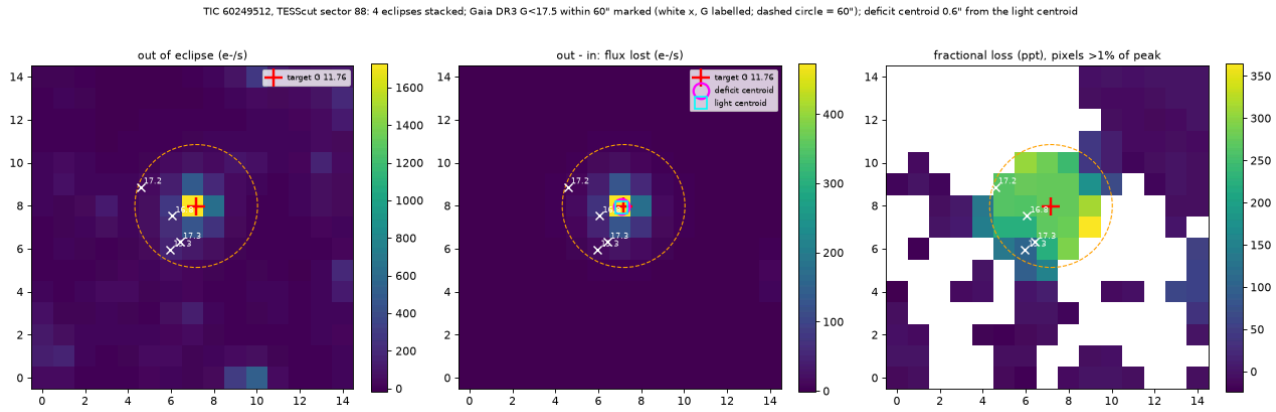
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1497.6546055	0.010417	295.120	usable; same-kind selection applies
7	1	1505.2723818	0.010417	300.513	usable; same-kind selection applies
7	2	1512.9545493	0.010417	294.334	usable; same-kind selection applies
61	192	2966.5277090	0.001157	310.399	usable; same-kind selection applies
61	193	2974.1458765	0.001157	312.867	usable; same-kind selection applies
61	194	2981.8283020	0.001157	309.391	usable; same-kind selection applies
88	287	3693.2815932	0.001157	314.161	usable; same-kind selection applies
88	288	3700.9644077	0.001157	310.212	usable; same-kind selection applies
88	289	3708.5822465	0.001157	314.693	usable; same-kind selection applies
88	290	3716.2650033	0.001157	309.125	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s88.json, historical/diff_s88.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R223. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R223; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R223; diff_s88.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R224. TIC 250562568

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 250562568: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[18, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=4.8539745097 ±0.0000139746d, reference eclipse epoch BTJD1793.7829364, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S18 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.867arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.317. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths61.057/62.870/63.120ppt, with 4 stacked events. QLP selected-kind fitted median60.978ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3359013773655012096, P=4.8534475881±0.0003560668d, e=0.073594, omega=200.0311deg. Closest edge-on conjunction offset0.023951d (0.004935phase), full-SB1-covariance linear propagated Gaia uncertainty0.099201d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.128329d (photometric period projection allowance0.004039d). The approximate central duration7.63h vs observed median6.42h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipse; alternative aperture supports target. Depth varies with aperture/dilution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

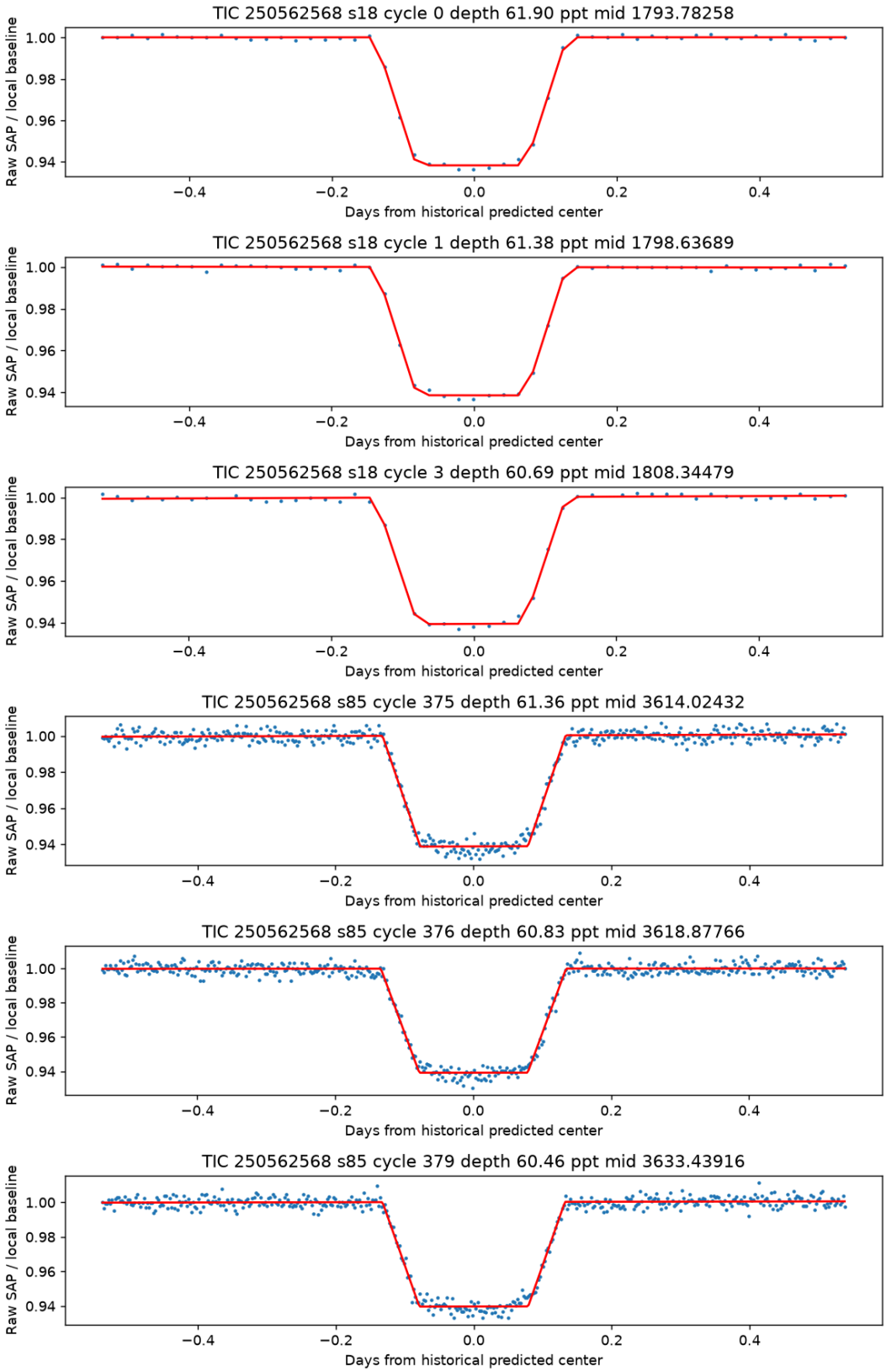
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

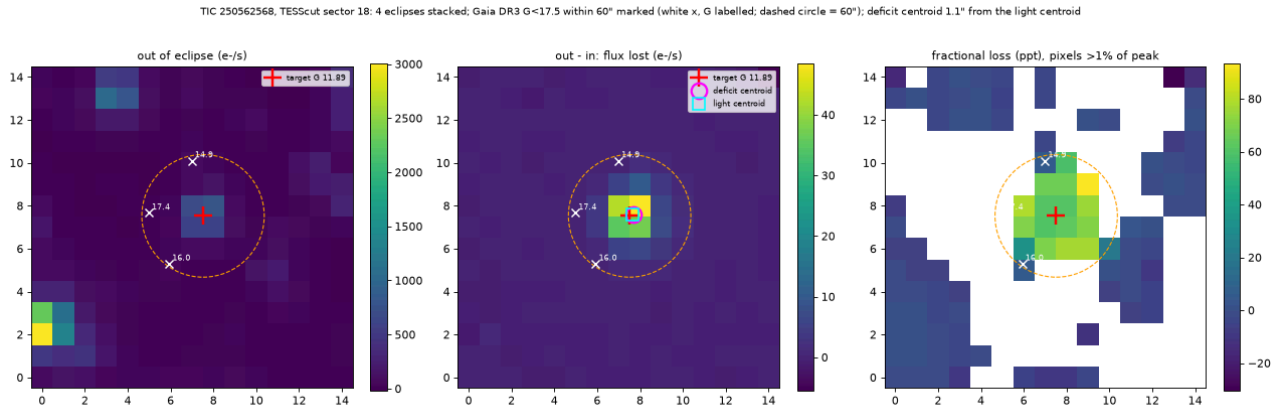
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
18	0	1793.7825787	0.010417	61.898	usable; same-kind selection applies
18	1	1798.6368853	0.010417	61.381	usable; same-kind selection applies
18	3	1808.3447864	0.010417	60.687	usable; same-kind selection applies
18	4	1813.1987722	0.010417	61.127	usable; same-kind selection applies
85	375	3614.0243173	0.001157	61.361	usable; same-kind selection applies
85	376	3618.8776580	0.001157	60.830	usable; same-kind selection applies
85	377	3623.7301999	0.001157	57.765	usable; same-kind selection applies
85	379	3633.4391631	0.001157	60.464	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s18.json, historical/diff_s18.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R224. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R224; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R224; diff_s18.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R225. TIC 325787864

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 325787864: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 21 distinct eclipse events in selected sectors[65, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 21 same-kind midtimes give P=1.9945846934 ±0.0000015903d, reference eclipse epoch BTJD3069.1499357, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S38 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.530arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.496. Capable neighbours: Gaia 5856732789162911104: 12.88 arcsec, G15.940, required fractional eclipse0.217, pixel fractional loss0.005861548844272418; Gaia 5856732789171468032: 14.25 arcsec, G16.231, required fractional eclipse0.281, pixel fractional loss0.004511681199667806; Gaia 5856735739766857984: 16.42 arcsec, G17.423, required fractional eclipse0.833, pixel fractional loss0.004408390932729964; Gaia 5856732784829357952: 18.94 arcsec, G15.771, required fractional eclipse0.186, pixel fractional loss0.005861548844272418; Gaia 5856732823522662912: 23.11 arcsec, G14.857, required fractional eclipse0.083, pixel fractional loss0.001221865944613266; Gaia 5856735744100421120: 28.93 arcsec, G16.951, required fractional eclipse0.541, pixel fractional loss0.004408390932729964; Gaia 5856732823522663808: 29.41 arcsec, G16.575, required fractional eclipse0.385, pixel fractional loss0.001221865944613266; Gaia 5856732754803167744: 29.83 arcsec, G14.363, required fractional eclipse0.055, pixel fractional loss0.004511681199667806; Gaia 5856732823501661056: 30.15 arcsec, G17.082, required fractional eclipse0.610, pixel fractional loss0.004054763655936261; Gaia 5856732823522662400: 33.07 arcsec, G16.742, required

fractional eclipse0.447, pixel fractional loss0.004054763655936261; Gaia 5856735774128949504: 33.91 arcsec, G16.709, required fractional eclipse0.434, pixel fractional loss0.004142405094476539; Gaia 5856735744100433280: 34.80 arcsec, G17.215, required fractional eclipse0.689, pixel fractional loss0.004142405094476539; Gaia 5856732789162900352: 35.87 arcsec, G16.610, required fractional eclipse0.397, pixel fractional loss0.0053843454605854955; Gaia 5856732716109755264: 37.41 arcsec, G16.857, required fractional eclipse0.497, pixel fractional loss0.003333548497386672; Gaia 5856732789162897920: 37.67 arcsec, G15.860, required fractional eclipse0.202, pixel fractional loss0.0053843454605854955; Gaia 5856732789171463552: 42.75 arcsec, G16.502, required fractional eclipse0.360, pixel fractional loss0.0014058986987312812; Gaia 5856735744100439296: 47.77 arcsec, G17.105, required fractional eclipse0.623, pixel fractional loss-0.010356956784428295; Gaia 5856735808486399104: 48.28 arcsec, G17.514, required fractional eclipse0.906, pixel fractional loss-0.008242876704880085; Gaia 5856732789162896384: 49.09 arcsec, G16.826, required fractional eclipse0.483, pixel fractional loss0.0017373637351093675; Gaia 5856732720443413504: 52.52 arcsec, G17.300, required fractional eclipse0.744, pixel fractional loss0.00310619209461405; Gaia 5856732754803157632: 53.84 arcsec, G16.946, required fractional eclipse0.539, pixel fractional loss0.0018103628451148776; Gaia 5856735778460177408: 53.87 arcsec, G16.443, required fractional eclipse0.341, pixel fractional loss-0.0015249522139335764; Gaia 5856735744100416000: 53.87 arcsec, G15.406, required fractional eclipse0.134, pixel fractional loss-0.002453045542999099; Gaia 5856735641021194624: 54.93 arcsec, G17.296, required fractional eclipse0.742, pixel fractional loss-0.002453045542999099; Gaia 5856732789162889344: 56.18 arcsec, G17.205, required fractional eclipse0.682, pixel fractional loss0.0014058986987312812; Gaia 5856732789162889856: 57.44 arcsec, G16.521, required fractional eclipse0.366, pixel fractional loss0.0017373637351093675; Gaia 5856735778460179712: 58.81 arcsec, G16.320, required fractional eclipse0.305, pixel fractional loss-0.0015249522139335764; Gaia 5856732823522666368: 59.26 arcsec, G17.500, required fractional eclipse0.894, pixel fractional loss0.0004835086646597644 6; Gaia 5856735778460182784: 59.40 arcsec, G15.023, required fractional eclipse0.096, pixel fractional loss-0.00106050945664324

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths5.862/4.971/4.297ppt, with 12 stacked events. QLP selected-kind fitted median6.201ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3585673278482718745 6, P=1.9946207355±0.0002572320d, e=0.157853, omega=318.6257deg. Closest edge-on conjunction offset0.027840d (0.013958phase), full-SB1-covariance linear propagated Gaia uncertainty0.370918d at the measured epoch. Near-Gaia-reference projected eclipse offset0.076281d (photometric period projection allowance0.002137d). The approximate central duration4.15h vs observed median1.28h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field:29 capable neighbors include nearest12.9/14.3arcsec. Deficit0.17pixel from target and target-centered lost-light core favor target; no calibrated centroid uncertainty. Shallow morphology repeats in newer sectors; first-sector windows did not satisfy fitting coverage cuts and are not nulls.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

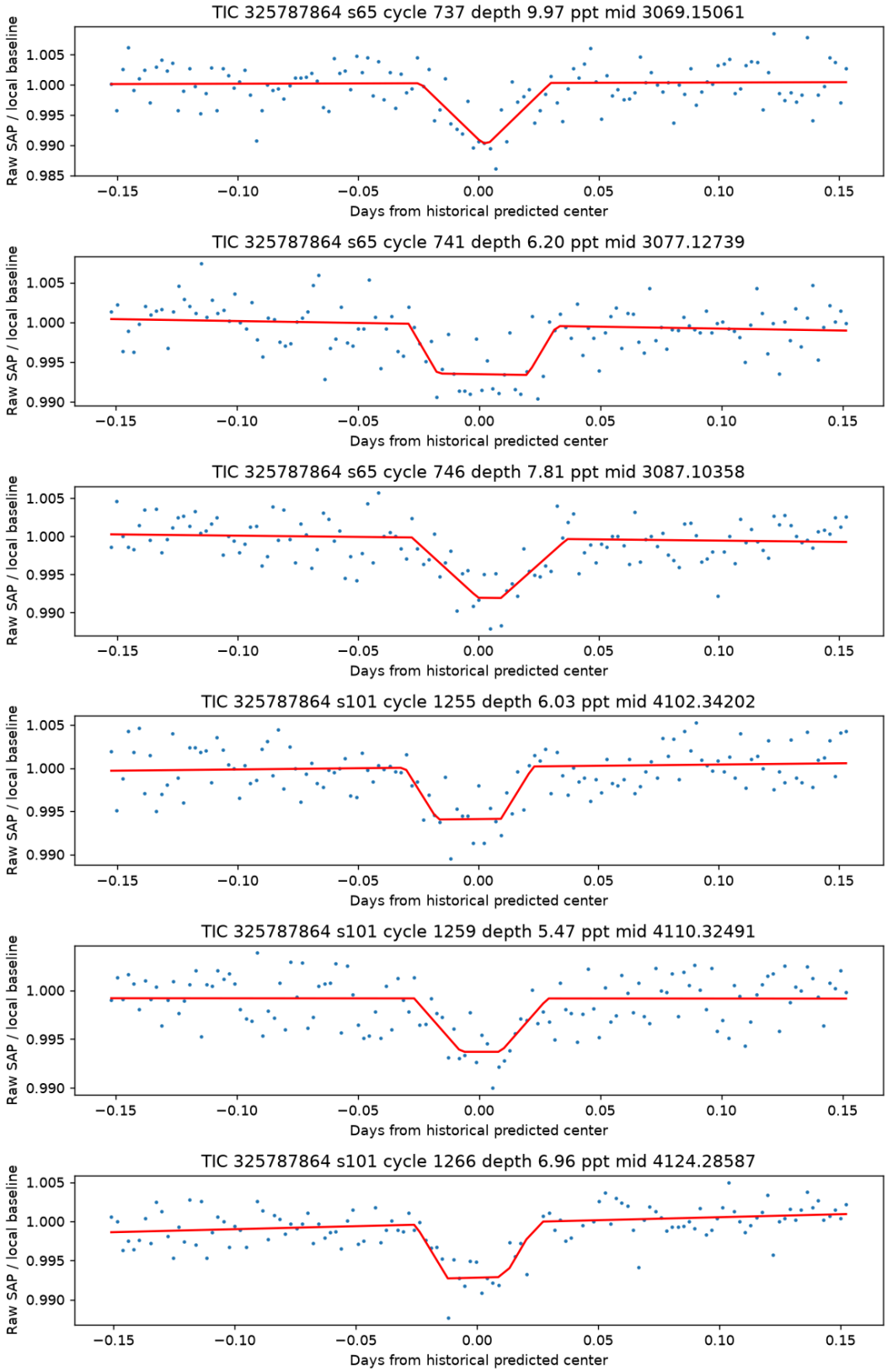
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

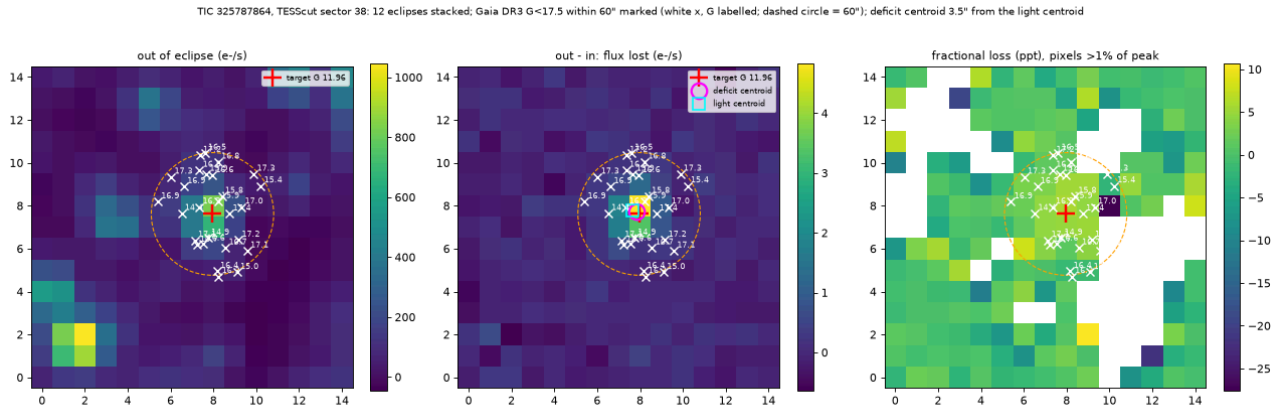
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
65	737	3069.1506115	0.001885	9.975	usable; same-kind selection applies
65	738	3071.1440916	0.001157	8.230	usable; same-kind selection applies
65	739	3073.1369987	0.001157	5.260	usable; same-kind selection applies
65	740	3075.1348121	0.002060	5.761	usable; same-kind selection applies
65	741	3077.1273875	0.001469	6.201	usable; same-kind selection applies
65	742	3079.1243802	0.002101	5.862	usable; same-kind selection applies
65	743	3081.1174595	0.002371	7.819	usable; same-kind selection applies
65	745	3085.1068536	0.001900	6.194	usable; same-kind selection applies
65	746	3087.1035783	0.001730	7.811	usable; same-kind selection applies
65	747	3089.0955579	0.001629	6.672	usable; same-kind selection applies
65	748	3091.0891981	0.002361	6.629	usable; same-kind selection applies
65	749	3093.0881669	0.001795	7.198	usable; same-kind selection applies
101	1255	4102.3420168	0.001816	6.031	usable; same-kind selection applies
101	1256	4104.3368302	0.001157	5.705	usable; same-kind selection applies
101	1257	4106.3367248	0.001795	4.626	usable; same-kind selection applies
101	1258	4108.3279610	0.001459	5.333	usable; same-kind selection applies
101	1259	4110.3249096	0.002249	5.473	usable; same-kind selection applies
101	1262	4116.3054534	0.001239	8.903	usable; same-kind selection applies
101	1263	4118.3043646	0.001662	6.641	usable; same-kind selection applies
101	1265	4122.2927310	0.001266	5.670	usable; same-kind selection applies
101	1266	4124.2858746	0.001299	6.961	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R225. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R225; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R225; diff_s38.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R226. TIC 180871891

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 180871891: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[29, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=13.0367888505 ±0.0000275594d, reference eclipse epoch BTJD2097.6610275, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S29 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation0.910arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.310. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths8.024/7.114/6.961ppt, with 2 stacked events. QLP selected-kind fitted median6.596ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3498809657429171148 8, P=13.0390915897±0.0016464671d, e=0.106260, omega=327.0889deg. Closest edge-on conjunction offset-0.327420d (-0.025111phase), full-SB1-covariance linear propagated Gaia uncertainty0.217962d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.025761d (photometric period projection allowance0.003610d). The approximate central duration6.79h vs observed median3.08h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four measured shallow eclipses across sectors29/104 establish repeatability and period. Target-centered deficit0.043pixel and no capable catalogued neighbor support attribution. Sector2 was downloaded but no event window passed coverage cuts; it is not a clean non-detection. No single-event availability gap remains.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

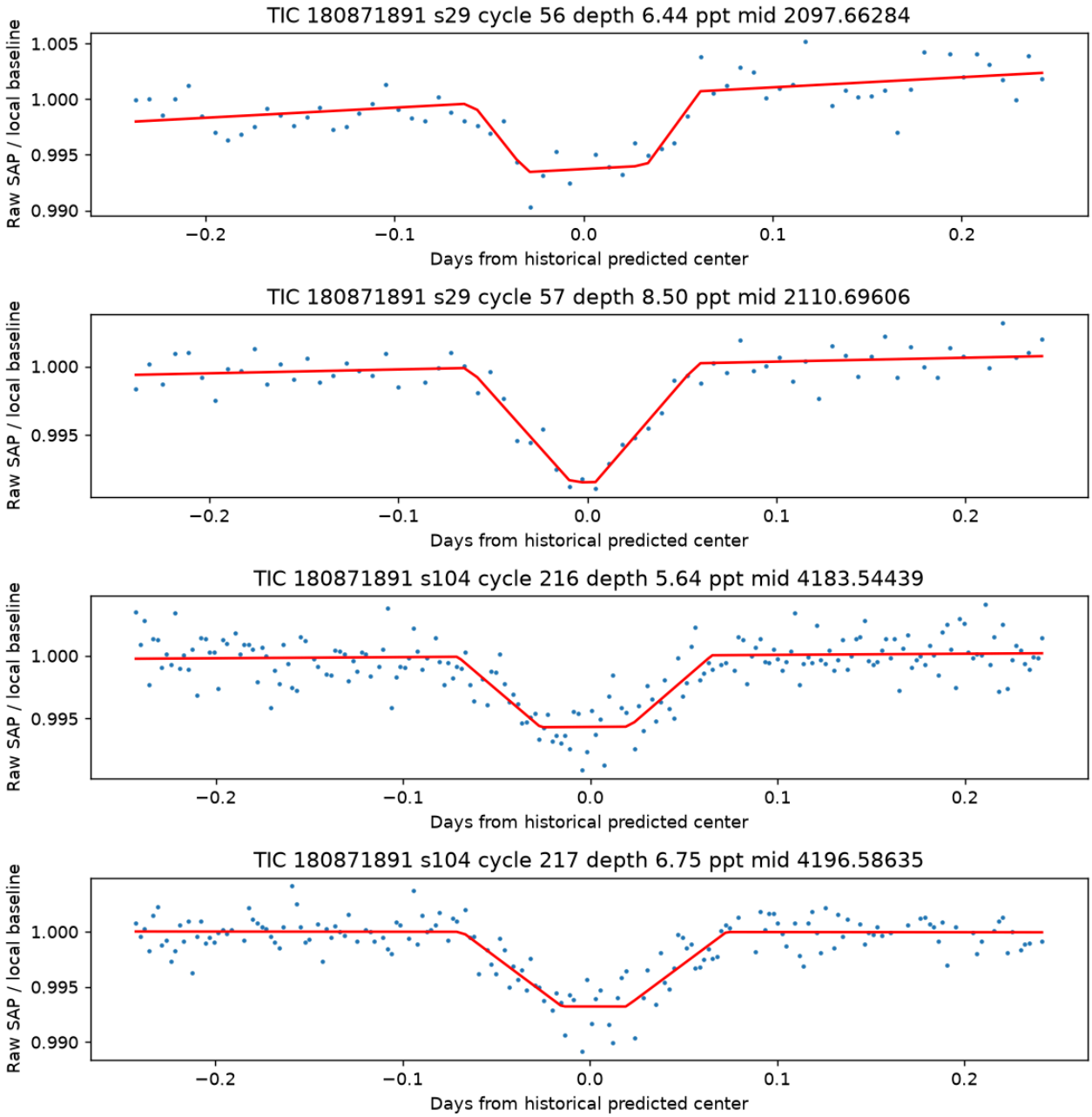
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

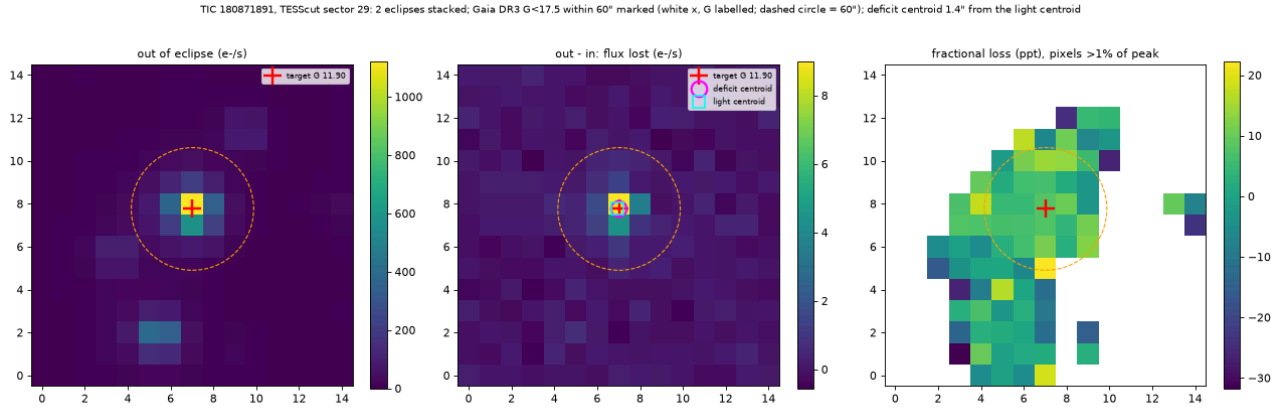
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
29	56	2097.6628371	0.003472	6.437	usable; same-kind selection applies
29	57	2110.6960642	0.003472	8.496	usable; same-kind selection applies
104	216	4183.5443929	0.001767	5.637	usable; same-kind selection applies
104	217	4196.5863524	0.001598	6.754	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s29.json, historical/diff_s29.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R226. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R226; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R226; diff_s29.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R227. TIC 129565419

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 129565419: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[57, 84]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=8.9837069710 ±0.0000373305d, fitted ephemeris intercept at earliest retained cycle BTJD2860.2271612, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S56 deficit peak[14, 12] vs target pixel[7, 8]; deficit-target separation0.846arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.968. Capable neighbours: Gaia 1931241188635723264: 38.91 arcsec, G14.422, required fractional eclipse0.353, pixel fractional loss0.012421269831853259
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths25.891/25.695/23.795ppt, with 3 stacked events. QLP selected-kind fitted median30.212ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3193124118863572569 6, P=8.9845539995±0.0011354221d, e=0.046905, omega=323.0527deg. Closest edge-on conjunction offset-0.297531d (-0.033116phase), full-SB1-covariance linear propagated Gaia uncertainty0.325225d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.064598d (photometric period projection allowance0.010266d). The approximate central

duration6.22h vs observed median3.23h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Remote bright region dominates historical unrestricted peak and PSF residual0.968. Independent helper local5x5 re-stack residual0.115 and target centroid offset(-0.02,+0.01)pixel recover the target eclipse; three aperture depths24.40/23.92/22.14ppt support it. Capable neighbor38.91arcsec away lies1.90pixel from original deficit vs0.040pixel target.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

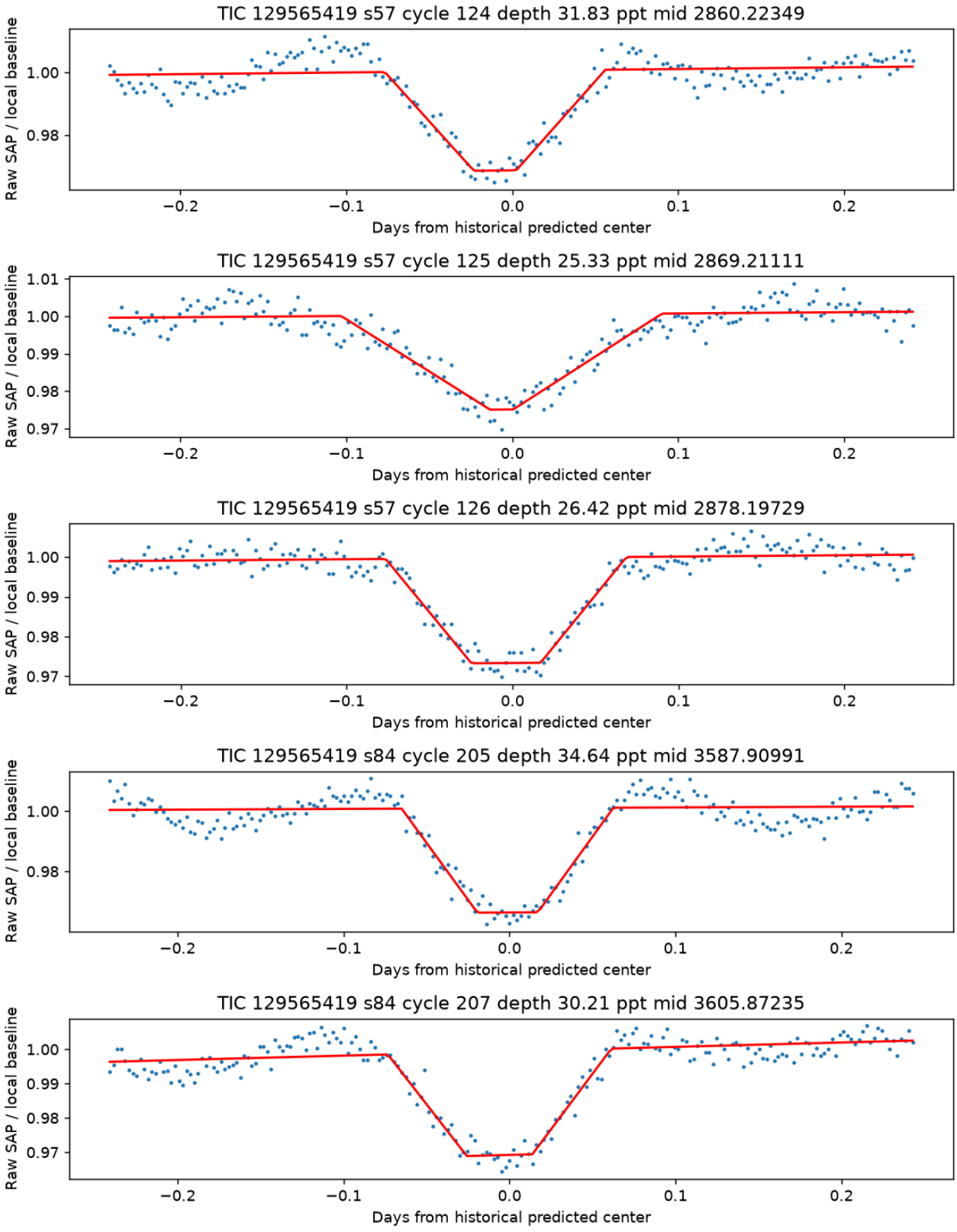
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

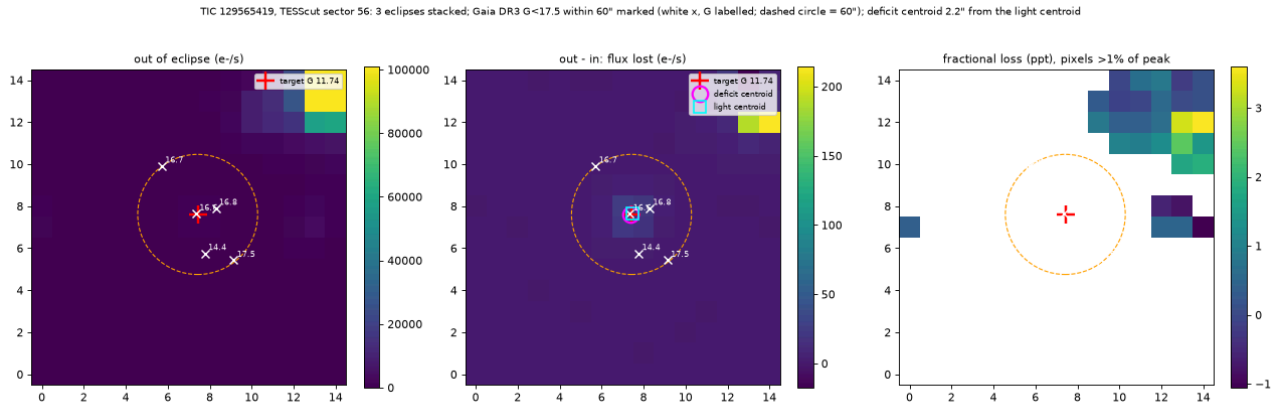
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
57	124	2860.2234867	0.001309	31.829	usable; same-kind selection applies
57	125	2869.2111081	0.001387	25.332	usable; same-kind selection applies
57	126	2878.1972885	0.001157	26.420	usable; same-kind selection applies
84	205	3587.9099069	0.001157	34.643	usable; same-kind selection applies
84	207	3605.8723510	0.001157	30.212	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s56.json, historical/diff_s56.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R227. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R227; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R227; diff_s56.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R228. TIC 125690413

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 125690413: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 18 distinct eclipse events in selected sectors[15, 41, 55]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 18 same-kind midtimes give P=3.6868965634 ±0.0000323523d, fitted ephemeris intercept at earliest retained cycle BTJD1712.7815721, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S41 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation7.799arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.454. Capable neighbours: Gaia 1841055774590654720: 24.35 arcsec, G15.751, required fractional eclipse0.759, pixel fractional loss0.016480613631840144; Gaia 1841055770295761664: 42.81 arcsec, G15.802, required fractional eclipse0.795, pixel fractional loss0.004229247182978673; Gaia 1841055740230914688: 50.54 arcsec, G15.518, required fractional eclipse0.617, pixel fractional loss0.0027038765085001747
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths22.331/22.134/21.692ppt, with 7 stacked events. QLP selected-kind fitted median21.909ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3184105649614516 3776, P=3.6869823318±0.0002444206d, e=0.064161, omega=270.3731deg. Closest edge-on conjunction

offset-0.058793d (-0.015946phase), full-SB1-covariance linear propagated Gaia uncertainty0.075668d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.028002d (photometric period projection allowance0.011614d). The approximate central duration8.23h vs observed median4.44h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Structured sector41 baseline gives broad34ppt cycle196, residual -0.038d and reduced chi square5.25. Retained event and inflated covariance document this; clean22ppt repeats occur in sectors15/55. Capable neighbors are24.35arcsec or farther,1.55pixel or farther from deficit vs0.371pixel target. Timing errors are not total uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

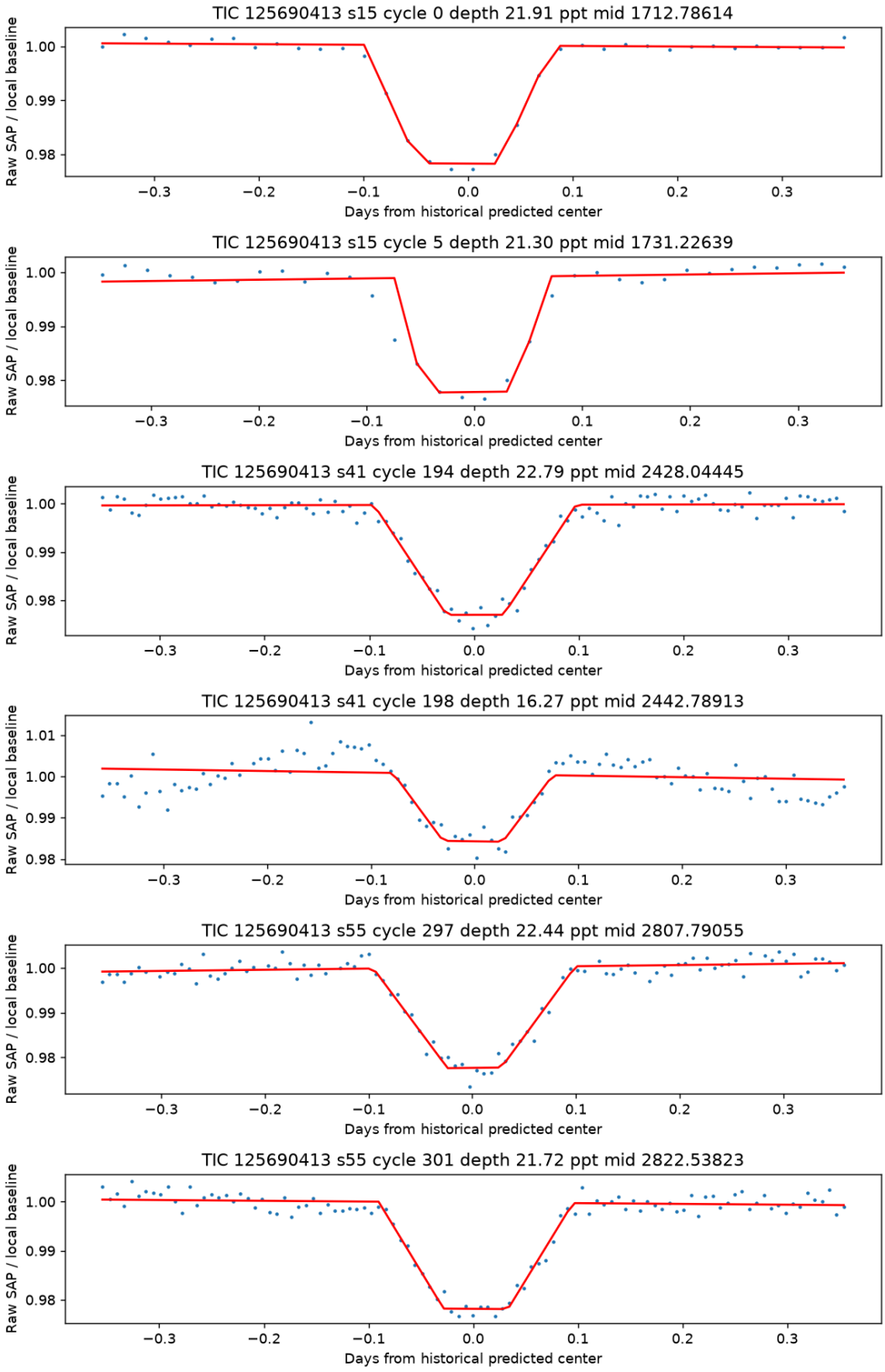
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1712.7861357	0.010417	21.906	usable; same-kind selection applies
15	1	1716.4713407	0.010417	21.202	usable; same-kind selection applies
15	4	1727.5374256	0.010417	21.256	usable; same-kind selection applies
15	5	1731.2263891	0.010417	21.295	usable; same-kind selection applies
41	192	2420.6701880	0.003472	22.428	usable; same-kind selection applies
41	193	2424.3549531	0.003472	23.614	usable; same-kind selection applies
41	194	2428.0444513	0.003472	22.786	usable; same-kind selection applies
41	195	2431.7296216	0.003472	22.501	usable; same-kind selection applies
41	196	2435.3752955	0.004378	33.997	usable; same-kind selection applies
41	197	2439.0933329	0.007020	20.348	usable; same-kind selection applies
41	198	2442.7891347	0.003607	16.274	usable; same-kind selection applies
55	295	2800.4166636	0.003472	21.750	usable; same-kind selection applies
55	296	2804.1036165	0.003472	22.065	usable; same-kind selection applies
55	297	2807.7905523	0.003472	22.439	usable; same-kind selection applies
55	298	2811.4786764	0.003472	22.534	usable; same-kind selection applies
55	299	2815.1641441	0.003472	21.911	usable; same-kind selection applies
55	300	2818.8511683	0.003472	21.302	usable; same-kind selection applies

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
55	301	2822.5382277	0.003472	21.720	usable; same-kind selection applies

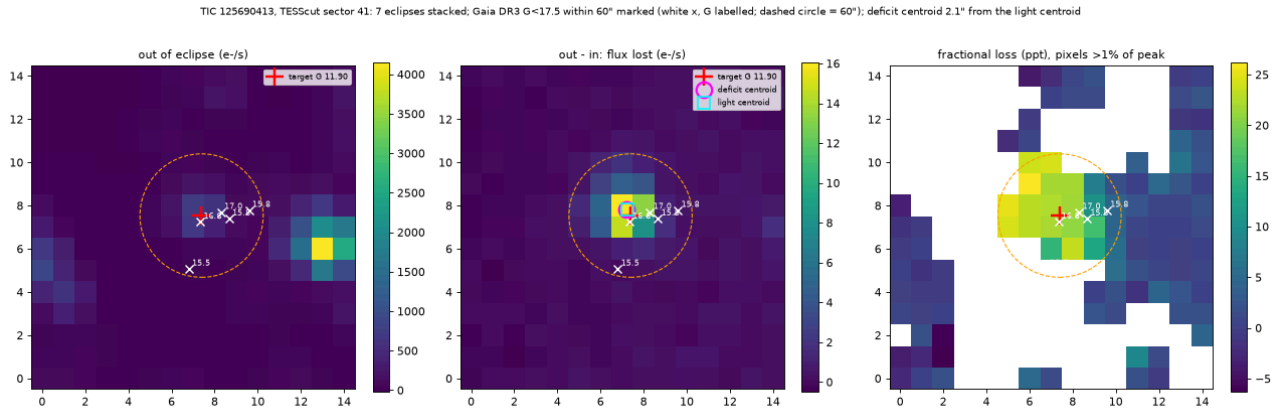
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s41.json, historical/diff_s41.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Timing sensitivity: omitting the complete structured-baseline sector41 gives P=3.6868774377158258d, only -0.00001913d from the adopted all-event ephemeris. See timing-sensitivity.json; this is a robustness diagnostic, not the adopted precision estimate.

Stable review key: R228. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R228; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R228; diff_s41.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R229. TIC 445045766

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 445045766: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[60, 74]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=5.0364803869 ±0.0000125874d, fitted ephemeris intercept at earliest retained cycle BTJD2939.7678090, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S47 deficit peak[8, 8] vs target pixel[7, 8]; deficit-target separation3.908arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.781. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths23.308/23.874/22.969ppt, with 5 stacked events. QLP selected-kind fitted median25.792ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3104084811688657612 8, P=5.0364132762±0.0002075065d, e=0.017166, omega=36.3338deg. Closest edge-on conjunction offset0.083686d (0.016616phase), full-SB1-covariance linear propagated Gaia uncertainty0.104481d at the measured epoch. Near-Gaia-reference projected eclipse offset0.049728d (photometric period projection allowance0.006369d). The approximate central duration4.07h vs observed median1.94h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: No capable catalogued neighbor within60arcsec. Deficit0.186pixel from target; three alternative aperture depths23.3-23.9ppt agree.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

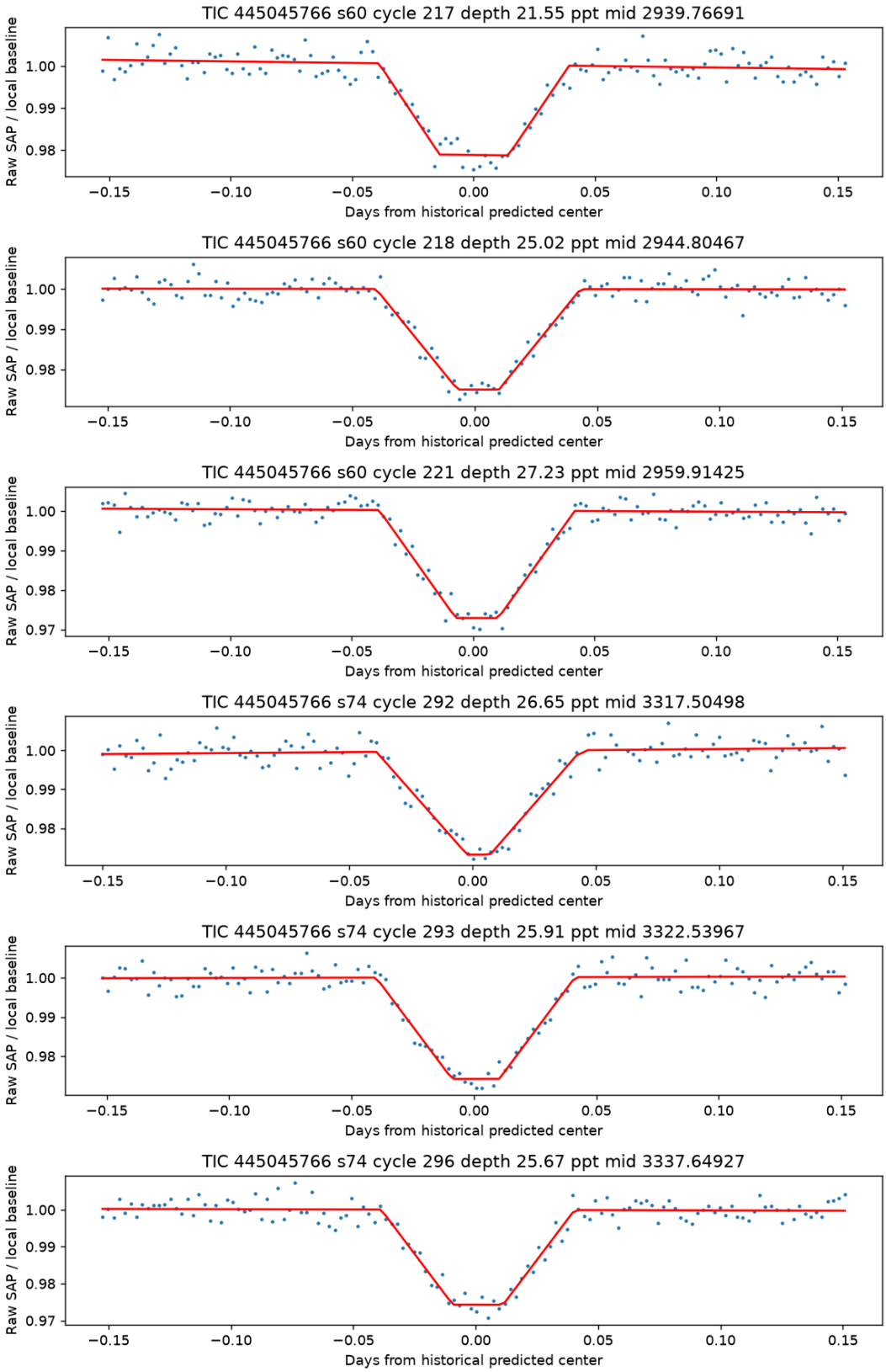
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

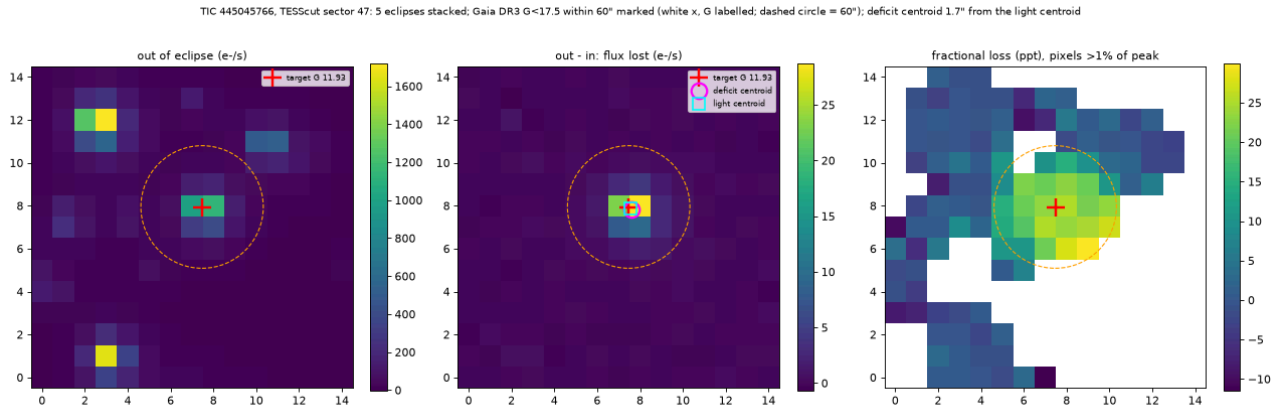
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
60	217	2939.7669090	0.001157	21.547	usable; same-kind selection applies
60	218	2944.8046715	0.001157	25.020	usable; same-kind selection applies
60	221	2959.9142466	0.001157	27.232	usable; same-kind selection applies
74	292	3317.5049774	0.001157	26.651	usable; same-kind selection applies
74	293	3322.5396657	0.001157	25.911	usable; same-kind selection applies
74	296	3337.6492749	0.001157	25.674	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s47.json, historical/diff_s47.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R229. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R229; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R229; diff_s47.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R230. TIC 120037640

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 120037640: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[93, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=3.6982624070 ±0.0000073790d, fitted ephemeris intercept at earliest retained cycle BTJD3833.6806832, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S66 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.736arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.697. Capable neighbours: Gaia 6651016827935020032: 34.86 arcsec, G14.449, required fractional eclipse0.290, pixel fractional loss0.016812257550119572; Gaia 6651016961075479552: 52.29 arcsec, G13.179, required fractional eclipse0.106, pixel fractional loss0.004252985656066122
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths22.768/22.273/20.123ppt, with 6 stacked events. QLP selected-kind fitted median24.000ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR36651016862294758784, P=3.6985144102±0.0001310459d, e=0.061638, omega=94.3358deg. Closest edge-on conjunction offset-0.189153d (-0.051143phase), full-SB1-covariance linear propagated Gaia uncertainty0.110953d at the measured epoch. Near-Gaia-reference projected eclipse offset0.045461d (photometric period

projection allowance0.006870d). The approximate central duration6.02h vs observed median3.18h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Actual-epoch Gaia offset -0.189d (phase -0.051) exceeds historical phase screen; near Gaia reference projected offset is +0.045d. Gaia period uncertainty accumulates over baseline. Low eccentricity0.062 and recurrence agreement support association without arbitrary sigma veto. Capable neighbors are34.9arcsec or farther and1.72pixel or farther from deficit vs0.178pixel target.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

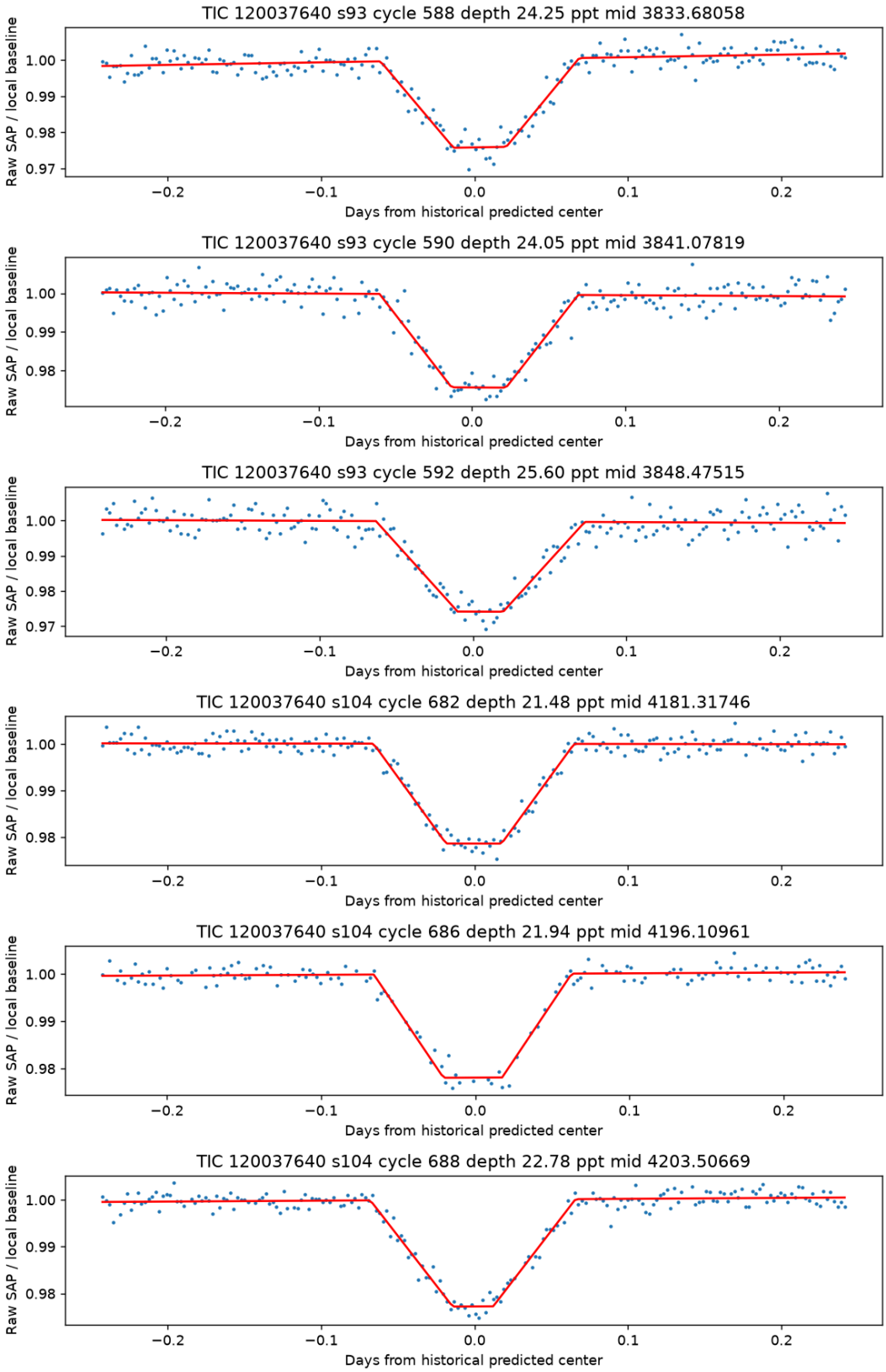
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

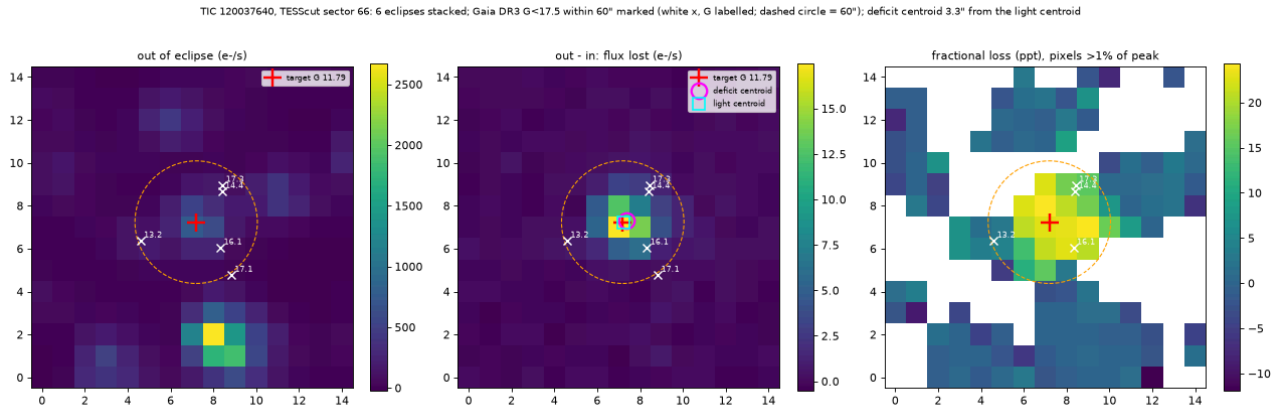
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
93	588	3833.6805823	0.001157	24.254	usable; same-kind selection applies
93	589	3837.3767157	0.001157	27.400	usable; same-kind selection applies
93	590	3841.0781928	0.001157	24.054	usable; same-kind selection applies
93	591	3844.7766053	0.001157	24.000	usable; same-kind selection applies
93	592	3848.4751515	0.001157	25.596	usable; same-kind selection applies
93	593	3852.1707996	0.001157	24.088	usable; same-kind selection applies
104	682	4181.3174585	0.001157	21.477	usable; same-kind selection applies
104	684	4188.7144386	0.001157	21.612	usable; same-kind selection applies
104	686	4196.1096105	0.001157	21.936	usable; same-kind selection applies
104	687	4199.8089998	0.001157	21.263	usable; same-kind selection applies
104	688	4203.5066893	0.001157	22.784	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s66.json, historical/diff_s66.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R230. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R230; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R230; diff_s66.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R231. TIC 448485777

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 448485777: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[11, 99, 102]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=5.8952882395 ±0.0000227545d, fitted ephemeris intercept at earliest retained cycle BTJD1603.4712444, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S11 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation5.303arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.300. Capable neighbours: Gaia 5847716106616806784: 14.44 arcsec, G15.332, required fractional eclipse0.976, pixel fractional loss0.0215936791346104; Gaia 5847716102285049344: 16.26 arcsec, G13.144, required fractional eclipse0.161, pixel fractional loss0.01979753444323785; Gaia 5847715728659680000: 20.52 arcsec, G14.358, required fractional eclipse0.419, pixel fractional loss0.03241426402958959; Gaia 5847715728659676800: 21.71 arcsec, G15.306, required fractional eclipse0.954, pixel fractional loss0.03276142993066764; Gaia 5847716102285056384: 25.76 arcsec, G15.235, required fractional eclipse0.896, pixel fractional loss0.0215936791346104; Gaia 5847716106616813440: 57.68 arcsec, G14.848, required fractional eclipse0.638, pixel fractional loss0.005557348234906238
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths33.268/29.822/29.128ppt, with 4 stacked events. QLP selected-kind fitted median37.981ppt. This separately extracted cutout is an alternative reduction of the same TESS

photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3584771572432743616O, P=5.8948949185±0.0003771119d, e=0.086403, omega=238.3671deg. Closest edge-on conjunction offset0.143279d (0.024306phase), full-SB1-covariance linear propagated Gaia uncertainty0.087194d at the measured epoch. Near-Gaia-reference projected eclipse offset0.062255d (photometric period projection allowance0.004687d). The approximate central duration11.63h vs observed median6.69h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field: G13.14 neighbor16.26arcsec away requires16.1% eclipse, deficit0.615pixel from neighbor vs0.252pixel target. Another14.44arcsec neighbor requires97.6%. Target is favored by central deficit and deeper1x1 aperture, but blending and uncalibrated localization leave attribution physically unresolved after completed checks.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

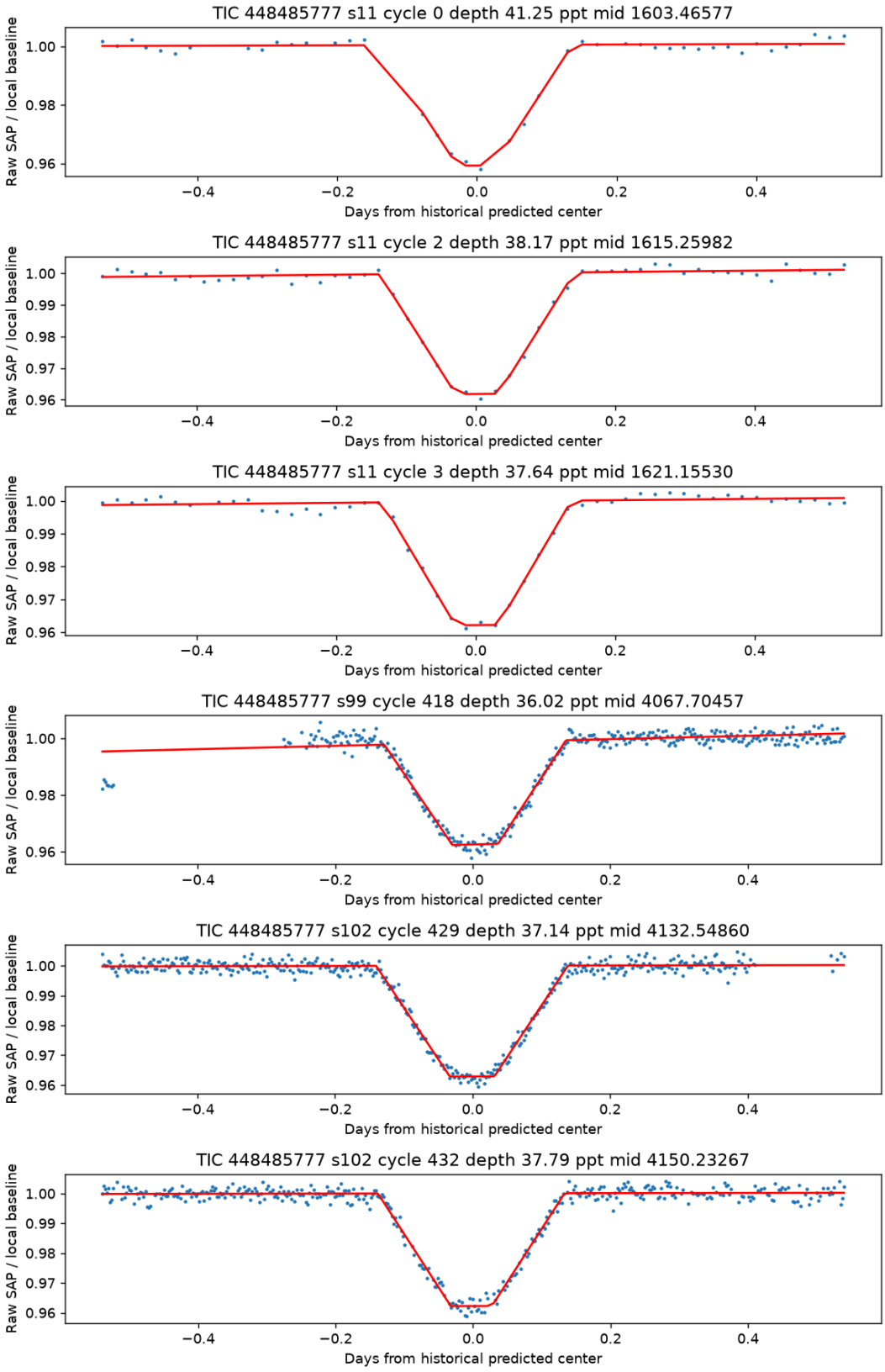
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

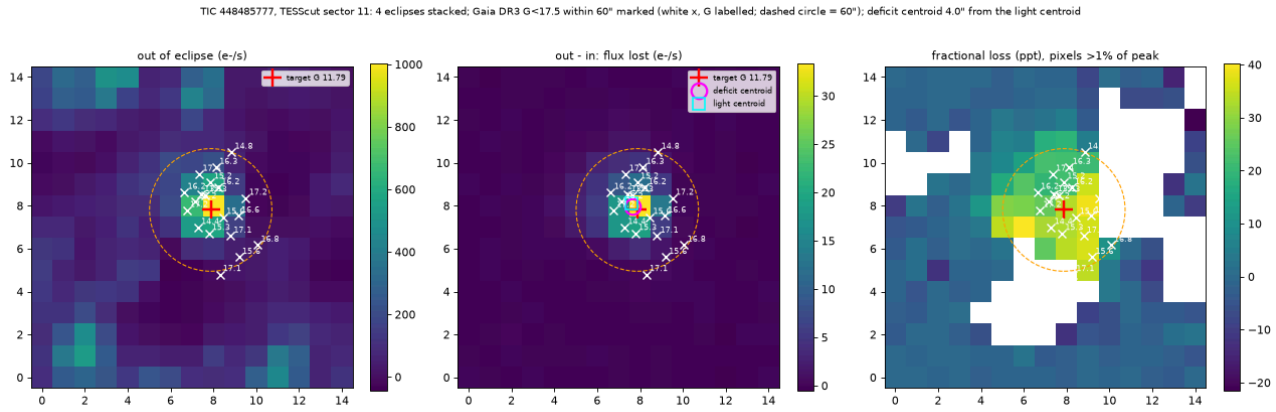
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
11	0	1603.4657715	0.010417	41.254	usable; same-kind selection applies
11	2	1615.2598235	0.010417	38.170	usable; same-kind selection applies
11	3	1621.1553002	0.010417	37.643	usable; same-kind selection applies
99	415	4050.0166038	0.001157	39.084	usable; same-kind selection applies
99	418	4067.7045691	0.001157	36.015	usable; same-kind selection applies
102	429	4132.5485969	0.001157	37.135	usable; same-kind selection applies
102	431	4144.3414105	0.001157	38.856	usable; same-kind selection applies
102	432	4150.2326655	0.001157	37.792	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s11.json, historical/diff_s11.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R231. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R231; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R231; diff_s11.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R232. TIC 158647370

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 158647370: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[69, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=6.0132323218 ±0.0000060378d, fitted ephemeris intercept at earliest retained cycle BTJD3189.6585657, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S29 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.428arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.024. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths87.134/86.012/85.155ppt, with 5 stacked events. QLP selected-kind fitted median85.396ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3491632155903871948 8, P=6.0135821955±0.0002440543d, e=0.037327, omega=306.8532deg. Closest edge-on conjunction offset-0.114751d (-0.019082phase), full-SB1-covariance linear propagated Gaia uncertainty0.124315d at the measured epoch. Near-Gaia-reference projected eclipse offset0.048291d (photometric period projection allowance0.002814d). The approximate central duration5.66h vs observed median3.38h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: No capable catalogued neighbor. Approximately85-95ppt eclipses repeat; custom aperture depths85-87ppt agree.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

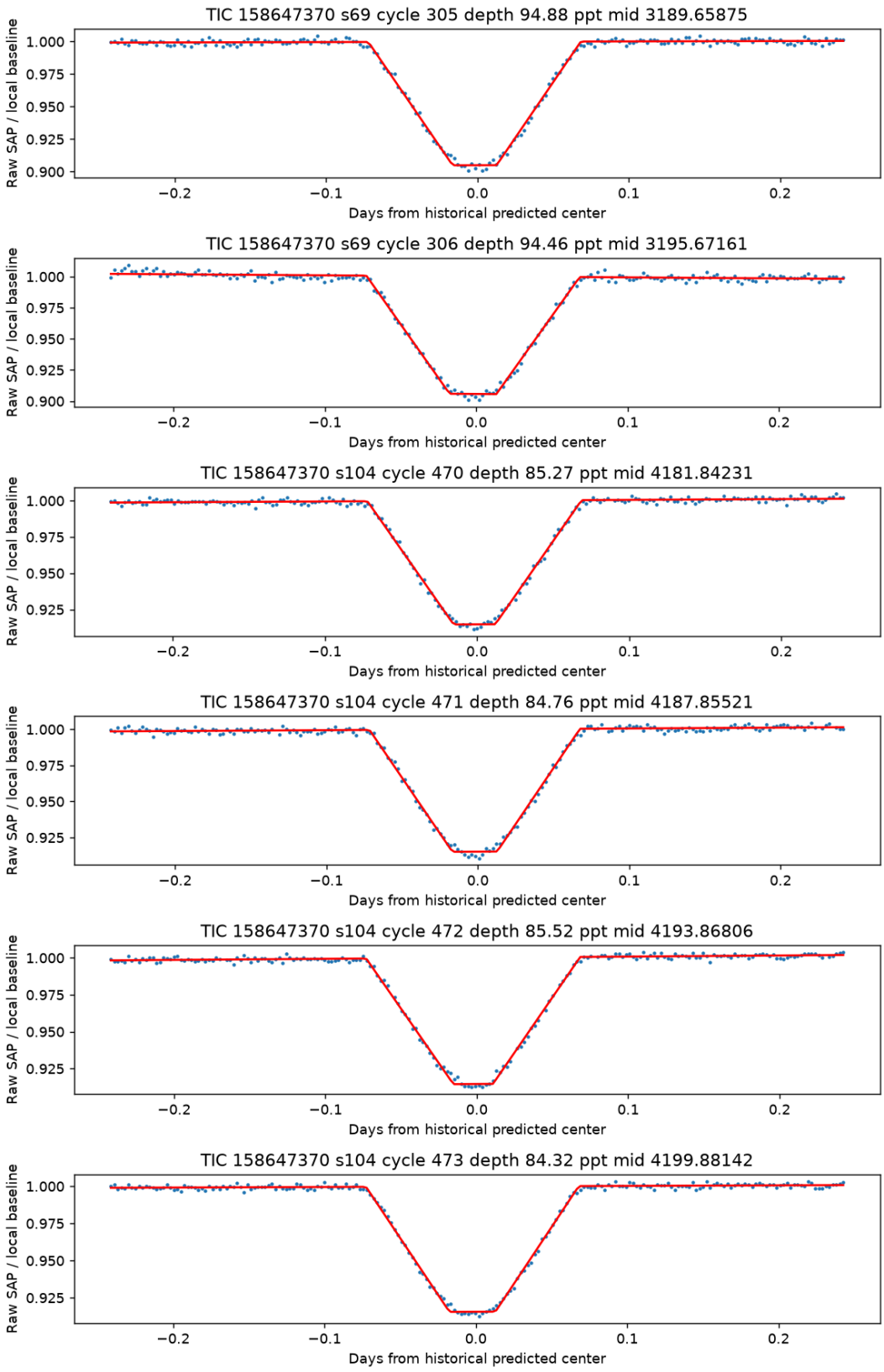
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

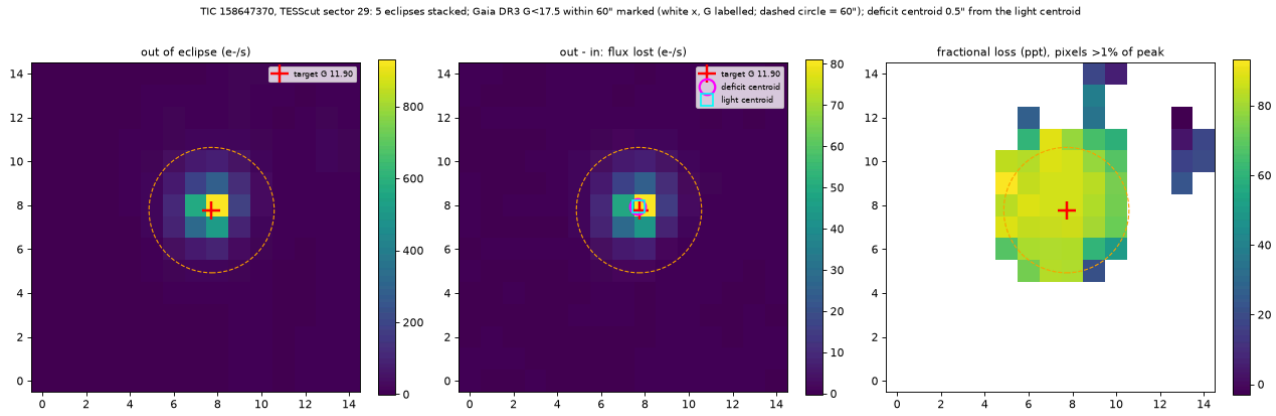
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
69	305	3189.6587472	0.001157	94.880	usable; same-kind selection applies
69	306	3195.6716091	0.001157	94.455	usable; same-kind selection applies
104	470	4181.8423089	0.001157	85.271	usable; same-kind selection applies
104	471	4187.8552064	0.001157	84.760	usable; same-kind selection applies
104	472	4193.8680622	0.001157	85.521	usable; same-kind selection applies
104	473	4199.8814193	0.001157	84.317	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s29.json, historical/diff_s29.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R232. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R232; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R232; diff_s29.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R233. TIC 302190527

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 302190527: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[42, 57, 84]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=9.4703064823 ±0.0000113926d, fitted ephemeris intercept at earliest retained cycle BTJD2450.9782406, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S57 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.058arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.114. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths263.443/258.276/255.104ppt, with 3 stacked events. QLP selected-kind fitted median281.535ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32779583640038802560, P=9.4713396104±0.0013422095d, e=0.105048, omega=22.4456deg. Closest edge-on conjunction offset-0.163176d (-0.017228phase), full-SB1-covariance linear propagated Gaia uncertainty0.259643d at the measured epoch. Near-Gaia-reference projected eclipse offset0.062045d (photometric period projection allowance0.002484d). The approximate central duration6.77h vs observed median5.09h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: No capable catalogued neighbor. Approximately275-283ppt raw eclipses repeat across widely separated sectors; alternative aperture depths255-263ppt are diluted extraction values.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

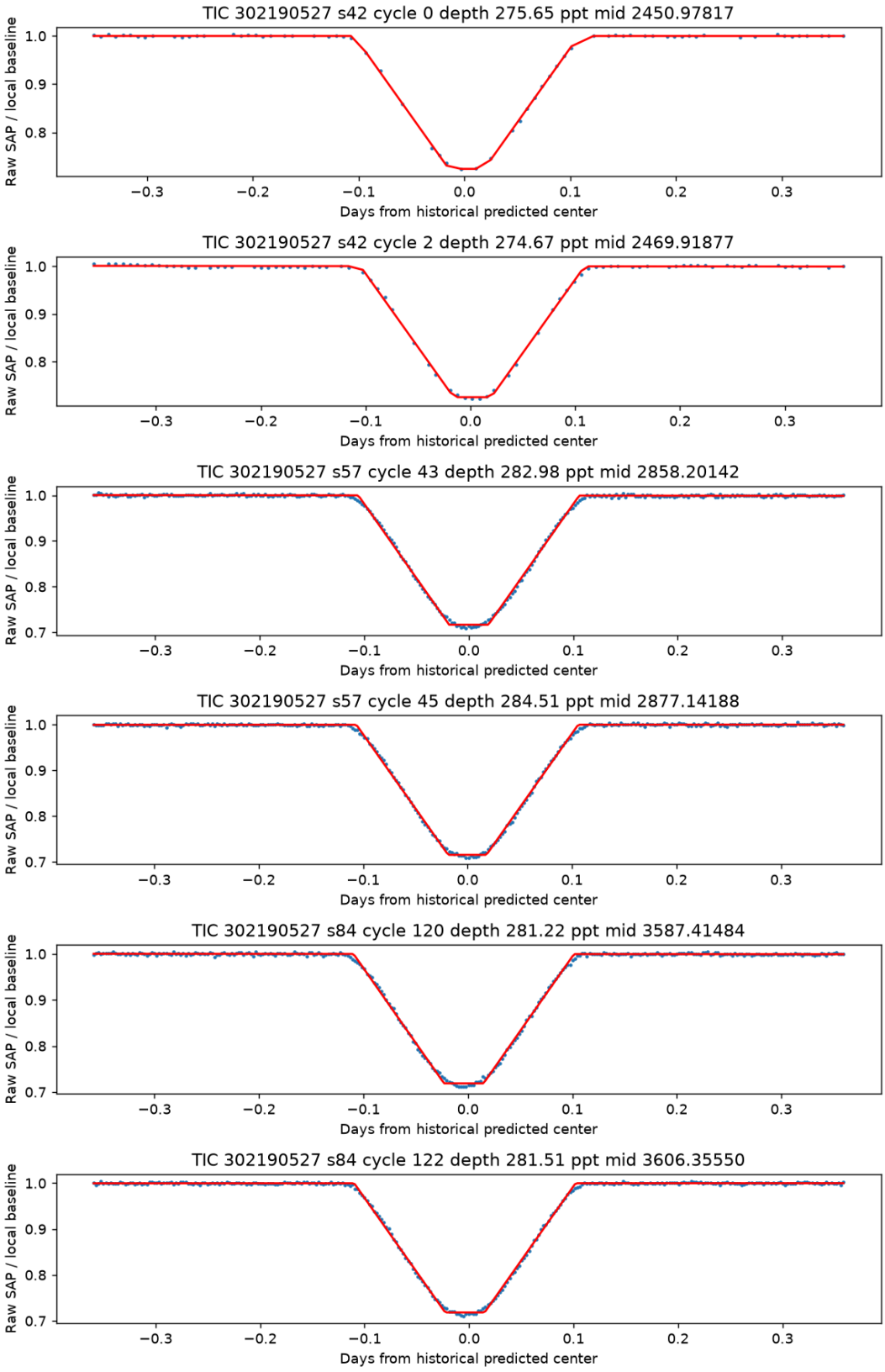
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

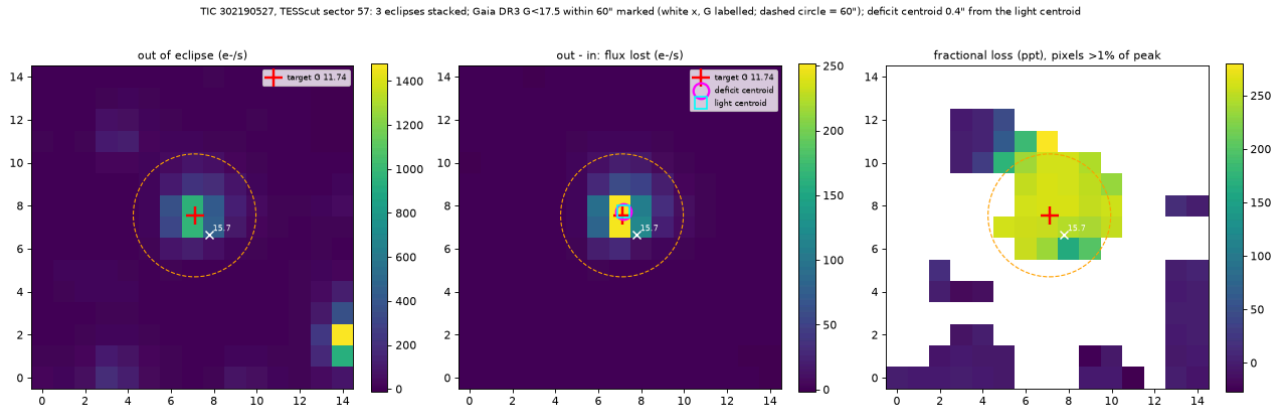
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
42	0	2450.9781740	0.003472	275.645	usable; same-kind selection applies
42	2	2469.9187712	0.003472	274.672	usable; same-kind selection applies
57	43	2858.2014181	0.001157	282.981	usable; same-kind selection applies
57	44	2867.6719084	0.001157	284.177	usable; same-kind selection applies
57	45	2877.1418754	0.001157	284.507	usable; same-kind selection applies
84	120	3587.4148446	0.001157	281.216	usable; same-kind selection applies
84	121	3596.8856220	0.001157	281.562	usable; same-kind selection applies
84	122	3606.3555006	0.001157	281.509	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s57.json, historical/diff_s57.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R233. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R233; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R233; diff_s57.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R234. TIC 138941853

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 138941853: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[6, 32, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.5985912130 ±0.0000160827d, fitted ephemeris intercept at earliest retained cycle BTJD1475.6404802, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S98 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.841arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.425. Capable neighbours: Gaia 3223511670208848896: 21.77 arcsec, G16.492, required fractional eclipse0.500, pixel fractional loss0.00821316823921336; Gaia 3223511734632170368: 33.94 arcsec, G16.100, required fractional eclipse0.351, pixel fractional loss0.005558516488320414; Gaia 3223511704568589952: 43.67 arcsec, G15.493, required fractional eclipse0.204, pixel fractional loss0.007256264006483505; Gaia 3223511429690683776: 44.80 arcsec, G13.282, required fractional eclipse0.033, pixel fractional loss0.0017700490677724006
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths9.200/8.300/6.809ppt, with 3 stacked events. QLP selected-kind fitted median9.255ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR33223511665912271232, P=11.6019567518±0.0050960393d, e=0.024700, omega=125.6521deg. Closest edge-on conjunction offset0.035580d (0.003067phase), full-SB1-covariance linear propagated Gaia uncertainty0.523683d at the measured epoch. Near-Gaia-reference projected eclipse offset0.351941d (photometric period projection allowance0.001512d). The approximate central duration7.90h vs observed median3.75h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Approximately8-10ppt eclipses repeat in sectors6/32/98. Deficit0.087pixel from target vs nearest capable source1.127pixel. Near-reference Gaia offset0.352d is descriptive; low-eccentricity orbit conjunction is poorly constrained. Actual epoch offset0.036d with0.524d propagated Gaia uncertainty is compatible.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

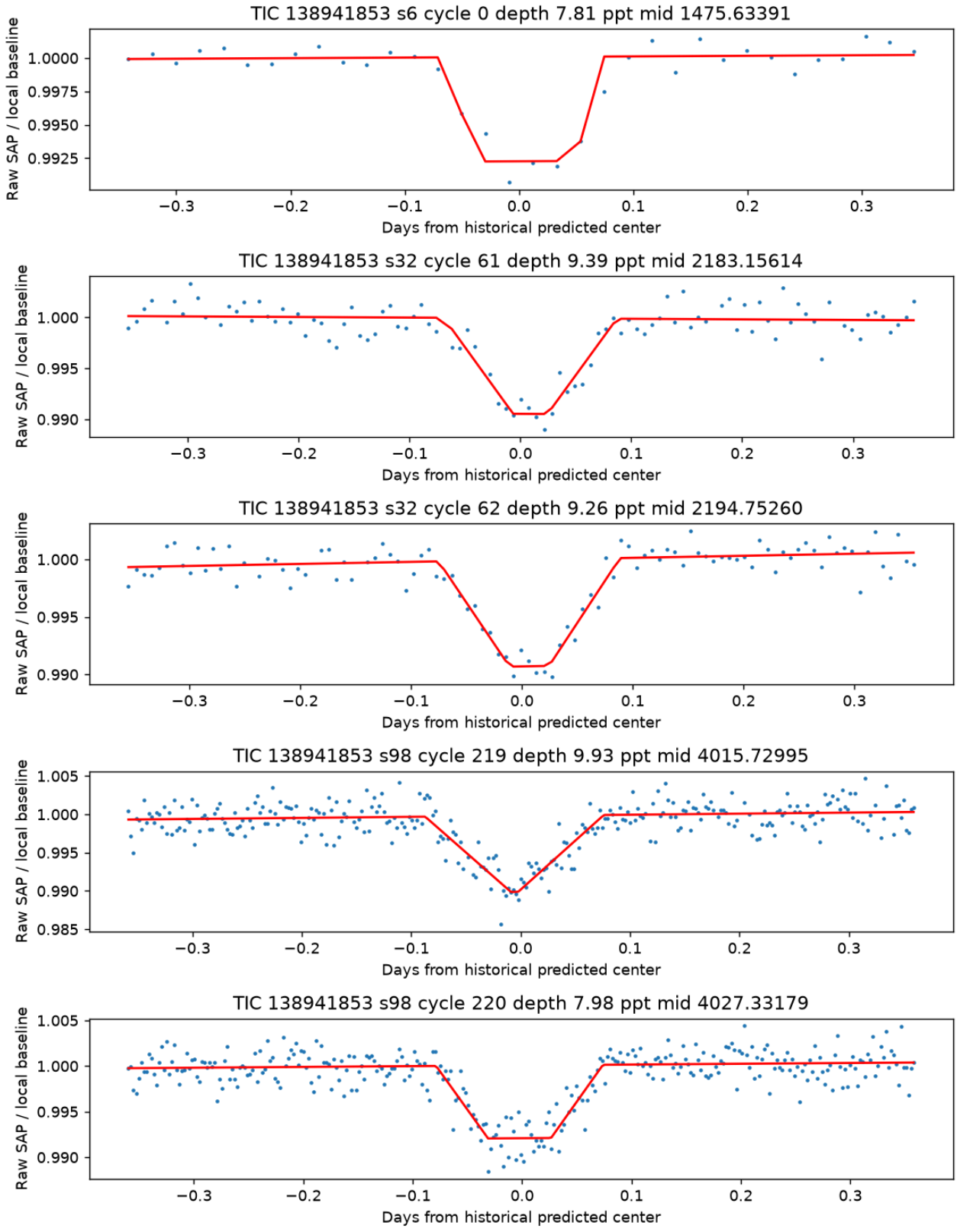
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

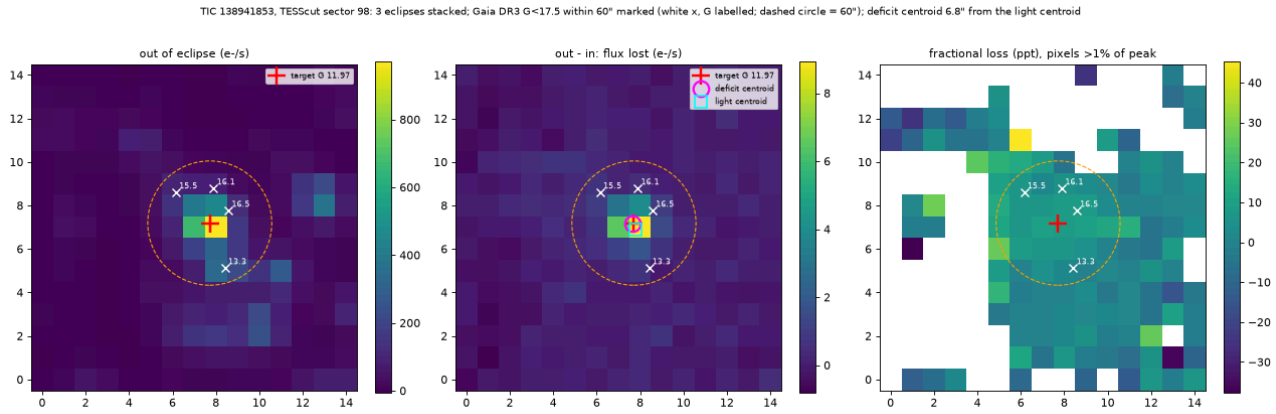
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
6	0	1475.6339074	0.010416	7.810	usable; same-kind selection applies
32	61	2183.1561435	0.003472	9.392	usable; same-kind selection applies
32	62	2194.7525984	0.003472	9.255	usable; same-kind selection applies
98	219	4015.7299466	0.001661	9.928	usable; same-kind selection applies
98	220	4027.3317908	0.001333	7.985	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s98.json, historical/diff_s98.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R234. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R234; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R234; diff_s98.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R235. TIC 256665207

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 256665207: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 19 distinct eclipse events in selected sectors[58, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 19 same-kind midtimes give P=2.7907385784 ±0.0000027067d, fitted ephemeris intercept at earliest retained cycle BTJD2884.6986301, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.481arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.177. Capable neighbours: Gaia 2238265718901540224: 32.78 arcsec, G16.300, required fractional eclipse0.384, pixel fractional loss0.005700579685187473; Gaia 2238265791919992576: 53.79 arcsec, G16.967, required fractional eclipse0.705, pixel fractional loss0.008951220717294421
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths7.140/6.683/6.612ppt, with 10 stacked events. QLP selected-kind fitted median6.603ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32238265757560253312, P=2.7907557184±0.0000952841d, e=0.051842, omega=220.0501deg. Closest edge-on conjunction offset-0.023180d (-0.008306phase), full-SB1-covariance linear propagated Gaia uncertainty0.096919d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.007857d (photometric period projection allowance0.002420d). The approximate central duration4.78h vs observed median1.41h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Approximately6-7ppt flat-bottom eclipses repeat. No compelling even/odd split establishes half/double alias; full Gaia period agrees with recurrence. Nearest capable source32.78arcsec away,1.70pixel from deficit vs0.166pixel target.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

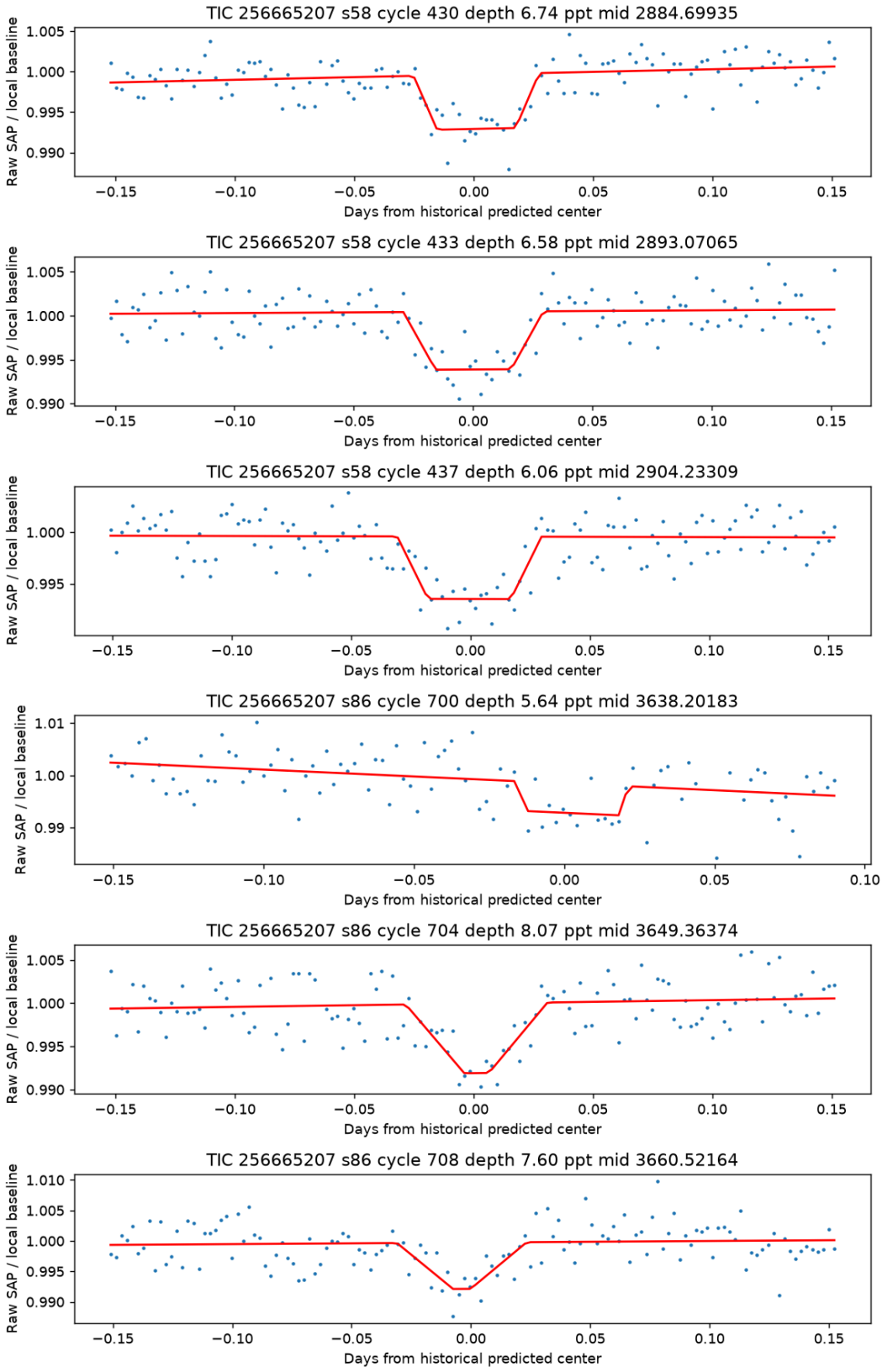
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

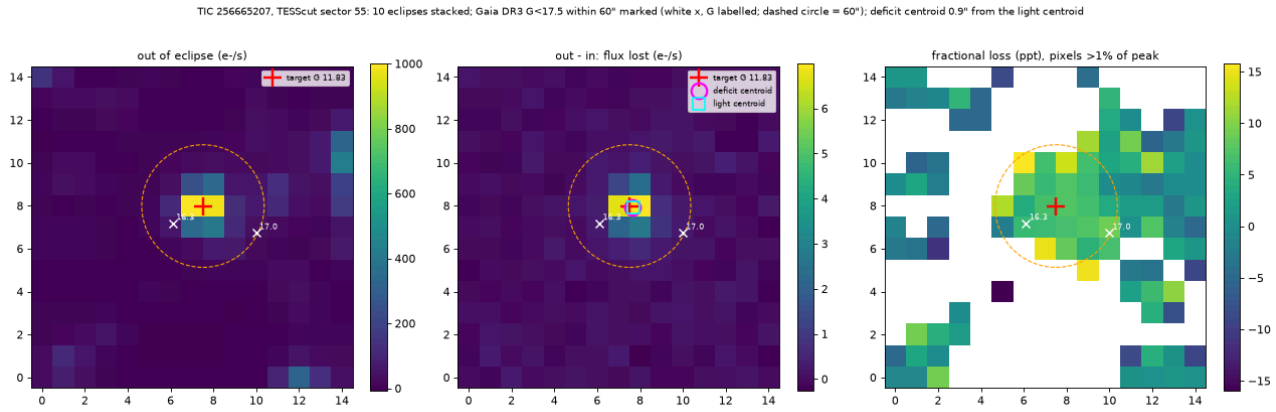
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
58	430	2884.6993485	0.001157	6.737	usable; same-kind selection applies
58	431	2887.4895215	0.001398	7.157	usable; same-kind selection applies
58	432	2890.2804994	0.001385	6.603	usable; same-kind selection applies
58	433	2893.0706498	0.001315	6.583	usable; same-kind selection applies
58	434	2895.8593776	0.001479	6.543	usable; same-kind selection applies
58	435	2898.6525323	0.001157	5.534	usable; same-kind selection applies
58	436	2901.4430937	0.001157	7.079	usable; same-kind selection applies
58	437	2904.2330928	0.001343	6.060	usable; same-kind selection applies
58	438	2907.0251760	0.001402	6.878	usable; same-kind selection applies
58	439	2909.8155043	0.001157	6.128	usable; same-kind selection applies
86	700	3638.2018317	0.002793	5.639	usable; same-kind selection applies
86	701	3640.9897766	0.001893	6.946	usable; same-kind selection applies
86	702	3643.7793978	0.001157	6.017	usable; same-kind selection applies
86	703	3646.5684652	0.001526	6.952	usable; same-kind selection applies
86	704	3649.3637364	0.001989	8.067	usable; same-kind selection applies
86	705	3652.1544705	0.006063	4.604	usable; same-kind selection applies
86	706	3654.9429783	0.002169	8.794	usable; same-kind selection applies
86	707	3657.7327724	0.001587	6.142	usable; same-kind selection applies
86	708	3660.5216443	0.002011	7.599	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R235. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R235; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R235; diff_s55.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R236. TIC 91628894

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 91628894: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[10, 64, 100]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=6.7661193197 ±0.0000092506d, fitted ephemeris intercept at earliest retained cycle BTJD1576.8187199, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S63 deficit peak[7, 7] vs target pixel[7, 8]; deficit-target separation7.925arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.290. Capable neighbours: Gaia 5348295453990085888: 19.20 arcsec, G15.582, required fractional eclipse0.381, pixel fractional loss0.01023010947109879; Gaia 5348295140443337472: 34.87 arcsec, G16.515, required fractional eclipse0.885, pixel fractional loss0.011668267374688467; Gaia 5348295453990088064: 37.41 arcsec, G16.480, required fractional eclipse0.857, pixel fractional loss0.011393153500120334; Gaia 5348295350910866688: 42.05 arcsec, G15.208, required fractional eclipse0.273, pixel fractional loss0.007769923584894719; Gaia 5348295350905264640: 43.46 arcsec, G16.548, required fractional eclipse0.912, pixel fractional loss0.007769923584894719; Gaia 5348295350910866176: 48.35 arcsec, G15.901, required fractional eclipse0.507, pixel fractional loss0.007769923584894719; Gaia 5348295282191399936: 56.55 arcsec, G16.151, required fractional eclipse0.636, pixel fractional loss0.0010946274488744217; Gaia 5348295144752436864: 58.14 arcsec, G16.385, required fractional eclipse0.786, pixel fractional loss0.004377720591664356

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths9.974/10.396/9.922ppt, with 4 stacked events. QLP selected-kind fitted median10.761ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3534829544968081920 0, P=6.7670124687±0.0024326020d, e=0.198278, omega=64.3659deg. Closest edge-on conjunction offset-0.195924d (-0.028953phase), full-SB1-covariance linear propagated Gaia uncertainty0.279206d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.038730d (photometric period projection allowance0.001628d). The approximate central duration8.09h vs observed median7.38h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deficit0.377pixel from target; light-centroid shift1.05arcsec is a different quantity. Nearest capable source19.20arcsec away requires38.1% eclipse and lies0.675pixel from deficit. Central lost-light peak and consistent10ppt aperture depths favor target; no calibrated subpixel uncertainty is inferred.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

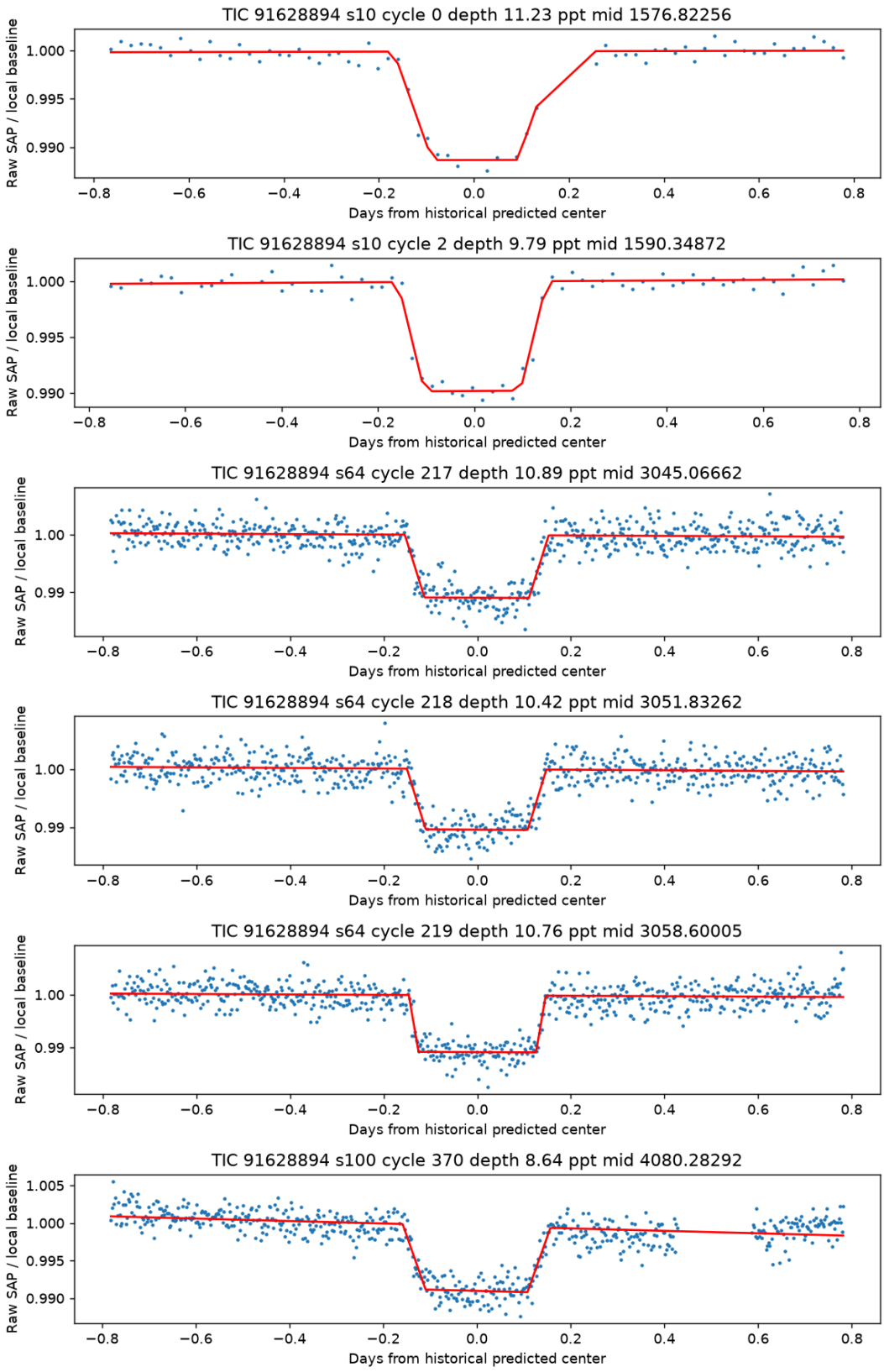
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

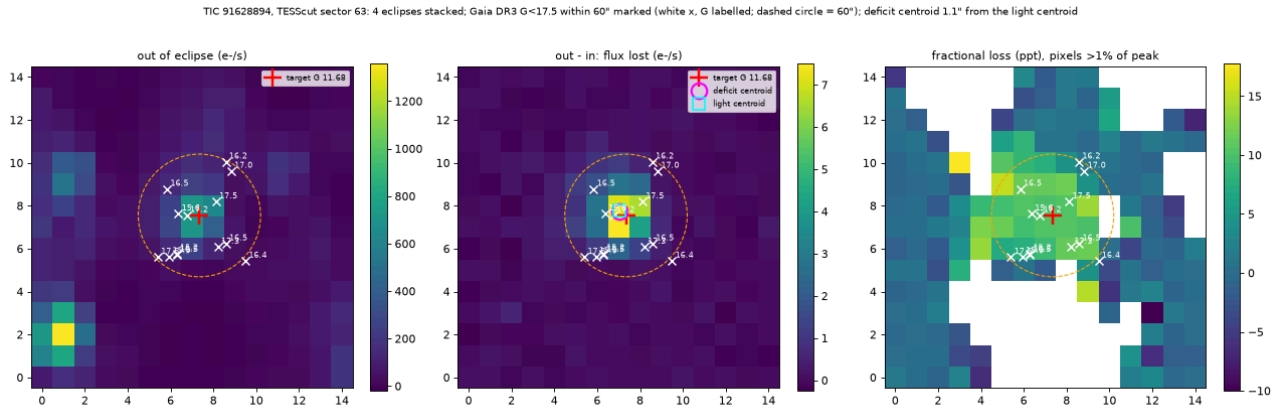
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
10	0	1576.8225588	0.010417	11.231	usable; same-kind selection applies
10	2	1590.3487237	0.010417	9.789	usable; same-kind selection applies
64	217	3045.0666167	0.001347	10.891	usable; same-kind selection applies
64	218	3051.8326234	0.001314	10.423	usable; same-kind selection applies
64	219	3058.6000453	0.001157	10.761	usable; same-kind selection applies
64	220	3065.3634430	0.001323	10.941	usable; same-kind selection applies
100	370	4080.2829159	0.001327	8.638	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s63.json, historical/diff_s63.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R236. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R236; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R236; diff_s63.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R237. TIC 252737077

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 252737077: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[16, 57, 84]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=10.2034445887 ±0.0000145011d, reference eclipse epoch BTJD1757.7979046, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S57 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation4.251arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.943. Capable neighbours: Gaia 1995327181638998016: 42.69 arcsec, G12.539, required fractional eclipse0.653, pixel fractional loss0.013331207987805165; Gaia 1995327525236380800: 53.79 arcsec, G9.660, required fractional eclipse0.237, pixel fractional loss0.00028959663111898507
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths160.420/95.526/43.208ppt, with 3 stacked events. QLP selected-kind fitted median208.935ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3199532725465834188 8, P=10.2047428233±0.0012404385d, e=0.066845, omega=343.3919deg. Closest edge-on conjunction offset-0.047365d (-0.004641phase), full-SB1-covariance linear propagated Gaia uncertainty0.203135d at the measured epoch. Near-Gaia-reference projected eclipse offset0.126599d (photometric period projection allowance0.001943d). The approximate central duration6.54h vs

observed median5.46h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Strong aperture dilution: raw QLP approximately209ppt versus cutout1/3/5 aperture160/96/43ppt. Deficit core is target-centered; capable neighbors42.7/53.8arcsec away are2.06/2.72pixels from the deficit centroid versus0.20pixel for target; blended local loss is not an intrinsic-source eclipse constraint. Poor total-light PSF residual0.943 reflects crowding; no intrinsic depth is inferred.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

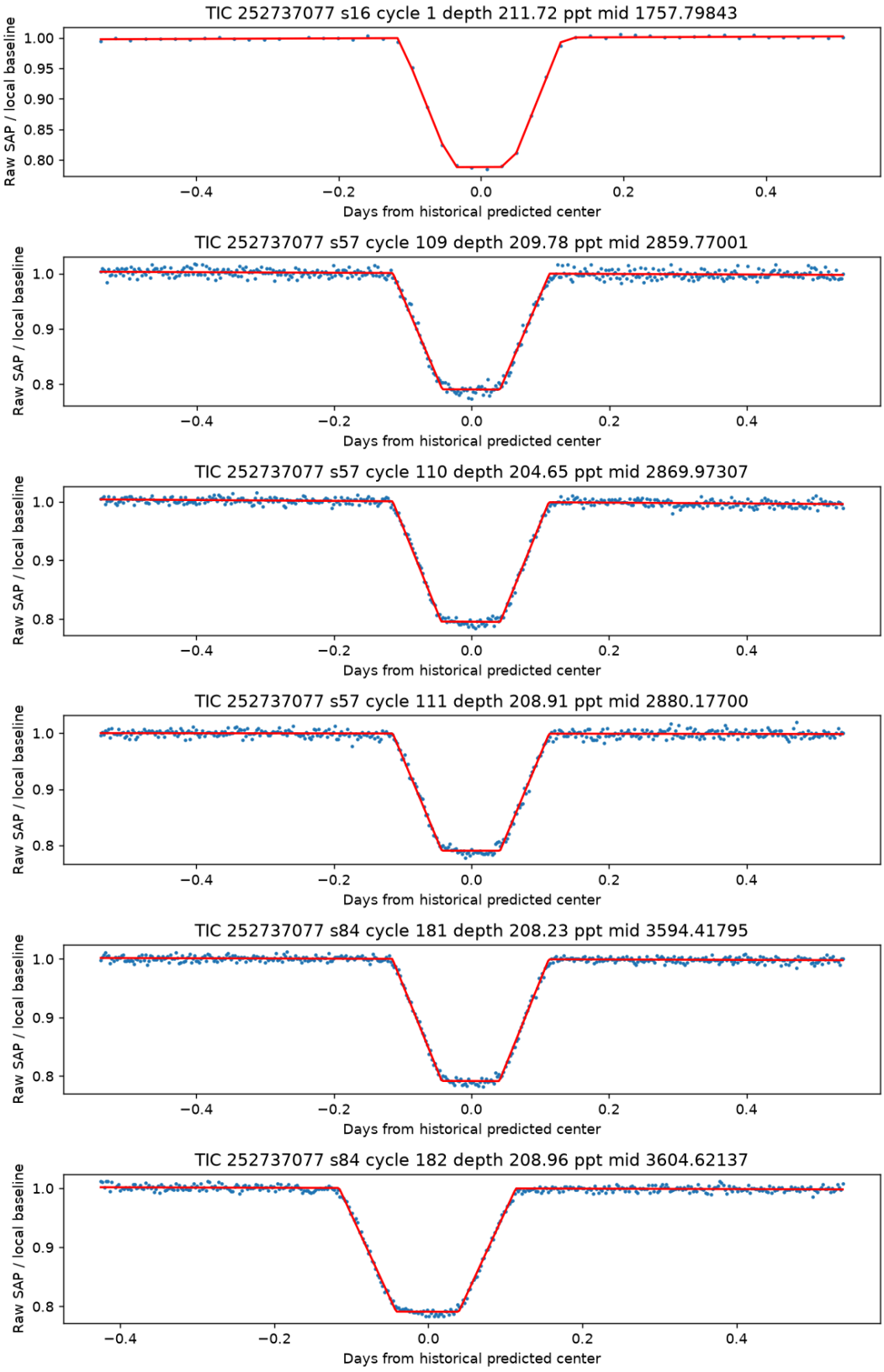
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

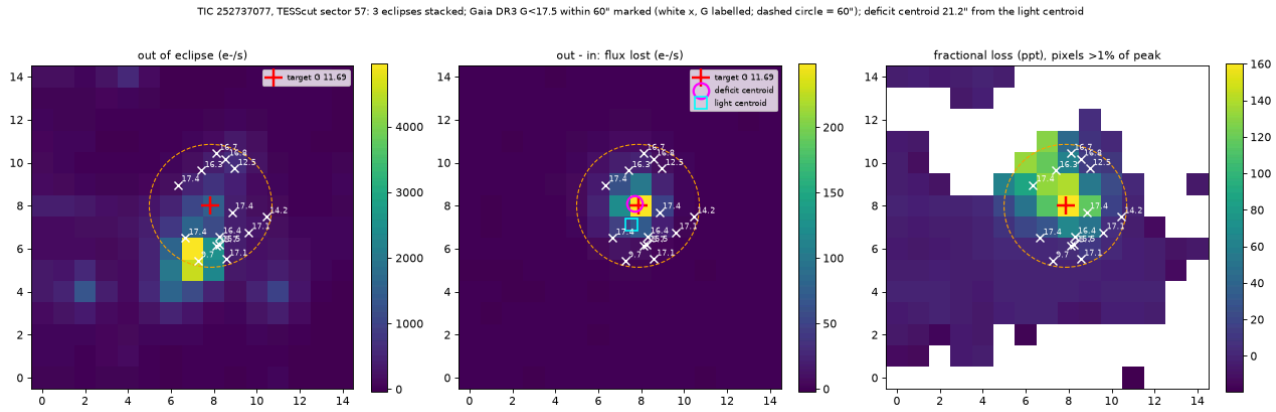
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
16	1	1757.7984314	0.010417	211.716	usable; same-kind selection applies
57	109	2859.7700065	0.001157	209.782	usable; same-kind selection applies
57	110	2869.9730680	0.001157	204.648	usable; same-kind selection applies
57	111	2880.1770048	0.001157	208.907	usable; same-kind selection applies
84	181	3594.4179451	0.001157	208.229	usable; same-kind selection applies
84	182	3604.6213692	0.001157	208.963	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s57.json, historical/diff_s57.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R237. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R237; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R237; diff_s57.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R238. TIC 240771237

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 240771237: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[84, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=5.4961100628 ±0.0001675565d, reference eclipse epoch BTJD3589.7561345, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S57 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.433arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.329. Capable neighbours: Gaia 415340212454228096: 31.45 arcsec, G14.768, required fractional eclipse0.620, pixel fractional loss0.015431025956409657
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths37.852/35.024/31.849ppt, with 4 stacked events. QLP selected-kind fitted median40.756ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3415340178091153024, P=5.4961262607±0.0001875488d, e=0.029793, omega=317.7859deg. Closest edge-on conjunction offset0.109935d (0.020002phase), full-SB1-covariance linear propagated Gaia uncertainty0.109819d at the measured epoch. Near-Gaia-reference projected eclipse offset0.119362d (photometric period projection allowance0.097518d). The approximate central duration5.44h vs observed median2.63h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Seven measured newer-sector repeats; historical sector57 did not pass present fitting window requirements and is not a null. Capable neighbor31.4arcsec away is separated from the target-centered deficit.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

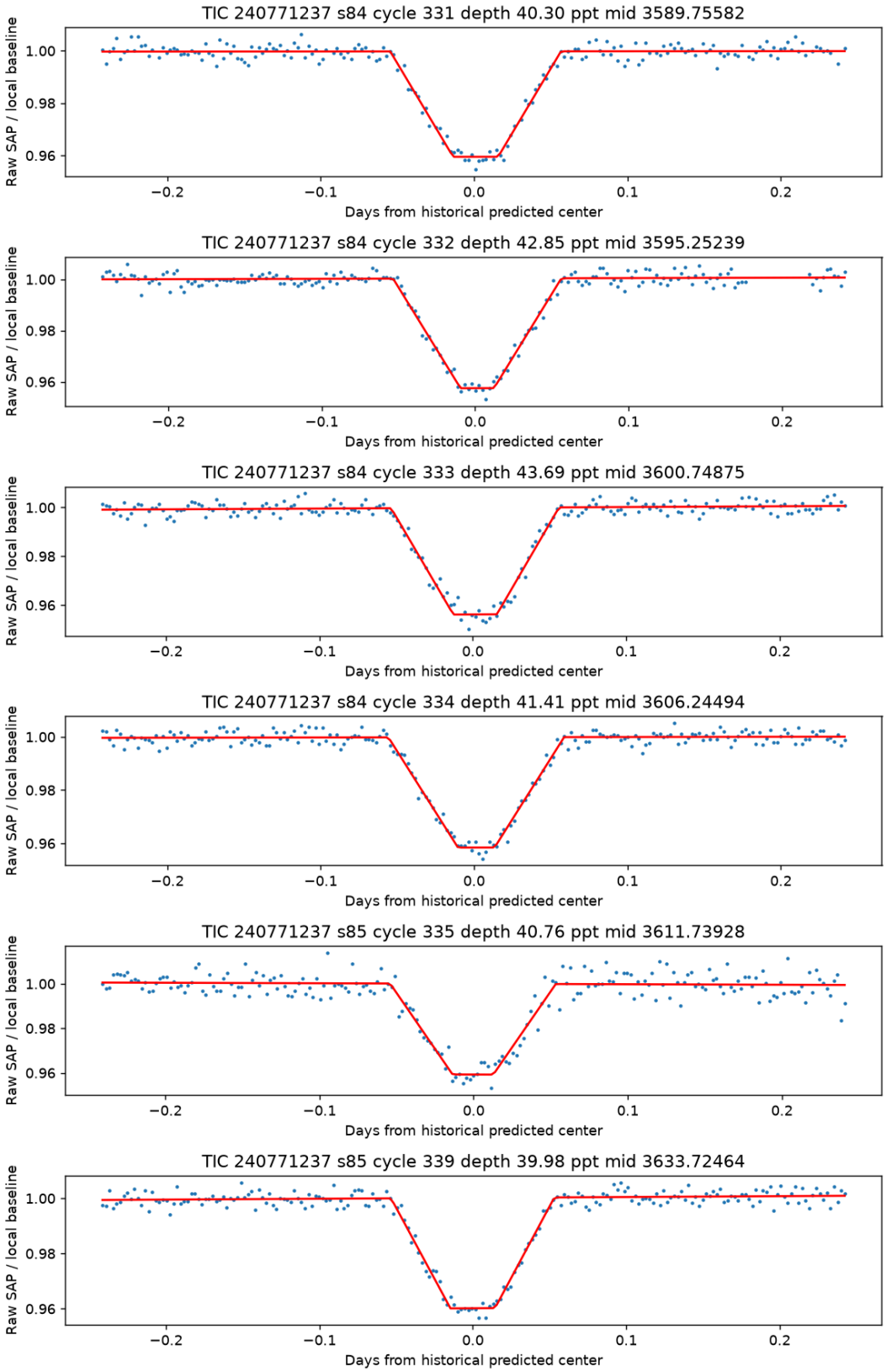
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

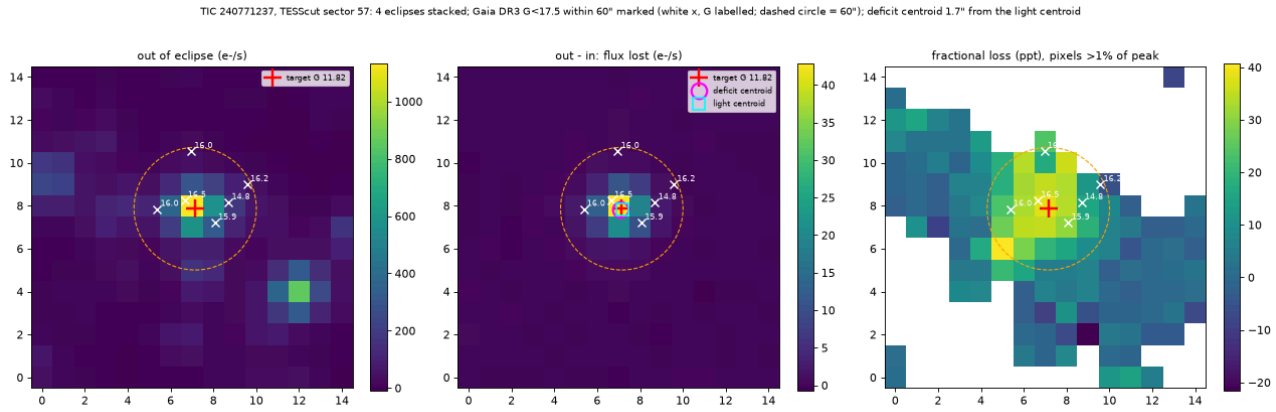
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
84	331	3589.7558172	0.001157	40.300	usable; same-kind selection applies
84	332	3595.2523916	0.001157	42.845	usable; same-kind selection applies
84	333	3600.7487471	0.001157	43.689	usable; same-kind selection applies
84	334	3606.2449399	0.001157	41.407	usable; same-kind selection applies
85	335	3611.7392832	0.001157	40.756	usable; same-kind selection applies
85	337	3622.7337666	0.001157	40.341	usable; same-kind selection applies
85	339	3633.7246371	0.001157	39.981	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s57.json, historical/diff_s57.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R238. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R238; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R238; diff_s57.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R239. TIC 369118947

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 369118947: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 9 distinct eclipse events in selected sectors[12, 93, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=5.3592734775 ±0.0000136631d, reference eclipse epoch BTJD1632.7557204, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S39 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.683arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.923. Capable neighbours: Gaia 5949510916010450048: 53.79 arcsec, G12.654, required fractional eclipse0.622, pixel fractional loss0.0011707993086129812
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths169.328/127.739/59.283ppt, with 4 stacked events. QLP selected-kind fitted median186.853ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3594951242783784614 4, P=5.3589663376±0.0003517770d, e=0.023477, omega=279.2669deg. Closest edge-on conjunction offset0.104460d (0.019493phase), full-SB1-covariance linear propagated Gaia uncertainty0.058542d at the measured epoch. Near-Gaia-reference projected eclipse offset0.033204d (photometric period projection allowance0.003170d). The approximate central duration6.24h vs observed median4.73h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Target-centered deficit despite poor0.923 total-light PSF residual in crowded field. Capable neighbor53.8arcsec away is well separated. Aperture-dependent diluted depth retained.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

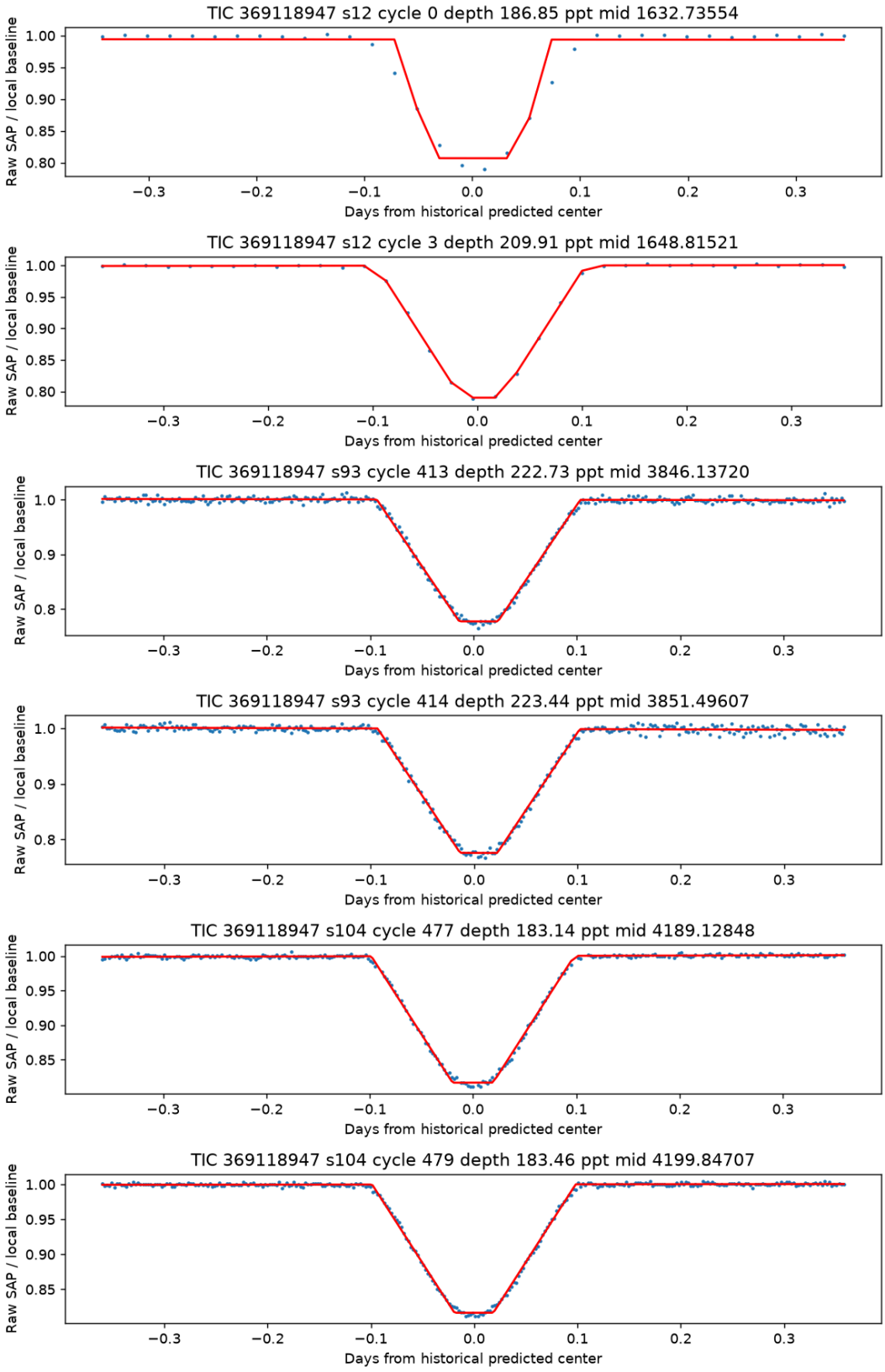
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

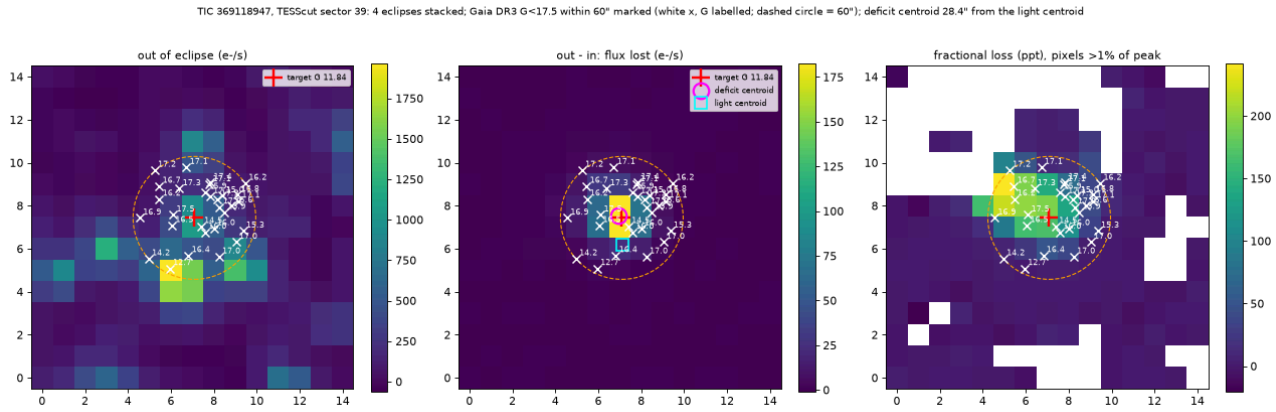
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
12	0	1632.7355450	0.010417	186.853	usable; same-kind selection applies
12	3	1648.8152110	0.010417	209.905	usable; same-kind selection applies
93	411	3835.4179474	0.001157	226.111	usable; same-kind selection applies
93	413	3846.1372029	0.001157	222.733	usable; same-kind selection applies
93	414	3851.4960744	0.001157	223.445	usable; same-kind selection applies
104	476	4183.7692321	0.001157	185.159	usable; same-kind selection applies
104	477	4189.1284814	0.001157	183.137	usable; same-kind selection applies
104	478	4194.4874165	0.001157	184.678	usable; same-kind selection applies
104	479	4199.8470706	0.001157	183.464	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s39.json, historical/diff_s39.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R239. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R239; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R239; diff_s39.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R240. TIC 399238907

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 399238907: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[80]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=4.1443054573 ±0.0002790706d, reference eclipse epoch BTJD3480.3126460, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S80 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.732arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.329. Capable neighbours: Gaia 4468583068251428736: 16.59 arcsec, G14.439, required fractional eclipse0.537, pixel fractional loss0.03270235223204699; Gaia 4468582965172213888: 22.12 arcsec, G13.831, required fractional eclipse0.324, pixel fractional loss0.016929322145052952; Gaia 4468582969467512192: 23.01 arcsec, G14.622, required fractional eclipse0.628, pixel fractional loss0.016929322145052952; Gaia 4468583106906461184: 52.56 arcsec, G14.077, required fractional eclipse0.396, pixel fractional loss0.010143271811782536
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths34.251/31.049/29.498ppt, with 5 stacked events. QLP selected-kind fitted median42.314ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3446858306825142864, P=4.1442204144±0.0002752424d, e=0.053485, omega=202.9714deg. Closest edge-on conjunction

offset0.076594d (0.018482phase), full-SB1-covariance linear propagated Gaia uncertainty0.214766d at the measured epoch. Near-Gaia-reference projected eclipse offset0.013152d (photometric period projection allowance0.208187d). The approximate central duration9.18h vs observed median7.03h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Only QLP sector80 is available in the shared availability receipt, containing five measured repeats. Queue4.14670d is the historical screening seed, not current Gaia period; full NSS4.144220414d agrees measured period. Sector80 pipeline timing offsets are an uncalibrated systematic. Four capable neighbors are offset from the target-centered lost-light core; closest16.6arcsec.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

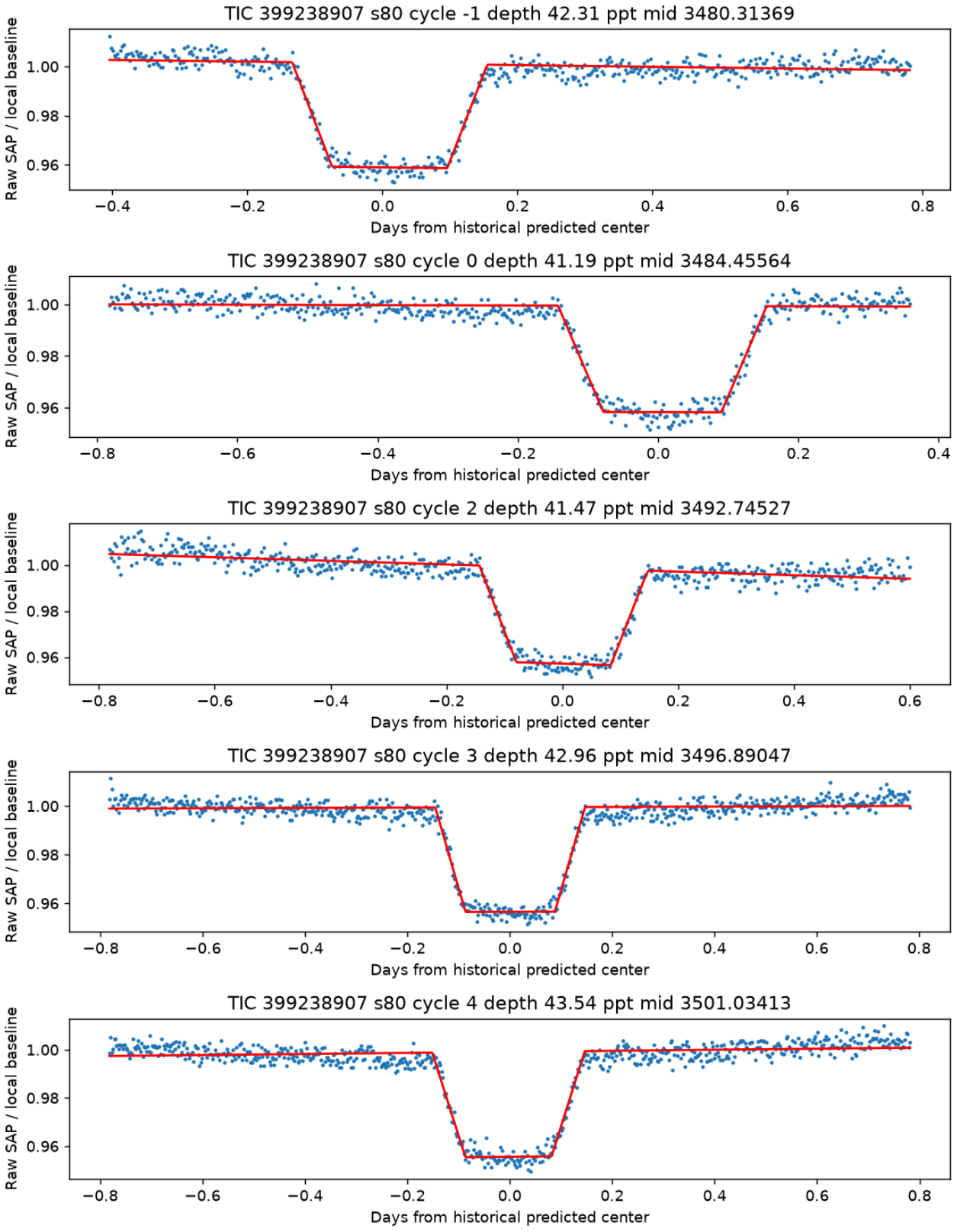
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

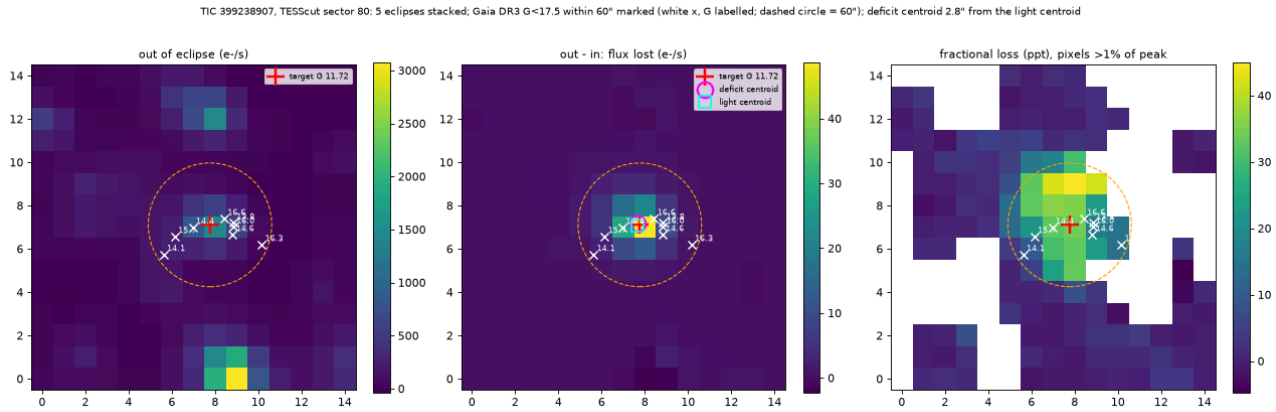
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
80	-1	3480.3136886	0.001157	42.314	usable; same-kind selection applies
80	0	3484.4556420	0.001157	41.193	usable; same-kind selection applies
80	2	3492.7452726	0.001157	41.471	usable; same-kind selection applies
80	3	3496.8904725	0.001157	42.956	usable; same-kind selection applies
80	4	3501.0341254	0.001157	43.541	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s80.json, historical/diff_s80.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R240. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R240; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R240; diff_s80.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R241. TIC 146319847

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 146319847: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[9, 63, 100]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=8.8705576715 ±0.0000110757d, reference eclipse epoch BTJD1548.3578550, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S63 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.042arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.927. Capable neighbours: Gaia 5392222730215912320: 33.84 arcsec, G9.668, required fractional eclipse0.139, pixel fractional loss0.0016831041794920934
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths107.695/54.423/20.351ppt, with 3 stacked events. QLP selected-kind fitted median133.766ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3539221980963815091 2, P=8.8705042457±0.0005204978d, e=0.018718, omega=121.6281deg. Closest edge-on conjunction offset-0.008879d (-0.001001phase), full-SB1-covariance linear propagated Gaia uncertainty0.078702d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.015877d (photometric period projection allowance0.001451d). The approximate central duration6.01h vs observed median4.02h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Capable neighbor33.8arcsec away is distinct from the target-centered deficit. Raw depths105-138ppt and cutout apertures108/54/20ppt reflect substantial dilution; total-light PSF residual0.927 is poor and not used as a sole attribution test.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

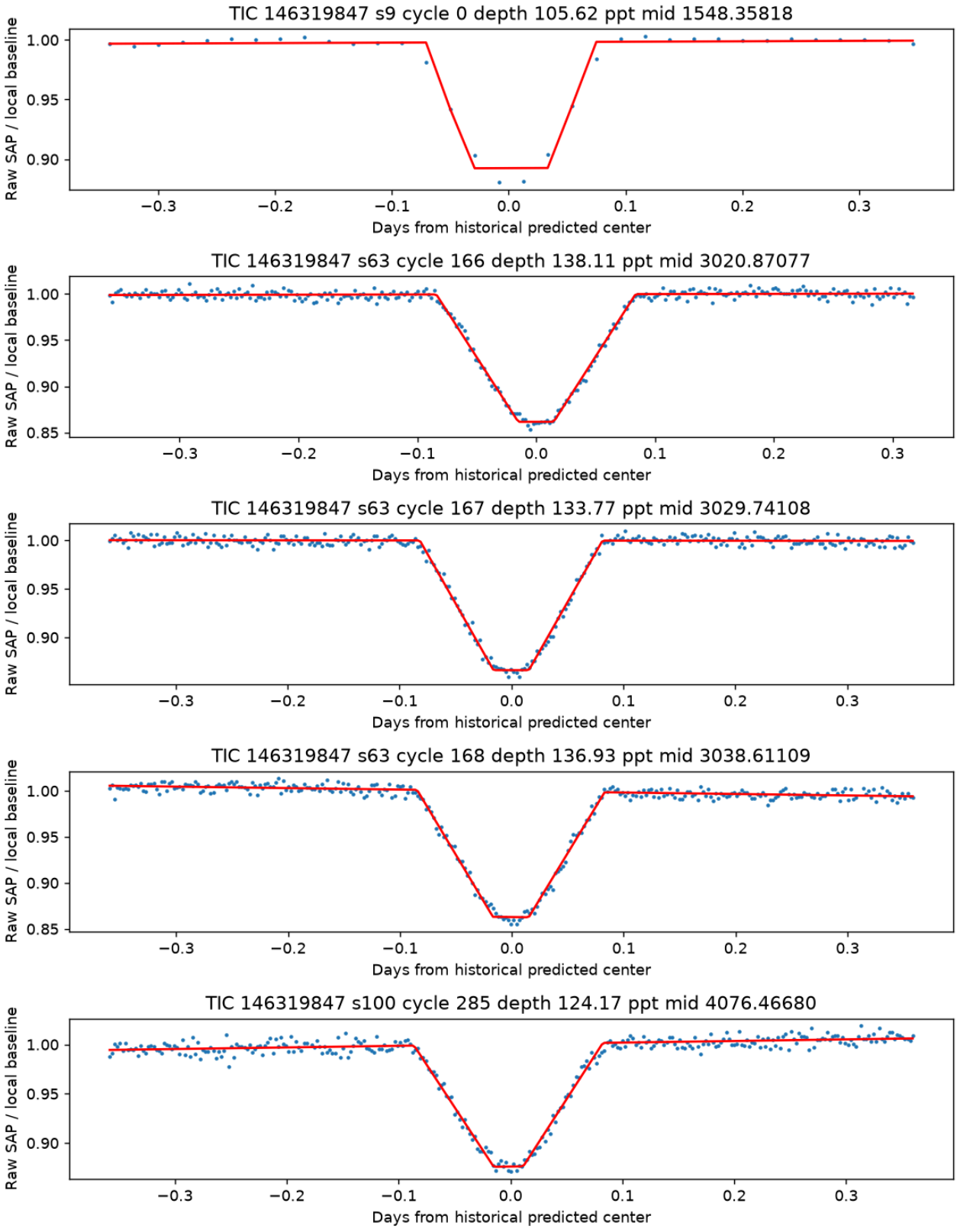
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

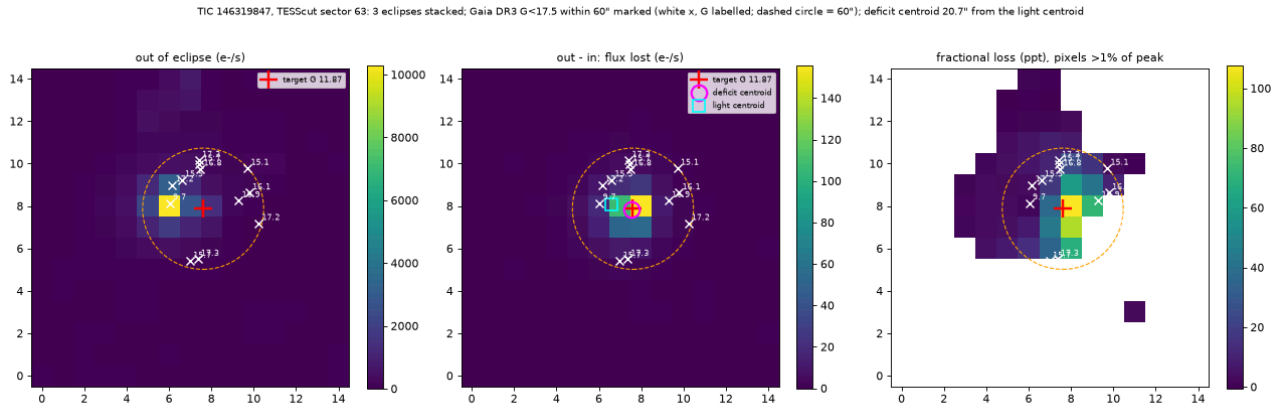
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	0	1548.3581812	0.010417	105.624	usable; same-kind selection applies
63	166	3020.8707657	0.001157	138.114	usable; same-kind selection applies
63	167	3029.7410830	0.001157	133.766	usable; same-kind selection applies
63	168	3038.6110933	0.001157	136.935	usable; same-kind selection applies
100	285	4076.4668038	0.001157	124.170	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s63.json, historical/diff_s63.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R241. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R241; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R241; diff_s63.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R242. TIC 466690912

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 466690912: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 32 distinct eclipse events in selected sectors[27, 94, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 32 same-kind midtimes give P=2.1159658358 ±0.0000011526d, reference eclipse epoch BTJD2037.8376563, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S66 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation6.475arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.677. Capable neighbours: Gaia 6366373024352389504: 49.13 arcsec, G15.303, required fractional eclipse0.883, pixel fractional loss0.011787977275723208
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths31.335/31.036/29.192ppt, with 12 stacked events. QLP selected-kind fitted median29.016ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR36366373058712130432, P=2.1159665560±0.0000929679d, e=0.040428, omega=2.4544deg. Closest edge-on conjunction offset0.022977d (0.010859phase), full-SB1-covariance linear propagated Gaia uncertainty0.073318d at the measured epoch. Near-Gaia-reference projected eclipse offset0.023538d (photometric period projection allowance0.000898d). The approximate central duration5.24h vs observed median5.25h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated flat-bottom eclipses with changing raw sector depths19-35ppt; aperture and baseline dependence is retained. Capable neighbor49.1arcsec away is distinct from the deficit core. Observed duration5.25h agrees approximate central estimate5.24h to the precision of its assumptions.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

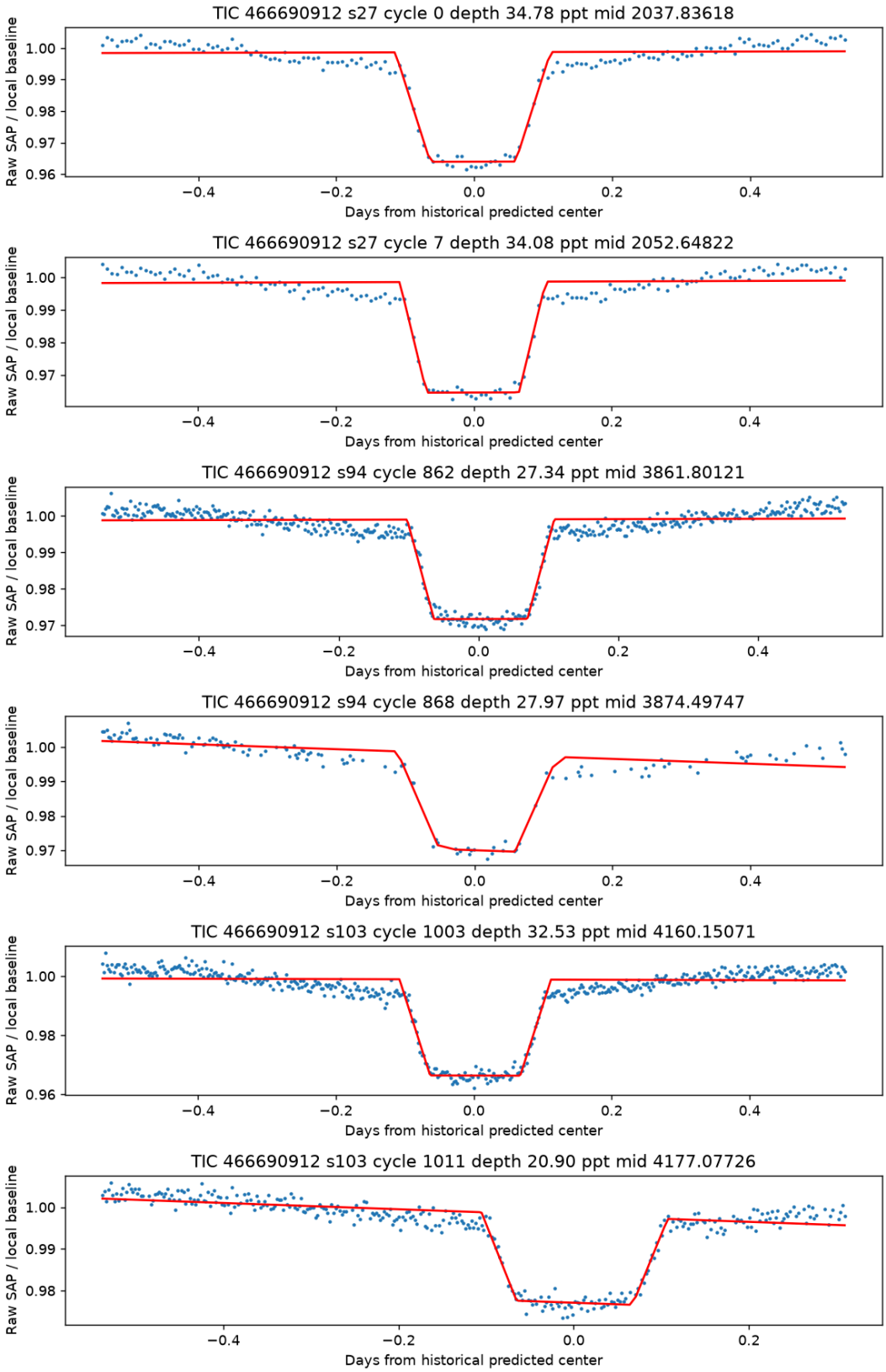
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
27	0	2037.8361783	0.003472	34.782	usable; same-kind selection applies
27	1	2039.9533091	0.003472	34.775	usable; same-kind selection applies
27	2	2042.0690102	0.003472	34.784	usable; same-kind selection applies
27	3	2044.1840321	0.003472	35.232	usable; same-kind selection applies
27	4	2046.3011911	0.003472	35.430	usable; same-kind selection applies
27	6	2050.5317487	0.003472	34.694	usable; same-kind selection applies
27	7	2052.6482193	0.003472	34.085	usable; same-kind selection applies
27	8	2054.7637686	0.003472	35.224	usable; same-kind selection applies
27	9	2056.8805472	0.003472	34.224	usable; same-kind selection applies
27	10	2058.9964185	0.003472	35.110	usable; same-kind selection applies
94	860	3857.5697483	0.001157	26.507	usable; same-kind selection applies
94	861	3859.6848160	0.001157	27.726	usable; same-kind selection applies
94	862	3861.8012077	0.001157	27.342	usable; same-kind selection applies
94	863	3863.9164637	0.001157	27.763	usable; same-kind selection applies
94	864	3866.0331157	0.001157	27.412	usable; same-kind selection applies
94	865	3868.1484324	0.001306	24.010	usable; same-kind selection applies
94	866	3870.2657924	0.001756	25.667	usable; same-kind selection applies
94	867	3872.3807083	0.001157	27.539	usable; same-kind selection applies
94	868	3874.4974663	0.002350	27.969	usable; same-kind selection applies
94	869	3876.6133249	0.001157	27.617	usable; same-kind selection applies

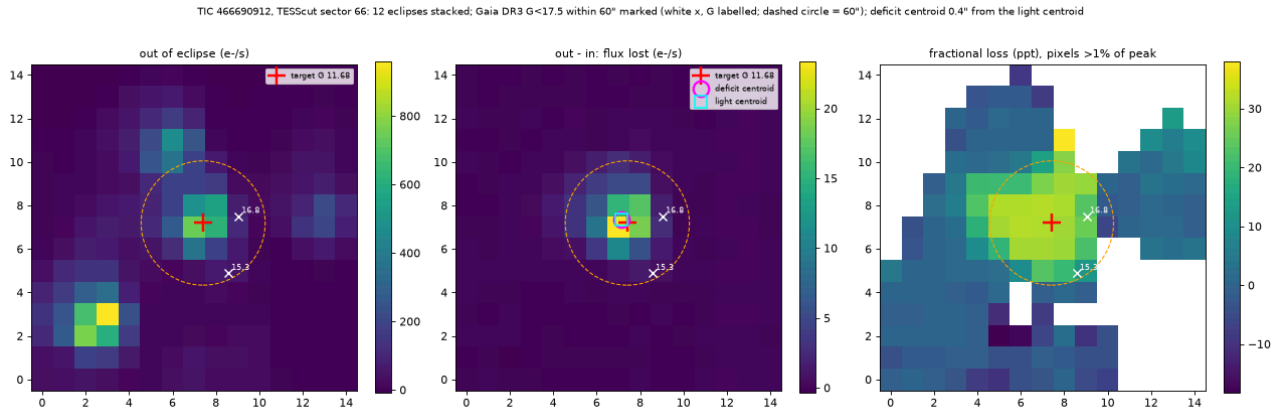
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
94	870	3878.7283753	0.001157	27.917	usable; same-kind selection applies
103	1000	4153.8028232	0.001521	30.062	usable; same-kind selection applies
103	1001	4155.9189734	0.001157	32.520	usable; same-kind selection applies
103	1002	4158.0361246	0.001157	32.474	usable; same-kind selection applies
103	1003	4160.1507115	0.001157	32.529	usable; same-kind selection applies
103	1004	4162.2671677	0.001157	32.057	usable; same-kind selection applies
103	1005	4164.3811602	0.001157	32.014	usable; same-kind selection applies
103	1006	4166.4984552	0.001996	19.166	usable; same-kind selection applies
103	1007	4168.6147203	0.001157	22.145	usable; same-kind selection applies
103	1009	4172.8466735	0.001157	22.417	usable; same-kind selection applies
103	1010	4174.9625320	0.001157	22.742	usable; same-kind selection applies
103	1011	4177.0772644	0.001157	20.903	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s66.json, historical/diff_s66.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R242. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R242; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R242; diff_s66.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R243. TIC 436092743

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 436092743: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[55, 82]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=13.1290724046 ±0.0000461943d, reference eclipse epoch BTJD2799.7762801, kind: Deeper primary (odd half-period cycles). Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.240arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.238. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths118.232/111.779/104.775ppt, with 4 stacked events. QLP selected-kind fitted median152.348ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3186033103376599488O, P=13.1281268605±0.0022765186d, e=0.011420, omega=39.9211deg. Closest edge-on conjunction offset0.178987d (0.013634phase), full-SB1-covariance linear propagated Gaia uncertainty0.443548d at the measured epoch. Near-Gaia-reference projected eclipse offset0.005007d (photometric period projection allowance0.008500d). The approximate central duration17.53h vs observed median13.88h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses approximately153/100ppt establish full orbital period13.12907d, twice historical SAP branch. Odd half-period cycles are deeper primary eclipses and alone define ephemeris. Historical cutout stack mixes eclipse kinds; no capable neighbor.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

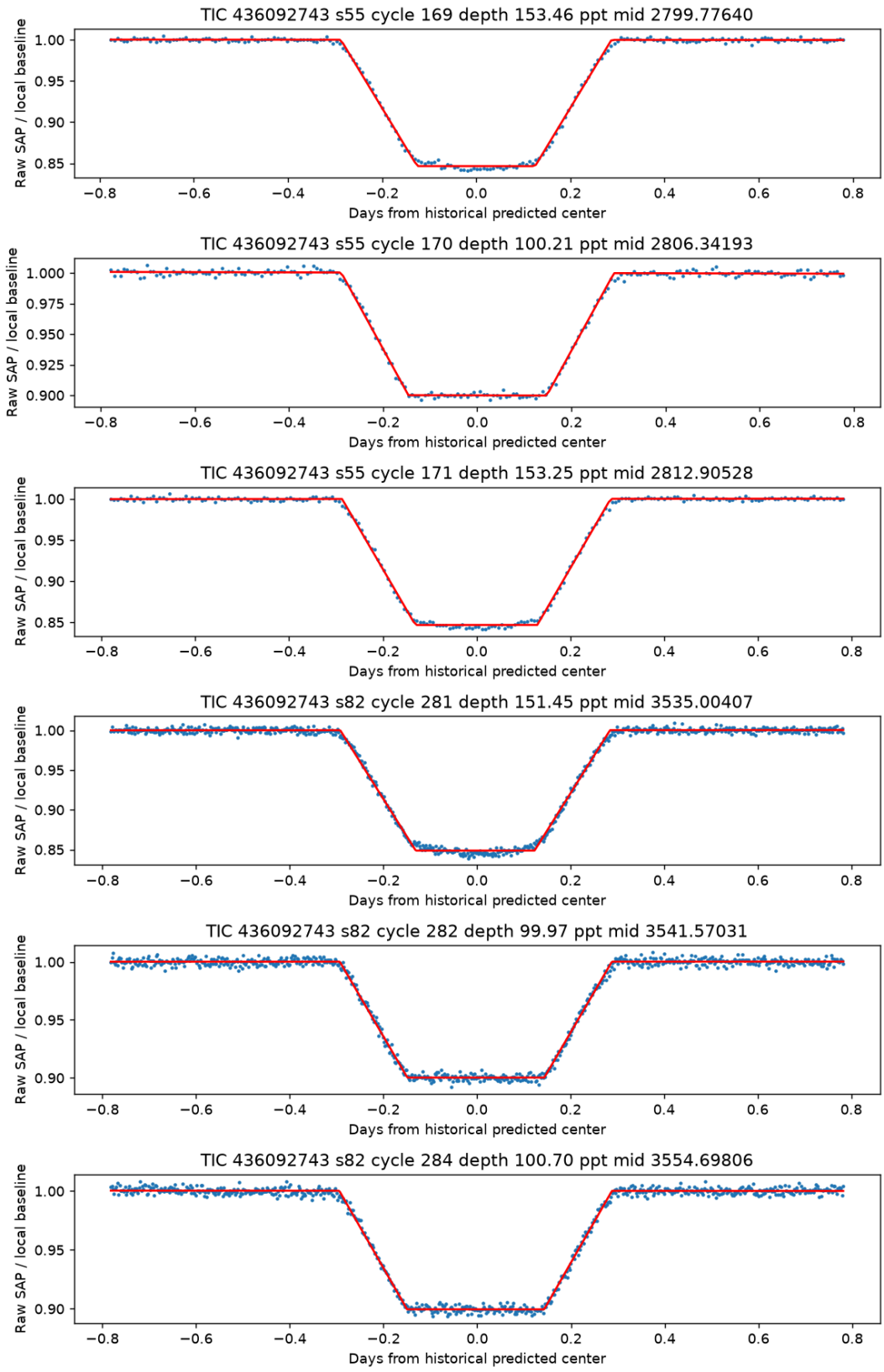
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

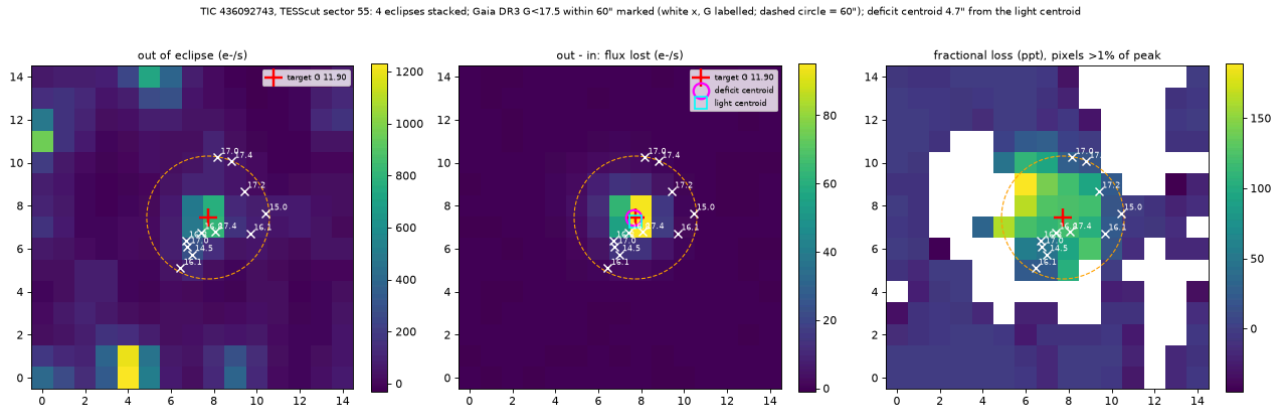
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
55	169	2799.7763956	0.003472	153.464	usable; same-kind selection applies
55	170	2806.3419285	0.003472	100.211	usable; same-kind selection applies
55	171	2812.9052776	0.003472	153.246	usable; same-kind selection applies
55	172	2819.4696319	0.003472	101.964	usable; same-kind selection applies
82	281	3535.0040697	0.001157	151.450	usable; same-kind selection applies
82	282	3541.5703121	0.001157	99.967	usable; same-kind selection applies
82	283	3548.1336678	0.001157	151.281	usable; same-kind selection applies
82	284	3554.6980615	0.001157	100.703	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R243. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R243; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R243; diff_s55.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R244. TIC 38850860

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 38850860: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[27, 68, 96]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=8.4145931953 ±0.0000081269d, reference eclipse epoch BTJD2042.7168402, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S27 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.214arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.190. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths31.055/30.538/29.006ppt, with 3 stacked events. QLP selected-kind fitted median30.158ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3470240352414200837 12, P=8.4129097571±0.0020308000d, e=0.169799, omega=71.4874deg. Closest edge-on conjunction offset-0.092244d (-0.010965phase), full-SB1-covariance linear propagated Gaia uncertainty0.475998d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.423882d (photometric period projection allowance0.001601d). The approximate central duration3.97h vs observed median3.76h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Eight measured repeats across three sectors. No capable catalogued neighbor; target-centered deficit. Raw depth26-31ppt varies by extraction/sector.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

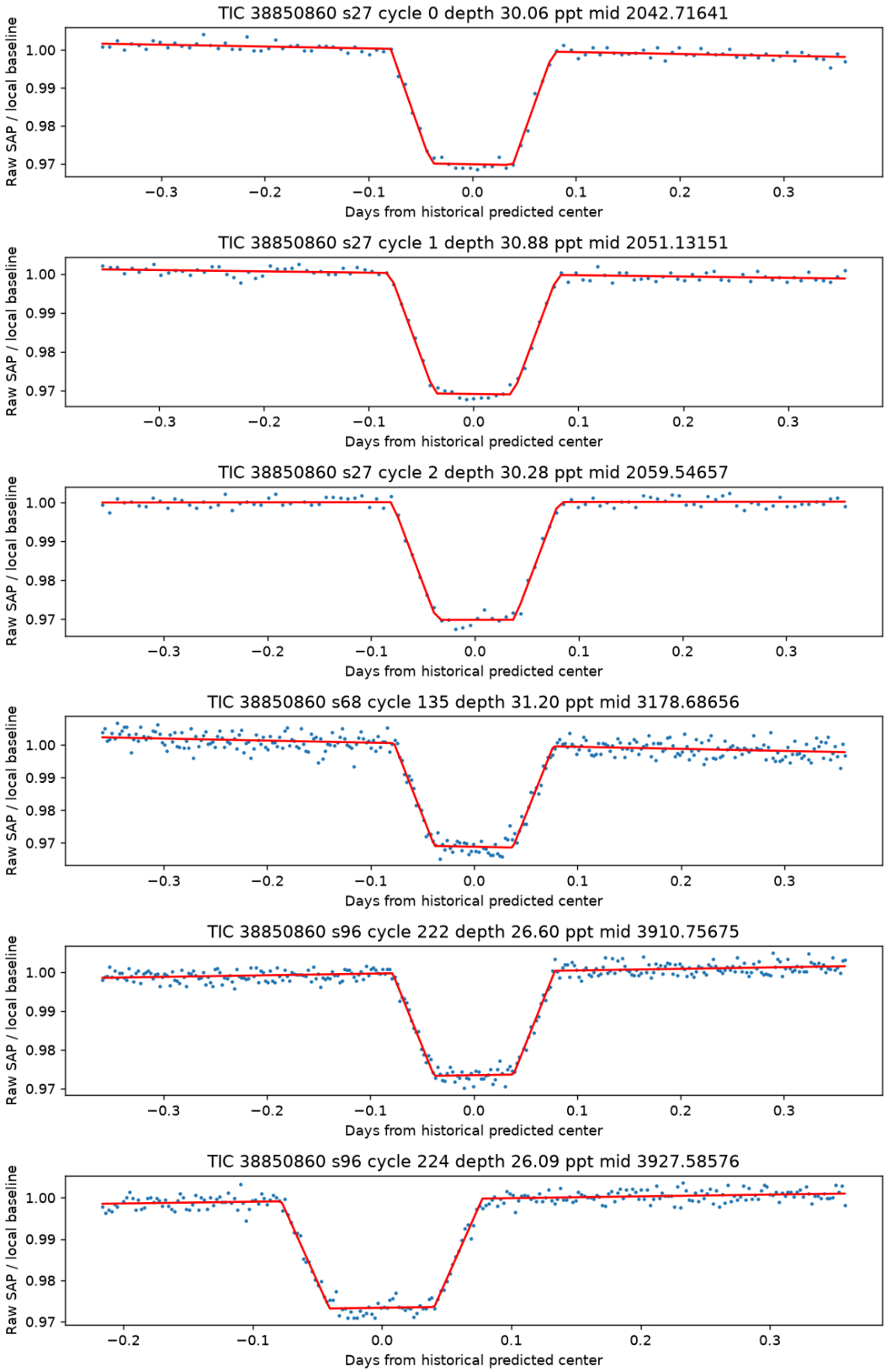
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

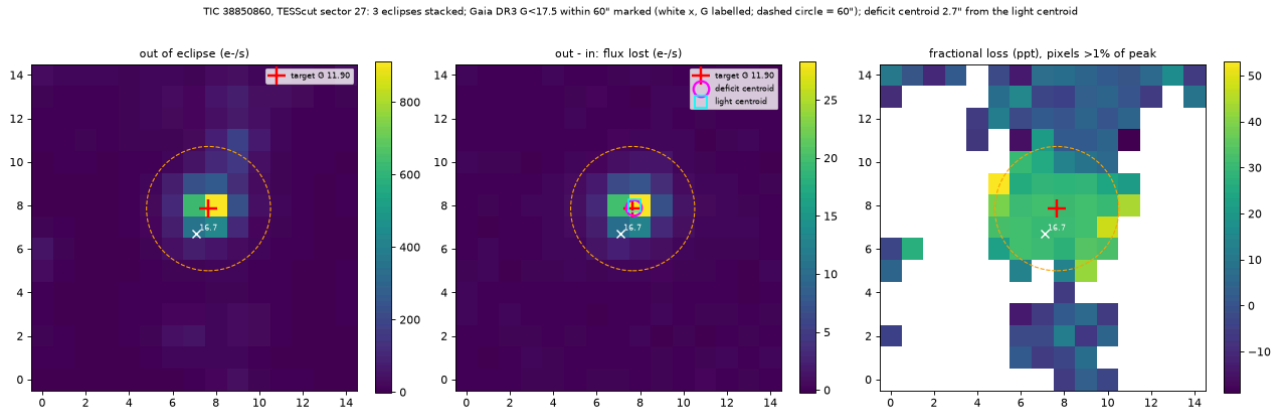
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
27	0	2042.7164086	0.003472	30.060	usable; same-kind selection applies
27	1	2051.1315095	0.003472	30.882	usable; same-kind selection applies
27	2	2059.5465702	0.003472	30.280	usable; same-kind selection applies
68	134	3170.2726330	0.001157	30.256	usable; same-kind selection applies
68	135	3178.6865601	0.001157	31.201	usable; same-kind selection applies
96	222	3910.7567478	0.001157	26.599	usable; same-kind selection applies
96	223	3919.1708937	0.001157	26.586	usable; same-kind selection applies
96	224	3927.5857628	0.001157	26.089	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s27.json, historical/diff_s27.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R244. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R244; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R244; diff_s27.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R245. TIC 140281693

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 140281693: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 15 distinct eclipse events in selected sectors[5, 33, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 15 same-kind midtimes give P=3.8719820163 ±0.0000030426d, reference eclipse epoch BTJD1440.1160470, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S32 deficit peak[8, 8] vs target pixel[8, 7]; deficit-target separation2.132arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual1.000. Capable neighbours: Gaia 2903798530942872704: 37.49 arcsec, G14.214, required fractional eclipse0.428, pixel fractional loss0.0062627425786385355; Gaia 2903798634022085504: 56.76 arcsec, G14.826, required fractional eclipse0.720, pixel fractional loss0.008418660852396504
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths40.235/37.239/29.706ppt, with 7 stacked events. QLP selected-kind fitted median42.031ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3290379856530260953 6, P=3.8720833801±0.0002902097d, e=0.100248, omega=219.1844deg. Closest edge-on conjunction offset-0.037247d (-0.009619phase), full-SB1-covariance linear propagated Gaia uncertainty0.069345d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.009778d (photometric period projection allowance0.000825d). The approximate central

duration4.57h vs observed median3.89h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Target-centered deficit despite poor0.9999 total-light PSF residual caused by unrelated bright light outside target core. Capable neighbors37.5/56.8arcsec are spatially separate. Raw depth34-45ppt depends on sector aperture.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

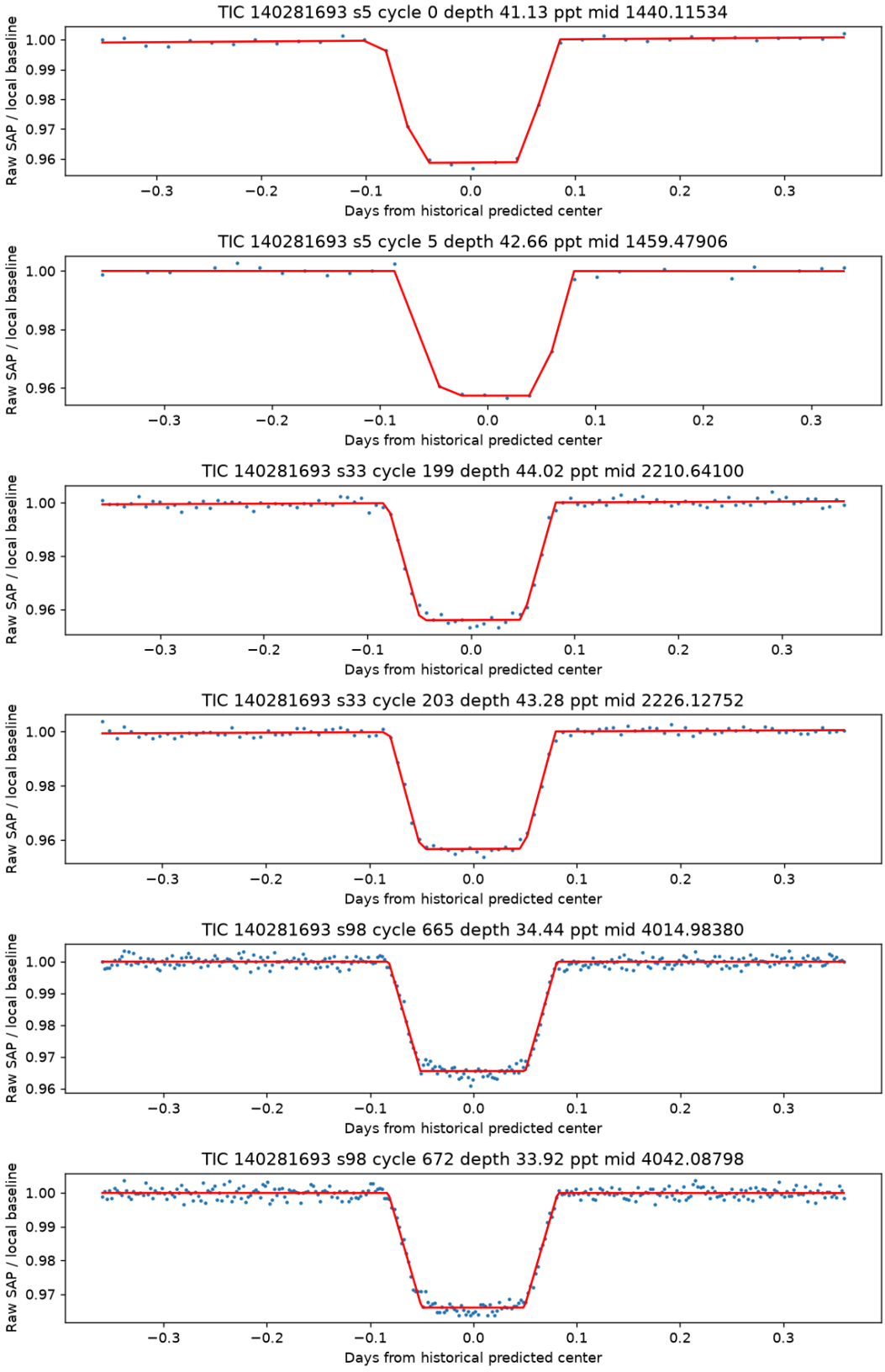
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

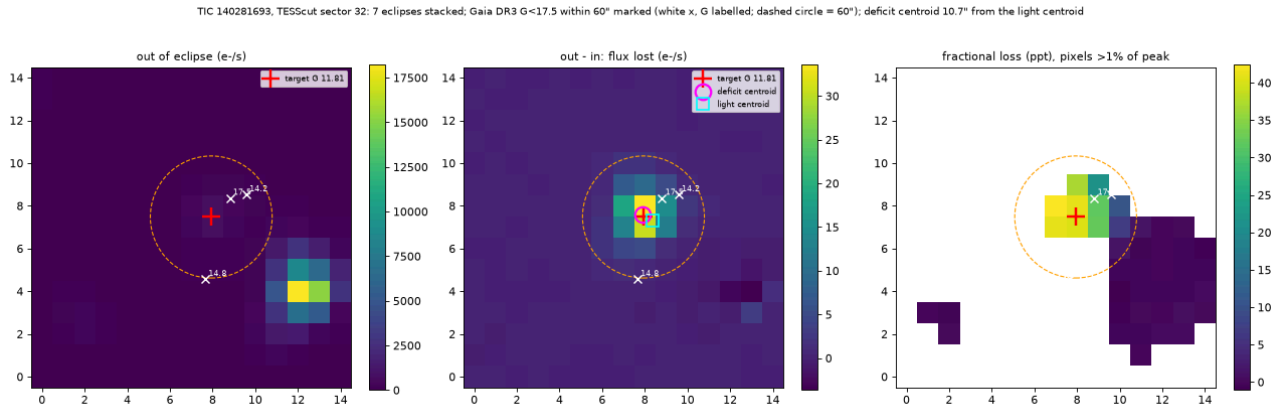
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
5	0	1440.1153374	0.010417	41.132	usable; same-kind selection applies
5	2	1447.8590467	0.010417	42.031	usable; same-kind selection applies
5	5	1459.4790570	0.010417	42.655	usable; same-kind selection applies
33	197	2202.8964232	0.003472	43.750	usable; same-kind selection applies
33	198	2206.7686474	0.003472	43.818	usable; same-kind selection applies
33	199	2210.6410010	0.003472	44.018	usable; same-kind selection applies
33	201	2218.3841162	0.003472	44.681	usable; same-kind selection applies
33	202	2222.2567471	0.003472	44.732	usable; same-kind selection applies
33	203	2226.1275199	0.003472	43.278	usable; same-kind selection applies
98	661	3999.4965366	0.001157	34.683	usable; same-kind selection applies
98	664	4011.1119223	0.001157	34.553	usable; same-kind selection applies
98	665	4014.9838019	0.001157	34.442	usable; same-kind selection applies
98	668	4026.6000678	0.001157	34.935	usable; same-kind selection applies
98	669	4030.4720694	0.001157	34.958	usable; same-kind selection applies
98	672	4042.0879774	0.001157	33.925	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s32.json, historical/diff_s32.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R245. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R245; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R245; diff_s32.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R246. TIC 128460489

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 128460489: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[7, 34, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=6.3628732011 ±0.0000073187d, reference eclipse epoch BTJD1493.6722577, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S7 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation5.679arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.317. Capable neighbours: Gaia 5600259057572857856: 11.02 arcsec, G14.307, required fractional eclipse0.254, pixel fractional loss0.020584271802751925; Gaia 5600259091932595584: 37.61 arcsec, G13.736, required fractional eclipse0.160, pixel fractional loss0.00852574510837417; Gaia 5600258988853381376: 41.92 arcsec, G13.394, required fractional eclipse0.123, pixel fractional loss0.002302554160844853; Gaia 5600258885774167040: 59.41 arcsec, G15.688, required fractional eclipse0.845, pixel fractional loss0.007786925055711405
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths20.584/19.674/17.212ppt, with 4 stacked events. QLP selected-kind fitted median26.324ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35600259057572857984, P=6.3633869971±0.0004569123d, e=0.017282, omega=333.8953deg. Closest edge-on conjunction

offset-0.069923d (-0.010988phase), full-SB1-covariance linear propagated Gaia uncertainty0.086227d at the measured epoch. Near-Gaia-reference projected eclipse offset0.019477d (photometric period projection allowance0.001273d). The approximate central duration8.56h vs observed median3.68h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unresolved attribution after completed check: capable Gaia5600259057572857856 neighbor11.02arcsec away requires25.4percent intrinsic eclipse and shares rounded TESS pixel with target. Deficit centroid is0.270pixel from target and0.516pixel from neighbor, favoring target but without calibrated uncertainty sufficient to exclude neighbor. Local blended pixel loss cannot exclude that neighbor. Retain probable for target pending resolved photometry or multi-source PRF localization.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

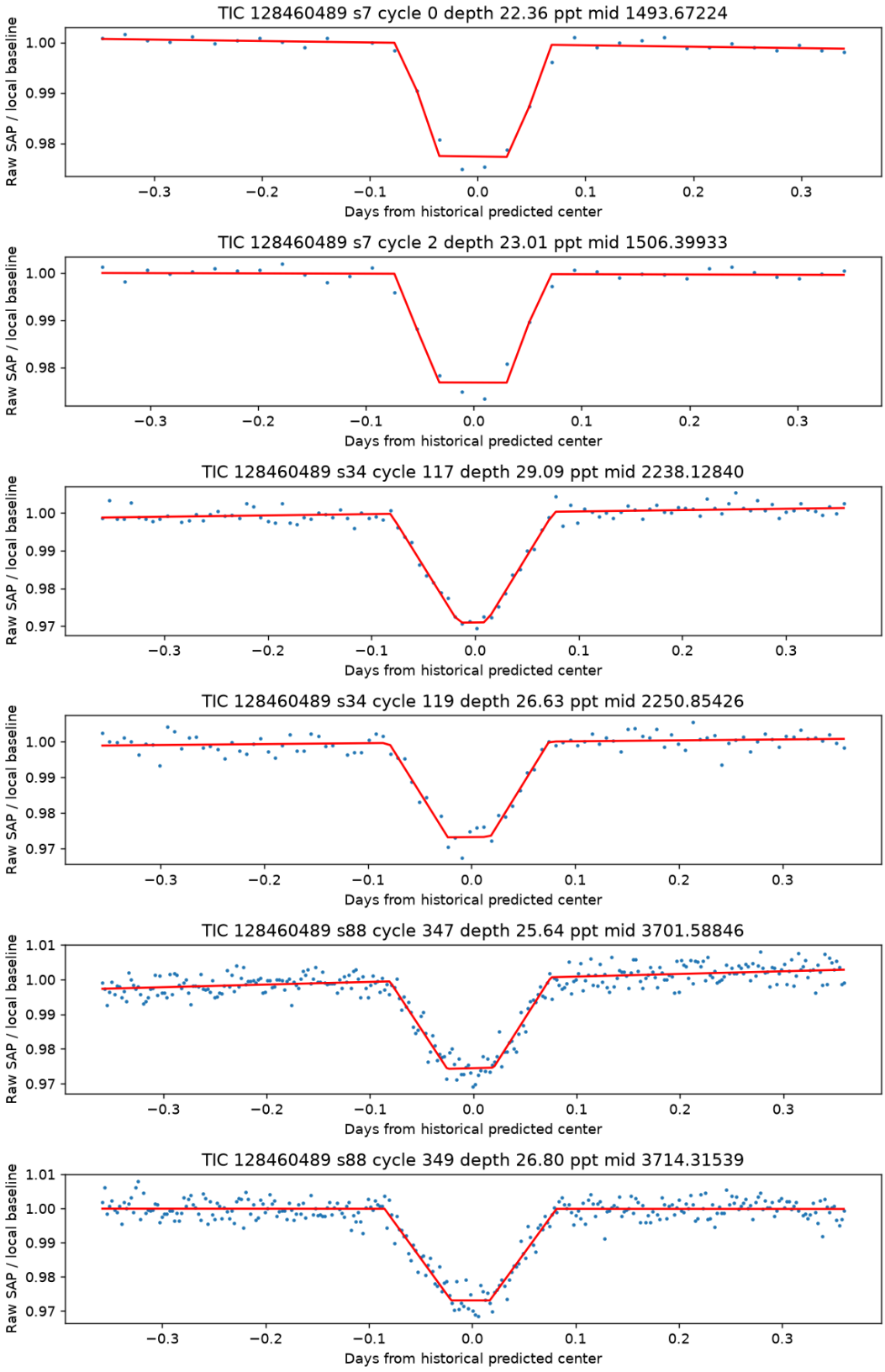
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

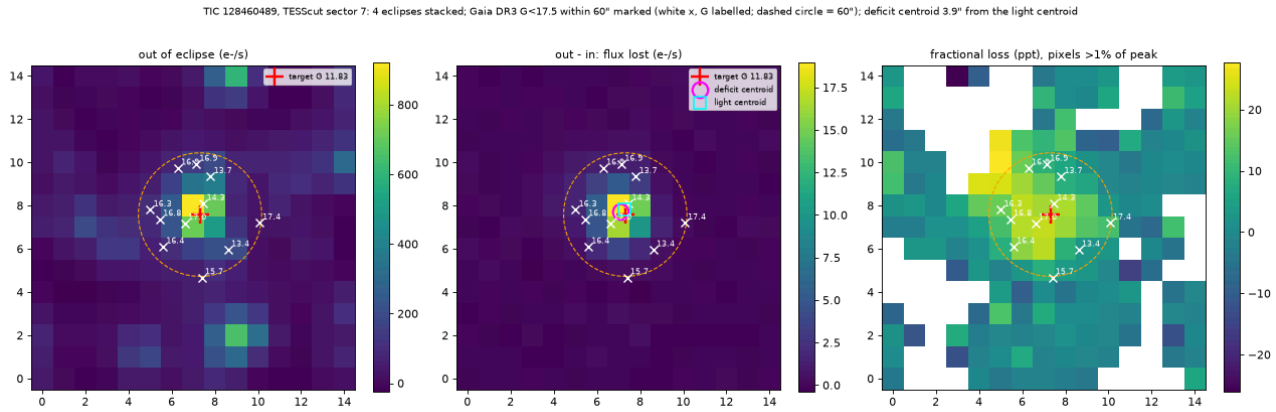
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1493.6722444	0.010417	22.360	usable; same-kind selection applies
7	1	1500.0353892	0.010417	24.308	usable; same-kind selection applies
7	2	1506.3993270	0.010417	23.012	usable; same-kind selection applies
34	116	2231.7657675	0.003472	25.015	usable; same-kind selection applies
34	117	2238.1283990	0.003472	29.085	usable; same-kind selection applies
34	118	2244.4907224	0.003472	27.235	usable; same-kind selection applies
34	119	2250.8542640	0.003472	26.625	usable; same-kind selection applies
88	346	3695.2272142	0.001157	26.324	usable; same-kind selection applies
88	347	3701.5884641	0.001157	25.638	usable; same-kind selection applies
88	348	3707.9517291	0.001157	27.144	usable; same-kind selection applies
88	349	3714.3153850	0.001157	26.796	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s07.json, historical/diff_s07.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R246. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R246; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R246; diff_s07.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R247. TIC 379054533

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 379054533: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 13 distinct eclipse events in selected sectors[14, 55, 82]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 13 same-kind midtimes give P=4.4428656929 ±0.0000195958d, reference eclipse epoch BTJD1686.0962455, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S82 deficit peak[11, 14] vs target pixel[8, 8]; deficit-target separation3.485arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.837. Capable neighbours: Gaia 1861488445861008000: 6.95 arcsec, G17.068, required fractional eclipse0.595, pixel fractional loss0.005536889873247984; Gaia 1861488445861008384: 11.93 arcsec, G12.985, required fractional eclipse0.021, pixel fractional loss0.004008526707694567; Gaia 1861488445861007232: 12.35 arcsec, G16.932, required fractional eclipse0.526, pixel fractional loss0.00610347105169857; Gaia 1861488480220748928: 18.63 arcsec, G16.932, required fractional eclipse0.526, pixel fractional loss0.0035429415590126586; Gaia 1861488441550000896: 27.13 arcsec, G16.963, required fractional eclipse0.541, pixel fractional loss0.004249971627127366; Gaia 1861488514580486016: 28.13 arcsec, G13.346, required fractional eclipse0.026, pixel fractional loss0.0020125508599080602; Gaia 1861488445861005696: 29.78 arcsec, G17.337, required fractional eclipse0.761, pixel fractional loss0.004249971627127366; Gaia 1861488480220750720: 29.95 arcsec, G16.582, required fractional eclipse0.383, pixel fractional loss0.0035429415590126586; Gaia 1861488480220750464: 30.68 arcsec, G17.125, required

fractional eclipse0.627, pixel fractional loss0.0035429415590126586; Gaia 1861488548940228352: 32.07 arcsec, G17.191, required fractional eclipse0.666, pixel fractional loss0.0035429415590126586; Gaia 1861488514580489600: 33.12 arcsec, G17.008, required fractional eclipse0.564, pixel fractional loss0.0030123842254599752; Gaia 1861488514580487424: 46.11 arcsec, G16.447, required fractional eclipse0.339, pixel fractional loss-0.0003105896797276555; Gaia 1861488548940229888: 47.60 arcsec, G15.629, required fractional eclipse0.163, pixel fractional loss-0.00041136836152545964; Gaia 1861488411501266688: 48.52 arcsec, G16.747, required fractional eclipse0.445, pixel fractional loss-0.0003528013488482508; Gaia 1861488235391555456: 49.23 arcsec, G17.402, required fractional eclipse0.807, pixel fractional loss0.0023865347480355233; Gaia 1861488514580486912: 50.11 arcsec, G15.726, required fractional eclipse0.178, pixel fractional loss-0.0003105896797276555; Gaia 1861488475909766528: 52.03 arcsec, G17.592, required fractional eclipse0.960, pixel fractional loss0.0047536375912856565; Gaia 1861488235391555328: 52.46 arcsec, G17.043, required fractional eclipse0.582, pixel fractional loss0.0009135255125798424; Gaia 1861488548940230784: 55.41 arcsec, G17.350, required fractional eclipse0.770, pixel fractional loss0.003415755548098546; Gaia 1861488239702690944: 58.13 arcsec, G17.405, required fractional eclipse0.809, pixel fractional loss0.00019595392004733594

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths5.337/4.380/3.702ppt, with 6 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median8.985ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3186148844586100825 6, P=4.4421085480±0.0015018636d, e=0.092033, omega=131.7871deg. Closest edge-on conjunction offset0.070473d (0.015865phase), full-SB1-covariance linear propagated Gaia uncertainty0.342971d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.150613d (photometric period projection allowance0.005722d). The approximate central duration11.53h vs observed median9.76h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field with capable bright neighbor at11.93arcsec requiring only2.1% eclipse; target-centered deficit favors target but uncalibrated localization cannot securely distinguish close blended contributors. Attribution remains uncertain.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

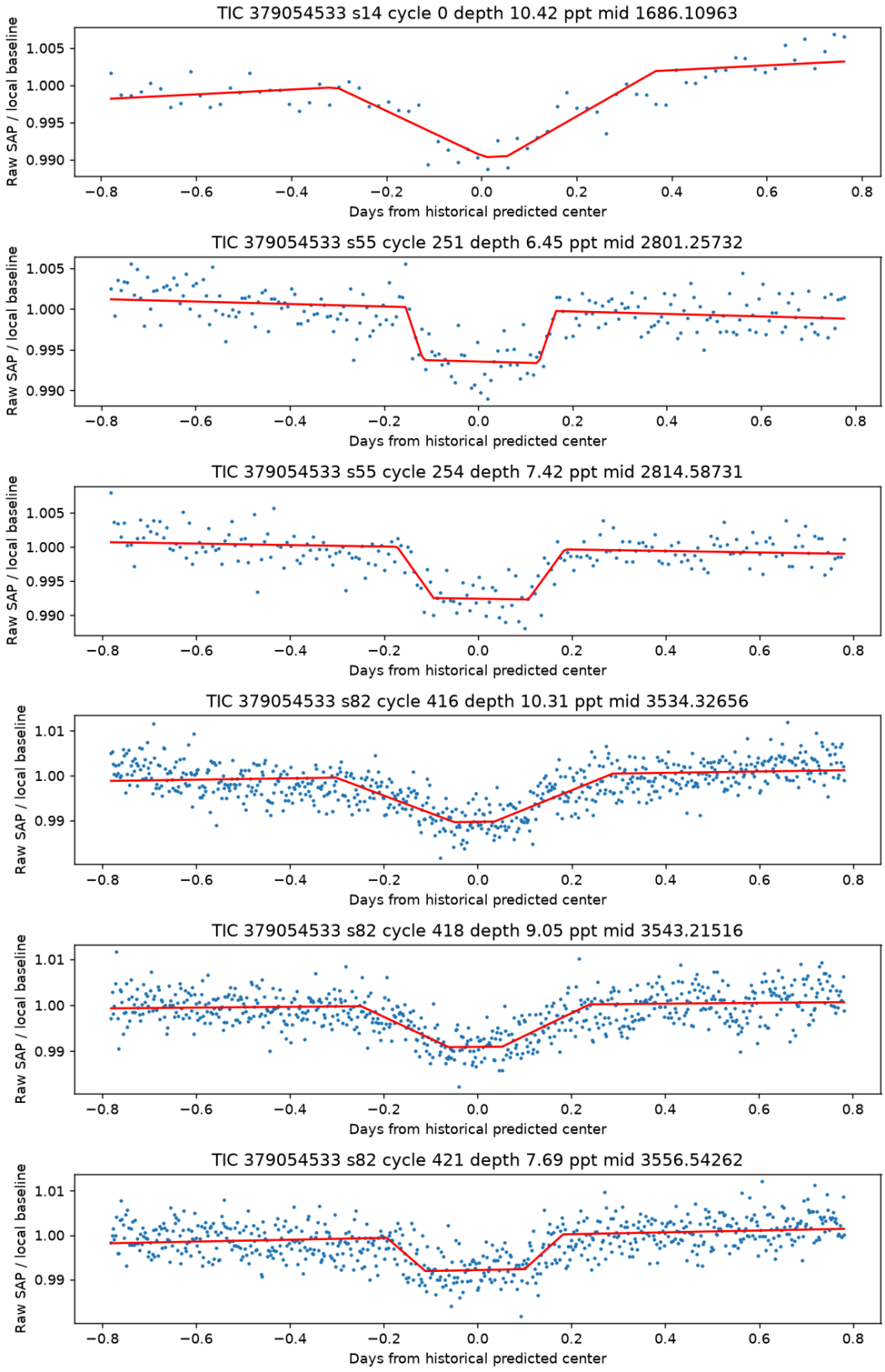
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	0	1686.1096297	0.016228	10.417	usable; same-kind selection applies
14	3	1699.4104001	0.010417	9.382	usable; same-kind selection applies
55	251	2801.2573159	0.004272	6.453	usable; same-kind selection applies
55	252	2805.6847193	0.004997	8.351	usable; same-kind selection applies

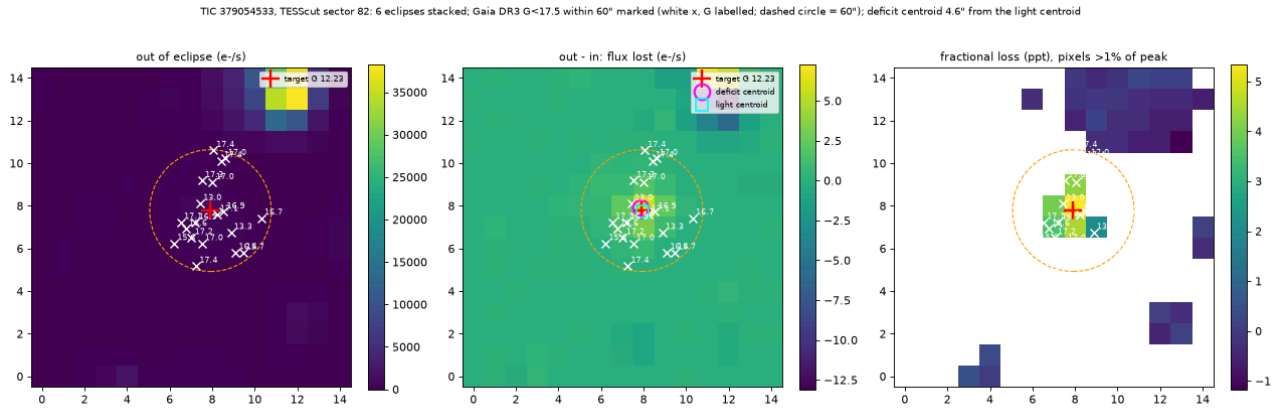
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
55	254	2814.5873124	0.004381	7.424	usable; same-kind selection applies
55	255	2819.0322955	0.003472	8.426	usable; same-kind selection applies
55	256	2823.4736602	0.007813	6.168	usable; same-kind selection applies
82	416	3534.3265638	0.006287	10.307	usable; same-kind selection applies
82	417	3538.7659109	0.007079	9.135	usable; same-kind selection applies
82	418	3543.2151644	0.006051	9.055	usable; same-kind selection applies
82	419	3547.6517174	0.007935	9.203	usable; same-kind selection applies
82	420	3552.1137455	0.013553	8.985	usable; same-kind selection applies
82	421	3556.5426194	0.004581	7.685	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s82.json, historical/diff_s82.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R247. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R247; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R247; diff_s82.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R248. TIC 362822264

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 362822264: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 2 distinct eclipse events in selected sectors[100, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 2 same-kind midtimes give P=18.1224025318 ±0.0004427023d, reference eclipse epoch BTJD4088.4759074, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S39 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation4.079arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.876. Capable neighbours: Gaia 5809488458121122688: 25.90 arcsec, G16.725, required fractional eclipse0.916, pixel fractional loss0.006363474205114983; Gaia 5809488458121125376: 27.52 arcsec, G14.468, required fractional eclipse0.129, pixel fractional loss0.006363474205114983; Gaia 5809489179675649024: 35.87 arcsec, G14.902, required fractional eclipse0.184, pixel fractional loss0.0015995434889155829; Gaia 5809489076596420224: 36.21 arcsec, G15.634, required fractional eclipse0.346, pixel fractional loss0.009451136465246602; Gaia 5809488458121137920: 38.81 arcsec, G14.778, required fractional eclipse0.166, pixel fractional loss0.007186929794530505; Gaia 5809489179675647104: 40.29 arcsec, G14.576, required fractional eclipse0.141, pixel fractional loss0.004786218431079745; Gaia 5809489076596419584: 42.94 arcsec, G16.597, required fractional eclipse0.816, pixel fractional loss0.000594145698815468; Gaia 5809488458121120128: 44.18 arcsec, G15.916, required fractional eclipse0.444, pixel fractional loss-0.00032071891081499976; Gaia 5809489076596420992: 48.38 arcsec, G16.615, required fractional

eclipse0.830, pixel fractional loss0.000594145698815468; Gaia 5809488458121518336: 50.76 arcsec, G16.488, required fractional eclipse0.740, pixel fractional loss-0.00032071891081499976; Gaia 5809489214035390848: 56.68 arcsec, G13.380, required fractional eclipse0.058, pixel fractional loss0.0001064147781249135; Gaia 5809489209723092096: 58.25 arcsec, G12.624, required fractional eclipse0.037, pixel fractional loss0.0013891626777344018

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths17.021/13.194/6.730ppt, with 2 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median17.722ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35809488423761389440, P=18.1258785067±0.0037695277d, e=0.396175, omega=68.1766deg. Closest edge-on conjunction offset-0.448298d (-0.024732phase), full-SB1-covariance linear propagated Gaia uncertainty0.750828d at the measured epoch. Near-Gaia-reference projected eclipse offset0.260801d (photometric period projection allowance0.090311d). The approximate central duration6.61h vs observed median3.80h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Two QLP measured events separated by four cycles give a conditional ephemeris. Independent raw cutout S39 centers2366.85039+/-0.00592 and2384.98788+/-0.00879 BTJD, separation18.13749+/-0.01060d, support same-kind adjacent-cycle recurrence near18.1224d; see helper/same-sector-refits.json. Alternative sector39 cutout shows target-centered deficit; nearest capable neighbor25.9arcsec. Alias search is bounded rather than exhaustive.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

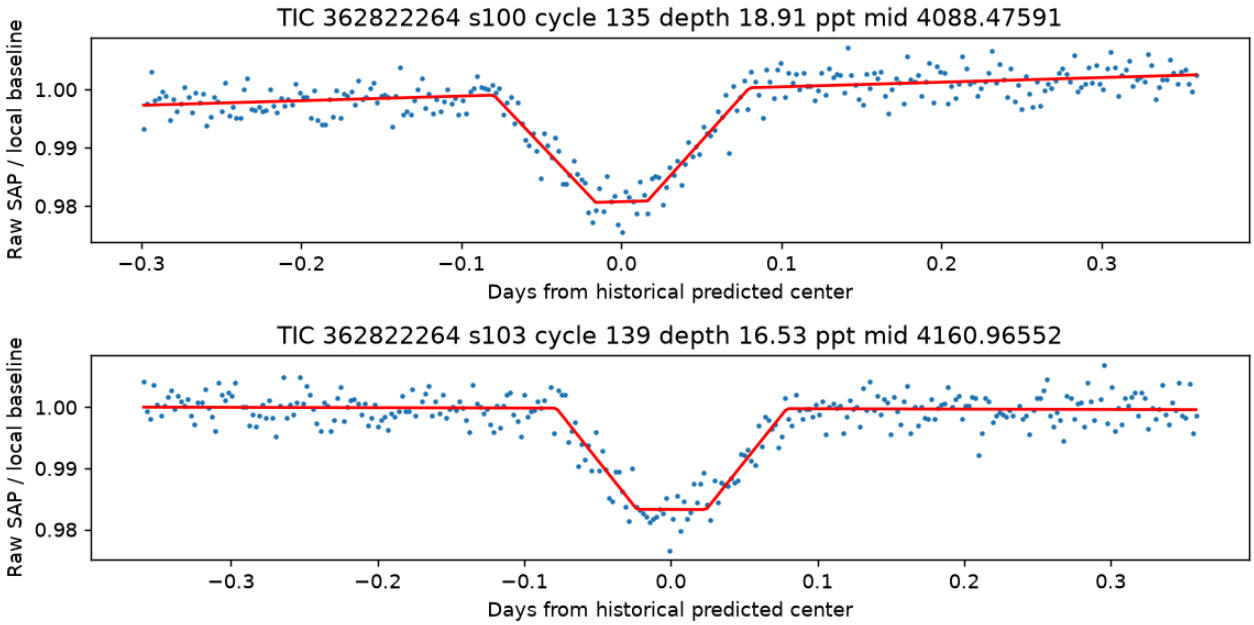
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

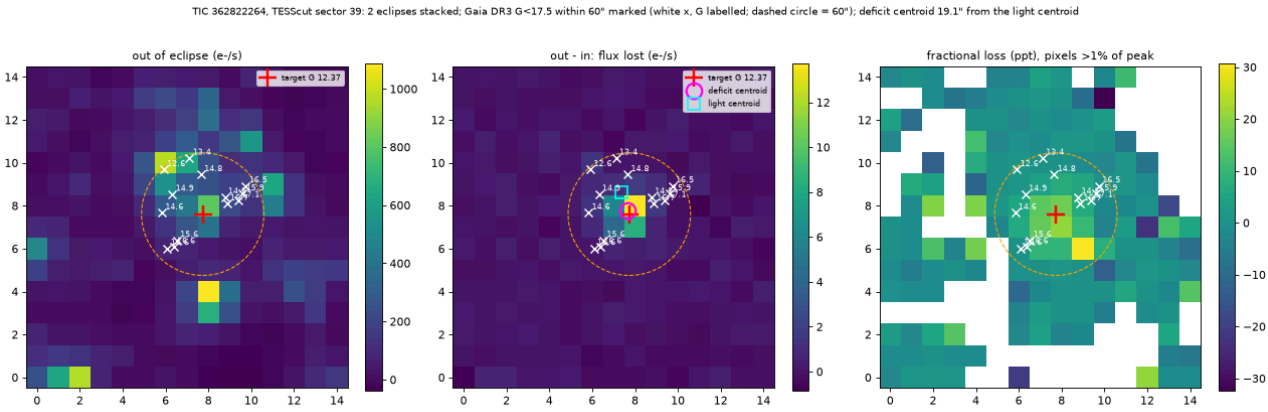
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
100	135	4088.4759074	0.001340	18.914	usable; same-kind selection applies
103	139	4160.9655175	0.001157	16.530	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s39.json, historical/diff_s39.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R248. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R248; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R248; diff_s39.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R249. TIC 314047448

physical_verdict: PLAUSIBLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Consecutive same-kind resolved eclipse timings; source attribution

Archived source: sanitized report text.

TIC 314047448: PLAUSIBLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 2 distinct eclipse events in selected sectors[77, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: conditional ephemeris; period not independently tested.** 2 same-kind midtimes give P=16.3753656517 ±0.0004502238d, reference eclipse epoch BTJD3397.5479750, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S24 deficit peak[11, 8] vs target pixel[8, 8]; deficit-target separation13.232arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual1.000. Capable neighbours: Gaia 2013332341680377600: 27.20 arcsec, G14.774, required fractional eclipse0.083, pixel fractional loss0.00021165848446181757; Gaia 2013332513479061504: 34.61 arcsec, G16.402, required fractional eclipse0.348, pixel fractional loss-0.00028384734199898555; Gaia 2013332307320743040: 40.41 arcsec, G16.114, required fractional eclipse0.269, pixel fractional loss-0.0026636487312149573; Gaia 2013332513479058560: 45.31 arcsec, G14.753, required fractional eclipse0.081, pixel fractional loss-0.0030179173295920513; Gaia 2013332513479057792: 45.81 arcsec, G17.240, required fractional eclipse0.745, pixel fractional loss-0.0030179173295920513; Gaia 2013332238601177728: 47.17 arcsec, G16.121, required fractional eclipse0.270, pixel fractional loss-0.008904227173039583; Gaia 2013331517046673536: 50.32 arcsec, G15.943, required fractional eclipse0.230, pixel fractional loss-0.004864827679816797; Gaia 2013331517046676992: 58.04 arcsec, G15.929, required fractional eclipse0.227, pixel fractional loss-0.004864827679816797

4. **Alternative reduction: inconclusive after completed check.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths4.502/3.227/1.299ppt, with 2 stacked events. Poor S24 extraction for this target is explicitly inconclusive rather than a validated alternative eclipse recovery. QLP selected-kind fitted median8.026ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3201333234168038092 8, P=16.3692406983±0.0073099680d, e=0.373760, omega=242.9939deg. Closest edge-on conjunction offset1.057420d (0.064598phase), full-SB1-covariance linear propagated Gaia uncertainty1.336052d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.069571d (photometric period projection allowance0.082841d). The approximate central duration4.49h vs observed median4.08h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Historical sector24 deficit has large PSF residual and displaced bright-source structure; target attribution remains unresolved after completed independent helper audit. Only two long-separated raw events have usable timings, with no independent adjacent same-kind interval; the period is untested against integer-cycle alternatives, so the physical tier is PLAUSIBLE. Two measured QLP events fourteen historical cycles apart support timing conditional on integer-cycle assignment; alternative cycle aliases are not independently eliminated. The helper S24 cutout center fits failed (bounds/zero depth), not valid timings or nulls. Gaia conjunction offset0.0646phase has propagated uncertainty1.34d, not an arbitrary inconsistency veto.

Next check: Measure consecutive same-kind eclipses with resolved photometry to test the period and source attribution.

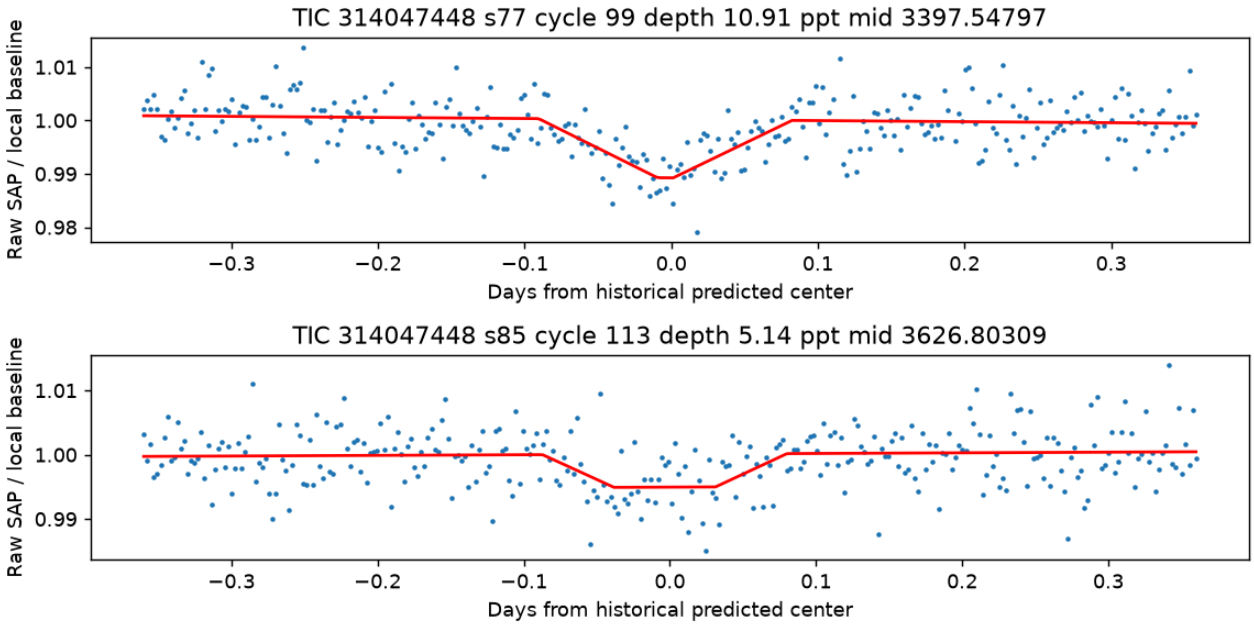
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

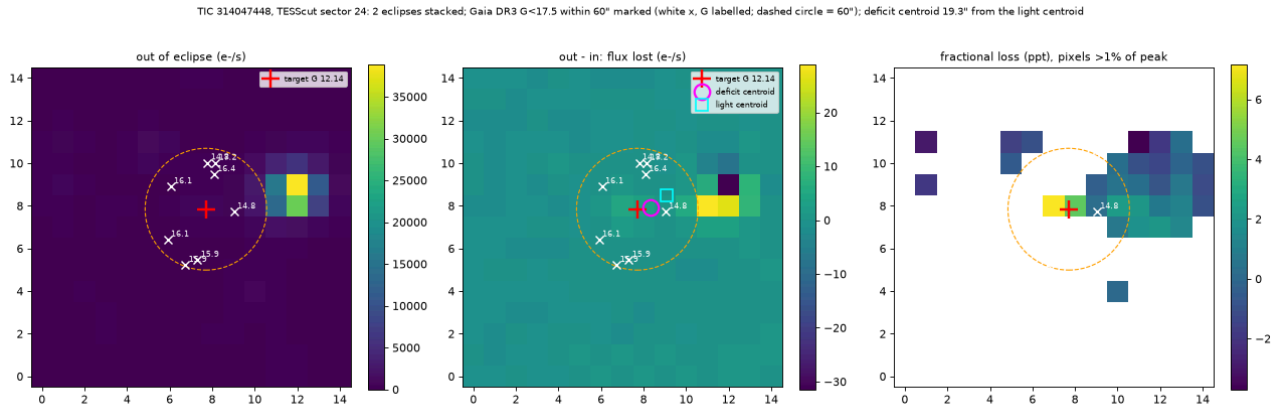
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
77	99	3397.5479750	0.003385	10.914	usable; same-kind selection applies
85	113	3626.8030941	0.005317	5.138	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s24.json, historical/diff_s24.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R249. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R249; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R249; diff_s24.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R250. TIC 373621318

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 373621318: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[58, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 11 same-kind midtimes give P=4.1104208467 ±0.0000052818d, reference eclipse epoch BTJD2883.5677296, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S58 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.280arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.941. Capable neighbours: Gaia 511214808904243584: 30.05 arcsec, G14.117, required fractional eclipse0.169, pixel fractional loss0.004656160732442441; Gaia 511214052989995264: 33.49 arcsec, G13.916, required fractional eclipse0.145, pixel fractional loss0.0054985862487090705; Gaia 511214430947407872: 34.68 arcsec, G15.661, required fractional eclipse0.614, pixel fractional loss0.005682766497404662; Gaia 511214774544514560: 34.80 arcsec, G15.160, required fractional eclipse0.397, pixel fractional loss0.004714198414336461; Gaia 511214430947117824: 41.03 arcsec, G14.835, required fractional eclipse0.302, pixel fractional loss0.005682766497404662; Gaia 511214430947115264: 45.57 arcsec, G15.688, required fractional eclipse0.629, pixel fractional loss0.005682766497404662; Gaia 511214362227651712: 51.37 arcsec, G15.873, required fractional eclipse0.741, pixel fractional loss0.008128411725709894
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths26.264/17.422/6.989ppt, with 7 stacked events. Historical outputs are retained

as measured pipeline evidence. QLP selected-kind fitted median29.444ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3511214048684134528, P=4.1103587301±0.0002578027d, e=0.020847, omega=29.4745deg. Closest edge-on conjunction offset0.061699d (0.015011phase), full-SB1-covariance linear propagated Gaia uncertainty0.158656d at the measured epoch. Near-Gaia-reference projected eclipse offset0.023994d (photometric period projection allowance0.003206d). The approximate central duration6.16h vs observed median3.62h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated narrow eclipses and target-centered deficit favor target; bright field source affects whole-image PSF residual and dilution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

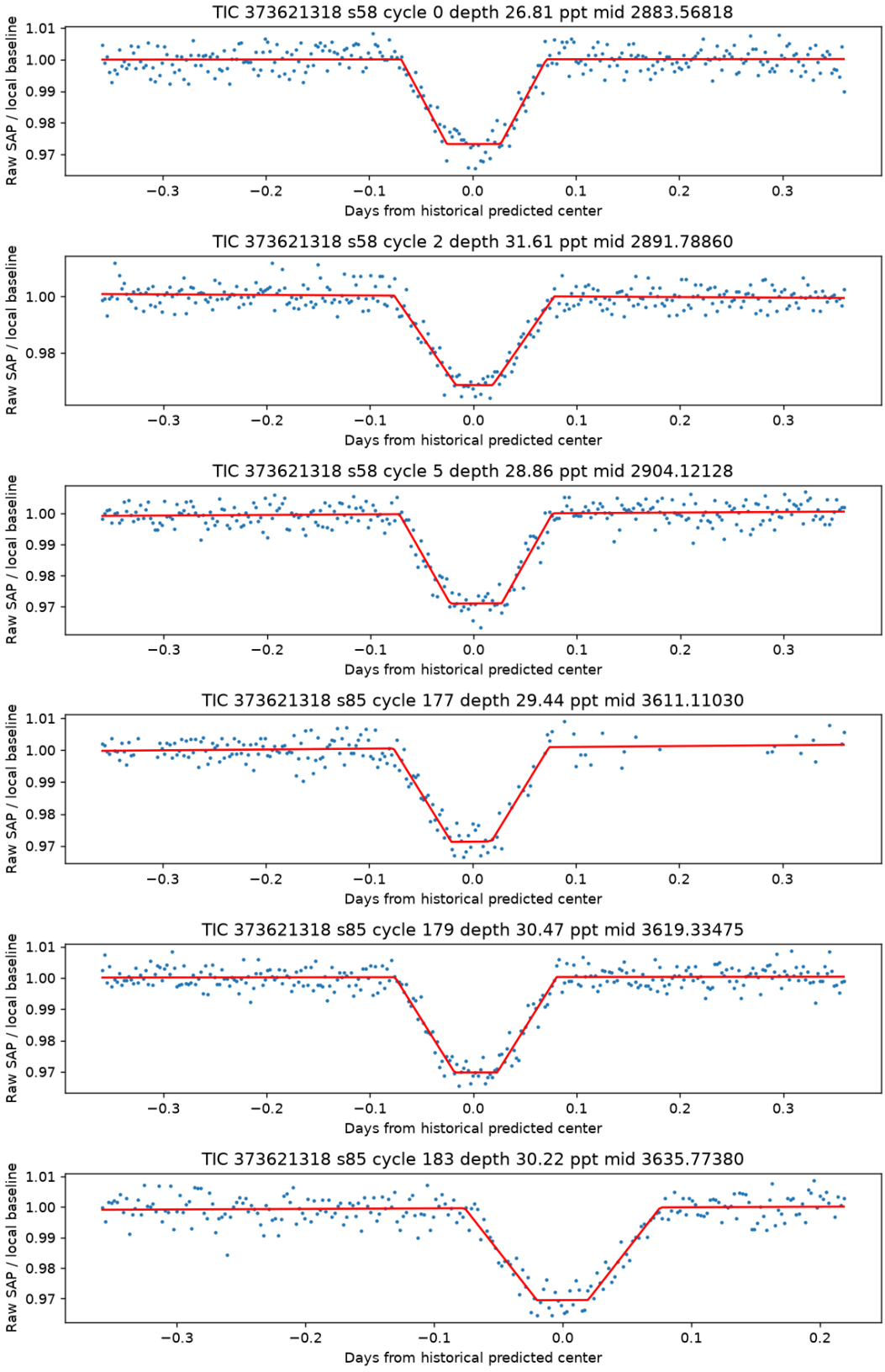
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

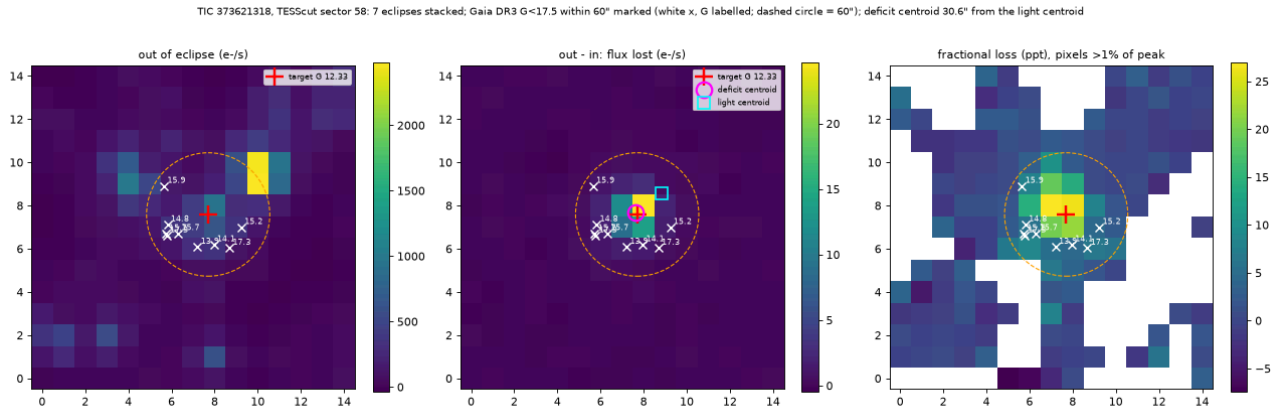
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
58	0	2883.5681774	0.001157	26.815	usable; same-kind selection applies
58	1	2887.6759071	0.001157	29.832	usable; same-kind selection applies
58	2	2891.7886027	0.001157	31.608	usable; same-kind selection applies
58	4	2900.0092722	0.001157	32.323	usable; same-kind selection applies
58	5	2904.1212799	0.001157	28.857	usable; same-kind selection applies
58	6	2908.2307234	0.001157	27.630	usable; same-kind selection applies
85	177	3611.1102951	0.001157	29.444	usable; same-kind selection applies
85	178	3615.2249131	0.001157	29.014	usable; same-kind selection applies
85	179	3619.3347543	0.001157	30.474	usable; same-kind selection applies
85	182	3631.6632148	0.001157	28.443	usable; same-kind selection applies
85	183	3635.7738019	0.001157	30.219	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s58.json, historical/diff_s58.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R250. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R250; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R250; diff_s58.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R251. TIC 129765199

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 129765199: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[3, 97, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 5 same-kind midtimes give P=18.2494315475 ±0.0000459899d, reference eclipse epoch BTJD1385.4630846, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S3 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.721arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.152. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths124.676/125.418/125.076ppt, with 2 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median108.159ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3496399428727525900 8, P=18.2510587758±0.0033978994d, e=0.275285, omega=357.8710deg. Closest edge-on conjunction offset0.029584d (0.001621phase), full-SB1-covariance linear propagated Gaia uncertainty0.275262d at the measured epoch. Near-Gaia-reference projected eclipse offset0.119082d (photometric period projection allowance0.002529d). The approximate central duration10.13h vs observed median8.54h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeating eclipses, target-centered deficit, no capable catalogued neighbor. Sector-dependent depth is dilution or extraction dependent.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

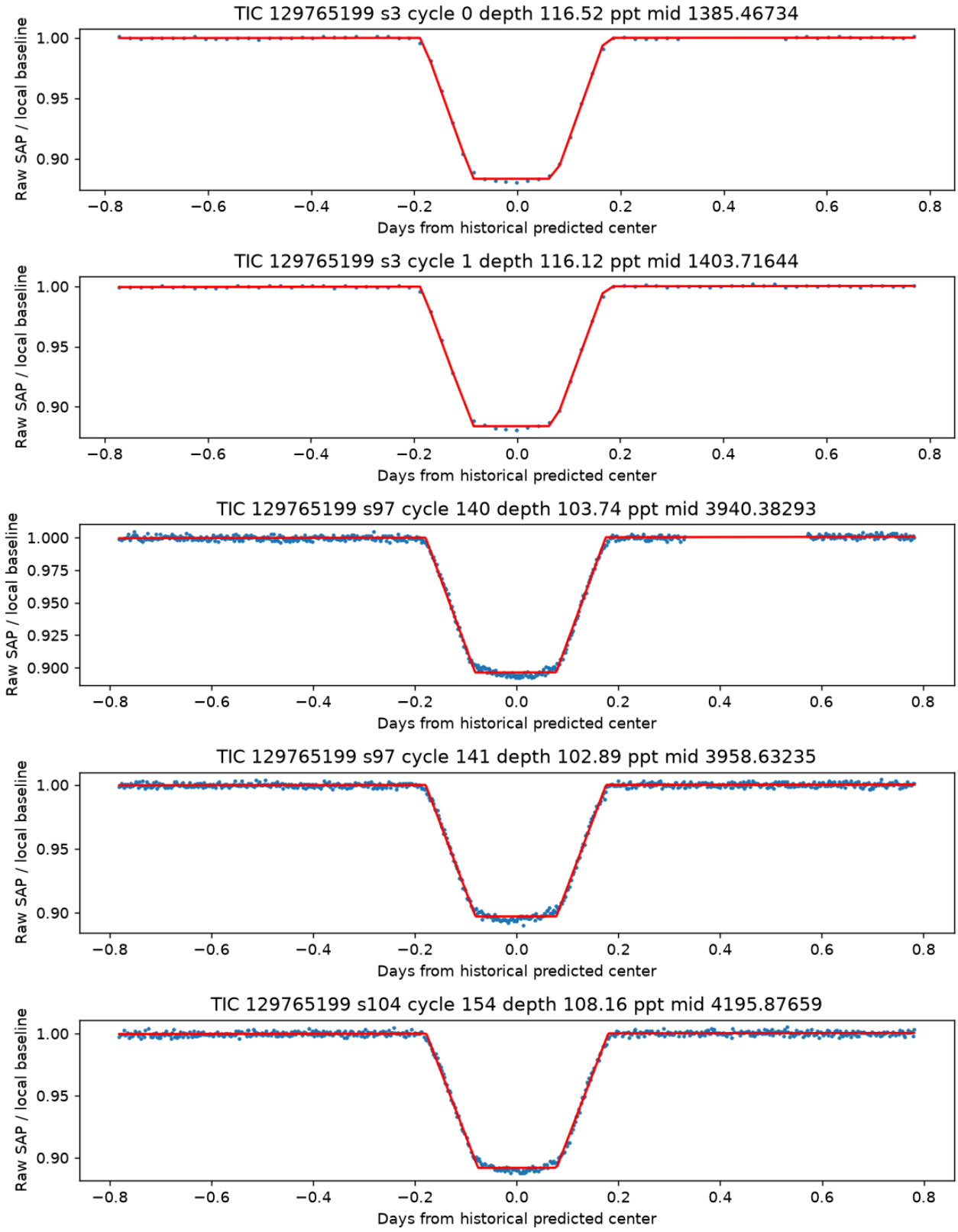
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

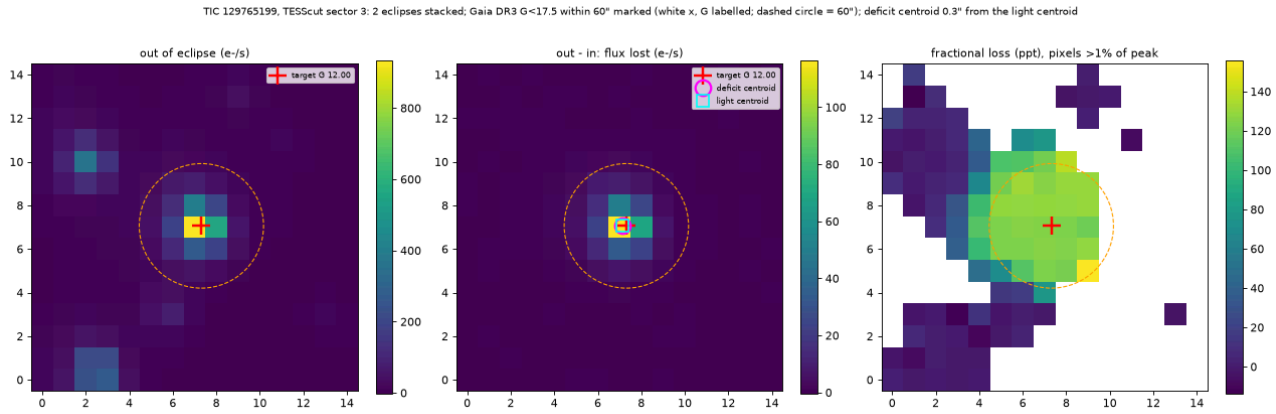
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
3	0	1385.4673363	0.010417	116.516	usable; same-kind selection applies
3	1	1403.7164435	0.010417	116.116	usable; same-kind selection applies
97	140	3940.3829309	0.001157	103.742	usable; same-kind selection applies
97	141	3958.6323546	0.001157	102.892	usable; same-kind selection applies
104	154	4195.8765905	0.001157	108.159	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s03.json, historical/diff_s03.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R251. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R251; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R251; diff_s03.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R252. TIC 309558046

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 309558046: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 9 distinct eclipse events in selected sectors[1, 37, 90]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 6 same-kind midtimes give P=13.0830020618 ±0.0000240012d, reference eclipse epoch BTJD1331.6469978, kind: Deeper primary (even historical half-period cycles). Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S9 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.695arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.165. Capable neighbours: Gaia 5277988591691031808: 34.66 arcsec, G15.095, required fractional eclipse0.291, pixel fractional loss0.00740682275584591; Gaia 5277988660410508160: 41.05 arcsec, G16.170, required fractional eclipse0.751, pixel fractional loss0.00817955277592489; Gaia 5277988660410508928: 54.10 arcsec, G16.233, required fractional eclipse0.794, pixel fractional loss0.007053373948642565
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths13.627/14.520/14.475ppt, with 4 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median26.962ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35277988587391544320, P=13.0837683530±0.0023013833d, e=0.018445, omega=309.8190deg. Closest edge-on conjunction

offset0.062058d (0.004743phase), full-SB1-covariance linear propagated Gaia uncertainty0.209431d at the measured epoch. Near-Gaia-reference projected eclipse offset0.117231d (photometric period projection allowance0.001728d). The approximate central duration17.77h vs observed median14.32h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses approximately27/17ppt establish orbital period twice historical SAP half period. Even half-period cycles are deeper primaries and alone define ephemeris. Historical cutout stacks mixed eclipse kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

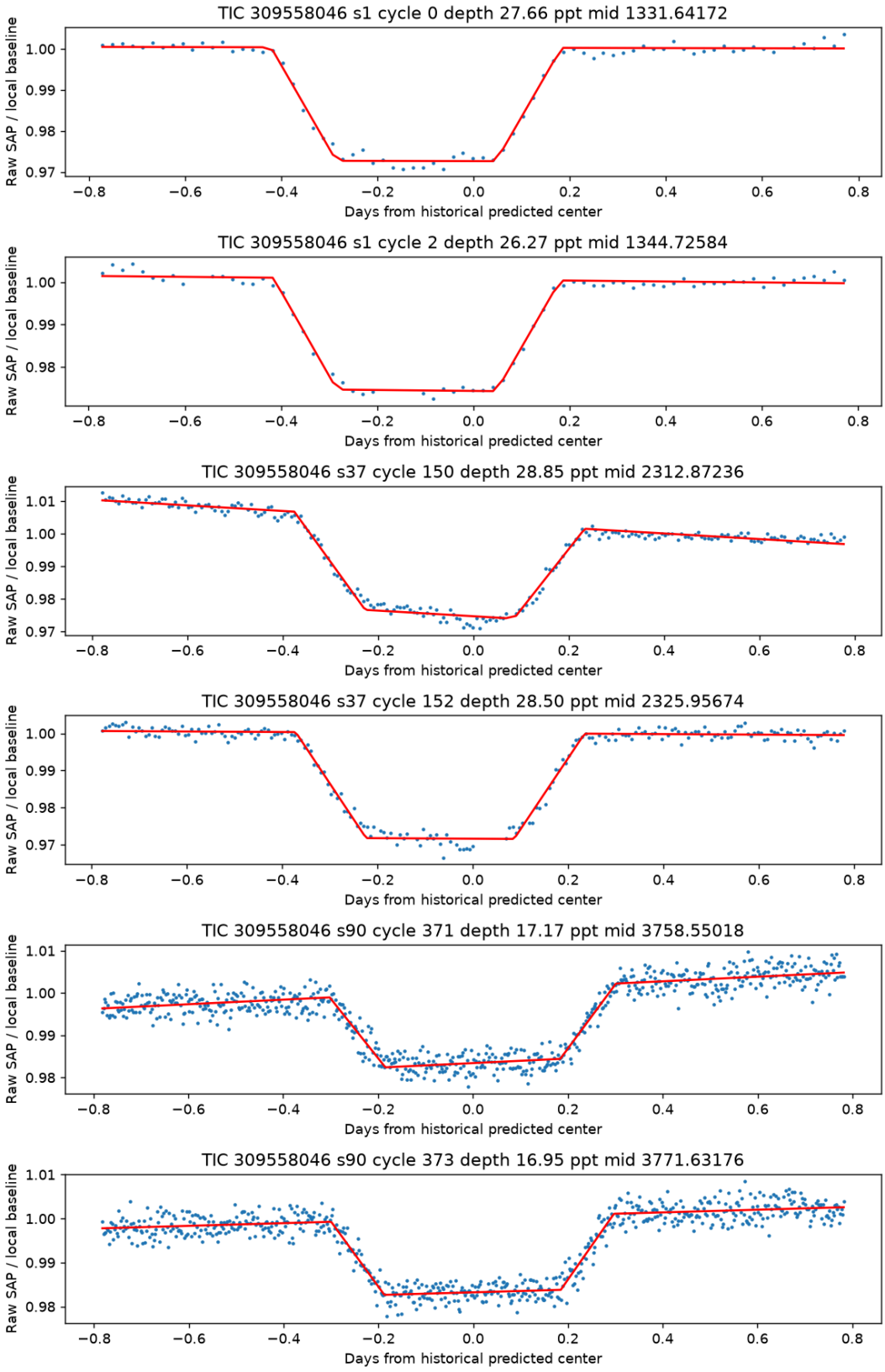
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

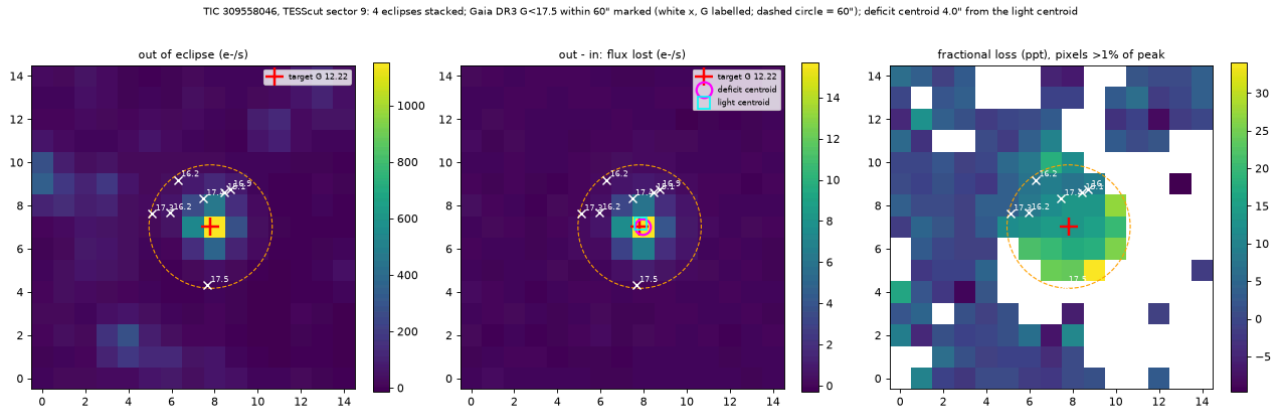
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1331.6417226	0.010416	27.658	usable; same-kind selection applies
1	2	1344.7258353	0.010416	26.267	usable; same-kind selection applies
1	3	1351.2669693	0.024253	16.592	usable; same-kind selection applies
37	150	2312.8723630	0.003472	28.847	usable; same-kind selection applies
37	152	2325.9567397	0.003472	28.504	usable; same-kind selection applies
90	370	3752.0006882	0.002552	24.395	usable; same-kind selection applies
90	371	3758.5501810	0.003074	17.174	usable; same-kind selection applies
90	372	3765.0864708	0.002348	25.496	usable; same-kind selection applies
90	373	3771.6317568	0.002008	16.951	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s09.json, historical/diff_s09.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R252. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R252; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R252; diff_s09.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R253. TIC 419573851

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 419573851: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[11, 64, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 11 same-kind midtimes give P=7.0626836551 ±0.0000742402d, reference eclipse epoch BTJD1601.4763953, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S11 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.538arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.888. Capable neighbours: Gaia 6060988067803130240: 4.34 arcsec, G16.724, required fractional eclipse0.491, pixel fractional loss0.007795596643938603; Gaia 6060988067816452608: 10.69 arcsec, G16.788, required fractional eclipse0.520, pixel fractional loss0.007058937863968618; Gaia 6060988067816321920: 21.99 arcsec, G16.902, required fractional eclipse0.577, pixel fractional loss0.010065040351534618; Gaia 6060985215958167296: 22.39 arcsec, G16.828, required fractional eclipse0.539, pixel fractional loss0.004818000551885144; Gaia 6060988170896549888: 29.59 arcsec, G16.492, required fractional eclipse0.398, pixel fractional loss0.010402585441476401; Gaia 6060988136535806976: 31.34 arcsec, G14.923, required fractional eclipse0.100, pixel fractional loss0.002591926435788933; Gaia 6060985112878951296: 33.95 arcsec, G15.650, required fractional eclipse0.187, pixel fractional loss0.0001420925382612005; Gaia 6060985108554678400: 34.31 arcsec, G17.379, required fractional eclipse0.891, pixel fractional loss0.0001420925382612005; Gaia 6060985112878816512: 34.73 arcsec, G17.411, required

fractional eclipse0.917, pixel fractional loss0.0001420925382612005; Gaia 6060985112878815744: 37.79 arcsec, G13.849, required fractional eclipse0.042, pixel fractional loss0.0001420925382612005; Gaia 6060988170895551104: 40.59 arcsec, G17.116, required fractional eclipse0.700, pixel fractional loss0.0043683648521924716; Gaia 6060985215958039680: 43.67 arcsec, G16.978, required fractional eclipse0.618, pixel fractional loss0.006510666471249432; Gaia 6060988166571446528: 45.38 arcsec, G17.450, required fractional eclipse0.950, pixel fractional loss0.00018582566195307051; Gaia 6060985215958037760: 48.87 arcsec, G17.432, required fractional eclipse0.935, pixel fractional loss0.006510666471249432; Gaia 6060988239615031296: 51.63 arcsec, G15.706, required fractional eclipse0.197, pixel fractional loss0.0010582906118086985; Gaia 6060988132211716096: 52.31 arcsec, G17.486, required fractional eclipse0.982, pixel fractional loss0.002060095278231686; Gaia 6060985215958031744: 53.53 arcsec, G17.041, required fractional eclipse0.655, pixel fractional loss0.0006973246049324945; Gaia 6060985147238555648: 57.91 arcsec, G14.228, required fractional eclipse0.056, pixel fractional loss0.00035066955549270946; Gaia 6060985142918003456: 58.65 arcsec, G17.448, required fractional eclipse0.949, pixel fractional loss-0.0003788039727050195; Gaia 6060988033456574976: 58.72 arcsec, G17.158, required fractional eclipse0.728, pixel fractional loss-0.011902421354625275; Gaia 6060988166571437696: 59.06 arcsec, G17.244, required fractional eclipse0.787, pixel fractional loss-0.0009683391516432161

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths7.796/6.963/6.200ppt, with 4 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median9.231ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3606098806349131980 8, P=7.0622964355±0.0030493697d, e=0.116094, omega=132.3120deg. Closest edge-on conjunction offset-0.135760d (-0.019223phase), full-SB1-covariance linear propagated Gaia uncertainty0.531776d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.202362d (photometric period projection allowance0.012769d). The approximate central duration7.50h vs observed median8.33h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Capable neighbor at4.34arcsec needs49% eclipse. Deficit0.073pixel from target favors target, but close separation and uncalibrated centroid cannot securely exclude neighbor. Completed review retains probable attribution.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

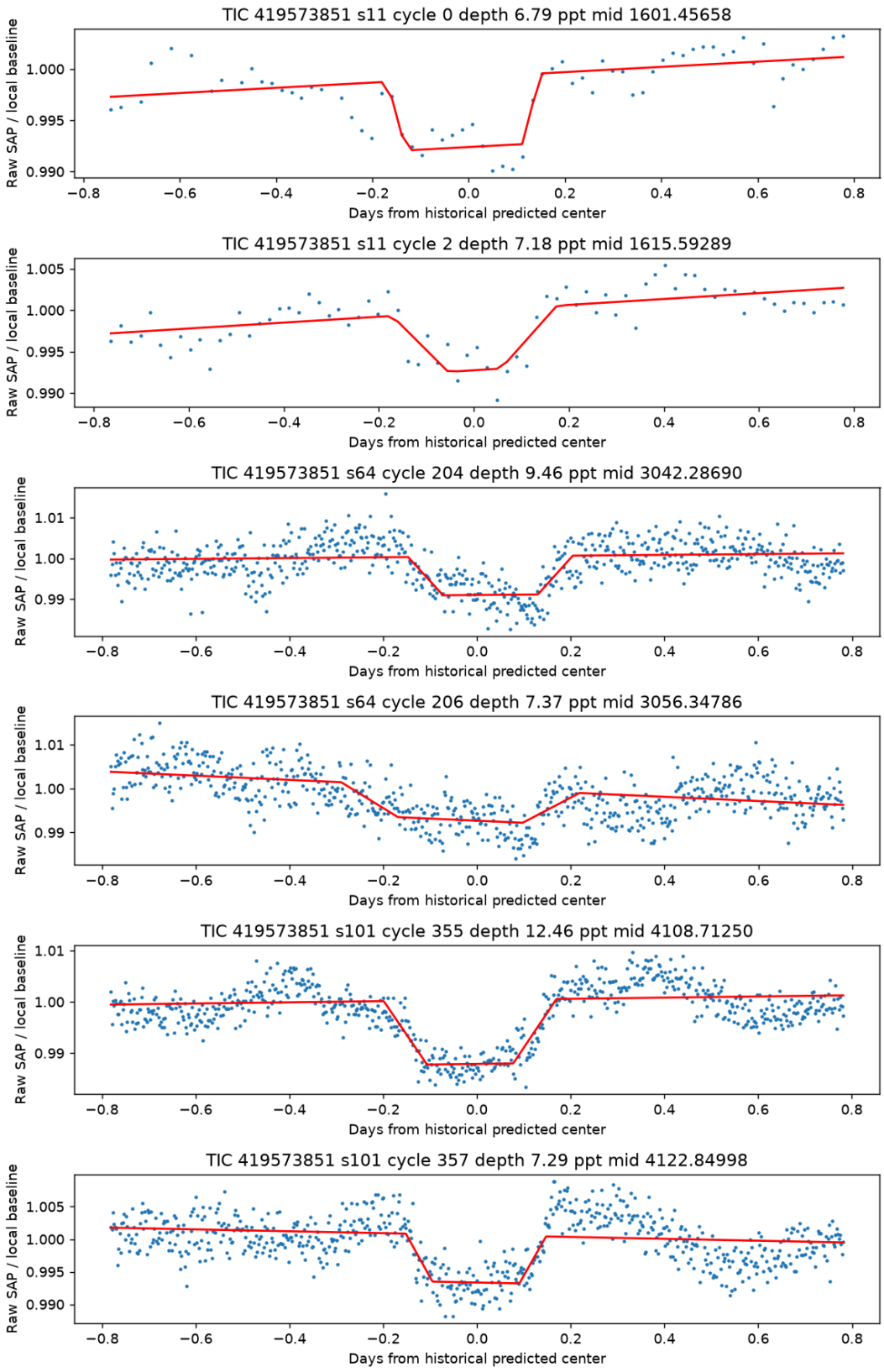
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
11	0	1601.4565824	0.010417	6.789	usable; same-kind selection applies
11	1	1608.5251846	0.015491	8.996	usable; same-kind selection applies

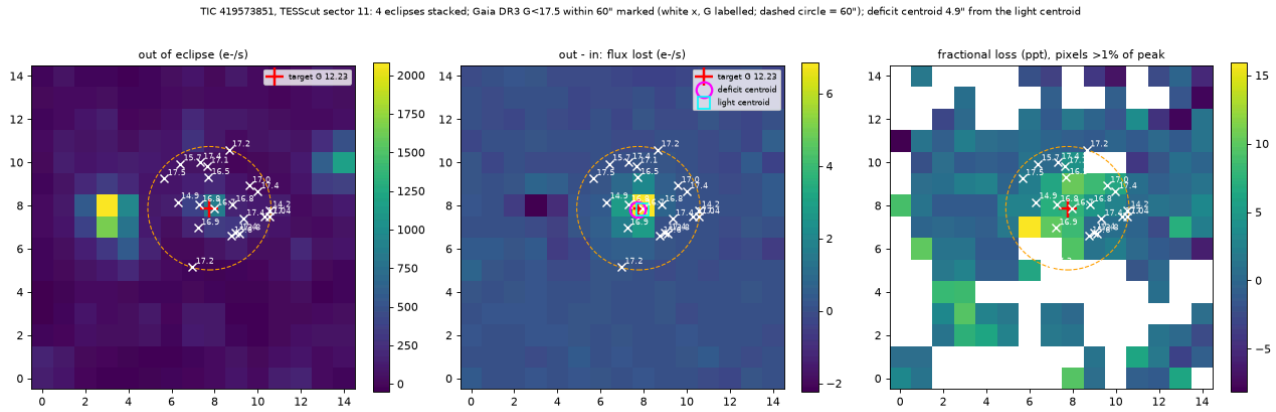
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
11	2	1615.5928942	0.013248	7.180	usable; same-kind selection applies
11	3	1622.6684678	0.010417	9.272	usable; same-kind selection applies
64	204	3042.2868973	0.003795	9.459	usable; same-kind selection applies
64	205	3049.3220088	0.002420	9.866	usable; same-kind selection applies
64	206	3056.3478614	0.008488	7.365	usable; same-kind selection applies
64	207	3063.4579946	0.004179	9.740	usable; same-kind selection applies
101	355	4108.7124966	0.002954	12.460	usable; same-kind selection applies
101	356	4115.8634803	0.006060	9.231	usable; same-kind selection applies
101	357	4122.8499761	0.003370	7.289	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s11.json, historical/diff_s11.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R253. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R253; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R253; diff_s11.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R254. TIC 124007093

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 124007093: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[7, 61, 89]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 7 same-kind midtimes give P=10.0645223865 ±0.0000700239d, reference eclipse epoch BTJD1500.7378621, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S62 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.581arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.749. Capable neighbours: Gaia 5535503079732178432: 18.57 arcsec, G13.455, required fractional eclipse0.083, pixel fractional loss0.013268088594969267; Gaia 5535503281596254080: 19.22 arcsec, G15.535, required fractional eclipse0.449, pixel fractional loss0.014072376440044656; Gaia 5535526543139207808: 41.15 arcsec, G16.294, required fractional eclipse0.882, pixel fractional loss0.0017686528954465618; Gaia 5535526478714025856: 49.25 arcsec, G14.345, required fractional eclipse0.163, pixel fractional loss0.009145411171091576; Gaia 5535503045372439808: 50.59 arcsec, G16.074, required fractional eclipse0.724, pixel fractional loss0.007014614314219571; Gaia 5535503320250307712: 54.31 arcsec, G16.374, required fractional eclipse0.948, pixel fractional loss-0.0010512912208078477
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths16.996/15.975/13.665ppt, with 3 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median21.330ppt. This separately extracted

cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3553503075428606976, P=10.0674356008±0.0026207857d, e=0.038103, omega=258.4752deg. Closest edge-on conjunction offset-0.334392d (-0.033215phase), full-SB1-covariance linear propagated Gaia uncertainty0.335271d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.013938d (photometric period projection allowance0.007703d). The approximate central duration13.86h vs observed median13.00h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Long eclipses coexist with baseline variability; free local linear fits and conservative timing errors preserve its contribution. Target-centered cutout deficit0.17pixel from target favors target, but nearest capable neighbor18.57arcsec needs only8.3% eclipse, and PSF residual0.75 without calibrated centroid uncertainty leaves attribution unresolved. Independent optimizer timing centers differ by up to0.0152d; model systematics are not total-error calibrated.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

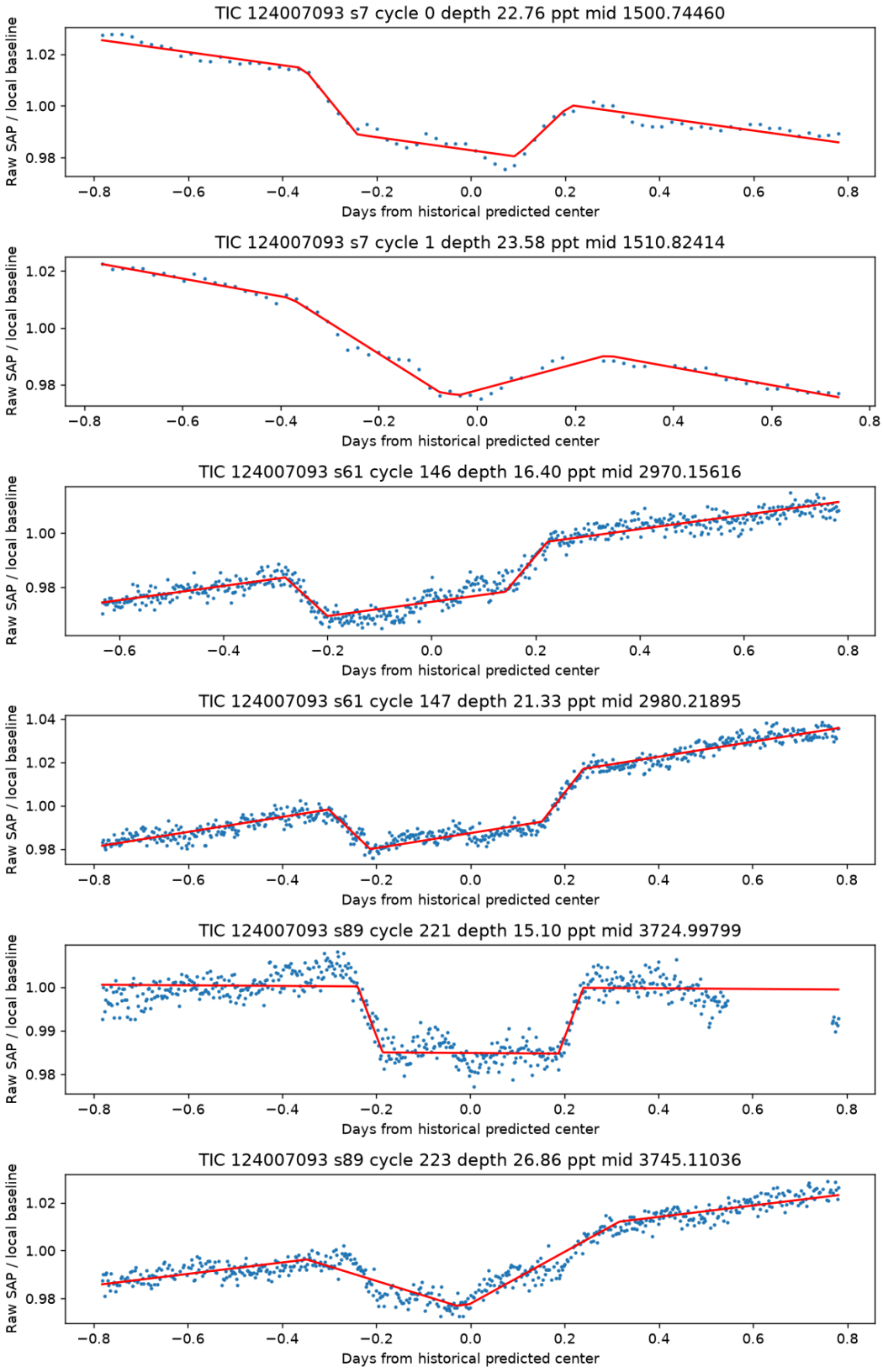
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

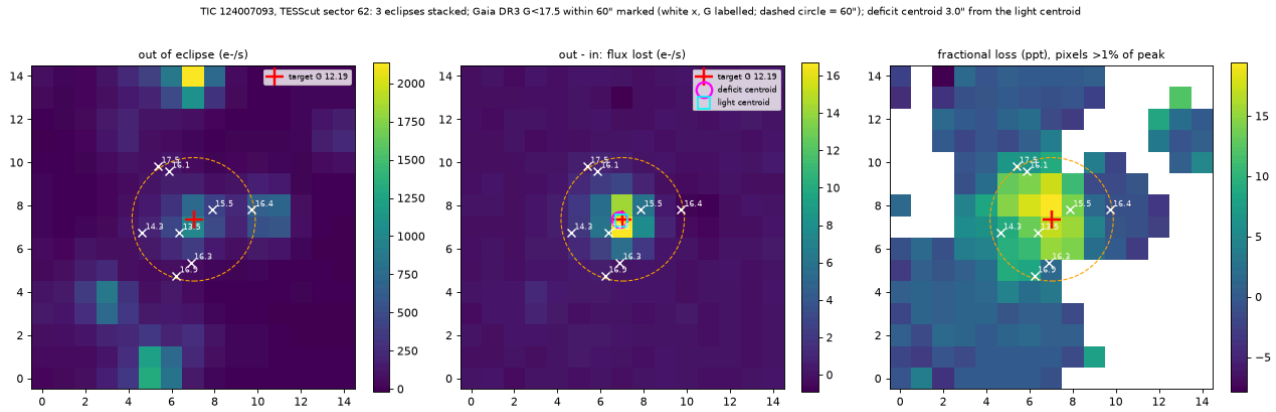
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1500.7445967	0.024319	22.761	usable; same-kind selection applies
7	1	1510.8241398	0.065925	23.579	usable; same-kind selection applies
61	146	2970.1561640	0.008230	16.398	usable; same-kind selection applies
61	147	2980.2189489	0.012409	21.330	usable; same-kind selection applies
89	221	3724.9979898	0.001534	15.103	usable; same-kind selection applies
89	222	3735.0507811	0.008169	16.790	usable; same-kind selection applies
89	223	3745.1103635	0.013566	26.858	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R254. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R254; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R254; diff_s62.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R255. TIC 30753325

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 30753325: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[64, 102]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 8 same-kind midtimes give P=3.8144612029 ±0.0000028664d, reference eclipse epoch BTJD3043.2158652, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S37 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.566arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.353. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths58.070/57.472/55.477ppt, with 6 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median58.989ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3616566906087424998 4, P=3.8146327900±0.0002474851d, e=0.061242, omega=70.4217deg. Closest edge-on conjunction offset-0.055060d (-0.014434phase), full-SB1-covariance linear propagated Gaia uncertainty0.155049d at the measured epoch. Near-Gaia-reference projected eclipse offset0.064365d (photometric period projection allowance0.001995d). The approximate central

duration4.07h vs observed median2.74h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Clean triangular repeating eclipses, target-centered deficit and no capable catalogued neighbor. Depth changes between sectors; no intrinsic undiluted depth inferred.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

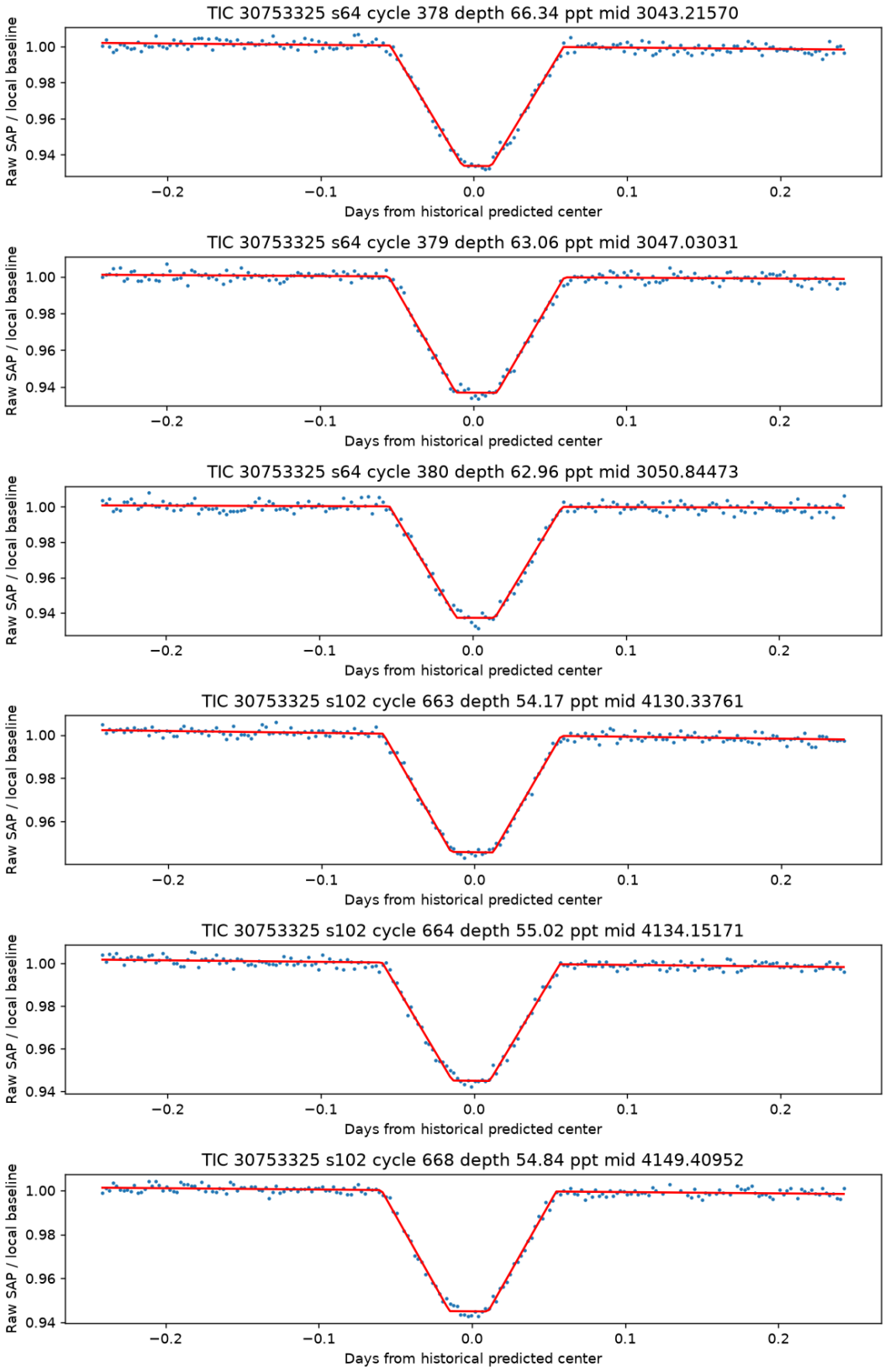
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

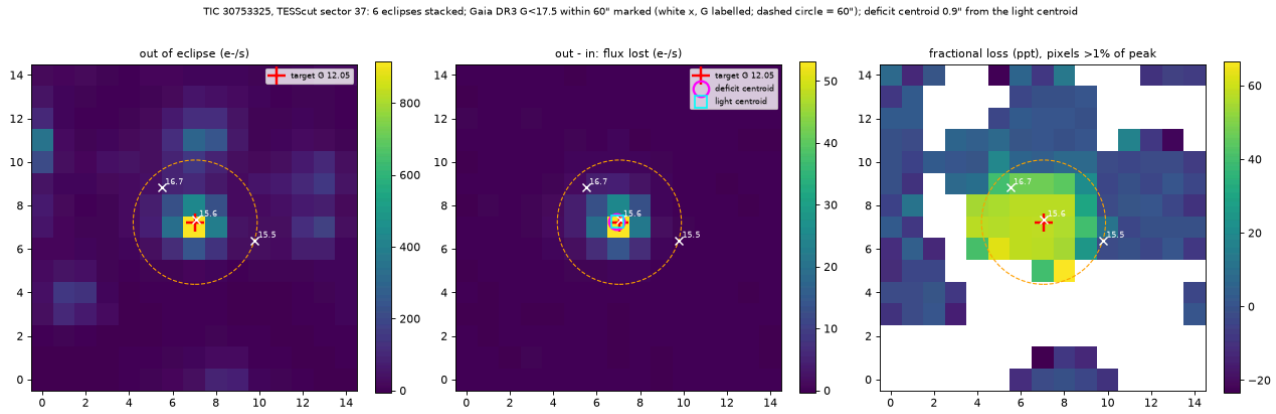
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
64	378	3043.2156965	0.001157	66.336	usable; same-kind selection applies
64	379	3047.0303056	0.001157	63.061	usable; same-kind selection applies
64	380	3050.8447263	0.001157	62.955	usable; same-kind selection applies
64	382	3058.4739603	0.001157	64.027	usable; same-kind selection applies
102	663	4130.3376092	0.001157	54.165	usable; same-kind selection applies
102	664	4134.1517113	0.001157	55.024	usable; same-kind selection applies
102	666	4141.7805440	0.001157	54.718	usable; same-kind selection applies
102	668	4149.4095188	0.001157	54.840	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s37.json, historical/diff_s37.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R255. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R255; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R255; diff_s37.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R256. TIC 48108063

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 48108063: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[35, 62, 89]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass with disclosed cycle assumptions.** 10 same-kind midtimes give P=5.7972370881 ±0.0000058204d, reference eclipse epoch BTJD2272.5117326, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S62 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation0.764arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.508. Capable neighbours: Gaia 5439164283215218688: 15.01 arcsec, G11.053, required fractional eclipse0.183, pixel fractional loss0.03502751096712482
4. **Alternative reduction: eclipse recovered with depth discrepancy.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths35.028/31.419/31.304ppt, with 4 stacked events. Historical outputs are retained as measured pipeline evidence. QLP selected-kind fitted median139.304ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3543916428321802188 8, P=5.7969445240±0.0002979833d, e=0.091414, omega=170.9390deg. Closest edge-on conjunction offset0.077140d (0.013307phase), full-SB1-covariance linear propagated Gaia uncertainty0.096630d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.017943d (photometric period

projection allowance0.001892d). The approximate central duration5.41h vs observed median4.53h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep raw eclipses approximately134-143ppt. Independent measured-ephemeris S62 cutout re-extraction gives3x3 core deficit26.5ppt and1x1 32.1ppt; the factor4-5 contrast persists. Dilution and extraction differences are possible, unestablished explanations. Deficit0.036pixel from target vs neighbor0.72pixel favors target, but15arcsec capable neighbor and uncalibrated centroid leave interpretation unresolved.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

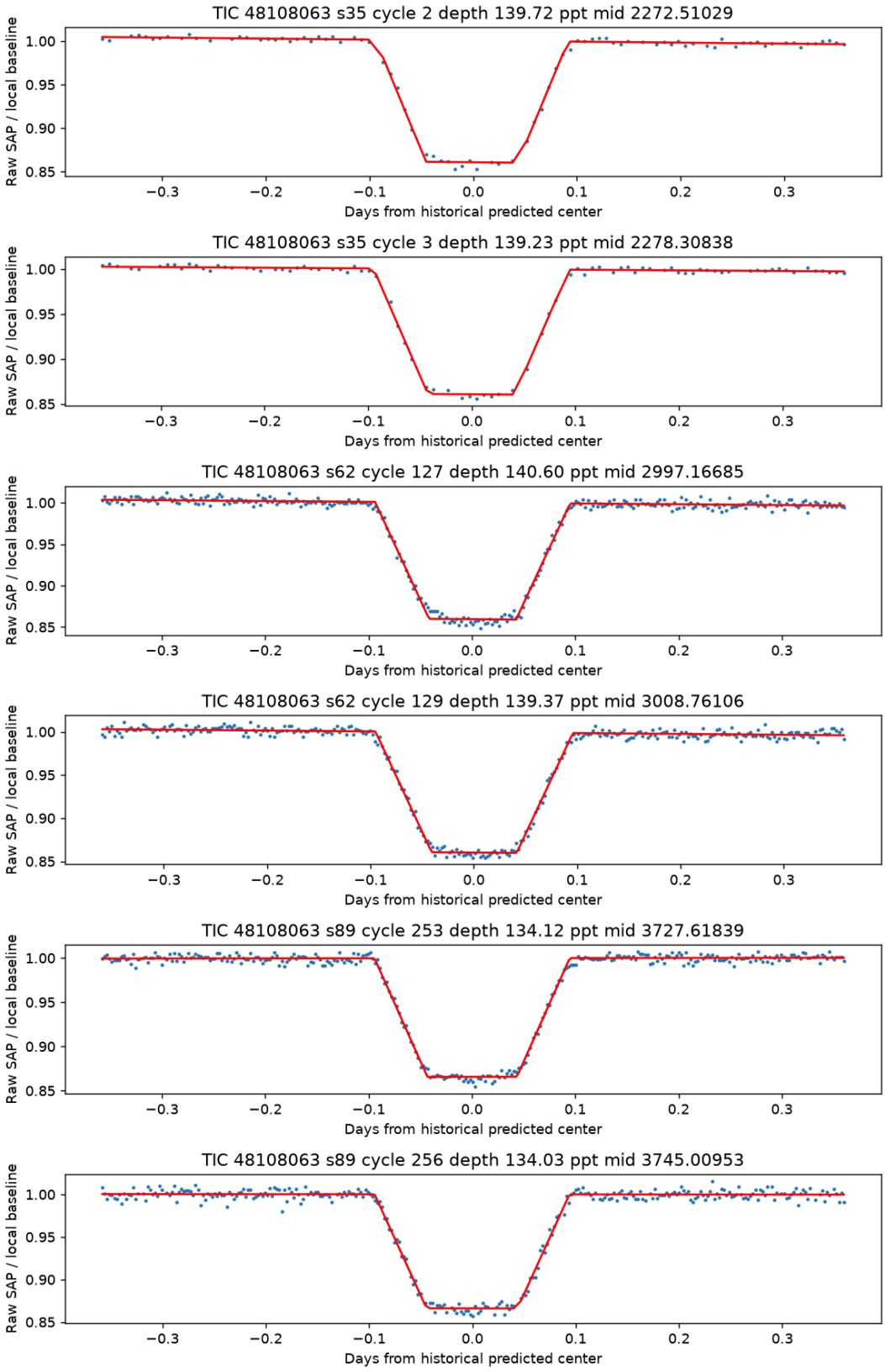
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

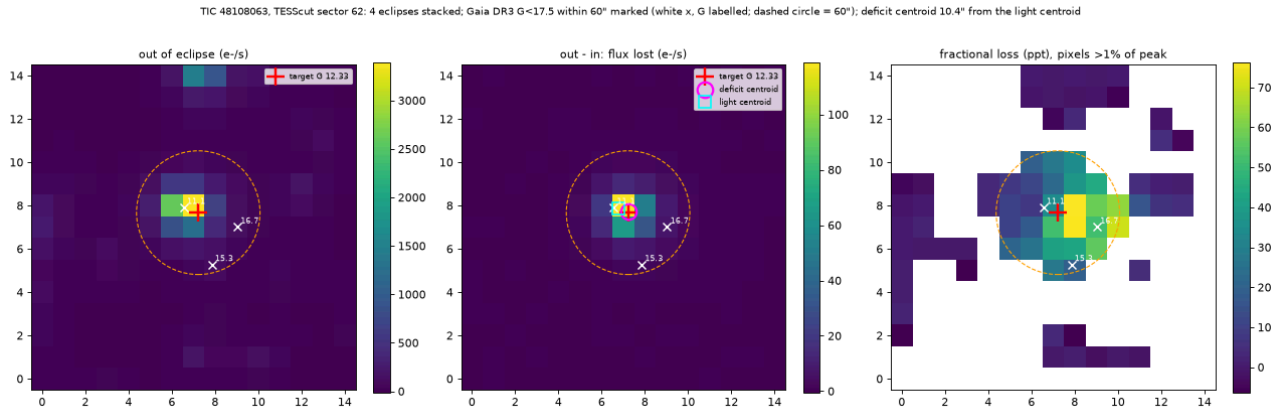
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
35	2	2272.5102891	0.003472	139.721	usable; same-kind selection applies
35	3	2278.3083799	0.003472	139.233	usable; same-kind selection applies
62	126	2991.3686717	0.001157	141.236	usable; same-kind selection applies
62	127	2997.1668504	0.001157	140.595	usable; same-kind selection applies
62	128	3002.9638159	0.001157	142.585	usable; same-kind selection applies
62	129	3008.7610555	0.001157	139.374	usable; same-kind selection applies
89	252	3721.8212328	0.001157	133.229	usable; same-kind selection applies
89	253	3727.6183945	0.001157	134.116	usable; same-kind selection applies
89	254	3733.4152996	0.001157	134.076	usable; same-kind selection applies
89	256	3745.0095320	0.001157	134.026	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): ../helper/pixel-identity-audit.json, ../helper/same-sector-refits.json, ../helper/314047448-measured-diff.json, ../helper/48108063-measured-diff.json; measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R256. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R256; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R256; diff_s62.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R257. TIC 377389892

physical_verdict: CONFIRMED

novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 377389892: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[21, 48, 75]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=4.5881927790 +/-0.0000096916d, reference eclipse epoch BTJD1875.0768278, kind: Deeper primary; shallower secondary at orbital phase0.5. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S21 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.123arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.044. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths228.894/227.798/226.259ppt, with 4 stacked events. QLP selected-kind fitted median227.447ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3854441274902991616, P=4.5880638447+/-0.0001891317d, e=0.018288, omega=77.4620deg. Closest edge-on conjunction offset0.028670d (0.006249phase), full-SB1-covariance linear propagated Gaia uncertainty0.068238d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.013105d (photometric period projection allowance0.003140d). The approximate central duration5.18h vs observed median4.54h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Known VSX EA within0.02arcsec; excluded from new-system count. Catalogue1.147d is a quarter-period alias: raw fold shows deeper primary each4.588d and shallower secondary at phase0.5. Fold at1.147d mixes out-of-eclipse and unequal eclipse points. Primary epoch identifies deeper eclipse; physical Gaia orbit association retained.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

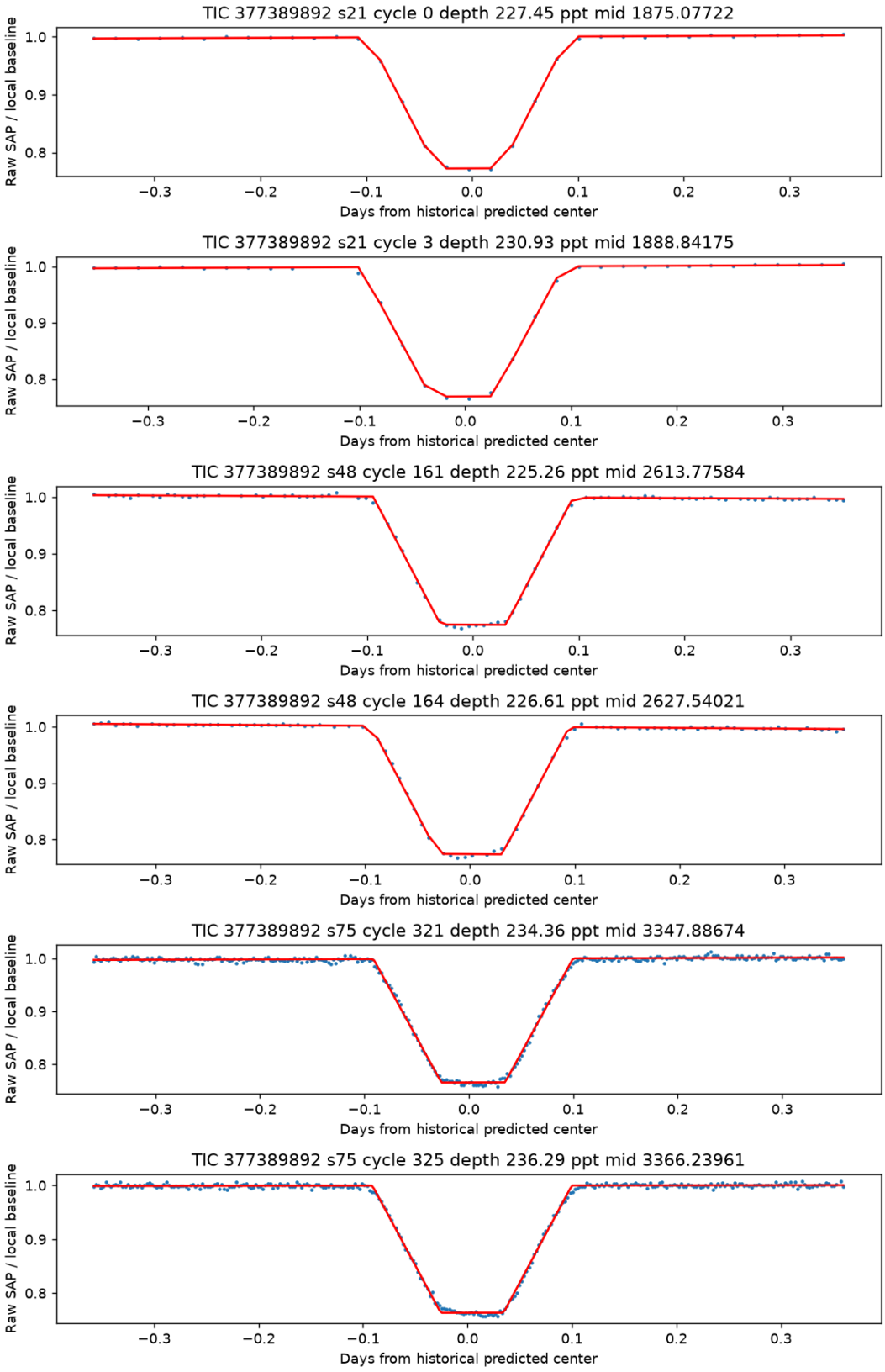
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

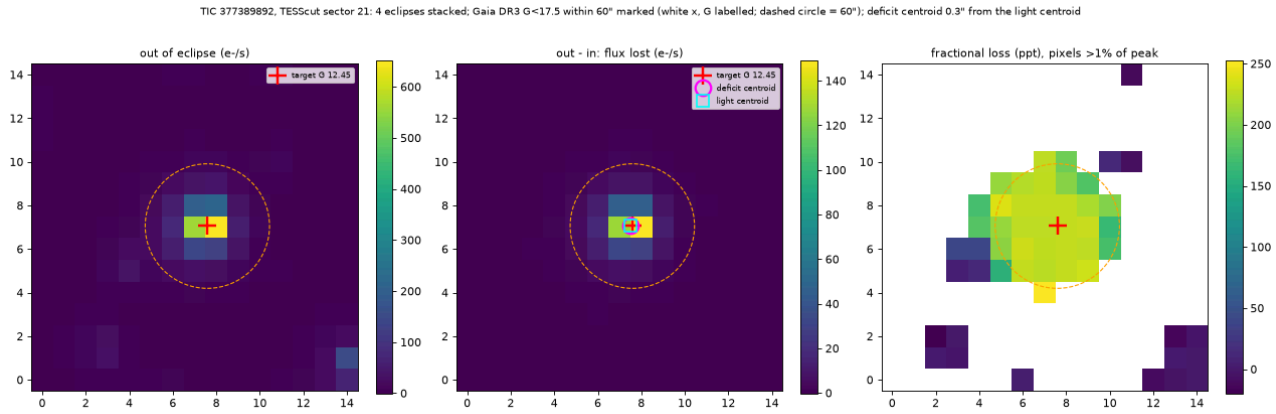
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
21	0	1875.0772236	0.010417	227.447	usable; same-kind selection applies
21	1	1879.6688126	0.010417	214.370	usable; same-kind selection applies
21	3	1888.8417538	0.010417	230.928	usable; same-kind selection applies
21	4	1893.4307987	0.010417	218.572	usable; same-kind selection applies
48	161	2613.7758350	0.003472	225.265	usable; same-kind selection applies
48	162	2618.3634915	0.003472	226.593	usable; same-kind selection applies
48	164	2627.5402139	0.003472	226.614	usable; same-kind selection applies
48	165	2632.1281960	0.003472	229.113	usable; same-kind selection applies
75	321	3347.8867371	0.001157	234.356	usable; same-kind selection applies
75	322	3352.4748173	0.001157	235.871	usable; same-kind selection applies
75	325	3366.2396089	0.001157	236.295	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s21.json, historical/diff_s21.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R257. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R257; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R257; diff_s21.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R258. TIC 88811793

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 88811793: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[13, 93, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 10 same-kind midtimes give P=7.4357138600 +/-0.0000172221d, reference eclipse epoch BTJD1660.0553916, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Deciding S93 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.829arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.732. Capable neighbours: Gaia 6721228062775457920: 14.54 arcsec, G12.652, required fractional eclipse0.161, pixel fractional loss0.024512582723309626; Gaia 6721225146508733184: 27.33 arcsec, G10.653, required fractional eclipse0.078, pixel fractional loss0.002358223522502439
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths44.826/23.910/14.579ppt, with 4 stacked events. QLP selected-kind fitted median63.740ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3672122806277545830 4, P=7.4358869328+/-0.0013790350d, e=0.068823, omega=310.9380deg. Closest edge-on conjunction offset0.081898d (0.011014phase), full-SB1-covariance linear propagated Gaia uncertainty0.348216d at the measured epoch. Near-Gaia-reference projected eclipse offset0.111493d (photometric period projection allowance0.002945d). The approximate central duration9.19h vs

observed median7.58h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unresolved close-neighbor attribution after completed check. Capable neighbor14.54arcsec away lies0.75pixel from deficit centroid; target is0.087pixel away. Another capable neighbor27.33arcsec away is spatially distinct. Deficit favors target, but no calibrated multi-source PRF localization excludes the closest source. Strong aperture dilution: cutout45/24/15ppt versus QLP approximately60ppt. Pixel fractional loss is blended light, not source-intrinsic required eclipse.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

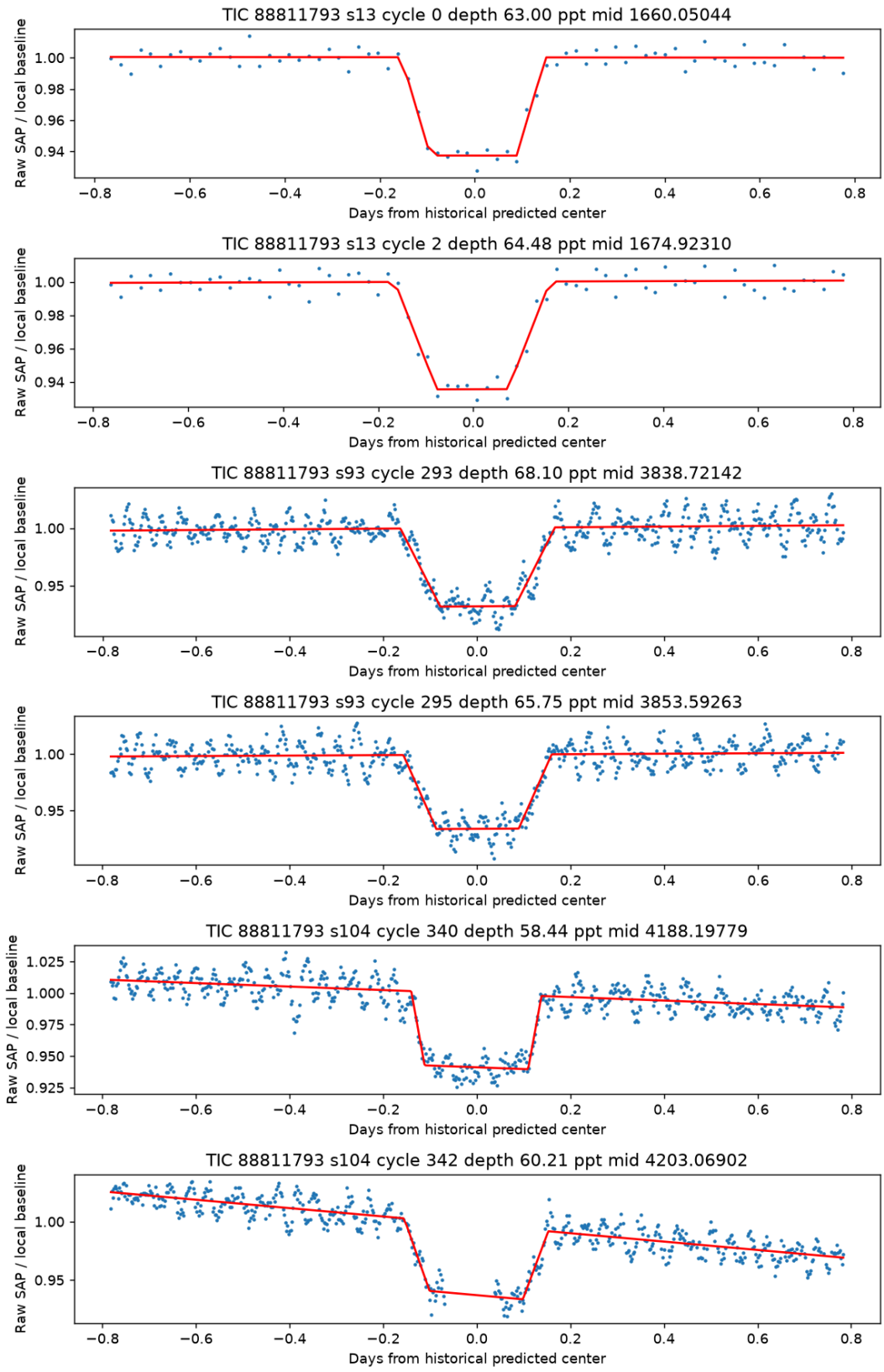
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

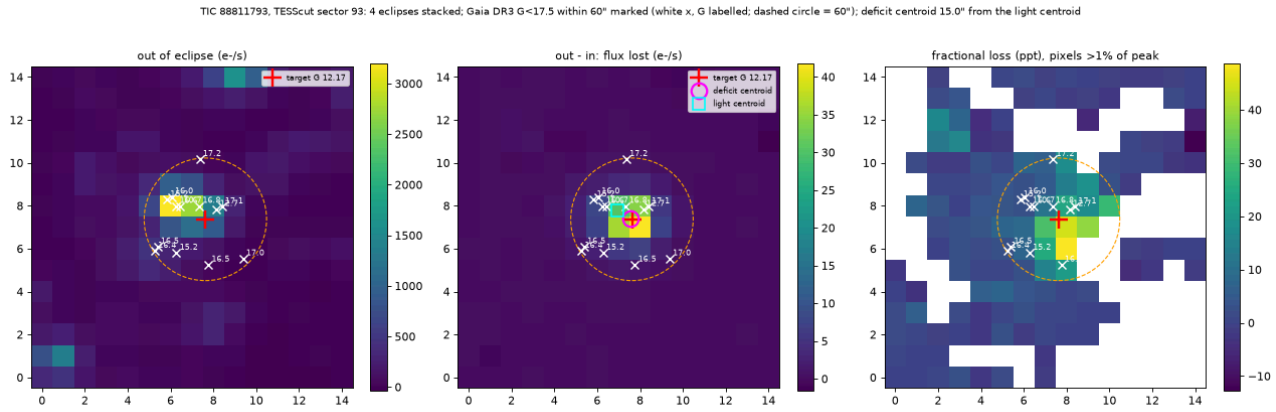
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
13	0	1660.0504423	0.010416	63.001	usable; same-kind selection applies
13	2	1674.9231024	0.010416	64.479	usable; same-kind selection applies
93	292	3831.2811996	0.001388	68.170	usable; same-kind selection applies
93	293	3838.7214210	0.001701	68.095	usable; same-kind selection applies
93	294	3846.1566901	0.001436	67.941	usable; same-kind selection applies
93	295	3853.5926349	0.001559	65.753	usable; same-kind selection applies
104	339	4180.7615783	0.002137	57.446	usable; same-kind selection applies
104	340	4188.1977932	0.001313	58.441	usable; same-kind selection applies
104	341	4195.6333412	0.001633	55.911	usable; same-kind selection applies
104	342	4203.0690200	0.004059	60.207	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s93.json, historical/diff_s93.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R258. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R258; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R258; diff_s93.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R259. TIC 467753507

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 467753507: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[59, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=9.9441928263 +/-0.0000158531d, reference eclipse epoch BTJD2917.6341831, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S43 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.668arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.200. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths29.800/29.222/27.872ppt, with 2 stacked events. QLP selected-kind fitted median30.155ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3169109080240209408, P=9.9436342684+/-0.0008415654d, e=0.604192, omega=278.8133deg. Closest edge-on conjunction offset0.155462d (0.015634phase), full-SB1-covariance linear propagated Gaia uncertainty0.239090d at the measured epoch. Near-Gaia-reference projected eclipse offset0.013589d (photometric period projection allowance0.004027d). The approximate central duration3.75h vs observed median2.49h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four measured eclipses in two representative sectors; no capable neighbor. Alternative cutout approximately29ppt agrees raw26-31ppt. Epoch identifies the repeating selected eclipse, deeper-secondary identity not established.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

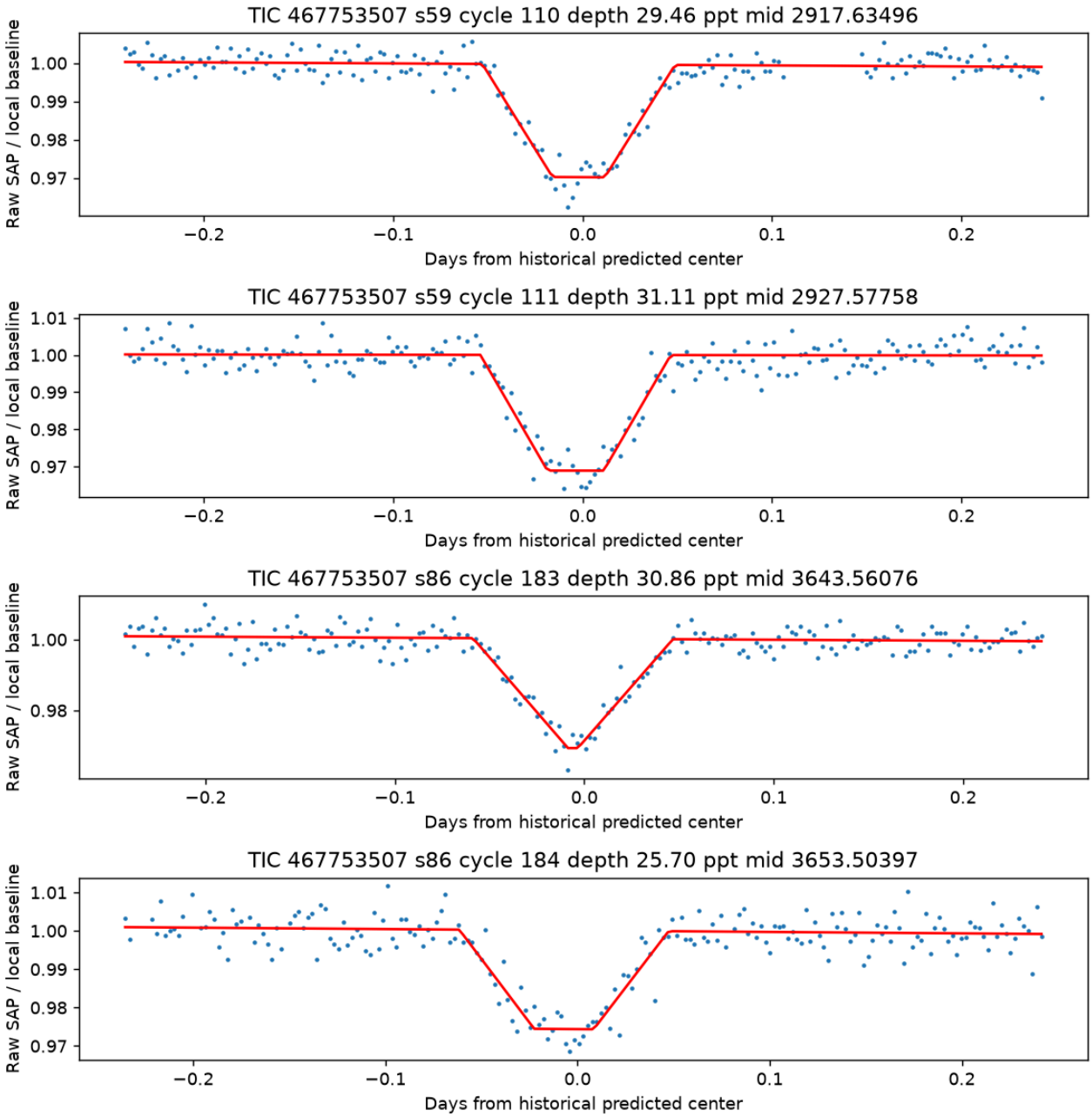
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

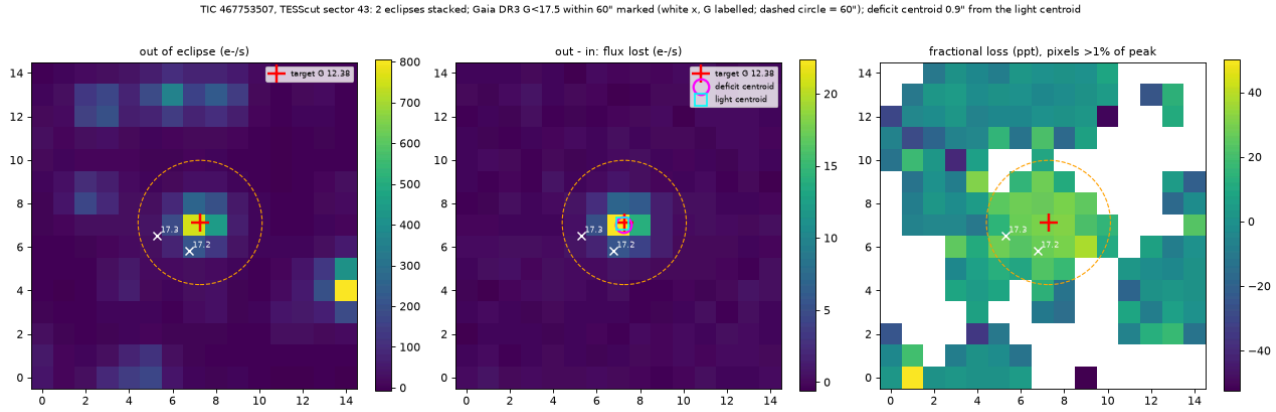
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
59	110	2917.6349616	0.001157	29.455	usable; same-kind selection applies
59	111	2927.5775799	0.001157	31.113	usable; same-kind selection applies
86	183	3643.5607559	0.001157	30.856	usable; same-kind selection applies
86	184	3653.5039731	0.001157	25.697	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s43.json, historical/diff_s43.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R259. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R259; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R259; diff_s43.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R260. TIC 350760292

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 350760292: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[92]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=4.6960724668 +/-0.0003660296d, reference eclipse epoch BTJD3805.7766806, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S92 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation5.064arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.936. Capable neighbours: Gaia 4100891781366788480: 26.14 arcsec, G15.354, required fractional eclipse0.898, pixel fractional loss0.06433309025869284; Gaia 4100894839383543040: 50.66 arcsec, G15.125, required fractional eclipse0.738, pixel fractional loss0.000569337846151796; Gaia 4100894701944580480: 53.65 arcsec, G15.307, required fractional eclipse0.862, pixel fractional loss0.004018742576455143; Gaia 4100894701944566784: 55.28 arcsec, G15.175, required fractional eclipse0.770, pixel fractional loss0.035653750928356334
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths67.342/67.856/63.124ppt, with 4 stacked events. QLP selected-kind fitted median62.873ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3410089473630430536, P=4.6961323973+/-0.0007029011d, e=0.098920, omega=100.7554deg. Closest edge-on conjunction

offset0.039875d (0.008491phase), full-SB1-covariance linear propagated Gaia uncertainty0.540721d at the measured epoch. Near-Gaia-reference projected eclipse offset0.083505d (photometric period projection allowance0.266470d). The approximate central duration5.17h vs observed median3.28h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four raw repeats in available representative sector92; historical sector39 cutout independently reuses earlier observing occasions. Measured4.69607d differs screening4.69710d; short baseline and timing covariance retained. Capable nearest neighbor26.14arcsec away is1.45pixels from deficit centroid versus0.24pixel for target; other capable neighbors are50-55arcsec away. Poor total-light PSF residual0.936 reflects crowded light and is not sole attribution evidence.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

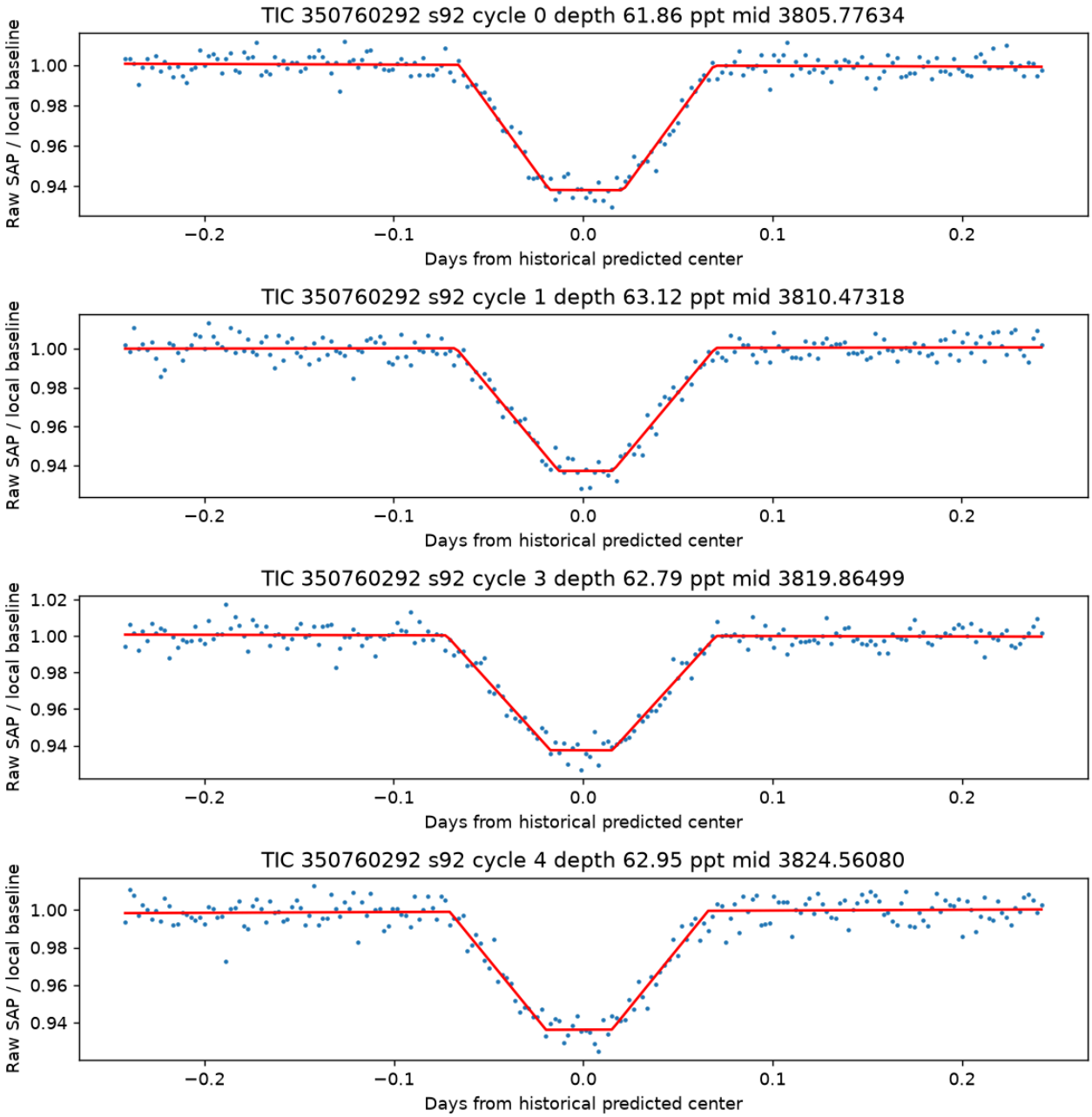
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

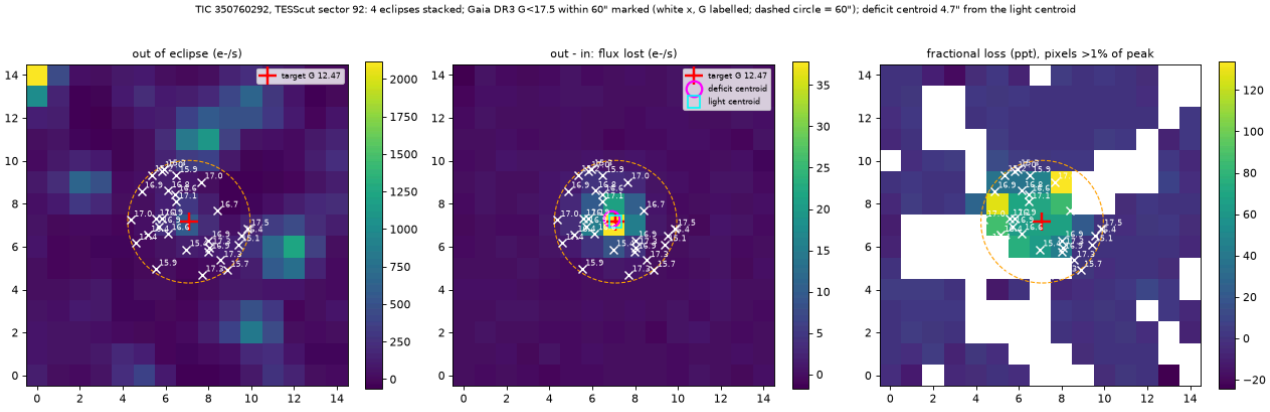
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
92	0	3805.7763399	0.001157	61.862	usable; same-kind selection applies
92	1	3810.4731779	0.001157	63.121	usable; same-kind selection applies
92	3	3819.8649857	0.001157	62.791	usable; same-kind selection applies
92	4	3824.5607984	0.001157	62.955	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s92.json, historical/diff_s92.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R260. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R260; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R260; diff_s92.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R261. TIC 371241950

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 371241950: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 13 distinct eclipse events in selected sectors[10, 38, 90]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 13 same-kind midtimes give P=5.8085988833 +/-0.0000066634d, reference eclipse epoch BTJD1575.3856200, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S90 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.325arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.363. Capable neighbours: Gaia 5243963620257188736: 45.52 arcsec, G14.028, required fractional eclipse0.536, pixel fractional loss0.017013504837050105
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths80.317/75.007/63.791ppt, with 4 stacked events. QLP selected-kind fitted median83.558ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35243963688976670592, P=5.8081279932+/-0.0004474477d, e=0.047392, omega=328.0911deg. Closest edge-on conjunction offset0.172284d (0.029663phase), full-SB1-covariance linear propagated Gaia uncertainty0.095187d at the measured epoch. Near-Gaia-reference projected eclipse offset0.076223d (photometric period projection allowance0.001359d). The approximate central duration7.35h vs observed median4.34h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Thirteen measured repeats span three sectors; capable neighbor45.52arcsec is spatially separated from deficit core. Gaia phase offset0.030 is discussed with covariance rather than arbitrary cutoff; observed4.3h duration can be noncentral.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

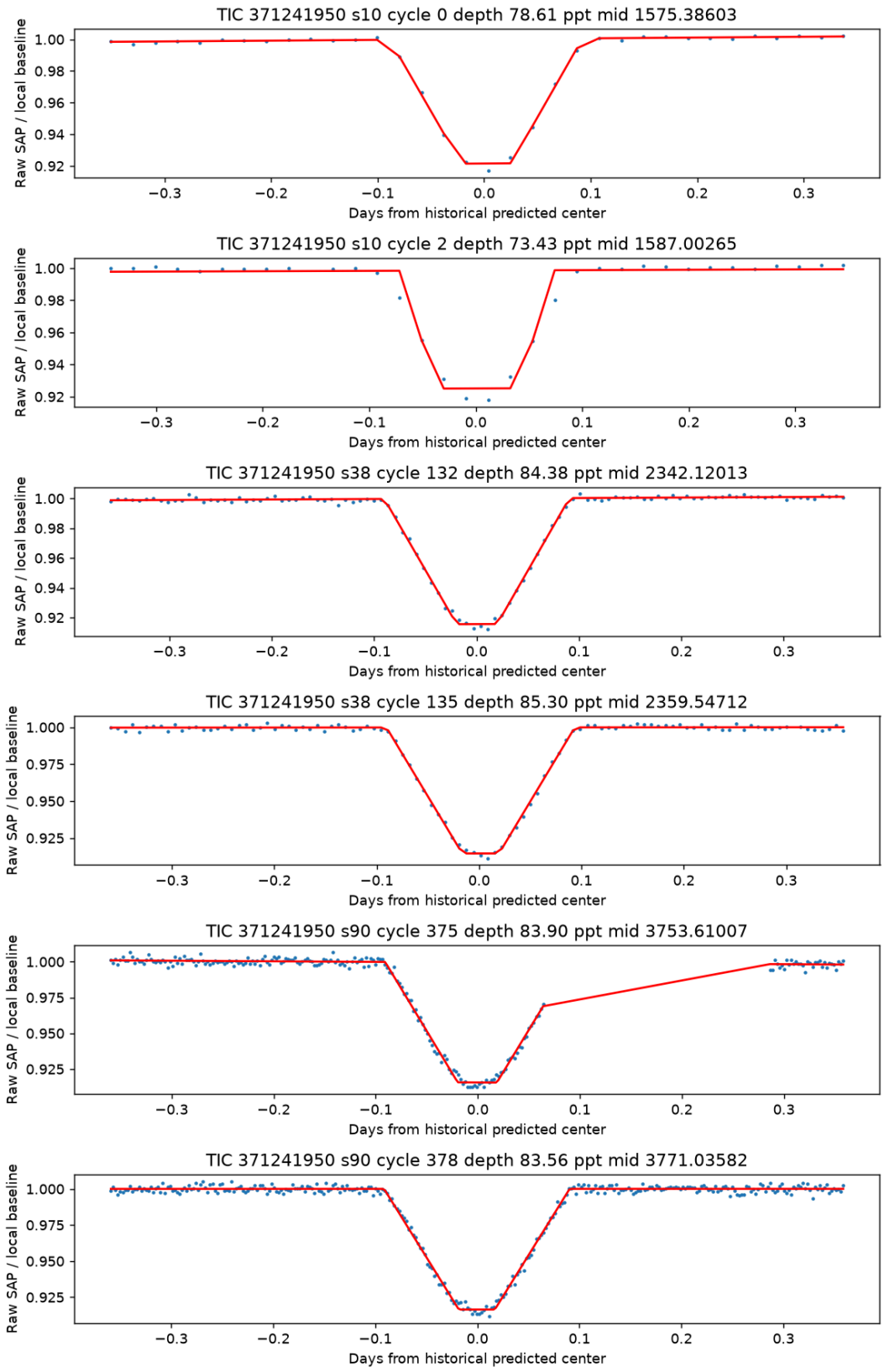
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

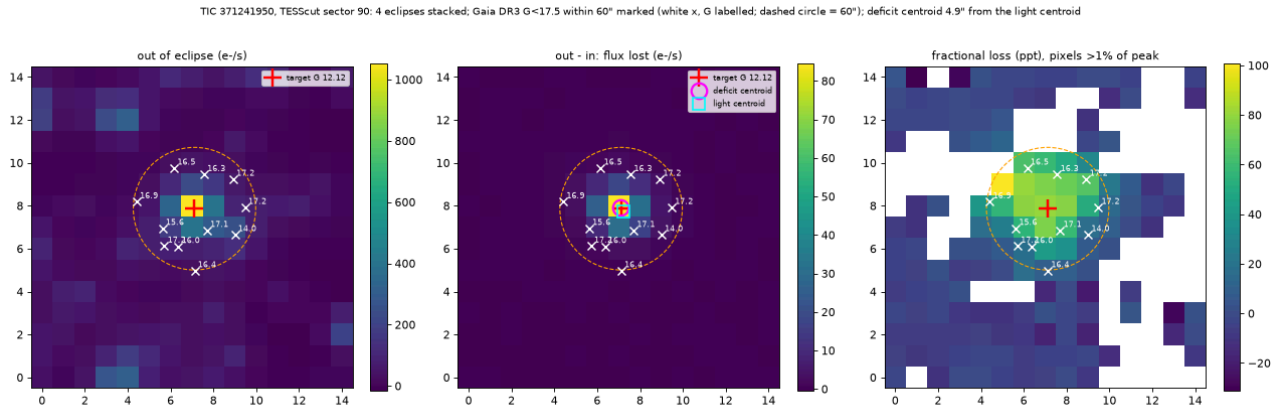
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
10	0	1575.3860340	0.010417	78.605	usable; same-kind selection applies
10	1	1581.1912528	0.010417	73.981	usable; same-kind selection applies
10	2	1587.0026511	0.010417	73.432	usable; same-kind selection applies
10	3	1592.8118838	0.010417	82.230	usable; same-kind selection applies
38	132	2342.1201329	0.003472	84.376	usable; same-kind selection applies
38	133	2347.9294237	0.003472	84.047	usable; same-kind selection applies
38	134	2353.7379595	0.003472	83.362	usable; same-kind selection applies
38	135	2359.5471217	0.003472	85.296	usable; same-kind selection applies
90	374	3747.8019640	0.001157	84.022	usable; same-kind selection applies
90	375	3753.6100691	0.001157	83.899	usable; same-kind selection applies
90	376	3759.4188695	0.001157	84.903	usable; same-kind selection applies
90	377	3765.2272652	0.001157	82.496	usable; same-kind selection applies
90	378	3771.0358210	0.001157	83.558	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s90.json, historical/diff_s90.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R261. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R261; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R261; diff_s90.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R262. TIC 407207285

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 407207285: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[17, 25, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=12.8149004912 +/-0.0000392303d, reference eclipse epoch BTJD1767.6750368, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S17 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation2.991arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.606. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths69.282/68.374/67.489ppt, with 2 stacked events. QLP selected-kind fitted median67.187ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3529579745461206016, P=12.8145405975+/-0.0008878672d, e=0.052468, omega=134.5472deg. Closest edge-on conjunction offset0.164534d (0.012840phase), full-SB1-covariance linear propagated Gaia uncertainty0.104311d at the measured epoch. Near-Gaia-reference projected eclipse offset0.125665d (photometric period projection allowance0.004237d). The approximate central duration36.58h vs observed median21.65h permits noncentral geometry; this radius/minimum-mass estimate is not a

detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Historical0.18d duration placed out-of-eclipse windows inside the approximately0.90d eclipse and suppressed cutout depth to6ppt. Revised21.6h extraction yields69/68/67ppt versus raw median67ppt. Updated flux budget makes all neighbors incapable (closest18.27arcsec requires253percent intrinsic eclipse). Target-centered revised deficit passes attribution; original extraction preserved. Raw fit uses0.65d baseline exclusion and free duration; no unperformed check is hidden.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

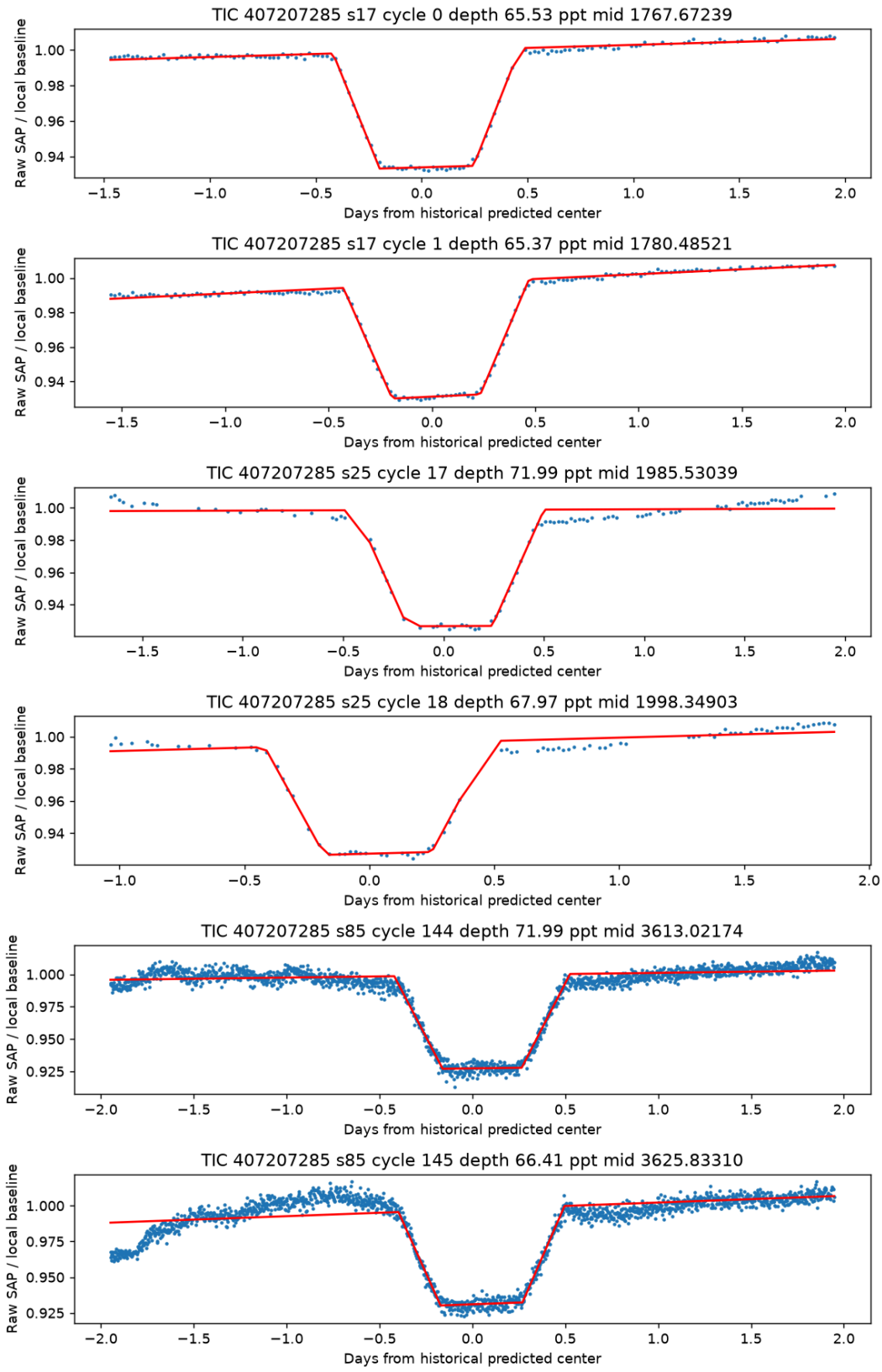
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

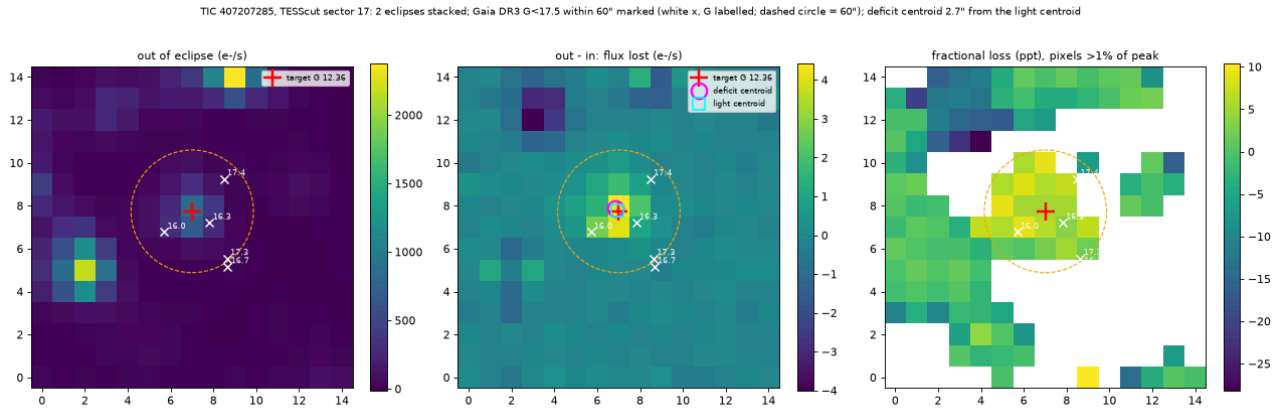
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
17	0	1767.6723916	0.010417	65.533	usable; same-kind selection applies
17	1	1780.4852109	0.010417	65.367	usable; same-kind selection applies
25	17	1985.5303882	0.010417	71.993	usable; same-kind selection applies
25	18	1998.3490303	0.010417	67.967	usable; same-kind selection applies
85	144	3613.0217424	0.001285	71.986	usable; same-kind selection applies
85	145	3625.8330986	0.001994	66.407	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s17.json, historical/diff_s17.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R262. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R262; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R262; diff_s17.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R263. TIC 177257211

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 177257211: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 17 distinct eclipse events in selected sectors[1, 38, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 17 same-kind midtimes give P=3.5443290949 +/-0.0000030739d, reference eclipse epoch BTJD1329.0245577, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S97 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.827arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.403. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths62.101/59.341/55.456ppt, with 16 stacked events. QLP selected-kind fitted median55.820ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3526087417350849088O, P=3.5442836835+/-0.0003211078d, e=0.026413, omega=251.8290deg. Closest edge-on conjunction offset0.042646d (0.012032phase), full-SB1-covariance linear propagated Gaia uncertainty0.104328d at the measured epoch. Near-Gaia-reference projected eclipse offset0.030612d (photometric period projection allowance0.000815d). The approximate central duration5.70h vs observed median3.57h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Seventeen raw repeated events across three sectors; no capable neighbor. Target-centered deficit and56-62ppt cutout/raw amplitudes. Historical253 events are screening statistics, not measured timing count.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

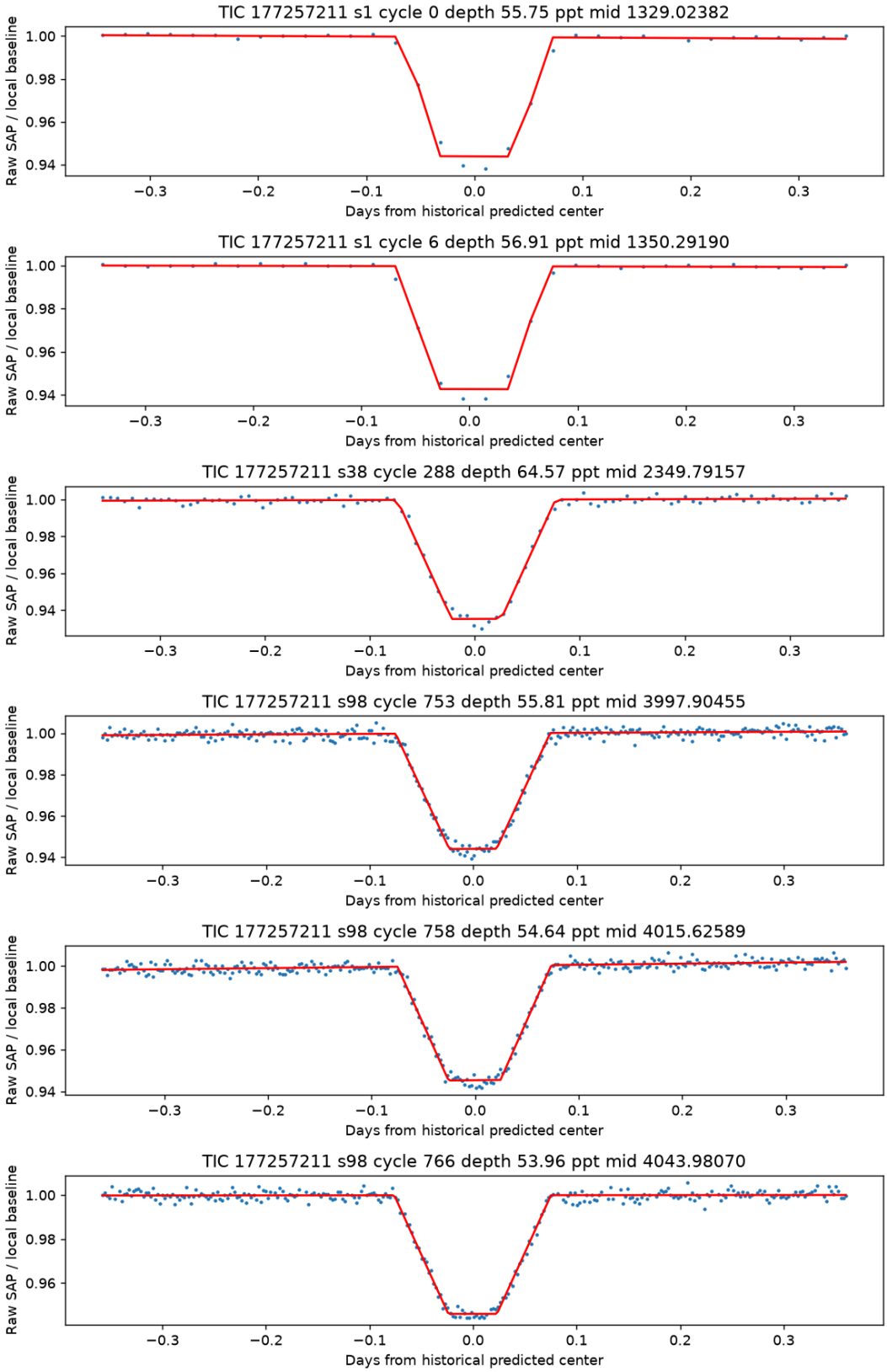
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

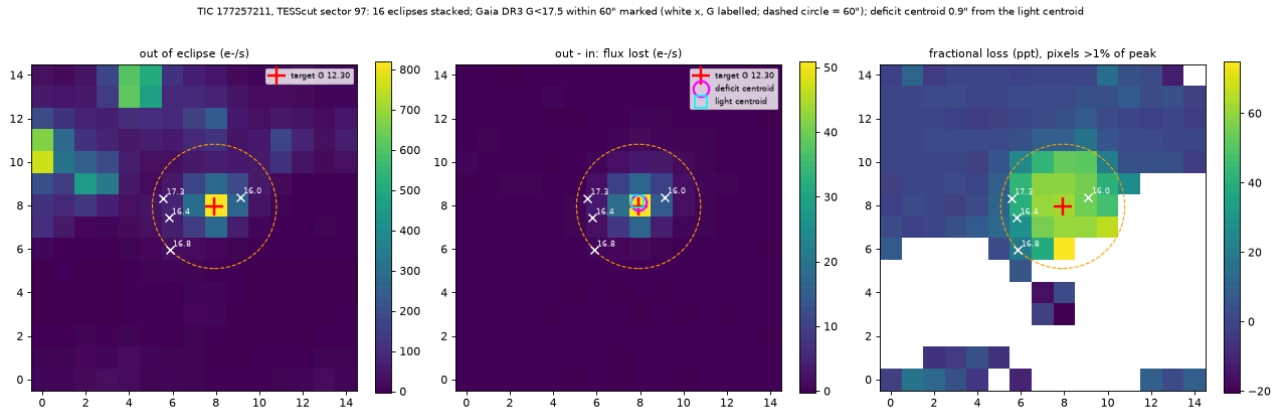
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1329.0238243	0.010417	55.747	usable; same-kind selection applies
1	1	1332.5665308	0.010417	56.768	usable; same-kind selection applies
1	2	1336.1134253	0.010417	60.495	usable; same-kind selection applies
1	6	1350.2918981	0.010417	56.906	usable; same-kind selection applies
38	285	2339.1581439	0.003472	65.499	usable; same-kind selection applies
38	286	2342.7023435	0.003472	64.357	usable; same-kind selection applies
38	288	2349.7915677	0.003472	64.572	usable; same-kind selection applies
38	289	2353.3359850	0.003472	64.276	usable; same-kind selection applies
38	290	2356.8802599	0.003472	65.126	usable; same-kind selection applies
98	753	3997.9045538	0.001157	55.813	usable; same-kind selection applies
98	754	4001.4485625	0.001157	55.587	usable; same-kind selection applies
98	757	4012.0816396	0.001157	54.990	usable; same-kind selection applies
98	758	4015.6258855	0.001157	54.638	usable; same-kind selection applies
98	761	4026.2591449	0.001157	54.557	usable; same-kind selection applies
98	762	4029.8033523	0.001157	53.978	usable; same-kind selection applies
98	765	4040.4361945	0.001157	55.820	usable; same-kind selection applies
98	766	4043.9806979	0.001157	53.961	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R263. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R263; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R263; diff_s97.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R264. TIC 29114881

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 29114881: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[43, 45, 72]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=12.6025319623 +/-0.0000314390d, reference eclipse epoch BTJD2478.4844056, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S43 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.217arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.315. Capable neighbours: Gaia 3403151219984383616: 53.54 arcsec, G14.899, required fractional eclipse0.579, pixel fractional loss0.010934079114229483
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths57.519/56.424/51.492ppt, with 2 stacked events. QLP selected-kind fitted median57.390ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3340314534446912678 4, P=12.5997639477+/-0.0076575857d, e=0.081743, omega=206.9858deg. Closest edge-on conjunction offset0.028161d (0.002235phase), full-SB1-covariance linear propagated Gaia uncertainty1.025084d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.431329d (photometric period projection allowance0.005219d). The approximate central duration6.14h vs observed median5.33h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Six raw events across three sectors. Capable neighbor53.54arcsec is distinct from target-centered deficit. Raw56-59ppt and cutout57/56/51ppt depend on aperture.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

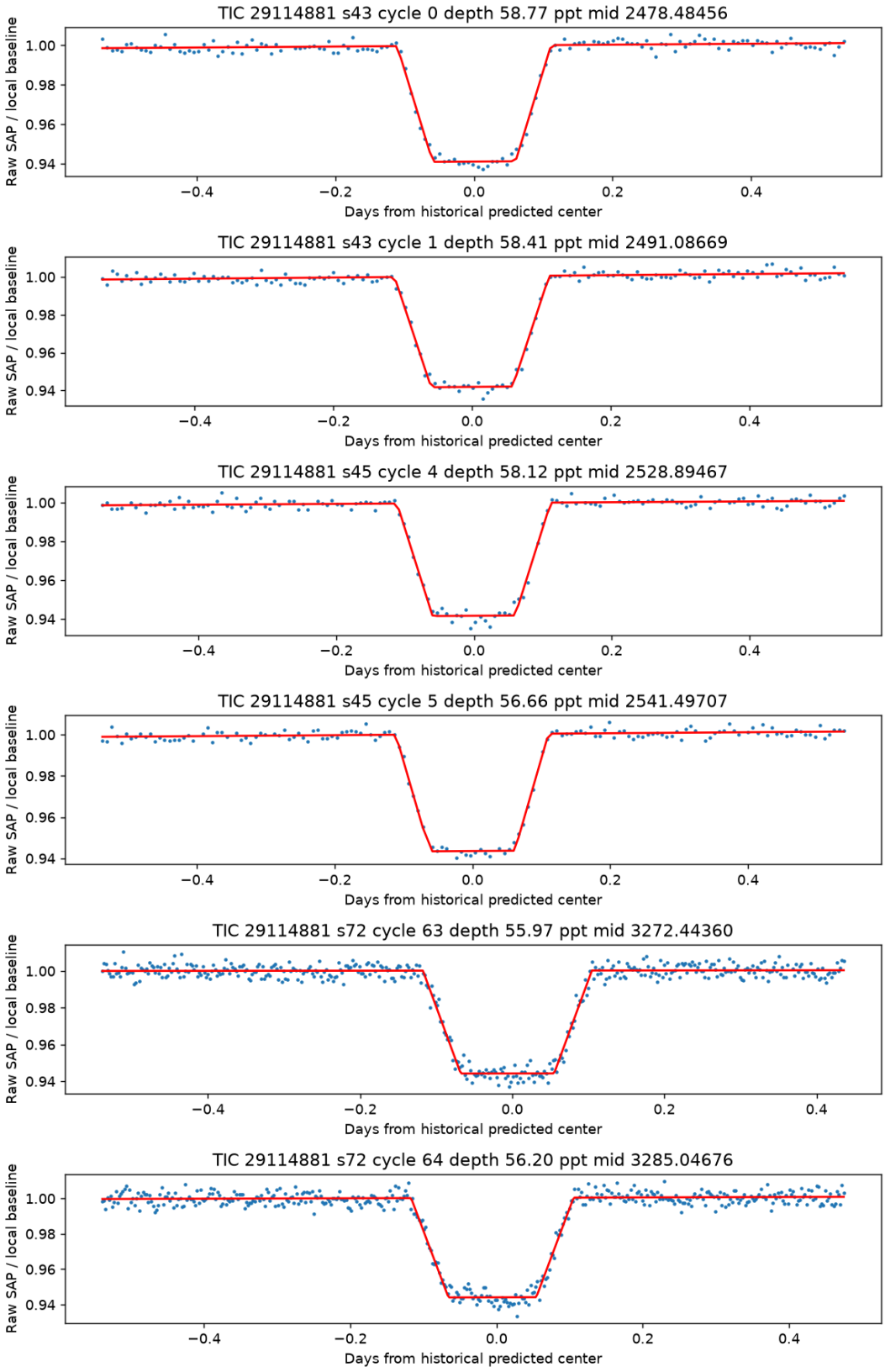
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

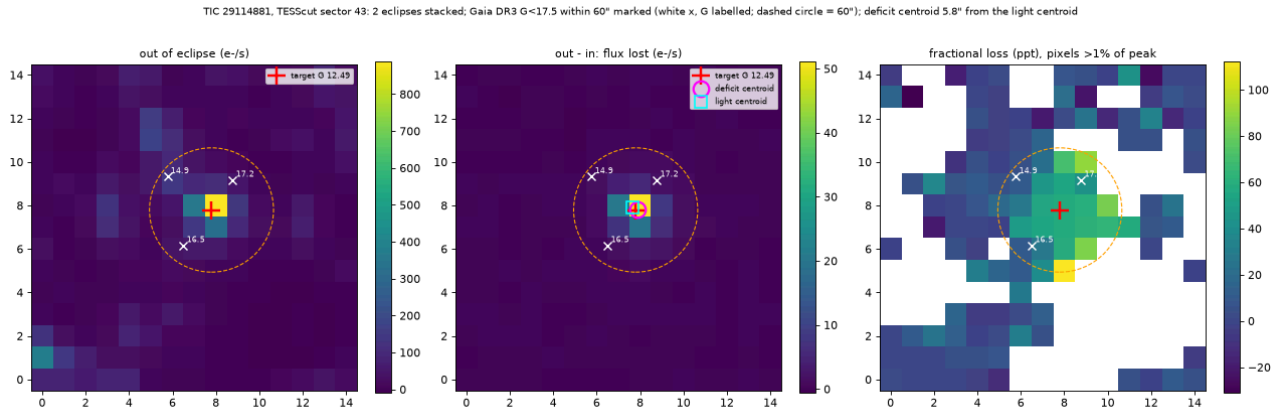
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
43	0	2478.4845644	0.003473	58.767	usable; same-kind selection applies
43	1	2491.0866878	0.003473	58.407	usable; same-kind selection applies
45	4	2528.8946730	0.003472	58.122	usable; same-kind selection applies
45	5	2541.4970666	0.003472	56.659	usable; same-kind selection applies
72	63	3272.4436002	0.001157	55.969	usable; same-kind selection applies
72	64	3285.0467647	0.001157	56.204	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s43.json, historical/diff_s43.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R264. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R264; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R264; diff_s43.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R265. TIC 77605102

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 77605102: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[9, 62, 99]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.2688089976 +/-0.0000140630d, reference eclipse epoch BTJD1548.9000776, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S62 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation4.232arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.760. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths146.819/143.733/133.360ppt, with 3 stacked events. QLP selected-kind fitted median167.856ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3542419968514814924 8, P=11.2714186464+/-0.0039065215d, e=0.591502, omega=296.0774deg. Closest edge-on conjunction offset-0.297978d (-0.026437phase), full-SB1-covariance linear propagated Gaia uncertainty0.256542d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.029184d (photometric period projection allowance0.001448d). The approximate central duration4.83h vs observed median4.63h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five repeated raw eclipses across three sectors; no capable neighbor. Raw145-169ppt and cutout147/144/133ppt show extraction dependence. Geometry offset minus0.026phase retained with propagated Gaia covariance; not sigma veto.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

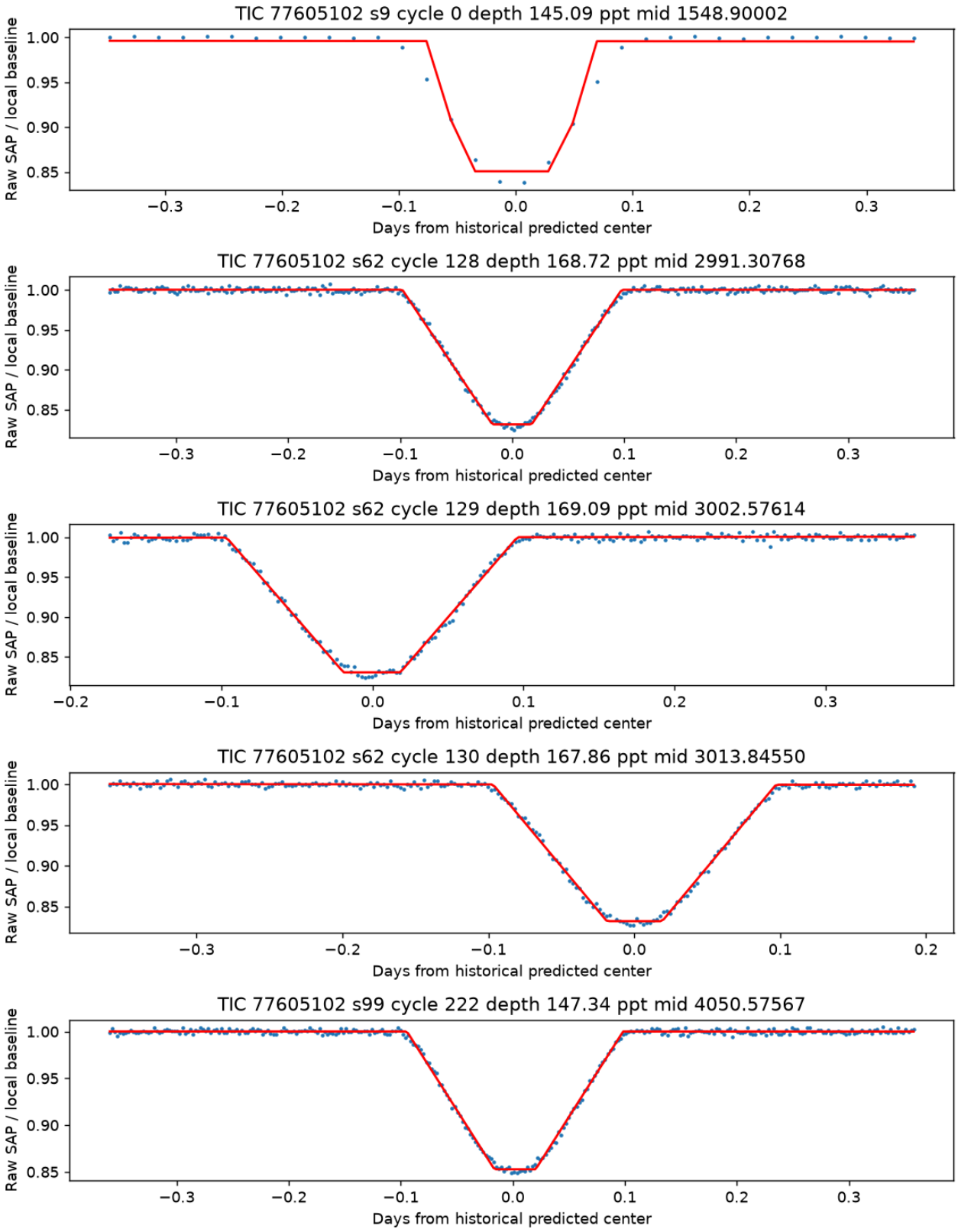
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

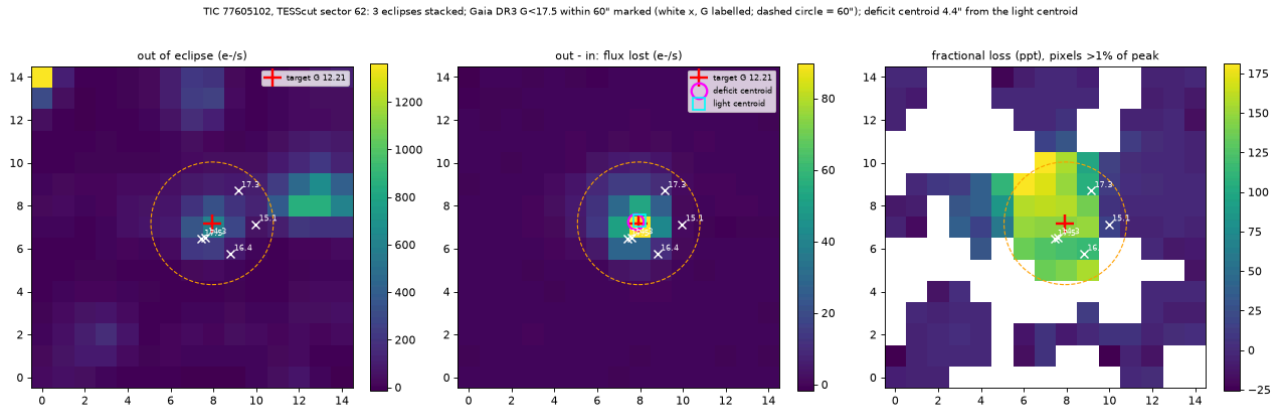
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	0	1548.9000213	0.010417	145.092	usable; same-kind selection applies
62	128	2991.3076783	0.001157	168.725	usable; same-kind selection applies
62	129	3002.5761416	0.001157	169.090	usable; same-kind selection applies
62	130	3013.8454989	0.001157	167.856	usable; same-kind selection applies
99	222	4050.5756719	0.001157	147.340	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R265. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R265; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R265; diff_s62.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R266. TIC 35635884

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 35635884: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[33, 87]. The raw-event plot and numeric raw-windows.csv were inspected. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=5.5005487862 +/-0.0000107720d, reference eclipse epoch BTJD2203.8207426, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Deciding S33 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.630arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.454. Capable neighbours: Gaia 2946315752238013696: 47.20 arcsec, G13.768, required fractional eclipse0.566, pixel fractional loss0.009677394392040716
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths116.027/108.058/81.954ppt, with 4 stacked events. QLP selected-kind fitted median117.741ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3294631536998117478 4, P=5.5005719282+/-0.0003916036d, e=0.166765, omega=78.6462deg. Closest edge-on conjunction offset-0.008812d (-0.001602phase), full-SB1-covariance linear propagated Gaia uncertainty0.131948d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.001175d (photometric period projection allowance0.003555d). The approximate central duration4.36h vs observed median3.34h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Eight repeated raw eclipses across three sectors; capable neighbor47.20arcsec is distinct from deficit core. Raw114-129ppt vs cutout116/108/82ppt show aperture and sector dilution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

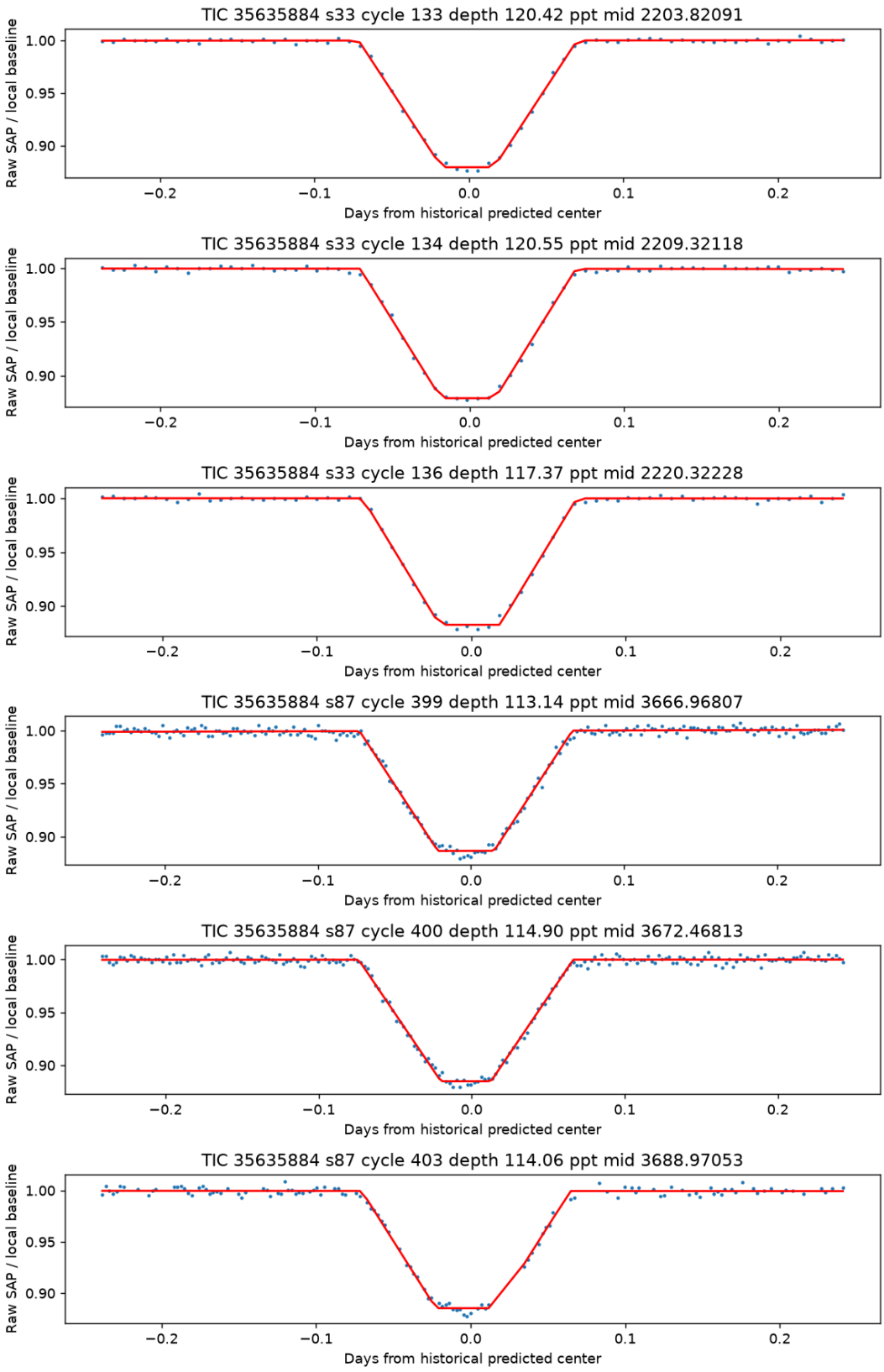
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

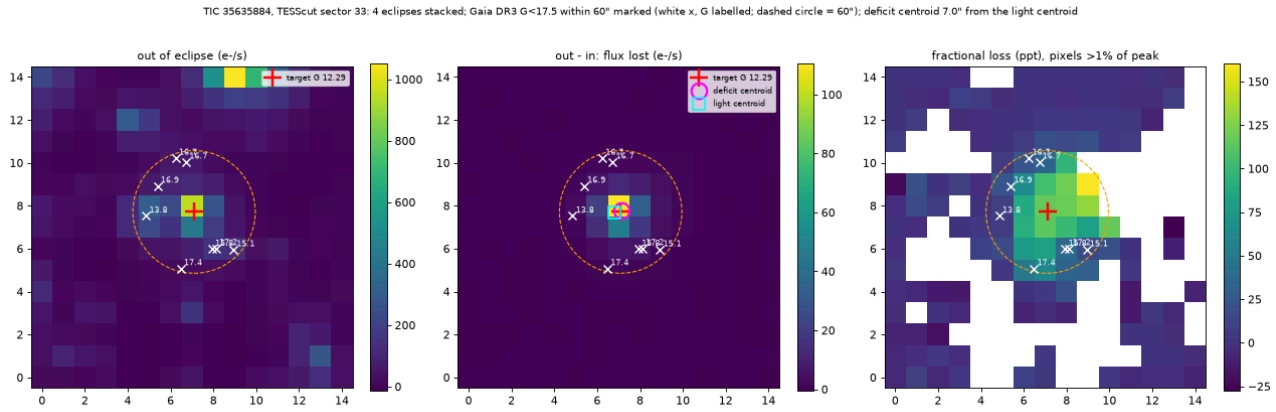
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	133	2203.8209106	0.003472	120.424	usable; same-kind selection applies
33	134	2209.3211757	0.003472	120.554	usable; same-kind selection applies
33	136	2220.3222756	0.003472	117.370	usable; same-kind selection applies
33	137	2225.8228566	0.003472	120.238	usable; same-kind selection applies
87	399	3666.9680740	0.001157	113.143	usable; same-kind selection applies
87	400	3672.4681274	0.001157	114.903	usable; same-kind selection applies
87	402	3683.4645546	0.001157	118.111	usable; same-kind selection applies
87	403	3688.9705292	0.001157	114.055	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, ../identity-check.json, shared-qlp QLP FITS; historical/diff_s33.json, historical/diff_s33.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] For TIC407207285 the deciding revised cutout is revised-cutout/diff_revised-long-window.json and .png; historical extraction remains for comparison. Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R266. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R266; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R266; diff_s33.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R267. TIC 396859601

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 396859601: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 3 directly measured raw QLP SAP events in sectors[5, 32]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 2 same-kind measured times give P=16.77770105414 +/-0.00491037467d, reference eclipse epoch BTJD2179.72210646; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS32 image peak[7, 8] matches target pixel[7, 8]; target-deficit offset0.112pixel, light-deficit offset1.611arcsec; PSF total-light scaled residual0.138. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths18.583/18.905/18.782ppt, 2 stacked events, versus selected-kind raw QLP median18.238ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR33258578909949773056; P=16.7849273877 +/-0.0081647730d, e=0.285924, omega=191.4077deg. Closest edge-on conjunction offset-0.376136d (-0.022409phase), full-SB1-covariance linear Gaia uncertainty1.029050d at measured epoch. Approximate central duration7.28h vs measured median2.24h permits noncentral eclipses. Mass function0.027170solar, assumed primary mass1.05,

minimum companion0.382solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Two S32 well-sampled events determine the reported period. A third real S5 eclipse tests recurrence across 44 cycles; its zero/singular covariance is excluded from timing. No newer timeseries sectors in current MAST query beyond5/32. Period error remains short-baseline and conditional on tested cycle branch, not precision long-baseline characterization.

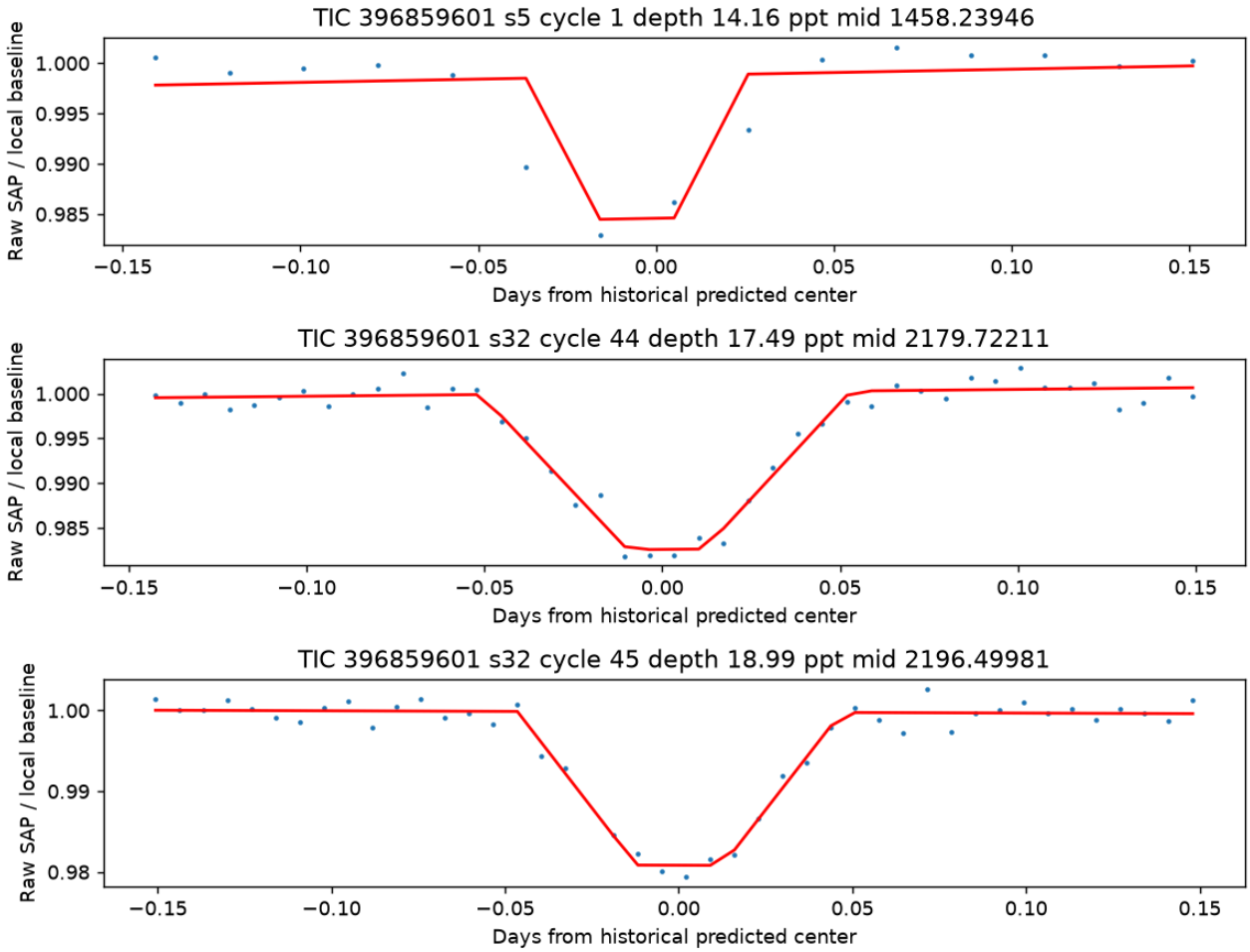
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
5	1	1458.23946377	0.0104166	14.162	False
32	44	2179.72210646	0.0034722	17.486	True
32	45	2196.49980751	0.0034722	18.990	True

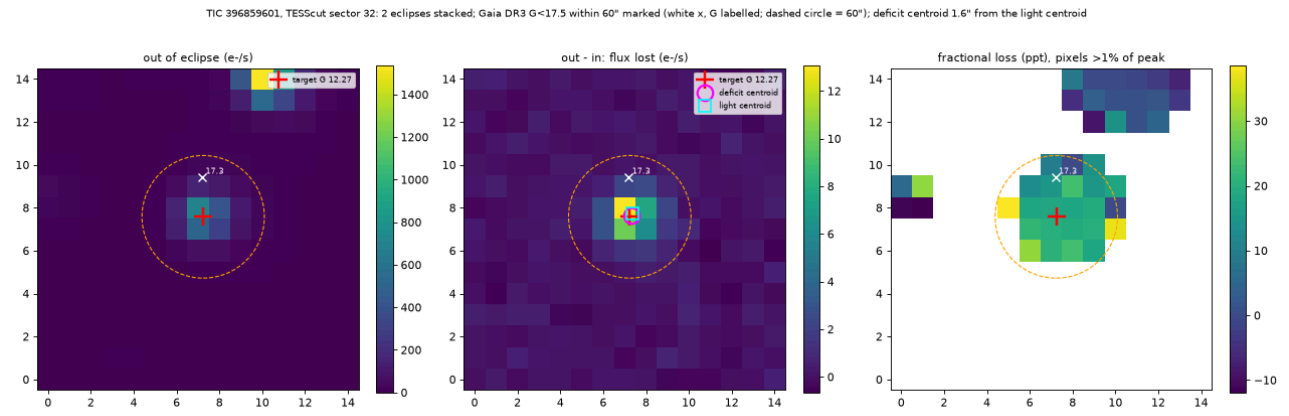
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
None				

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s32.json, historical/diff_s32.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R267. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R267; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R267; diff_s32.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R268. TIC 115106052

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 115106052: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 15 directly measured raw QLP SAP events in sectors[1, 68, 95]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 10 same-kind measured times give P=7.64111642841 +/-0.00000850037d, reference eclipse epoch BTJD1329.80183865; epoch kind: Deeper primary: odd half-period cycles. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS27 image peak[7, 8] matches target pixel[7, 8]; target-deficit offset0.199pixel, light-deficit offset0.464arcsec; PSF total-light scaled residual0.237. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths64.129/63.115/61.825ppt, 5 stacked events, versus selected-kind raw QLP median69.362ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3677443219349133504O; P=7.6419883813 +/-0.0004853004d, e=0.050448, omega=186.4256deg. Closest edge-on conjunction offset0.054355d (0.007113phase), full-SB1-covariance linear Gaia uncertainty0.075422d at measured epoch. Approximate central duration9.39h vs measured median5.84h permits noncentral eclipses. Mass function0.174560solar, assumed primary mass1.1, minimum companion0.882solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Deeper odd half-period cycles selected:10 primary times, with independent even-cycle timing agreement. Alternating depth and center residuals support doubled orbital period; historical half-period stack mixes unequal eclipses.

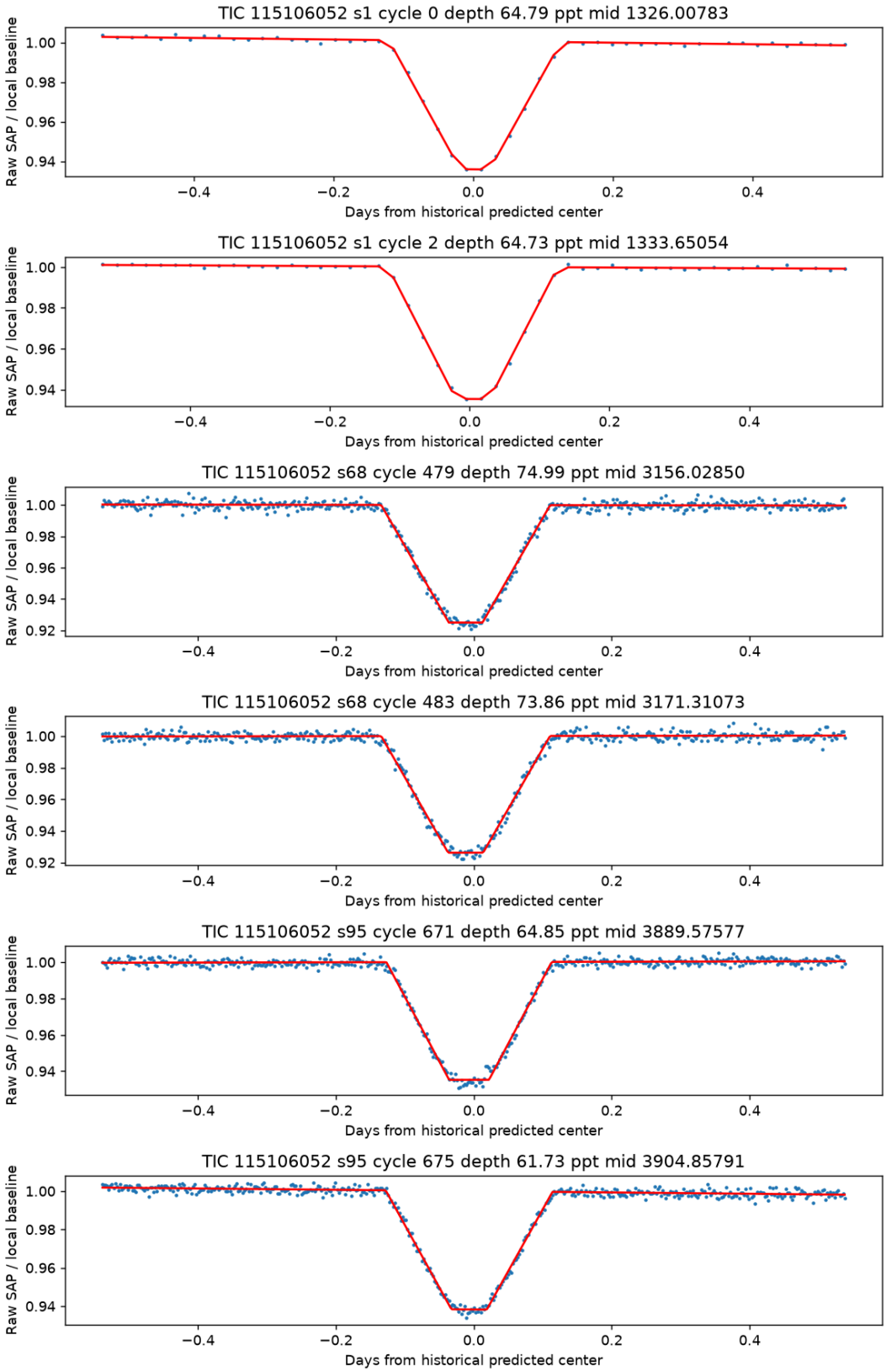
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
1	0	1326.00783212	0.0104166	64.792	True
1	1	1329.80077528	0.0104166	69.532	True
1	2	1333.65053740	0.0104166	64.734	True
1	3	1337.44218674	0.0104166	69.191	True
1	5	1345.08288030	0.0104166	68.195	True
68	479	3156.02850104	0.0011574	74.991	True
68	480	3159.87771782	0.0011574	67.889	True
68	481	3163.67014168	0.0011574	73.933	True
68	483	3171.31072552	0.0011574	73.858	True
68	485	3178.95212182	0.0011574	76.796	True
95	670	3885.78441073	0.0011574	57.862	True
95	671	3889.57577389	0.0011574	64.850	True
95	672	3893.42574517	0.0011574	57.533	True
95	673	3897.21709914	0.0011574	61.629	True
95	675	3904.85790871	0.0011574	61.732	True

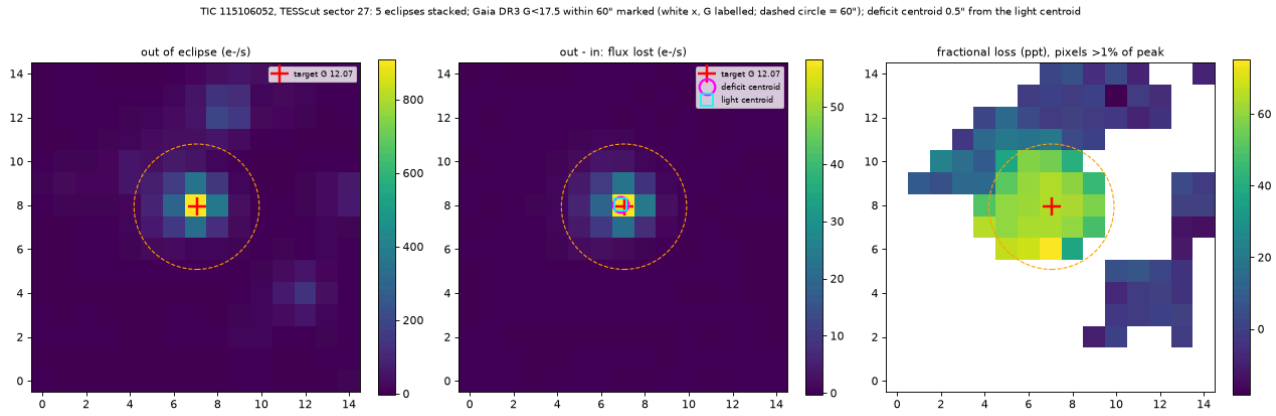
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
None				

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s27.json, historical/diff_s27.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R268. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R268; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R268; diff_s27.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R269. TIC 443911710

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 443911710: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 9 directly measured raw QLP SAP events in sectors[7, 46, 88]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 9 same-kind measured times give P=8.34083548141 +/-0.00001306561d, reference eclipse epoch BTJD1492.60574892; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS7 image peak[8, 8] matches target pixel[8, 8]; target-deficit offset0.253pixel, light-deficit offset0.485arcsec; PSF total-light scaled residual0.355. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths41.234/39.862/39.295ppt, 3 stacked events, versus selected-kind raw QLP median40.506ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3649656340876678272; P=8.3417793302 +/-0.0013516296d, e=0.051939, omega=330.7882deg. Closest edge-on conjunction offset0.029366d (0.003520phase), full-SB1-covariance linear Gaia uncertainty0.210879d at measured epoch. Approximate central duration8.65h vs measured median5.97h permits noncentral eclipses. Mass function0.024608solar, assumed primary mass1.24,

minimum companion0.405solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Close2.88arcsec neighbor G16.14 requires129percent intrinsic loss under approximate Gaia-G flux budget; excluded as sole source under that assumption. TESS/G color differences are not modeled. Nine measured repeats, consistent target-centered cutout.

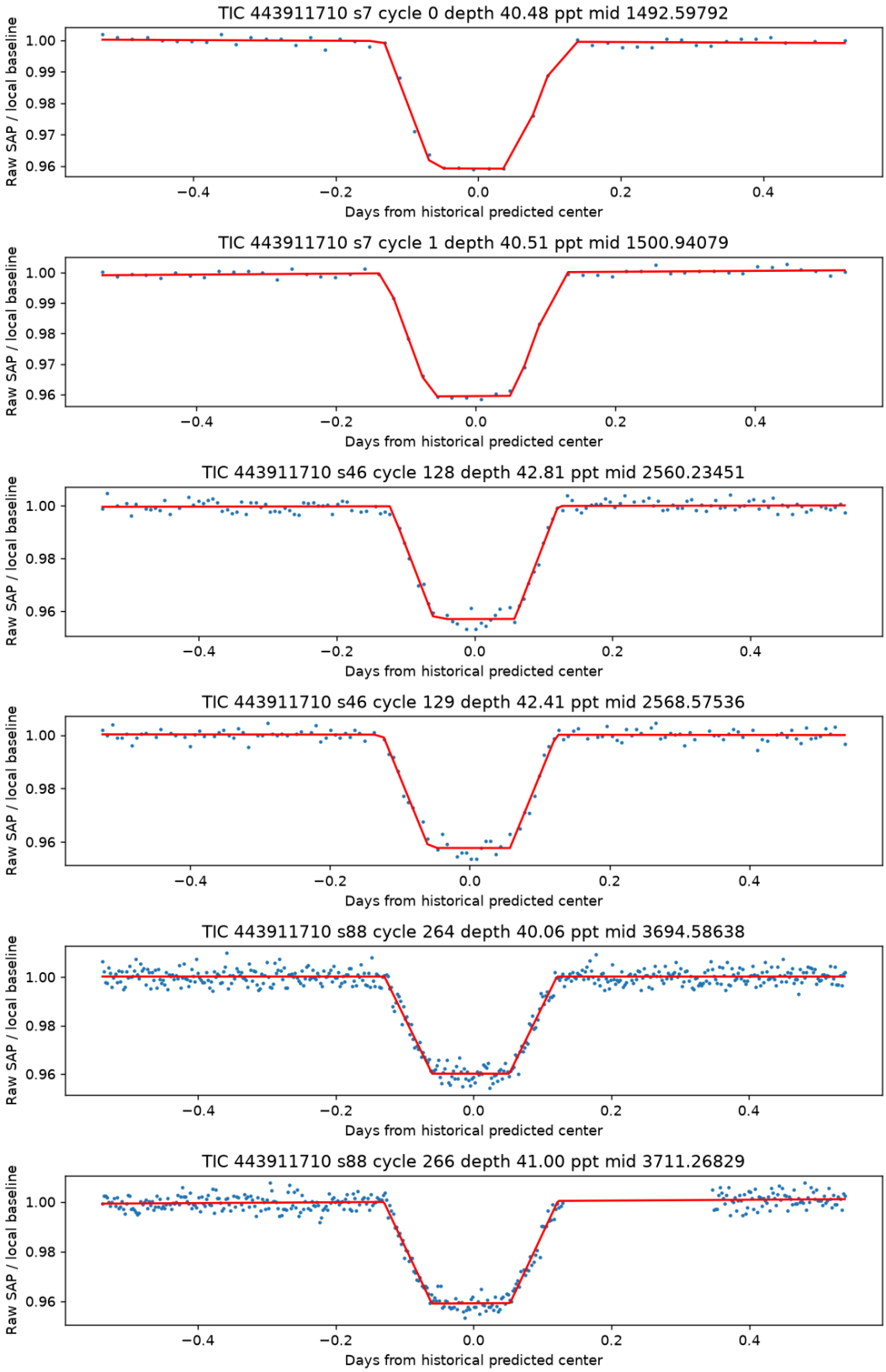
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
7	0	1492.59791575	0.0104167	40.481	True
7	1	1500.94079005	0.0104167	40.506	True
7	2	1509.28073502	0.0104167	39.159	True
46	128	2560.23450630	0.0034724	42.814	True
46	129	2568.57535963	0.0034724	42.410	True
46	130	2576.91510282	0.0034724	42.053	True
88	264	3694.58637853	0.0011574	40.061	True
88	265	3702.92654723	0.0011574	39.902	True
88	266	3711.26829171	0.0011574	40.997	True

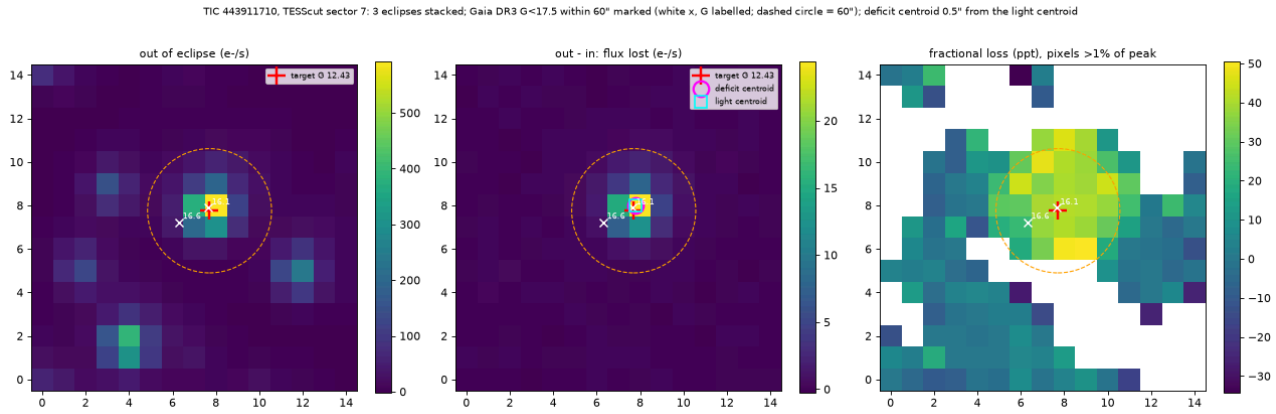
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
None				

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s07.json, historical/diff_s07.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R269. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R269; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R269; diff_s07.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R270. TIC 144113939

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 144113939: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 4 directly measured raw QLP SAP events in sectors[64, 101]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 4 same-kind measured times give P=11.78879640721 +/-0.00001285928d, reference eclipse epoch BTJD3046.67692883; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS37 image peak[8, 7] matches target pixel[8, 7]; target-deficit offset0.135pixel, light-deficit offset2.607arcsec; PSF total-light scaled residual0.442. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths36.457/35.553/33.835ppt, 2 stacked events, versus selected-kind raw QLP median35.558ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3614687627685199104; P=11.7889746135 +/-0.0024534753d, e=0.061265, omega=283.7125deg. Closest edge-on conjunction offset-0.011443d (-0.000971phase), full-SB1-covariance linear Gaia uncertainty0.581526d at measured epoch. Approximate central duration5.96h vs measured median3.13h permits noncentral eclipses. Mass function0.108096solar, assumed primary mass0.96,

minimum companion0.656solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: One capable neighbor38.54arcsec away is separated from target-centered deficit. Four raw repeated events span sectors64/101; alternative historicalS37 provides earlier reduction evidence.

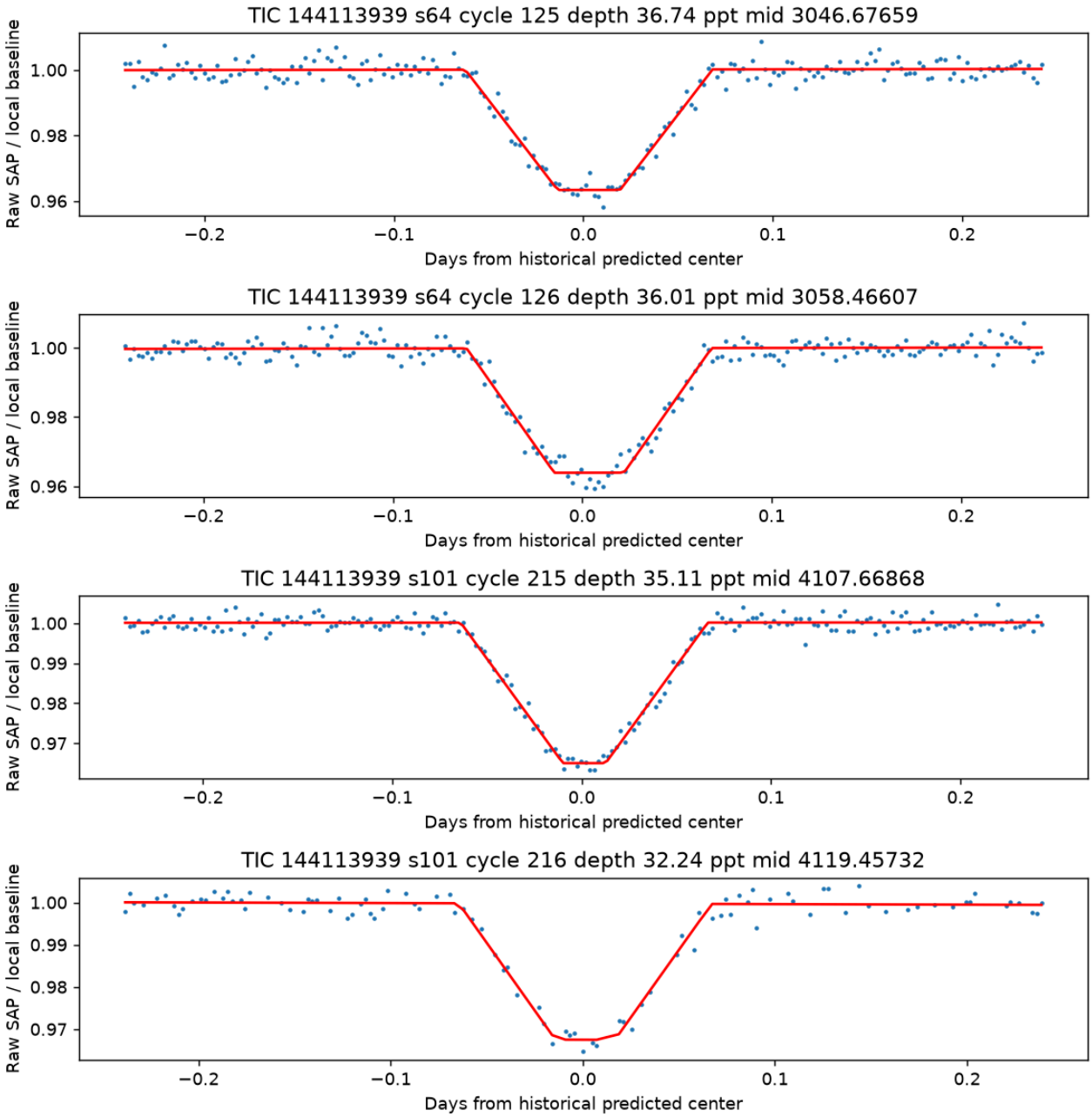
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
64	125	3046.67659151	0.0011574	36.741	True
64	126	3058.46606543	0.0011574	36.006	True
101	215	4107.66868332	0.0011574	35.110	True
101	216	4119.45732116	0.0011574	32.241	True

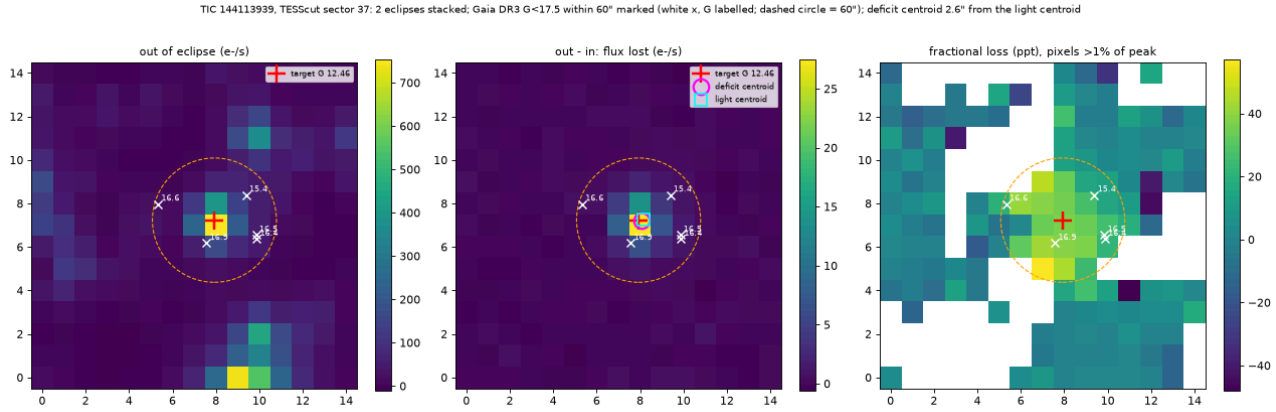
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
6146876242492663808	38.543	15.351	0.5114	1.787

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s37.json, historical/diff_s37.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R270. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R270; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R270; diff_s37.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R271. TIC 339578643

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 339578643: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 4 directly measured raw QLP SAP events in sectors[95, 103]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 4 same-kind measured times give P=7.13460231510 +/-0.00003044773d, reference eclipse epoch BTJD3886.00718979; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS102 image peak[8, 7] matches target pixel[8, 7]; target-deficit offset0.168pixel, light-deficit offset1.974arcsec; PSF total-light scaled residual0.382. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths31.120/30.815/29.768ppt, 4 stacked events, versus selected-kind raw QLP median27.000ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3649438507436423257 6; P=7.1343937492 +/-0.0007271545d, e=0.040070, omega=215.9747deg. Closest edge-on conjunction offset0.004981d (0.000698phase), full-SB1-covariance linear Gaia uncertainty0.330956d at measured epoch. Approximate central duration5.85h vs measured median2.83h permits noncentral eclipses. Mass function0.055553solar, assumed primary mass1.03, minimum companion0.509solar.

This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Four raw events in sectors95/103; target-centeredS102 cutout supports separate reduction. No capable neighbor within60arcsec.

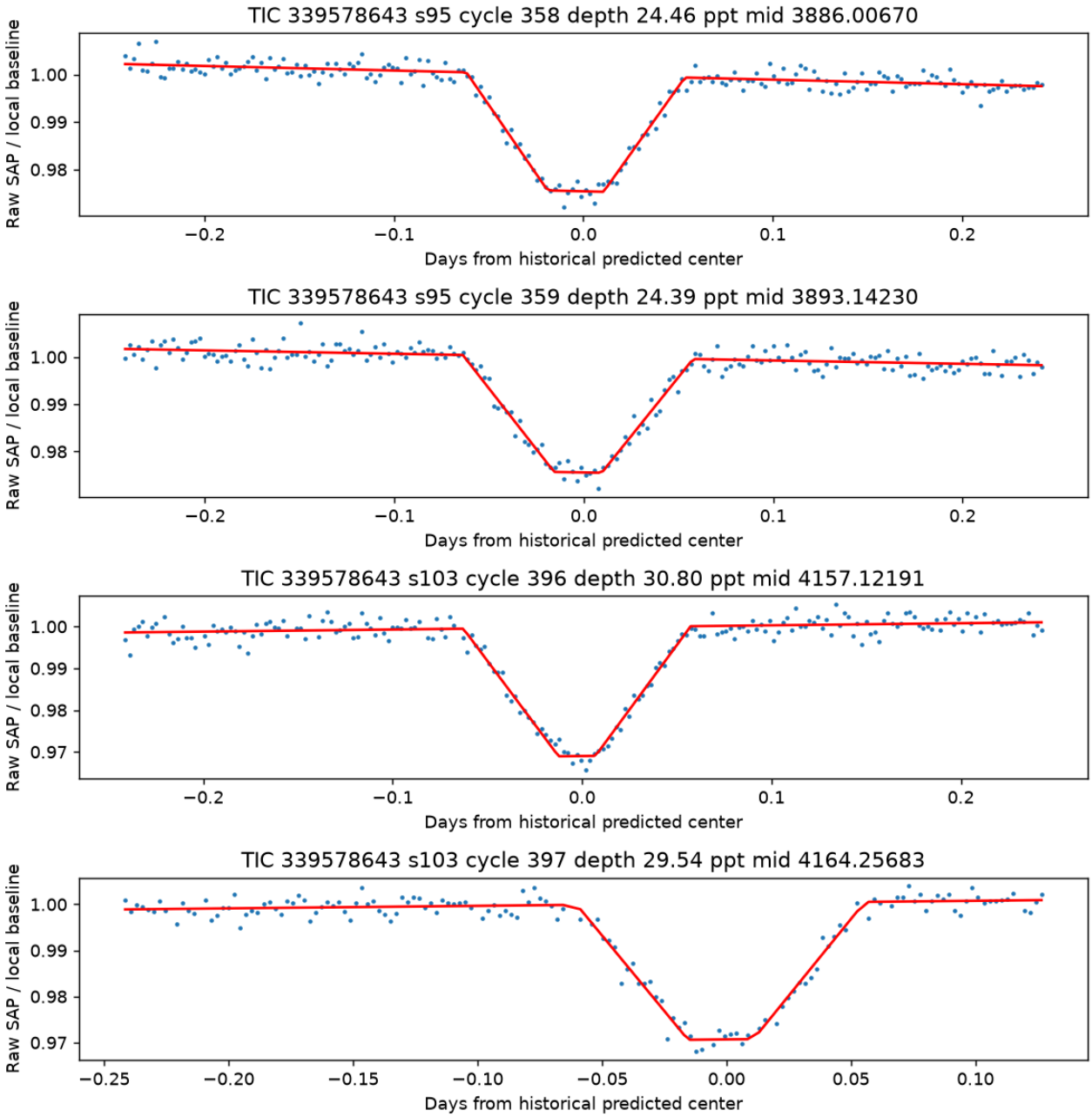
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
95	358	3886.00669743	0.0011574	24.460	True
95	359	3893.14230193	0.0011574	24.386	True
103	396	4157.12190663	0.0011574	30.801	True
103	397	4164.25683375	0.0011574	29.539	True

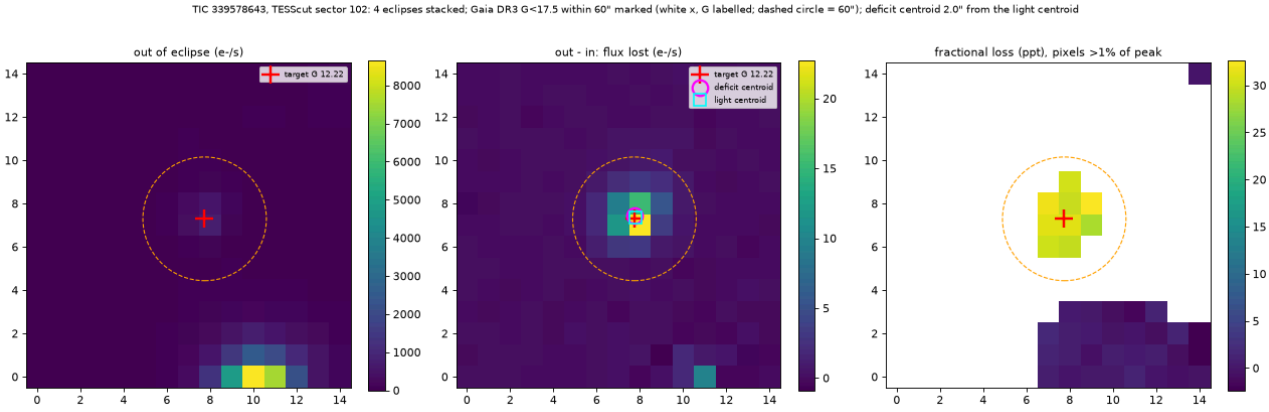
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
None				

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s102.json, historical/diff_s102.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R271. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R271; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R271; diff_s102.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R272. TIC 207813753

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or calibrated multisource PRF localization.

Archived source: sanitized report text.

TIC 207813753: PROBABLE

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 9 directly measured raw QLP SAP events in sectors[12, 93, 103]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 9 same-kind measured times give P=5.40796620007 +/-0.00005411377d, reference eclipse epoch BTJD1648.61982916; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: unresolved after performed check.** DecidingS39 image peak[7, 7] matches target pixel[7, 7]; target-deficit offset0.057pixel, light-deficit offset19.218arcsec; PSF total-light scaled residual0.994. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths29.314/16.217/3.548ppt, 5 stacked events, versus selected-kind raw QLP median32.832ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3583521895442183782 4; P=5.4075148223 +/-0.0012803780d, e=0.091258, omega=234.4079deg. Closest edge-on conjunction offset-0.005827d (-0.001078phase), full-SB1-covariance linear Gaia uncertainty0.330001d at measured epoch. Approximate central duration8.60h vs measured median5.26h permits noncentral eclipses. Mass function0.025727solar, assumed primary mass1.03,

minimum companion0.369solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Completed attribution check remains unresolved:16 capable neighbors, including12.72–18.90arcsec sources. Target-centered deficit favors target but lacks calibrated multistar localization to rule out close sources. Aperture depths differ strongly in crowded field. Real repeating eclipses retained probable.

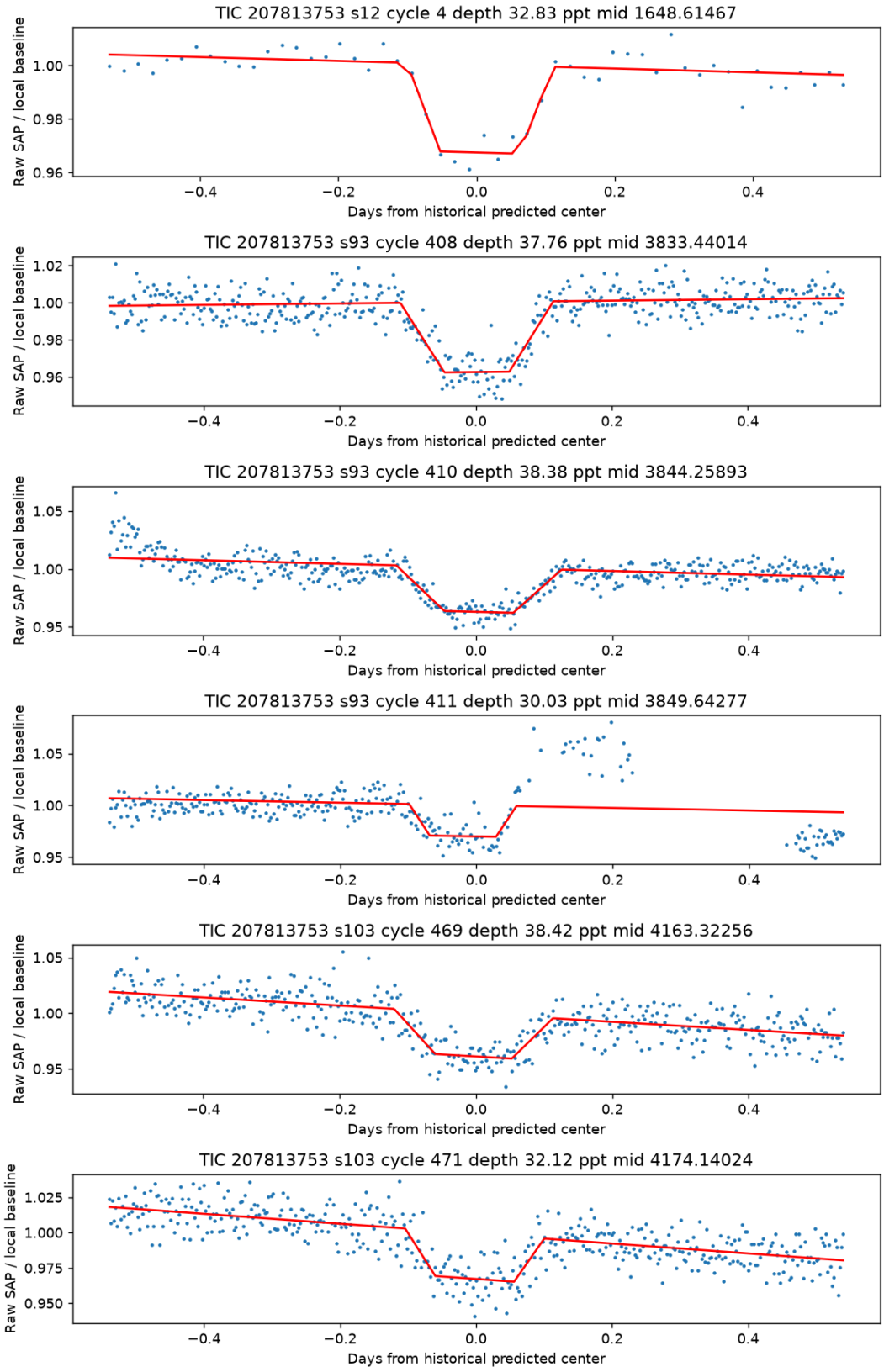
Next check: Resolved eclipse photometry or calibrated multisource PRF localization.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
12	4	1648.61467421	0.0104166	32.832	True
93	408	3833.44014443	0.0018734	37.764	True
93	409	3838.85041011	0.0019785	36.879	True
93	410	3844.25892757	0.0021579	38.382	True
93	411	3849.64276890	0.0028419	30.033	True
103	468	4157.91521042	0.0029628	28.652	True
103	469	4163.32256182	0.0042963	38.422	True
103	470	4168.73180569	0.0019699	31.421	True
103	471	4174.14024031	0.0042243	32.124	True

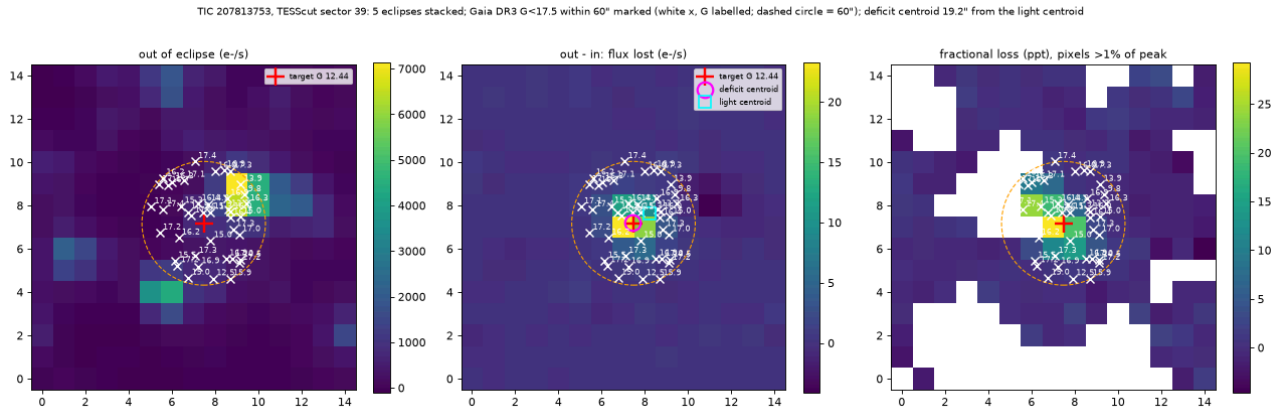
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
5835218920062098176	12.720	15.251	0.4643	0.623
5835218954421849472	17.561	14.998	0.3747	0.899
5835218920062083456	18.497	15.926	0.8368	0.849
5835218920062093440	18.902	14.770	0.3098	0.905
5835218920062078080	26.681	15.336	0.4998	1.251
5835218920062107904	27.183	13.722	0.1382	1.339
5835218920029036544	28.032	15.993	0.8885	1.386
5835219121848729088	35.372	14.990	0.3722	1.751
5835220363171449344	45.581	15.547	0.5997	2.174
5835219126220540672	48.304	9.754	0.0352	2.347
5835219160580307712	48.550	14.465	0.2420	2.429
5835218816982865024	51.062	15.970	0.8698	2.459
5835218816982888832	51.759	13.948	0.1626	2.503
5835220431896165376	53.212	12.547	0.0683	2.623
5835220431896233728	54.327	15.011	0.3788	2.630
5835218954421880192	57.862	15.895	0.8141	2.873

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s39.json, historical/diff_s39.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R272. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R272; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R272; diff_s39.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R273. TIC 173040554

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or calibrated multisource PRF localization.

Archived source: sanitized report text.

TIC 173040554: PROBABLE

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 36 directly measured raw QLP SAP events in sectors[33, 87]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 17 same-kind measured times give P=2.35330880019 +/-0.00000483332d, reference eclipse epoch BTJD2203.63257783; epoch kind: Selected even half-period conjunction; usually deeper, identity provisional. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: unresolved after performed check.** DecidingS33 image peak[8, 8] matches target pixel[8, 8]; target-deficit offset0.369pixel, light-deficit offset17.224arcsec; PSF total-light scaled residual0.818. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths3.864/2.872/1.354ppt, 20 stacked events, versus selected-kind raw QLP median6.434ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3292178275730627008O; P=2.3535060399 +/-0.0001121901d, e=0.044708, omega=342.5383deg. Closest edge-on conjunction offset-0.094204d (-0.040027phase), full-SB1-covariance linear Gaia uncertainty0.086885d at measured epoch. Approximate central duration6.29h vs measured median1.82h permits noncentral eclipses. Mass function0.115351solar, assumed primary mass1.77, minimum companion0.948solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Seventeen selected even-half-cycle times; deeper ordering usually favors even but noisy shapes make primary identity provisional. Thirteen capable neighbors include13.75arcsec G13.70; centroid0.37pixel from target cannot exclude blended sources. Real raw alternating dips and orbital-period fit retained probable. Geometry phase offset-0.040 is discussed with Gaia covariance, not an automatic veto.

Next check: Resolved eclipse photometry or calibrated multisource PRF localization.

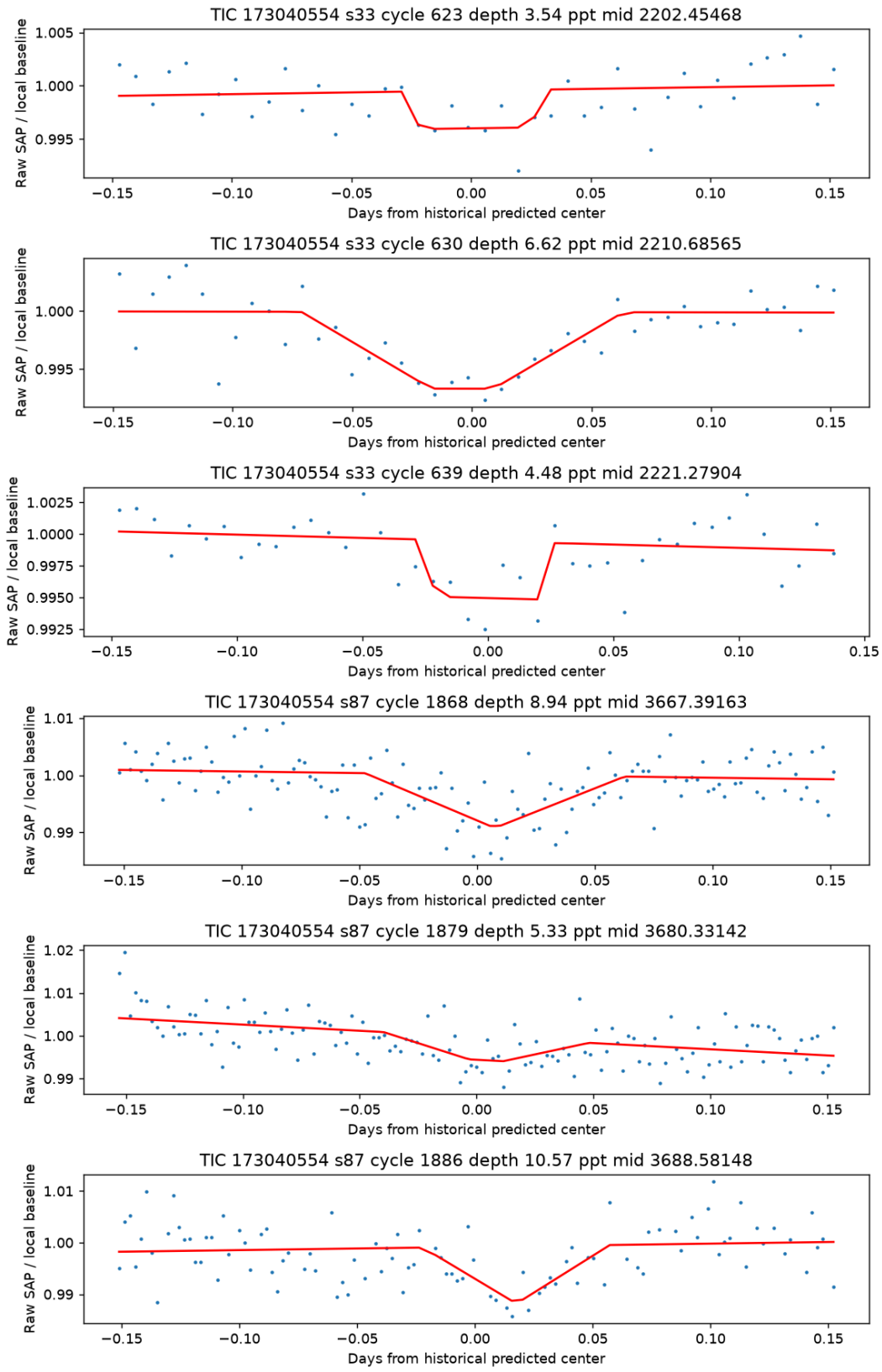
Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
33	623	2202.45467722	0.0034723	3.536	True
33	624	2203.63153055	0.0034723	5.934	True
33	625	2204.80777236	0.0034723	5.208	True
33	626	2205.98544864	0.0042922	7.216	True
33	627	2207.16618782	0.0056696	5.103	True
33	628	2208.34073568	0.0034723	6.580	True
33	629	2209.50507888	0.0034723	4.916	True
33	630	2210.68565454	0.0051993	6.622	True
33	631	2211.86553740	0.0050984	4.987	True
33	632	2213.04956265	0.0034723	6.138	True
33	635	2216.57968852	0.0073950	5.106	True
33	636	2217.74892481	0.0034723	6.434	True
33	637	2218.93131282	0.0036913	3.924	True
33	638	2220.10382583	0.0064483	7.427	True
33	639	2221.27903720	0.0034723	4.483	True
33	640	2222.45910772	0.0042123	7.719	True
33	641	2223.63627837	0.0034723	3.959	True
33	642	2224.81109345	0.0069973	6.010	True
33	643	2225.99251302	0.0036874	7.494	True
33	644	2227.16911949	0.0034723	5.938	True
87	1867	3666.21007917	0.0060446	3.930	True
87	1868	3667.39163276	0.0032113	8.945	True
87	1869	3668.58315954	0.0102858	3.139	True
87	1871	3670.92014296	0.0047045	3.912	True
87	1872	3672.10710251	0.0020568	7.175	True
87	1873	3673.27522041	0.0060850	3.876	True
87	1874	3674.44442136	0.0015527	4.656	True
87	1875	3675.62415262	0.0034833	4.023	True
87	1879	3680.33141713	0.0063028	5.331	True
87	1880	3681.50391813	0.0039498	4.884	True
87	1881	3682.69630068	0.0050018	6.465	True
87	1882	3683.86279221	0.0019153	5.799	True
87	1883	3685.06527153	0.0056078	5.264	True
87	1884	3686.21837377	0.0033512	4.302	True
87	1885	3687.39761630	0.0029788	8.967	True
87	1886	3688.58147585	0.0039129	10.567	True

Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
2921782761599913984	12.797	16.964	0.3031	0.954
2921782761604874368	13.754	13.698	0.0185	0.992
2921782692885397120	43.176	14.306	0.0296	2.471
2921782692885398016	44.143	17.230	0.3863	2.514
2921782795964604544	48.051	16.143	0.1443	2.341

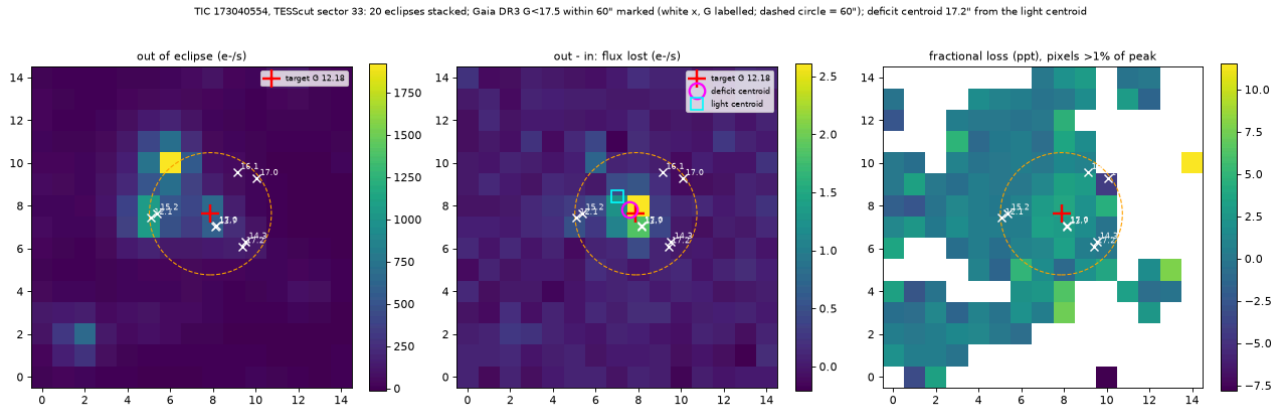
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
2921782658525666944	49.379	17.681	0.5835	2.250
2921785613463163520	51.960	15.245	0.0652	2.199
2921782795964603392	54.451	17.694	0.5907	2.595
2921782727245128704	56.367	18.225	0.9606	2.946
2921782658525668736	56.924	18.166	0.9100	2.822
2921782795964603136	56.940	17.049	0.3277	2.878
2921785609164554752	57.993	12.077	0.0070	2.502
2921782692885396224	59.934	17.654	0.5691	3.287

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R273. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R273; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R273; diff_s33.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R274. TIC 67300566

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or calibrated multisource PRF localization.

Archived source: sanitized report text.

TIC 67300566: PROBABLE

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 6 directly measured raw QLP SAP events in sectors[73, 86]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 6 same-kind measured times give P=4.42341934165 +/-0.00002389186d, reference eclipse epoch BTJD3288.44495593; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: unresolved after performed check.** DecidingS59 image peak[7, 7] matches target pixel[7, 7]; target-deficit offset0.051pixel, light-deficit offset6.735arcsec; PSF total-light scaled residual0.272. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths6.926/6.929/6.498ppt, 5 stacked events, versus selected-kind raw QLP median12.142ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3189461074831319680; P=4.4238367396 +/-0.0002841869d, e=0.083592, omega=61.1479deg. Closest edge-on conjunction offset-0.147851d (-0.033421phase), full-SB1-covariance linear Gaia uncertainty0.180326d at measured epoch. Approximate central duration3.99h vs measured median1.51h permits noncentral eclipses. Mass function0.076596solar, assumed primary mass1.145, minimum

companion0.620solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Real repeated raw dips; capable G12.84 neighbor18.26arcsec away lies0.84pixel from deficit versus0.05pixel target offset. Uncalibrated close-source localization prevents definitive attribution; completed review retains probable.

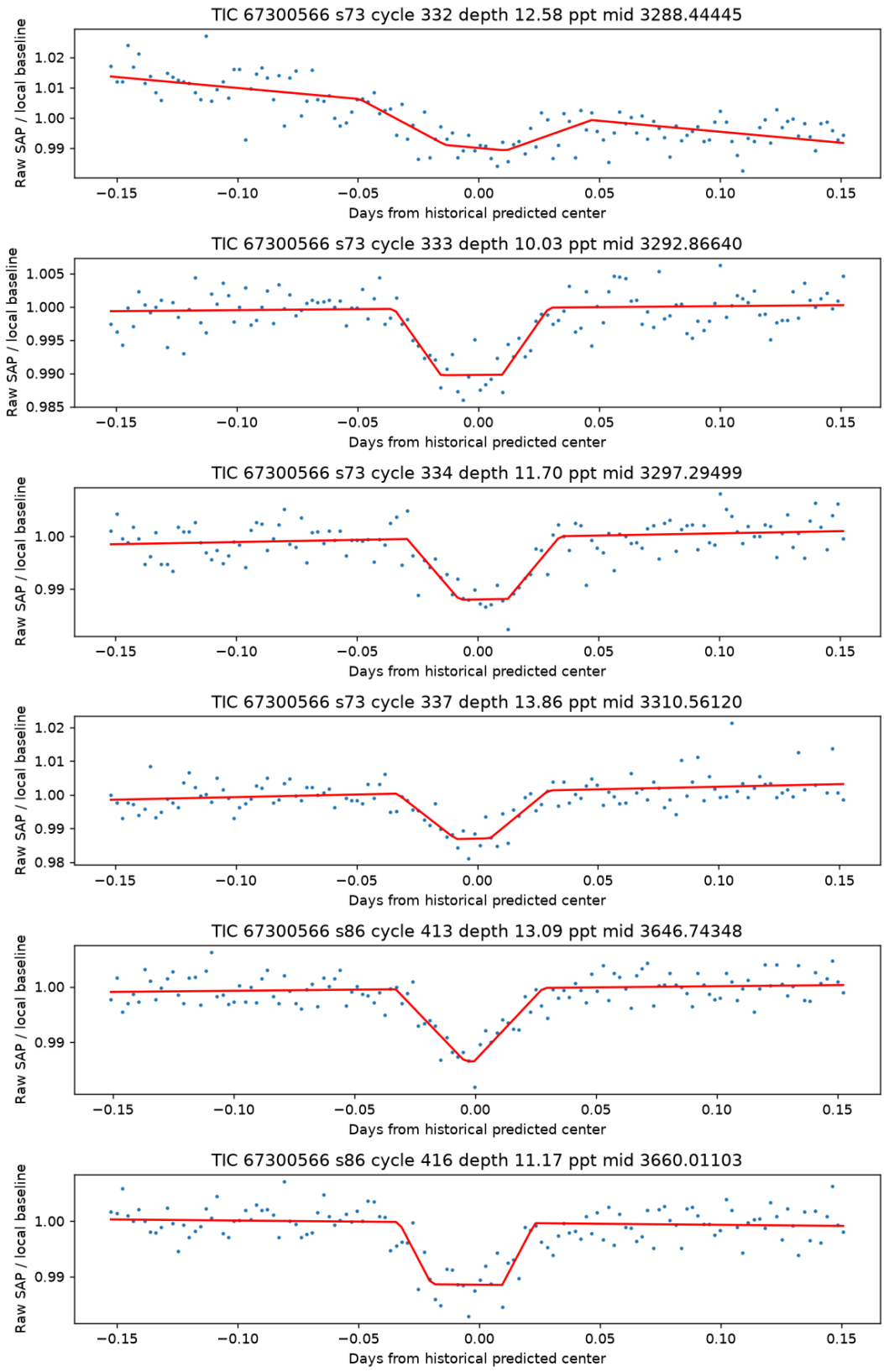
Next check: Resolved eclipse photometry or calibrated multisource PRF localization.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
73	332	3288.44445394	0.0052524	12.584	True
73	333	3292.86640317	0.0012004	10.033	True
73	334	3297.29498816	0.0013114	11.699	True
73	337	3310.56119791	0.0013023	13.865	True
86	413	3646.74348143	0.0013213	13.087	True
86	416	3660.01103005	0.0011574	11.173	True

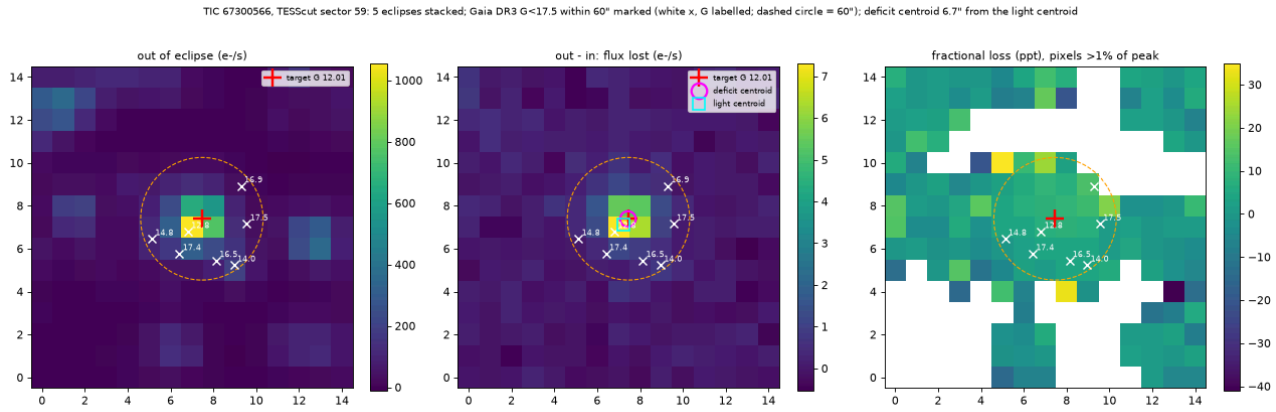
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
189461044769259648	18.262	12.839	0.0336	0.836
189461010409521408	41.798	16.454	0.6516	2.058
189461109191746432	49.582	16.879	0.9583	2.432
189461044769263232	51.897	14.831	0.1544	2.462
189461006112498688	53.184	14.006	0.0779	2.644

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R274. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R274; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R274; diff_s59.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R275. TIC 20176241

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 20176241: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 3 directly measured raw QLP SAP events in sectors[21, 48]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 3 same-kind measured times give P=13.51921255715 +/-0.00106281621d, reference eclipse epoch BTJD1890.22784388; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS48 image peak[8, 7] matches target pixel[8, 7]; target-deficit offset0.095pixel, light-deficit offset1.009arcsec; PSF total-light scaled residual0.045. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths80.163/80.642/79.239ppt, 2 stacked events, versus selected-kind raw QLP median85.111ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3819970107903053312; P=13.5172019155 +/-0.0032729947d, e=0.059889, omega=16.4528deg. Closest edge-on conjunction offset0.152987d (0.011318phase), full-SB1-covariance linear Gaia uncertainty0.321871d at measured epoch. Approximate central duration6.98h vs measured median3.89h permits noncentral eclipses. Mass function0.115116solar, assumed primary mass1.08, minimum companion0.720solar. This is

not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Two well-sampledS48 events and third clear sparseS21 recurrence test period across55cycles. S21 missing egress/sparse core gets duration/2 conservative timing allowance, not calibrated confidence. Current MAST query finds only21/48, no newer sector.

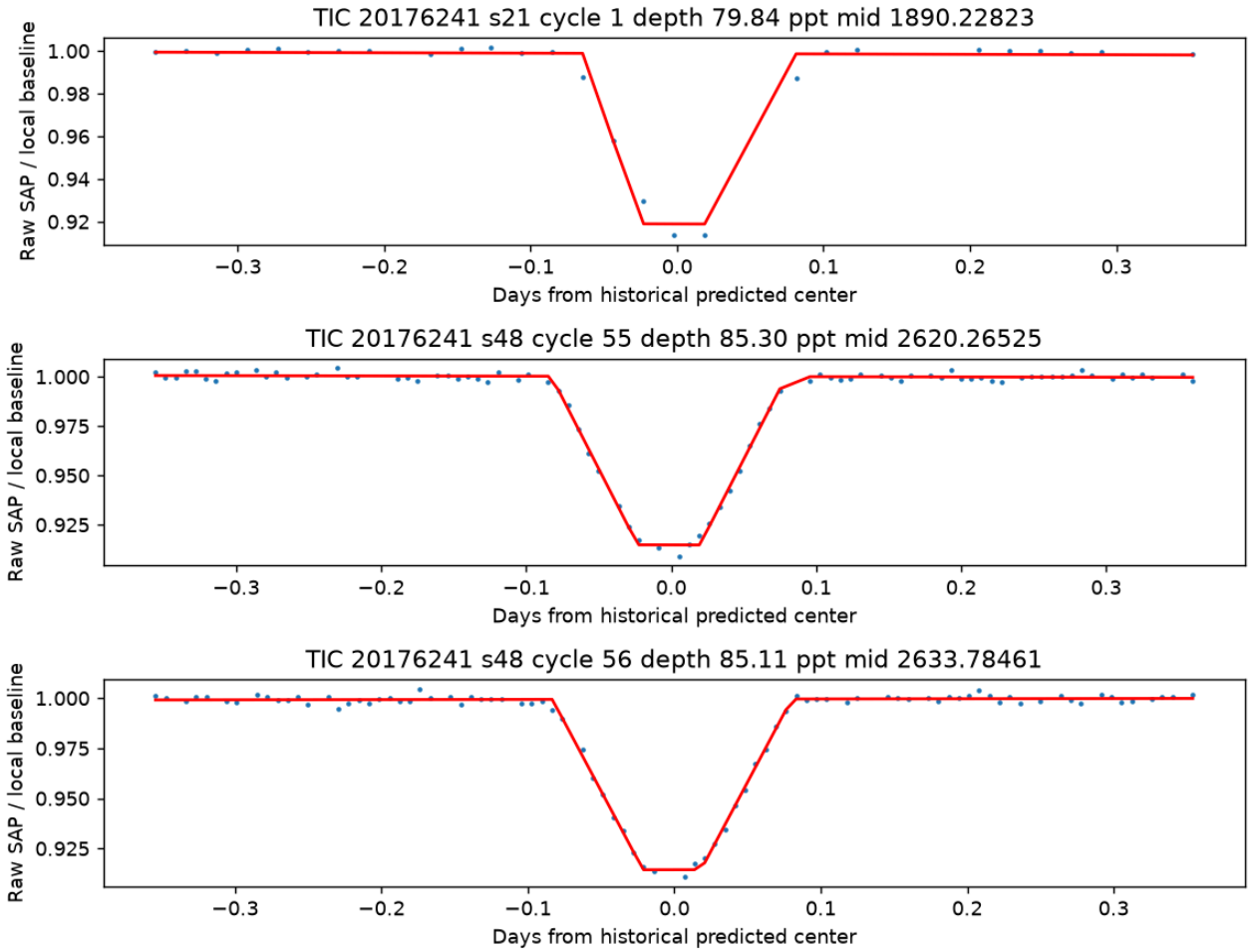
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
21	1	1890.22823249	0.0592790	79.842	True
48	55	2620.26524864	0.0034722	85.301	True
48	56	2633.78460652	0.0034722	85.111	True

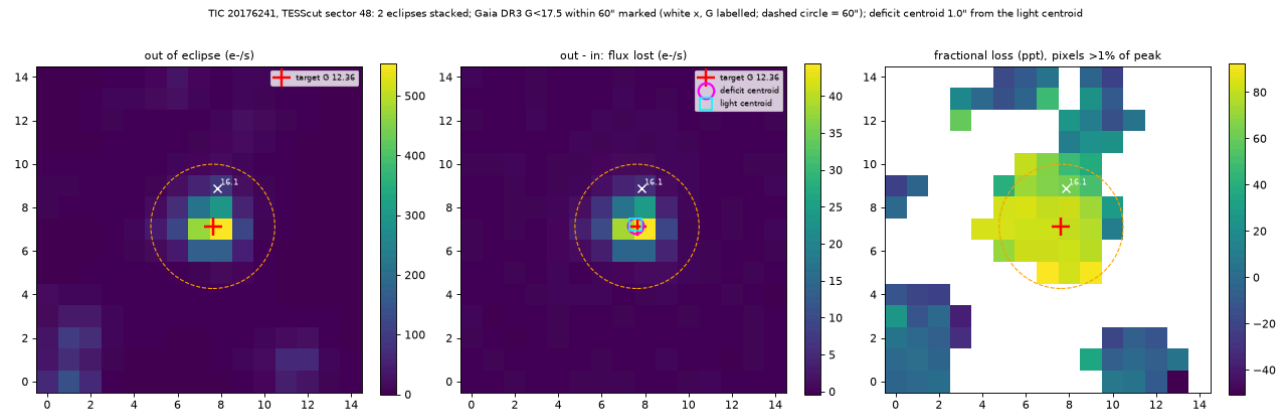
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
None				

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s48.json, historical/diff_s48.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R275. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R275; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R275; diff_s48.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R276. TIC 387275908

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization; no missing bounded admission check.

Archived source: sanitized report text.

TIC 387275908: CONFIRMED

Full bounded review COMPLETE under current paper section2.7. Historical screened status was not an admission. Catalogue novelty is separate; central seven-group coverage applies, see ../../catalogues/coverage-matrix.csv. No exhaustive priority claim.

1. **Raw repeated eclipses: pass.** 10 directly measured raw QLP SAP events in sectors[46, 87]. Representative raw-event plots and numeric windows inspected. A local linear baseline plus free trapezoid was fitted to quality-zero cadences. Predicted centers only guide windows. Flagged/uncovered windows are not clean non-detections. At most three spread QLP sectors used; no mandatory all-event/all-sector fitting.
2. **Timing: pass with limitations below.** 10 same-kind measured times give P=4.39055204155 +/-0.00001165698d, reference eclipse epoch BTJD2556.19633037; epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Historical photometric cycle branch guides association, not a fixed Gaia period fit. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. SparseS21 TIC20176241 instead uses duration/2 conservative allowance. Zero/singular timing fits are excluded; their real raw detections remain evidence. This is not a blind all-alias search.
3. **Attribution: pass at TESS image resolution with limitations below.** DecidingS72 image peak[8, 8] matches target pixel[8, 8]; target-deficit offset0.097pixel, light-deficit offset6.110arcsec; PSF total-light scaled residual0.823. Gaia identifiers/WCS validated in identity-check.json (zero pixel mismatch). Capable neighbors listed below; required eclipse is approximate Gaia-G flux budget and pixel fractional losses are blended, not intrinsic source eclipses.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut yields1x1/3x3/5x5 depths7.102/7.464/6.359ppt, 6 stacked events, versus selected-kind raw QLP median8.059ppt. Aperture dilution and mixed primary/secondary stacks may change depth. Reused numeric image and plots, cutout identity and source algorithm checked; same TESS photons test reduction robustness, not independent observing occasions.
5. **Gaia geometry: consistent within stated model limitations.** DR3316613880557800601 6; P=4.3913179458 +/-0.0006353543d, e=0.075001, omega=354.8170deg. Closest edge-on conjunction offset-0.257200d (-0.058570phase), full-SB1-covariance linear Gaia uncertainty0.254946d at measured epoch. Approximate central duration6.45h vs measured median2.15h permits noncentral eclipses. Mass function0.058270solar, assumed primary mass1.17,

minimum companion0.558solar. This is not a detailed binary fit or an arbitrary sigma veto; correlation convention and near-reference sensitivity recorded in ../gaia-geometry.json.

Deciding limitations: Capable neighbors27.34 and36.11arcsec are spatially separated from deficit0.10pixel from target. High total-light PSF residual0.823 reflects crowding and is not used alone as attribution evidence. Gaia phase offset-0.059 has covariance uncertainty0.255d; duration can be noncentral.

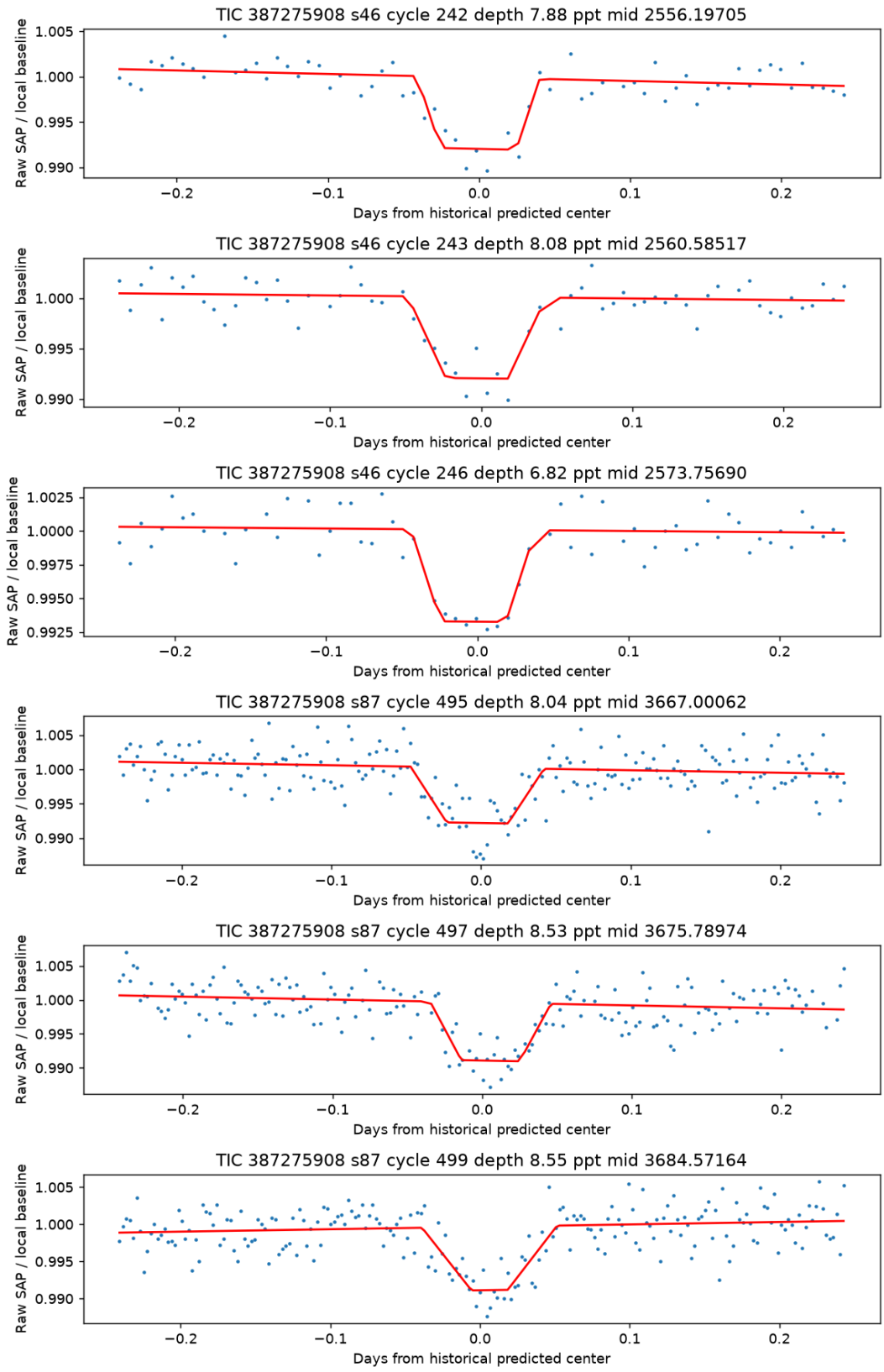
Next check: Optional precision characterization; no missing bounded admission check.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Depth ppt	Usable timing before same-kind selection
46	242	2556.19705006	0.0034724	7.880	True
46	243	2560.58516895	0.0034724	8.081	True
46	245	2569.37245514	0.0034724	7.755	True
46	246	2573.75689717	0.0034724	6.815	True
46	247	2578.14744976	0.0034724	8.541	True
87	495	3667.00062154	0.0016082	8.037	True
87	496	3671.39959704	0.0018368	8.965	True
87	497	3675.78973554	0.0017435	8.526	True
87	498	3680.17549709	0.0020934	7.428	True
87	499	3684.57163960	0.0020927	8.546	True

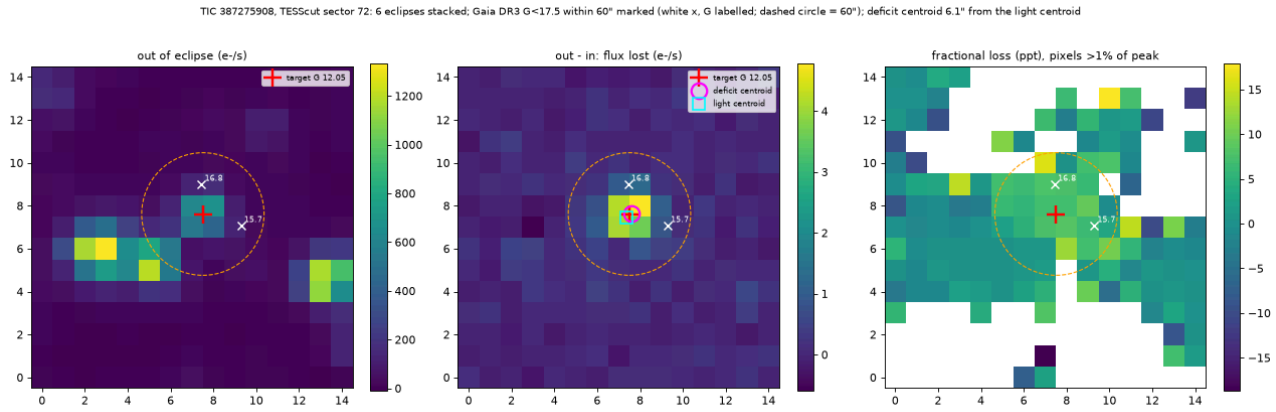
Capable Gaia neighbor	Separation arcsec	G	Required intrinsic fractional eclipse	Distance from deficit pixel
3166139119111943424	27.344	16.766	0.5694	1.397
3166138741154825728	36.107	15.699	0.2177	1.758

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s72.json, historical/diff_s72.png, historical/gaia.json; ../identity-check.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Original cutout: [local path omitted] QLP FITS and source paths with hashes retained in manifest. Difference events' JSON coordinates are predictions, never timing measurements. Interpretation uses bounded representative evidence rather than automatic historical screening.

Stable review key: R276. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R276; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R276; diff_s72.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

Reserved and cumulative magnitude comparison. The second reserved sixty: At G = 11.5-12 (actual DR3 range 11.6648-11.9955), 29/30 are physically confirmed (96.7%; descriptive Wilson 95% interval 83.3-99.4%), 1 probable and 0 plausible; 30 new counted additions. At G = 12-12.5 (actual DR3 range 12.0036-12.4872), 26/30 are physically confirmed (86.7%; descriptive Wilson 95% interval 70.3-94.7%), 2 probable and 2 plausible; 28 new counted additions. Both rounds together: At G = 11.5-12 (actual DR3 range 11.6648-11.9955), 56/60 are physically confirmed (93.3%; descriptive Wilson 95% interval 84.1-97.4%), 4 probable and 0 plausible; 60 new counted additions. At G = 12-12.5 (actual DR3 range 12.0029-12.4879), 47/60 are physically confirmed (78.3%; descriptive Wilson 95% interval 66.4-86.9%), 10 probable and 3 plausible; 56 new counted additions. All 120 selected targets completed the bounded review and remain in their selected denominators. Known recoveries enter physical confirmation but add nothing; probable and plausible outcomes remain separate. The first-round results above are retained unchanged. These are descriptive remaining-screened candidate comparisons, not blind discovery yield, completeness or a causal magnitude effect. Wilson intervals are binomial model summaries despite finite historical strata; case mix, depletion, crowding, coverage and selection bookkeeping limit interpretation.

R277. TIC 336856559

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 336856559: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 13 distinct eclipse events in selected sectors[66, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 13 same-kind midtimes give P=2.9097059074 ±0.0000018140d, reference eclipse epoch BTJD3098.6256467, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S103 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.441arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.937. Capable neighbours: Gaia 5969863533437354240: 45.91 arcsec, G9.994, required fractional eclipse0.170
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths99.668/54.231/27.297ppt, with 8 stacked events. QLP selected-kind fitted median124.260ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3596986352907839848, P=2.9097381651±0.0001997714d, e=0.101456, omega=131.0731deg. Closest edge-on conjunction offset-0.035368d (-0.012155phase), full-SB1-covariance linear propagated Gaia uncertainty0.195472d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.005336d (photometric period projection allowance0.001689d). The approximate central duration4.67h vs observed median3.37h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Strong aperture dilution: cutout1/3/5 depths99.7/54.2/27.3ppt; QLP sectors66/103 differ. Deficit0.211pixel from target, bright capable neighbor2.26pixels from deficit favors target. Depths are pipeline/aperture dependent, not intrinsic.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

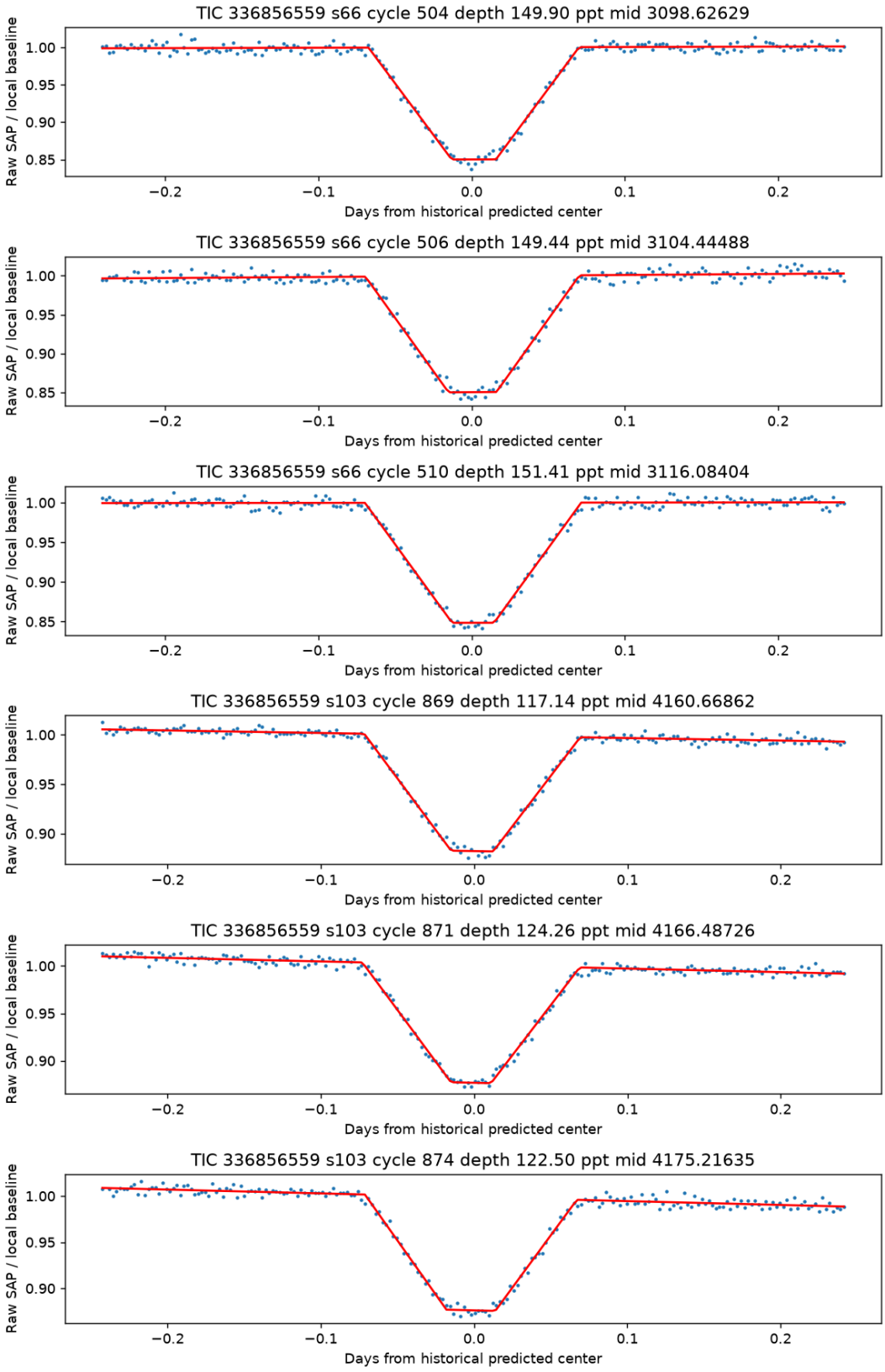
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

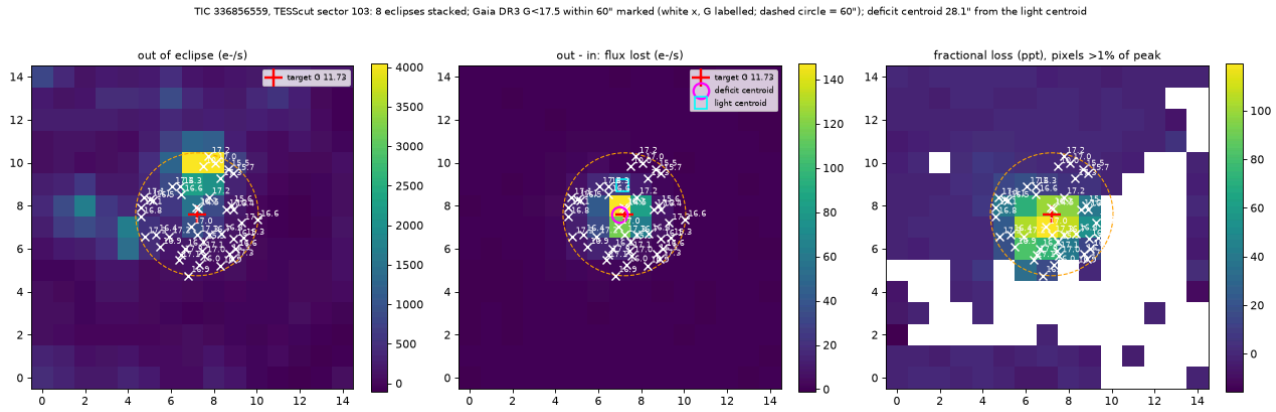
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
66	504	3098.6262889	0.001157	149.901	usable; same-kind selection applies
66	505	3101.5347270	0.001157	152.075	usable; same-kind selection applies
66	506	3104.4448759	0.001157	149.438	usable; same-kind selection applies
66	509	3113.1741657	0.001157	149.144	usable; same-kind selection applies
66	510	3116.0840417	0.001157	151.414	usable; same-kind selection applies
103	867	4154.8493451	0.001157	122.891	usable; same-kind selection applies
103	868	4157.7592234	0.001157	123.057	usable; same-kind selection applies
103	869	4160.6686202	0.001157	117.136	usable; same-kind selection applies
103	870	4163.5779190	0.001157	131.159	usable; same-kind selection applies
103	871	4166.4872588	0.001157	124.260	usable; same-kind selection applies
103	872	4169.3972923	0.001157	123.971	usable; same-kind selection applies
103	873	4172.3068993	0.001157	115.232	usable; same-kind selection applies
103	874	4175.2163536	0.001157	122.496	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s103.json, historical/diff_s103.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R277. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R277; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R277; diff_s103.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R278. TIC 146878534

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 146878534: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[33, 87]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=8.9801438437 ±0.0000144498d, reference eclipse epoch BTJD2207.2276616, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S33 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.872arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.982. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths140.094/127.643/68.760ppt, with 3 stacked events. QLP selected-kind fitted median156.945ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3294612244435067865 6, P=8.9793296680±0.0004372996d, e=0.037114, omega=168.8437deg. Closest edge-on conjunction offset0.112481d (0.012527phase), full-SB1-covariance linear propagated Gaia uncertainty0.109595d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.051982d (photometric period projection allowance0.002919d). The approximate central duration5.91h vs observed median3.52h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Clean repeated sector33 eclipses150-174ppt; sector87 depths113-277ppt have changing steep baselines/noise. Real morphology and timing recur; apparent depth variability is not established astrophysically. Local-model timing systematics remain.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

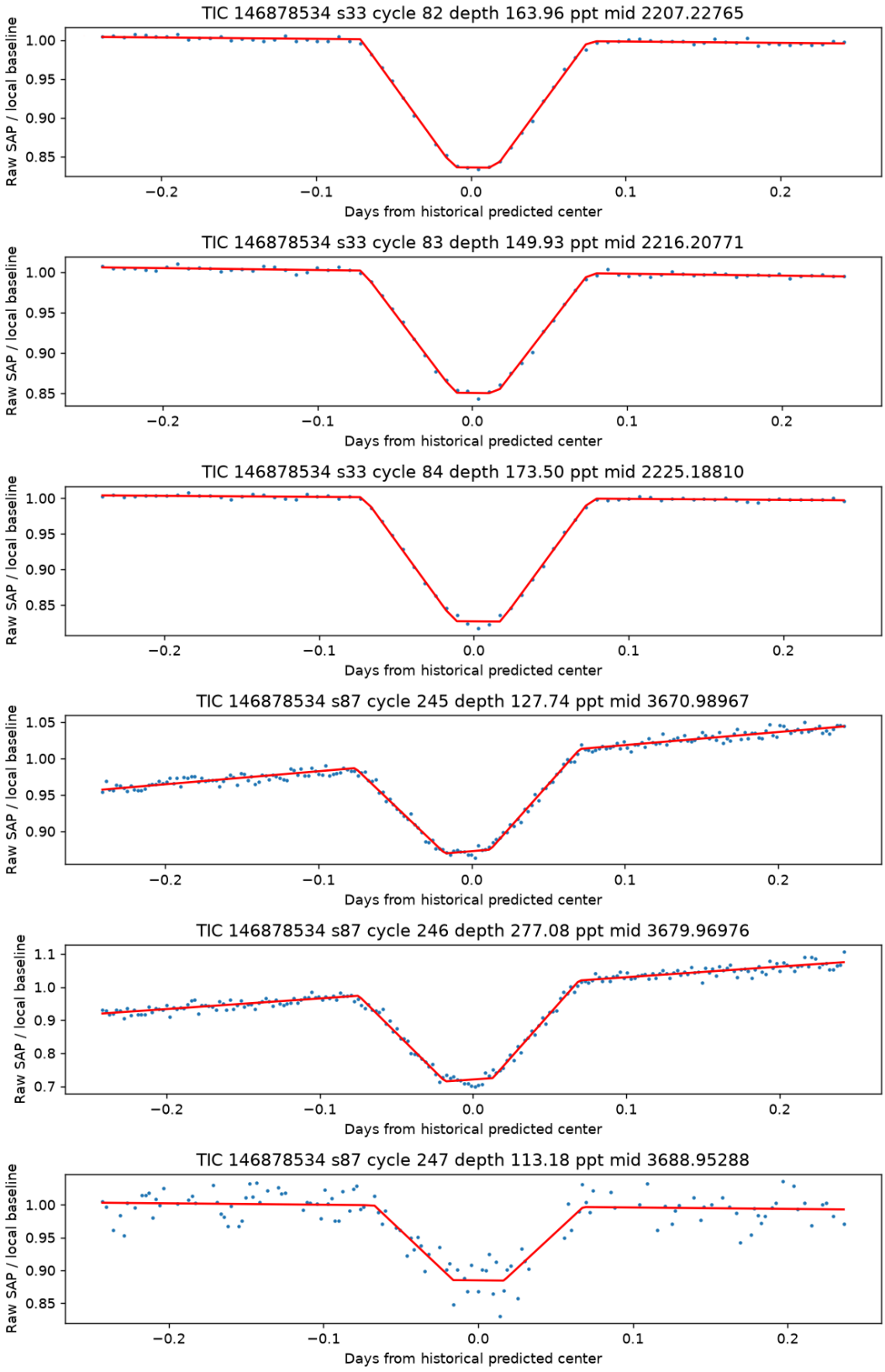
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

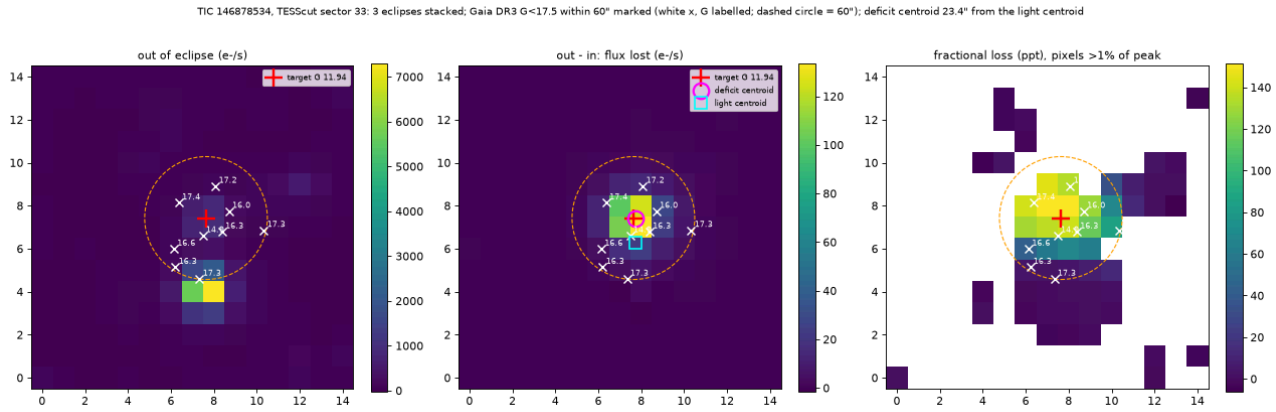
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	82	2207.2276538	0.003472	163.964	usable; same-kind selection applies
33	83	2216.2077107	0.003472	149.927	usable; same-kind selection applies
33	84	2225.1881019	0.003472	173.500	usable; same-kind selection applies
87	245	3670.9896667	0.002756	127.744	usable; same-kind selection applies
87	246	3679.9697552	0.002272	277.076	usable; same-kind selection applies
87	247	3688.9528822	0.001768	113.176	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s33.json, historical/diff_s33.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R278. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R278; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R278; diff_s33.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R279. TIC 123220126

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 123220126: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 12 distinct eclipse events in selected sectors[8, 62, 99]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=7.9647365731 ±0.0000076939d, reference eclipse epoch BTJD1519.0816018, kind: Deeper repeating eclipse; even half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S62 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.449arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.558. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths105.928/103.207/96.327ppt, with 6 stacked events. QLP selected-kind fitted median120.347ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR33824101617499474304, P=7.9648895117±0.0004901494d, e=0.017204, omega=144.7292deg. Closest edge-on conjunction offset-0.099881d (-0.012540phase), full-SB1-covariance linear propagated Gaia uncertainty0.050834d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.078164d (photometric period projection allowance0.001093d). The approximate central duration6.89h vs observed median4.58h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipse depths establish doubled orbital period. Even historical half-period cycles alone define deeper-eclipse ephemeris; historical cutout stack mixes kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

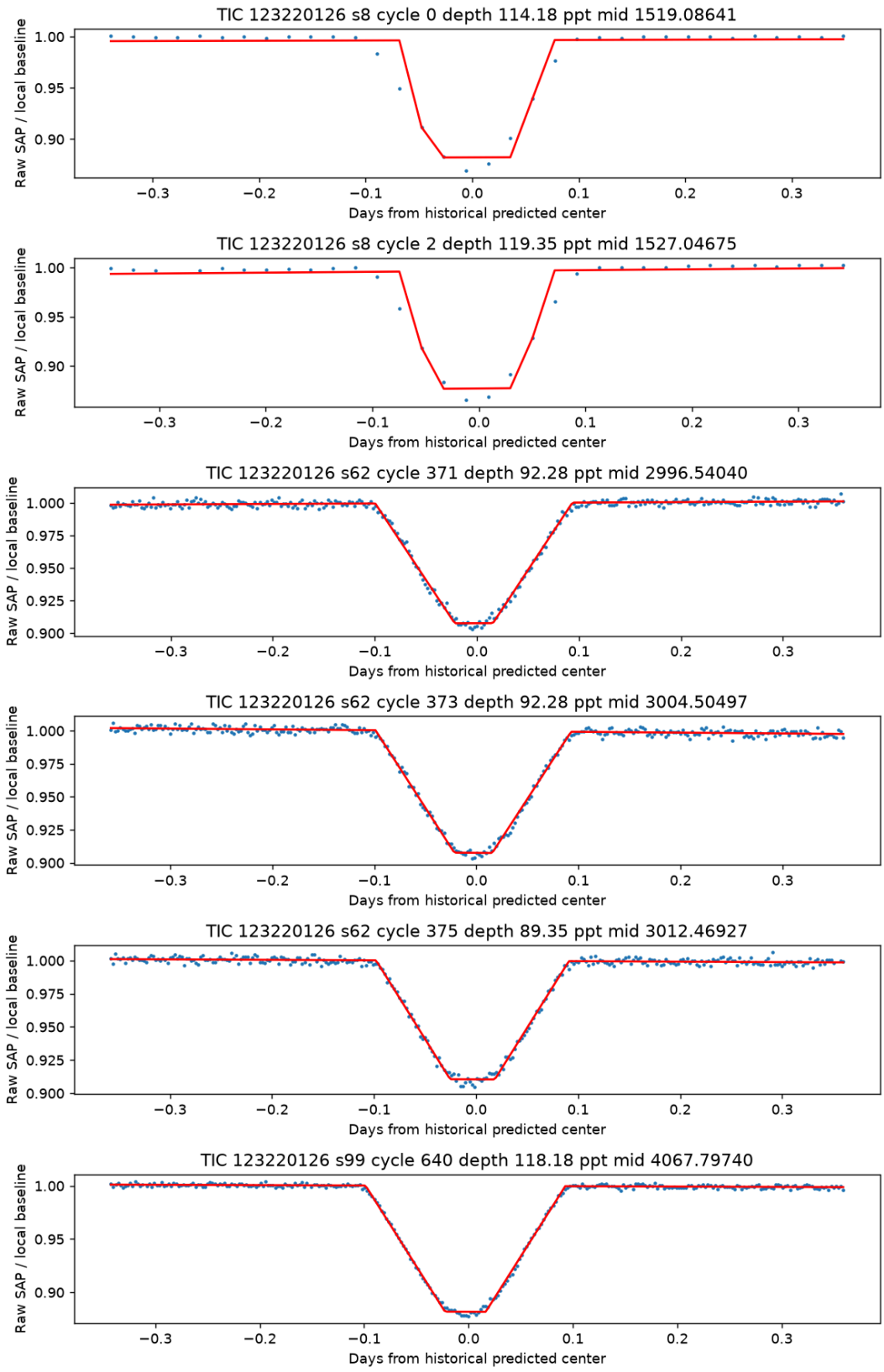
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

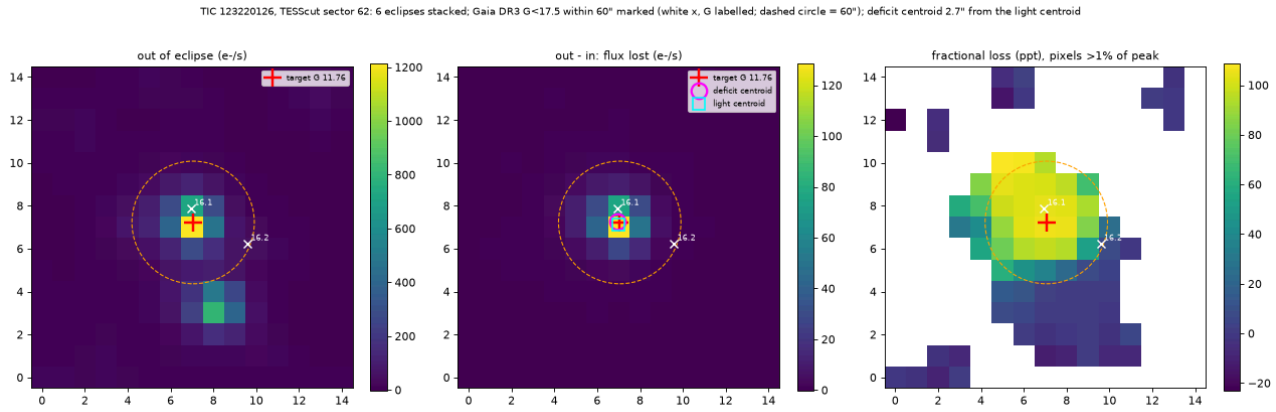
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
8	0	1519.0864143	0.010417	114.184	usable; same-kind selection applies
8	1	1523.0667550	0.010417	79.475	usable; same-kind selection applies
8	2	1527.0467454	0.010417	119.345	usable; same-kind selection applies
62	370	2992.5576999	0.001157	136.052	usable; same-kind selection applies
62	371	2996.5404004	0.001157	92.284	usable; same-kind selection applies
62	372	3000.5228394	0.001157	137.775	usable; same-kind selection applies
62	373	3004.5049693	0.001157	92.278	usable; same-kind selection applies
62	374	3008.4871198	0.001157	137.932	usable; same-kind selection applies
62	375	3012.4692650	0.001157	89.354	usable; same-kind selection applies
99	636	4051.8678251	0.001157	120.347	usable; same-kind selection applies
99	639	4063.8153617	0.001157	79.716	usable; same-kind selection applies
99	640	4067.7974018	0.001157	118.182	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R279. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R279; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R279; diff_s62.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R280. TIC 42960522

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 42960522: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 20 distinct eclipse events in selected sectors[12, 99, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 20 same-kind midtimes give P=2.5438060407 ±0.0000075069d, reference eclipse epoch BTJD1627.7359239, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S38 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation6.451arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.589. Capable neighbours: Gaia 5886342431117033728: 29.06 arcsec, G12.760, required fractional eclipse0.072; Gaia 5886336555601769088: 34.16 arcsec, G13.662, required fractional eclipse0.137; Gaia 5886336521242033536: 58.54 arcsec, G13.330, required fractional eclipse0.107; Gaia 5886342431117035648: 58.97 arcsec, G15.833, required fractional eclipse0.875; Gaia 5886336585650415360: 59.38 arcsec, G15.897, required fractional eclipse0.927
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths20.934/15.800/12.060ppt, with 10 stacked events. QLP selected-kind fitted median22.061ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35886336555601771392, P=2.5437447918±0.0002679482d, e=0.084019, omega=95.1819deg. Closest edge-on conjunction

offset0.042597d (0.016746phase), full-SB1-covariance linear propagated Gaia uncertainty0.140235d at the measured epoch. Near-Gaia-reference projected eclipse offset0.012769d (photometric period projection allowance0.003656d). The approximate central duration3.20h vs observed median3.27h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deficit0.307pixel from target; nearest capable neighbor1.55pixels from deficit. Spatial core supports target at TESS resolution; no calibrated centroid errors.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

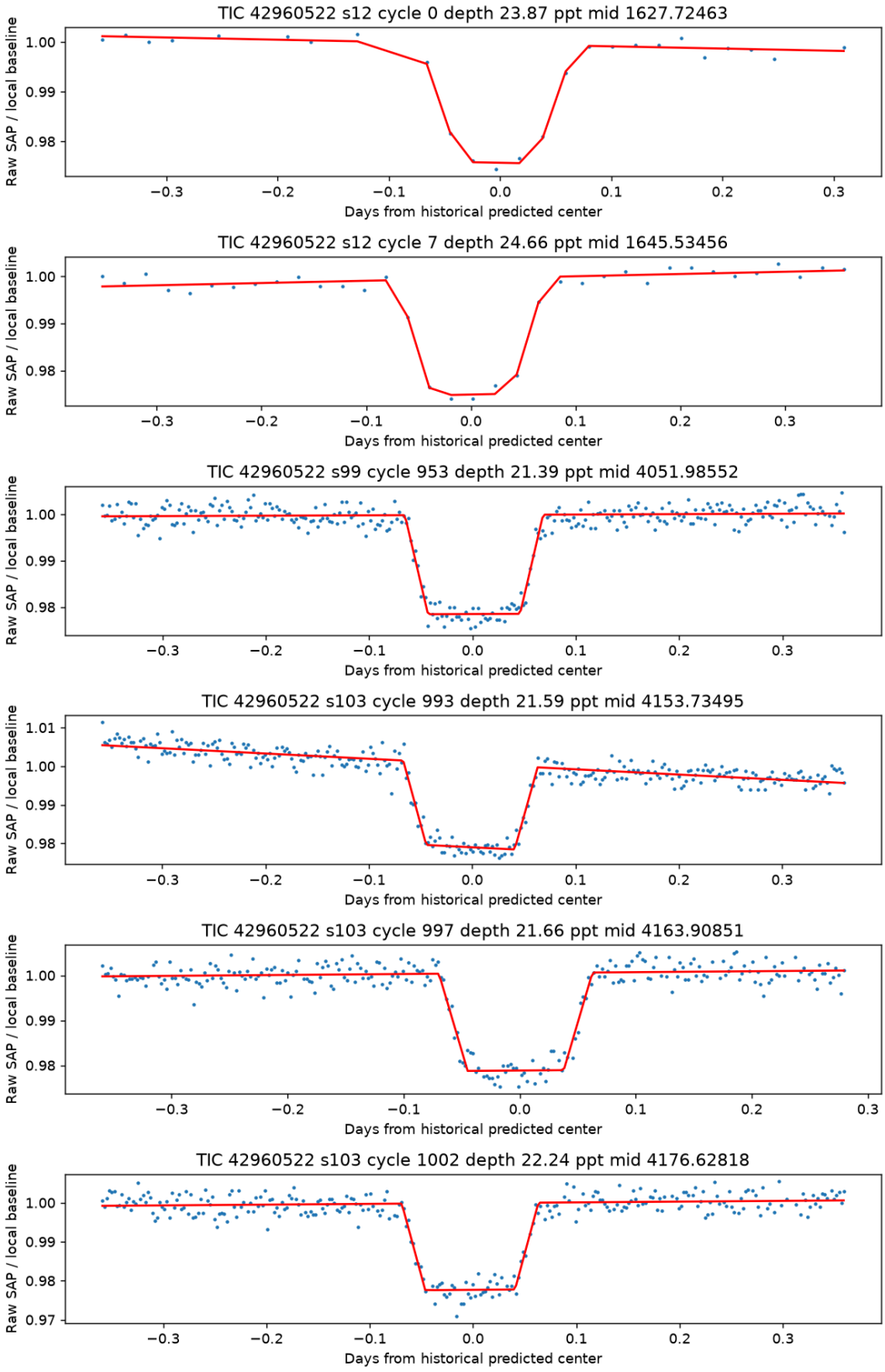
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
12	0	1627.7246309	0.010416	23.866	usable; same-kind selection applies
12	5	1640.4445722	0.010416	21.960	usable; same-kind selection applies
12	6	1642.9885060	0.010416	23.265	usable; same-kind selection applies
12	7	1645.5345634	0.010416	24.656	usable; same-kind selection applies
12	9	1650.6206212	0.010416	24.767	usable; same-kind selection applies
99	951	4046.8975859	0.001157	22.161	usable; same-kind selection applies
99	952	4049.4411514	0.001157	22.265	usable; same-kind selection applies
99	953	4051.9855243	0.001157	21.386	usable; same-kind selection applies
99	954	4054.5294988	0.001157	22.438	usable; same-kind selection applies
99	958	4064.7051281	0.001157	21.113	usable; same-kind selection applies
99	960	4069.7916359	0.001157	20.224	usable; same-kind selection applies
103	993	4153.7349545	0.001157	21.587	usable; same-kind selection applies
103	994	4156.2784550	0.001157	21.849	usable; same-kind selection applies
103	995	4158.8220809	0.001157	24.906	usable; same-kind selection applies
103	996	4161.3646923	0.001157	21.138	usable; same-kind selection applies
103	997	4163.9085124	0.001157	21.659	usable; same-kind selection applies
103	998	4166.4525329	0.001614	23.179	usable; same-kind selection applies
103	999	4168.9956876	0.001157	21.428	usable; same-kind selection applies
103	1001	4174.0831874	0.001157	21.791	usable; same-kind selection applies

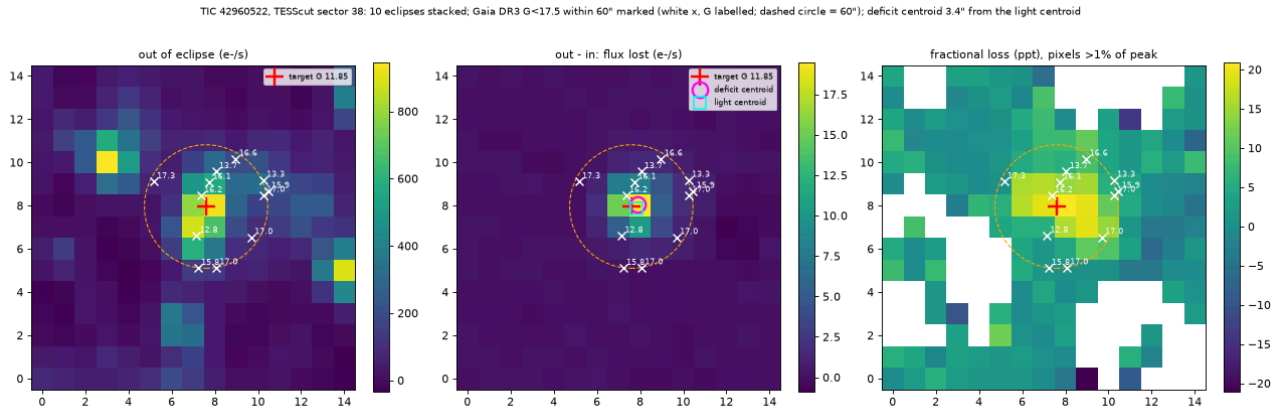
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
103	1002	4176.6281771	0.001157	22.242	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R280. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R280; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R280; diff_s38.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R281. TIC 374589618

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 374589618: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 23 distinct eclipse events in selected sectors[54, 81]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=3.1662972967 ±0.0000098424d, reference eclipse epoch BTJD2778.4906828, kind: Deeper repeating eclipse; odd half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S81 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.081arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.658. Capable neighbours: Gaia 1802168316060661376: 13.53 arcsec, G17.280, required fractional eclipse0.790; Gaia 1802168311763607936: 24.73 arcsec, G17.409, required fractional eclipse0.889; Gaia 1802168419139874048: 31.65 arcsec, G15.749, required fractional eclipse0.197; Gaia 1802165464202369664: 42.66 arcsec, G16.941, required fractional eclipse0.580; Gaia 1802168384780141824: 43.76 arcsec, G15.783, required fractional eclipse0.203; Gaia 1802168483562365568: 47.58 arcsec, G16.950, required fractional eclipse0.585; Gaia 1802168380483110784: 52.04 arcsec, G16.353, required fractional eclipse0.340; Gaia 1802168449202616960: 53.79 arcsec, G17.242, required fractional eclipse0.763
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths5.646/5.783/5.584ppt, with 14 stacked events. QLP selected-kind fitted median8.042ppt. This separately extracted cutout is an alternative reduction of the same TESS

photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3180216841913987328O, P=3.1662003845±0.0001950480d, e=0.028319, omega=52.2631deg. Closest edge-on conjunction offset0.107914d (0.034083phase), full-SB1-covariance linear propagated Gaia uncertainty0.120399d at the measured epoch. Near-Gaia-reference projected eclipse offset0.034746d (photometric period projection allowance0.007431d). The approximate central duration5.86h vs observed median1.60h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated shallow alternating eclipses are real; odd half-period cycles define deeper-eclipse timing. Deficit0.099pixel from target favors target, but capable neighbor13.5arcsec away is only0.67pixel from deficit and needs approximately79percent intrinsic eclipse. An undersampled sector54 fit can exaggerate depth (cycle2 fitted15.37ppt versus observed minimum about7-8ppt); that even-parity fit is excluded from the deeper odd-parity ephemeris. Shallow noisy difference image lacks calibrated localization; source attribution remains unresolved after bounded check.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

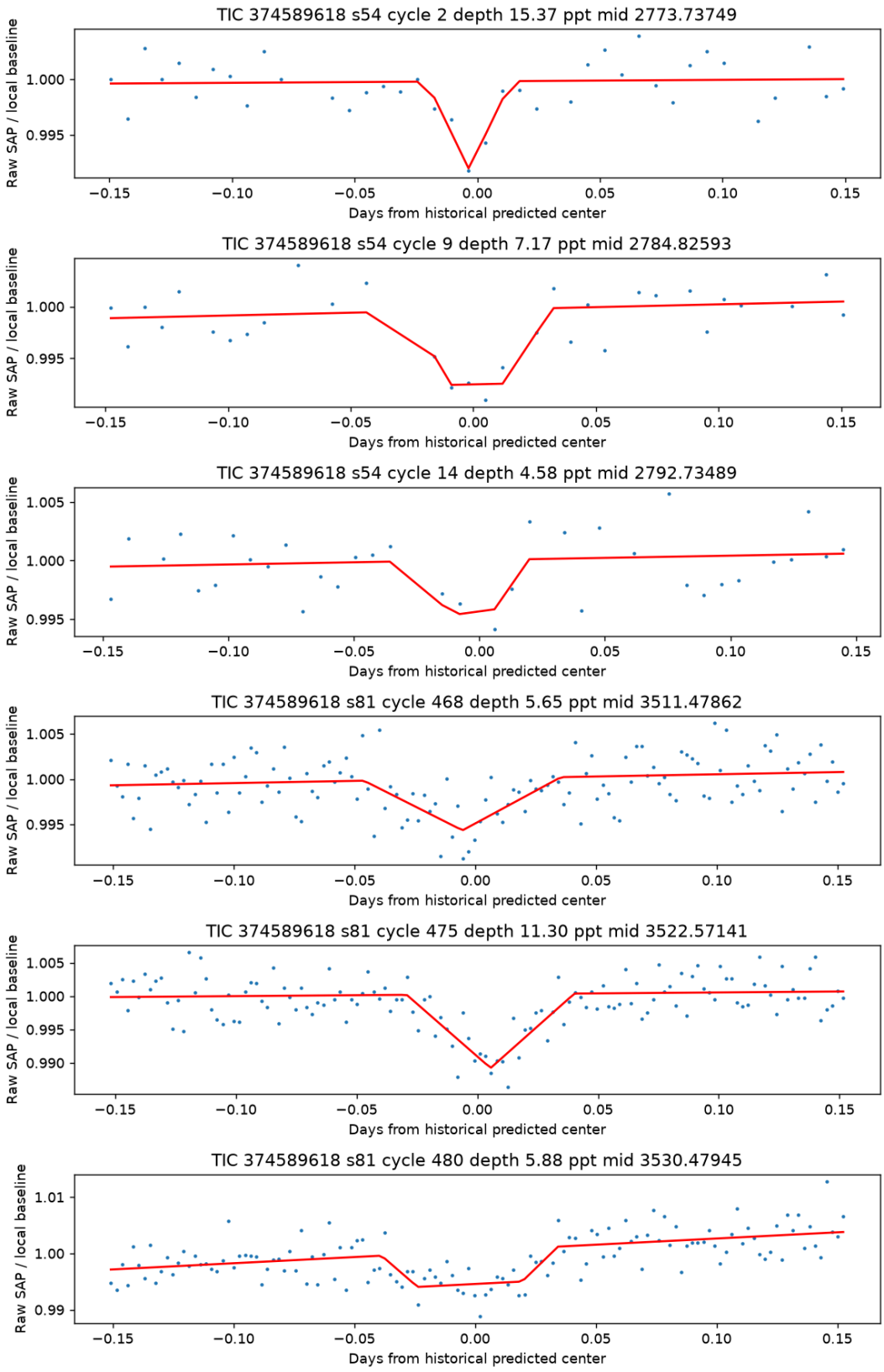
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
54	2	2773.7374851	0.003472	15.365	usable; same-kind selection applies
54	4	2776.8974321	0.005716	6.642	usable; same-kind selection applies
54	5	2778.4855848	0.003959	8.042	usable; same-kind selection applies
54	6	2780.0708114	0.005585	8.551	usable; same-kind selection applies
54	9	2784.8259336	0.003472	7.175	usable; same-kind selection applies
54	10	2786.4000831	0.005845	3.812	usable; same-kind selection applies
54	11	2787.9910721	0.003472	7.679	usable; same-kind selection applies
54	13	2791.1539523	0.003472	5.668	excluded singular timing fit
54	14	2792.7348870	0.006090	4.583	usable; same-kind selection applies
54	16	2795.9030916	4.191226	18.031	excluded singular timing fit
81	465	3506.7395943	0.002124	8.472	usable; same-kind selection applies
81	466	3508.3230973	0.002226	6.562	usable; same-kind selection applies
81	467	3509.9039935	0.001199	9.904	usable; same-kind selection applies
81	468	3511.4786201	0.003212	5.652	usable; same-kind selection applies
81	469	3513.0721228	0.001496	9.469	usable; same-kind selection applies

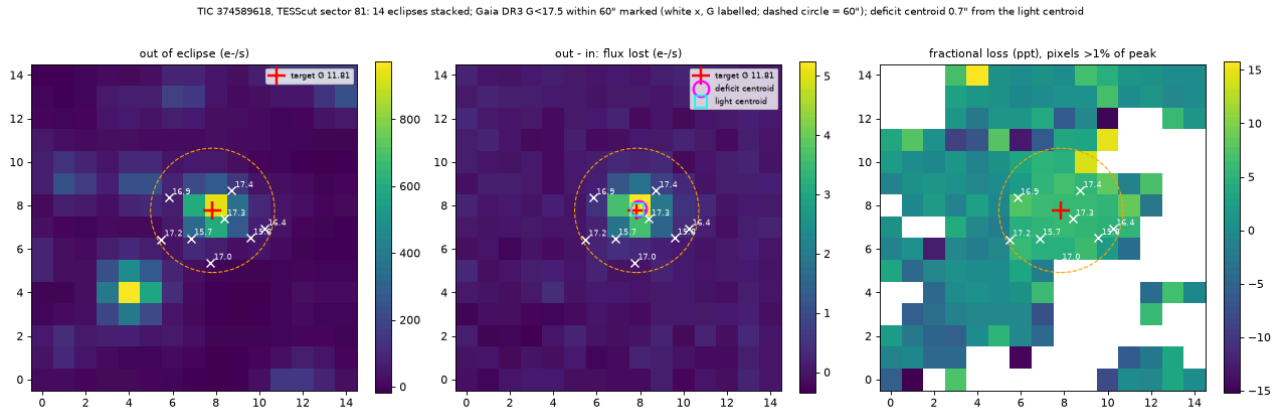
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
81	472	3517.8136376	0.003690	9.062	usable; same-kind selection applies
81	474	3520.9864498	0.001157	5.197	usable; same-kind selection applies
81	475	3522.5714060	0.001663	11.300	usable; same-kind selection applies
81	476	3524.1545767	0.001247	5.174	usable; same-kind selection applies
81	477	3525.7353536	0.001639	7.462	usable; same-kind selection applies
81	478	3527.3206150	0.002551	5.028	usable; same-kind selection applies
81	479	3528.9062454	0.001913	7.414	usable; same-kind selection applies
81	480	3530.4794510	0.002681	5.876	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s81.json, historical/diff_s81.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R281. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R281; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R281; diff_s81.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R282. TIC 355885228

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 355885228: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[24, 51, 78]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=12.2236227521 ±0.0000760471d, reference eclipse epoch BTJD1959.6132243, kind: Deeper repeating eclipse; even half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S24 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.151arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.060. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths21.673/20.791/20.470ppt, with 4 stacked events. QLP selected-kind fitted median26.934ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3121446711464822361 6, P=12.2241552395±0.0009737126d, e=0.003387, omega=37.0094deg. Closest edge-on conjunction offset-0.119695d (-0.009792phase), full-SB1-covariance linear propagated Gaia uncertainty0.140237d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.051537d (photometric period projection allowance0.009734d). The approximate central duration16.72h vs observed median10.74h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses support doubled orbital period; three deeper even-half-period events include a sector24 same-kind repeat and sector78 recurrence. Historical cutout mixes kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

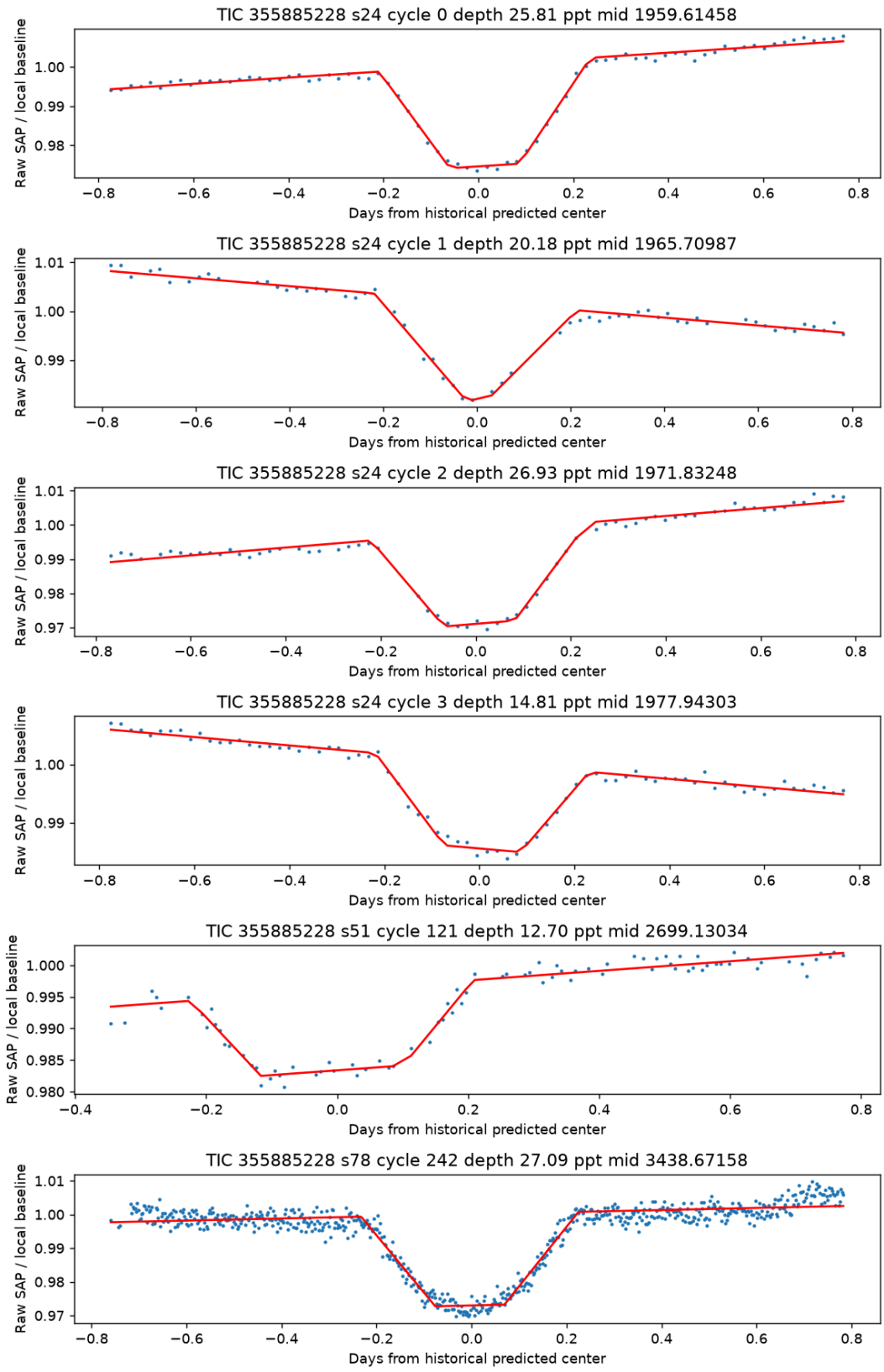
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

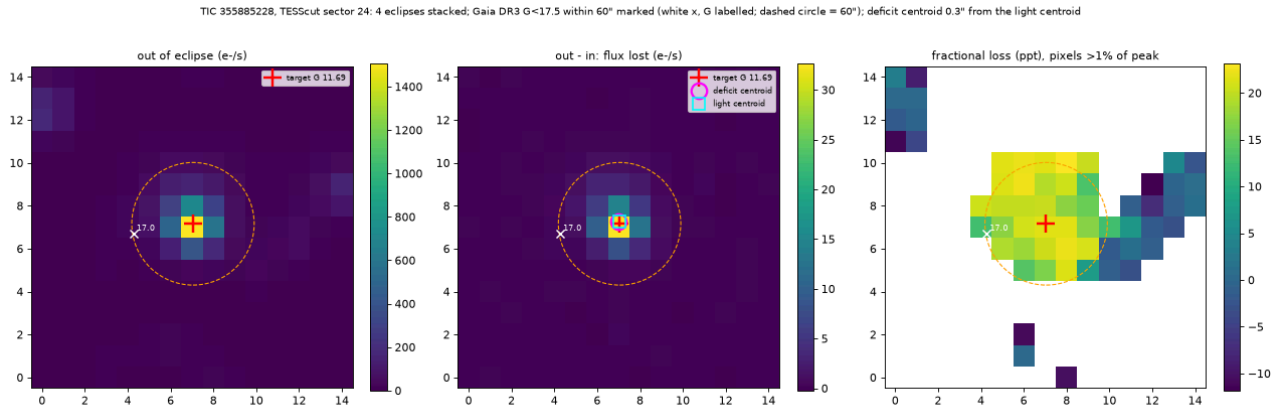
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
24	0	1959.6145829	0.010417	25.806	usable; same-kind selection applies
24	1	1965.7098686	0.016196	20.182	usable; same-kind selection applies
24	2	1971.8324839	0.018590	26.934	usable; same-kind selection applies
24	3	1977.9430294	0.014045	14.810	usable; same-kind selection applies
51	121	2699.1303431	0.003472	12.696	usable; same-kind selection applies
78	242	3438.6715775	0.001327	27.086	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s24.json, historical/diff_s24.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R282. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R282; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R282; diff_s24.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R283. TIC 169303527

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 169303527: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[29, 69, 96]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 10 same-kind midtimes give P=5.4456847708 ±0.0000051113d, reference eclipse epoch BTJD2091.7221616, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S69 deficit peak[8, 8] vs target pixel[8, 7]; deficit-target separation1.330arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.165. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths131.025/128.228/120.717ppt, with 4 stacked events. QLP selected-kind fitted median132.330ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3232644655574370611 2, P=5.4456409811±0.0011930950d, e=0.165642, omega=93.6455deg. Closest edge-on conjunction offset0.022408d (0.004115phase), full-SB1-covariance linear propagated Gaia uncertainty0.264837d at the measured epoch. Near-Gaia-reference projected eclipse offset0.008702d (photometric period projection allowance0.001600d). The approximate central duration3.96h vs observed median3.17h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses show sector-dependent dilution/depth. No capable catalogued neighbor under approximate Gaia brightness screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

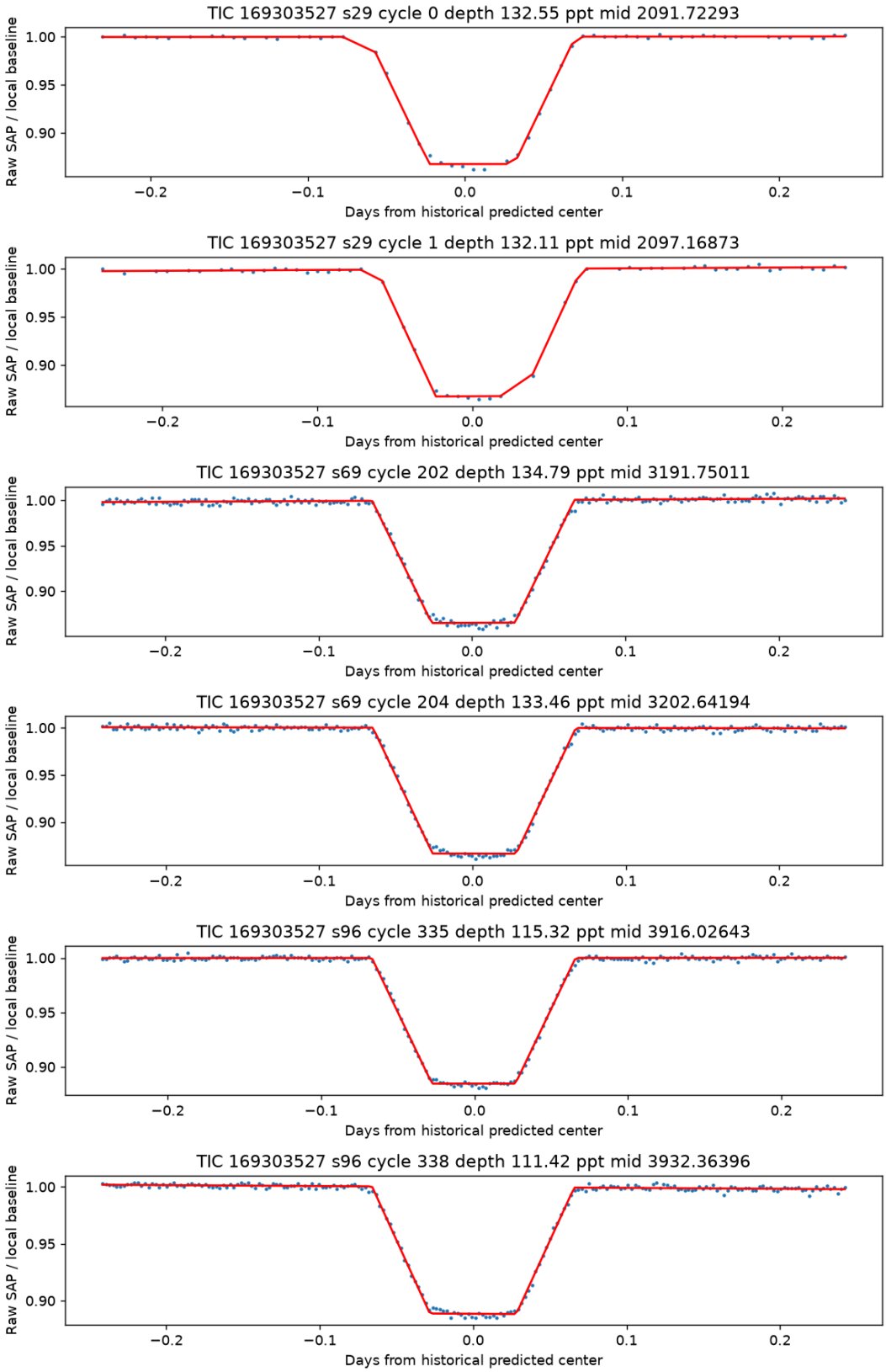
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

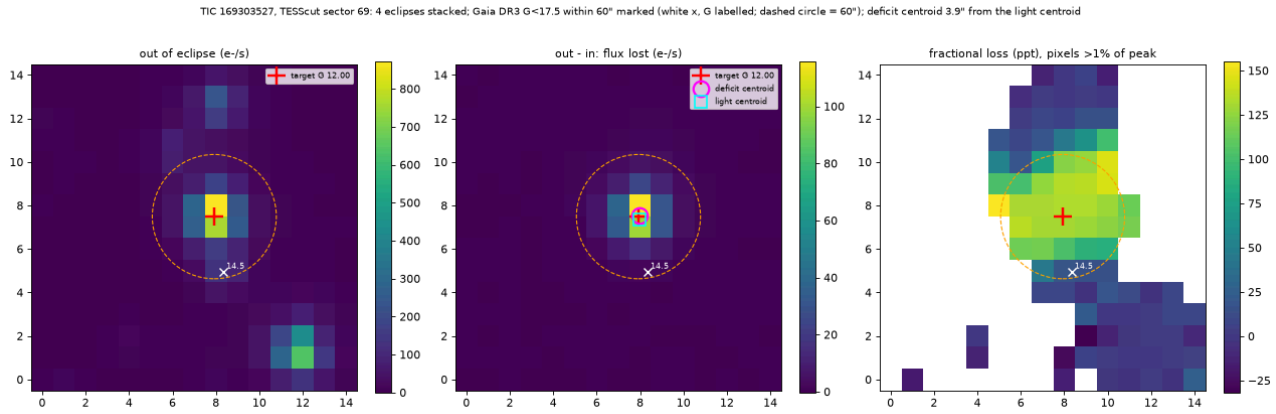
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
29	0	2091.7229281	0.003472	132.553	usable; same-kind selection applies
29	1	2097.1687307	0.003472	132.107	usable; same-kind selection applies
69	201	3186.3046405	0.001157	133.930	usable; same-kind selection applies
69	202	3191.7501075	0.001157	134.793	usable; same-kind selection applies
69	203	3197.1961780	0.001157	133.455	usable; same-kind selection applies
69	204	3202.6419381	0.001157	133.461	usable; same-kind selection applies
96	334	3910.5807085	0.001157	115.495	usable; same-kind selection applies
96	335	3916.0264305	0.001157	115.316	usable; same-kind selection applies
96	337	3926.9181463	0.001157	112.695	usable; same-kind selection applies
96	338	3932.3639560	0.001157	111.422	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s69.json, historical/diff_s69.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R283. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R283; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R283; diff_s69.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R284. TIC 94541119

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 94541119: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[18, 44, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=10.8933276600 ±0.0000227964d, reference eclipse epoch BTJD1797.8362059, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S18 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.970arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.063. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths109.574/108.335/107.029ppt, with 2 stacked events. QLP selected-kind fitted median109.533ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3170384999063309440, P=10.8955400325±0.0012192578d, e=0.271372, omega=337.0819deg. Closest edge-on conjunction offset-0.172802d (-0.015860phase), full-SB1-covariance linear propagated Gaia uncertainty0.156102d at the measured epoch. Near-Gaia-reference projected eclipse offset0.112595d (photometric period projection allowance0.002941d). The approximate central duration5.18h vs observed median4.53h permits noncentral geometry; this radius/minimum-mass estimate is not a

detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five measured eclipses in sectors18/44/86 test period recurrence. No capable neighbor under approximate Gaia brightness screen; adopted depth is diluted.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

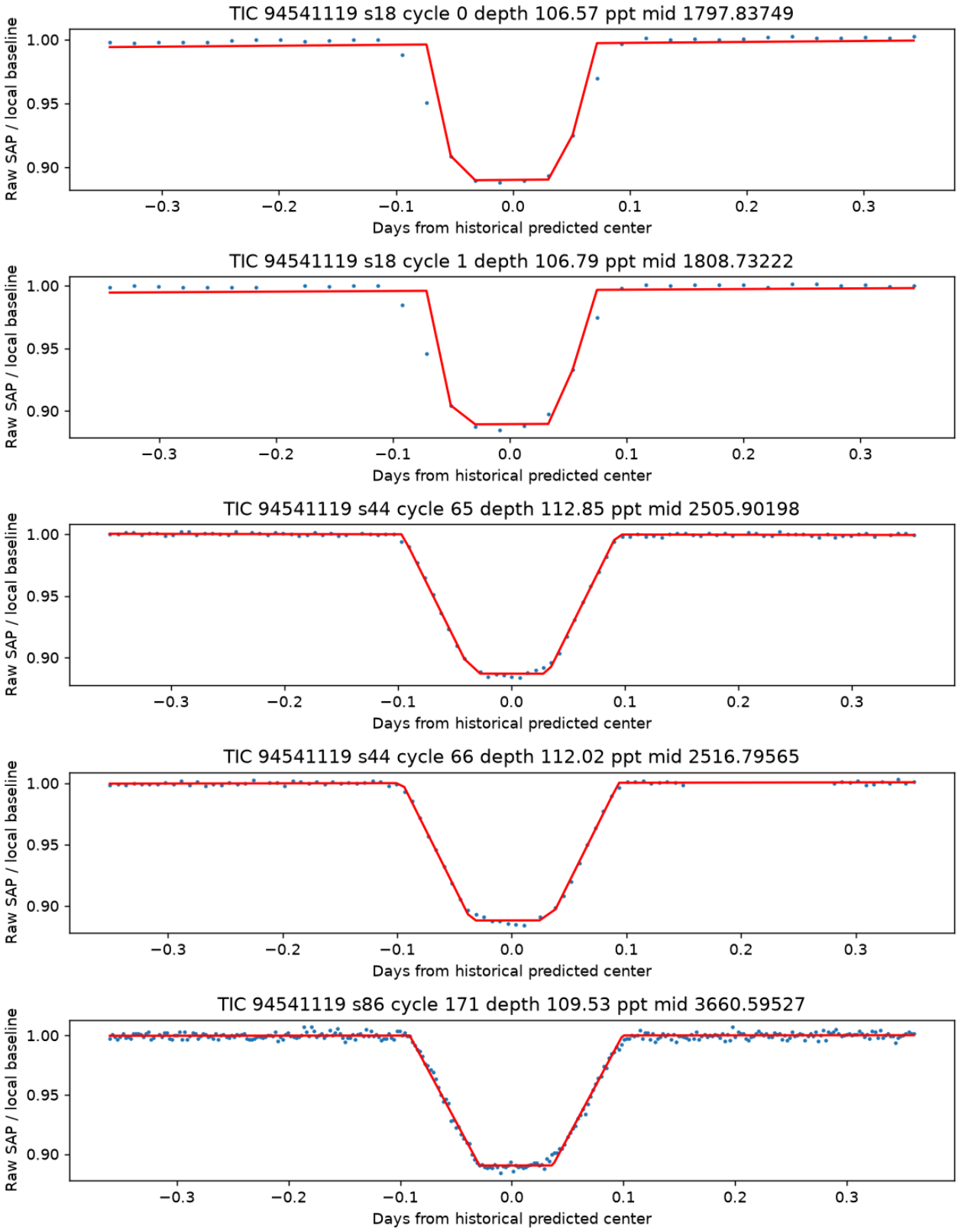
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

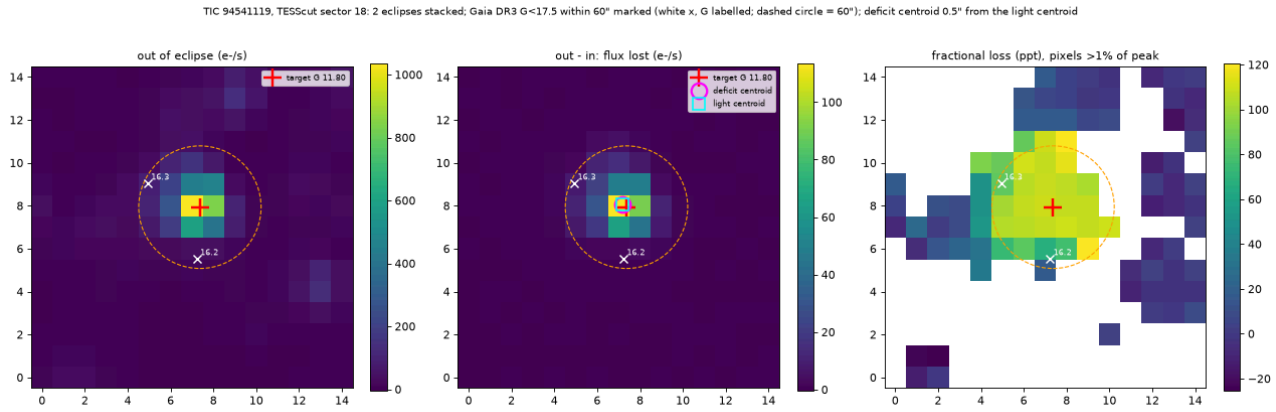
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
18	0	1797.8374939	0.010417	106.574	usable; same-kind selection applies
18	1	1808.7322153	0.010417	106.790	usable; same-kind selection applies
44	65	2505.9019779	0.003472	112.850	usable; same-kind selection applies
44	66	2516.7956468	0.003472	112.023	usable; same-kind selection applies
86	171	3660.5952657	0.001157	109.533	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s18.json, historical/diff_s18.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R284. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R284; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R284; diff_s18.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R285. TIC 409935170

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 409935170: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 16 distinct eclipse events in selected sectors[13, 94, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=8.7039814271 ±0.0000183962d, reference eclipse epoch BTJD1655.8339615, kind: Deeper repeating eclipse; even half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S94 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.121arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.184. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths129.075/126.963/119.540ppt, with 6 stacked events. QLP selected-kind fitted median135.909ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3642238084100151884 8, P=8.7056187340±0.0007330769d, e=0.021485, omega=165.0639deg. Closest edge-on conjunction offset-0.194825d (-0.022379phase), full-SB1-covariance linear propagated Gaia uncertainty0.106030d at the measured epoch. Near-Gaia-reference projected eclipse offset0.044222d (photometric period projection allowance0.002686d). The approximate central duration9.09h vs observed median7.60h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating depths within sectors establish doubled orbital period even though across-sector pooled depths overlap. Even half-period cycles alone define deeper-eclipse timing. Historical cutout mixes kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

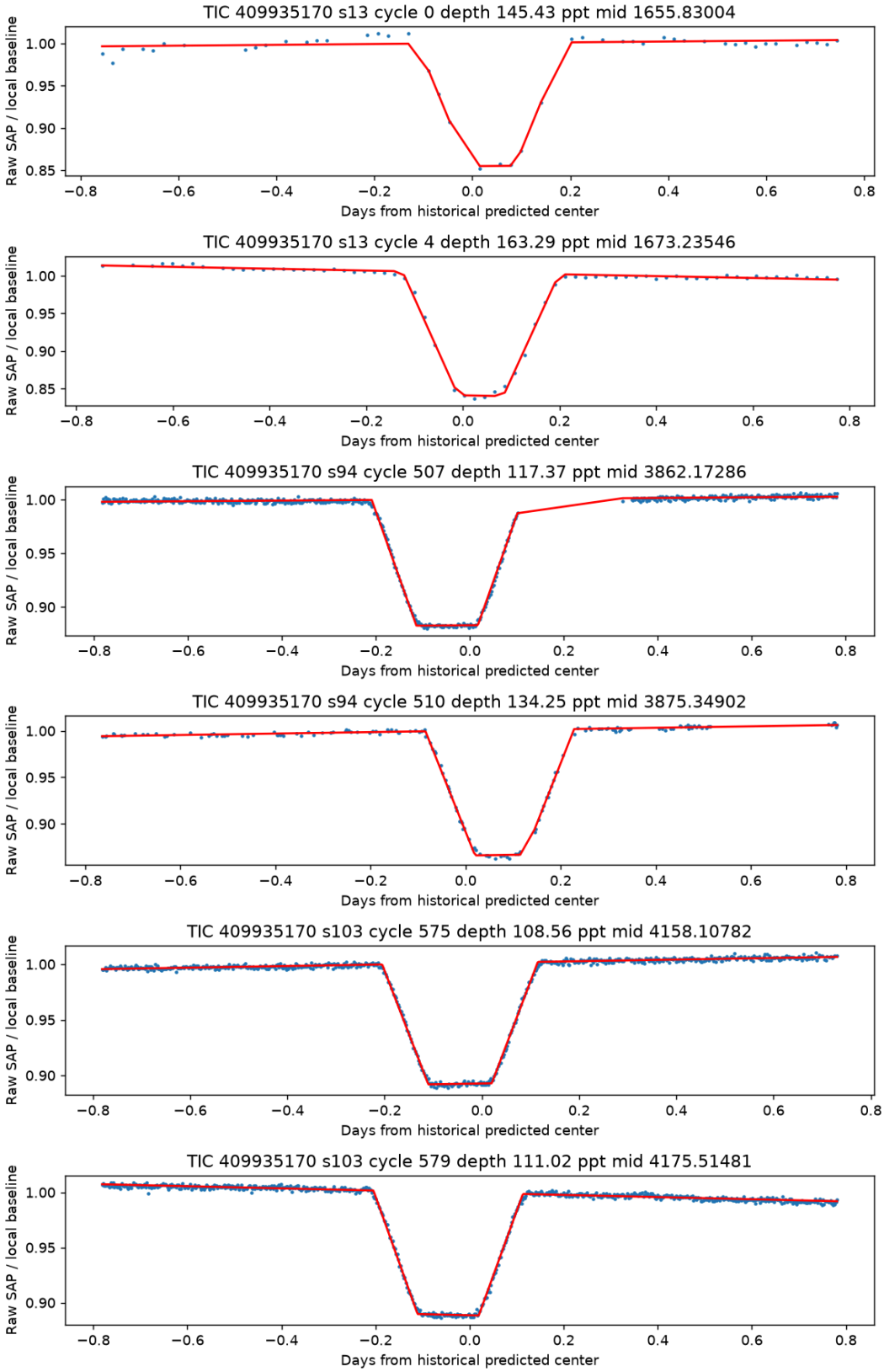
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

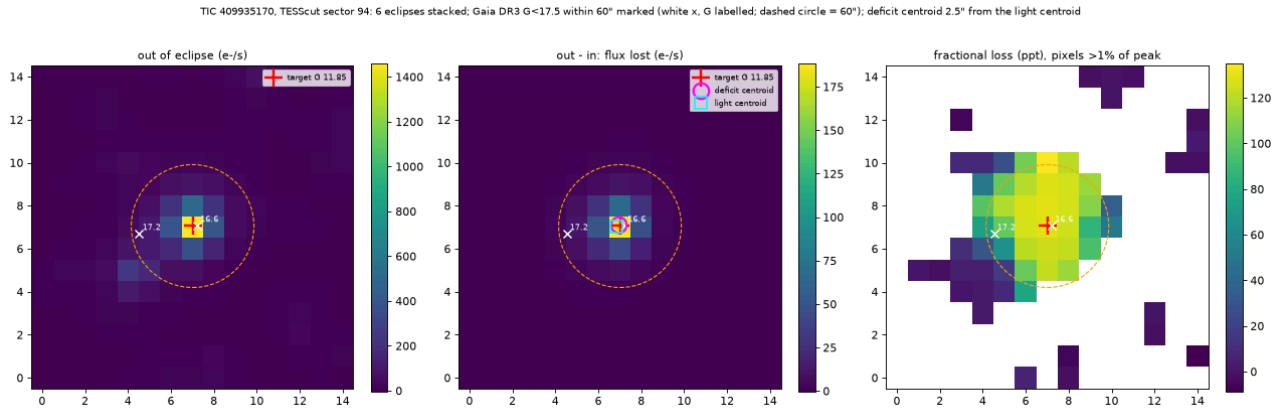
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
13	0	1655.8300360	0.010417	145.432	usable; same-kind selection applies
13	1	1660.0610314	0.010417	121.233	usable; same-kind selection applies
13	2	1664.5343687	0.010417	160.466	usable; same-kind selection applies
13	4	1673.2354553	0.010417	163.291	usable; same-kind selection applies
13	5	1677.4663082	0.010417	134.616	usable; same-kind selection applies
94	506	3857.9434991	0.001157	134.902	usable; same-kind selection applies
94	507	3862.1728628	0.001157	117.366	usable; same-kind selection applies
94	508	3866.6445909	0.001157	136.915	usable; same-kind selection applies
94	509	3870.8770563	0.001157	117.821	usable; same-kind selection applies
94	510	3875.3490234	0.001157	134.251	usable; same-kind selection applies
94	511	3879.5810360	0.001157	114.162	usable; same-kind selection applies
103	574	4153.8766805	0.001157	127.455	usable; same-kind selection applies
103	575	4158.1078217	0.001157	108.562	usable; same-kind selection applies
103	576	4162.5793538	0.001157	129.196	usable; same-kind selection applies
103	577	4166.8113257	0.001157	108.638	usable; same-kind selection applies
103	579	4175.5148085	0.001157	111.024	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s94.json, historical/diff_s94.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R285. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R285; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R285; diff_s94.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R286. TIC 365864542

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 365864542: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 6 distinct eclipse events in selected sectors[15, 75, 83]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=13.3838603016 ±0.0000467881d, reference eclipse epoch BTJD1717.1172335, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S15 deficit peak[7, 8] vs target pixel[8, 8]; deficit-target separation2.180arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.482. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths257.577/247.957/195.923ppt, with 2 stacked events. QLP selected-kind fitted median277.197ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3216806596022264563 2, P=13.3807392140±0.0042403250d, e=0.382153, omega=217.8118deg. Closest edge-on conjunction offset0.192270d (0.014369phase), full-SB1-covariance linear propagated Gaia uncertainty0.445137d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.116718d (photometric period projection allowance0.004632d). The approximate central duration13.08h vs observed median9.73h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated isolated eclipses across three sectors; one singular center-covariance fit excluded. Aperture/sector depth differences retained; no intrinsic undiluted depth inferred.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

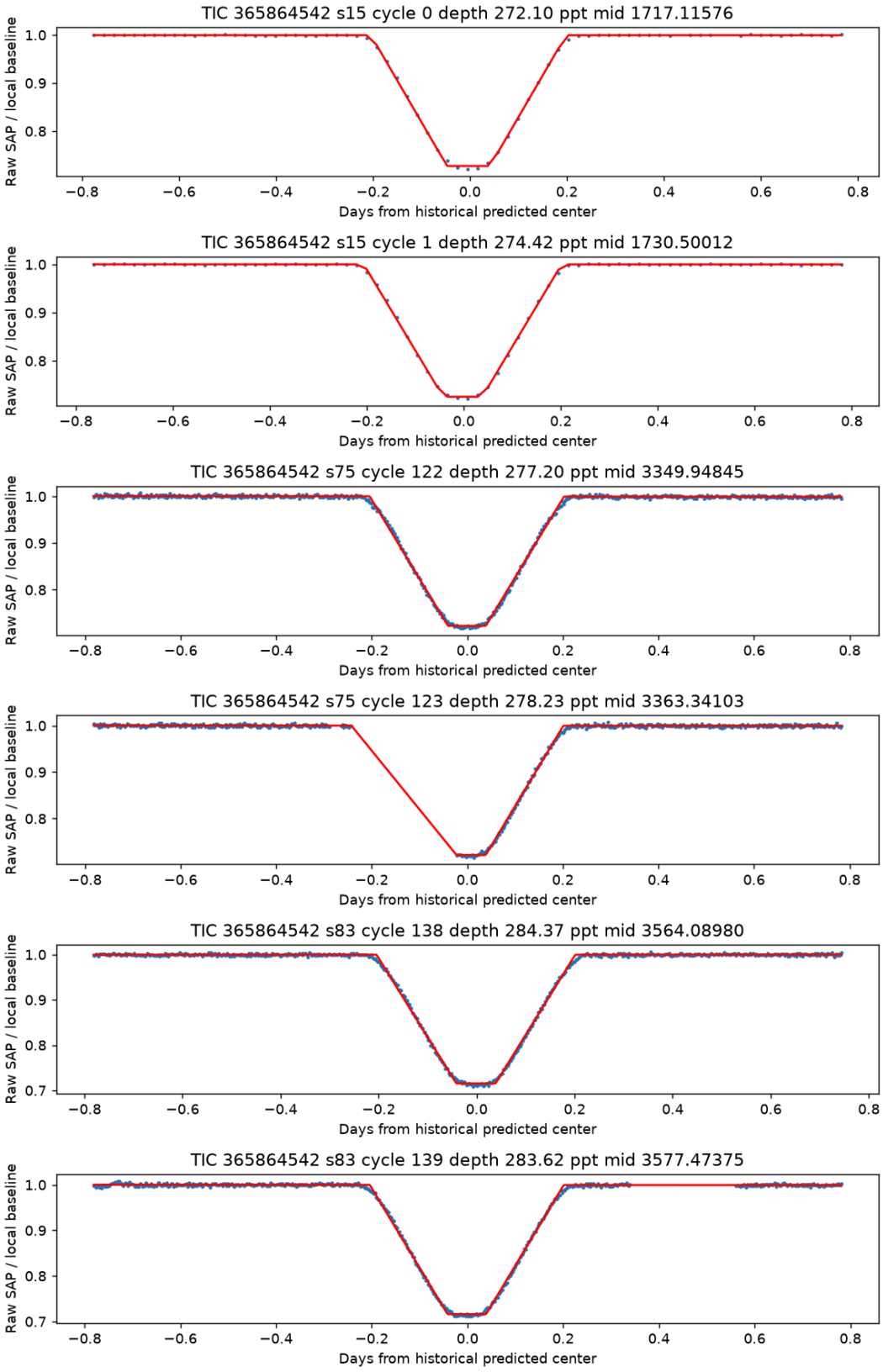
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

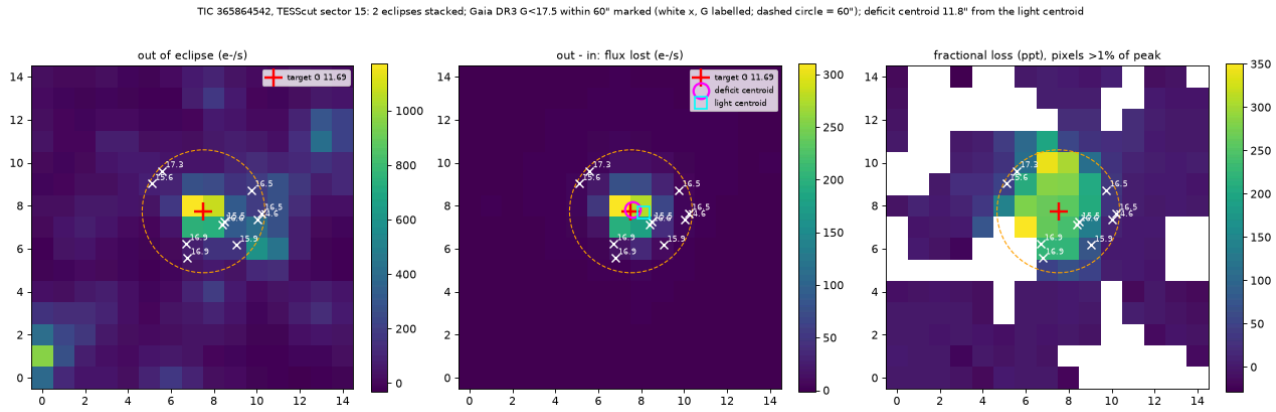
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1717.1157569	0.010417	272.104	usable; same-kind selection applies
15	1	1730.5001172	0.010417	274.416	usable; same-kind selection applies
75	122	3349.9484463	0.001157	277.197	usable; same-kind selection applies
75	123	3363.3410291	45629.070605	278.227	excluded singular timing fit
83	138	3564.0897989	0.001157	284.372	usable; same-kind selection applies
83	139	3577.4737458	0.001157	283.622	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s15.json, historical/diff_s15.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R286. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R286; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R286; diff_s15.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R287. TIC 143375995

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 143375995: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 9 distinct eclipse events in selected sectors[7, 34, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=8.3596843935 ±0.0000123196d, reference eclipse epoch BTJD1492.9984843, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S88 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.472arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.856. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths107.960/102.672/81.871ppt, with 4 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median113.726ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3570040092735898316 8, P=8.3599586040±0.0006591686d, e=0.210170, omega=8.4067deg. Closest edge-on conjunction offset0.060141d (0.007194phase), full-SB1-covariance linear propagated Gaia uncertainty0.114884d at the measured epoch. Near-Gaia-reference projected eclipse offset0.096337d (photometric period projection allowance0.001626d). The approximate central duration8.53h vs observed median6.22h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep isolated eclipses recur across three sectors. Aperture depths vary with dilution; no capable catalogued neighbor within60arcsec under approximate Gaia G brightness test. PSF-scaled residual0.856 warns against interpreting this simple PSF fit quantitatively; deficit core is centered on target.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

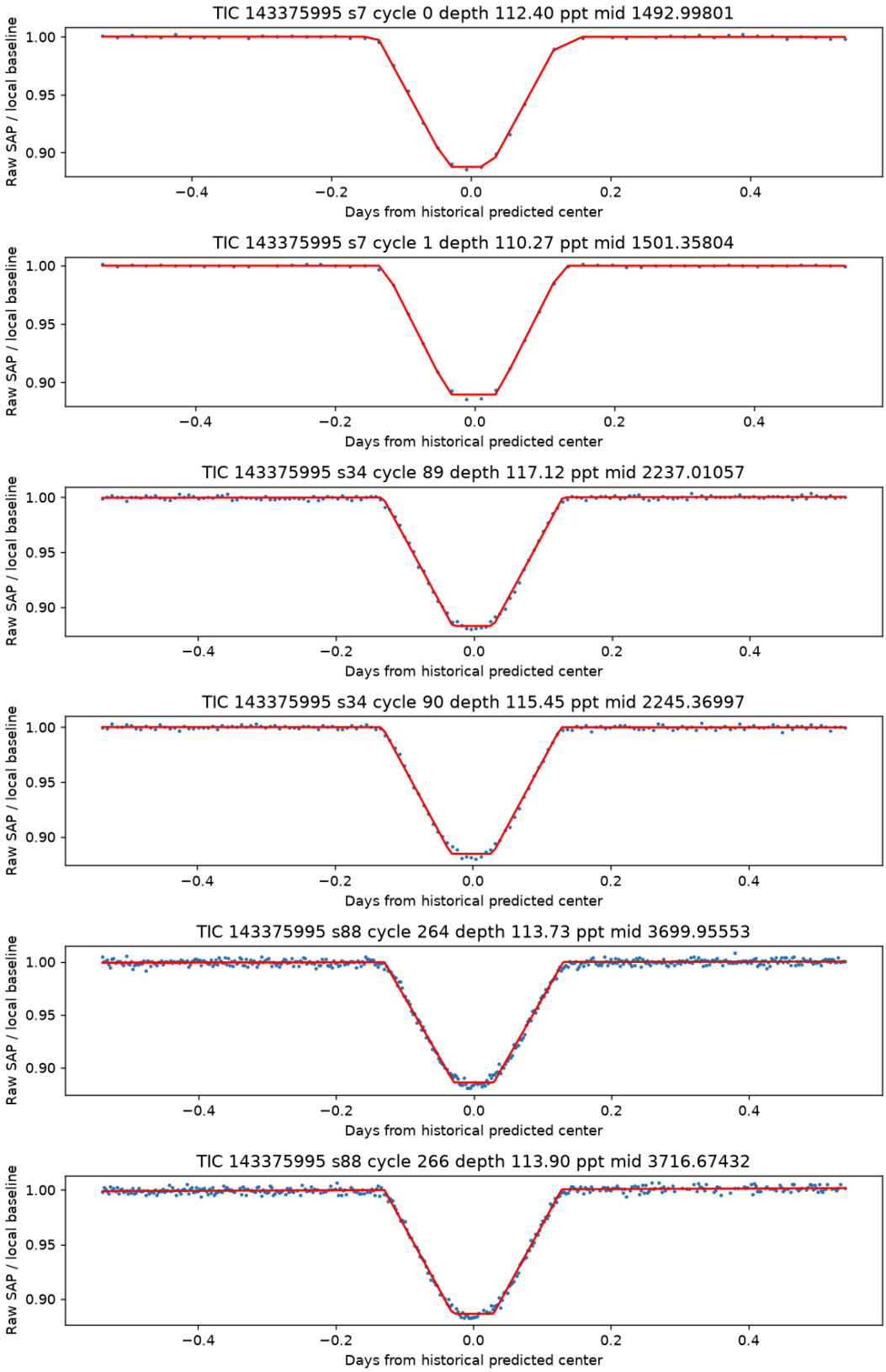
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

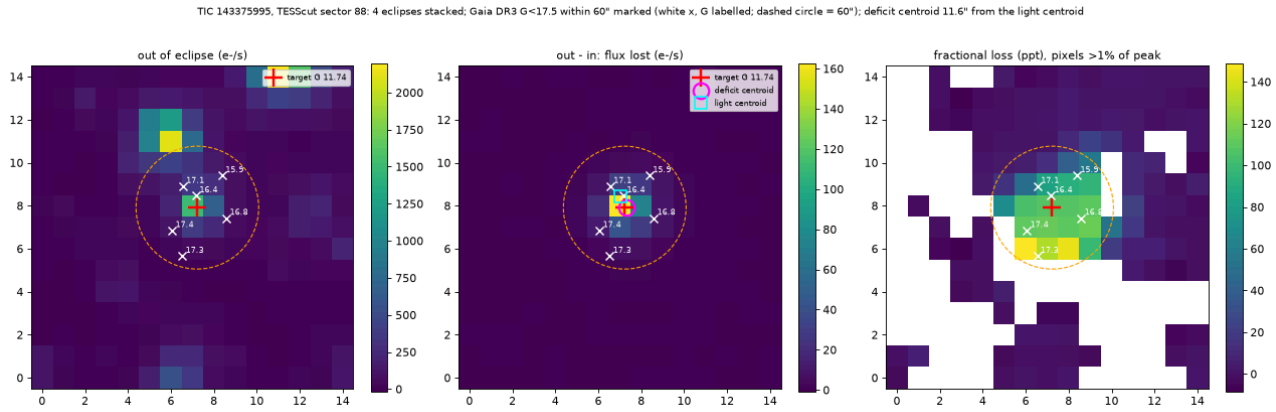
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1492.9980107	0.010417	112.398	usable; same-kind selection applies
7	1	1501.3580369	0.010417	110.266	usable; same-kind selection applies
7	2	1509.7179946	0.010417	112.749	usable; same-kind selection applies
34	89	2237.0105695	0.003472	117.121	usable; same-kind selection applies
34	90	2245.3699671	0.003472	115.447	usable; same-kind selection applies
88	263	3691.5953460	0.001157	114.711	usable; same-kind selection applies
88	264	3699.9555267	0.001157	113.726	usable; same-kind selection applies
88	265	3708.3148285	0.001157	113.351	usable; same-kind selection applies
88	266	3716.6743233	0.001157	113.901	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s88.json, historical/diff_s88.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R287. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R287; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R287; diff_s88.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R288. TIC 122506579

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 122506579: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[97]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=6.9345089138 ±0.0002217831d, reference eclipse epoch BTJD3939.4721930, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation0.972arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.187. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths14.430/13.648/13.226ppt, with 7 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median12.939ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3506714875020586624O, P=6.9335988748±0.0005002026d, e=0.004443, omega=329.5125deg. Closest edge-on conjunction offset0.607534d (0.087622phase), full-SB1-covariance linear propagated Gaia uncertainty0.259233d at the measured epoch. Near-Gaia-reference projected eclipse offset0.141594d (photometric period projection allowance0.113553d). The approximate central duration5.63h vs observed median2.26h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated shallow eclipses in sector97. Gaia orbit is nearly circular (e0.00444), so periastron/omega parameterization is weak; measured eclipse falls0.60753d (phase0.08762) from nearer nominal edge-on conjunction versus full-covariance linear allowance0.25923d. Other conjunction is2.87617d away. Near-reference projected offset0.14159d with photometric period projection allowance0.11355d; photometric P differs from Gaia by0.000910d versus Gaia error0.000500d. Retain mild orbit/timing tension, no arbitrary sigma veto or forced alias. Approximate central duration5.63h vs observed2.26h permits grazing/noncentral eclipse. Updated orbital solution would sharpen association.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

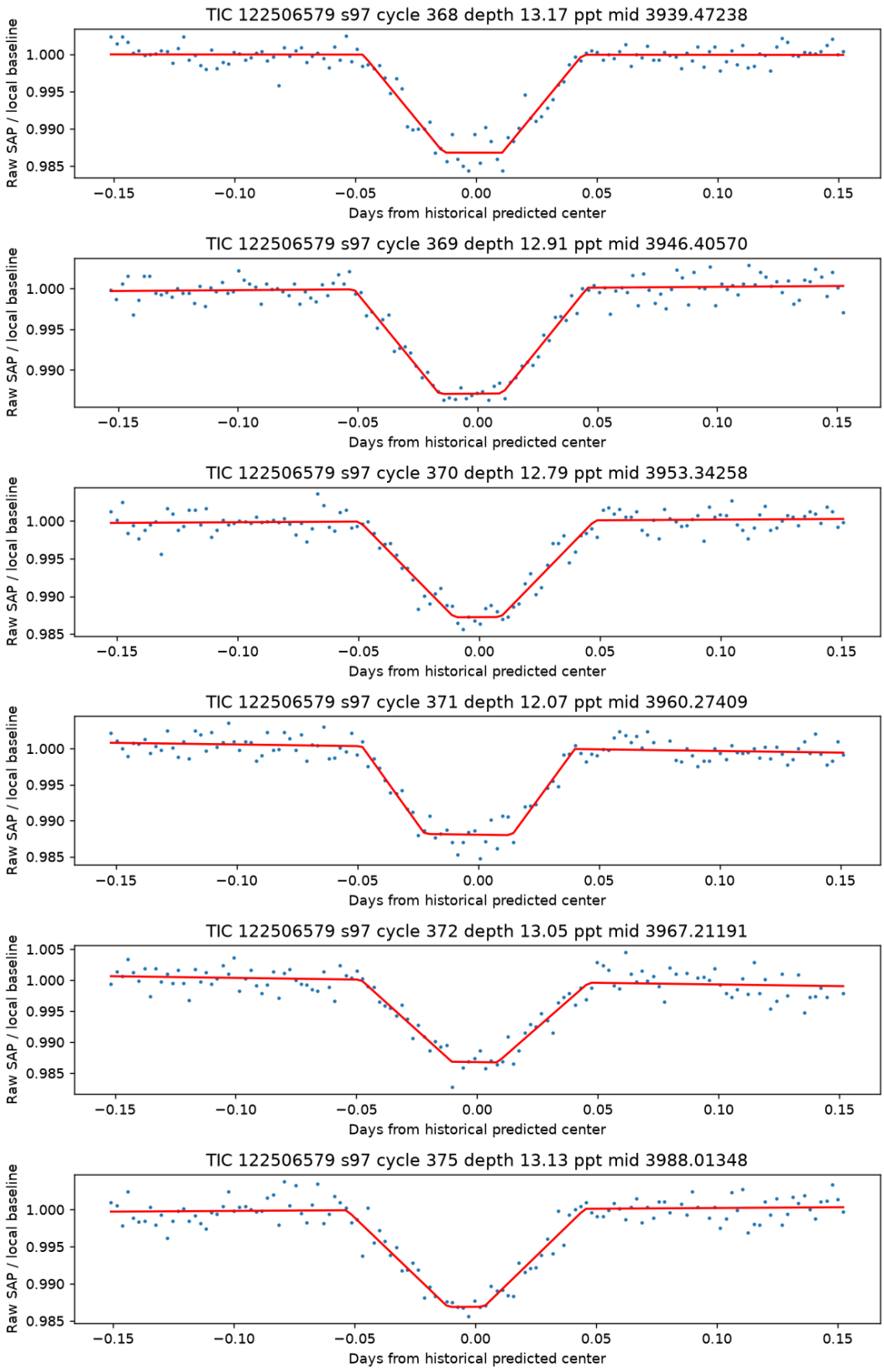
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

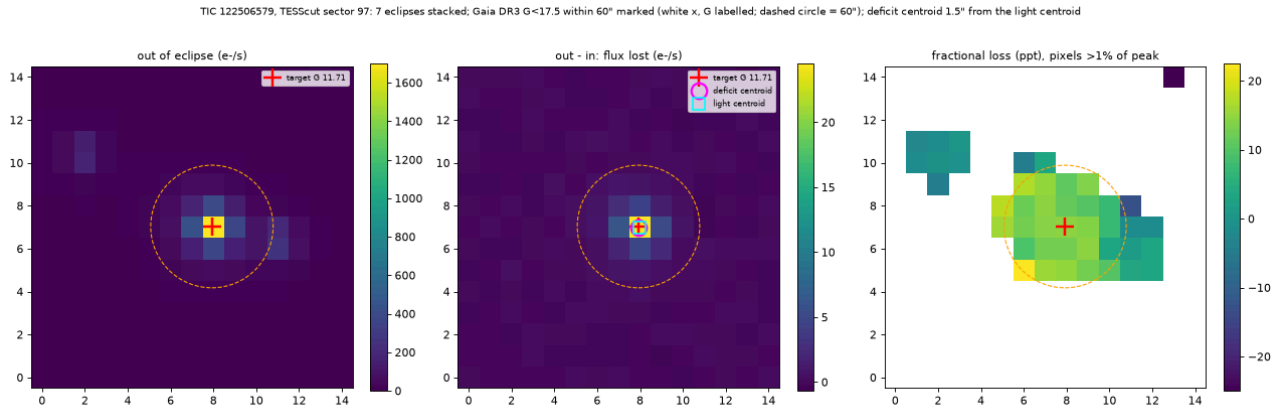
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
97	368	3939.4723816	0.001157	13.165	usable; same-kind selection applies
97	369	3946.4056957	0.001157	12.910	usable; same-kind selection applies
97	370	3953.3425790	0.001157	12.788	usable; same-kind selection applies
97	371	3960.2740933	0.001157	12.069	usable; same-kind selection applies
97	372	3967.2119089	0.001157	13.053	usable; same-kind selection applies
97	373	3974.1444079	0.001157	12.939	usable; same-kind selection applies
97	375	3988.0134806	0.001157	13.126	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R288. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R288; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R288; diff_s97.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R289. TIC 201598744

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 201598744: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 14 distinct eclipse events in selected sectors[93, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 14 same-kind midtimes give P=3.4086667035 ±0.0000083995d, reference eclipse epoch BTJD3830.7318892, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S93 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation7.215arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.803. Capable neighbours: Gaia 6633731921030220032: 37.11 arcsec, G15.886, required fractional eclipse0.468; Gaia 6633638045927526528: 37.76 arcsec, G16.525, required fractional eclipse0.833; Gaia 6633684985627605248: 46.68 arcsec, G13.068, required fractional eclipse0.046; Gaia 6633684985627605376: 48.13 arcsec, G13.843, required fractional eclipse0.081; Gaia 6633638118945259264: 48.84 arcsec, G16.471, required fractional eclipse0.793; Gaia 6633731955389963776: 53.75 arcsec, G13.807, required fractional eclipse0.079
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths12.147/10.609/8.018ppt, with 8 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median12.606ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3663373188237256857 6, P=3.4088296693±0.0003292291d, e=0.056446, omega=38.6510deg. Closest edge-on conjunction offset-0.084458d (-0.024776phase), full-SB1-covariance linear propagated Gaia uncertainty0.308497d at the measured epoch. Near-Gaia-reference projected eclipse offset0.080138d (photometric period projection allowance0.008483d). The approximate central duration3.94h vs observed median1.96h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Raw shallow eclipses recur in sectors93/104. Strong aperture dilution:1x1/3x3/5x5 depths12.15/10.61/8.02ppt. Deficit0.344pixel from target and closest capable catalogued neighbor1.63pixels from deficit favor target at TESS resolution. PSF-scaled residual0.803 is not a calibrated localization uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

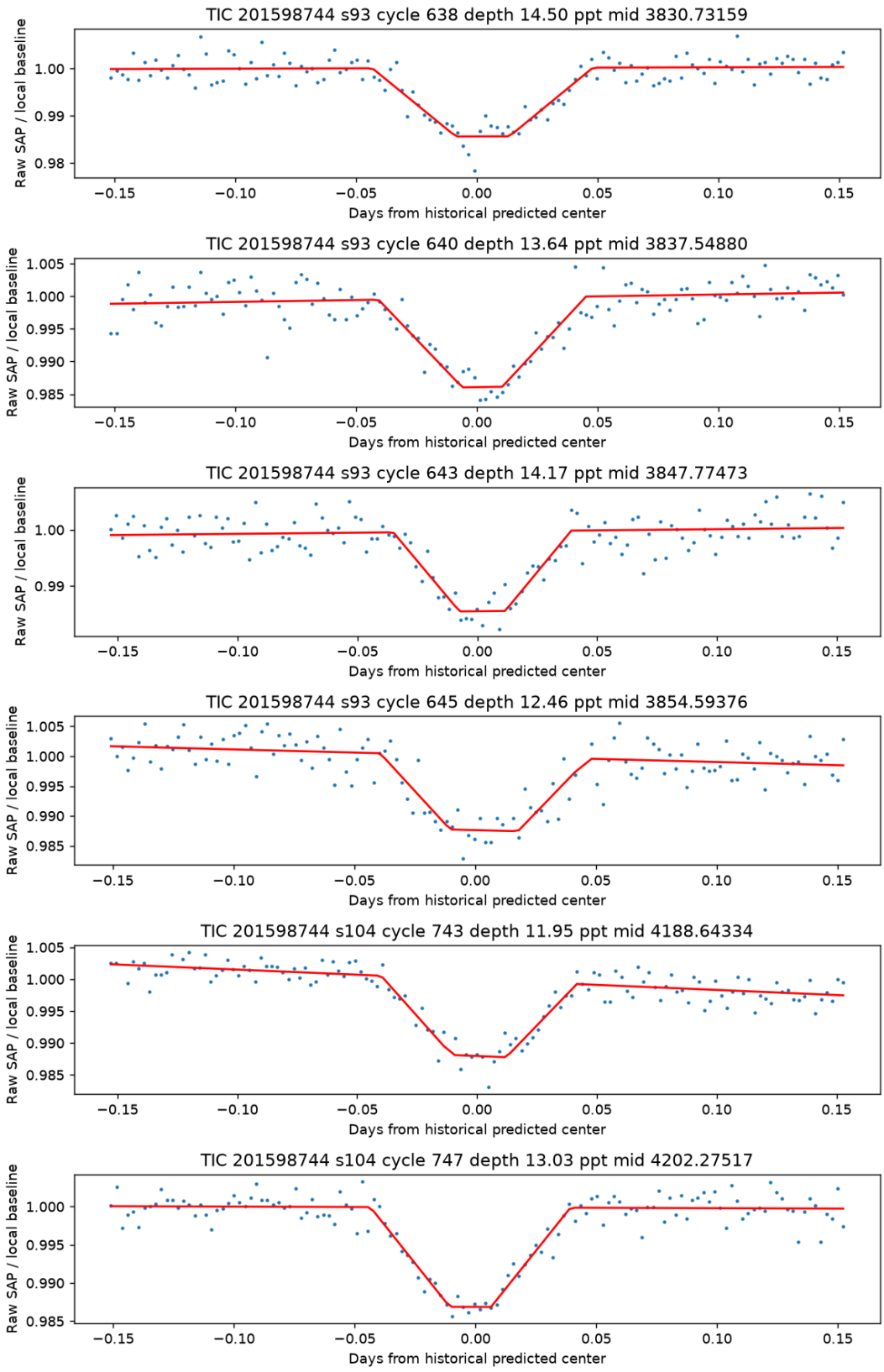
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
93	638	3830.7315945	0.001262	14.498	usable; same-kind selection applies
93	639	3834.1379008	0.001323	12.730	usable; same-kind selection applies
93	640	3837.5487973	0.001157	13.644	usable; same-kind selection applies
93	641	3840.9594548	0.001397	12.346	usable; same-kind selection applies
93	642	3844.3667858	0.001165	12.686	usable; same-kind selection applies
93	643	3847.7747315	0.001265	14.174	usable; same-kind selection applies
93	644	3851.1850801	0.001157	12.366	usable; same-kind selection applies
93	645	3854.5937586	0.001633	12.465	usable; same-kind selection applies
104	741	4181.8262152	0.001157	12.422	usable; same-kind selection applies
104	742	4185.2324238	0.001622	12.527	usable; same-kind selection applies
104	743	4188.6433377	0.001435	11.949	usable; same-kind selection applies
104	745	4195.4610324	0.001157	11.376	usable; same-kind selection applies
104	746	4198.8653100	0.001157	13.308	usable; same-kind selection applies
104	747	4202.2751695	0.001157	13.029	usable; same-kind selection applies

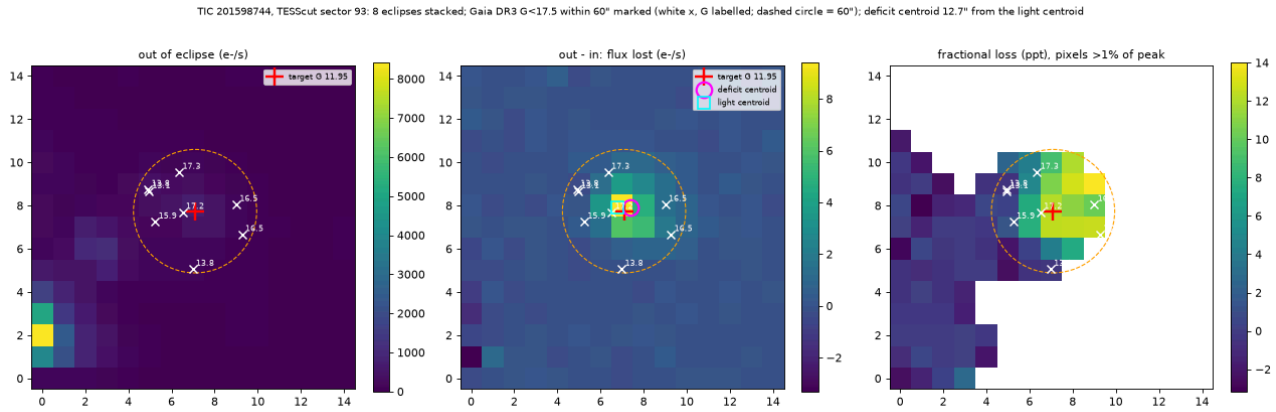
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s93.json, historical/diff_s93.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md;

../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R289. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R289; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R289; diff_s93.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R290. TIC 267094867

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 267094867: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 13 distinct eclipse events in selected sectors[1, 67, 96]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 12 same-kind midtimes give P=3.4708302926 ±0.0000031344d, reference eclipse epoch BTJD1345.8329151, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S66 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.558arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.036. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths67.600/68.139/67.682ppt, with 8 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median65.115ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3463989931929362816O, P=3.4710058980±0.0001561795d, e=0.015578, omega=37.0096deg. Closest edge-on conjunction offset-0.037602d (-0.010833phase), full-SB1-covariance linear propagated Gaia uncertainty0.038192d at the measured epoch. Near-Gaia-reference projected eclipse offset0.010865d (photometric period projection allowance0.000865d). The approximate central

duration6.12h vs observed median4.02h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep eclipses recur in sectors1/67/96; sector-dependent depths and shapes retained. Singular sector67 cycle520 fit excluded from timing. No capable catalogued neighbor in approximate brightness test.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

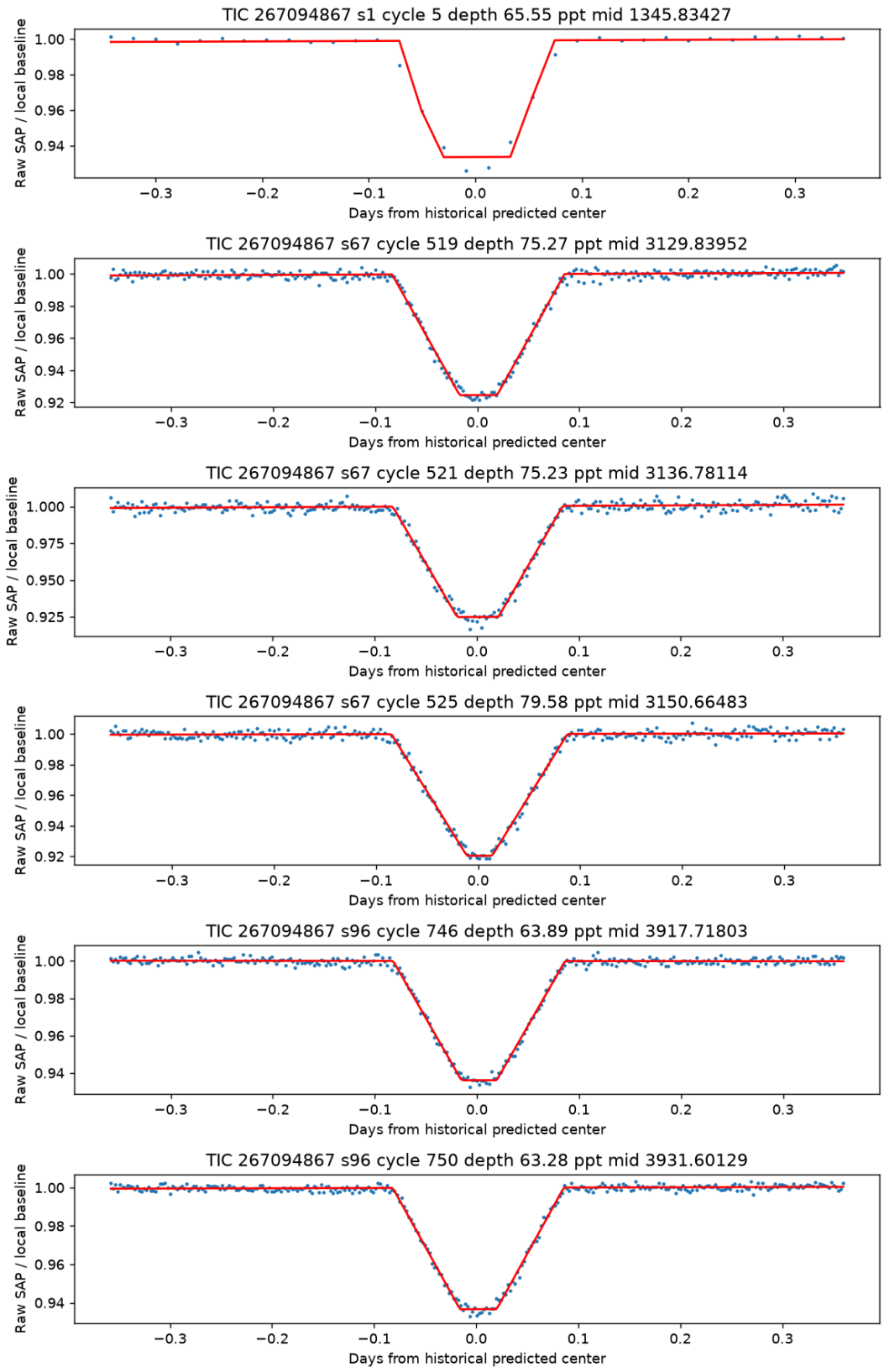
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

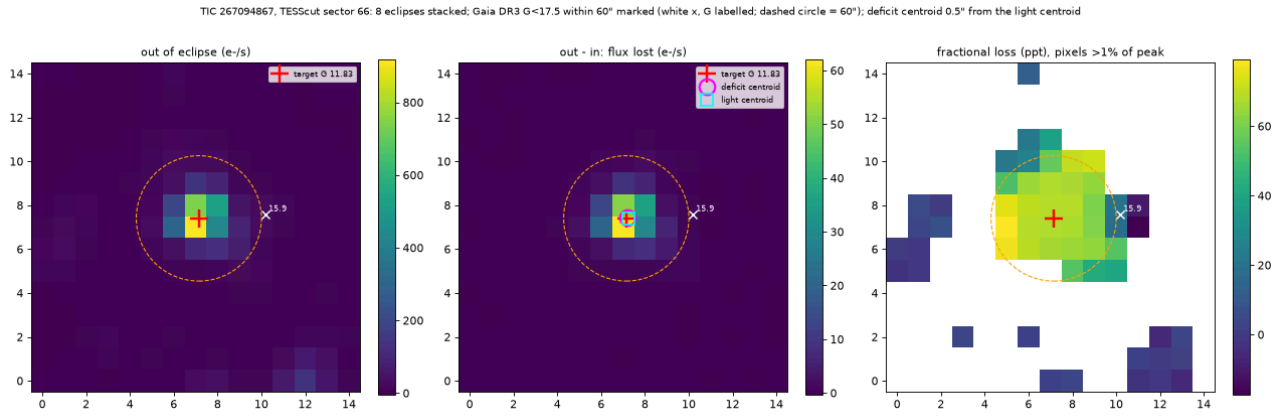
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	5	1345.8342689	0.010417	65.545	usable; same-kind selection applies
1	7	1352.7735477	0.010417	64.684	usable; same-kind selection applies
67	519	3129.8395208	0.001157	75.267	usable; same-kind selection applies
67	520	3133.3214697	118740.580542	65.654	excluded unreliable timing fit
67	521	3136.7811360	0.001157	75.230	usable; same-kind selection applies
67	523	3143.7229409	0.001157	75.953	usable; same-kind selection applies
67	524	3147.1941038	0.001157	75.635	usable; same-kind selection applies
67	525	3150.6648322	0.001157	79.582	usable; same-kind selection applies
96	744	3910.7766784	0.001157	62.725	usable; same-kind selection applies
96	746	3917.7180340	0.001157	63.889	usable; same-kind selection applies
96	748	3924.6597605	0.001157	62.610	usable; same-kind selection applies
96	749	3928.1308625	0.001157	62.576	usable; same-kind selection applies
96	750	3931.6012908	0.001157	63.284	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s66.json, historical/diff_s66.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R290. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R290; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R290; diff_s66.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R291. TIC 156985493

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional short-cadence resolved photometry and revised spectroscopy

Archived source: sanitized report text.

TIC156985493: CONFIRMED

Full bounded review: COMPLETE. Physical classification is separate from central catalogue novelty. Root scientific acceptance is required before integration.

1. **Raw repeats: pass.** Two actual unflagged narrow sector22 SAP eclipses and two sector49 eclipses are measured. The generic25-point fit gate missed short events at30minute cadence; its empty result remains pipeline_measurements.json. Independent helper examined actual SAP samples and plots, measuring quadratic trough centers from3-4 samples in sector22 and free Gaussian plus linear baseline centers in sector49. Historical predicted epochs drifted and were not used as measured timings. See helper/independent_raw.png and helper/156-alias-windows.png.
2. **Measured same-kind timing and aliases: pass.** Four measured centers at cycles0,1,74,76 give P=9.9472023084 +/-0.0001041208d and selected-eclipse epoch BTJD1908.15679456. Kind: Selected repeating narrow eclipse; primary by convention, relative secondary depth unestablished. Sector22 adjacent spacing9.94631d and sector49 two-cycle spacing19.89509d test recurrence across the gap. Half-period alternatives at BTJD2649.223 and2659.171 have15/17 and14/17 unflagged cadences and minima0.99764/0.99915 of baseline, excluding comparable12ppt intermediate eclipses. A separate shallower secondary remains possible. This is a bounded alias check, not exhaustive all-period search. Error meaning: Weighted linear fit to two sparse quadratic trough centers at sector22 and two free Gaussian plus linear centers at sector49. Inputs max(formal covariance where available, half cadence); sector22 half cadence is a coarse allowance. Covariance inflated if reduced chi square exceeds one. Mixed-model, cadence-smearing, baseline and pipeline systematics are not calibrated, not total uncertainty or coverage.
3. **Target attribution: pass at TESS resolution.** Newly fetched sector22 TESScut difference has positive-deficit5x5 centroid(7.7458,7.7067) vs WCS target(7.797,7.723), separation0.054pixel. No calibrated centroid uncertainty. Successful current MAST TIC60arcsec cone finds two Gaia-linked neighbors at39.92/54.44arcsec, G20.231/20.323. Producing11ppt target loss would require approximately26/28 times their entire Gaia G flux, so neither is capable under that approximate color/aperture brightness proxy. This is not proof against uncatalogued unresolved companions. Receipt helper/156-tic-cone.json; cutout image helper/156-cutout-difference.png.
4. **Alternative reduction: pass.** Independently extracted TESScut pixel-aperture flux reproduces both sector22 dips. Event1 1x1/3x3/5x5 average depths5.93/6.66/4.95ppt; event2 9.39/5.42/2.93ppt. Apertures center on brightest pixel(8,8), without background subtraction, in-event +/-0.027d and out-of-event0.065-0.17d. These smeared average depths differ from QLP trough/model depths, as expected from cadence, extraction and dilution. Same TESS photons, not independent observations. Source FITS, script and hash in helper/156-independent.json and helper/tesscut156.

5. **Full NSS Gaia geometry: pass with limitations.** DR34000328107088372352; GaiaP=9.9466027631+/-0.0012787442d, e=0.151498, omega=132.2616deg. Closest edge-on conjunction offset0.101824d (0.010237phase), full SB1 covariance linear allowance0.191661d at measured epoch. Mass function0.042525solar masses, assumed primary1.090, minimum companion0.469. Approximate central duration5.46h vs observed approximate sector22 cutout FWHM1.2h allows grazing/noncentral geometry. This is descriptive edge-on/minimum-mass comparison, not detailed binary modeling; no arbitrary sigma veto. Full inputs/covariance assumptions in gaia-geometry.json.

Queue-preparation QLP availability product list (targets-selected.json, hashed in helper/156-checks.json) contains sectors22 and49, both investigated. This reuses the campaign product list rather than asserting an independent fresh helper query. Sector49 previously flagged predicted windows were not clean nulls; shifted actual eclipse windows reveal repeats. No replacement or denominator exclusion.

Limitations: Narrow eclipses at30minute cadence in sector22 have only3-4 trough samples. Mixed quadratic/Gaussian timing model and cadence floors are descriptive allowances, not total uncertainty. Sector49 cycle75 and sector22 half-period midpoint are flagged/gaps, not clean nulls; other sector49 half-period windows have usable flat samples. Same photons, aperture dilution and uncalibrated centroid uncertainty remain. Approximate1.2h observed FWHM differs from modeled central duration; grazing geometry possible, no detailed binary fit.

Next check: Optional short-cadence resolved photometry and revised spectroscopy for sharper eclipse morphology/geometry; no missing bounded admission check.

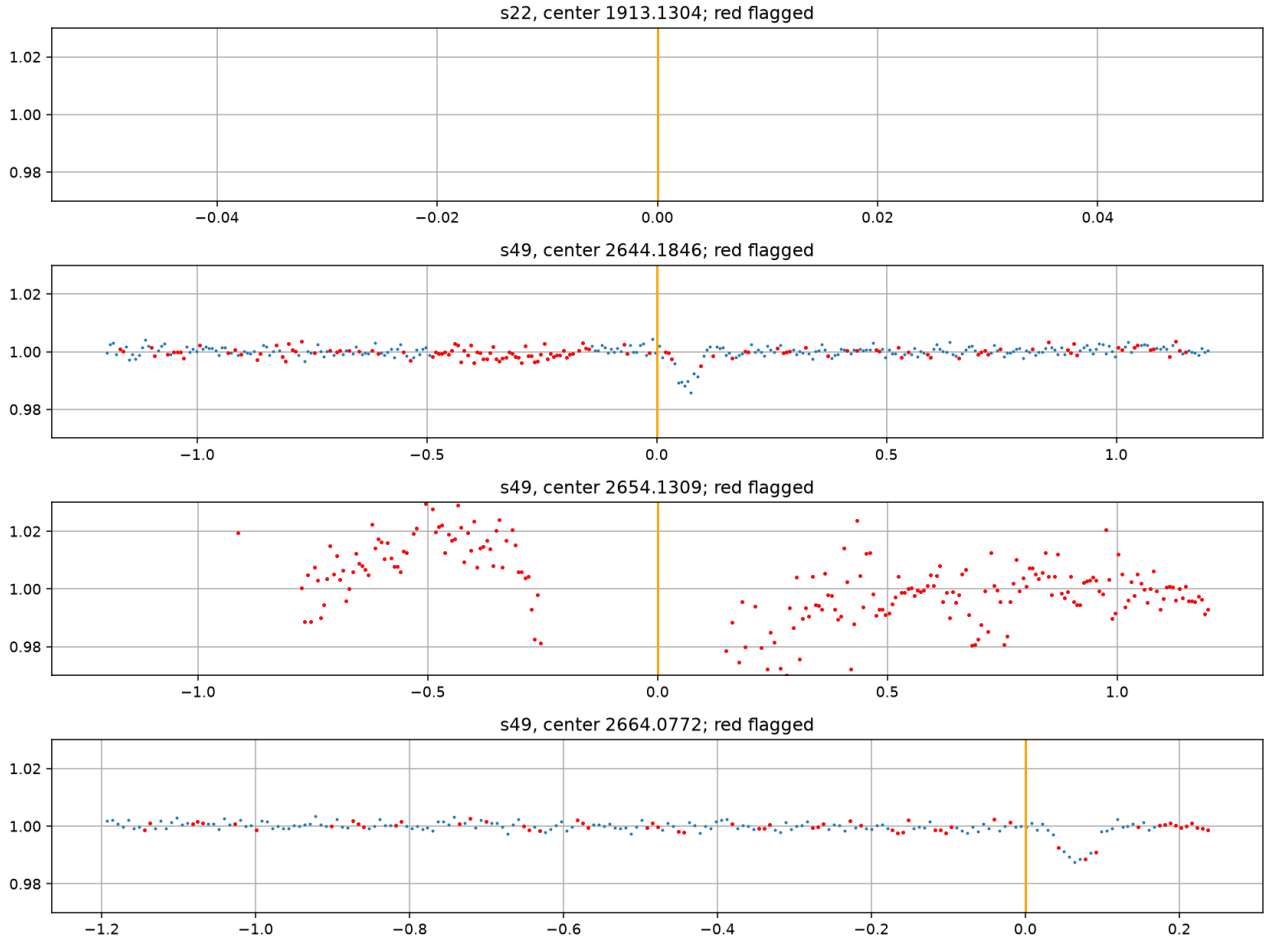
Catalogue novelty is supplied separately by ../../catalogues/coverage-matrix.csv. This physical report does not claim exhaustive literature priority.

Sector	Cycle	Measured midpoint BTJD	Timing input d	Depth ppt	Method
22	0	1908.15727703	0.01041675	11.524	quadratic through 3-4 trough samples, cadence/2 allowance
22	1	1918.10359176	0.01041675	11.950	quadratic through 3-4 trough samples, cadence/2 allowance
49	74	2644.24941621	0.00347220	13.284	Gaussian plus linear baseline, unflagged SAP, sigma max(formal, cadence/2)
49	76	2664.14451057	0.00347220	12.603	Gaussian plus linear baseline, unflagged SAP, sigma max(formal, cadence/2)

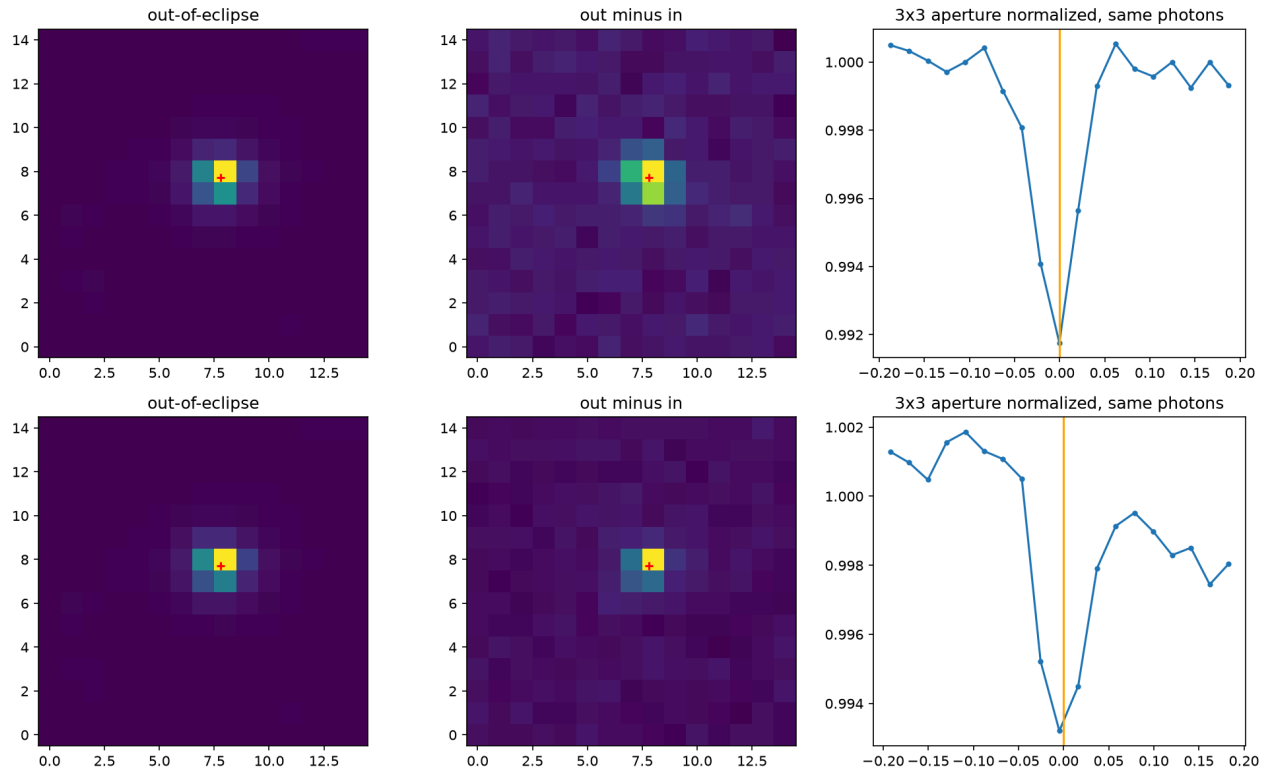
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json; pipeline_measurements.json; ../helper/156-independent.json; ../helper/156-independent.py; ../helper/156-sector49.py; ../helper/156-alias-windows.png; ../helper/156-cutout-difference.png; ../helper/156-tic-cone.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Shared QLP source hashes in ../download-receipts.json. Public identity boundary respected.

Stable review key: R291. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Editorial update after review: the coordinator accepted this completed review for integration. The original pre-acceptance wording above records the lead handoff. Direct deciding evidence: raw sector-49 eclipse and coverage windows, sector-22 eclipse extractions and deficit images, and independent measurements. Orange guide lines are predictions, not measured timing points; flagged or empty windows are not clean nulls.



R291; 156-alias-windows.png. Actual sector-49 raw-flux windows include two shifted eclipses and empty or flagged intermediate windows. Orange predicted lines are not measured timings; the report gives measured centers and alias qualifications.



R291; 156-cutout-difference.png. Sector-22 actual-event cutout and pixel-aperture alternative reduction; centroid and angular-resolution qualifications remain in the full report.

R292. TIC 189664666

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 189664666: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[91]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=3.4725104030 ±0.0005545785d, reference eclipse epoch BTJD3778.6307516, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S91 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.658arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.056. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths43.381/43.134/42.826ppt, with 2 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median43.073ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3630082350835706867 2, P=3.4727173222±0.0003480687d, e=0.061438, omega=290.4505deg. Closest edge-on conjunction offset0.002382d (0.000686phase), full-SB1-covariance linear propagated Gaia uncertainty0.359401d at the measured epoch. Near-Gaia-reference projected eclipse offset0.204335d (photometric period projection allowance0.541269d). The approximate central

duration5.39h vs observed median5.01h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four real raw eclipses in sector91; discontinuous egress/baseline ramp in cycle5 makes its trapezoid center unreliable and excluded. Three other same-kind timings fit the period. No capable catalogued neighbor in approximate brightness test. Timing baseline is only14d, not precision long-baseline characterization.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

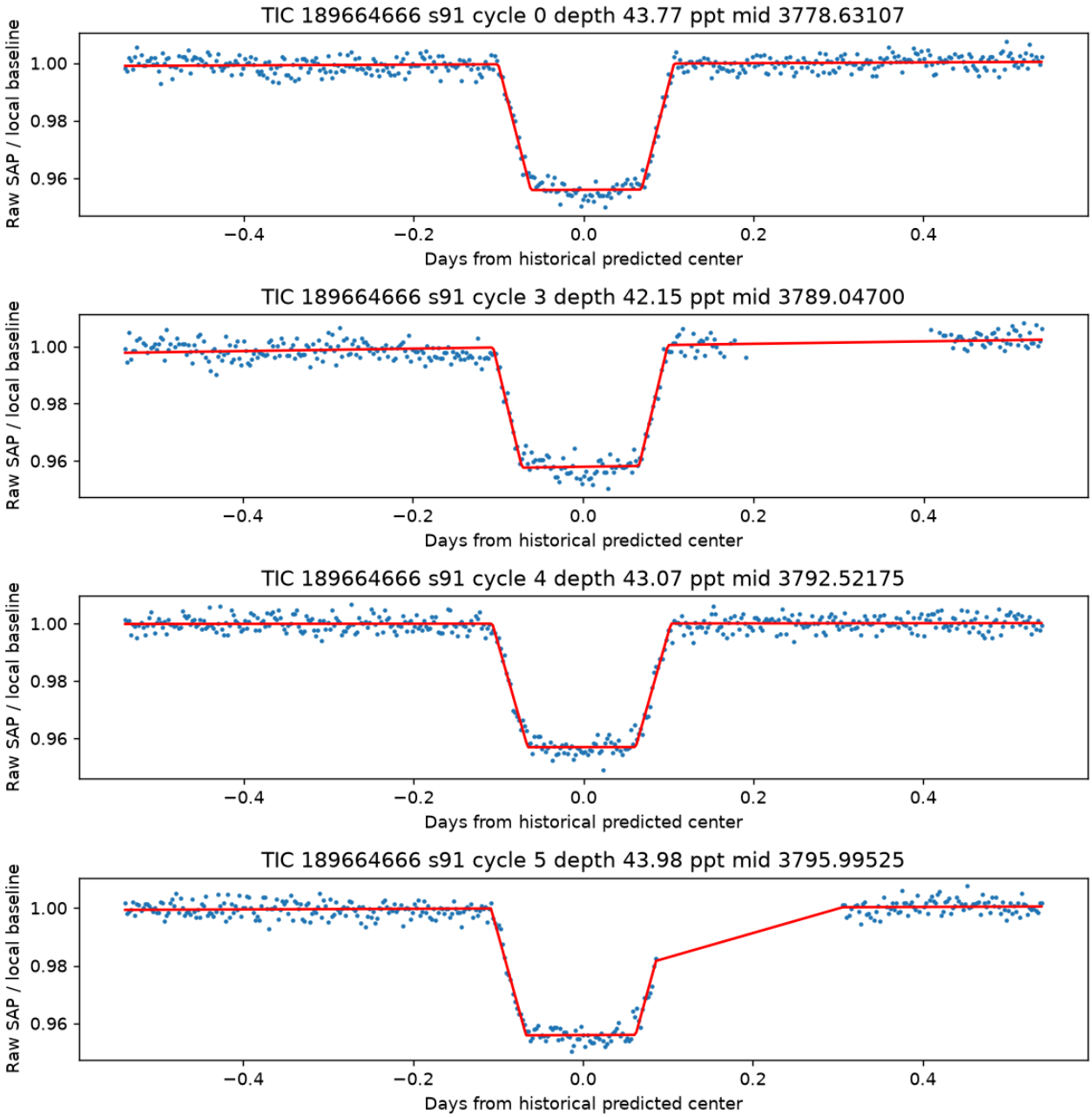
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

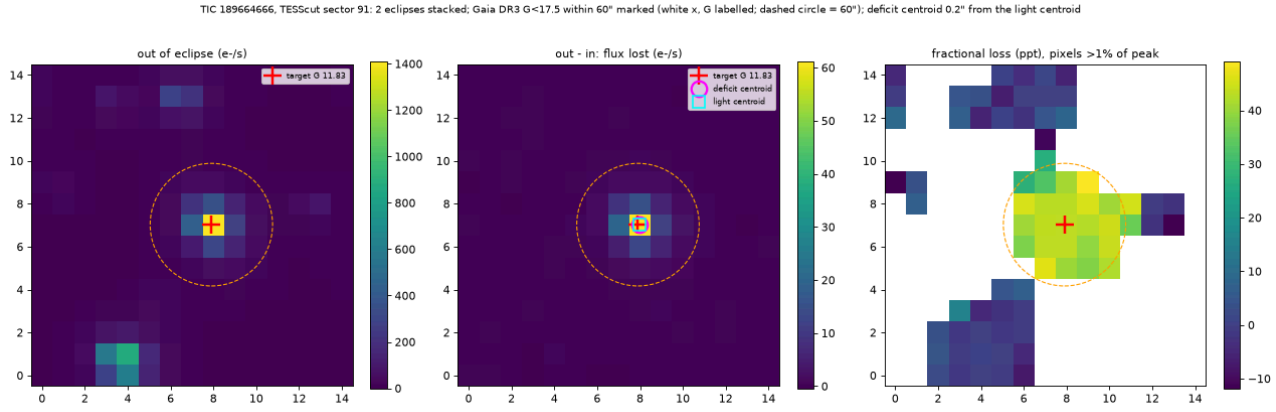
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
91	0	3778.6310718	0.001157	43.774	usable; same-kind selection applies
91	3	3789.0470020	0.001157	42.146	usable; same-kind selection applies
91	4	3792.5217538	0.001157	43.073	usable; same-kind selection applies
91	5	3795.9952516	0.001157	43.976	excluded unreliable timing fit

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s91.json, historical/diff_s91.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R292. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R292; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R292; diff_s91.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R293. TIC 309308701

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 309308701: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[73, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=18.3848616420 ±0.0001132702d, reference eclipse epoch BTJD3291.8509571, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S73 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.247arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.312. Capable neighbours: Gaia 261872853348447872: 27.98 arcsec, G15.831, required fractional eclipse0.708
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths20.421/18.408/17.309ppt, with 2 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median20.689ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3261872780332852736, P=18.3808647997±0.0048918530d, e=0.311976, omega=136.7904deg. Closest edge-on conjunction offset0.655409d (0.035657phase), full-SB1-covariance linear propagated Gaia uncertainty0.842000d at the measured epoch. Near-Gaia-reference projected eclipse offset0.023907d (photometric period projection allowance0.017897d). The approximate central

duration11.27h vs observed median5.50h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four shallow isolated eclipses in sectors73/86, including adjacent-cycle pairs, test recurrence. Target-deficit0.155pixel vs nearest capable neighbor1.50pixels favors target; no calibrated centroid errors. Gaia nominal conjunction offset0.6554d but full-covariance linear allowance0.8420d and near-reference offset0.0239d support compatible geometry.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

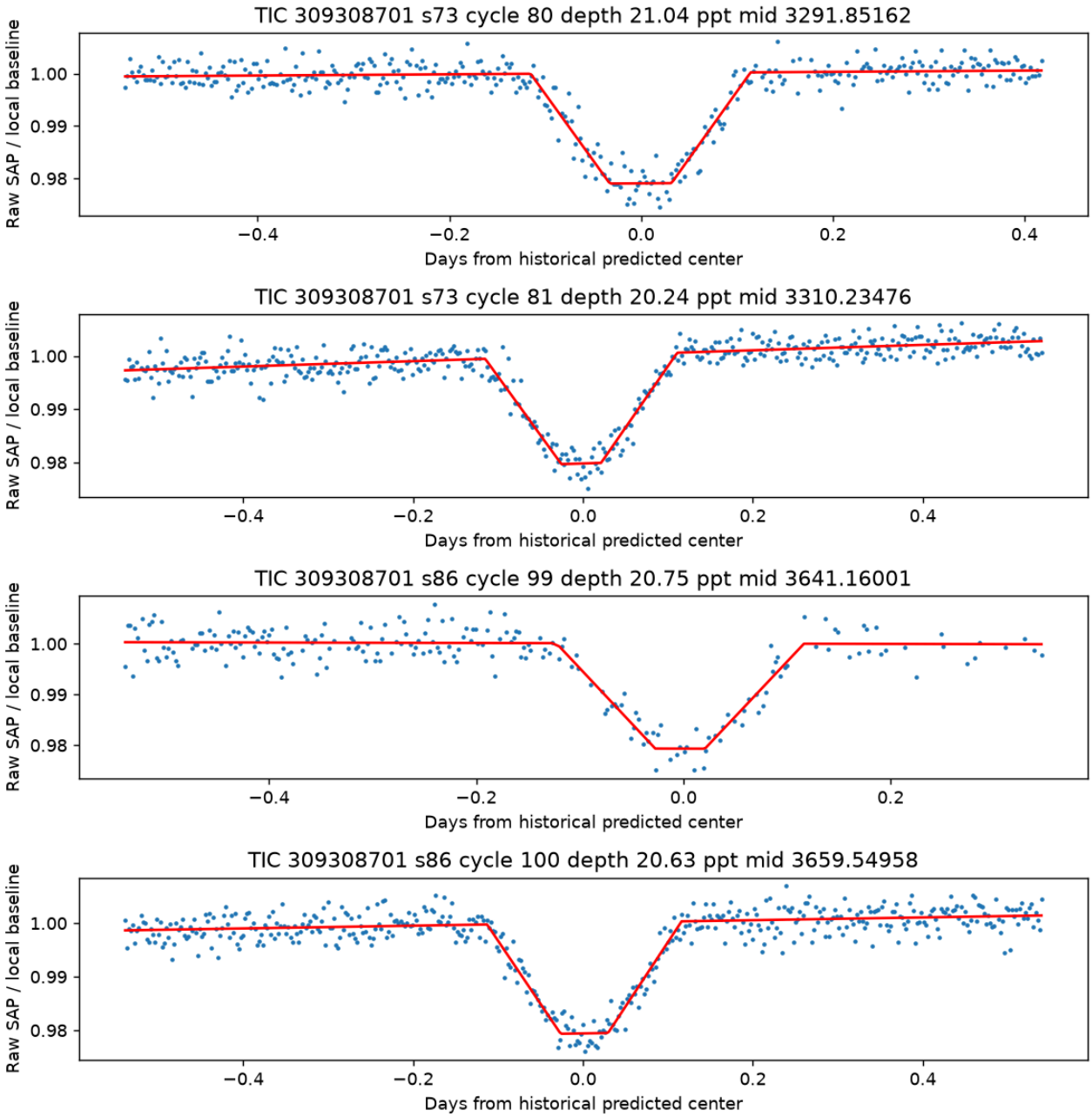
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

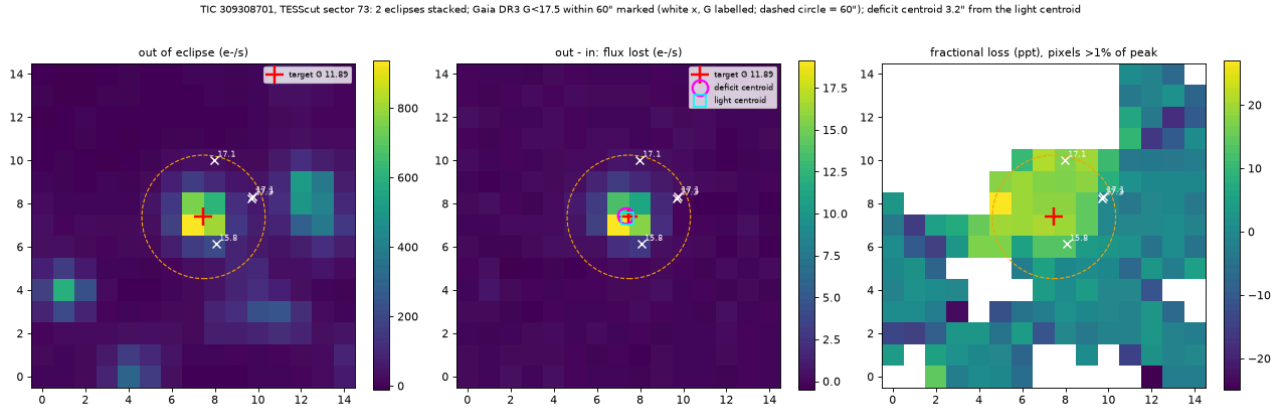
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
73	80	3291.8516219	0.001157	21.044	usable; same-kind selection applies
73	81	3310.2347606	0.001487	20.239	usable; same-kind selection applies
86	99	3641.1600099	0.001993	20.748	usable; same-kind selection applies
86	100	3659.5495792	0.001303	20.629	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s73.json, historical/diff_s73.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R293. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R293; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R293; diff_s73.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R294. TIC 141811748

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 141811748: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[1, 37, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=6.8389313923 ±0.0000094980d, reference eclipse epoch BTJD1334.0304484, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation3.935arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.218. Capable neighbours: Gaia 5260245291514395136: 38.08 arcsec, G14.242, required fractional eclipse0.394
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths37.325/34.581/32.925ppt, with 7 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median31.682ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3526024560075204 3648, P=6.8386133503±0.0007182882d, e=0.029587, omega=203.5107deg. Closest edge-on conjunction offset0.001233d (0.000180phase), full-SB1-covariance linear propagated Gaia uncertainty0.123473d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.042657d (photometric period projection allowance0.001311d). The approximate central duration8.07h vs observed median4.78h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses across three sectors; sector/aperture-dependent dilution retained. Target-deficit0.187pixel vs closest capable neighbor1.67pixels favors target; no calibrated centroid errors. Sector37 actual adjacent cycles144/145/146 show raw core depths38.52/37.30/37.68ppt (11/11/12 in-event cadences); cycle145 fails fit side-window threshold but tests recurrence. Missing even sector98 cycles have no raw coverage. Sparse sector1 cycles0/1/2 also recur. Thus apparent alternating-sector98 pattern is coverage, not doubled period; see coverage-alias.json.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

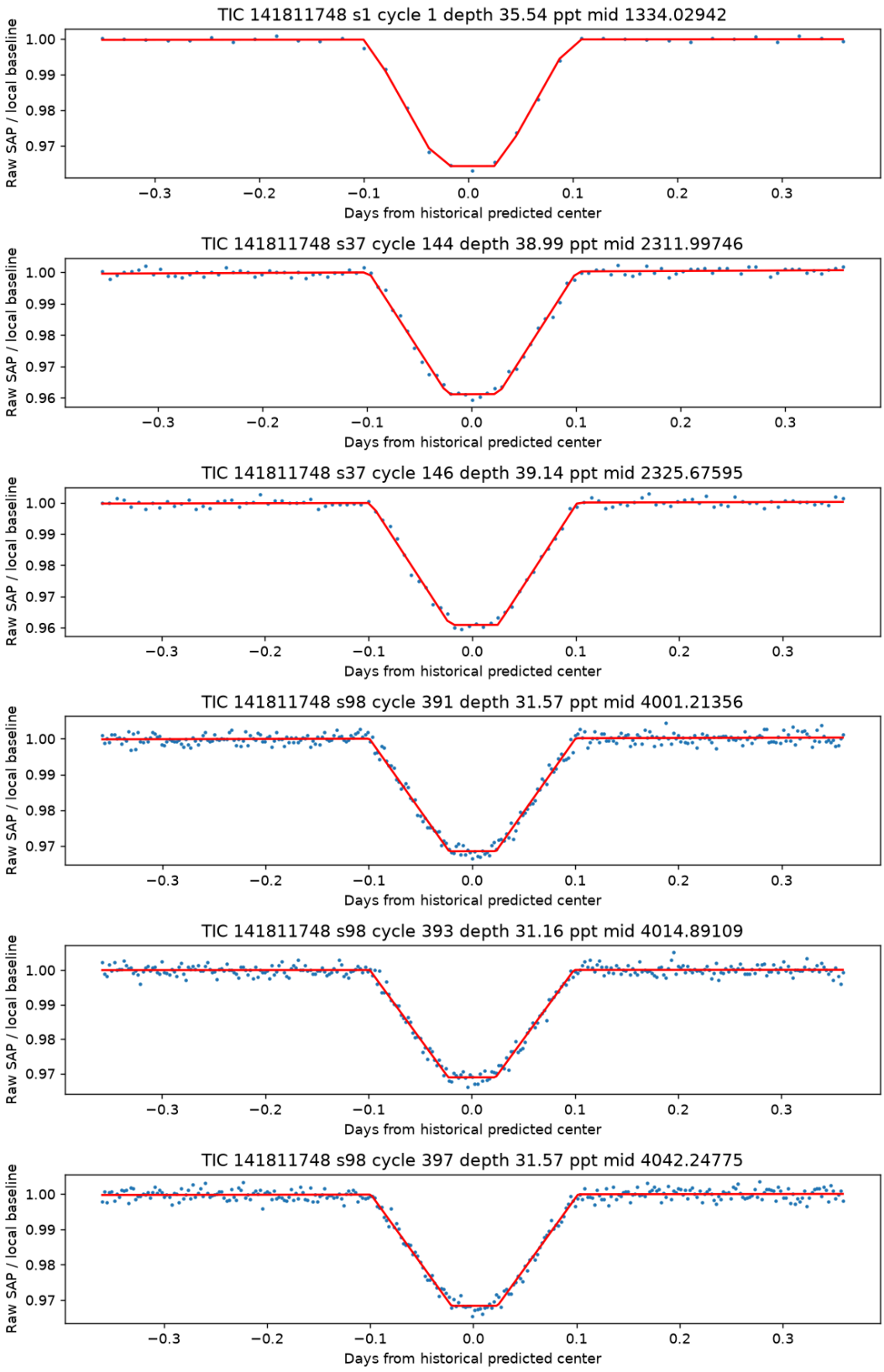
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

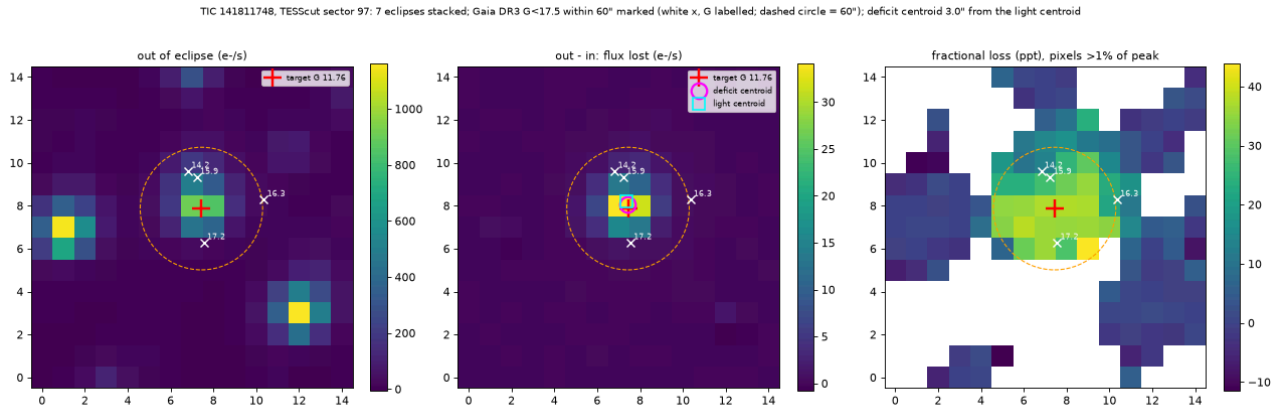
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	1	1334.0294197	0.010417	35.535	usable; same-kind selection applies
37	144	2311.9974617	0.003472	38.985	usable; same-kind selection applies
37	146	2325.6759452	0.003472	39.138	usable; same-kind selection applies
98	391	4001.2135603	0.001157	31.574	usable; same-kind selection applies
98	393	4014.8910898	0.001157	31.163	usable; same-kind selection applies
98	395	4028.5695202	0.001157	31.682	usable; same-kind selection applies
98	397	4042.2477547	0.001157	31.567	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R294. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R294; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R294; diff_s97.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R295. TIC 5781196

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 5781196: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 2 distinct eclipse events in selected sectors[36, 99]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 2 same-kind midtimes give P=11.0017719770 ±0.0000228750d, reference eclipse epoch BTJD2289.0187466, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S89 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation5.093arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.441. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths215.498/201.534/157.504ppt, with 3 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median202.989ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3543584746895666856, P=11.0028295172±0.0015099322d, e=0.570356, omega=115.9229deg. Closest edge-on conjunction offset-0.189598d (-0.017232phase), full-SB1-covariance linear propagated Gaia uncertainty0.272145d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.006644d (photometric period projection allowance0.003957d). The approximate central

duration2.79h vs observed median2.30h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Two precisely fitted deep events in sectors36/99 are supplemented by helper raw sector89 THREE adjacent repeated eclipses near BTJD3719/3730/3741 separated by11.0018d. Those actual raw repeats test long-baseline integer-cycle aliases; they are not predicted centers used as measured timings. Strong aperture/sector dilution differences remain; no capable catalogued neighbor within60arcsec.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

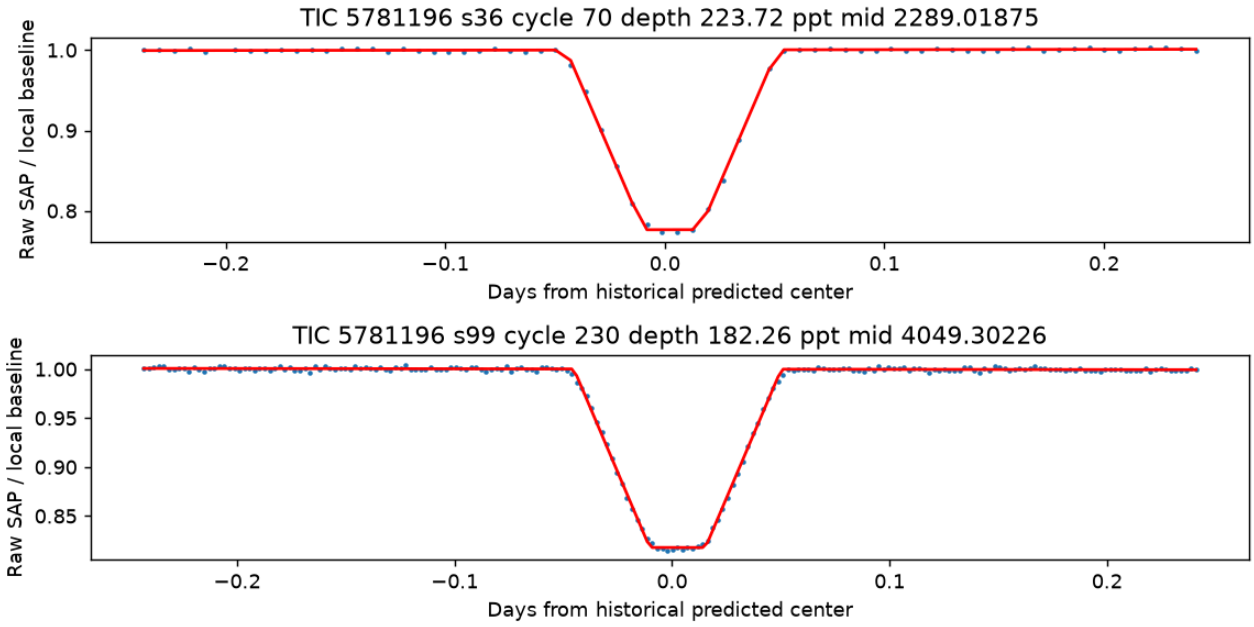
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

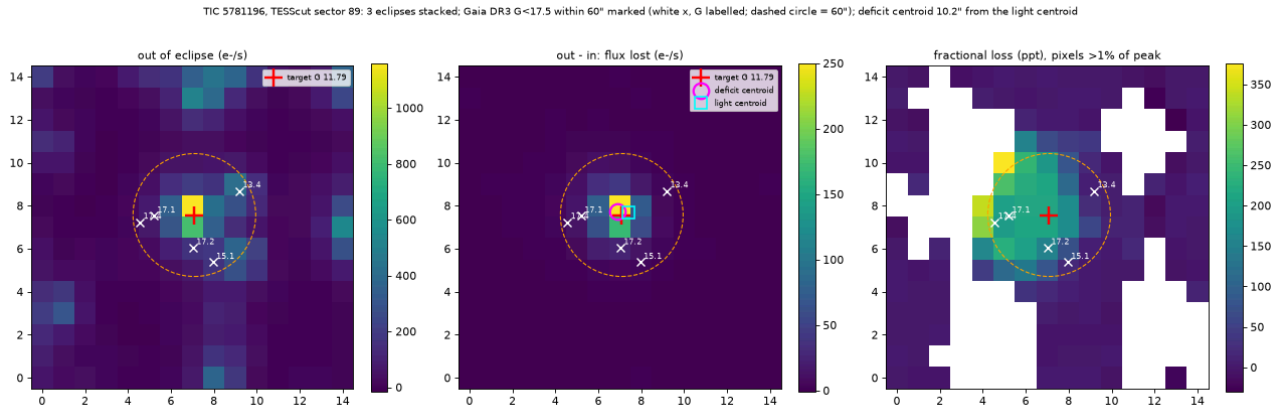
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
36	70	2289.0187466	0.003472	223.722	usable; same-kind selection applies
99	230	4049.3022629	0.001157	182.256	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s89.json, historical/diff_s89.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R295. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R295; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R295; diff_s89.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R296. TIC 370329832

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 370329832: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[38, 90]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=9.4634975241 ±0.0000170165d, reference eclipse epoch BTJD2344.0833839, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Unreliable fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S63 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.495arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.582. Capable neighbours: Gaia 5219338717319681536: 35.44 arcsec, G15.976, required fractional eclipse0.980; Gaia 5219338511161254272: 38.70 arcsec, G12.998, required fractional eclipse0.084; Gaia 5219340263507912448: 47.07 arcsec, G15.901, required fractional eclipse0.916; Gaia 5219338751679414528: 59.13 arcsec, G15.279, required fractional eclipse0.527
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths22.301/21.219/17.450ppt, with 3 stacked events. Historical event coordinates guide these stacks and are not newly measured midtimes. QLP selected-kind fitted median24.210ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3521933871301999936O, P=9.4629554699±0.0011615825d, e=0.014692, omega=346.6392deg. Closest edge-on conjunction

offset0.099963d (0.010564phase), full-SB1-covariance linear propagated Gaia uncertainty0.214371d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.012242d (photometric period projection allowance0.003522d). The approximate central duration8.06h vs observed median3.32h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five measured shallow eclipses include adjacent repeated pairs in sectors38/90, testing period recurrence. Target-deficit0.071pixel vs closest capable neighbor1.86pixels favors target. Sector63 alternative extraction depths22.30/21.22/17.45ppt show dilution. No calibrated localization error.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

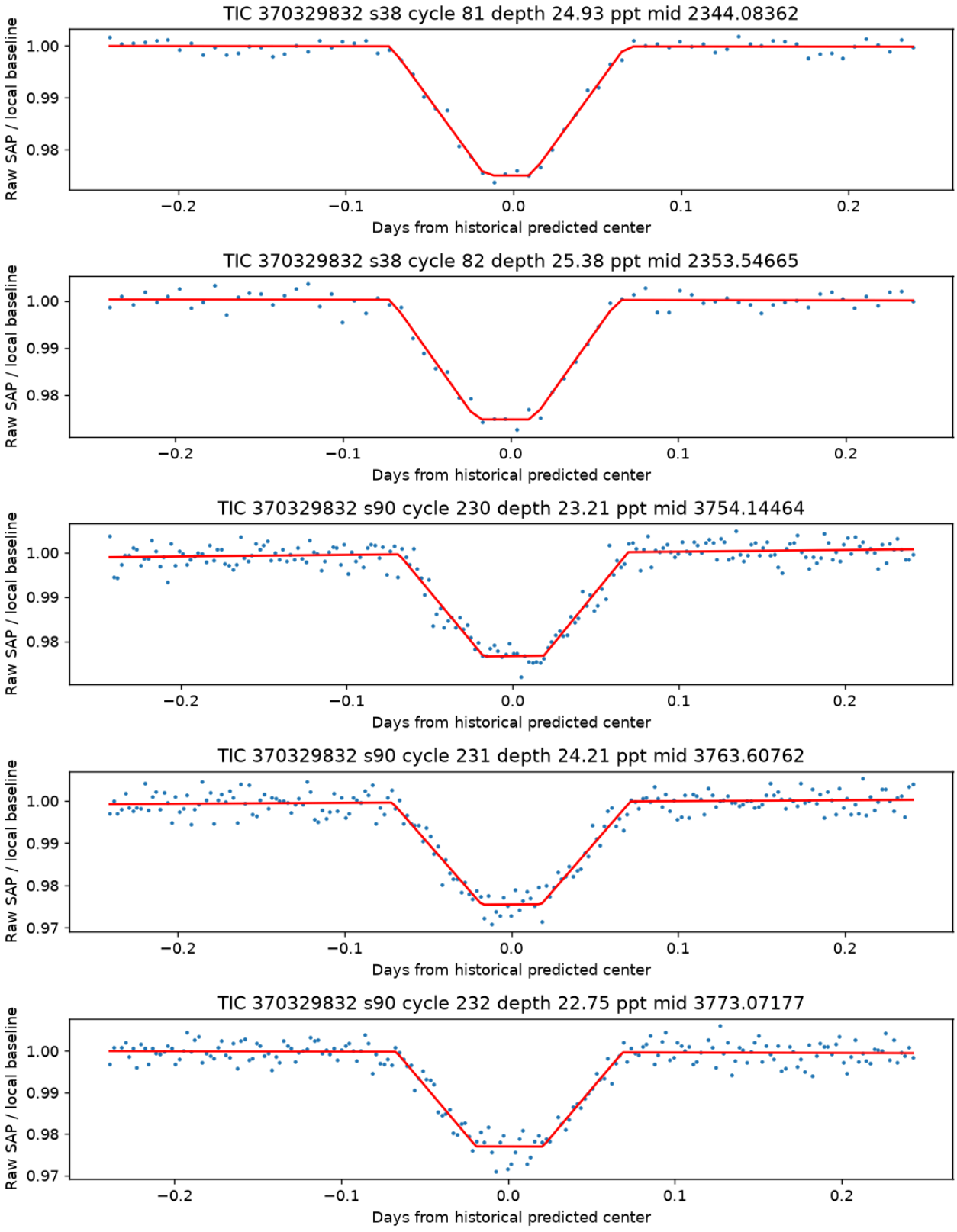
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

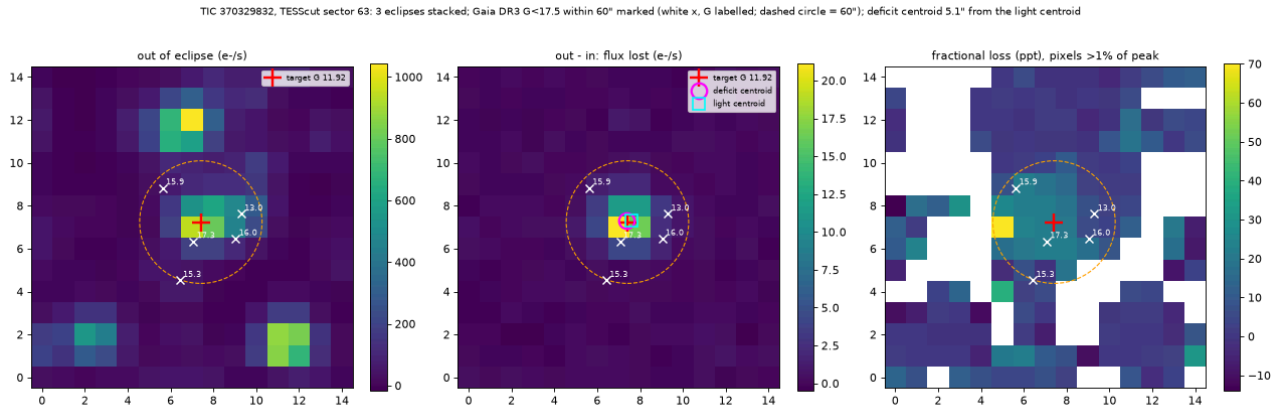
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
38	81	2344.0836175	0.003472	24.926	usable; same-kind selection applies
38	82	2353.5466544	0.003472	25.382	usable; same-kind selection applies
90	230	3754.1446441	0.001157	23.211	usable; same-kind selection applies
90	231	3763.6076196	0.001157	24.210	usable; same-kind selection applies
90	232	3773.0717732	0.001157	22.746	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s63.json, historical/diff_s63.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R296. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R296; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R296; diff_s63.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R297. TIC 379550793

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 379550793: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 19 distinct eclipse events in selected sectors[64, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 19 same-kind midtimes give P=2.4798111572 +/-0.0000014767d, reference eclipse epoch BTJD3041.9007652, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S38 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation6.620arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.700. Capable neighbours: Gaia 6058502724870374912: 15.61 arcsec, G16.097, required fractional eclipse0.529; Gaia 6058502793589851136: 20.31 arcsec, G15.796, required fractional eclipse0.404; Gaia 6058502724870373888: 21.42 arcsec, G13.912, required fractional eclipse0.081; Gaia 6058502724870375552: 27.27 arcsec, G14.352, required fractional eclipse0.115; Gaia 6058502724870373760: 29.18 arcsec, G16.482, required fractional eclipse0.749; Gaia 6058502754891546752: 36.05 arcsec, G16.282, required fractional eclipse0.625; Gaia 6057752102023464064: 38.96 arcsec, G16.156, required fractional eclipse0.558; Gaia 6058502793589852544: 40.20 arcsec, G16.063, required fractional eclipse0.513; Gaia 6058502690510531712: 42.43 arcsec, G16.447, required fractional eclipse0.726; Gaia 6057752136383209344: 42.52 arcsec, G15.938, required fractional eclipse0.458; Gaia 6058502690510531456: 45.24 arcsec, G15.898, required fractional eclipse0.442; Gaia 6058502759230114816: 46.85 arcsec, G14.905, required fractional eclipse0.184; Gaia

6057752136383211136: 48.56 arcsec, G15.402, required fractional eclipse0.284; Gaia 6058502793555875456: 51.23 arcsec, G16.436, required fractional eclipse0.718; Gaia 6058502793589748352: 51.39 arcsec, G14.227, required fractional eclipse0.104; Gaia 6058502621791051776: 53.01 arcsec, G13.381, required fractional eclipse0.054; Gaia 6058502793589748992: 54.98 arcsec, G16.251, required fractional eclipse0.608; Gaia 6058502621791051904: 55.79 arcsec, G15.214, required fractional eclipse0.241; Gaia 6057752136383212800: 56.20 arcsec, G14.549, required fractional eclipse0.136

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths12.036/10.485/7.514ppt, with 11 stacked events. QLP selected-kind fitted median11.453ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR36058502724870374272, P=2.4797362150+/-0.0001521287d, e=0.018215, omega=160.3186deg. Closest edge-on conjunction offset0.076701d (0.030931phase), full-SB1-covariance linear propagated Gaia uncertainty0.172965d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.003488d (photometric period projection allowance0.001580d). The approximate central duration4.42h vs observed median1.79h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field: deficit0.315pixel from target; closest capable neighbor0.856pixel from deficit (G13.91,21.4arcsec). Peak/core favors target, with no calibrated centroid errors; aperture depths12.04/10.48/7.51ppt and sector depths differ through dilution. Intrinsic neighbor eclipse requirements are not directly compared with blended pixel losses.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

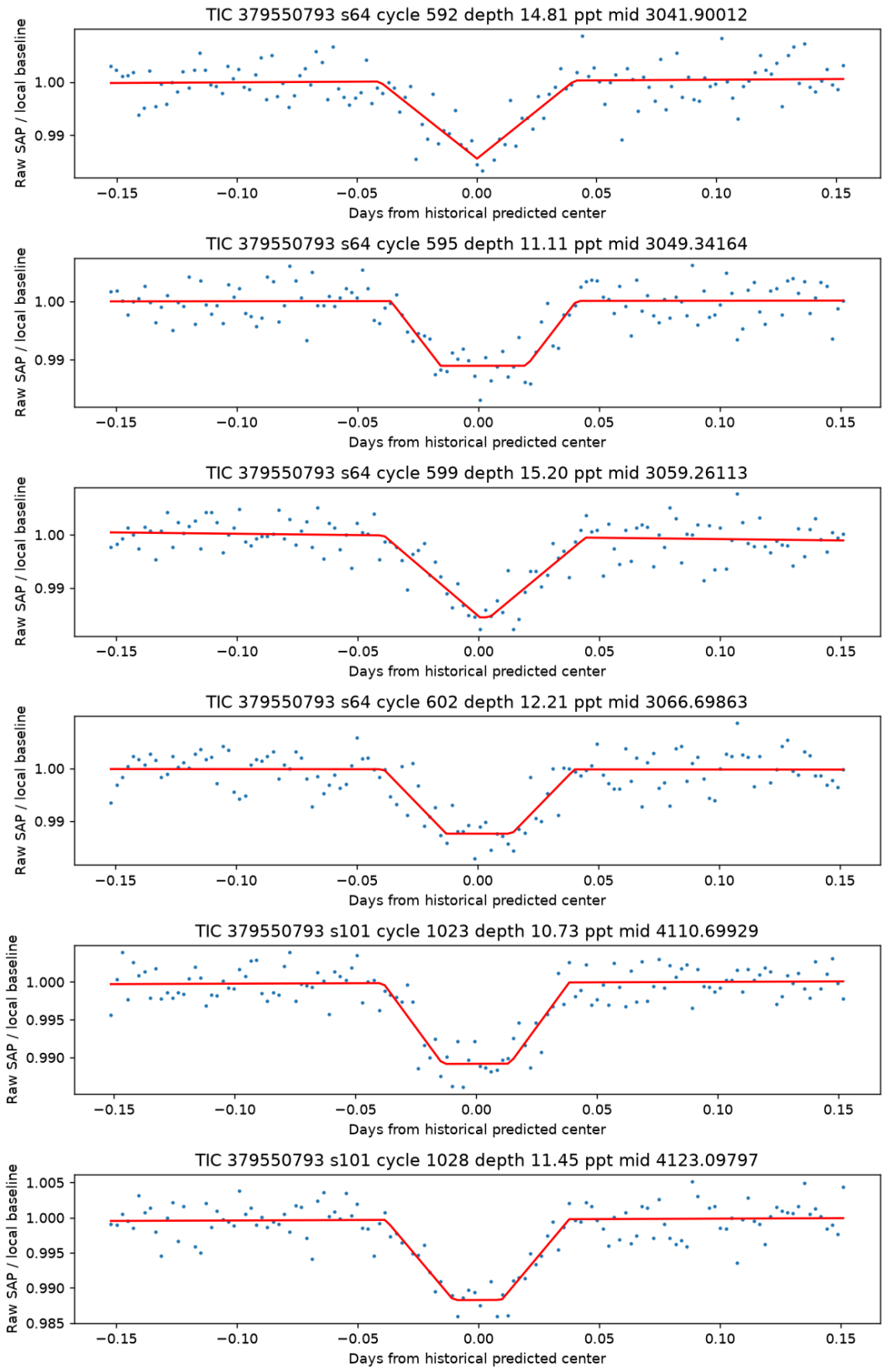
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
64	592	3041.9001207	0.001605	14.812	usable; same-kind selection applies
64	593	3044.3767653	0.001420	12.393	usable; same-kind selection applies
64	594	3046.8611067	0.001157	17.351	usable; same-kind selection applies
64	595	3049.3416450	0.001511	11.108	usable; same-kind selection applies
64	596	3051.8199952	0.001522	13.523	usable; same-kind selection applies
64	597	3054.2992656	0.001450	12.556	usable; same-kind selection applies
64	598	3056.7792143	0.001157	10.596	usable; same-kind selection applies
64	599	3059.2611263	0.001408	15.204	usable; same-kind selection applies
64	600	3061.7411370	0.001418	14.044	usable; same-kind selection applies

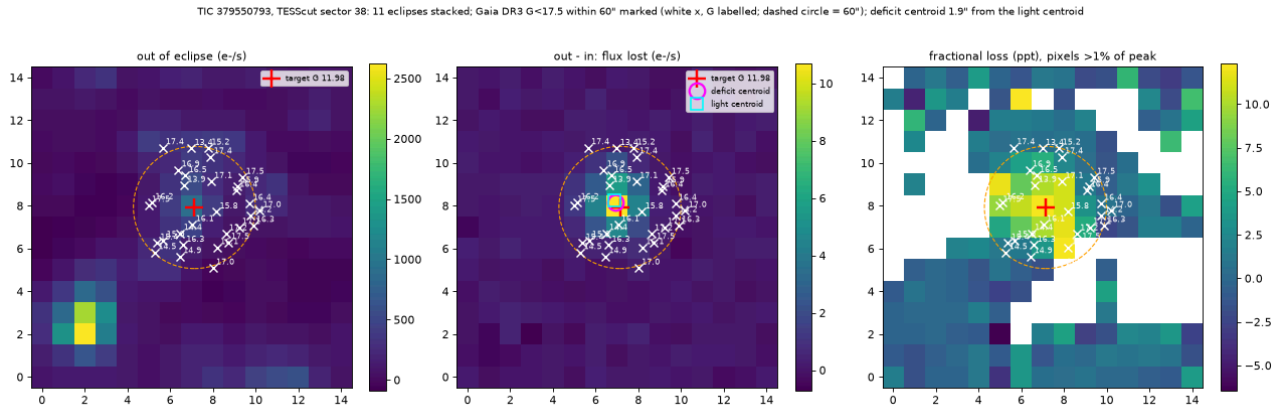
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
64	601	3064.2188907	0.001157	14.765	usable; same-kind selection applies
64	602	3066.6986276	0.001582	12.214	usable; same-kind selection applies
101	1020	4103.2606006	0.001178	9.125	usable; same-kind selection applies
101	1021	4105.7396162	0.001157	11.128	usable; same-kind selection applies
101	1022	4108.2214580	0.001157	11.305	usable; same-kind selection applies
101	1023	4110.6992867	0.001157	10.728	usable; same-kind selection applies
101	1025	4115.6582089	0.001157	9.993	usable; same-kind selection applies
101	1026	4118.1363897	0.001318	10.210	usable; same-kind selection applies
101	1027	4120.6194115	0.001157	10.500	usable; same-kind selection applies
101	1028	4123.0979682	0.001163	11.453	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R297. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R297; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R297; diff_s38.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R298. TIC 161823655

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 161823655: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[15, 51, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.6070592589 +/-0.0000733037d, reference eclipse epoch BTJD1713.7278770, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S16 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.147arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.153. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths64.990/64.779/63.789ppt, with 2 stacked events. QLP selected-kind fitted median67.139ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3162598549340050764 8, P=11.6063345473+/-0.0017583299d, e=0.067125, omega=257.8654deg. Closest edge-on conjunction offset0.085945d (0.007405phase), full-SB1-covariance linear propagated Gaia uncertainty0.256699d at the measured epoch. Near-Gaia-reference projected eclipse offset0.003328d (photometric period projection allowance0.008357d). The approximate central duration7.97h vs observed median7.18h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five measured events over three sectors, including adjacent same-kind repeats. Raw sector85 has changing baselines but clear narrow core; depth variation alone is not established astrophysically. No capable neighbor under approximate Gaia flux-budget screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

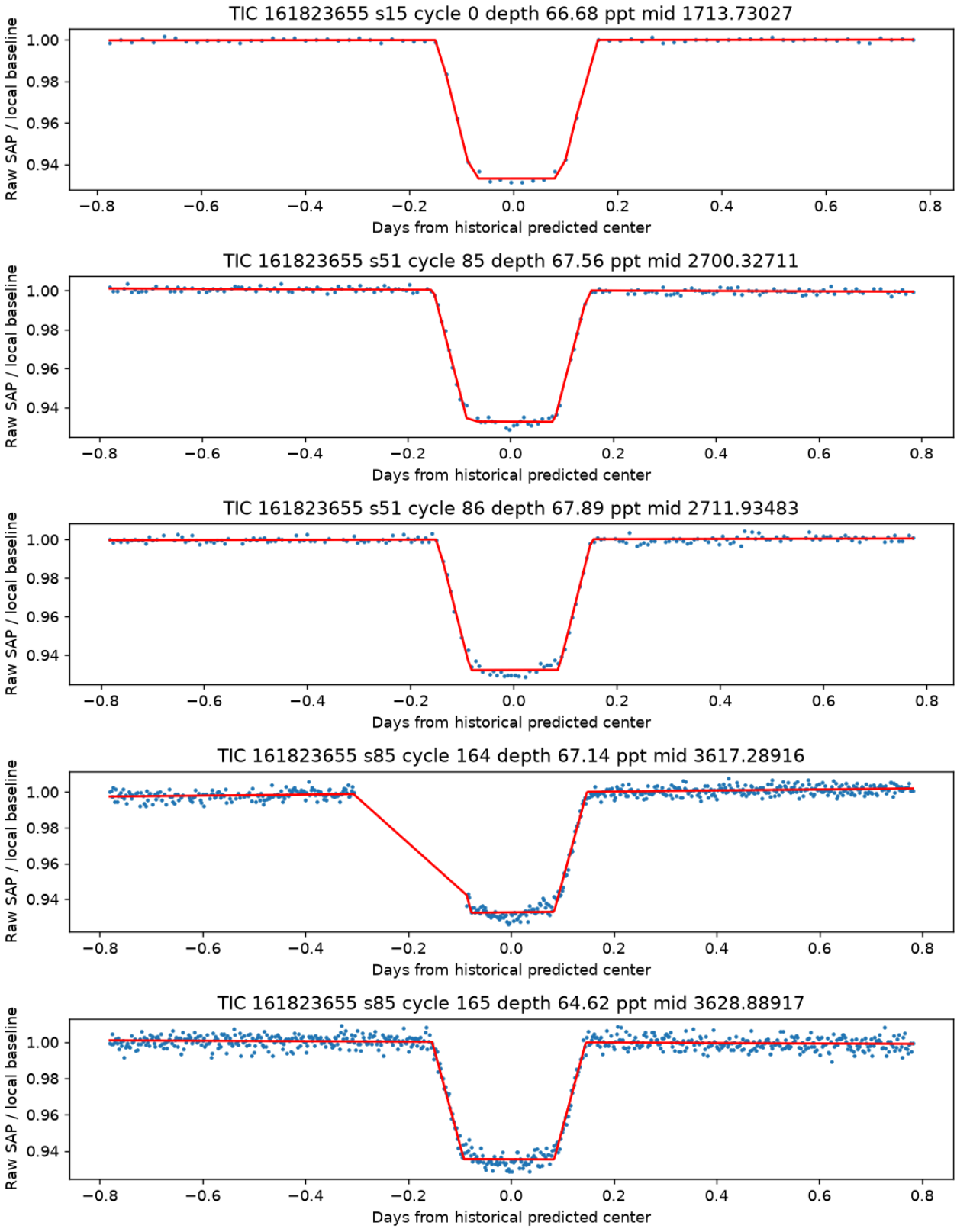
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

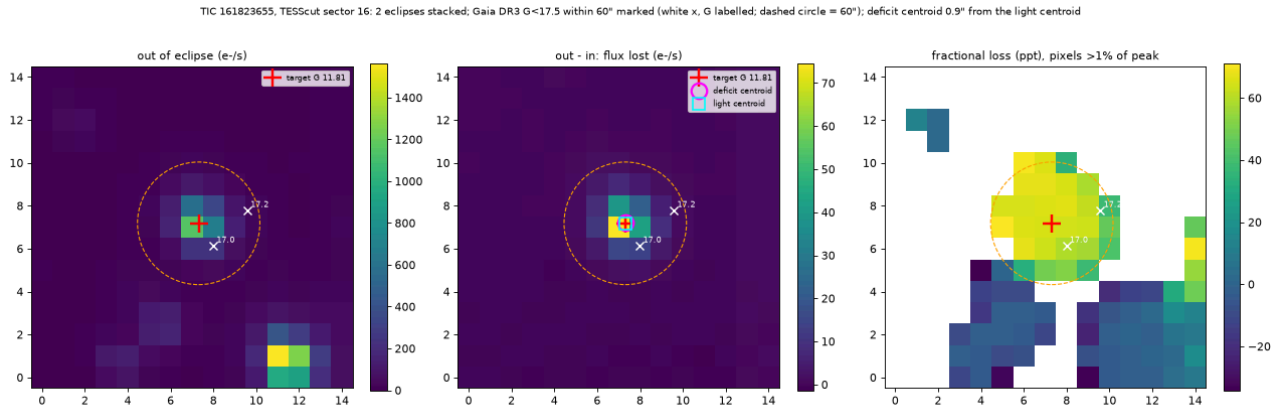
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1713.7302715	0.010416	66.681	usable; same-kind selection applies
51	85	2700.3271109	0.003472	67.562	usable; same-kind selection applies
51	86	2711.9348250	0.003472	67.887	usable; same-kind selection applies
85	164	3617.2891575	0.001157	67.139	usable; same-kind selection applies
85	165	3628.8891688	0.001157	64.618	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s16.json, historical/diff_s16.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R298. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R298; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R298; diff_s16.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R299. TIC 341118777

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 341118777: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[13, 101, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=6.8474689646 +/-0.0000169388d, reference eclipse epoch BTJD1665.0429604, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S67 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation0.609arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.056. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths112.012/112.820/109.723ppt, with 4 stacked events. QLP selected-kind fitted median95.174ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3645427248070527884 8, P=6.8477864674+/-0.0007556392d, e=0.070259, omega=102.3940deg. Closest edge-on conjunction offset0.013863d (0.002024phase), full-SB1-covariance linear propagated Gaia uncertainty0.171384d at the measured epoch. Near-Gaia-reference projected eclipse offset0.072919d (photometric period projection allowance0.003151d). The approximate central duration4.74h vs observed median3.81h permits noncentral geometry; this radius/minimum-mass estimate is not a

detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated events in sectors13/101. Sector104 cycle371 singular center covariance is excluded from timing; its real partial/noisy morphology is retained as documentation. No capable neighbor under approximate flux-budget screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

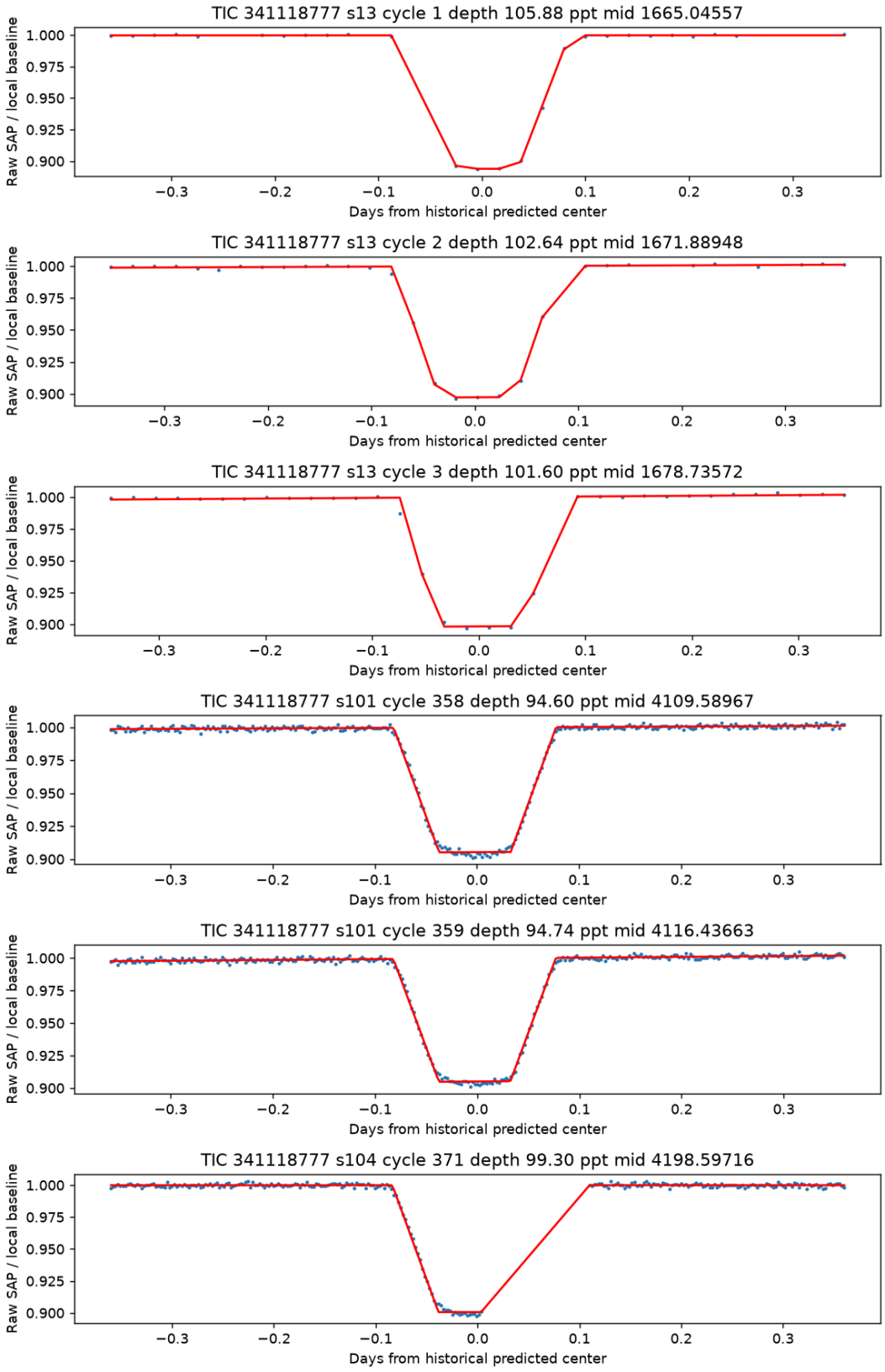
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

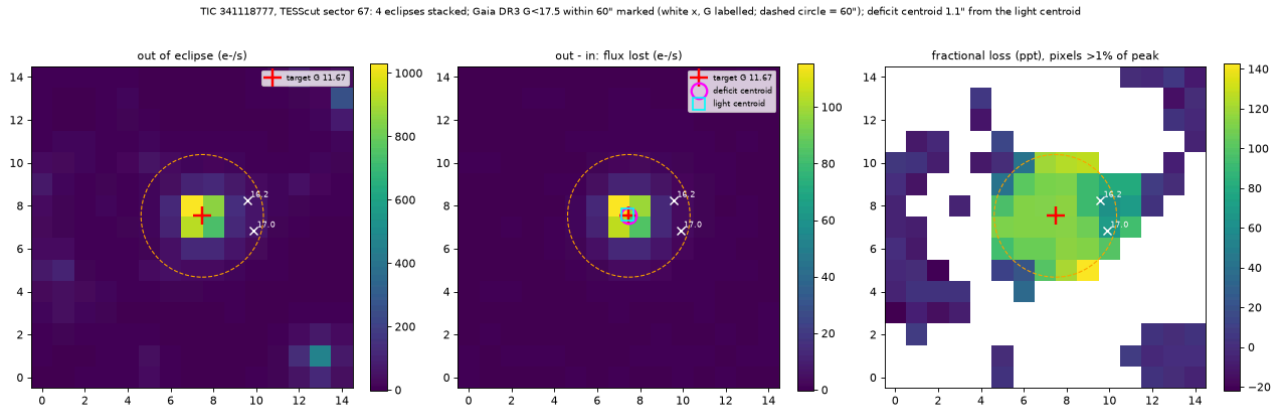
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
13	1	1665.0455748	0.010417	105.879	usable; same-kind selection applies
13	2	1671.8894812	0.010417	102.644	usable; same-kind selection applies
13	3	1678.7357199	0.010417	101.600	usable; same-kind selection applies
101	357	4102.7425263	0.001157	95.174	usable; same-kind selection applies
101	358	4109.5896726	0.001157	94.603	usable; same-kind selection applies
101	359	4116.4366276	0.001157	94.736	usable; same-kind selection applies
101	360	4123.2836410	0.001157	95.114	usable; same-kind selection applies
104	371	4198.5971566	41218.230390	99.299	excluded singular timing fit

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s67.json, historical/diff_s67.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R299. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R299; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R299; diff_s67.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R300. TIC 21952826

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 21952826: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[62, 99]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=9.7685847110 +/-0.0000106665d, reference eclipse epoch BTJD2995.8445787, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S35 deficit peak[8, 8] vs target pixel[8, 7]; deficit-target separation7.214arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.137. Capable neighbours: Gaia 5659918043089101184: 40.76 arcsec, G15.240, required fractional eclipse0.520; Gaia 5659918008729365120: 55.61 arcsec, G15.689, required fractional eclipse0.774
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths24.848/24.899/23.814ppt, with 2 stacked events. QLP selected-kind fitted median25.067ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3565992099802659936, P=9.7676391567+/-0.0023223683d, e=0.018212, omega=79.9643deg. Closest edge-on conjunction offset0.238592d (0.024427phase), full-SB1-covariance linear propagated Gaia uncertainty0.691423d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.013871d (photometric period projection allowance0.002848d). The approximate central duration4.73h vs observed median2.44h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four measured repeats, with consecutive events in sector62. Deficit0.344pixel from target; nearest capable neighbor2.08pixels from deficit. Sector-dependent depth may reflect aperture dilution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

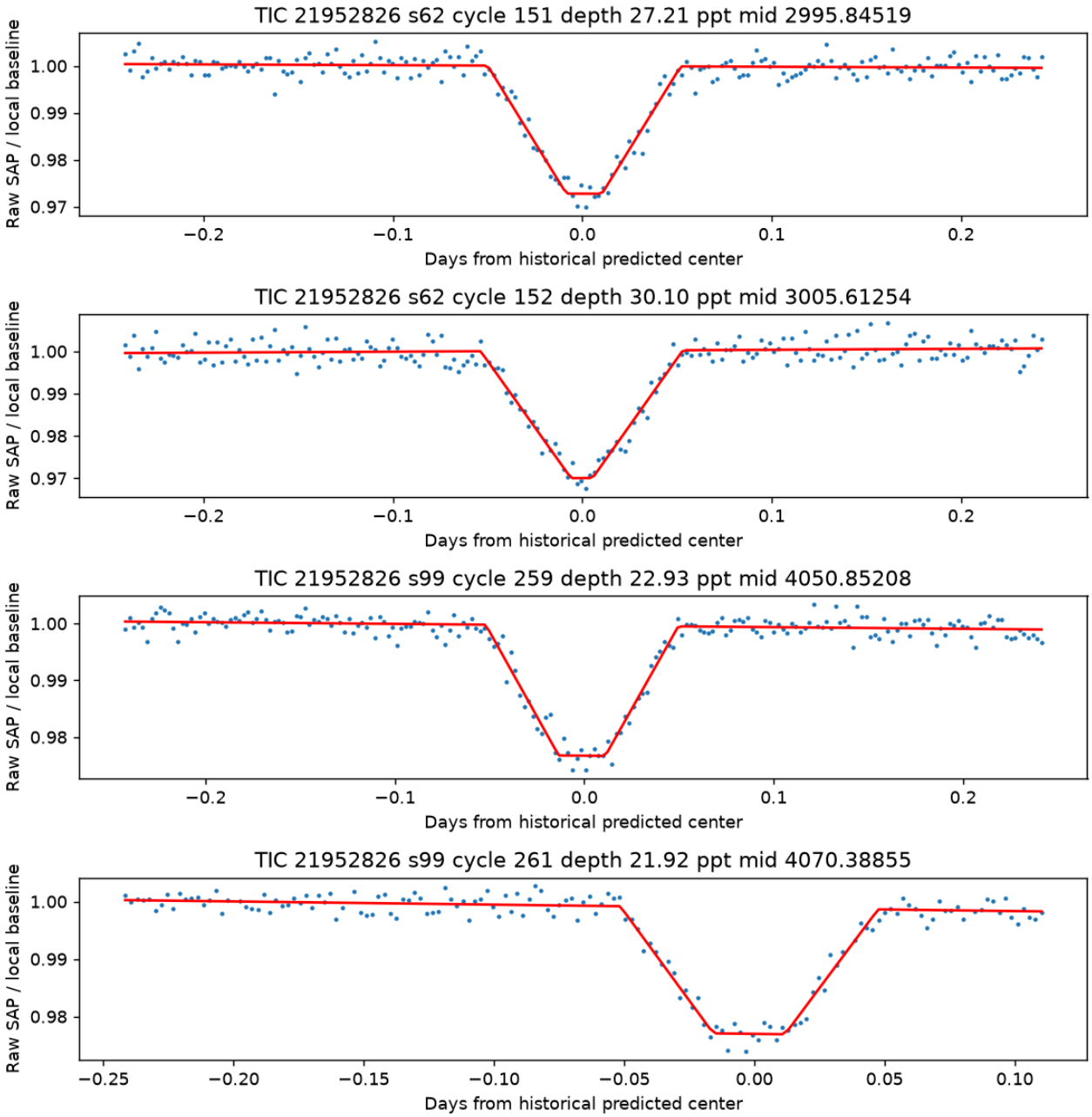
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

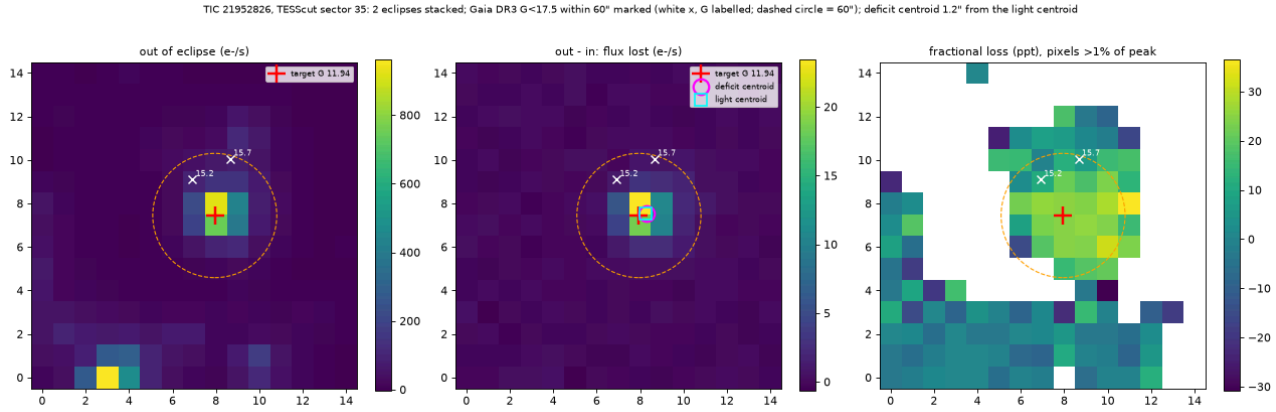
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
62	151	2995.8451919	0.001157	27.208	usable; same-kind selection applies
62	152	3005.6125381	0.001157	30.099	usable; same-kind selection applies
99	259	4050.8520827	0.001157	22.925	usable; same-kind selection applies
99	261	4070.3885538	0.001157	21.919	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s35.json, historical/diff_s35.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R300. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R300; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R300; diff_s35.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R301. TIC 308986181

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 308986181: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[9, 89, 100]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=5.4022635217 +/-0.0000095649d, reference eclipse epoch BTJD1548.8549832, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S89 deficit peak[7, 8] vs target pixel[7, 7]; deficit-target separation3.686arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.246. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths32.580/32.803/31.486ppt, with 4 stacked events. QLP selected-kind fitted median31.450ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3565980650708206720O, P=5.4023202539+/-0.0006108168d, e=0.039226, omega=160.9576deg. Closest edge-on conjunction offset-0.103122d (-0.019088phase), full-SB1-covariance linear propagated Gaia uncertainty0.138927d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.090925d (photometric period projection allowance0.002056d). The approximate central duration5.81h vs observed median4.85h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Eleven clear measured eclipses in three widely separated sectors. No capable neighbor under approximate Gaia flux-budget screen; target deficit0.176pixel from target.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

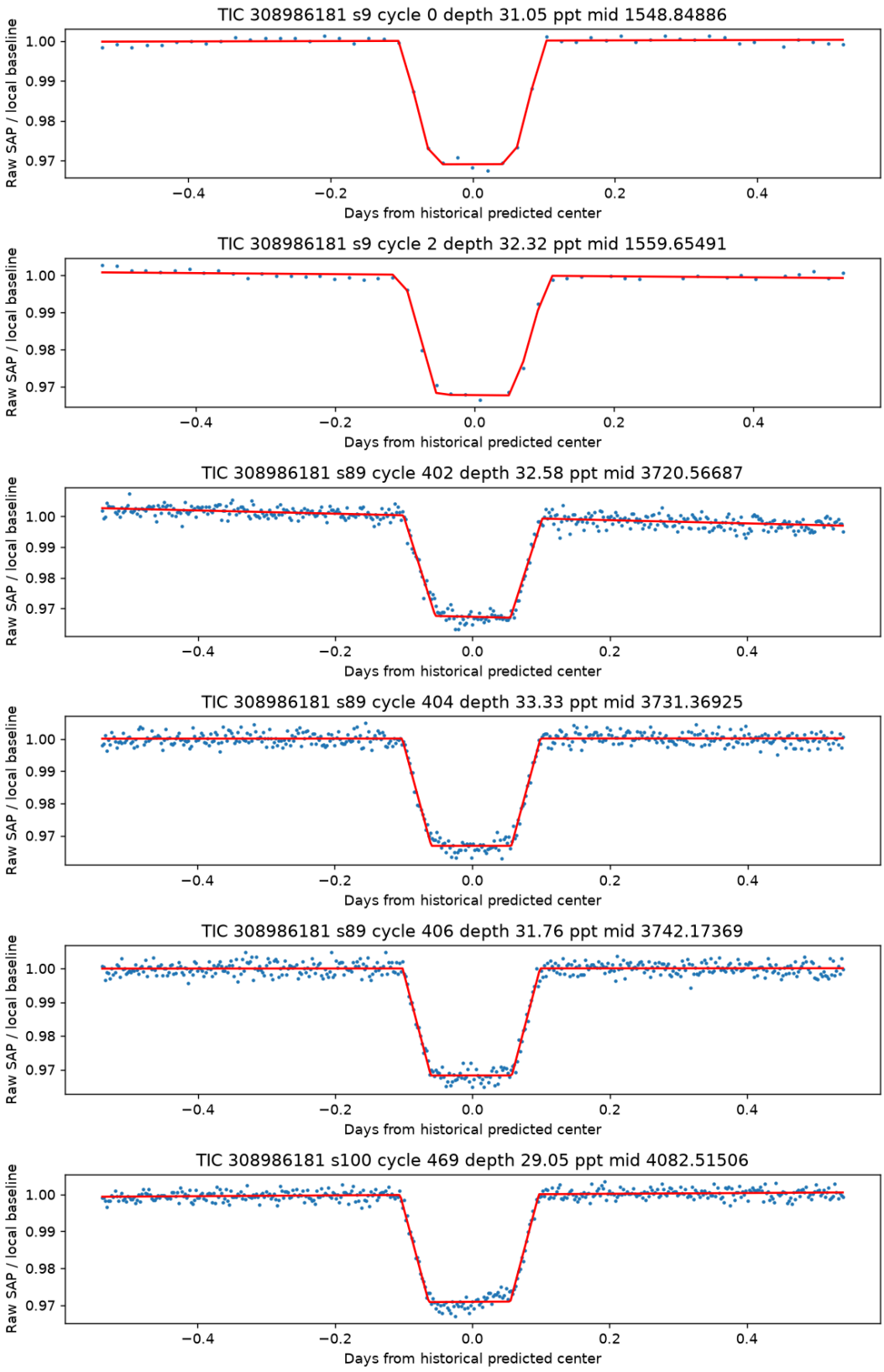
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

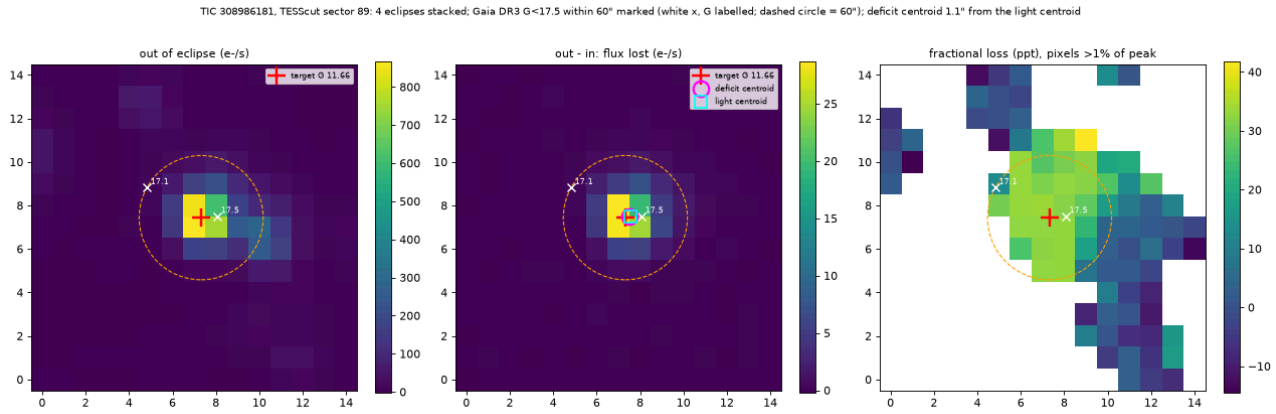
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	0	1548.8488599	0.010417	31.049	usable; same-kind selection applies
9	1	1554.2519120	0.010417	30.490	usable; same-kind selection applies
9	2	1559.6549102	0.010417	32.323	usable; same-kind selection applies
9	3	1565.0562199	0.010417	31.312	usable; same-kind selection applies
89	402	3720.5668740	0.001157	32.581	usable; same-kind selection applies
89	403	3725.9678623	0.001157	33.241	usable; same-kind selection applies
89	404	3731.3692487	0.001157	33.325	usable; same-kind selection applies
89	405	3736.7713877	0.001157	31.450	usable; same-kind selection applies
89	406	3742.1736932	0.001157	31.756	usable; same-kind selection applies
100	468	4077.1142603	0.001157	30.305	usable; same-kind selection applies
100	469	4082.5150569	0.001157	29.049	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s89.json, historical/diff_s89.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R301. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R301; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R301; diff_s89.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R302. TIC 262082690

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 262082690: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 4 distinct eclipse events in selected sectors[99, 102]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=8.9388713010 +/-0.0002620764d, reference eclipse epoch BTJD4046.7449950, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S38 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation0.127arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.770. Capable neighbours: Gaia 5824713631846649600: 15.10 arcsec, G14.737, required fractional eclipse0.238; Gaia 5824713631839658496: 18.02 arcsec, G13.778, required fractional eclipse0.108; Gaia 5824713533092293376: 25.66 arcsec, G12.694, required fractional eclipse0.050; Gaia 5824713498732554368: 50.69 arcsec, G15.928, required fractional eclipse0.679; Gaia 5824713464372814080: 50.83 arcsec, G15.472, required fractional eclipse0.452; Gaia 5824713567452030208: 51.09 arcsec, G13.914, required fractional eclipse0.120; Gaia 5824713842341027456: 51.58 arcsec, G15.649, required fractional eclipse0.529; Gaia 5824713567452030080: 54.31 arcsec, G13.998, required fractional eclipse0.129; Gaia 5824713567452031488: 56.02 arcsec, G14.401, required fractional eclipse0.179
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths14.101/12.255/9.816ppt, with 3 stacked events. QLP selected-kind fitted median16.691ppt. This separately extracted cutout is an alternative reduction of the same TESS

photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3582471353309229552, P=8.9404820160+/-0.0006222623d, e=0.422046, omega=315.8129deg. Closest edge-on conjunction offset-0.440570d (-0.049278phase), full-SB1-covariance linear propagated Gaia uncertainty0.230013d at the measured epoch. Near-Gaia-reference projected eclipse offset0.218213d (photometric period projection allowance0.107189d). The approximate central duration5.73h vs observed median3.25h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four real measured events in sectors99/102; adjacent cycles280/281 test recurrence. Historical sector38 adds independently extracted repeated events. Target deficit0.006pixel from target; nearest capable neighbor0.74pixel from deficit. Crowding and aperture dilution remain, but exact target-centered deficit/core favors target without a calibrated localization error.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

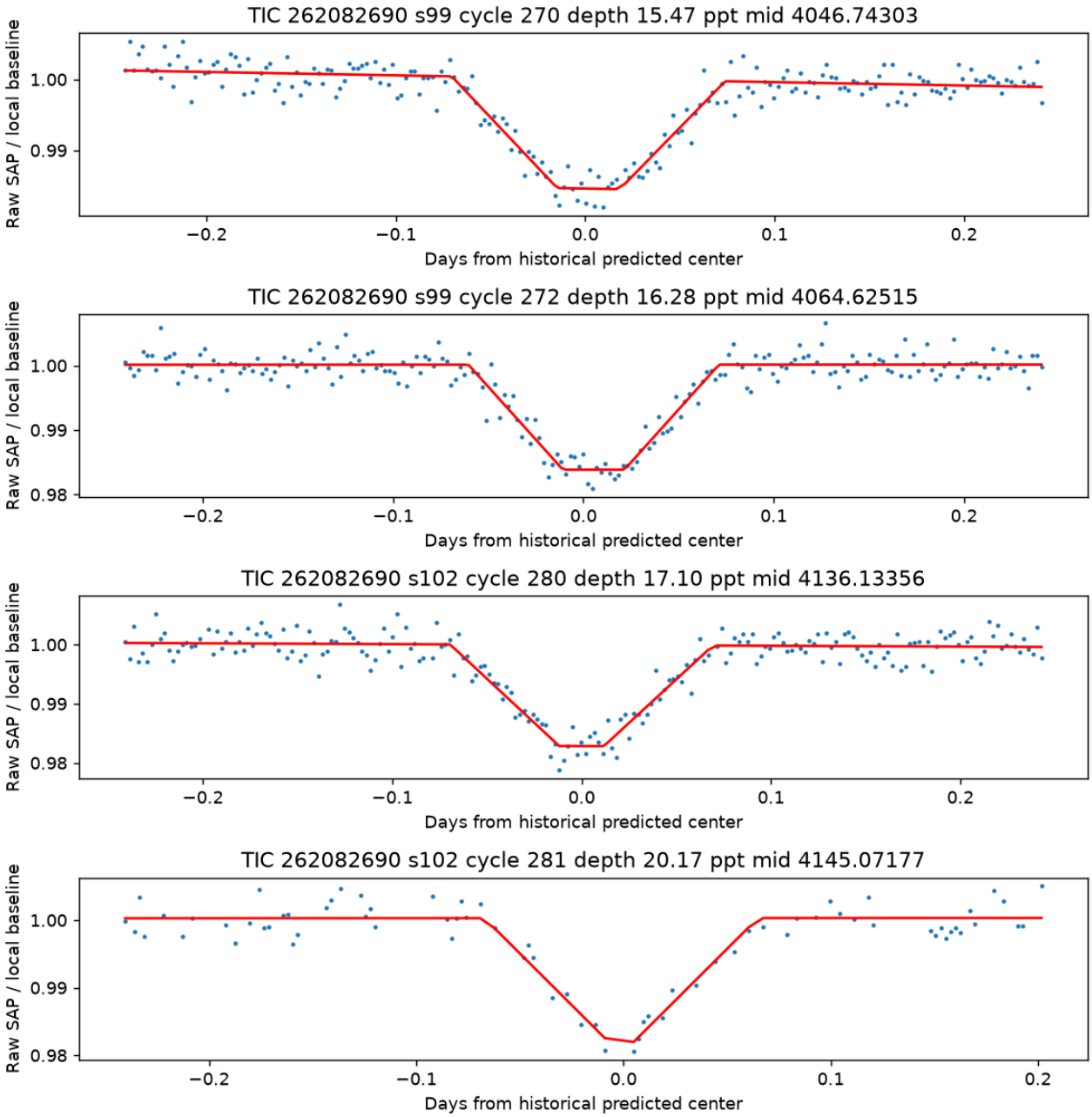
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

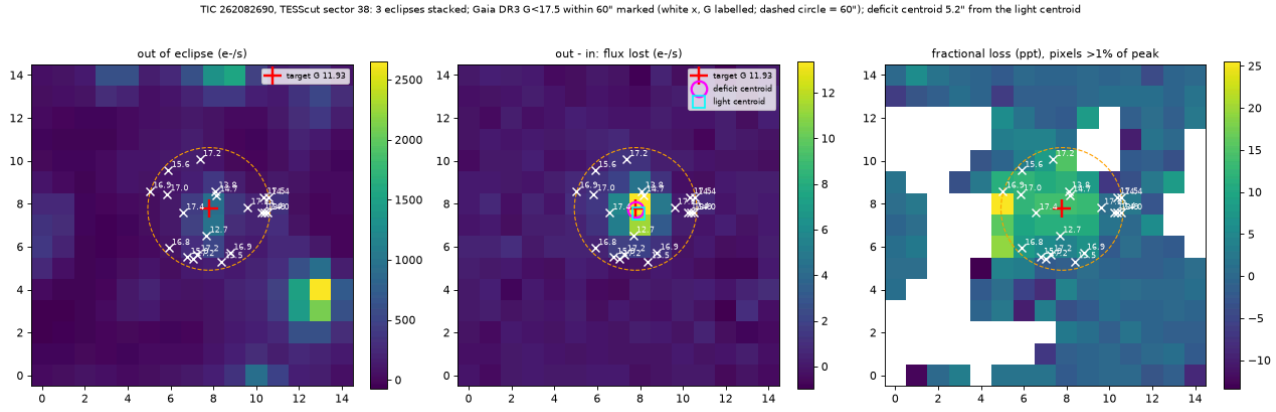
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
99	270	4046.7430330	0.001157	15.474	usable; same-kind selection applies
99	272	4064.6251520	0.001157	16.281	usable; same-kind selection applies
102	280	4136.1335612	0.001157	17.101	usable; same-kind selection applies
102	281	4145.0717693	0.001885	20.165	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R302. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R302; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R302; diff_s38.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R303. TIC 220568520

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 220568520: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[1, 30, 97]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=18.5572206314 +/-0.0000259597d, reference eclipse epoch BTJD1337.7203277, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S2 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.121arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.083. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths19.851/19.780/19.532ppt, with 2 stacked events. QLP selected-kind fitted median19.549ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3472072559677245440, P=18.5622394903+/-0.0104861800d, e=0.056200, omega=167.8786deg. Closest edge-on conjunction offset-0.168961d (-0.009102phase), full-SB1-covariance linear propagated Gaia uncertainty0.673811d at the measured epoch. Near-Gaia-reference projected eclipse offset0.087001d (photometric period projection allowance0.001324d). The approximate central duration5.56h vs observed median4.93h permits noncentral geometry; this radius/minimum-mass estimate is not a

detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated flat-bottomed raw eclipses over sectors1/30/97. Near-gap asymmetric sector97 cycle141 has singular center covariance and is excluded from timing; sector1 cycle0 and clean sector97 cycle142 are retained. No capable neighbor under approximate flux-budget screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

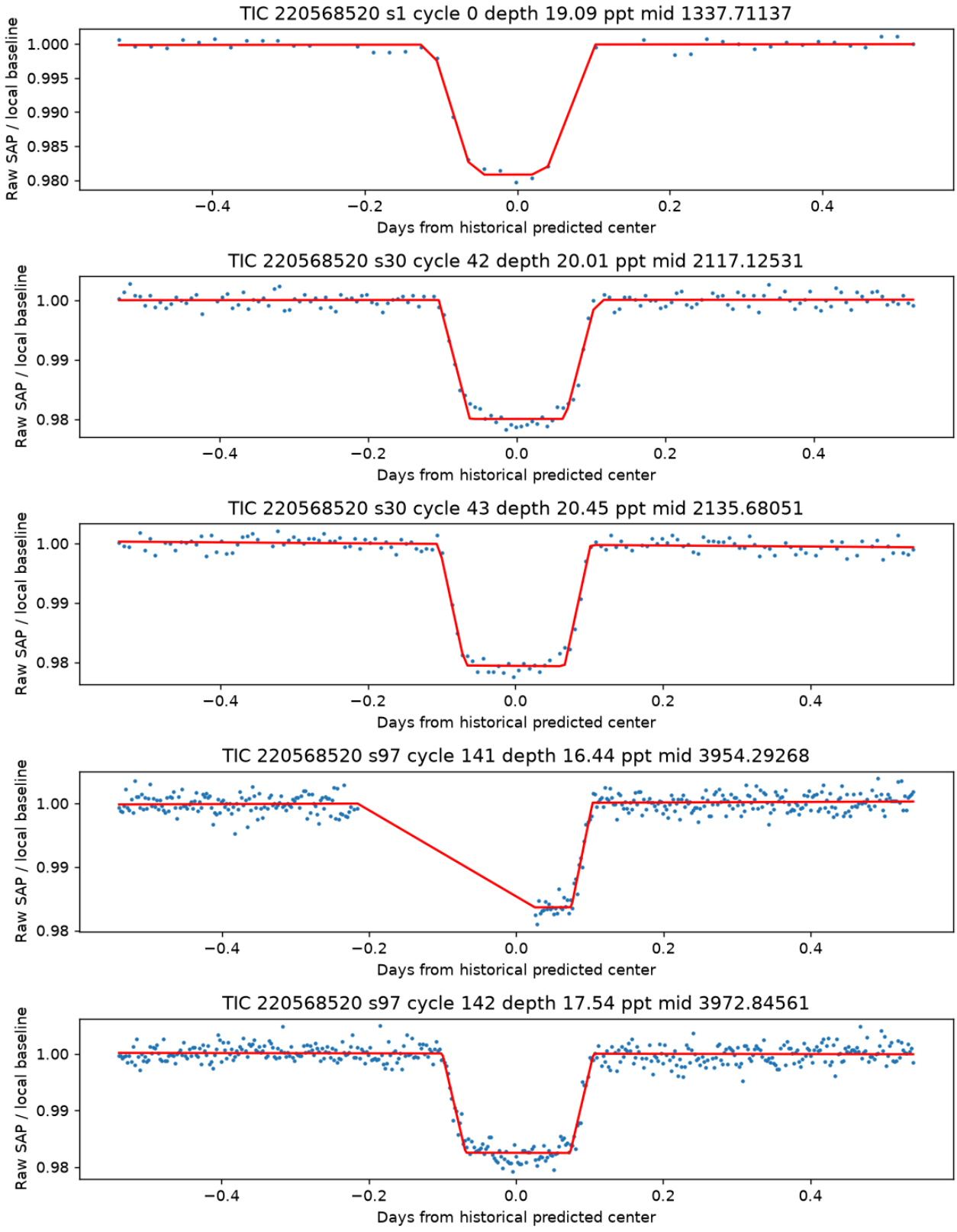
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

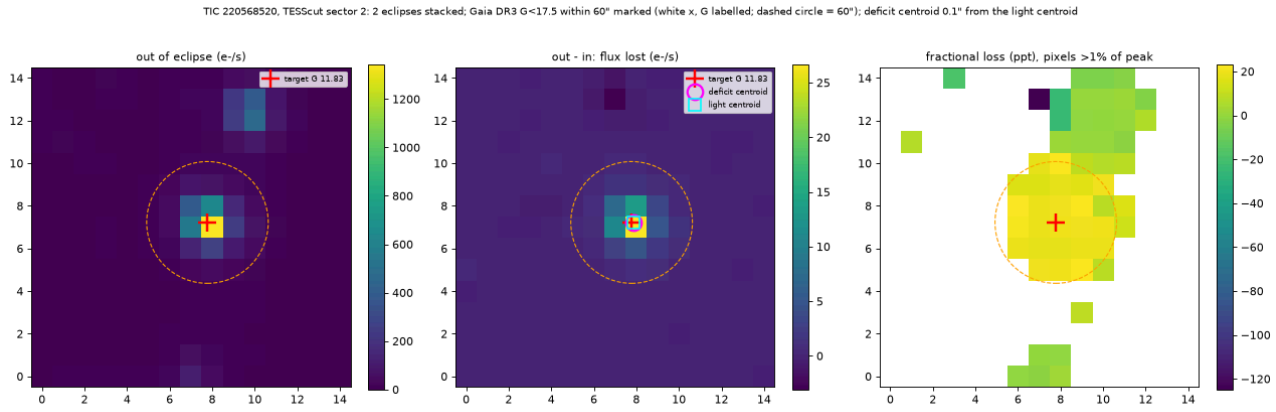
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1337.7113666	0.010417	19.091	usable; same-kind selection applies
30	42	2117.1253104	0.003472	20.007	usable; same-kind selection applies
30	43	2135.6805095	0.003472	20.447	usable; same-kind selection applies
97	141	3954.2926804	73390.275727	16.439	excluded singular timing fit
97	142	3972.8456113	0.001157	17.540	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s02.json, historical/diff_s02.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R303. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R303; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R303; diff_s02.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R304. TIC 276277940

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 276277940: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 9 distinct eclipse events in selected sectors[15, 76, 83]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=6.8645586129 +/-0.0000230538d, reference eclipse epoch BTJD1717.7413603, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[7, 8] vs target pixel[7, 7]; deficit-target separation4.635arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.931. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths109.693/109.506/90.147ppt, with 4 stacked events. QLP selected-kind fitted median150.628ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3216338427408009561 6, P=6.8634769946+/-0.0015596212d, e=0.171838, omega=314.6172deg. Closest edge-on conjunction offset0.141873d (0.020671phase), full-SB1-covariance linear propagated Gaia uncertainty0.270100d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.067961d (photometric period projection allowance0.004472d). The approximate central duration6.68h vs observed median4.75h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep raw eclipses recur across three sectors, with sector-dependent depths130-160ppt. Sector83 cycle271 singular center covariance is excluded from timing. Historical cutout1/3/5 depths109.7/109.5/90.1ppt show aperture dilution; no capable catalogued neighbor under approximate brightness screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

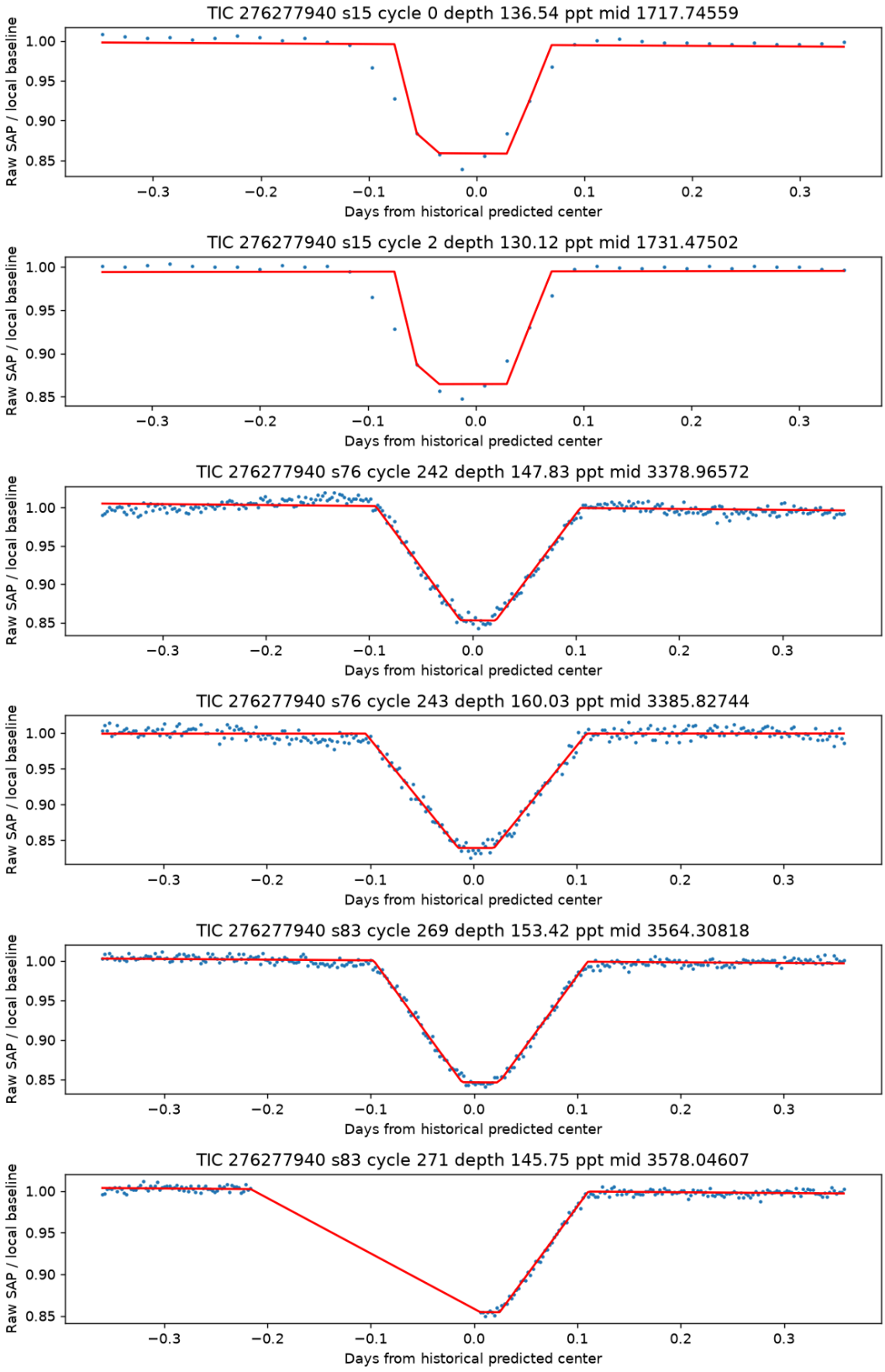
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

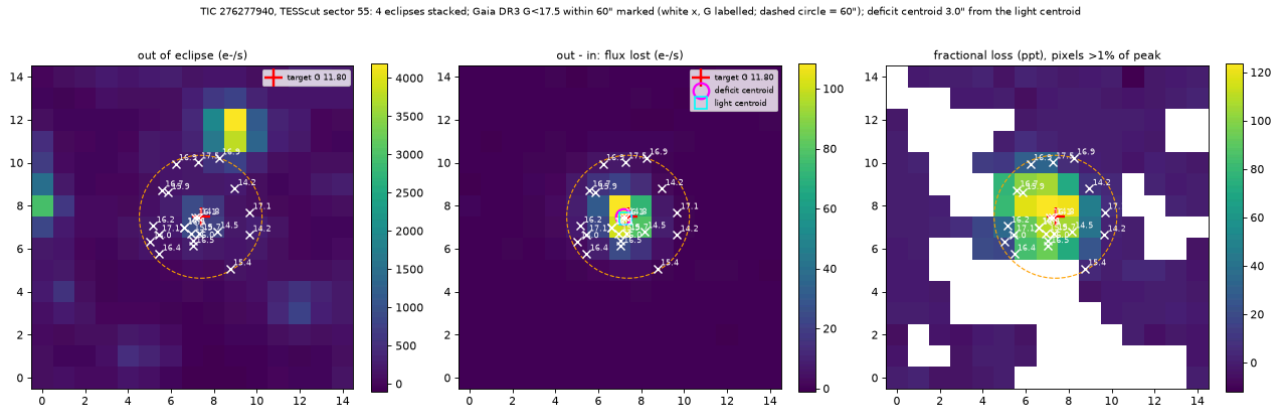
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1717.7455895	0.010417	136.535	usable; same-kind selection applies
15	2	1731.4750235	0.010417	130.121	usable; same-kind selection applies
76	241	3372.1000060	0.001157	146.002	usable; same-kind selection applies
76	242	3378.9657159	0.001157	147.832	usable; same-kind selection applies
76	243	3385.8274449	0.001157	160.030	usable; same-kind selection applies
76	244	3392.6929583	0.001157	156.758	usable; same-kind selection applies
83	269	3564.3081827	0.001157	153.424	usable; same-kind selection applies
83	270	3571.1726924	0.001157	155.728	usable; same-kind selection applies
83	271	3578.0460711	194277.187178	145.755	excluded singular timing fit

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R304. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R304; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R304; diff_s55.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R305. TIC 406863699

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 406863699: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 3 distinct eclipse events in selected sectors[66, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=16.7492445898 +/-0.0000238215d, reference eclipse epoch BTJD3105.4389091, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S65 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.749arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.153. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths32.830/32.671/33.367ppt, with 2 stacked events. QLP selected-kind fitted median34.841ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3577230606293467084 8, P=16.7429244065+/-0.0036265093d, e=0.221229, omega=96.3987deg. Closest edge-on conjunction offset0.881505d (0.052649phase), full-SB1-covariance linear propagated Gaia uncertainty0.520887d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.142364d (photometric period projection allowance0.003859d). The approximate central duration6.57h vs observed median3.05h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Three clear measured eclipses, including consecutive cycles88/89 in sector66 and recurrence in101; historical cutout65 also has two events. No capable neighbor under approximate flux-budget screen. Gaia conjunction offset0.053phase has propagated epoch uncertainty and approximate geometry, not an arbitrary significance veto.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

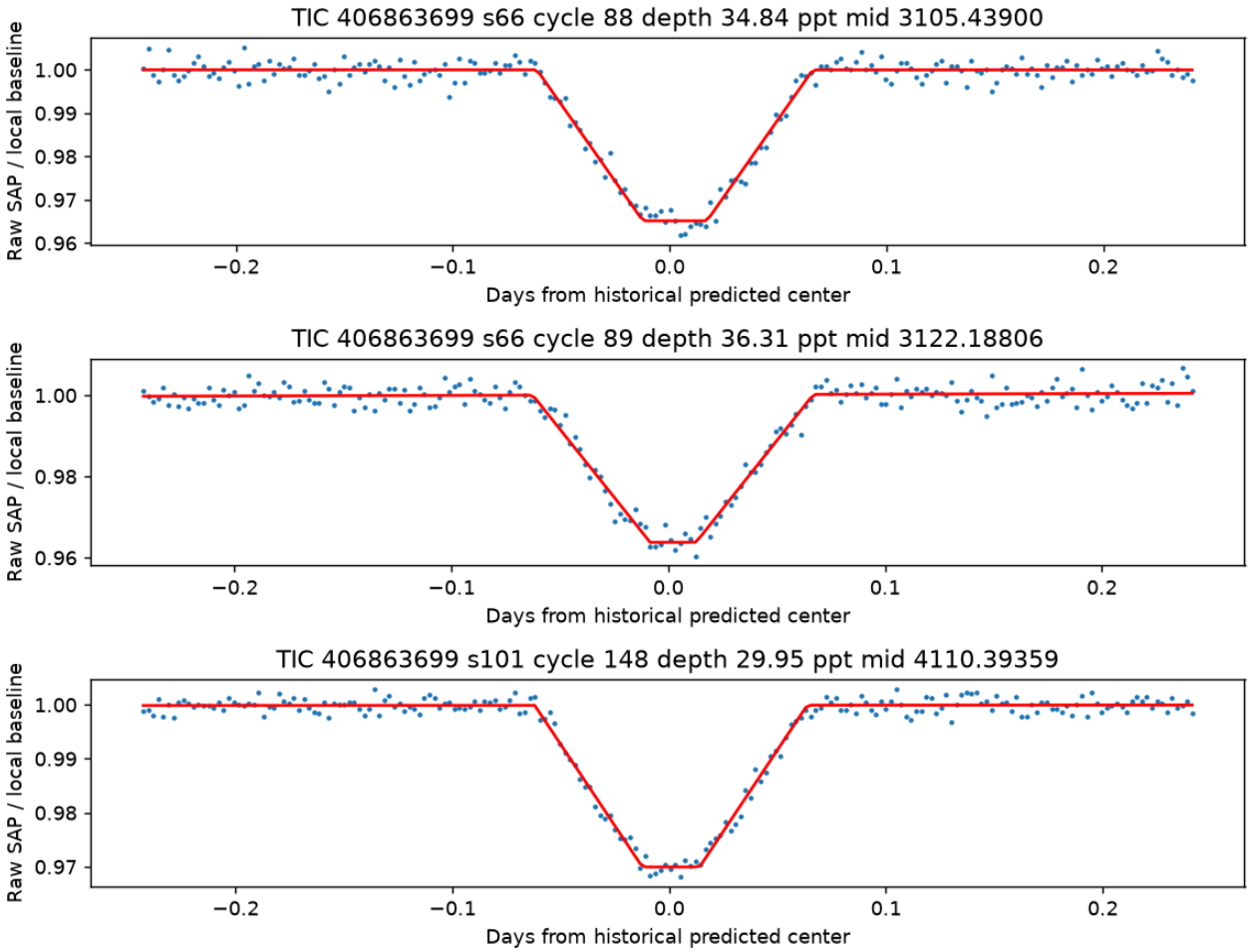
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

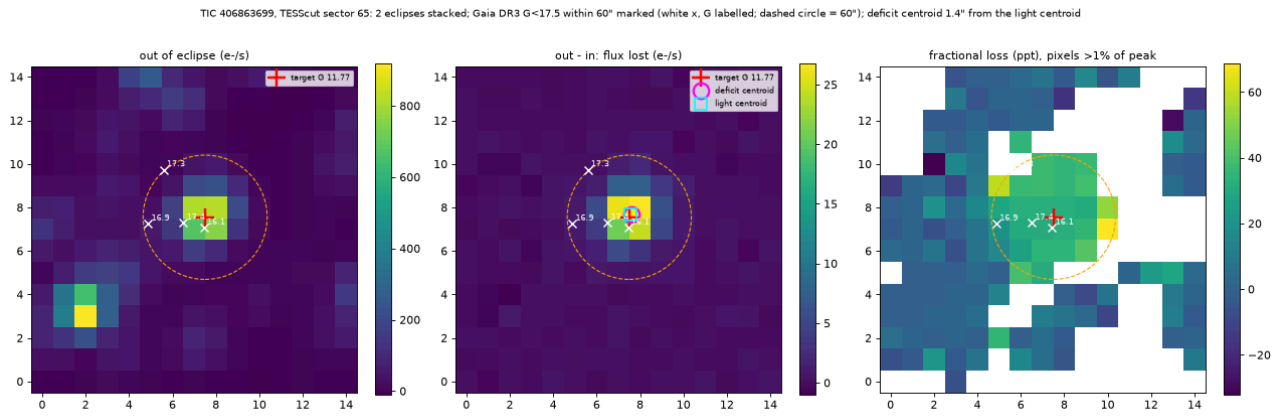
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
66	88	3105.4390017	0.001157	34.841	usable; same-kind selection applies
66	89	3122.1880594	0.001157	36.314	usable; same-kind selection applies
101	148	4110.3935860	0.001157	29.952	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s65.json, historical/diff_s65.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R305. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R305; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R305; diff_s65.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R306. TIC 207148517

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 207148517: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[69, 97]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=7.7713883517 +/-0.0000320555d, reference eclipse epoch BTJD3187.7124533, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[8, 8] vs target pixel[7, 8]; deficit-target separation2.732arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual1.000. Capable neighbours: Gaia 4728689624881108864: 40.99 arcsec, G17.789, required fractional eclipse0.736
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths2.980/3.417/3.404ppt, with 8 stacked events. QLP selected-kind fitted median3.405ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3472868635582532352, P=7.7705559611+/-0.0010075925d, e=0.050853, omega=206.9657deg. Closest edge-on conjunction offset0.315934d (0.040658phase), full-SB1-covariance linear propagated Gaia uncertainty0.422468d at the measured epoch. Near-Gaia-reference projected eclipse offset0.016274d (photometric period projection allowance0.011540d). The approximate central duration5.68h vs observed median1.86h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Shallow repeat signal is measured in sectors69/97; individual fits vary and broad sector69 cycle236 is baseline-sensitive. Median depth is diluted. Historical PSF-scaled residual0.9997 is poor because bright off-center source dominates wide fit; local deficit core0.130pixel from target versus capable neighbor1.84pixels away supports target. Neighbor intrinsic requirement0.736 is not compared directly with its blended pixel fractional loss. Precision timing/shape is limited by low signal and baseline model.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

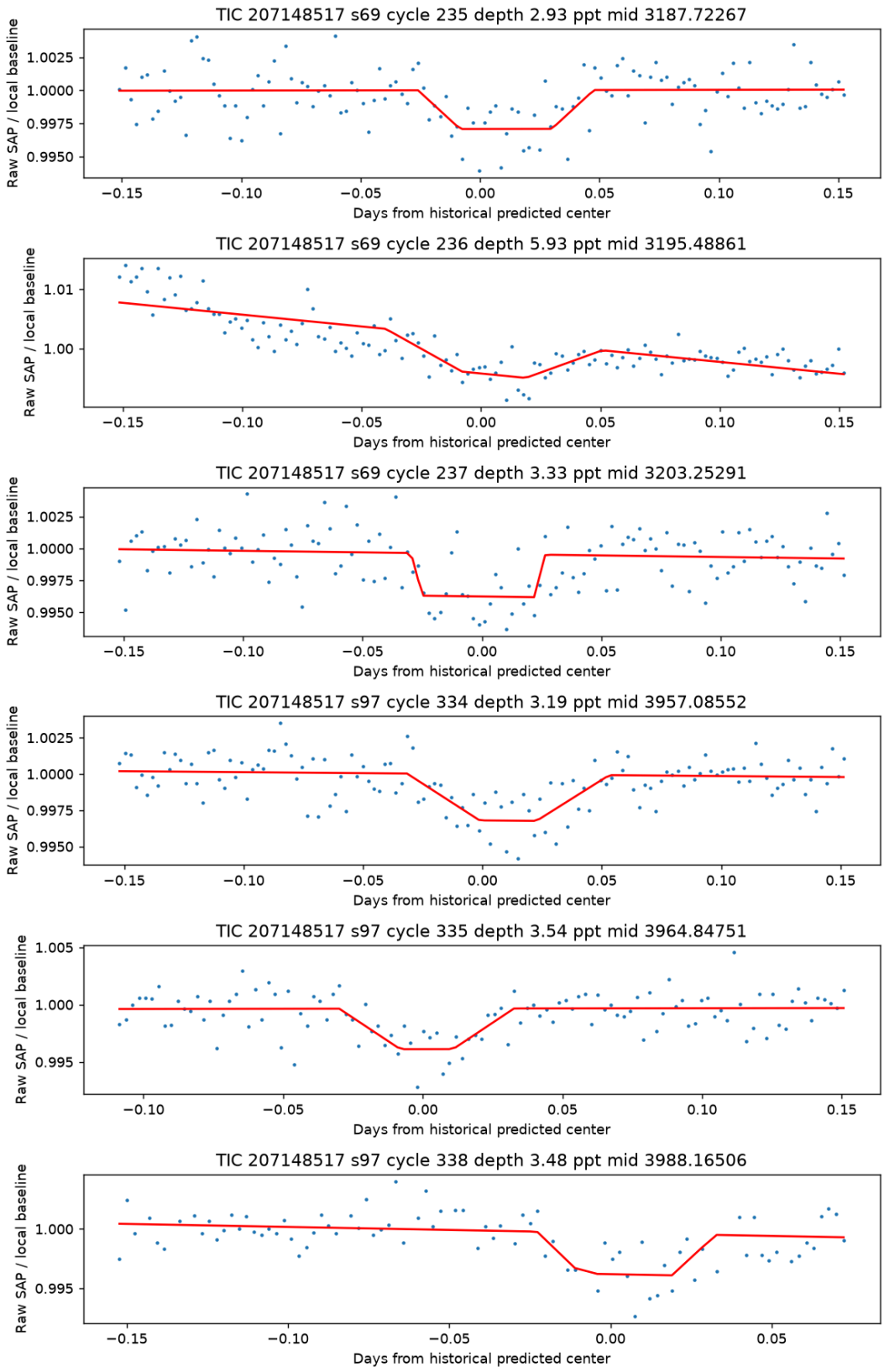
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

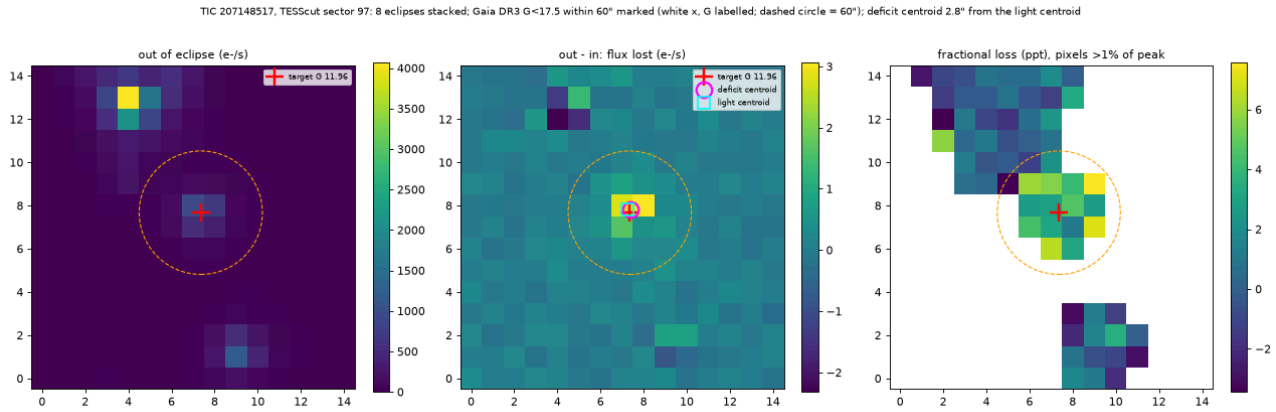
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
69	235	3187.7226688	0.002686	2.930	usable; same-kind selection applies
69	236	3195.4886084	0.004264	5.930	usable; same-kind selection applies
69	237	3203.2529147	0.001157	3.334	usable; same-kind selection applies
97	332	3941.5374232	0.001836	5.363	usable; same-kind selection applies
97	334	3957.0855244	0.002183	3.195	usable; same-kind selection applies
97	335	3964.8475080	0.001907	3.537	usable; same-kind selection applies
97	336	3972.6218020	0.003256	3.247	usable; same-kind selection applies
97	338	3988.1650616	0.002008	3.476	usable; same-kind selection applies

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R306. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R306; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R306; diff_s97.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R307. TIC 169657557

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 169657557: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 29 distinct eclipse events in selected sectors[61, 88]. The raw-event plot and numeric raw-windows.csv were inspected. The raw-windows.csv retains the initial broad linear model; deciding narrow-core fits and timings are in quadratic-core.json/png and measurements.json. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a quadratic baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 29 same-kind midtimes give P=1.4973843851 +/-0.0000030799d, reference eclipse epoch BTJD2965.1431096, kind: Narrow eclipse core atop broad phase modulation; physical primary identity unestablished. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S88 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation2.086arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.505. Capable neighbours: Gaia 3036481615590722176: 29.04 arcsec, G12.512, required fractional eclipse0.037; Gaia 3036481615590722304: 30.71 arcsec, G16.598, required fractional eclipse0.957; Gaia 3036528443615800960: 49.79 arcsec, G14.030, required fractional eclipse0.104; Gaia 3036481512511513216: 55.32 arcsec, G15.668, required fractional eclipse0.415; Gaia 3036481615590721408: 55.33 arcsec, G16.556, required fractional eclipse0.920; Gaia 3036481478151775744: 56.40 arcsec, G13.060, required fractional eclipse0.051; Gaia 3036481443792033920: 57.17 arcsec, G14.265, required fractional eclipse0.125; Gaia 3036481409432298368: 57.87 arcsec, G16.577, required fractional eclipse0.938
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths17.702/13.538/9.801ppt, with 15 stacked events (quality-zero data). QLP selected-kind fitted median16.335ppt. This separately extracted cutout is an alternative reduction of

the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3303648151251150963 2, P=1.4974373957+/-0.0000740271d, e=0.111978, omega=21.7640deg. Closest edge-on conjunction offset-0.042512d (-0.028390phase), full-SB1-covariance linear propagated Gaia uncertainty0.137602d at the measured epoch. Near-Gaia-reference projected eclipse offset0.048667d (photometric period projection allowance0.005297d). The approximate central duration4.45h vs observed median2.41h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Broad phase modulation surrounds a sharper eclipse core. Quadratic-baseline narrow-core fit replaces the broad linear-baseline timing/depth model; core depth16.34ppt and duration2.41h are model-dependent. Earlier broad-model period agrees within formal errors. Target deficit0.099pixel, closest capable neighbor1.41pixels; all capable sources checked geometrically. No calibrated centroid uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

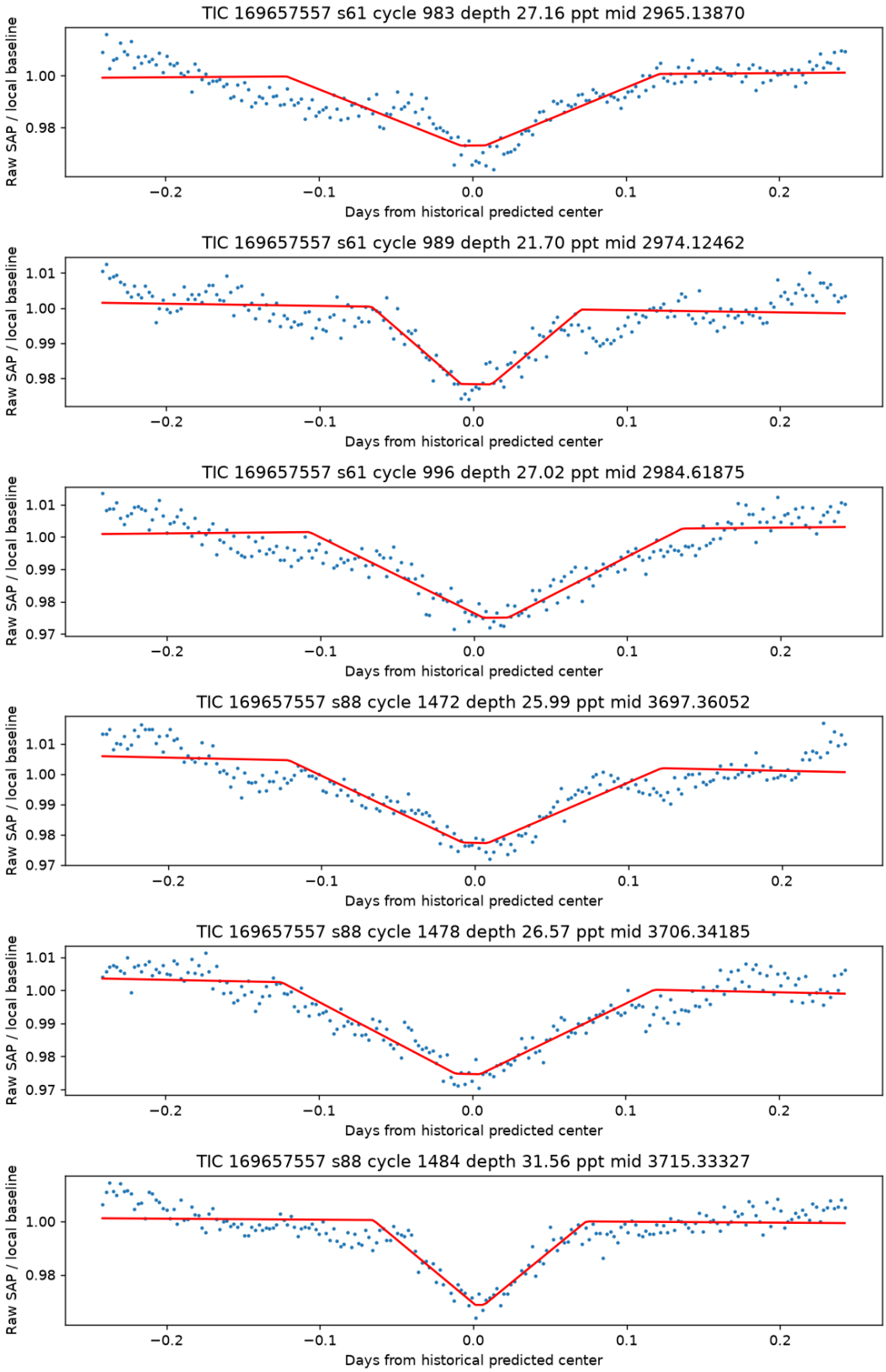
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
61	983	2965.1476881	0.002419	18.733	selected
61	984	2966.6365462	0.002810	14.874	selected
61	985	2968.1431215	0.002184	15.142	selected
61	987	2971.1281557	0.003943	18.583	selected
61	988	2972.6309835	0.002054	16.335	selected
61	989	2974.1244091	0.002184	16.403	selected
61	991	2977.1249005	0.004182	16.513	selected
61	992	2978.6114077	0.004062	14.097	selected
61	993	2980.1139928	0.003655	16.080	selected
61	994	2981.6145236	0.003681	18.270	selected
61	995	2983.1078798	0.003846	16.361	selected
61	996	2984.6198193	0.004447	14.869	selected
61	997	2986.1067020	0.001876	20.601	selected
61	998	2987.5907552	0.005203	9.619	selected
88	1470	3694.3723254	0.003593	12.287	selected
88	1471	3695.8593781	0.003299	13.535	selected
88	1472	3697.3658519	0.002383	15.136	selected
88	1473	3698.8589370	0.002788	17.388	selected
88	1474	3700.3562855	0.002571	15.531	selected
88	1475	3701.8600297	0.003675	14.452	selected
88	1476	3703.3513601	0.001696	19.675	selected
88	1477	3704.8579194	0.003987	13.193	selected
88	1478	3706.3428245	0.004063	17.047	selected
88	1479	3707.8463010	0.001981	18.762	selected
88	1480	3709.3408970	0.004171	15.776	selected
88	1481	3710.8508522	0.003222	16.539	selected

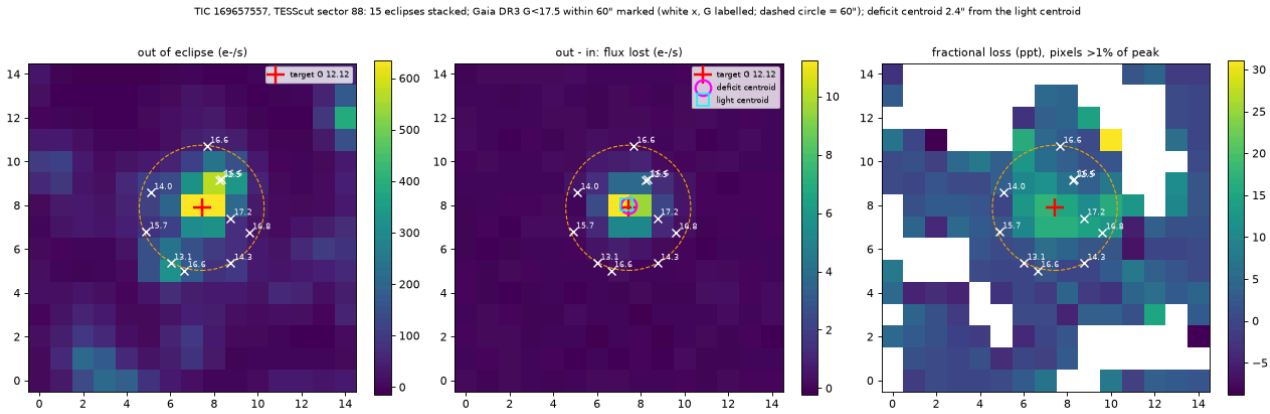
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
88	1482	3712.3382359	0.002425	20.943	selected
88	1483	3713.8337482	0.004714	14.454	selected
88	1484	3715.3332414	0.001475	22.483	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; quadratic-core.json/png and broad-linear-model.json; shared-qlp QLP FITS; historical/diff_s88.json, historical/diff_s88.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R307. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R307; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R307; diff_s88.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R308. TIC 363534683

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 363534683: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[9, 36, 90]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 10 same-kind midtimes give P=5.2864321424 +/-0.0000069133d, reference eclipse epoch BTJD1544.7583809, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S64 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.618arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.574. Capable neighbours: Gaia 5249644487609641344: 35.96 arcsec, G14.329, required fractional eclipse0.979
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths141.125/129.722/106.222ppt, with 5 stacked events (quality-zero data). QLP selected-kind fitted median144.520ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35249644448937154816, P=5.2864401901+/-0.0005018875d, e=0.048633, omega=239.4062deg. Closest edge-on conjunction offset-0.037278d (-0.007052phase), full-SB1-covariance linear propagated Gaia uncertainty0.097256d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.035516d (photometric period projection allowance0.001514d). The approximate central duration5.35h vs observed median4.27h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Ten raw eclipses across three sectors include adjacent repeats. Target deficit0.125pixel versus capable neighbor1.78pixels. Sector depths vary through aperture dilution; no claim of intrinsic constancy.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

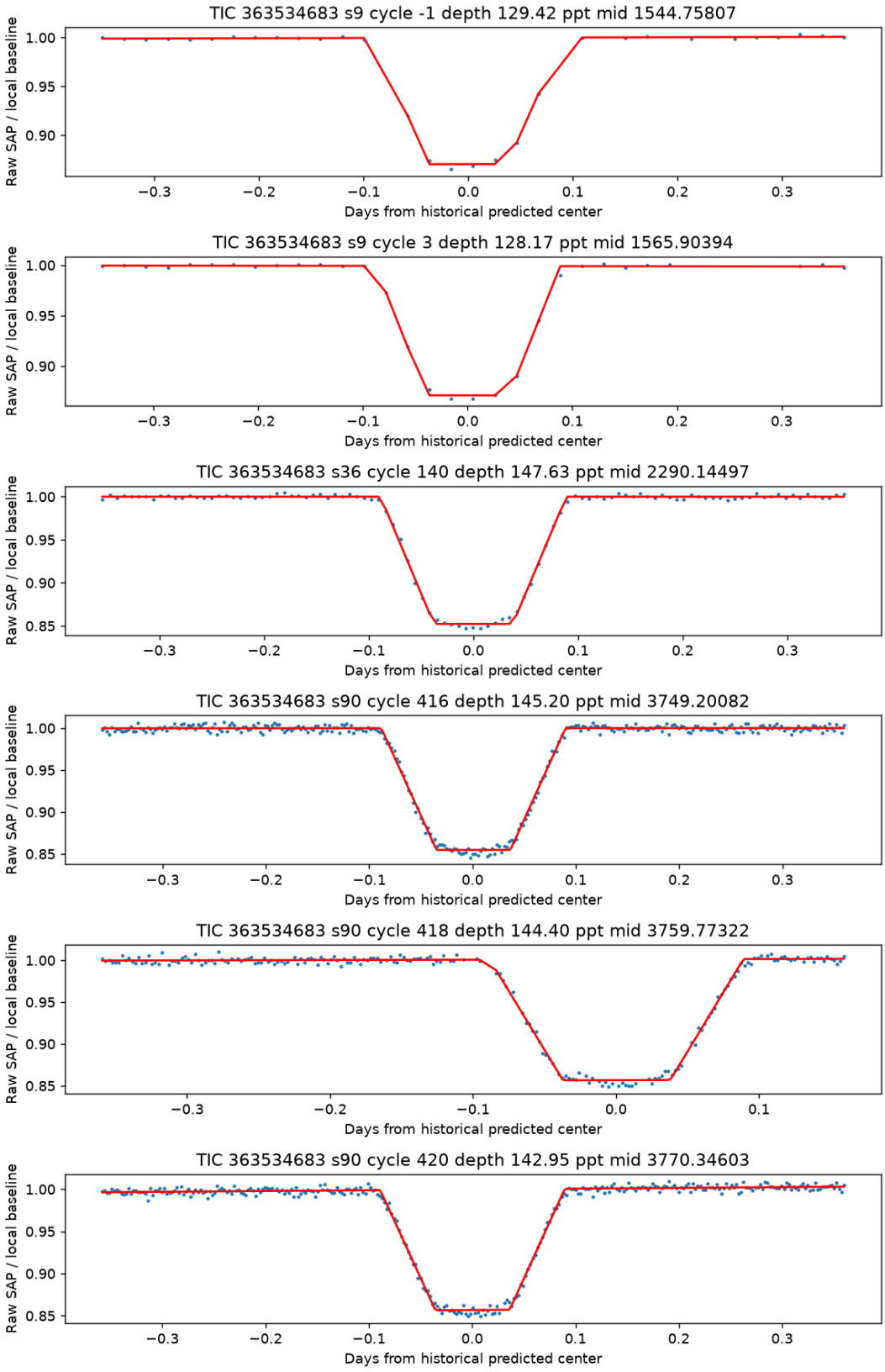
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

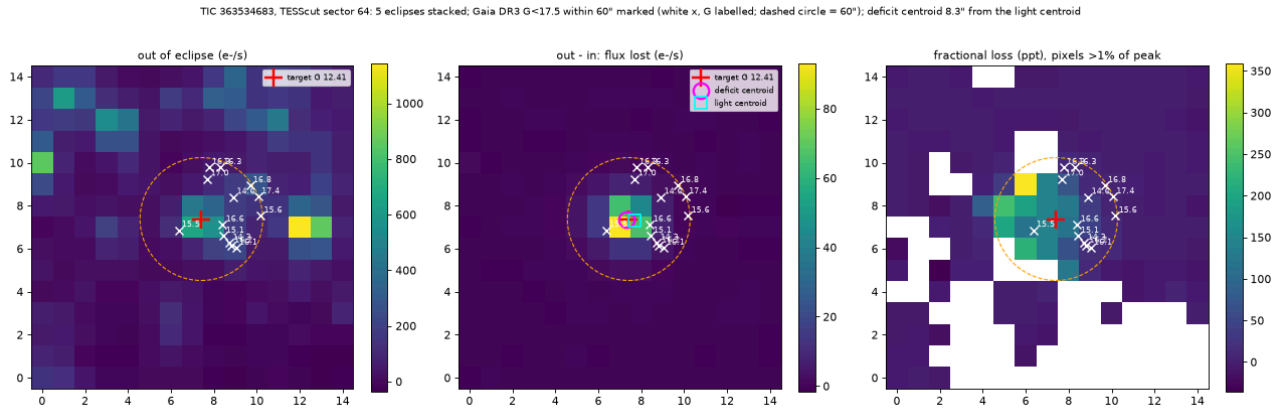
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	-1	1544.7580741	0.010417	129.425	selected
9	3	1565.9039421	0.010417	128.174	selected
36	139	2284.8591393	0.003472	148.482	selected
36	140	2290.1449691	0.003472	147.631	selected
36	142	2300.7183070	0.003472	143.991	selected
90	416	3749.2008153	0.001157	145.199	selected
90	417	3754.4871767	0.001157	144.947	selected
90	418	3759.7732248	0.001157	144.401	selected
90	419	3765.0599998	0.001157	144.640	selected
90	420	3770.3460272	0.001157	142.955	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s64.json, historical/diff_s64.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R308. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R308; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R308; diff_s64.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R309. TIC 21295315

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 21295315: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[20, 47, 60]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.6068097551 +/-0.0000652497d, reference eclipse epoch BTJD1861.1294409, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S47 deficit peak[7, 7] vs target pixel[8, 7]; deficit-target separation0.471arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.288. Capable neighbours: Gaia 895666127653255552: 58.00 arcsec, G13.835, required fractional eclipse0.428
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths84.424/81.615/70.247ppt, with 3 stacked events (quality-zero data). QLP selected-kind fitted median87.584ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3895666402531207936, P=11.6050108508+/-0.0044449065d, e=0.299861, omega=90.5800deg. Closest edge-on conjunction offset0.251296d (0.021654phase), full-SB1-covariance linear propagated Gaia uncertainty0.732493d at the measured epoch. Near-Gaia-reference projected eclipse offset0.022835d (photometric period projection allowance0.008287d). The approximate central duration3.85h vs observed median3.24h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five raw eclipses include consecutive cycles63/64/65 in sector47, testing recurrence. Target deficit0.022pixel versus capable neighbor2.73pixels. Gaia conjunction linear uncertainty0.73d limits phase inference.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

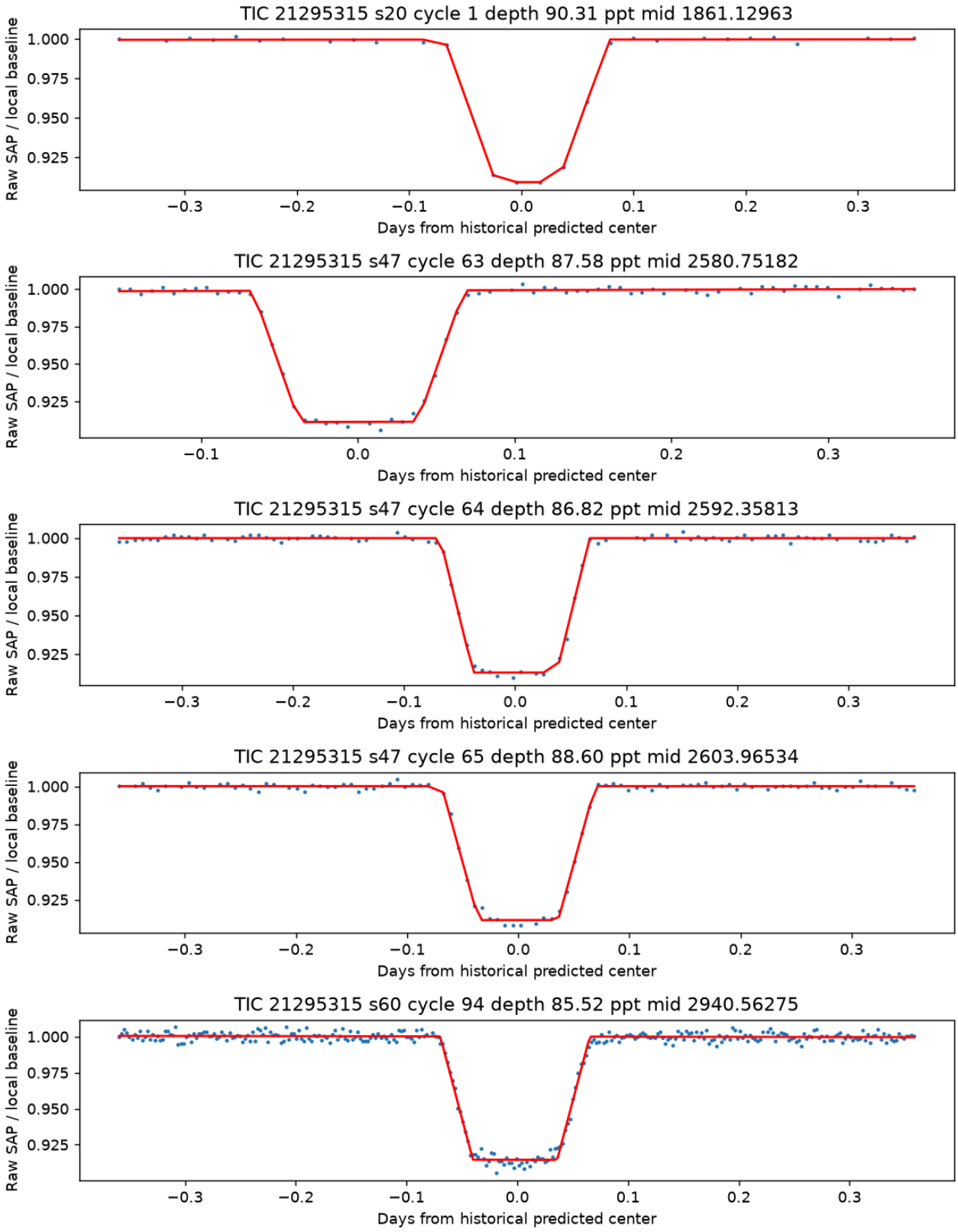
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

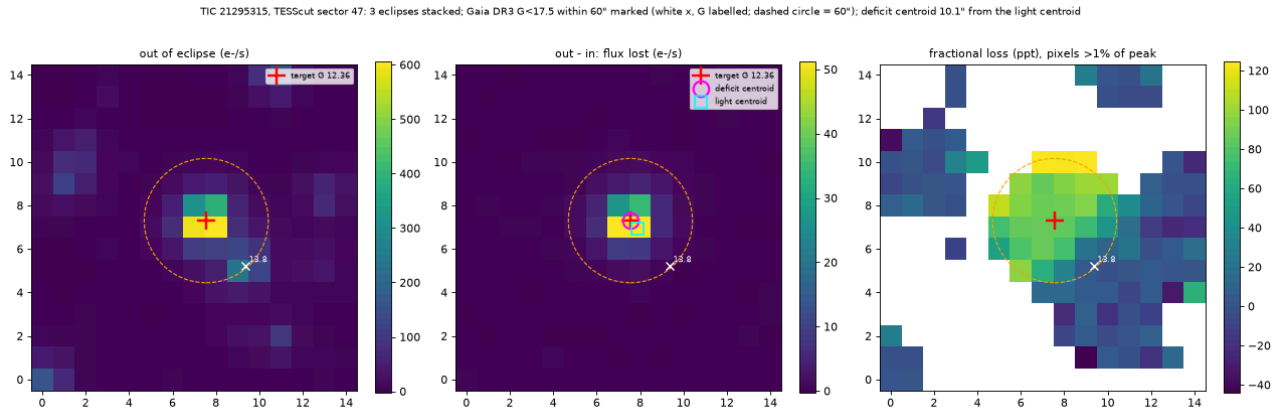
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
20	1	1861.1296303	0.010417	90.311	selected
47	63	2580.7518230	0.003472	87.584	selected
47	64	2592.3581302	0.003472	86.818	selected
47	65	2603.9653446	0.003472	88.603	selected
60	94	2940.5627534	0.001157	85.522	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s47.json, historical/diff_s47.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R309. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R309; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R309; diff_s47.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R310. TIC 445438807

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 445438807: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 12 distinct eclipse events in selected sectors[18, 58, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 12 same-kind midtimes give P=3.6681590097 +/-0.0000037951d, reference eclipse epoch BTJD1795.6720263, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S58 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation3.948arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.712. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths84.862/73.669/64.838ppt, with 5 stacked events (quality-zero data). QLP selected-kind fitted median92.316ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3505891592075928320, P=3.6680210933+/-0.0002846168d, e=0.007003, omega=146.5116deg. Closest edge-on conjunction offset0.043915d (0.011973phase), full-SB1-covariance linear propagated Gaia uncertainty0.122864d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.008907d (photometric period projection allowance0.001454d). The approximate central duration3.83h vs observed median3.18h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Twelve quality-zero measured raw events over three sectors test recurrence. No capable neighbor within60arcsec under approximate Gaia flux budget; target deficit0.188pixel. Finite-cadence trapezoid and aperture dilution limit precision.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

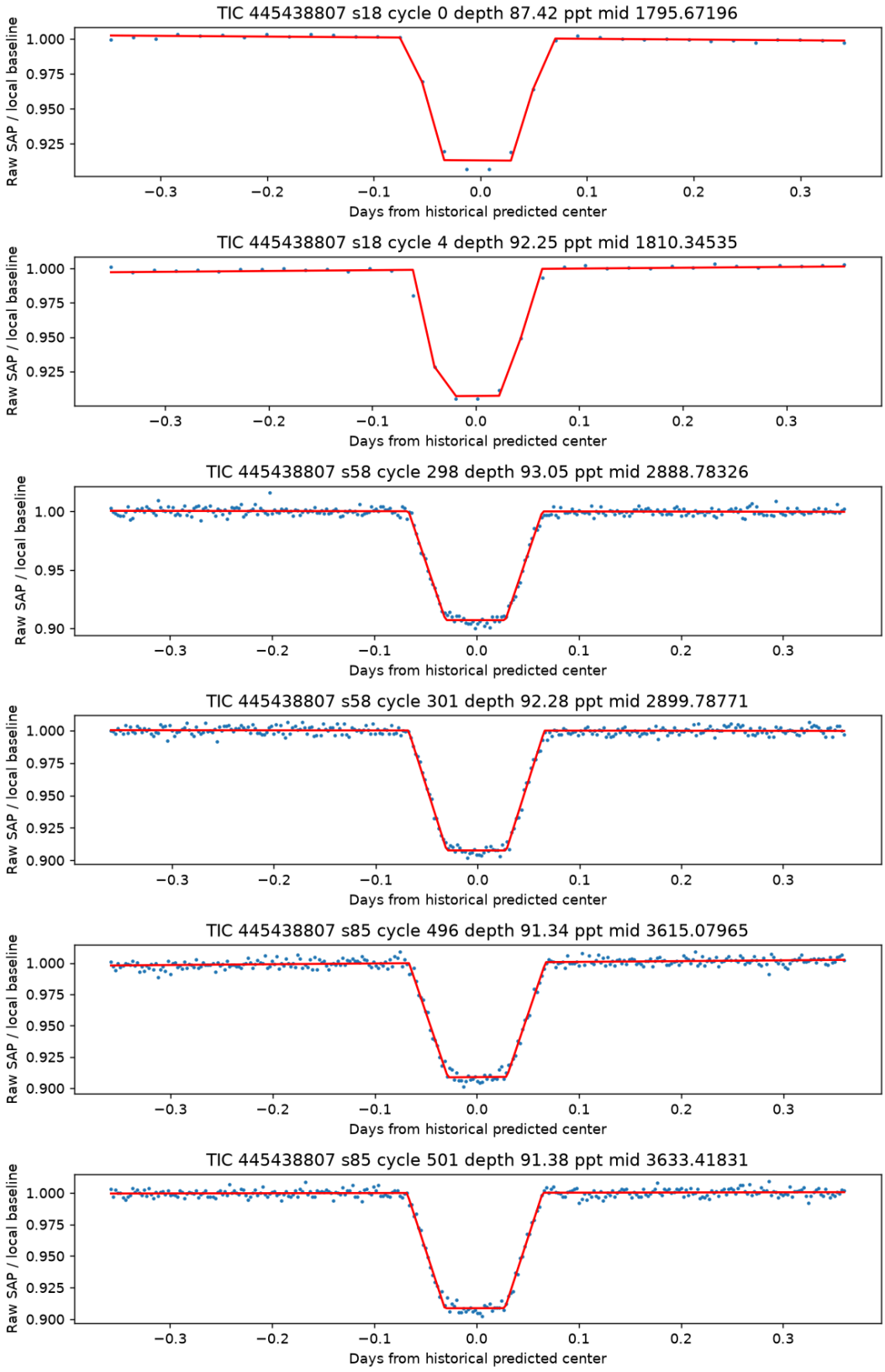
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

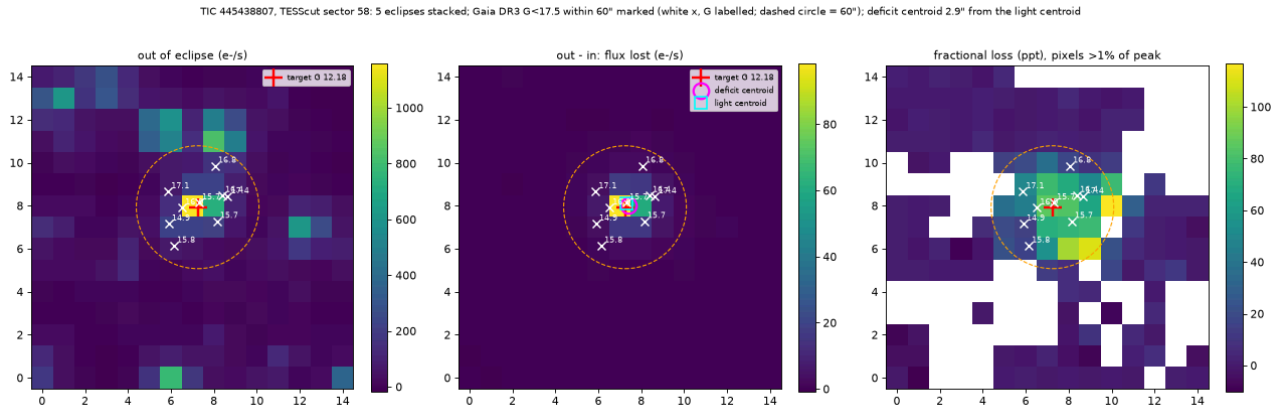
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
18	0	1795.6719555	0.010417	87.415	selected
18	1	1799.3391082	0.010417	88.036	selected
18	4	1810.3453508	0.010417	92.246	selected
58	297	2885.1151565	0.001157	93.175	selected
58	298	2888.7832647	0.001157	93.054	selected
58	299	2892.4516689	0.001157	92.363	selected
58	301	2899.7877145	0.001157	92.278	selected
58	303	2907.1245149	0.001157	92.354	selected
85	496	3615.0796458	0.001157	91.341	selected
85	497	3618.7471859	0.001157	92.723	selected
85	498	3622.4157207	0.001157	92.426	selected
85	501	3633.4183143	0.001157	91.382	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s58.json, historical/diff_s58.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R310. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R310; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R310; diff_s58.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R311. TIC 394386248

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 394386248: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 2 distinct eclipse events in selected sectors[81]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 2 same-kind midtimes give P=13.5773902434 +/-0.0016368077d, reference eclipse epoch BTJD3509.3712123, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S40 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.197arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.657. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths81.223/81.035/77.614ppt, with 2 stacked events (flag2048 cores). QLP selected-kind fitted median77.740ppt. Sector40 is a separate TESS observing occasion from the measured sector81 QLP events; its custom extraction is not an independent telescope. Alternative reductions of any one exposure share photons. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3204583764630507 5072, P=13.5745185100+/-0.0023485600d, e=0.333088, omega=30.2864deg. Closest edge-on conjunction offset0.613972d (0.045230phase), full-SB1-covariance linear propagated Gaia uncertainty0.499331d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.046526d (photometric period projection allowance0.376466d). The approximate central duration6.18h vs observed median4.46h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Two consecutive quality-zero sector81 eclipses measure only a13.58d timing baseline, with formal/cadence period error0.00164d; no long-baseline precision asserted. Custom TESScut sector40 contains two further13.5769d-spaced raw repeats, all cores flag2048; these are flagged supportive recurrence, not clean nulls or precise timing points. Flagged difference image target offset0.15pixel with no capable neighbor under approximate flux screen. Flagged-image systematic and 9x9 residual0.657 limit localization; raw sector81 eclipses are unflagged.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

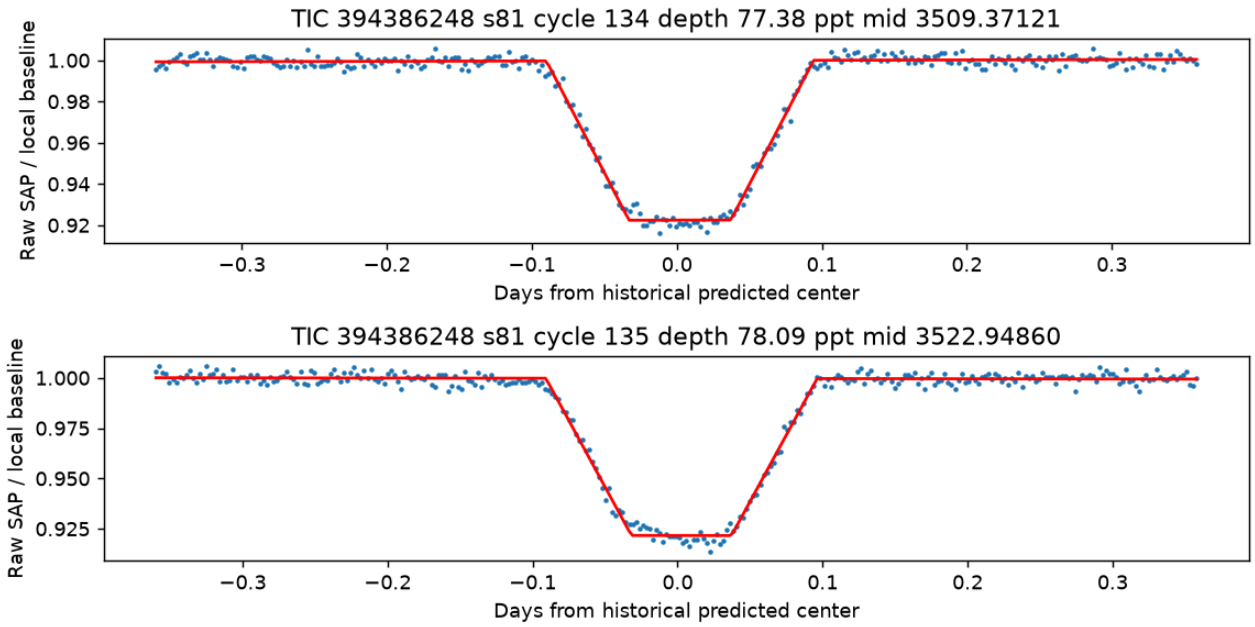
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

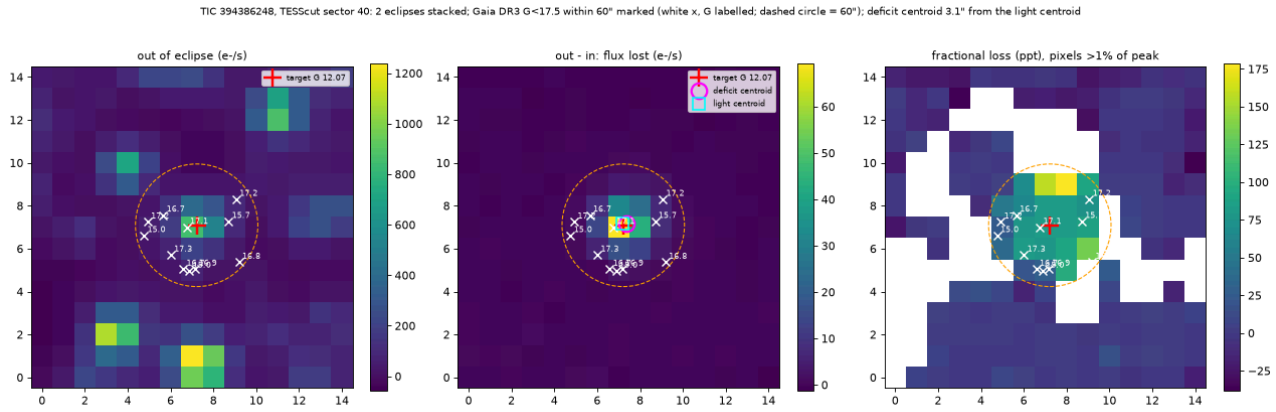
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
81	134	3509.3712123	0.001157	77.385	selected
81	135	3522.9486025	0.001157	78.095	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s40_flag2048.json, historical/diff_s40_flag2048.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R311. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R311; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R311; diff_s40_flag2048.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R312. TIC 403154475

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 403154475: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 15 distinct eclipse events in selected sectors[102, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=5.0696716413 +/-0.0001354749d, reference eclipse epoch BTJD4132.0979108, kind: Even historical half-period eclipse parity; slightly deeper by sample median, physical primary unestablished. No center-covariance fits were excluded. Odd historical half-period cycles are omitted by event kind, not failed timing covariance. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S95 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.031arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.332. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths7.303/7.076/7.059ppt, with 10 stacked events (quality-zero data). QLP selected-kind fitted median7.954ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3650830636904073292 8, P=5.0699604109+/-0.0003054710d, e=0.011145, omega=306.9271deg. Closest edge-on conjunction offset0.021105d (0.004163phase), full-SB1-covariance linear propagated Gaia uncertainty0.221143d at the measured epoch. Near-Gaia-reference projected eclipse offset0.234217d (photometric period projection allowance0.099981d). The approximate central duration6.00h vs observed median1.96h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Raw dips recur near2.535d, but Gaia orbit is5.06996d. Fit uses even half-period cycles1106,1108,1110,1112,1126,1128 as same-kind events; even median7.954ppt versus odd7.308ppt is not decisive physical-primary identification. Odd-only independent period5.0695421d agrees. Orbital5.0696716d interpretation uses same-kind parity and Gaia geometry; half-period dip recurrence is not silently called orbital period. No capable neighbor under flux-budget screen.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

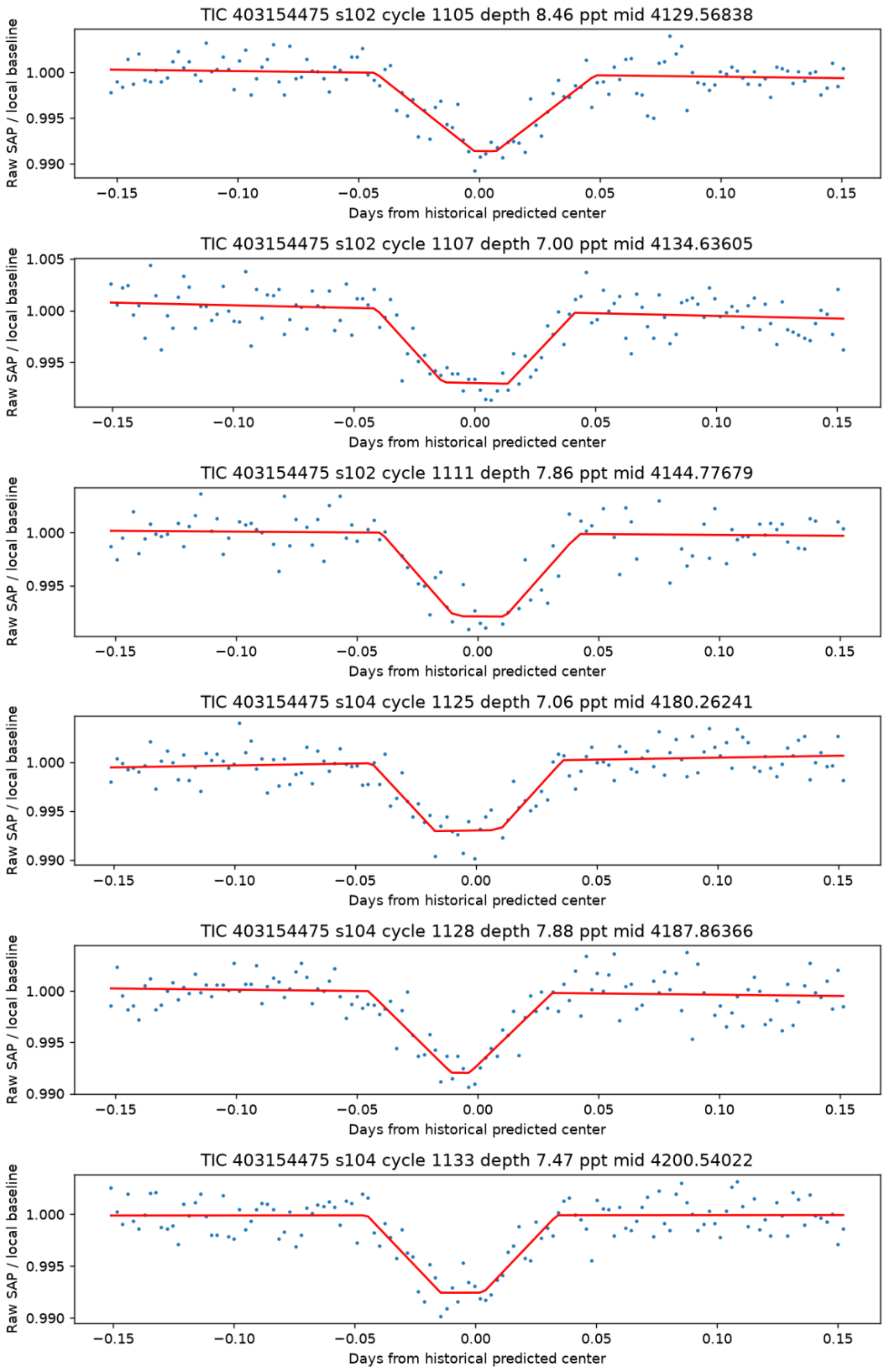
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

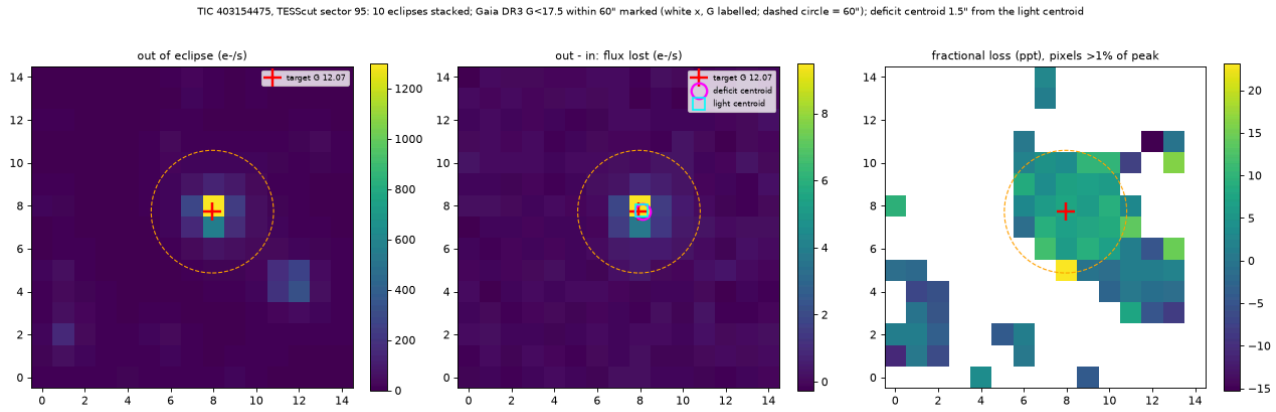
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
102	1105	4129.5683846	0.001285	8.460	usable, not selected (odd parity)
102	1106	4132.0969361	0.001533	8.039	selected (even parity)
102	1107	4134.6360507	0.001819	7.003	usable, not selected (odd parity)
102	1108	4137.1682844	0.001325	7.527	selected (even parity)
102	1110	4142.2381538	0.001662	7.120	selected (even parity)
102	1111	4144.7767894	0.001526	7.857	usable, not selected (odd parity)
102	1112	4147.3060842	0.001587	8.399	selected (even parity)
102	1113	4149.8468381	0.001535	7.673	usable, not selected (odd parity)
104	1125	4180.2624123	0.001186	7.056	usable, not selected (odd parity)
104	1126	4182.7951689	0.001329	8.033	selected (even parity)
104	1127	4185.3338158	0.002088	6.968	usable, not selected (odd parity)
104	1128	4187.8636609	0.001501	7.876	selected (even parity)
104	1129	4190.4064294	0.001544	7.308	usable, not selected (odd parity)
104	1131	4195.4712426	0.001595	5.839	usable, not selected (odd parity)
104	1133	4200.5402226	0.001414	7.466	usable, not selected (odd parity)

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s95.json, historical/diff_s95.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R312. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R312; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R312; diff_s95.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R313. TIC 137837801

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 137837801: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[28, 68, 96]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=7.1214903232 +/-0.0000068804d, reference eclipse epoch BTJD2062.5703486, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S28 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation3.658arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.035. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths292.097/293.233/291.190ppt, with 3 stacked events (quality-zero data). QLP selected-kind fitted median310.685ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3682166147923933248O, P=7.1214539995+/-0.0004287017d, e=0.024797, omega=143.1801deg. Closest edge-on conjunction offset-0.031138d (-0.004372phase), full-SB1-covariance linear propagated Gaia uncertainty0.107759d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.039674d (photometric period projection allowance0.001617d). The approximate central duration4.86h vs observed median3.67h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Eight deep raw eclipses over sectors28/68/96 include adjacent cycles0/1/2; aperture and sector depth differences reflect dilution/modeling. No capable neighbor under approximate Gaia flux budget.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

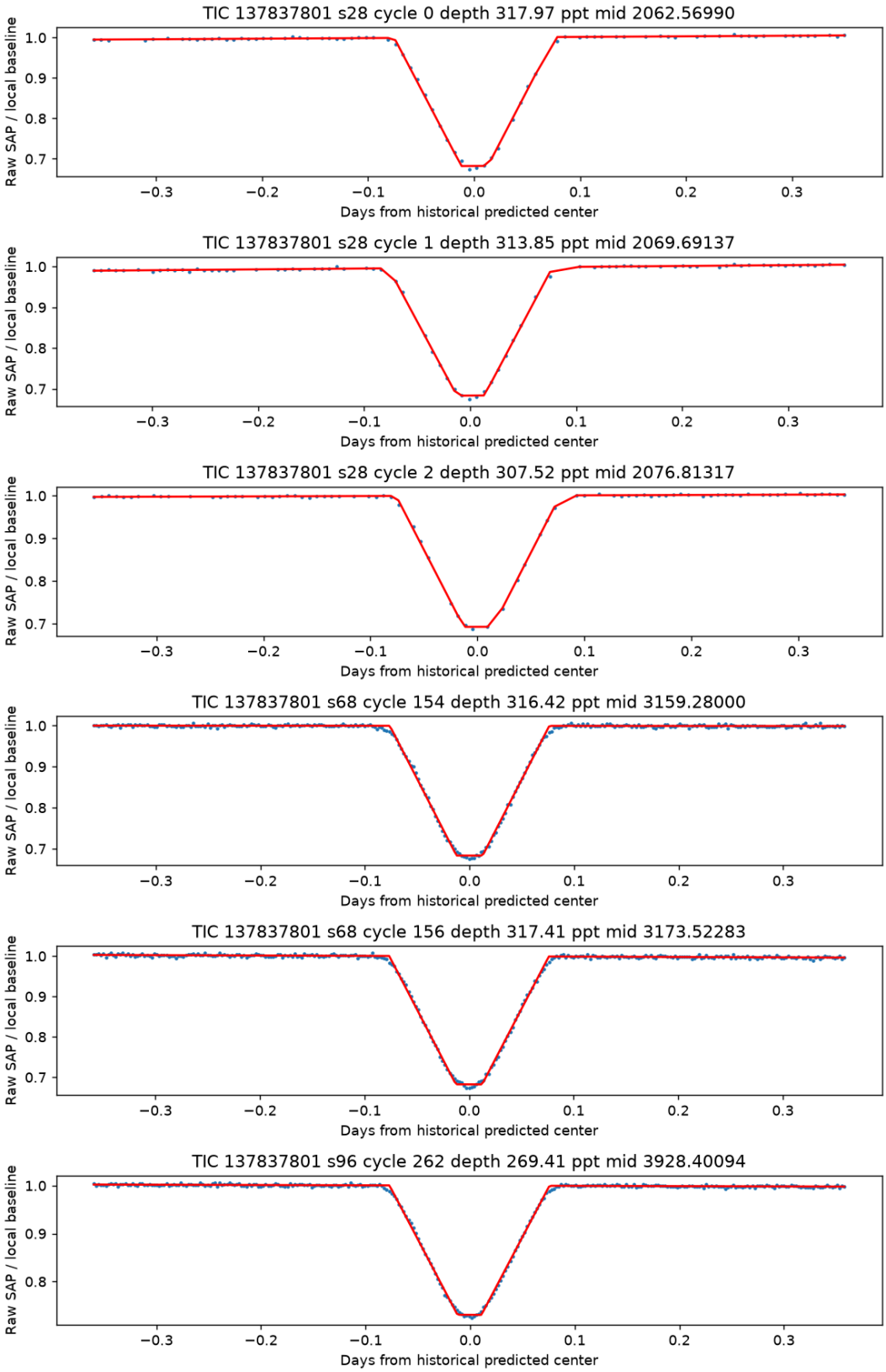
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

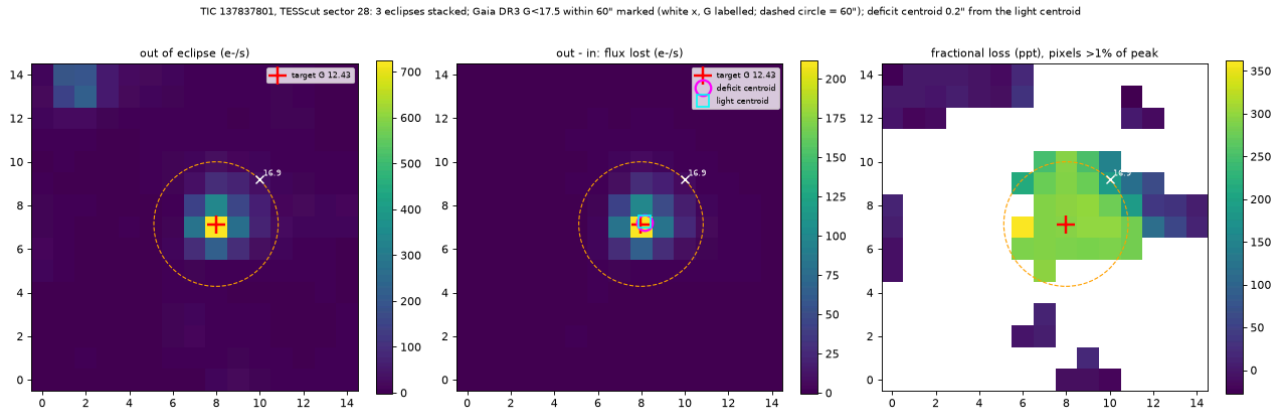
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
28	0	2062.5699010	0.003472	317.967	selected
28	1	2069.6913686	0.003472	313.846	selected
28	2	2076.8131743	0.003472	307.524	selected
28	3	2083.9354205	0.003472	300.114	selected
68	154	3159.2800028	0.001157	316.422	selected
68	156	3173.5228265	0.001157	317.412	selected
96	260	3914.1576245	0.001157	253.598	selected
96	262	3928.4009420	0.001157	269.409	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s28.json, historical/diff_s28.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R313. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R313; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R313; diff_s28.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R314. TIC 442606056

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 442606056: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[12, 65, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=5.3867042118 +/-0.0000046689d, reference eclipse epoch BTJD1651.0703266, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S65 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.973arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.905. Capable neighbours: Gaia 6015192755661126144: 8.81 arcsec, G12.576, required fractional eclipse0.165; Gaia 6015192790020864256: 50.94 arcsec, G13.336, required fractional eclipse0.259; Gaia 6015192790020865024: 56.30 arcsec, G14.423, required fractional eclipse0.581
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths51.676/42.364/34.810ppt, with 4 stacked events (quality-zero data). QLP selected-kind fitted median75.874ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR36015192755661126400, P=5.3867234906+/-0.0003950484d, e=0.026677, omega=105.4305deg. Closest edge-on conjunction offset-0.034817d (-0.006463phase), full-SB1-covariance linear propagated Gaia uncertainty0.107917d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.030306d (photometric period projection allowance0.001093d). The approximate central

duration5.75h vs observed median3.81h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded unresolved field: deficit0.189pixel from target versus closest capable G12.6neighbor0.62pixel. Deficit maximum is target pixel and centroid is displaced away from this neighbor, favoring target. No calibrated localization uncertainty, so resolved photometry remains useful. Intrinsic neighbor requirement is not compared directly to blended pixel loss.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

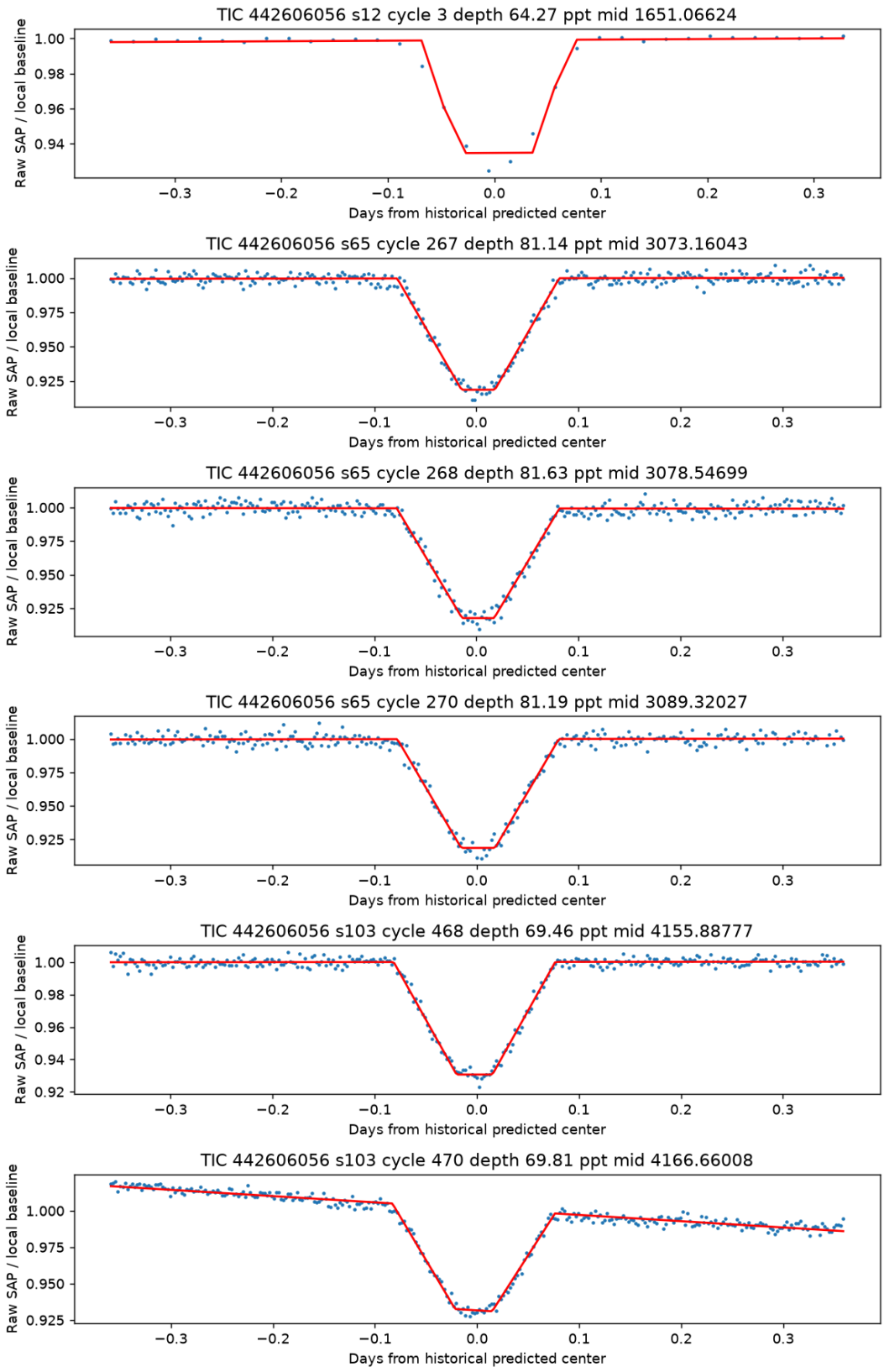
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

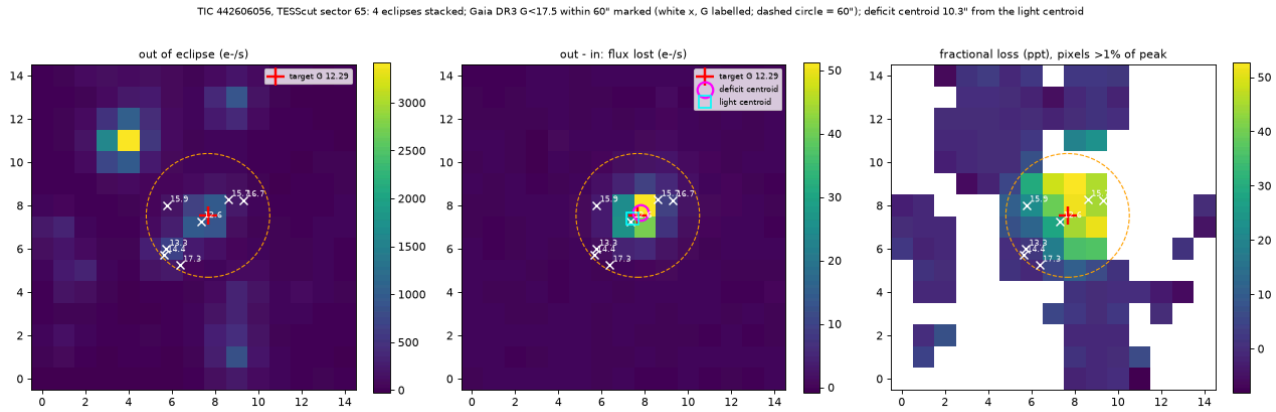
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
12	3	1651.0662448	0.010416	64.269	selected
65	267	3073.1604315	0.001157	81.142	selected
65	268	3078.5469895	0.001157	81.633	selected
65	269	3083.9336036	0.001157	80.791	selected
65	270	3089.3202674	0.001157	81.189	selected
103	468	4155.8877713	0.001157	69.463	selected
103	469	4161.2748774	0.001157	70.957	selected
103	470	4166.6600791	0.001849	69.807	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s65.json, historical/diff_s65.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R314. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R314; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R314; diff_s65.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R315. TIC 50993470

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 50993470: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[33, 87]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=9.1725585235 +/-0.0000429318d, reference eclipse epoch BTJD2209.6611630, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S87 deficit peak[7, 7] vs target pixel[7, 8]; deficit-target separation6.770arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.368. Capable neighbours: Gaia 3060034048014175872: 20.39 arcsec, G15.883, required fractional eclipse0.766; Gaia 3060033670057058304: 24.05 arcsec, G15.444, required fractional eclipse0.519; Gaia 3060033738776540032: 32.91 arcsec, G13.452, required fractional eclipse0.101; Gaia 3060033738776539648: 56.79 arcsec, G15.713, required fractional eclipse0.658; Gaia 3060033841855741696: 59.65 arcsec, G15.790, required fractional eclipse0.705
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths21.254/20.020/17.898ppt, with 3 stacked events (quality-zero data). QLP selected-kind fitted median25.260ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3306003367005705715 2, P=9.1714284019+/-0.0021381464d, e=0.183579, omega=151.0942deg. Closest edge-on conjunction offset0.179966d (0.019622phase), full-SB1-covariance linear propagated Gaia uncertainty0.476695d

at the measured epoch. Near-Gaia-reference projected eclipse offset-0.043798d (photometric period projection allowance0.008500d). The approximate central duration5.06h vs observed median2.31h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five measured raw eclipses in sectors33/87 include consecutive cycles78/79 and237/238/239, testing recurrence. Sector7 lacks qualifying fitted events and is not claimed a clean null. Target deficit0.322pixel versus nearest capable neighbor1.32pixels; centroid uncertainty uncalibrated. Baseline slope and dilution affect depth.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

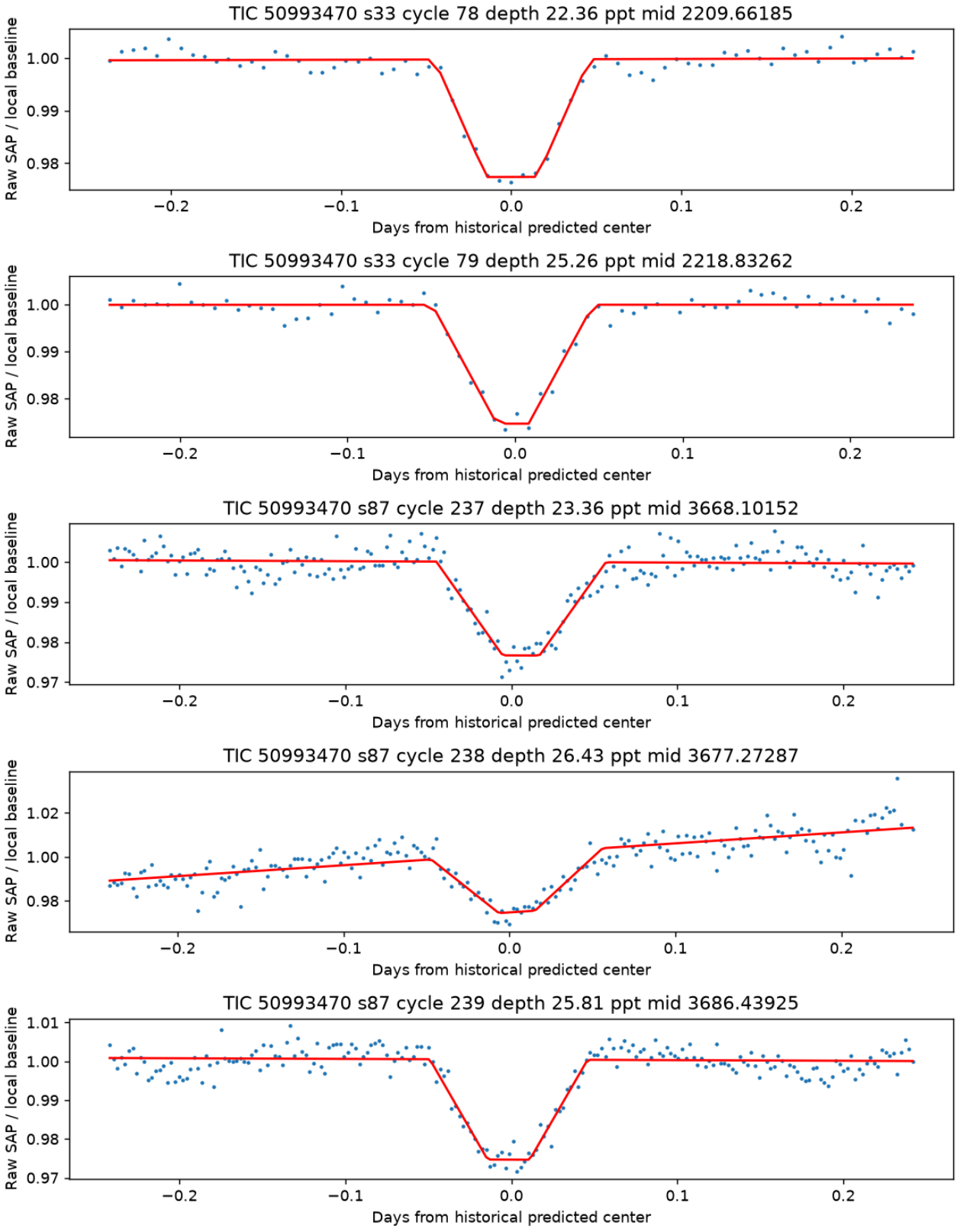
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

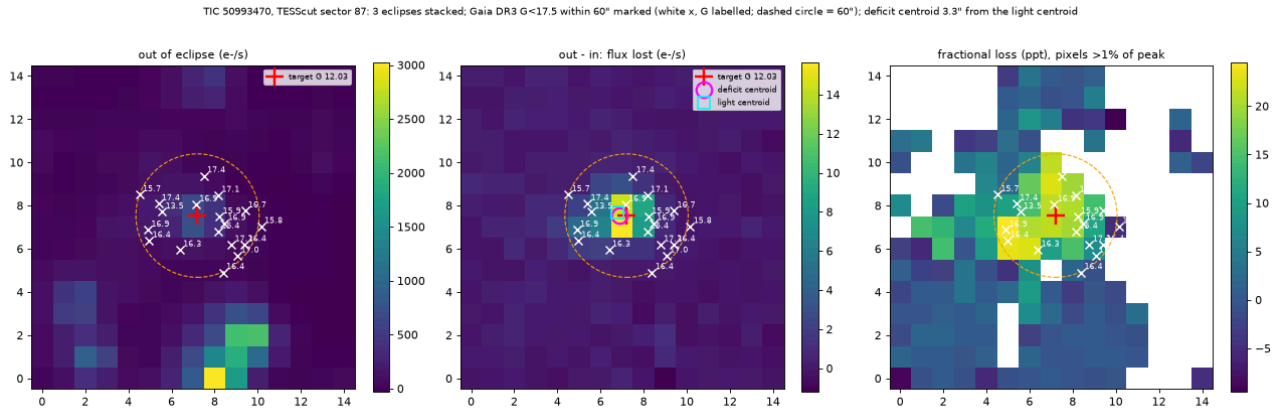
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	78	2209.6618452	0.003472	22.362	selected
33	79	2218.8326166	0.003472	25.260	selected
87	237	3668.1015216	0.001157	23.363	selected
87	238	3677.2728737	0.003074	26.427	selected
87	239	3686.4392463	0.001157	25.806	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s87.json, historical/diff_s87.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R315. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R315; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R315; diff_s87.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R316. TIC 367362188

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 367362188: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free eclipse-core fits to quality-zero raw QLP SAP_FLUX show 14 distinct eclipse events in selected sectors[19, 59, 86]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local baseline, free depth, center, duration and ingress were fitted with a linear baseline; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 14 same-kind midtimes give P=3.8453405155 +/-0.0000036346d, reference eclipse epoch BTJD1818.1697582, kind: Selected repeating eclipse; primary/secondary depth identity not established. No center-covariance fits were excluded. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Saved S44 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation4.492arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.754. Capable neighbours: Gaia 185367425540565760: 30.23 arcsec, G14.918, required fractional eclipse0.744
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths53.953/50.016/40.001ppt, with 6 stacked events (quality-zero data). QLP selected-kind fitted median54.935ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3185367421242800768, P=3.8451064321+/-0.0003851350d, e=0.091429, omega=237.6746deg. Closest edge-on conjunction offset-0.022555d (-0.005866phase), full-SB1-covariance linear propagated Gaia uncertainty0.145341d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.109634d (photometric period projection allowance0.001352d). The approximate central duration5.04h vs observed median3.89h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Fourteen raw eclipses across three sectors include consecutive cycles285/286 and multiple later repeats. Target deficit0.214pixel versus capable neighbor1.71pixels; depth and timing remain finite-cadence/model-dependent.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

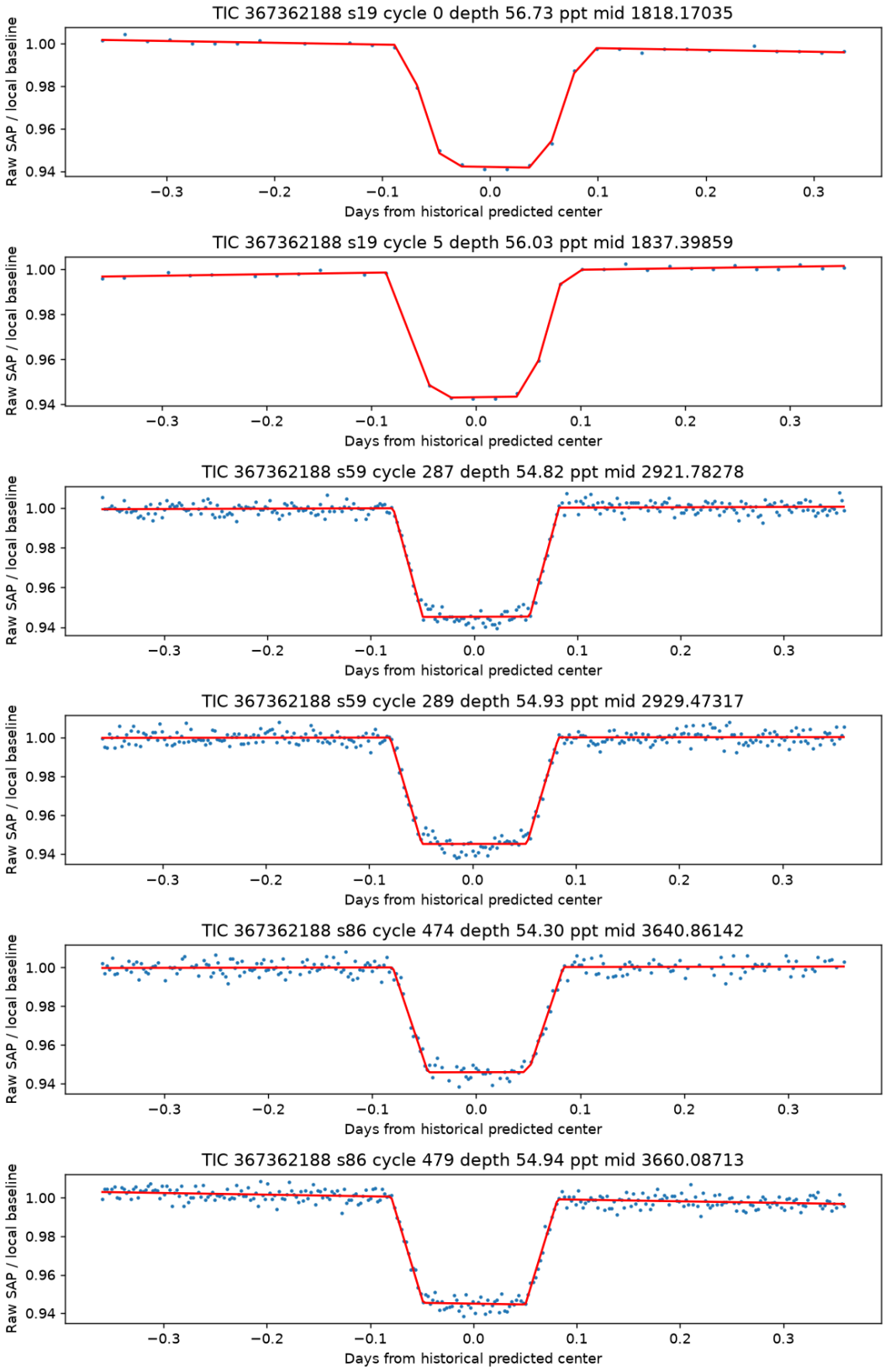
Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

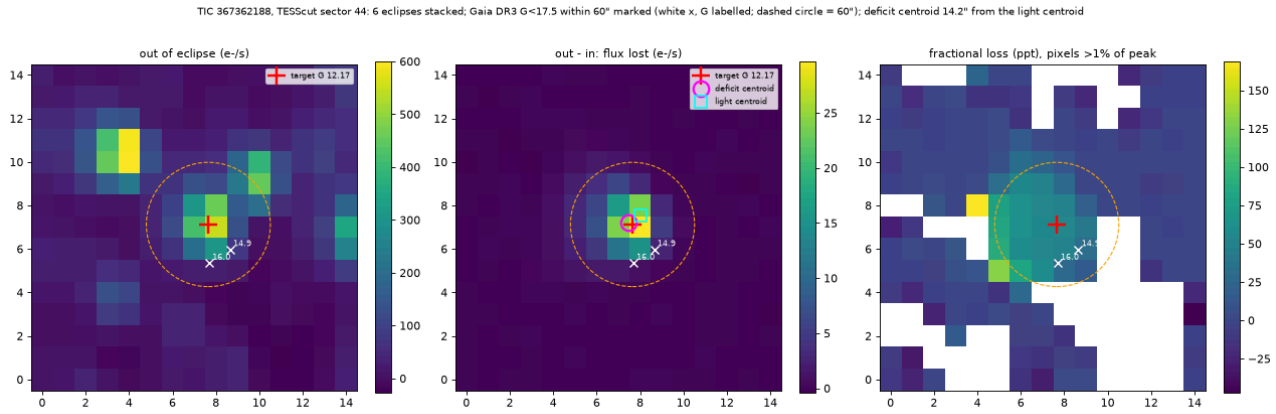
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
19	0	1818.1703500	0.010417	56.729	selected
19	1	1822.0151147	0.010417	54.705	selected
19	5	1837.3985928	0.010417	56.032	selected
59	285	2914.0918565	0.001157	55.760	selected
59	286	2917.9370382	0.001157	55.820	selected
59	287	2921.7827785	0.001157	54.819	selected
59	288	2925.6275834	0.001157	55.550	selected
59	289	2929.4731676	0.001157	54.927	selected
59	290	2933.3184088	0.001157	56.487	selected
86	473	3637.0160542	0.001157	54.286	selected
86	474	3640.8614157	0.001157	54.300	selected
86	475	3644.7065511	0.001157	53.565	selected
86	476	3648.5521141	0.001157	54.583	selected
86	479	3660.0871330	0.001157	54.943	selected

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv; shared-qlp QLP FITS; historical/diff_s44.json, historical/diff_s44.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R316. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R316; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R316; diff_s44.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R317. TIC 388152396

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 388152396: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 4 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[54, 81] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=10.9089333454 +/-0.0000386161d, epoch BTJD2780.52659071; epoch kind: Selected shallower eclipse. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Selected shallower eclipse at phase0; deeper distinct eclipse near phase0.34. S81 cycle67 unrelated local dip +0.20..0.45d masked from baseline fit; both sides retained. No half/double adoption. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR31807113502749250176; cutout WCS and selected target agree. Lost-light centroid is0.052px from target; peak[8, 7] versus target pixel[8, 7]. Historical PSF residual0.476. Capable-neighbour review: Gaia1807113953736371840 at37.90arcsec, required intrinsic fractional eclipse0.958, distance from deficit1.823px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S54 gives1x1/3x3/5x5 aperture depths103.520/97.563/84.866ppt from1 stacked event(s). QLP fitted selected-event median is108.190ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.
5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=10.9092234063+/-0.0025822378d, e=0.483069, omega=297.6635deg, periastron

BTJD391.840072. Measured-Gaia period difference-0.00029006d. Closest edge-on conjunction offset-0.170945d (-0.015670phase); full covariance linear propagated error0.591585d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.159138solar masses; assumed primary mass1.189, minimum companion mass0.880. Approximate central duration5.49h versus fitted median5.14h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

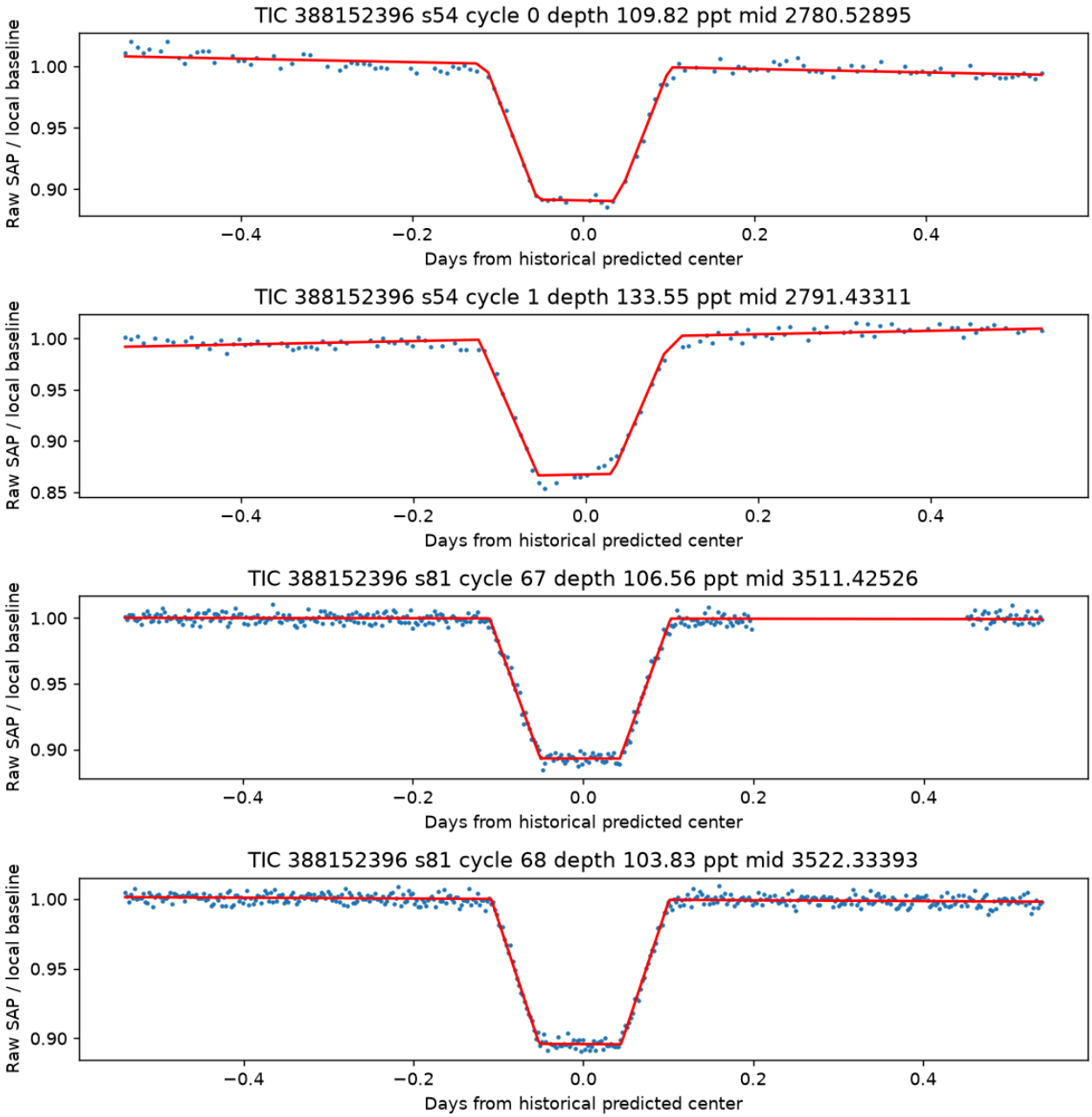
Limitations: Reference epoch is the selected shallower eclipse; a deeper eclipse near phase0.34 supports an eccentric binary at the retained period. S54 selected-event depth varies about110–134ppt; no common intrinsic depth is inferred. An unrelated local dip at +0.20..0.45d in S81 cycle67 was removed from baseline fitting, with baseline on both sides retained. The period changes by only0.00000312d relative to the unmasked fit; this is a model-sensitivity comparison, not a total uncertainty.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

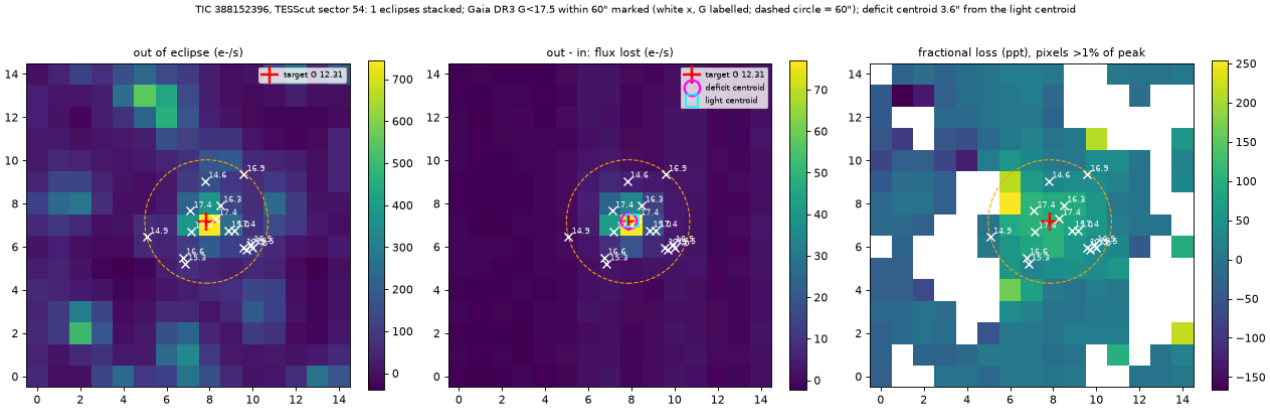
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
54	0	2780.52894938	0.00347227	109.824
54	1	2791.43311266	0.00347227	133.552
81	67	3511.42525520	0.00115740	106.556
81	68	3522.33393369	0.00115740	103.834

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s54.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R317. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R317; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R317; diff_s54.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R318. TIC 63577915

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 63577915: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 6 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[7, 34, 87] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=14.4103324466 +/-0.0000232127d, epoch BTJD1497.60282988; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Representative raw events support same event kind at retained photometric period; no persistent unequal alternating sequence seen. The measured period is seeded by the historical photometric branch, not an exhaustive alias search. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR3292128093763109644 8; cutout WCS and selected target agree. Lost-light centroid is0.145px from target; peak[8, 8] versus target pixel[8, 8]. Historical PSF residual0.902. Capable-neighbour review: No listed neighbour within60arcsec passes the approximate G-band flux-budget screen. Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S7 gives1x1/3x3/5x5 aperture depths224.852/220.968/205.231ppt from2 stacked event(s). QLP fitted selected-event median is226.649ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=14.4142632131+/-0.0018573652d, e=0.554808, omega=287.4635deg, periastron BTJD388.188059. Measured-Gaia period difference-0.00393077d. Closest edge-on conjunction offset-0.315064d (-0.021858phase); full covariance linear propagated error0.128221d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.204064solar masses; assumed primary mass1.260, minimum companion mass1.020. Approximate central duration8.75h versus fitted median7.08h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

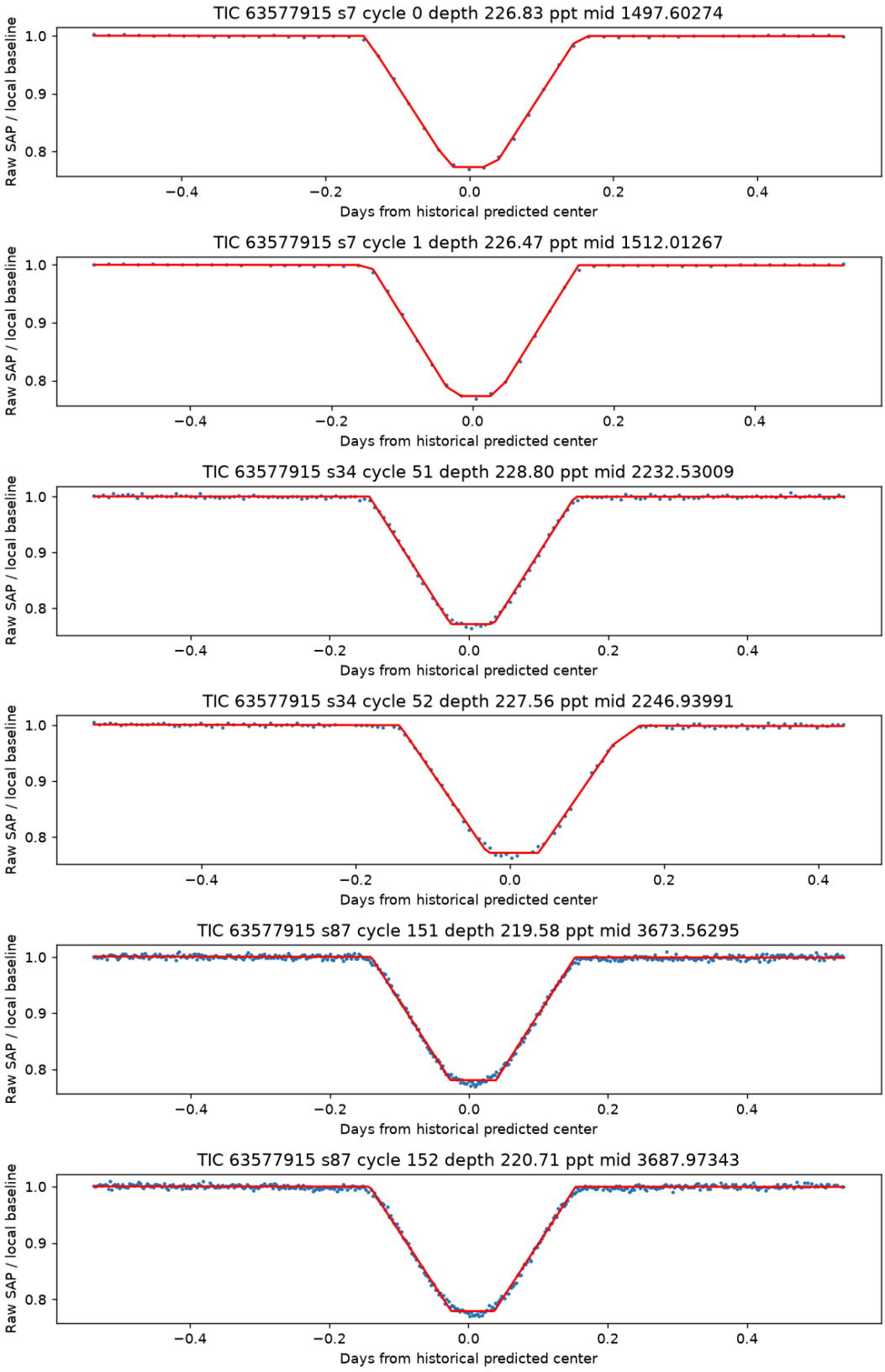
Limitations: Large scaled-out-image PSF residual0.902 is not alone a failed attribution check: deficit core is on target, and no listed Gaia neighbour within60arcsec can supply the eclipse under the approximate G-band flux budget. No calibrated centroid ellipse is available.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

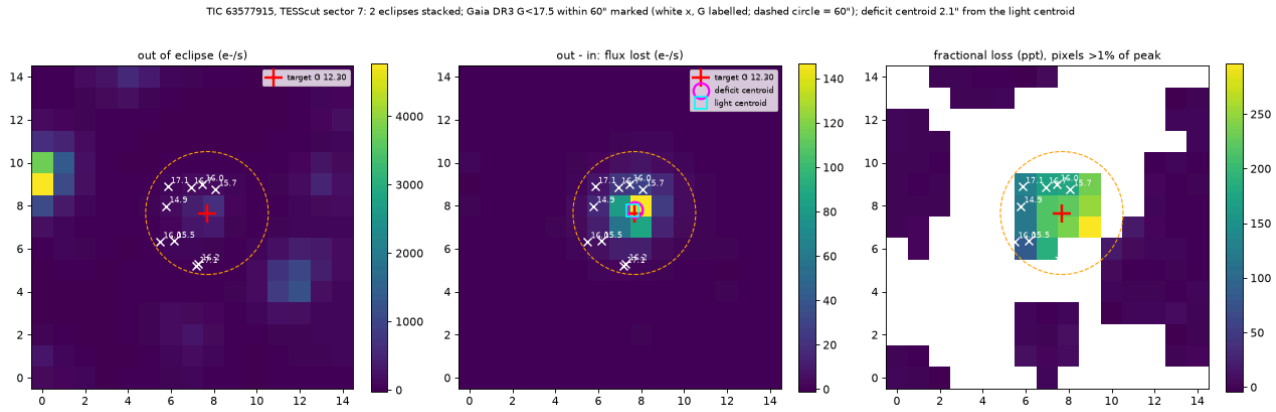
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
7	0	1497.60274145	0.01041656	226.826
7	1	1512.01267369	0.01041656	226.471
34	51	2232.53009473	0.00347216	228.801
34	52	2246.93990779	0.00347216	227.555
87	151	3673.56295216	0.00115742	219.578
87	152	3687.97343485	0.00115742	220.710

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s07.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R318. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R318; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R318; diff_s07.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R319. TIC 11249710

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 11249710: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 7 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[41, 74, 81] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=8.3357588781 +/-0.0000195304d, epoch BTJD2423.91922976; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Two distinct similar-depth eclipses at phases0 and0.34 establish retained eccentric orbital period; selected phase0 event kind, not claim deepest primary. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR32028395616657465856; cutout WCS and selected target agree. Lost-light centroid is0.306px from target; peak[7, 7] versus target pixel[7, 7]. Historical PSF residual0.595. Capable-neighbour review: Gaia2028394139188718336 at19.48arcsec, required intrinsic fractional eclipse0.827, distance from deficit1.283px; Gaia2028394139188714624 at26.32arcsec, required intrinsic fractional eclipse0.632, distance from deficit1.460px; Gaia2028394070469231616 at36.26arcsec, required intrinsic fractional eclipse0.458, distance from deficit1.777px; Gaia2028394070469227648 at56.99arcsec, required intrinsic fractional eclipse0.615, distance from deficit2.805px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S55 gives1x1/3x3/5x5 aperture depths79.059/62.084/47.787ppt from3 stacked event(s). QLP fitted selected-event median is97.705ppt. These are reductions of the same TESS photons, not

independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=8.3358909655+/-0.0014033620d, e=0.270374, omega=10.1155deg, periastron BTJD391.571708. Measured-Gaia period difference-0.00013209d. Closest edge-on conjunction offset-0.015004d (-0.001800phase); full covariance linear propagated error0.345895d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.221877solar masses; assumed primary mass1.000, minimum companion mass0.942. Approximate central duration7.99h versus fitted median8.60h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

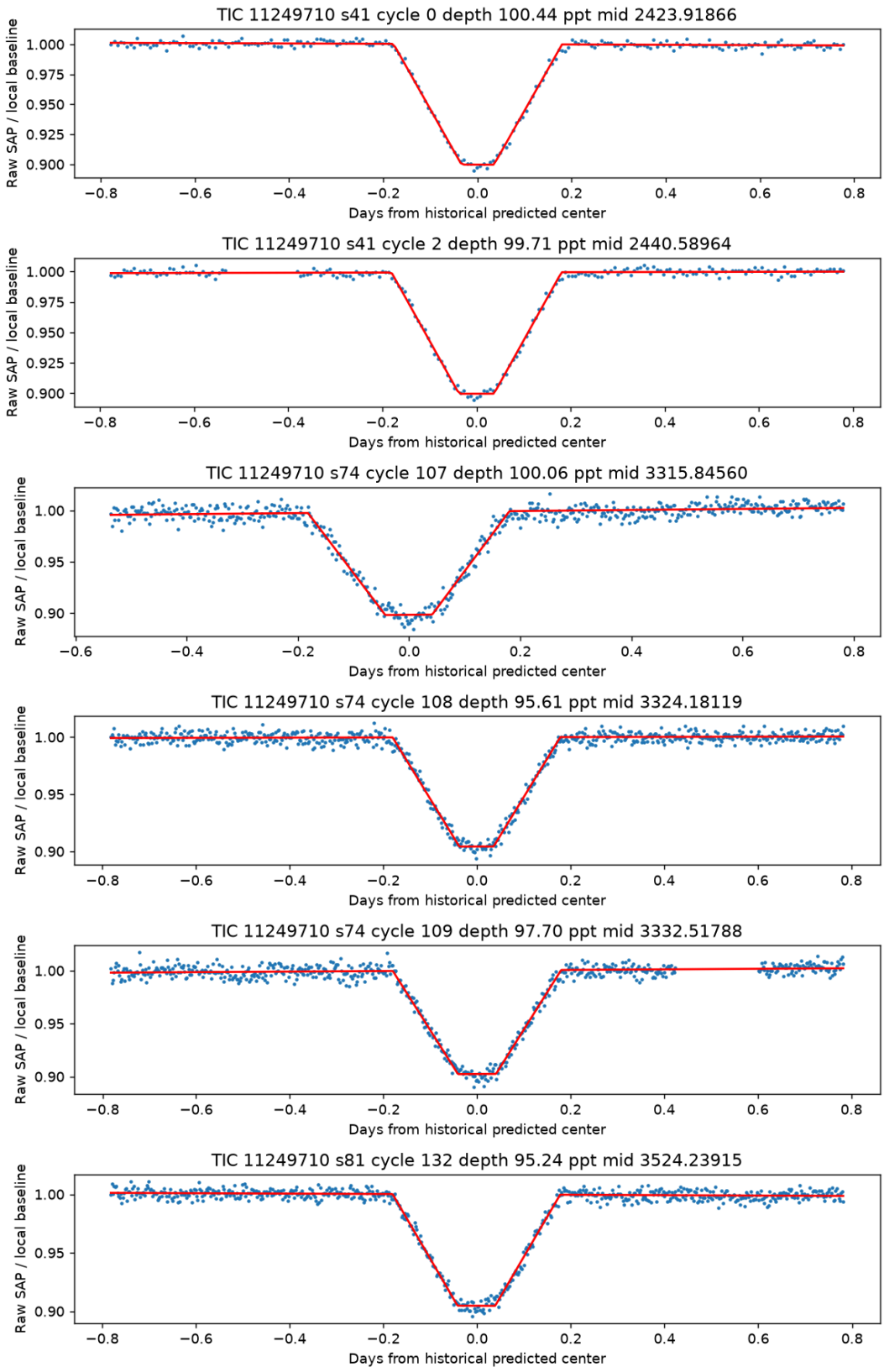
Limitations: Crowded field with four capable neighbours. The deficit is0.306pixel from target and1.283pixels from the closest capable neighbour, so the target is favored at TESS resolution; precise exclusion probabilities are unavailable. Alternative S55 target-centered aperture gives about62ppt in three events, versus QLP about100ppt; dilution and aperture differences prevent an intrinsic-depth equality claim. Second eclipse near phase0.34 supports retained orbital period.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

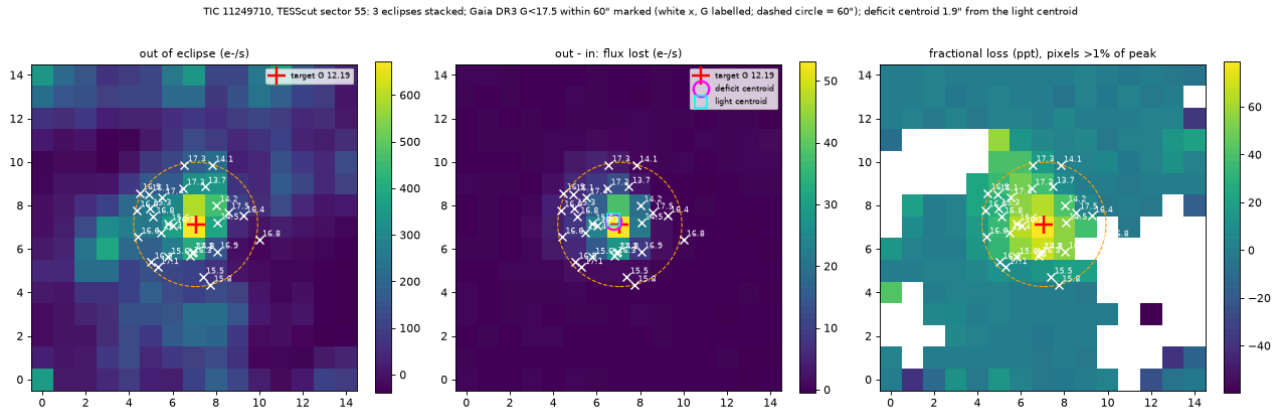
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
41	0	2423.91865809	0.00347218	100.444
41	2	2440.58964202	0.00347218	99.714
74	107	3315.84559621	0.00115739	100.060
74	108	3324.18119144	0.00115739	95.613
74	109	3332.51788005	0.00115739	97.705
81	130	3507.56722049	0.00115741	97.568
81	132	3524.23914951	0.00115741	95.238

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s55.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R319. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R319; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R319; diff_s55.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R320. TIC 150582131

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 150582131: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 4 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[25, 52, 79] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=17.2887175714 +/-0.0000991015d, epoch BTJD2003.32337520; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Representative raw events support same event kind at retained photometric period; no persistent unequal alternating sequence seen. The measured period is seeded by the historical photometric branch, not an exhaustive alias search. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR3456167294533319769 6; cutout WCS and selected target agree. Lost-light centroid is0.053px from target; peak[7, 8] versus target pixel[7, 7]. Historical PSF residual0.220. Capable-neighbour review: No listed neighbour within60arcsec passes the approximate G-band flux-budget screen. Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S79 gives1x1/3x3/5x5 aperture depths75.715/77.209/76.104ppt from1 stacked event(s). QLP fitted selected-event median is79.070ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=17.2926454274+/-0.0113118760d, e=0.110523, omega=31.1277deg, periastron BTJD392.576071. Measured-Gaia period difference-0.00392786d. Closest edge-on conjunction offset0.201874d (0.011674phase); full covariance linear propagated error1.022039d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0175455solar masses; assumed primary mass1.000, minimum companion mass0.311. Approximate central duration5.62h versus fitted median7.11h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

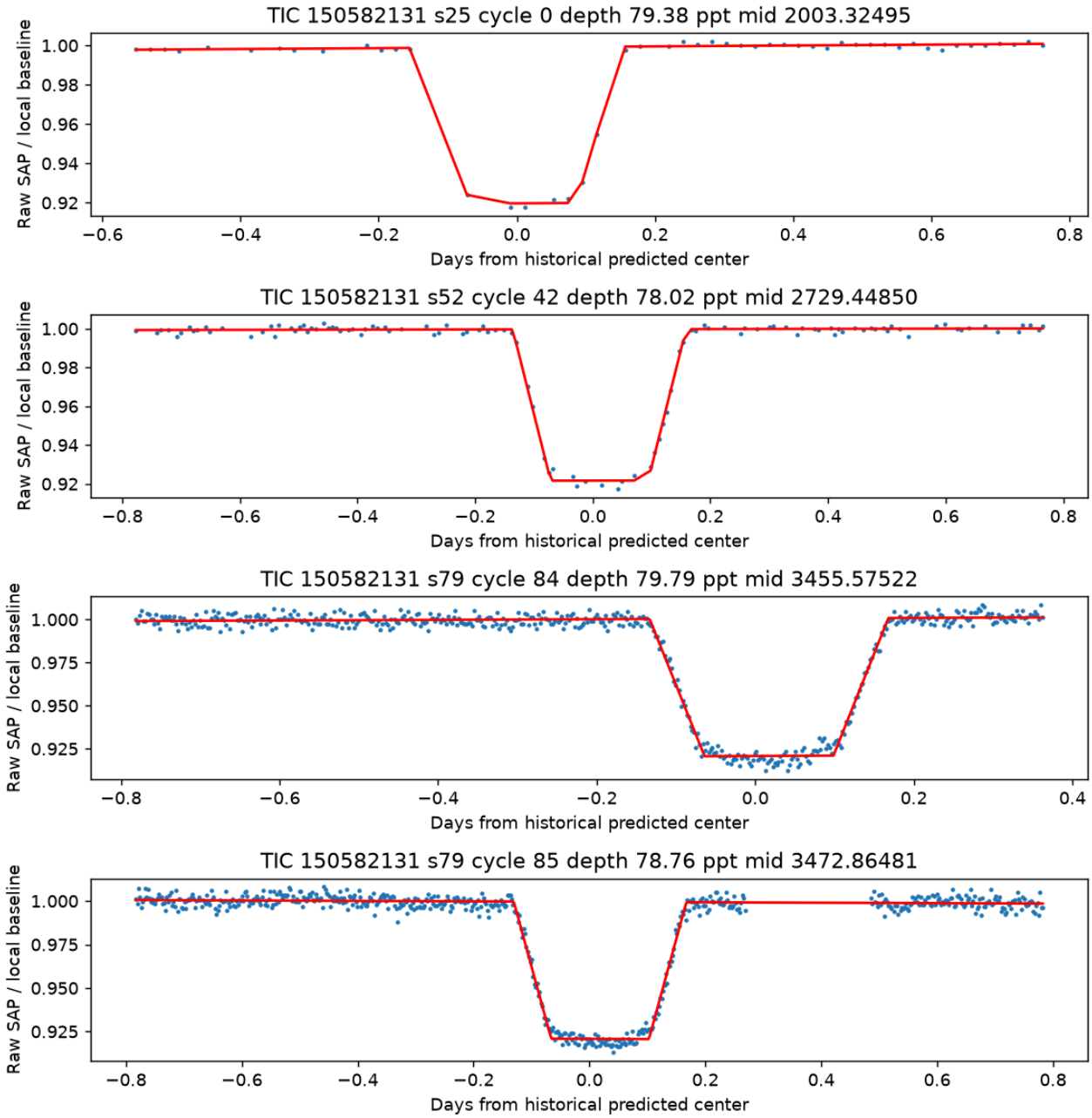
Limitations: Repeated deep events include consecutive S79 cycles84/85. Approximate central-duration estimate is shorter than the fitted width; TIC radii and assumed main-sequence minimum-mass companion are uncertain, so this is not a fitted binary-radius solution or a quantitative duration match.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

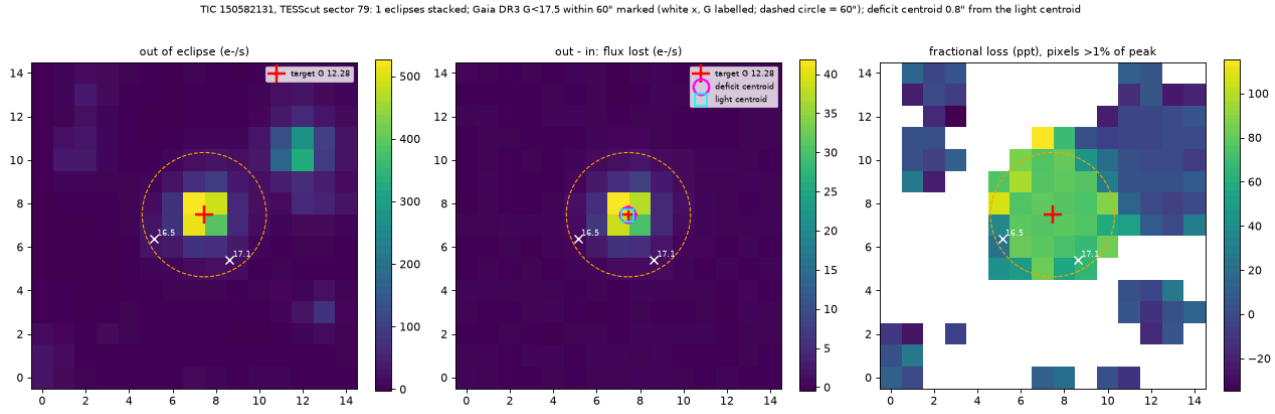
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
25	0	2003.32494860	0.01041680	79.378
52	42	2729.44849586	0.00694440	78.019
79	84	3455.57521531	0.00115740	79.785
79	85	3472.86481349	0.00115740	78.762

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s79.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R320. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R320; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R320; diff_s79.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R321. TIC 47653432

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 47653432: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 10 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[65, 102] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=4.1981909010 +/-0.0000097735d, epoch BTJD3071.05142551; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Representative raw events support same event kind at retained photometric period; no persistent unequal alternating sequence seen. The measured period is seeded by the historical photometric branch, not an exhaustive alias search. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR3621566090395219353 6; cutout WCS and selected target agree. Lost-light centroid is0.148px from target; peak[7, 7] versus target pixel[7, 7]. Historical PSF residual0.358. Capable-neighbour review: Gaia6215657949014886144 at43.41arcsec, required intrinsic fractional eclipse0.060, distance from deficit2.162px; Gaia6215657949014692480 at44.25arcsec, required intrinsic fractional eclipse0.395, distance from deficit2.101px; Gaia6215666779467456384 at48.79arcsec, required intrinsic fractional eclipse0.599, distance from deficit2.434px; Gaia6215663824529954048 at52.66arcsec, required intrinsic fractional eclipse0.368, distance from deficit2.626px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S38 gives1x1/3x3/5x5 aperture depths4.853/4.373/4.511ppt from6 stacked event(s). QLP

fitted selected-event median is5.090ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=4.1924134926+/-0.0013497127d, e=0.059512, omega=347.9732deg, periastron BTJD390.987237. Measured-Gaia period difference0.00577741d. Closest edge-on conjunction offset0.002164d (0.000516phase); full covariance linear propagated error0.827753d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0131992solar masses; assumed primary mass1.120, minimum companion mass0.298. Approximate central duration5.17h versus fitted median1.54h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

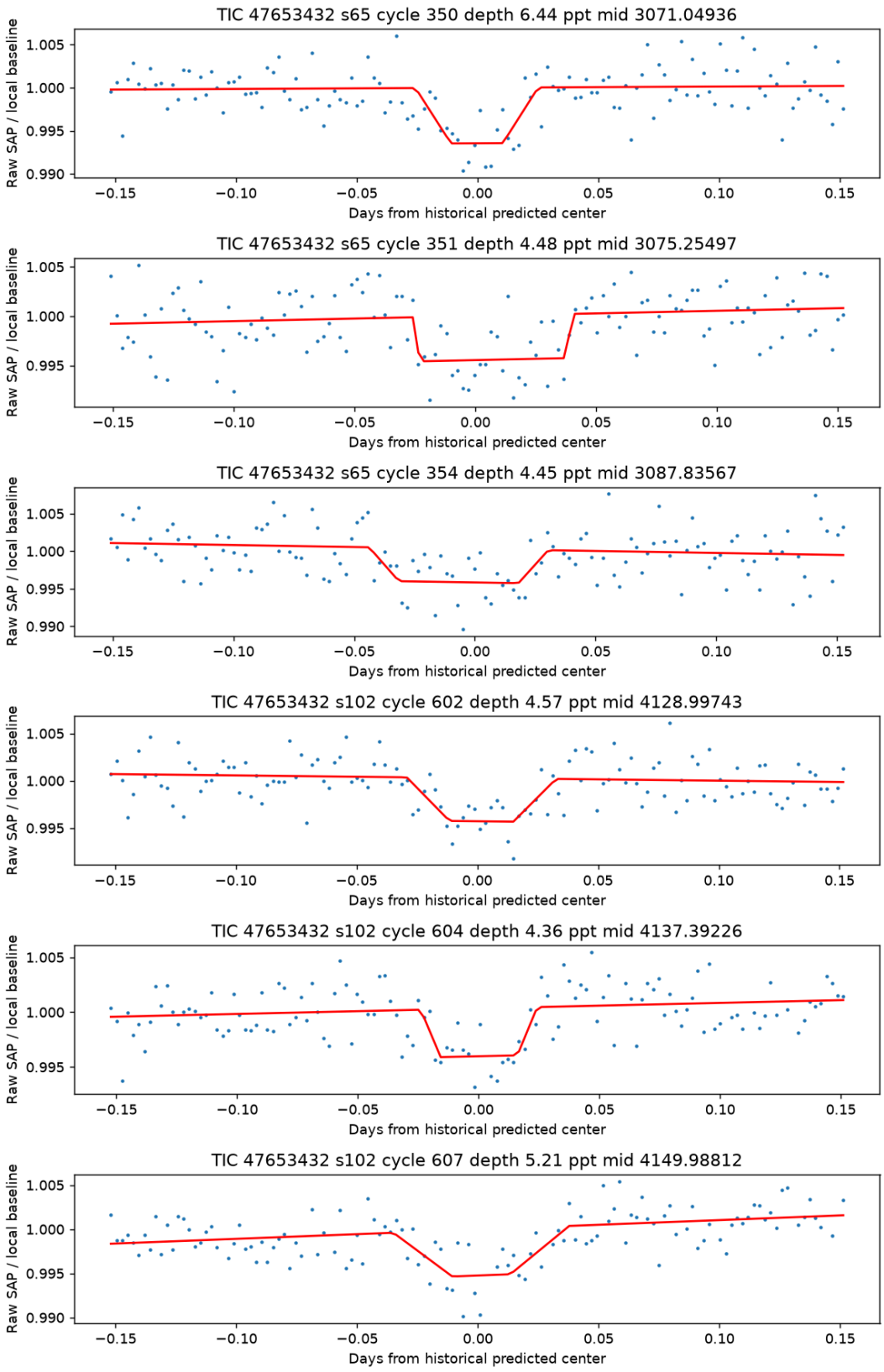
Limitations: Shallow4–7ppt locally normalized events recur with resolved localized shape; the whole-sector raw fold is dominated by baseline trends. Four capable neighbours are43arcsec or farther away from the compact target deficit. Observed duration about1.54h versus approximate central5.17h can arise from projected chord/radius assumptions; no measured central-duration match is claimed.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

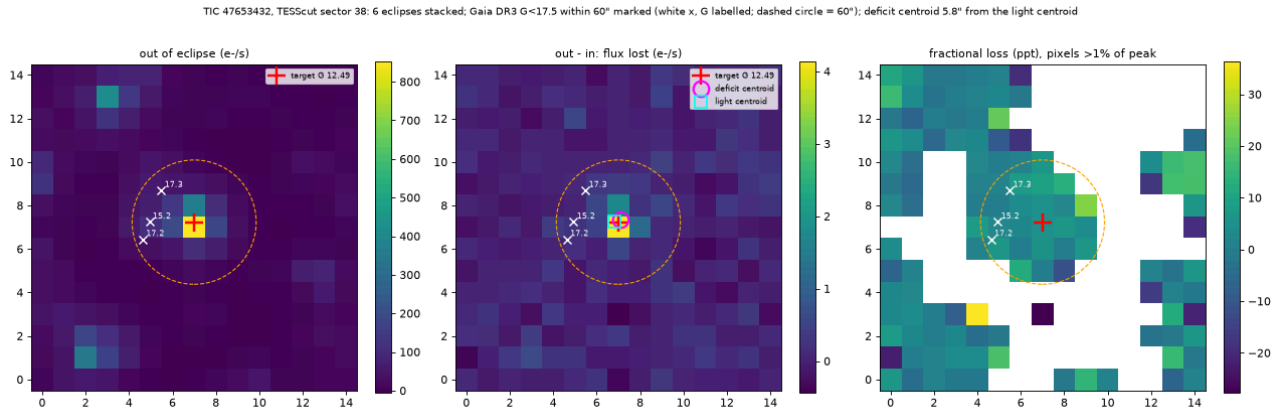
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
65	350	3071.04936256	0.00145452	6.443
65	351	3075.25496995	0.00144651	4.476
65	352	3079.44977713	0.00323568	5.037
65	354	3087.83567116	0.00255968	4.452
65	355	3092.04090879	0.00174376	6.818
102	602	4128.99742687	0.00211025	4.569
102	603	4133.18963558	0.00185280	5.143
102	604	4137.39225828	0.00153892	4.359
102	605	4141.59166940	0.00215224	6.258
102	607	4149.98812063	0.00235392	5.212

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s38.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R321. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R321; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R321; diff_s38.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R322. TIC 94113033

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 94113033: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 6 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[54, 81] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=4.9193469541 +/-0.0000326822d, epoch BTJD2773.32107341; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Representative raw events support same event kind at retained photometric period; no persistent unequal alternating sequence seen. The measured period is seeded by the historical photometric branch, not an exhaustive alias search. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR34289984996506672128; cutout WCS and selected target agree. Lost-light centroid is0.129px from target; peak[8, 7] versus target pixel[8, 7]. Historical PSF residual0.947. Capable-neighbour review: No listed neighbour within60arcsec passes the approximate G-band flux-budget screen. Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S81 gives1x1/3x3/5x5 aperture depths61.891/60.642/55.953ppt from3 stacked event(s). QLP fitted selected-event median is65.306ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=4.9192335218+/-0.0003838993d, e=0.256995, omega=223.8411deg, periastron BTJD389.127892. Measured-Gaia period difference0.00011343d. Closest edge-on conjunction offset-0.138537d (-0.028162phase); full covariance linear propagated error0.239835d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0286867solar masses; assumed primary mass1.150, minimum companion mass0.412. Approximate central duration6.89h versus fitted median5.29h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

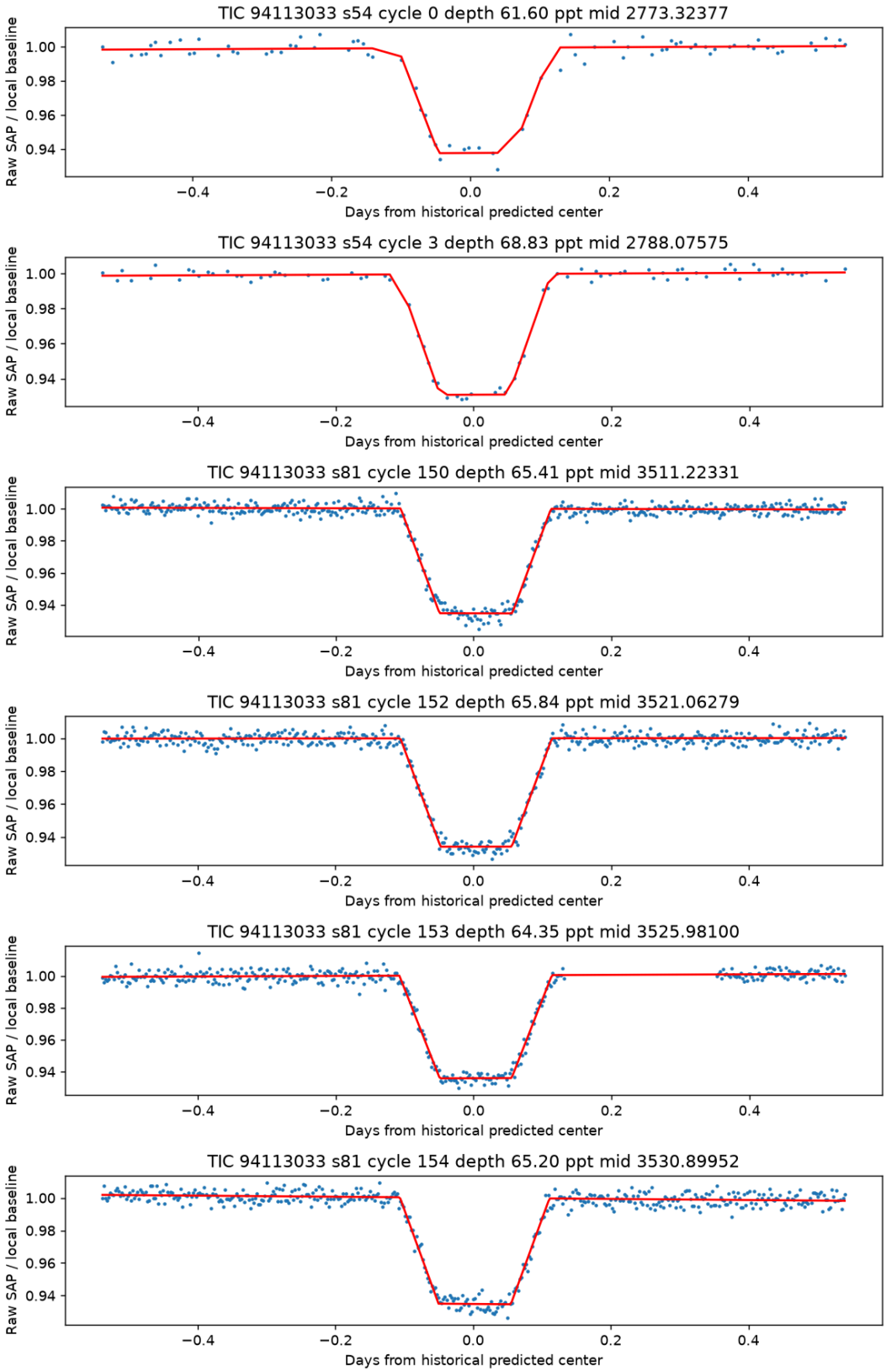
Limitations: Large scaled-out-image PSF residual0.947 is not alone a failed attribution check: deficit core is on target and no listed neighbour passes the approximate flux-budget screen. No calibrated centroid uncertainty is available.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

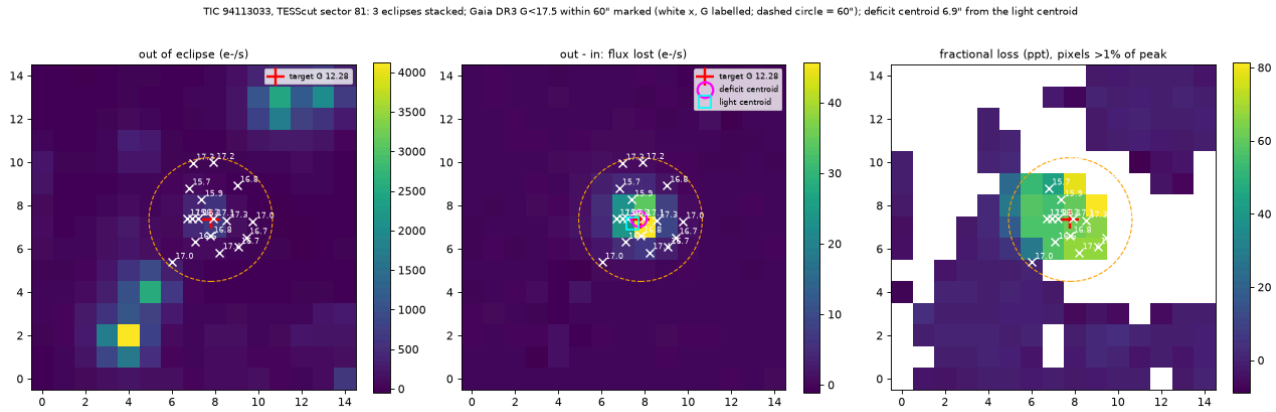
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
54	0	2773.32377472	0.00694434	61.597
54	3	2788.07575225	0.00694434	68.831
81	150	3511.22330501	0.00115739	65.407
81	152	3521.06278640	0.00115739	65.839
81	153	3525.98099772	0.00115739	64.352
81	154	3530.89951790	0.00115739	65.205

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s81.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R322. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R322; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R322; diff_s81.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R323. TIC 314991763

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 314991763: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 6 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[56, 83] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=7.7113830155 +/-0.0000099460d, epoch BTJD2833.13055669; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Deep primary phase0 and shallower secondary phase0.5 establish full orbital period. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR31777474556207679872; cutout WCS and selected target agree. Lost-light centroid is0.106px from target; peak[7, 7] versus target pixel[7, 7]. Historical PSF residual0.301. Capable-neighbour review: Gaia1777474590567419776 at56.26arcsec, required intrinsic fractional eclipse0.649, distance from deficit2.718px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S56 gives1x1/3x3/5x5 aperture depths124.344/116.460/102.295ppt from4 stacked event(s). QLP fitted selected-event median is148.371ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.
5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=7.7109095009+/-0.0008394808d, e=0.012496, omega=74.8256deg, periastron BTJD388.328610. Measured-Gaia period difference0.00047351d. Closest edge-on conjunction

offset0.126569d (0.016414phase); full covariance linear propagated error0.268207d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.101794solar masses; assumed primary mass0.965, minimum companion mass0.640. Approximate central duration6.40h versus fitted median4.56h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

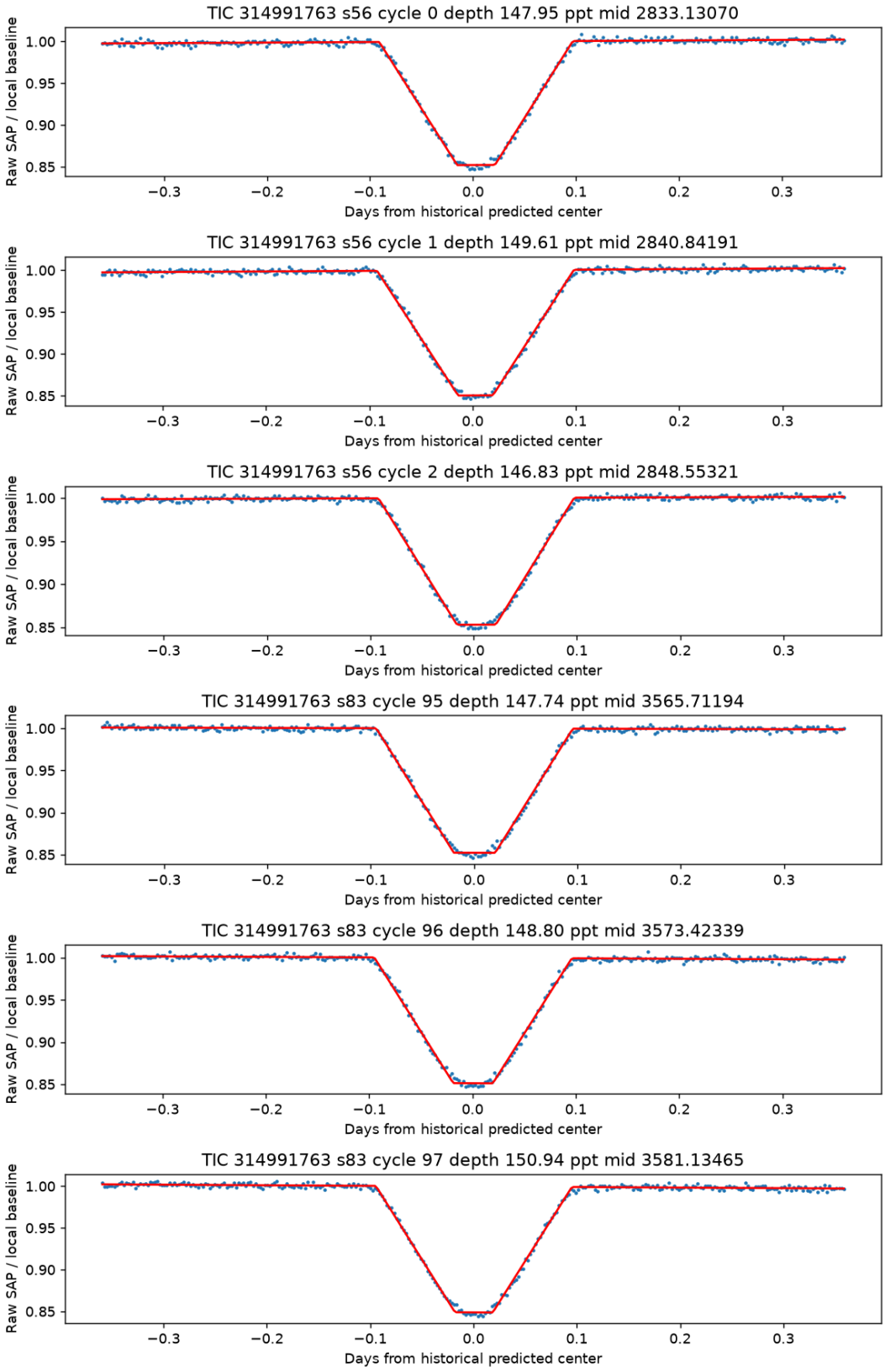
Limitations: Unequal phase-zero primary and phase-half secondary establish the retained full orbital period. One capable neighbour at56.26arcsec is separated from the compact target deficit.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

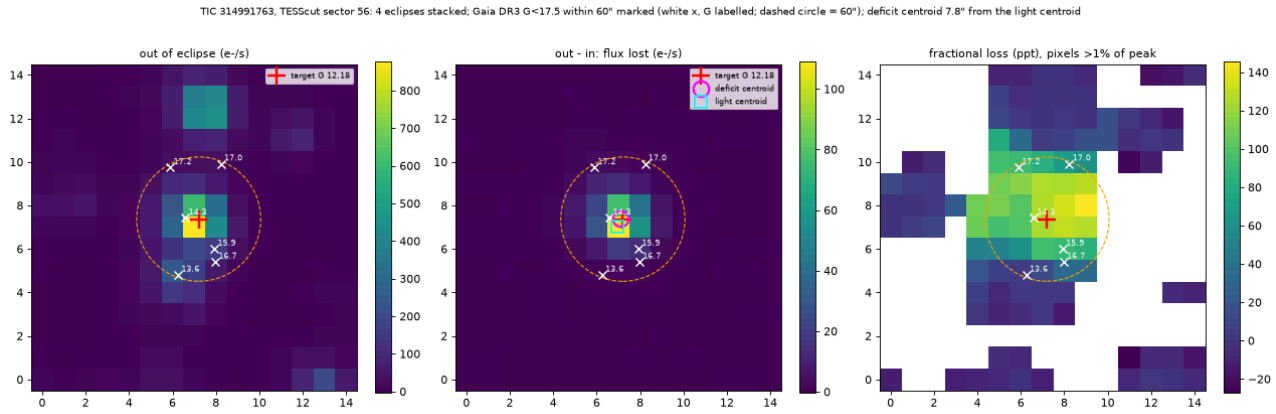
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
56	0	2833.13069709	0.00115739	147.947
56	1	2840.84190758	0.00115739	149.606
56	2	2848.55321125	0.00115739	146.828
83	95	3565.71193913	0.00115739	147.737
83	96	3573.42338948	0.00115739	148.795
83	97	3581.13465312	0.00115739	150.942

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s56.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R323. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R323; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R323; diff_s56.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R324. TIC 102716983

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization.

Archived source: sanitized report text.

TIC 102716983: PROBABLE

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 4 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[19, 59, 73] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=11.6869371250 +/-0.0000409829d, epoch BTJD1818.04578377; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Extra S59 QLP downloaded because initial three selected sectors yielded only two timed events. S59 cycles94,95 now provide consecutive recurrence and four timed eclipses total. Close capable neighbours14.39/14.45arcsec remain attribution concern. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after checking capable neighbours.** Identity: Gaia DR3490460942806688768; cutout WCS and selected target agree. Lost-light centroid is0.134px from target; peak[8, 7] versus target pixel[8, 7]. Historical PSF residual0.920. Capable-neighbour review: Gaia490460942808112640 at14.39arcsec, required intrinsic fractional eclipse0.323, distance from deficit0.665px; Gaia490460942803537408 at14.45arcsec, required intrinsic fractional eclipse0.263, distance from deficit0.660px; Gaia490460942808113408 at26.56arcsec, required intrinsic fractional eclipse0.702, distance from deficit1.389px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S59 gives1x1/3x3/5x5 aperture depths32.791/35.271/21.366ppt from1 stacked event(s). QLP fitted selected-event median is61.617ppt. These are reductions of the same TESS photons, not

independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=11.6869282788+/-0.0014018231d, e=0.104485, omega=170.1550deg, periastron BTJD394.383940. Measured-Gaia period difference0.00000885d. Closest edge-on conjunction offset0.081490d (0.006973phase); full covariance linear propagated error0.150298d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0465559solar masses; assumed primary mass1.000, minimum companion mass0.464. Approximate central duration10.99h versus fitted median6.52h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

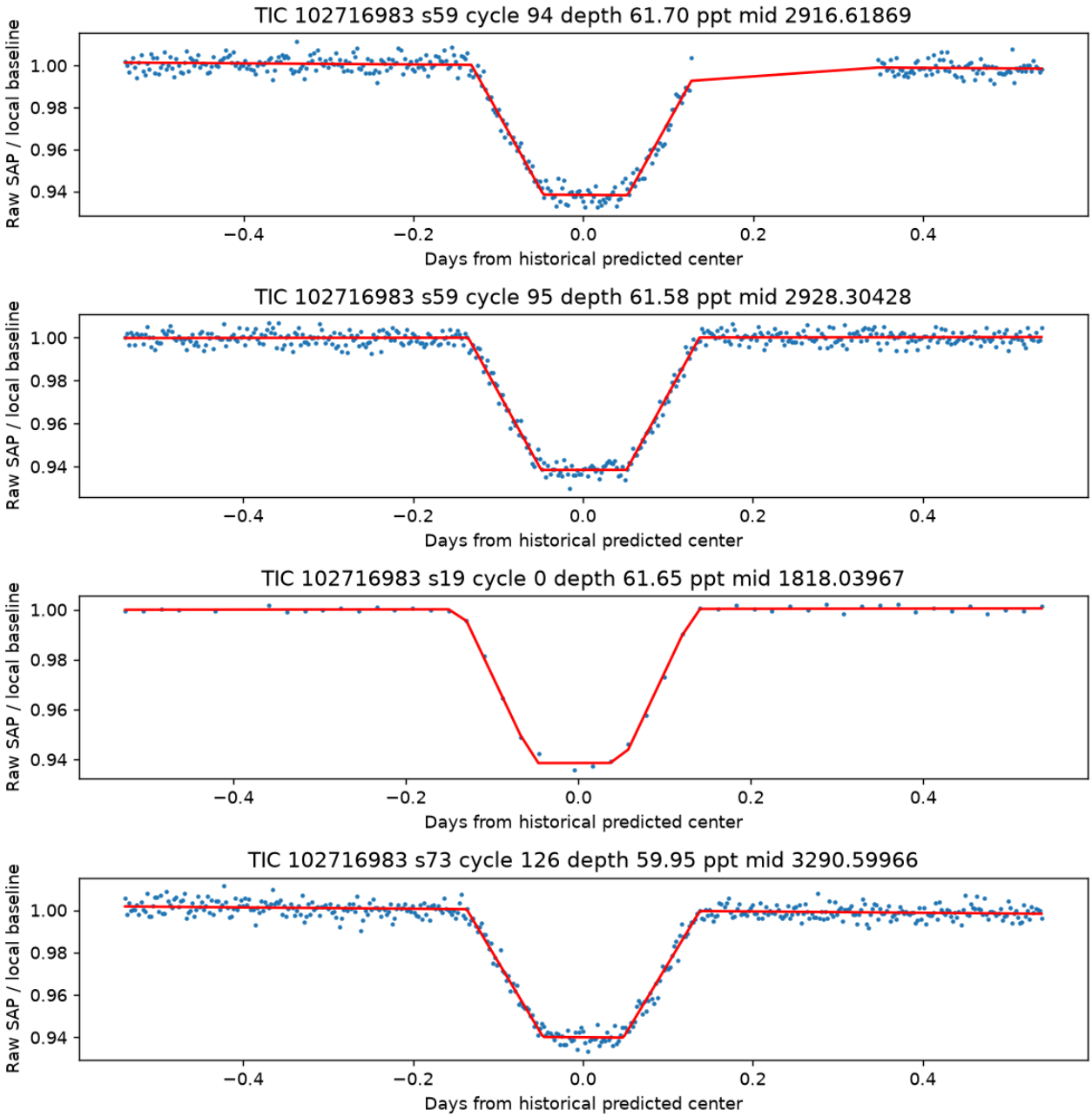
Limitations: Repeating eclipses are secure, including consecutive S59 cycles94/95 measured after supplementing the original selection. Attribution is unresolved: two capable neighbours at14.39/14.45arcsec lie within one TESS pixel, and the historical PSF residual is0.920. The third capable neighbour is26.56arcsec away. Source localization requires resolved imaging or multi-source PRF analysis.

Next check: Resolved eclipse photometry or multi-source PRF localization.

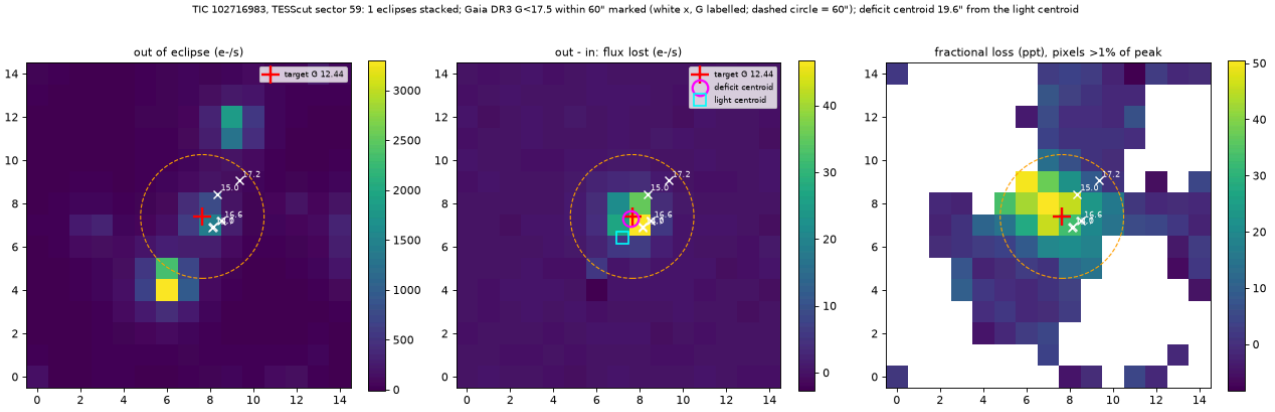
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
19	0	1818.03966588	0.01041664	61.652
59	94	2916.61868586	0.00115740	61.697
59	95	2928.30427910	0.00115740	61.581
73	126	3290.59965627	0.00115739	59.946

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s59.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R324. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R324; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R324; diff_s59.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R325. TIC 115048464

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization.

Archived source: sanitized report text.

TIC 115048464: CONFIRMED

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 7 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[19, 45, 86] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: pass at the retained photometric cycle assignment.** P=8.0142430574 +/-0.0000142842d, epoch BTJD1832.46809862; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Representative raw events support same event kind at retained photometric period; no persistent unequal alternating sequence seen. The measured period is seeded by the historical photometric branch, not an exhaustive alias search. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS resolution with limitations below.** Identity: Gaia DR3183073672486786048; cutout WCS and selected target agree. Lost-light centroid is0.201px from target; peak[7, 8] versus target pixel[7, 8]. Historical PSF residual0.791. Capable-neighbour review: Gaia183073775565996672 at51.03arcsec, required intrinsic fractional eclipse0.682, distance from deficit2.572px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S19 gives1x1/3x3/5x5 aperture depths37.833/39.394/37.099ppt from4 stacked event(s). QLP fitted selected-event median is37.720ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.

5. **Gaia orbital geometry: consistent within the stated bounded approximations.** Full NSS orbit: P=8.0157041322+/-0.0019466464d, e=0.050791, omega=286.7691deg, periastron BTJD386.221698. Measured-Gaia period difference-0.00146107d. Closest edge-on conjunction offset-0.176019d (-0.021959phase); full covariance linear propagated error0.411170d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0445999solar masses; assumed primary mass1.150, minimum companion mass0.494. Approximate central duration8.19h versus fitted median4.70h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

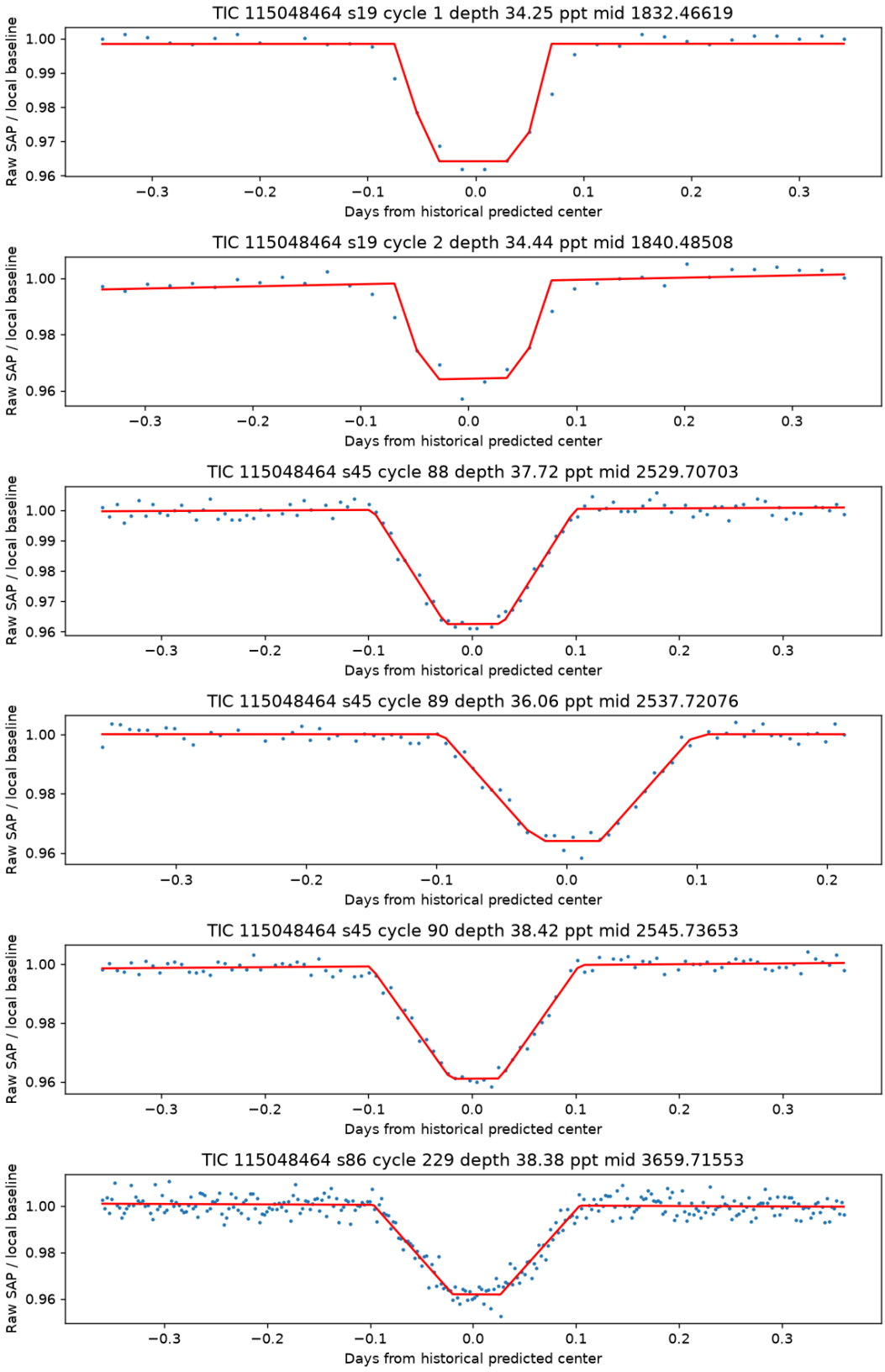
Limitations: Fitted duration and depth change across cadence/sector apertures; no precision intrinsic depth or common duration is claimed. The capable neighbour at51.03arcsec is separated from the target-core deficit.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

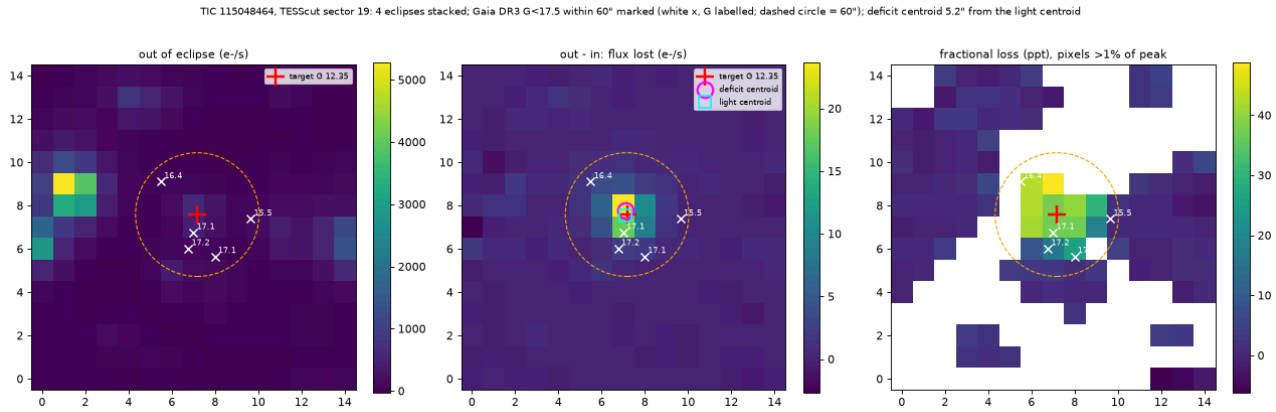
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
19	1	1832.46619041	0.01041680	34.249
19	2	1840.48508416	0.01041680	34.439
45	88	2529.70702596	0.00347237	37.720
45	89	2537.72076455	0.00347237	36.060
45	90	2545.73653228	0.00347237	38.423
86	227	3643.68701982	0.00115743	37.903
86	229	3659.71553079	0.00115743	38.378

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s19.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R325. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R325; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R325; diff_s19.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R326. TIC 331861613

physical_verdict: PLAUSIBLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve cycle-count aliases with an observed adjacent eclipse and resolve source localization.

Archived source: sanitized report text.

TIC 331861613: PLAUSIBLE

Full bounded review: COMPLETE. All five checks were performed; scientific uncertainty is retained separately from execution completeness. Catalogue novelty is supplied by the shared scoped catalogue receipts and is not inferred from this physical classification.

1. **Raw repeated events: pass.** 3 usable free-center trapezoid fits to quality-zero raw QLP SAP_FLUX in sectors[17, 18, 85] show localized eclipse morphology. Numeric raw-windows.csv and raw-events.png are deciding artifacts. Initial selection uses up to three widely separated available sectors; the two apparent two-event cases received bounded extra earlier-sector checks. Flagged and uncovered windows are never called clean negatives. A local linear baseline, depth, center, duration and ingress are fitted; no Gaia-fixed center is used as a measured midpoint.
2. **Eclipse timing: conditional; cycle aliases unresolved.** P=18.1534618869 +/-0.0000784945d, epoch BTJD1770.97977338; epoch kind: Selected phase-zero eclipse; not assumed deepest primary. The legacy CSV name primary_epoch_btjd does not assert this is the deeper eclipse. Cycle assignments start from the historical photometric branch; an independently recomputed weighted linear fit reproduces all numbers. Extra earlier S18 downloaded because initial three sectors yielded only two timed events. S17 cycle0,S18 cycle2,S85 cycle102 repeat at three epochs, but no adjacent-cycle pair; S58 cycle62 has zero quality-zero samples in +/-0.2d, not null. Integer/subharmonic ambiguity requires helper decision; full available QLP list17,18,58,85 checked. Close capable14arcsec neighbour also unresolved. This is a bounded recurrence/morphology audit, not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after checking capable neighbours.** Identity: Gaia DR3410380178061244416; cutout WCS and selected target agree. Lost-light centroid is0.049px from target; peak[8, 7] versus target pixel[8, 7]. Historical PSF residual0.538. Capable-neighbour review: Gaia410380178061244800 at13.99arcsec, required intrinsic fractional eclipse0.726, distance from deficit0.684px; Gaia410380173765981056 at23.03arcsec, required intrinsic fractional eclipse0.441, distance from deficit1.173px; Gaia410380139406239744 at40.59arcsec, required intrinsic fractional eclipse0.432, distance from deficit2.010px; Gaia410380105046671488 at45.04arcsec, required intrinsic fractional eclipse0.509, distance from deficit2.255px; Gaia410380109341769472 at45.33arcsec, required intrinsic fractional eclipse0.105, distance from deficit2.276px; Gaia410380276842680320 at45.50arcsec, required intrinsic fractional eclipse0.582, distance from deficit2.332px; Gaia410380349860139520 at46.18arcsec, required intrinsic fractional eclipse0.869,

distance from deficit2.297px; Gaia410380349859937920 at48.73arcsec, required intrinsic fractional eclipse0.528, distance from deficit2.435px; Gaia410380143701508736 at50.61arcsec, required intrinsic fractional eclipse0.617, distance from deficit2.518px; Gaia410380384219876096 at52.48arcsec, required intrinsic fractional eclipse0.170, distance from deficit2.586px; Gaia410380315500194944 at54.27arcsec, required intrinsic fractional eclipse0.035, distance from deficit2.746px; Gaia410380178061247488 at57.67arcsec, required intrinsic fractional eclipse0.840, distance from deficit2.834px Required intrinsic eclipse fractions and observed blended-pixel fractional losses have different dilution and are not compared as a veto. No calibrated centroid confidence ellipse or precise exclusion probability is claimed.

4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut S17 gives1x1/3x3/5x5 aperture depths10.223/9.849/7.372ppt from1 stacked event(s). QLP fitted selected-event median is11.888ppt. These are reductions of the same TESS photons, not independent observations. Different apertures, dilution and possible mixed eclipse-kind stacks preclude a claim of equal intrinsic depth.
5. **Gaia orbital geometry: conditional on the unresolved photometric cycle assignment.** Full NSS orbit: P=18.1514556913+/-0.0021802734d, e=0.451459, omega=336.4939deg, periastron BTJD388.294085. Measured-Gaia period difference0.00200620d. Closest edge-on conjunction offset0.124221d (0.006844phase); full covariance linear propagated error0.232960d at the measured epoch. Both conjunction branches and covariance details are retained in gaia-geometry.json. Mass function0.0485949solar masses; assumed primary mass1.050, minimum companion mass0.486. Approximate central duration13.51h versus fitted median5.61h uses TIC radius and an assumed main-sequence minimum-mass companion, not a fitted geometry. There is no arbitrary sigma veto, and neither this duration estimate nor the formal timing error establishes precision orbital association.

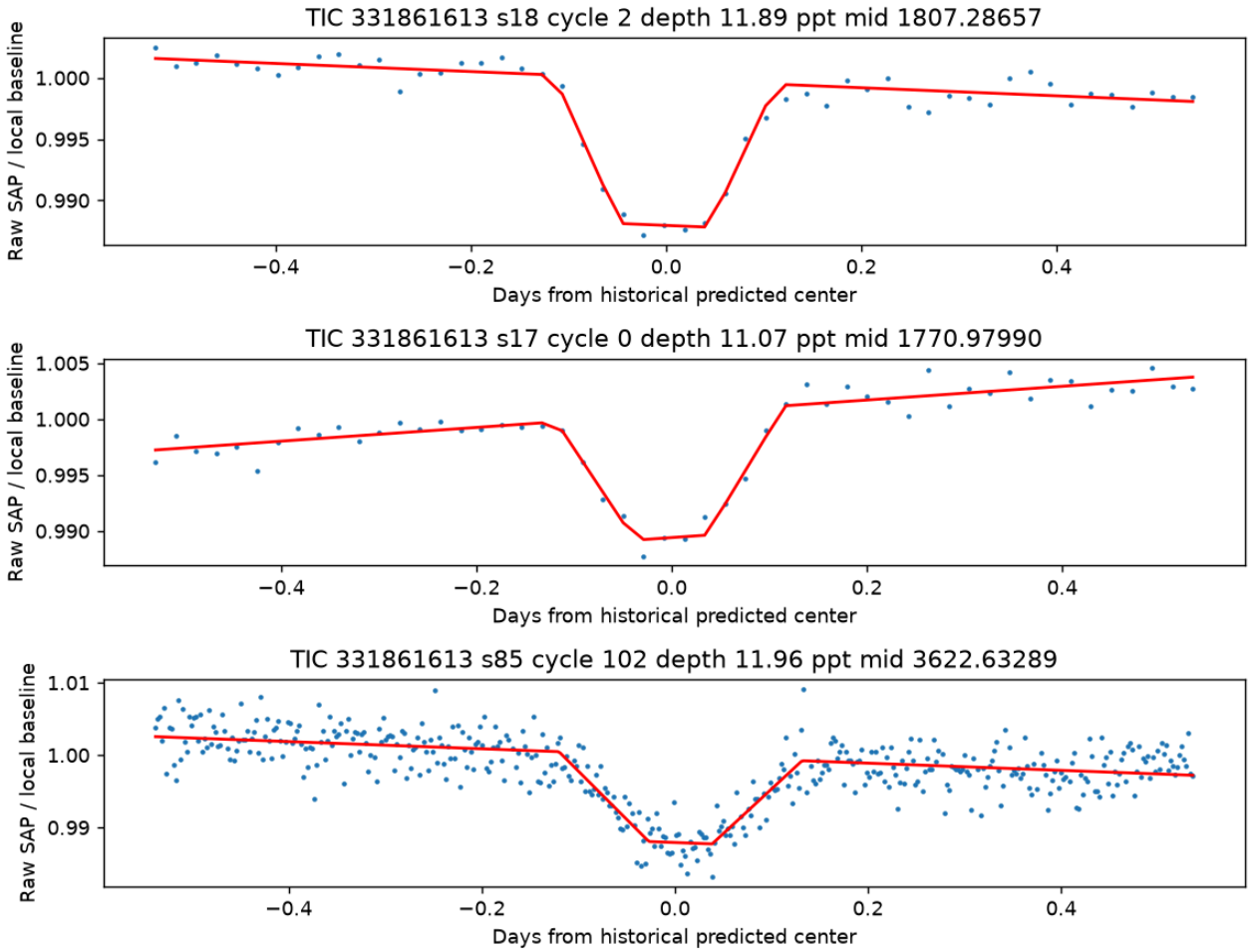
Limitations: Three measured events at historical cycles0,2,102 repeat, but no adjacent-cycle pair fixes the integer/subharmonic choice. The18.15346d ephemeris and its conditional fit error assume those cycle assignments; alternative periods remain unresolved. Available archived sectors17,18,58,85 were checked, including added earlierS18. S58 cycle62 has zero quality-zero samples within0.2d, so it is not a null. Twelve capable neighbours include one at13.99arcsec; attribution also remains unresolved.

Next check: Resolve cycle-count aliases with an observed adjacent eclipse and resolve source localization.

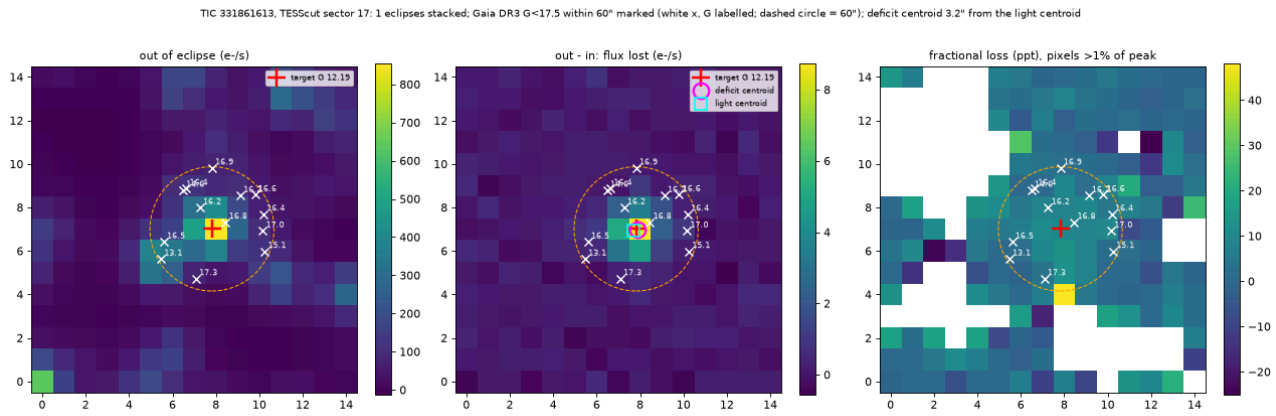
Sector	Historical photometric cycle	Measured midpoint BTJD	Fit-input error d	Depth ppt
17	0	1770.97990271	0.01041681	11.068
18	2	1807.28656523	0.01041651	11.888
85	102	3622.63288604	0.00293361	11.959

Provenance: measurements.json, raw-windows.csv, raw-events.png, historical/diff_s17.json and matching PNG/gaia.json; ../download-receipts.json, ../extra-downloads.json, ../alias-audit.json, ../recurrence-coverage.json, ../pixel-audit.json, ../gaia-geometry.json, ../HELPER-AUDIT.md, ../LEAD-AUDIT.md and ../SHA256SUMS.json. Original cutout: [local path omitted] Catalogue evidence: ../../catalogues/coverage-matrix.csv. Historical extraction definitions: [local path omitted] 5/made/sb1-scale/diffimg/diffstack.py. This reviewed evidence does not claim exhaustive literature priority.

Stable review key: R326. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R326; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R326; diff_s17.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R327. TIC 240786248

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 240786248

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.003646; inventory G=12.0119.

Raw events and timing: 7 accepted measured centers in sectors [57, 58, 84, 85], cycles [100, 102, 168, 170, 103, 105, 171]. Selected period/separation 10.7677990999 +/- 1.31461e-05 days; epoch BTJD 2854.3337057380 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 75.4ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector57 difference deficit lies 0.065px from target. WCS target identity agrees with Gaia 4151783093676200 96; coordinate check 2.68e-11arcsec. No capable Gaia neighbor within the historical60arcsec list. Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Historical image and capable-neighbor numeric audit support target attribution.

Alternative reduction: historical custom TESScut3x3 event depths [78.019, 77.5]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=10.76826206 +/- 0.0013796d, eccentricity 0.32715; nearest edge-on conjunction offset -0.062046d (-0.0057619phase), descriptive linear covariance error 0.32855d. Mass function 0.018245solar, approximate central duration 5.697h versus representative measured 4.762h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

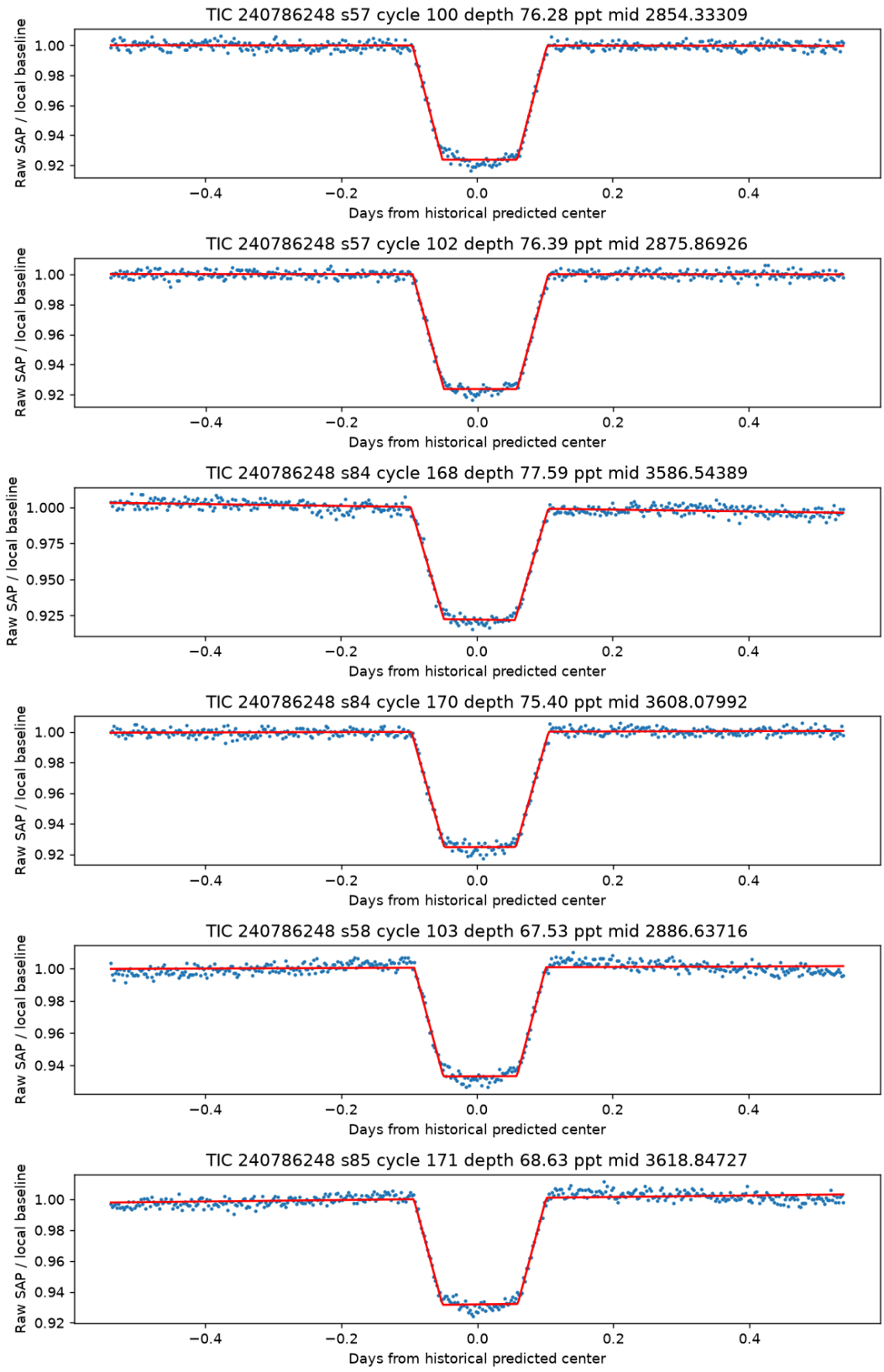
Limitations: Seven measured eclipses include consecutive cross-sector cycles102/103 and170/171 after checking all five available QLP sectors17,57,58,84,85. Even/odd depths are confounded with sectors/apertures; they do not independently exclude an orbital double alias. Repeated morphology, actual spacing and Gaia period support the retained association. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

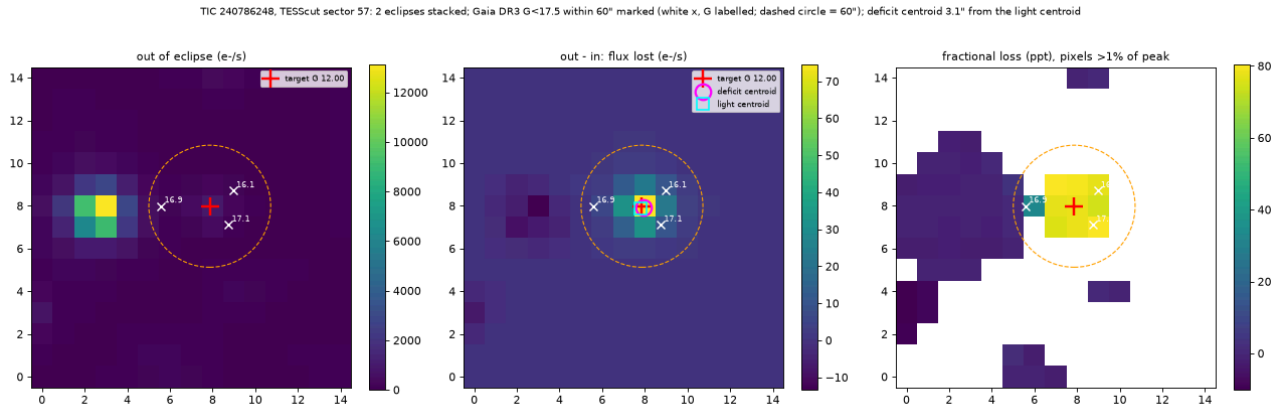
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s57.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R327. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R327; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R327; diff_s57.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R328. TIC 322048799

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 322048799

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.370095; inventory G=12.3826.

Raw events and timing: 4 accepted measured centers in sectors [96, 103], cycles [136, 137, 149, 150]. Selected period/separation 18.7267234834 +/- 8.87693e-05 days; epoch BTJD 3913.1527667597 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 241.6ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector96 difference deficit lies 0.156px from target. WCS target identity agrees with Gaia 4931814605587643008; coordinate check 0arcsec. No capable Gaia neighbor within the historical60arcsec list. Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Historical image and capable-neighbor numeric audit support target attribution.

Alternative reduction: historical custom TESScut3x3 event depths [265.515, 265.337]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=18.72837627 +/- 0.0038692d, eccentricity 0.53738; nearest edge-on conjunction offset -0.58466d (-0.031218phase), descriptive linear covariance error 0.75656d. Mass function 0.092767solar, approximate central duration 6.38h versus representative measured 5.197h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

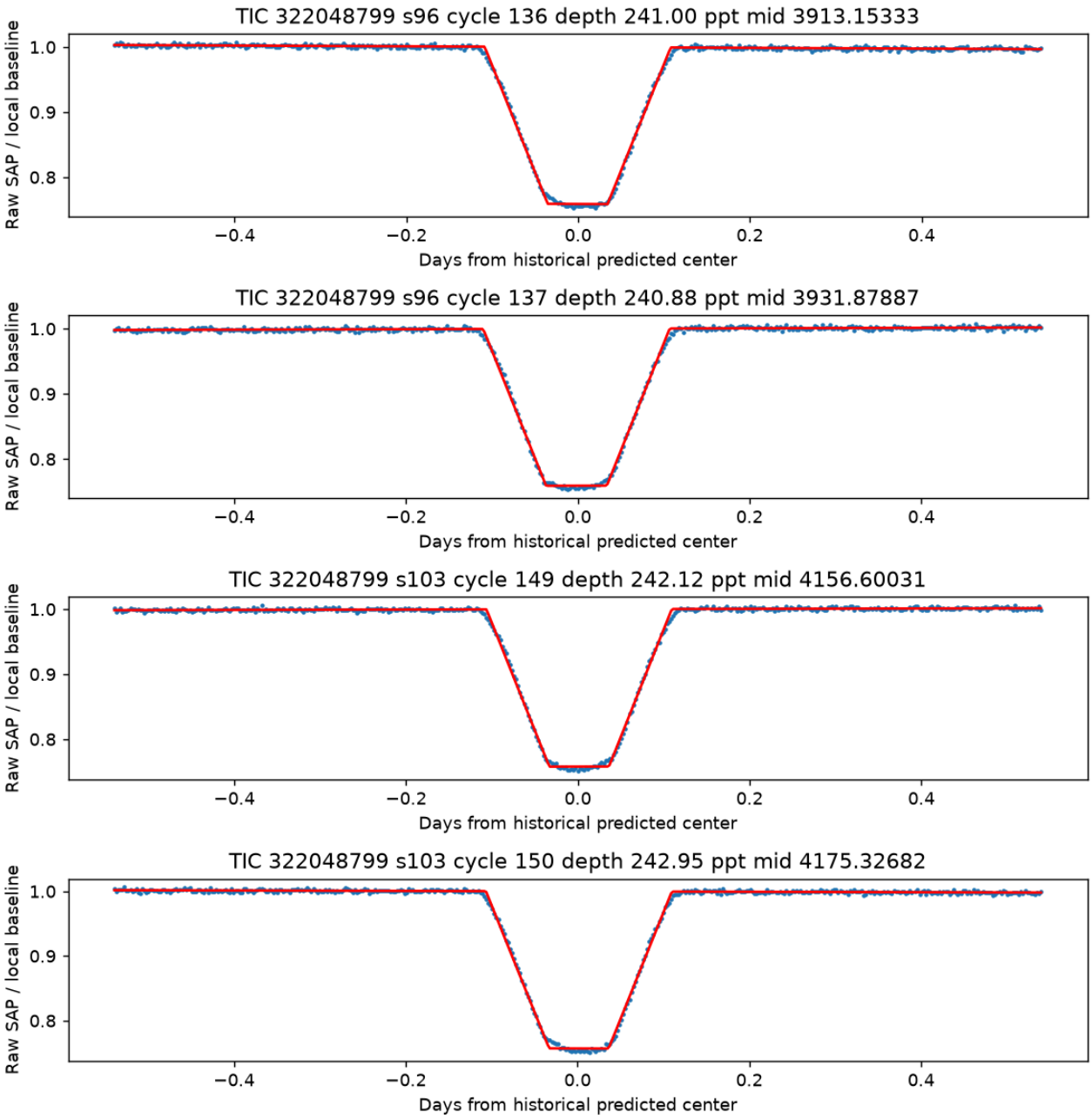
Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. A distinct secondary occurs near phase.20; it is not mixed into selected-event timings. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

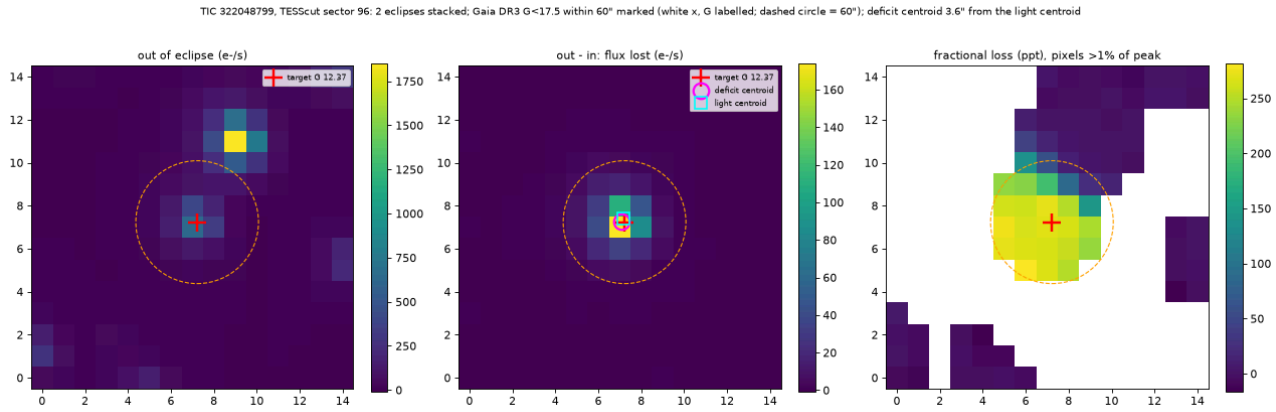
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s96.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R328. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R328; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R328; diff_s96.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R329. TIC 249834973

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 249834973

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.459992; inventory G=12.4686.

Raw events and timing: 10 accepted measured centers in sectors [18, 58, 85], cycles [-1, 0, 2, 189, 190, 191, 192, 193, 316, 319]. Selected period/separation 5.75740505436 +/- 7.35402e-06 days; epoch BTJD 1792.5373043261 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 65.17ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector58 difference deficit lies 0.144px from target. WCS target identity agrees with Gaia 450861236960143616; coordinate check 5.37e-11arcsec. Gaia 450861271320177536: separation 41.68arcsec, deficit distance 2.113px, required intrinsic fraction 0.191 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology.

Alternative reduction: historical custom TESScut3x3 event depths [57.13, 57.833, 57.149, 56.478]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=5.757135036 +/- 0.00061669d, eccentricity 0.082345; nearest edge-on conjunction offset -0.063223d (-0.010982phase), descriptive linear covariance error 0.1542d. Mass function 0.033643solar, approximate central duration 6.294h versus representative measured 5.582h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

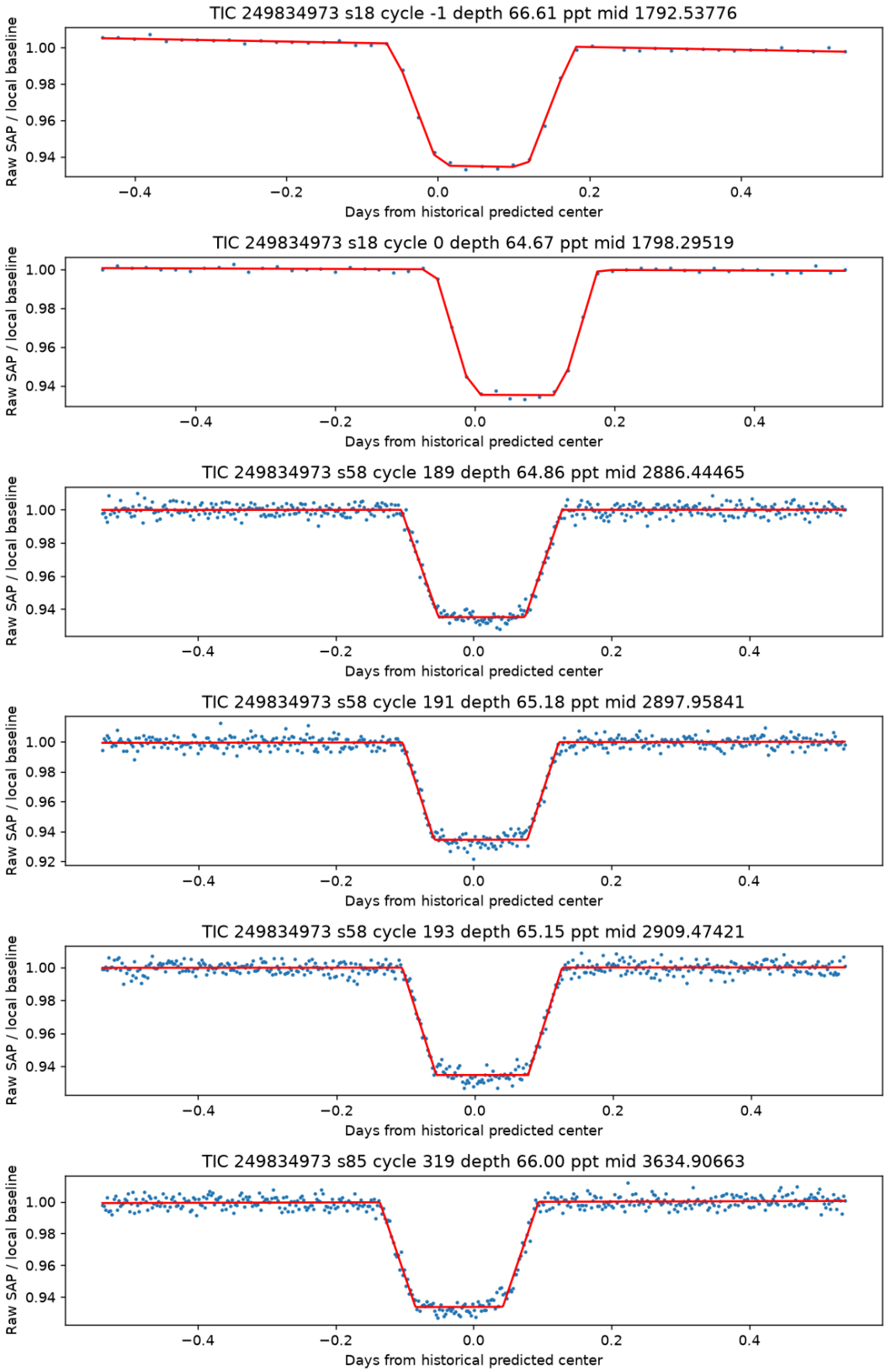
Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

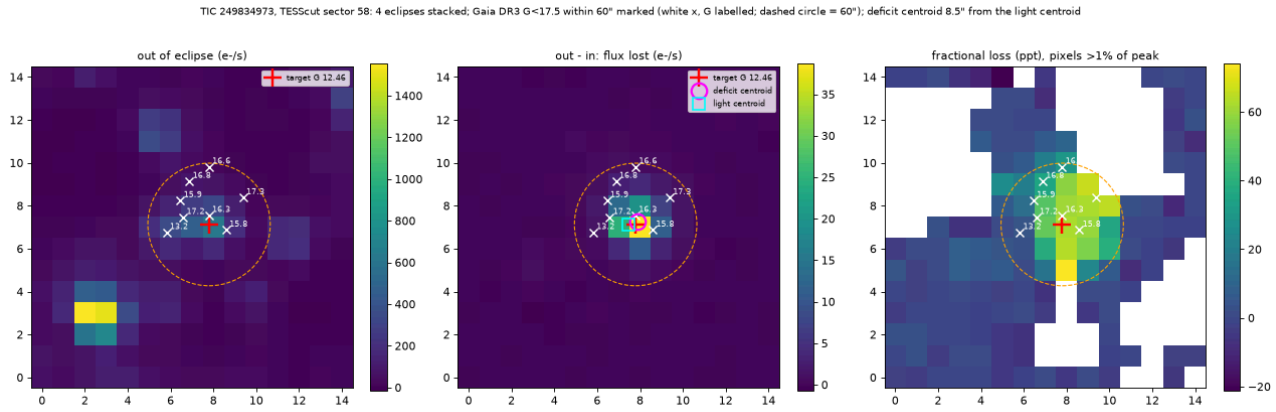
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s58.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R329. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R329; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R329; diff_s58.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R330. TIC 162289987

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 162289987

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.254248; inventory G=12.2645.

Raw events and timing: 4 accepted measured centers in sectors [56, 83], cycles [83, 84, 137, 138]. Selected period/separation 13.4287837332 +/- 2.14294e-05 days; epoch BTJD 2836.3581441376 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 102.4ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector56 difference deficit lies 0.074px from target. WCS target identity agrees with Gaia 1900057664482839168; coordinate check 2.56e-11arcsec. Gaia 1900057664482838272: separation 13.69arcsec, deficit distance 0.686px, required intrinsic fraction 0.797; Gaia 1900057767562052224: separation 27.24arcsec, deficit distance 1.304px, required intrinsic fraction 0.600; Gaia 1900057698843693312: separation 58.72arcsec, deficit distance 2.711px, required intrinsic fraction 0.161 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology.

Alternative reduction: historical custom TESScut3x3 event depths [72.609, 72.459]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

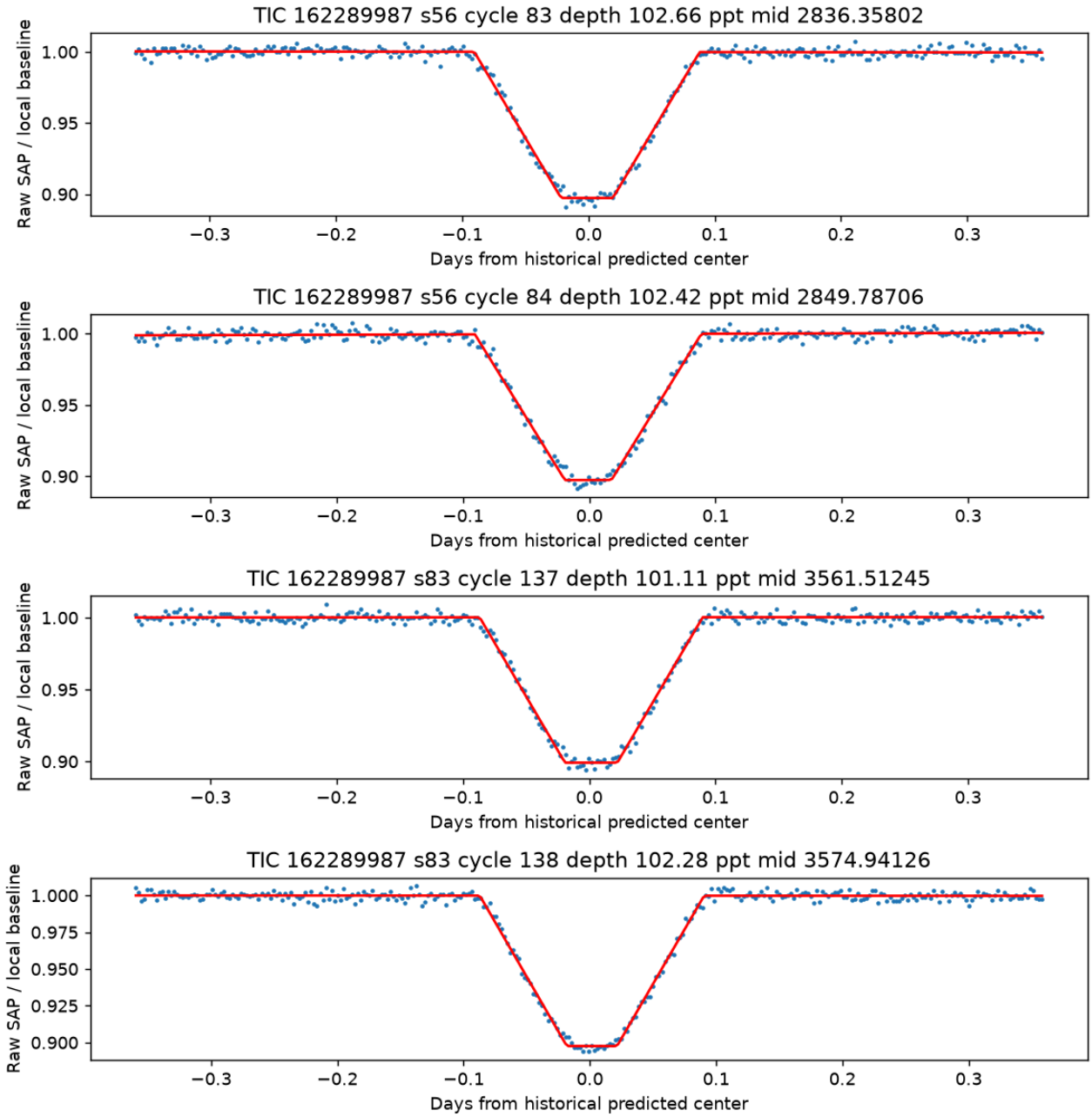
Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=13.42557836 +/- 0.0035549d, eccentricity 0.044707; nearest edge-on conjunction offset 0.46938d (0.034961phase), descriptive linear covariance error 0.62199d. Mass function 0.02195solar, approximate central duration 5.754h versus representative measured 4.275h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. A distinct secondary occurs near phase.53; it is not mixed into selected-event timings. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

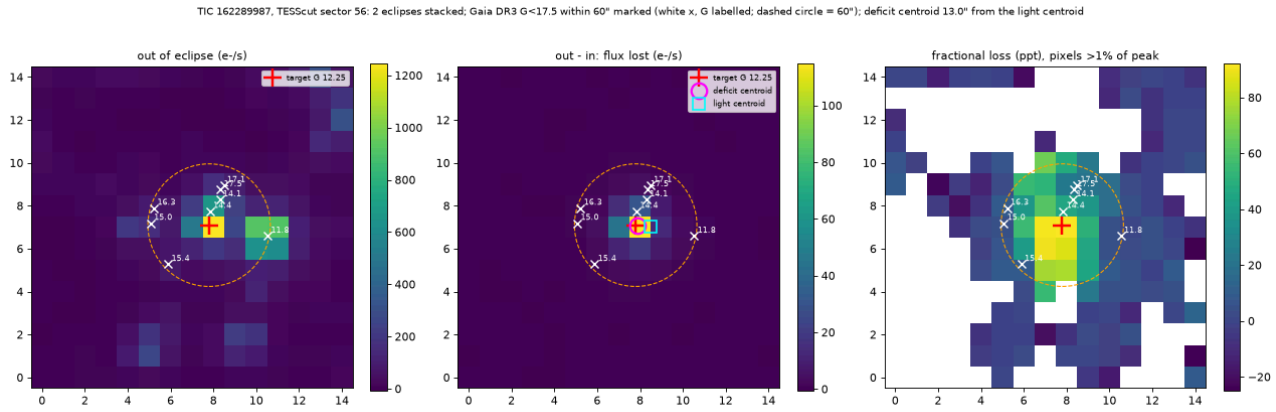
Next check: Optional precision characterization.

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s56.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R330. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R330; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R330; diff_s56.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R331. TIC 367461405

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 367461405

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.210865; inventory G=12.2195.

Raw events and timing: 6 accepted measured centers in sectors [18, 58, 85], cycles [0, 1, 67, 68, 112, 113]. Selected period/separation 16.2356660726 +/- 2.45504e-05 days; epoch BTJD 1796.1766804140 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 132ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector18 difference deficit lies 0.152px from target. WCS target identity agrees with Gaia 355352880968737536; coordinate check 5.12e-11arcsec. Gaia 355353091423471104: separation 32.16arcsec, deficit distance 1.651px, required intrinsic fraction 0.611; Gaia 355352919624776832: separation 39.73arcsec, deficit distance 1.793px, required intrinsic fraction 0.212 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology.

Alternative reduction: historical custom TESScut3x3 event depths [93.705, 91.84]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

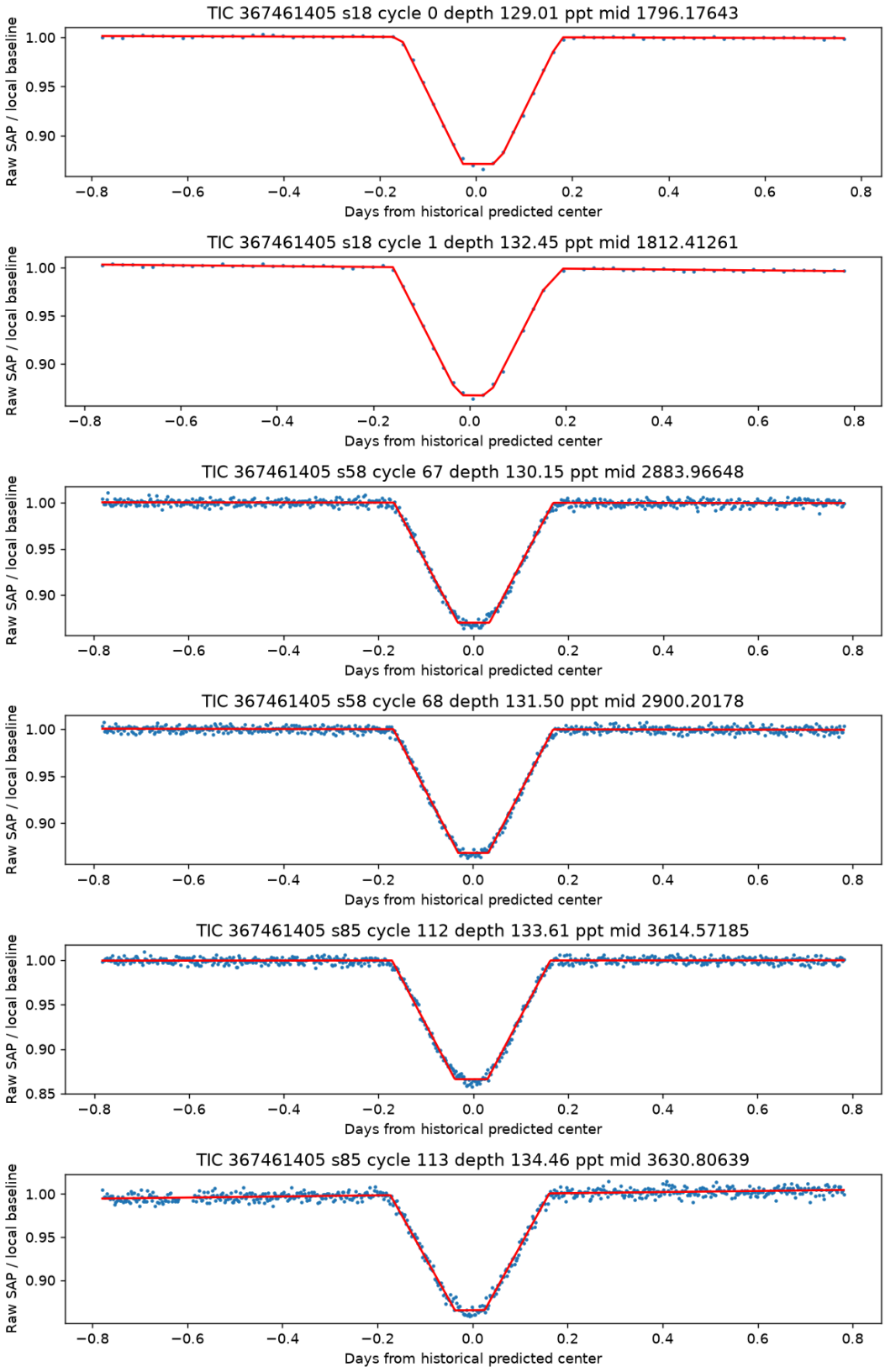
Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=16.23648148 +/- 0.008991d, eccentricity 0.28057; nearest edge-on conjunction offset 0.3967d (0.024433phase), descriptive linear covariance error 0.5521d. Mass function 0.19138solar, approximate central duration 11.62h versus representative measured 8.037h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. A distinct secondary occurs near phase.61; it is not mixed into selected-event timings. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

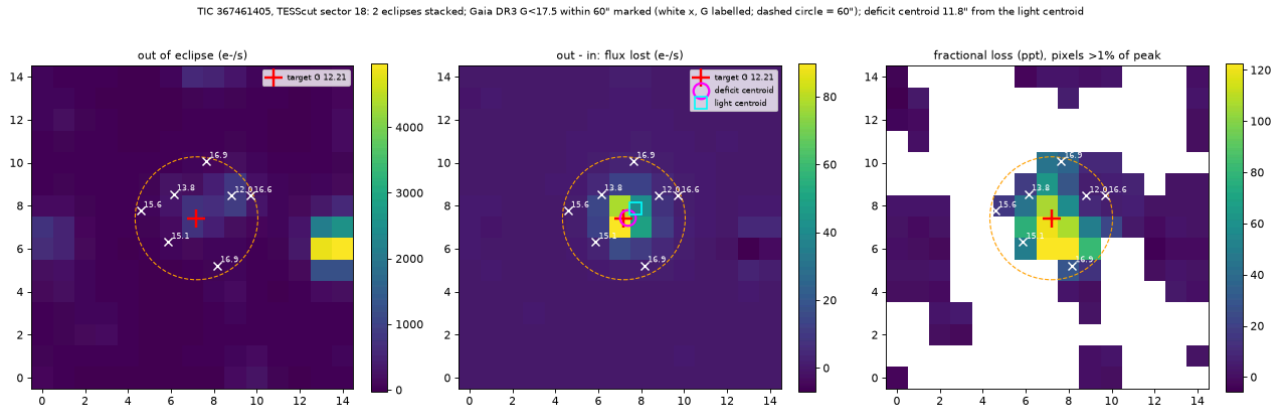
Next check: Optional precision characterization.

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s18.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R331. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R331; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R331; diff_s18.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R332. TIC 22026010

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 22026010

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.138945; inventory G=12.1474.

Raw events and timing: 9 accepted measured centers in sectors [9, 62, 99], cycles [0, 2, 3, 209, 210, 211, 212, 362, 365]. Selected period/separation 6.91281950163 +/- 6.30441e-06 days; epoch BTJD 1546.5402852917 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 244.4ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector9 difference deficit lies 0.084px from target. WCS target identity agrees with Gaia 5464923774774154496; coordinate check 1.28e-11arcsec. No capable Gaia neighbor within the historical60arcsec list. Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Historical image and capable-neighbor numeric audit support target attribution.

Alternative reduction: historical custom TESScut3x3 event depths [262.385, 260.366, 261.338, 257.946]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=6.912247787 +/- 0.00085065d, eccentricity 0.16776; nearest edge-on conjunction offset 0.0032664d (0.00047255phase), descriptive linear covariance error 0.16812d. Mass function 0.099224solar, approximate central duration 5.87h versus representative measured 4.757h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

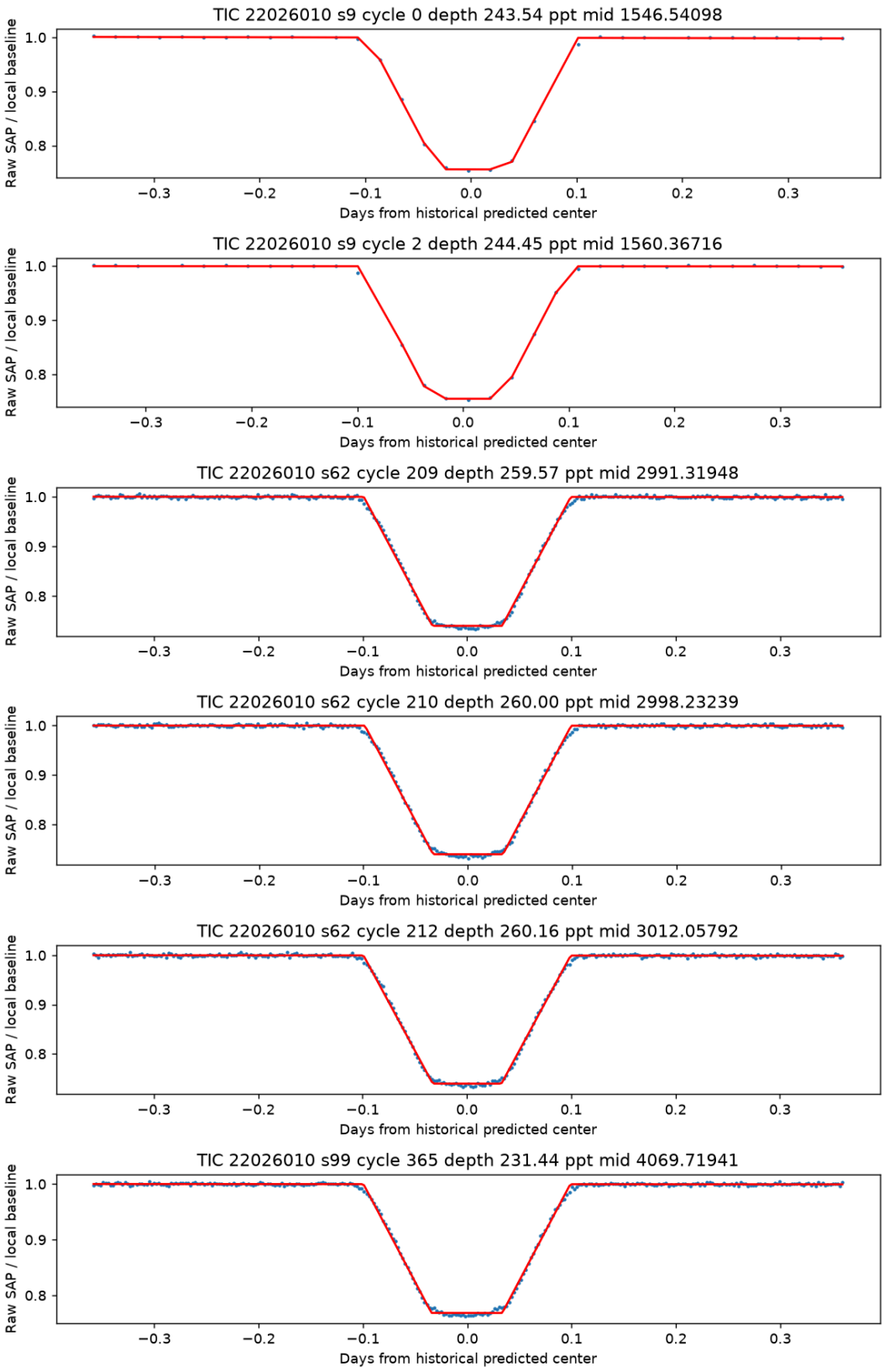
Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. A distinct secondary occurs near phase.60; it is not mixed into selected-event timings. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

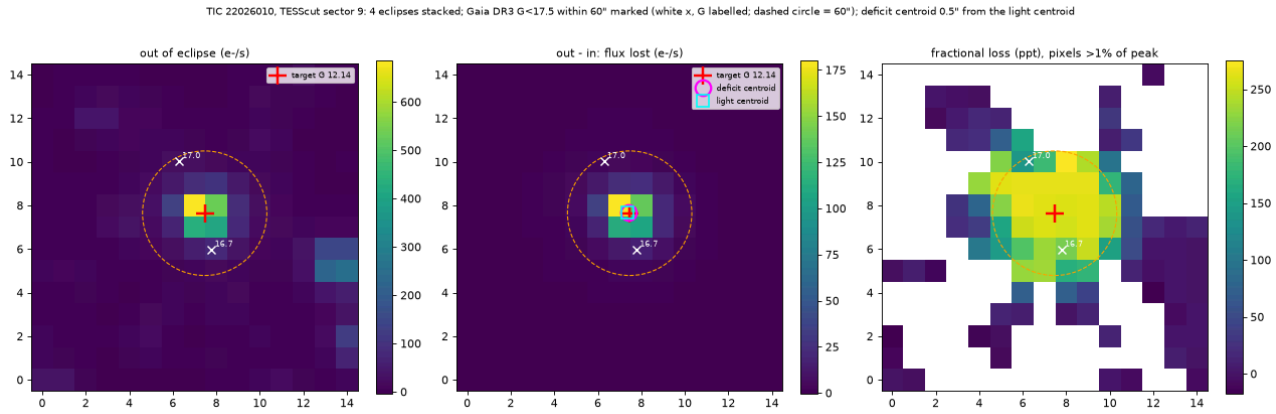
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s09.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R332. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R332; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R332; diff_s09.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R333. TIC 321044210

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 321044210

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.237933; inventory G=12.246.

Raw events and timing: 3 accepted measured centers in sectors [30, 104], cycles [43, 44, 163]. Selected period/separation 17.1621294823 +/- 2.27142e-05 days; epoch BTJD 2124.3460263829 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 18.25ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector30 difference deficit lies 0.166px from target. WCS target identity agrees with Gaia 4948003127599399936; coordinate check 0arcsec. Gaia 4948003127599400064: separation 26.21arcsec, deficit distance 1.287px, required intrinsic fraction 0.242 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology.

Alternative reduction: historical custom TESScut3x3 event depths [16.668, 15.479]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=17.15732289 +/- 0.0052721d, eccentricity 0.39269; nearest edge-on conjunction offset 0.27282d (0.015901phase), descriptive linear covariance error 0.42285d. Mass function 0.13212solar, approximate central duration 5.512h versus representative measured 2.204h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

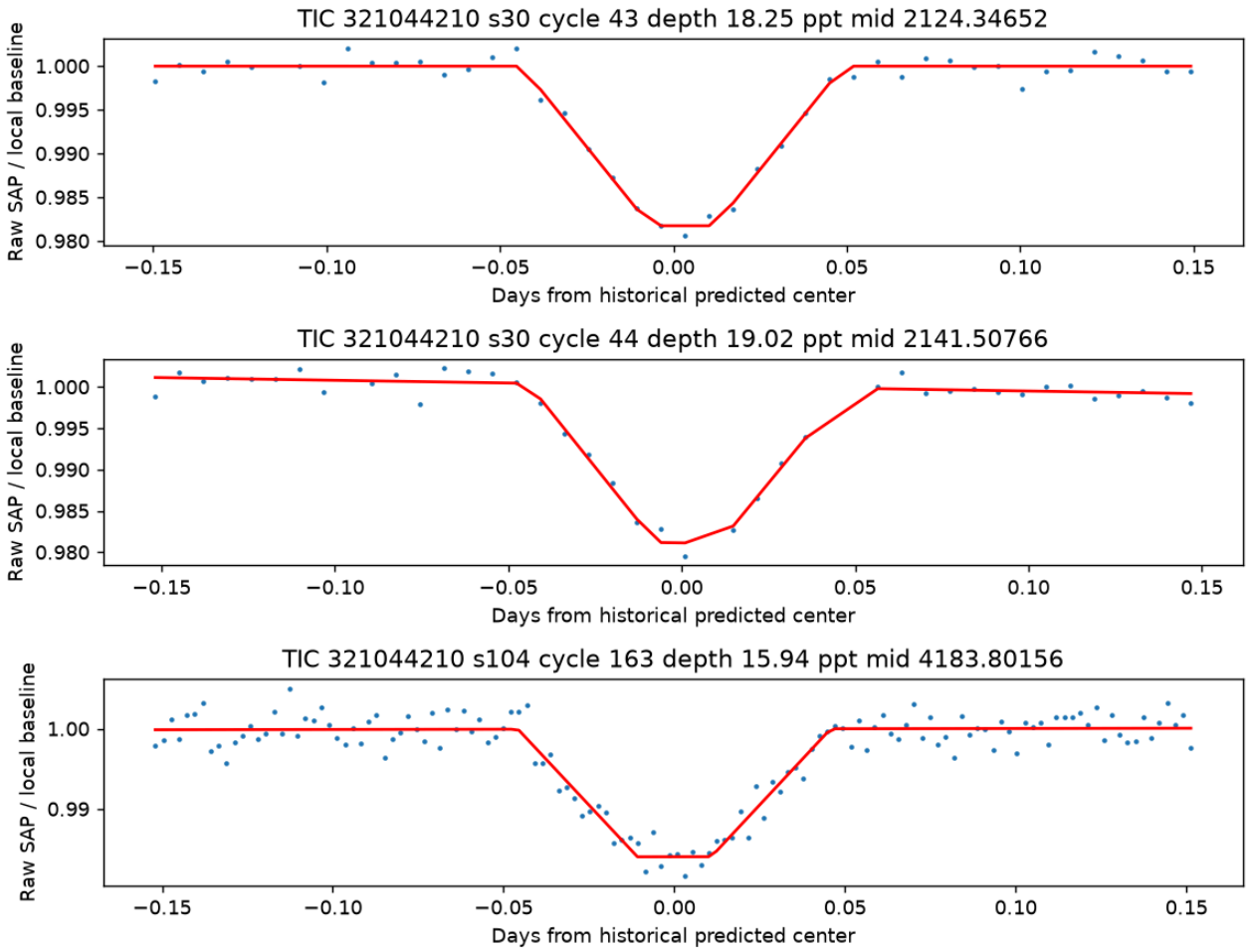
Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

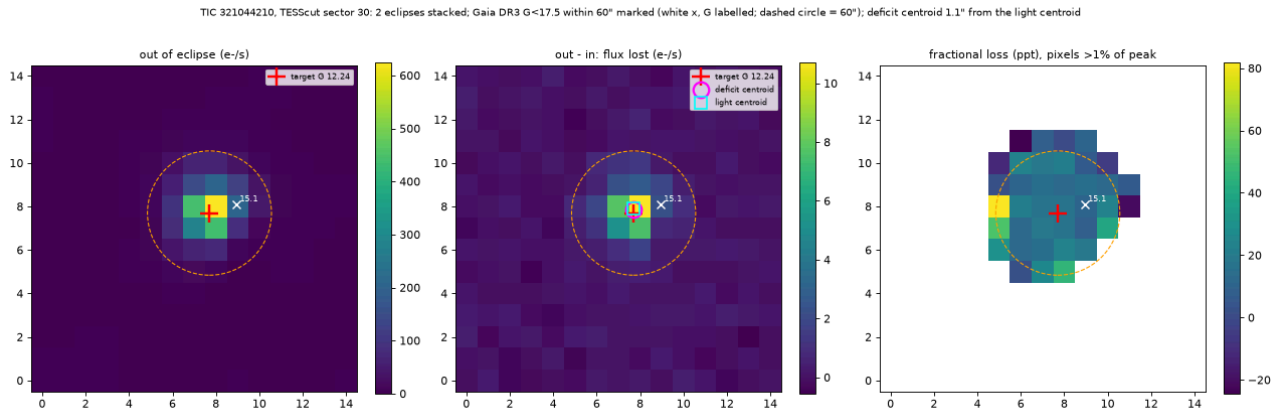
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s30.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R333. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R333; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R333; diff_s30.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R334. TIC 365325928

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 365325928

CONFIRMED; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.087224; inventory G=12.0981.

Raw events and timing: 6 accepted measured centers in sectors [13, 100, 103], cycles [0, 1, 175, 176, 181, 182]. Selected period/separation 13.7834802504 +/- 4.05533e-05 days; epoch BTJD 1665.4741263337 (Selected phase-zero eclipse; deepest primary not assumed). Median QLP raw depth 134.5ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector13 difference deficit lies 0.040px from target. WCS target identity agrees with Gaia 6629440217910875520; coordinate check 0arcsec. Gaia 6629441038245168000: separation 53.15arcsec, deficit distance 2.550px, required intrinsic fraction 0.339 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology.

Alternative reduction: historical custom TESScut3x3 event depths [144.955, 145.882]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=13.78076072 +/- 0.0018094d, eccentricity 0.4022; nearest edge-on conjunction offset -0.18529d (-0.013446phase), descriptive linear covariance error 0.21673d. Mass function 0.041406solar, approximate central duration 10.42h versus representative measured 8.812h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

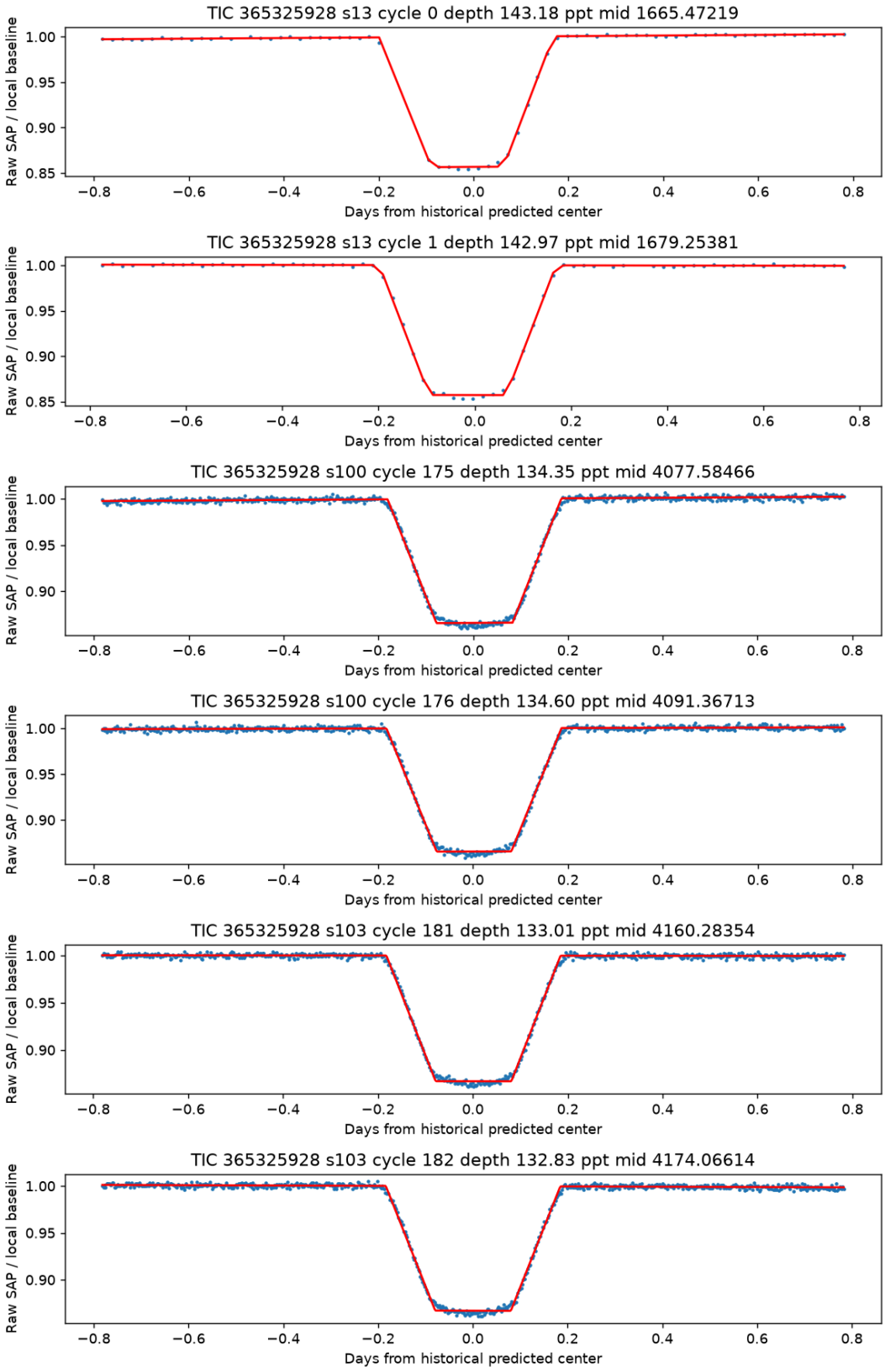
Limitations: Representative same-kind raw eclipses recur within a sector. Photometric period is fitted from measured centers; the historical photometric branch guides extraction. No exhaustive blind alias scan claimed. A distinct secondary occurs near phase.52; it is not mixed into selected-event timings. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

Next check: Optional precision characterization.

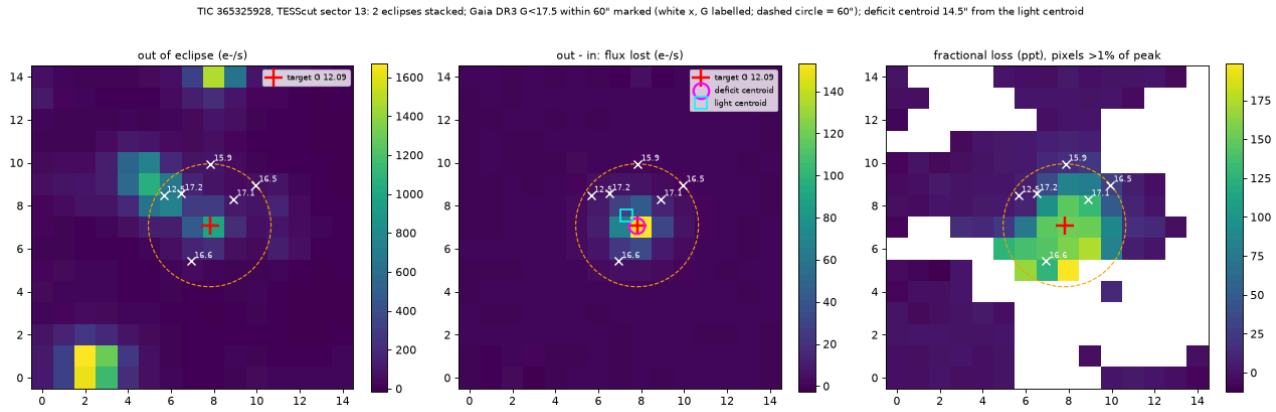
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s13.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and

SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R334. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R334; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R334; diff_s13.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R335. TIC 377129604

physical_verdict: PLAUSIBLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Obtain a third deep eclipse and test same-kind/double aliases

Archived source: sanitized report text.

TIC 377129604

PLAUSIBLE; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.135763; inventory G=12.1448.

Raw events and timing: 2 accepted measured centers in sectors [72], cycles [0, 1]. Selected period/separation 18.2332343366 +/- 0.00163699 days; epoch BTJD 3265.4274216896 (First of the two deep S72 events; primary kind unresolved). Median QLP raw depth 55.7ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector72 difference deficit lies 0.120px from target. WCS target identity agrees with Gaia 3850559470740449920; coordinate check 1.28e-11arcsec. No capable Gaia neighbor within the historical60arcsec list. Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Historical image and capable-neighbor numeric audit support target attribution.

Alternative reduction: historical custom TESScut3x3 event depths [46.792, 56.32]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

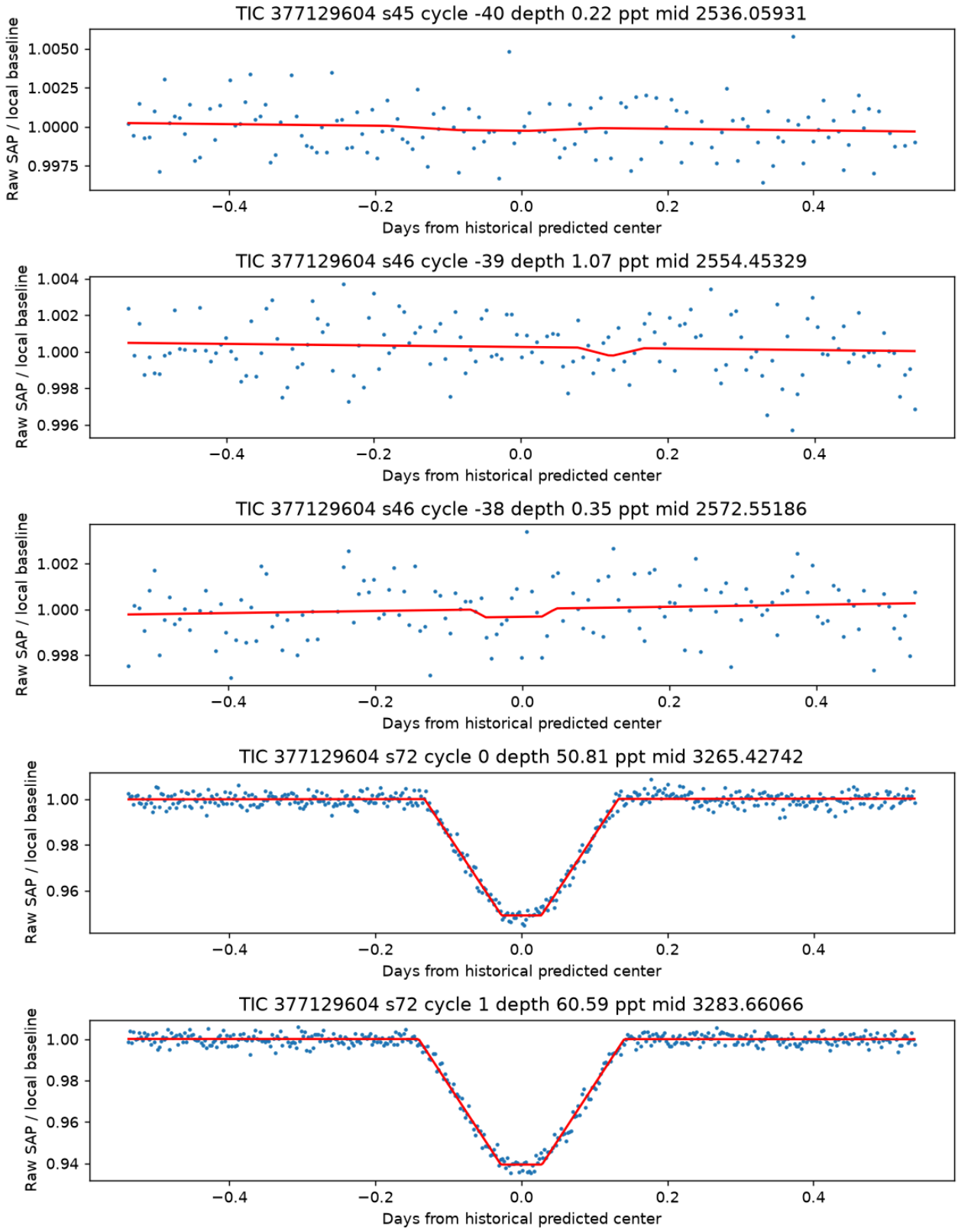
Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=18.24330138 +/- 0.0088054d, eccentricity 0.29798; nearest edge-on conjunction offset 0.6067d (0.033256phase), descriptive linear covariance error 1.3626d. Mass function 0.038711solar, approximate central duration 12.06h versus representative measured 6.567h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. Association remains uncertain with the unresolved event period; no arbitrary sigma veto applied.

Limitations: Only two deep S72 events are established. Their depths50.81/60.59ppt differ in QLP and custom aperture. The18.233234d value is a measured event separation, not an established same-kind orbital period;36.466469d double alias and eclipse-kind ambiguity remain. All available QLP sectors45,46,72 were inspected; earlier weak fits excluded, not used as timings. Earlier absence is not asserted as a clean null. Historical18.266535d period is not adopted. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

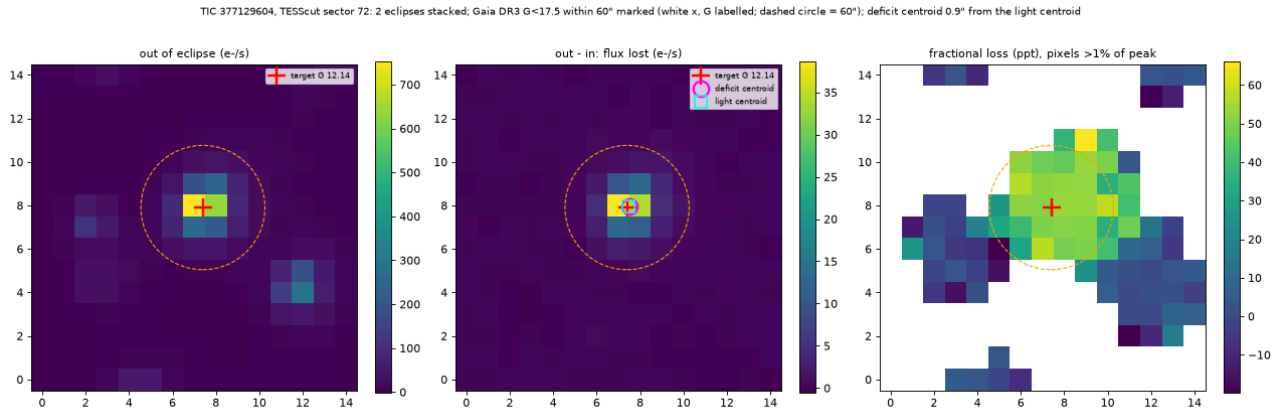
Next check: Obtain a third deep eclipse and test same-kind/double aliases.

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s72.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R335. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R335; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R335; diff_s72.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

R336. TIC 601646479

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve source with higher-resolution eclipse photometry

Archived source: sanitized report text.

TIC 601646479

PROBABLE; full_review_completion COMPLETE. The bounded five checks are complete; physical uncertainty is retained separately. Gaia DR3 G=12.4559765; inventory G=12.4571.

Raw events and timing: 13 accepted measured centers in sectors [57, 84], cycles [0, 2, 4, 8, 10, 12, 14, 414, 416, 418, 422, 424, 426]. Selected period/separation 3.53584755057 +/- 3.26717e-06 days; epoch BTJD 2855.0542777326 (Deeper alternating eclipse). Median QLP raw depth 73.38ppt. Predicted centers only guide windows; timings are fitted centers. Singular or zero-covariance fits are excluded. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Attribution: historical sector57 difference deficit lies 0.165px from target. WCS target identity agrees with Gaia 393144577378897536; coordinate check 2.12e-12arcsec. Gaia 393144577381887232: separation 1.89arcsec, deficit distance 0.137px, required intrinsic fraction 0.083; Gaia 393144611741629568: separation 54.13arcsec, deficit distance 2.688px, required intrinsic fraction 0.763 Intrinsic neighbor eclipse requirements and blended pixel losses have different dilution and are not compared as a veto. Helper inspected target/neighbor image morphology; the close source remains unresolved.

Alternative reduction: historical custom TESScut3x3 event depths [23.14, 14.397, 21.768, 13.414, 22.951, 23.163, 14.872, 22.965, 14.402, 21.746, 14.247, 22.908, 14.534]ppt. These repeat the eclipse pattern in a separate extraction of the same TESS photons; they are not independent observing occasions.

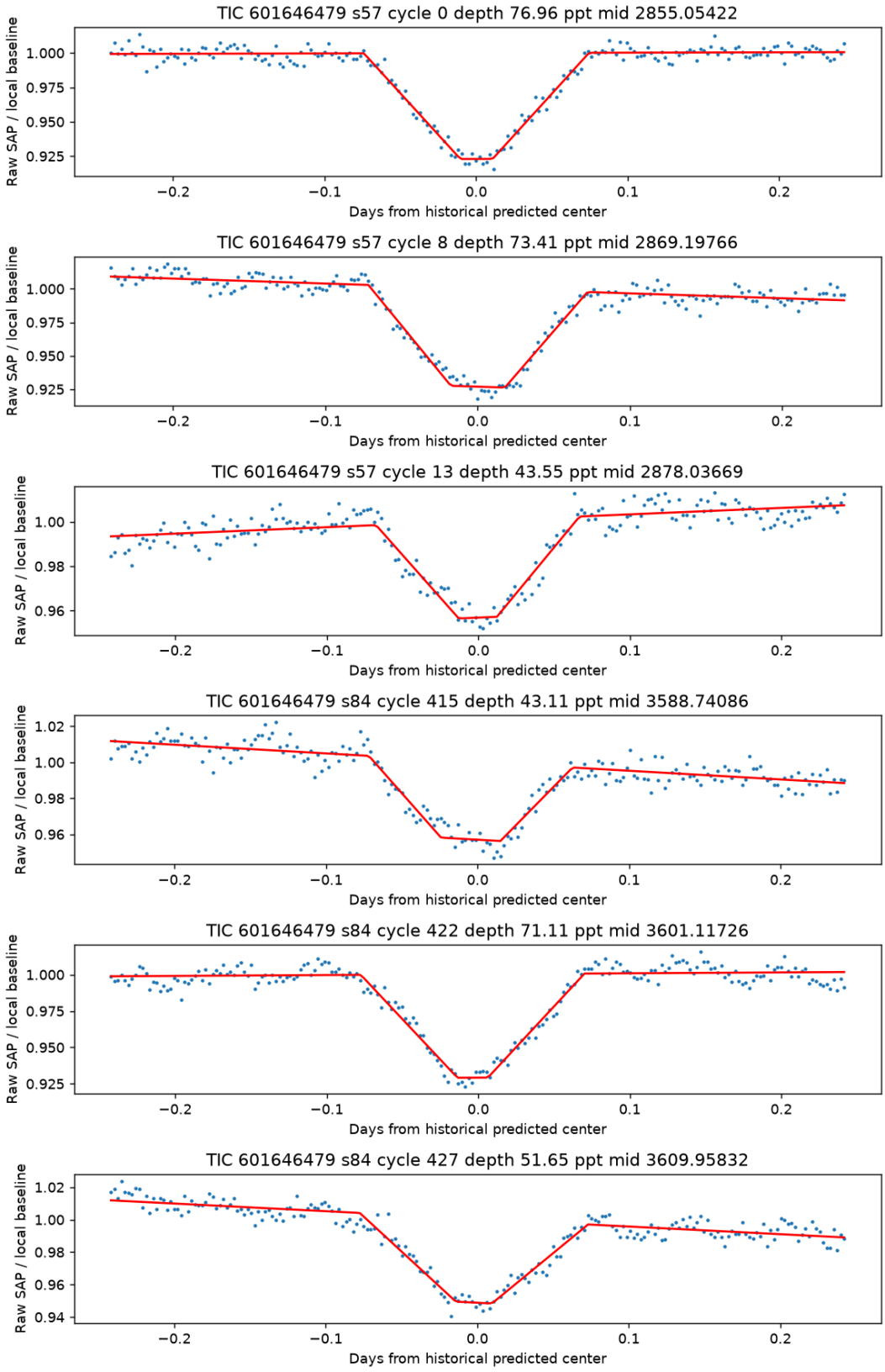
Gaia geometry: full NSS correlation vector propagated at observed epoch. Gaia P=3.535862158 +/- 0.00021477d, eccentricity 0.11456; nearest edge-on conjunction offset 0.084708d (0.023957phase), descriptive linear covariance error 0.15219d. Mass function 0.20385solar, approximate central duration 5.95h versus representative measured 3.485h. Duration uses TIC radius and main-sequence minimum-mass companion and is not a fitted physical solution. This supports physical plausibility without an arbitrary sigma veto.

Limitations: Unequal alternating eclipses establish full3.535848d orbit from13 deeper even historical half-period cycles, orbital_cycle=cycle/2. Attribution remains unresolved against the capable1.89arcsec neighbor; its deficit distance0.137px is slightly smaller than target0.165px. Image cannot resolve these sources. Depths depend on aperture dilution and local baseline. Pixel centroid uncertainties and timing systematics are not calibrated.

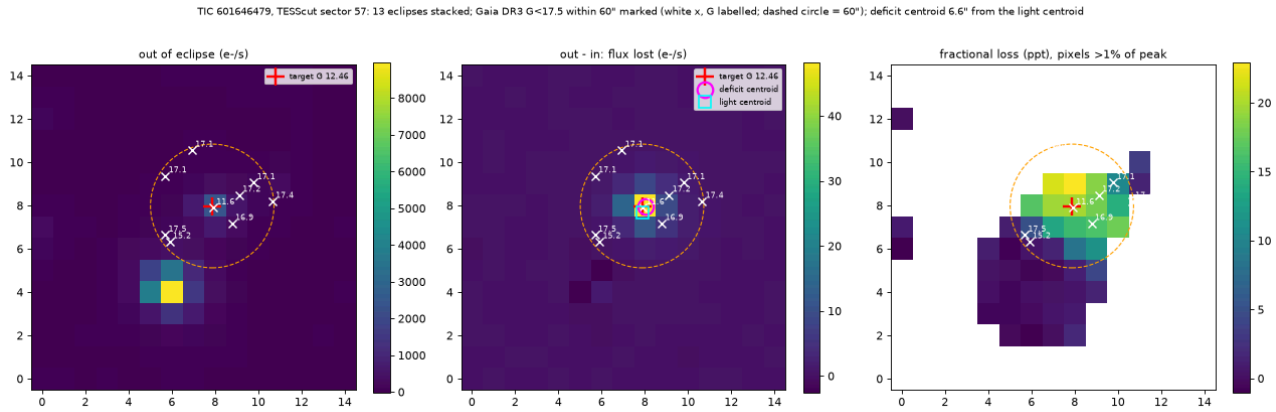
Next check: Resolve source with higher-resolution eclipse photometry.

Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-windows.csv, raw-events.png, historical/diff_s57.json; batch alias-audit.json, recurrence-coverage.json, pixel-audit.json, gaia-geometry.json, download-receipts.json and SHA256SUMS.json. Catalogue novelty is supplied separately by shared-catalogues and does not determine this physical verdict. No claim of exhaustive literature priority.

Stable review key: R336. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.



R336; raw-events.png. Representative raw-flux eclipse windows; timing and quality qualifications remain in the full report.



R336; diff_s57.png. Target-pixel eclipse-deficit evidence; capable-neighbour and angular-resolution limitations remain in the full report.

Spectroscopic-orbit period folds: remaining G 11.5-12 evidence

Completion of the bright screened stratum. The remaining 73 screened Gaia spectroscopic-orbit period-fold candidates at actual Gaia DR3 G = 11.5-12 (range 11.6578-11.9967) completed the same bounded review: 69 confirmed, 3 probable and 1 plausible; 72 are eligible unique additions. This follow-up completes that remaining bright stratum and is separate from the fixed balanced 120-target magnitude comparison, whose sample and descriptive results remain unchanged. Historical inventory-boundary exclusions that meet the actual DR3 interval retain their original substitution provenance. No targets were substituted in this follow-up. The remaining screened inventory contains 266 candidates: 90 at G 12-12.25, 96 at G 12.25-12.5 and 80 at G 12.5-12.75. Five weaker conjunction candidates, 58 lower-score candidates and 48 unexecuted exploratory appointment targets are separate populations. These outcomes do not measure blind discovery yield or completeness.

Period, epoch and attribution limits. TIC 16809274 retains a shallower-eclipse reference epoch. TIC 119193784 remains probable because integer-cycle aliases are unresolved; TIC 382631802 remains plausible and excluded because structured variability and broad-model duration tension leave the eclipse interpretation insecure; the pixel evidence favours the target. TIC 343340931 and TIC 363264601 remain probable because capable neighbours prevent definitive attribution. The equal-depth aliases of TIC 76020303 and TIC 168917716 are Gaia-informed interpretations; their eclipse timing uncertainties do not resolve the orbital half/double ambiguity. Full reports preserve measured event centers, quality flags, error meanings and source-specific limitations.

R337. TIC 135362193

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 135362193: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[53, 80]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=4.6690140750 ± 0.0000131496d, reference eclipse epoch BTJD2745.7473584, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S79 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation2.323arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.232. Capable neighbours: Gaia 4577603498730811008: 18.91 arcsec, G16.165, required fractional eclipse0.691; Gaia 4577603498730962560: 22.88 arcsec, G14.835, required fractional eclipse0.211
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths10.151/10.284/10.166ppt, with 4 stacked events. QLP selected-kind fitted median11.442ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3457603498730812672, P=4.6688816715 ± 0.0002208323d, e=0.038599, omega=117.5167deg. Closest edge-on conjunction offset0.046153d (0.009885phase), full-SB1-covariance linear propagated Gaia uncertainty0.103629d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.020711d (photometric period projection allowance0.006641d). The approximate central duration5.44h vs observed median2.09h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Shallow repeated eclipses across sectors53/80; capable neighbors at18.9/22.9arcsec lie0.91/1.06pixel from deficit versus target0.11pixel. Target-centered core supports attribution at TESS resolution; no calibrated localization errors. Sector80 pipeline offsets remain systematic uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

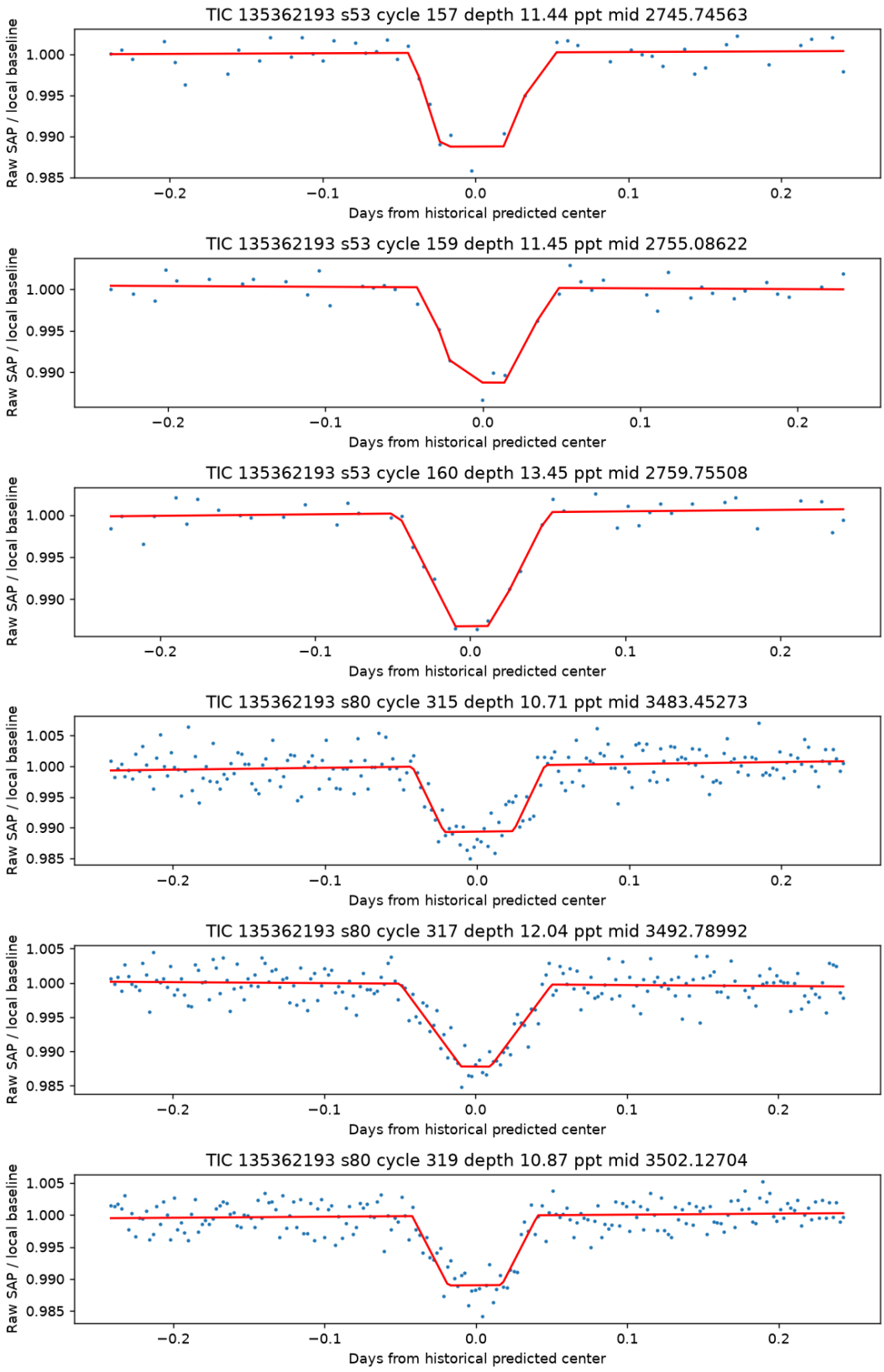
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
53	157	2745.7456293	0.003472	11.442	used in ephemeris
53	159	2755.0862173	0.003472	11.451	used in ephemeris
53	160	2759.7550782	0.003472	13.447	used in ephemeris
80	315	3483.4527319	0.001157	10.709	used in ephemeris
80	317	3492.7899207	0.001157	12.037	used in ephemeris
80	318	3497.4577896	0.001157	11.157	used in ephemeris
80	319	3502.1270379	0.001157	10.871	used in ephemeris

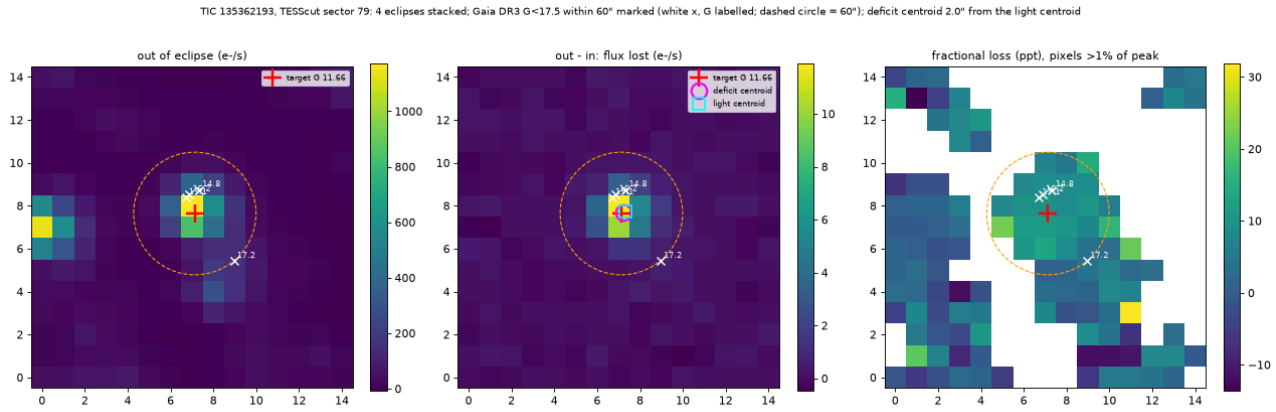
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s79.json, historical/diff_s79.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R337. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R337; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R337; diff_s79.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R338. TIC 346865542

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 346865542: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[42, 43, 70]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=12.0441549925 ± 0.0000365340d, reference eclipse epoch BTJD2449.5432269, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S42 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation0.666arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.080. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths60.315/59.572/57.677ppt, with 2 stacked events. QLP selected-kind fitted median67.432ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3256456430589727488, P=12.0444654688 ± 0.0006823424d, e=0.460253, omega=166.0696deg. Closest edge-on conjunction offset-0.081524d (-0.006769phase), full-SB1-covariance linear propagated Gaia uncertainty0.149158d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.028433d (photometric period projection allowance0.006247d). The approximate central

duration6.28h vs observed median3.82h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four deep isolated eclipses test within-sector recurrence and long-baseline cycle count. Aperture dilution retained; fitted depths are not intrinsic.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

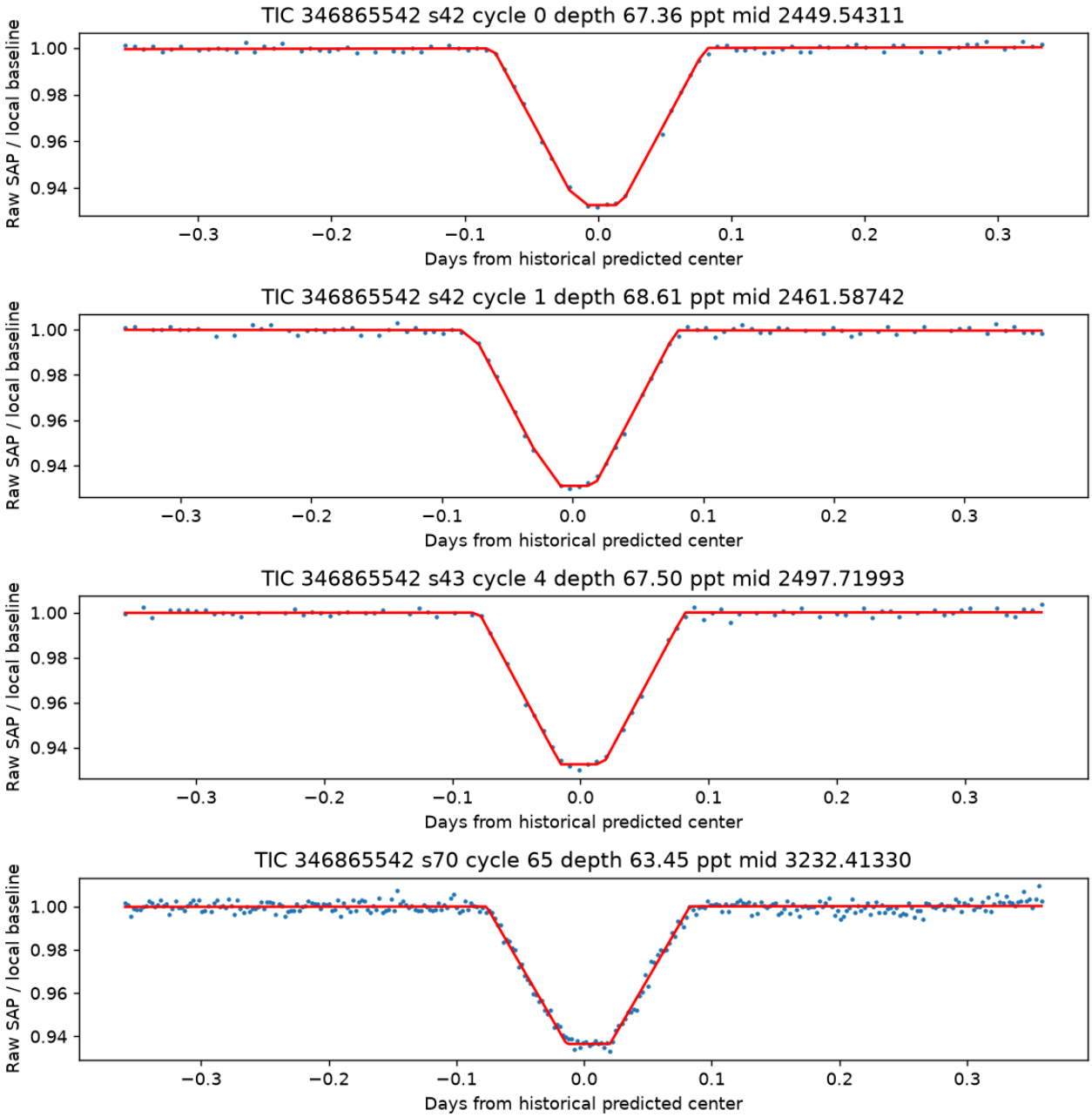
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
42	0	2449.5431115	0.003472	67.360	used in ephemeris
42	1	2461.5874220	0.003472	68.615	used in ephemeris
43	4	2497.7199278	0.003472	67.503	used in ephemeris
70	65	3232.4133008	0.001157	63.445	used in ephemeris

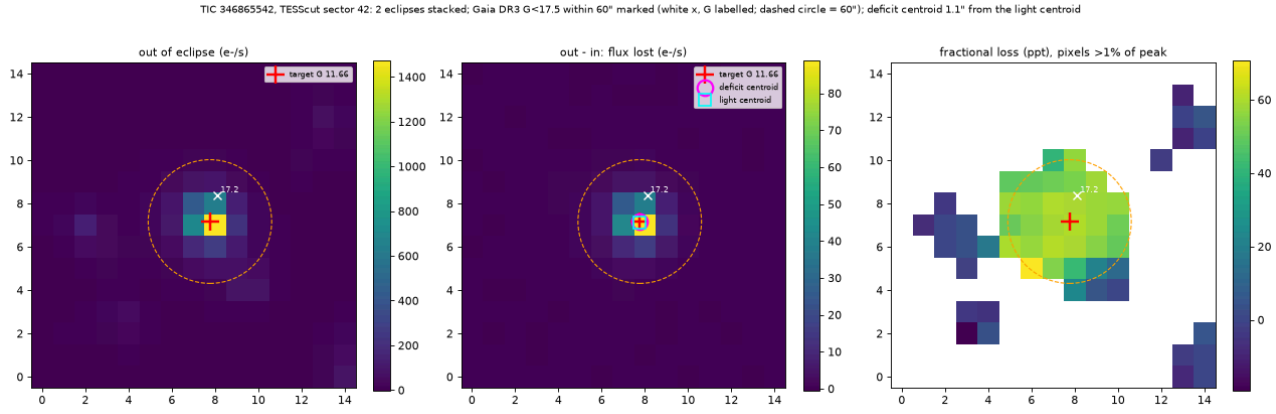
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s42.json, historical/diff_s42.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R338. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R338; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R338; diff_s42.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R339. TIC 131331287

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 131331287: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[37, 64, 102]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=6.8447853694 ± 0.0000054333d, reference eclipse epoch BTJD2313.5253293, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S64 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.201arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.046. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths121.693/118.758/116.981ppt, with 3 stacked events. QLP selected-kind fitted median135.367ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3616982634869970342 4, P=6.8443655076 ± 0.0009405552d, e=0.259533, omega=305.2292deg. Closest edge-on conjunction offset0.053274d (0.007784phase), full-SB1-covariance linear propagated Gaia uncertainty0.199939d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.064708d (photometric period projection allowance0.001527d). The approximate central

duration6.94h vs observed median5.70h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated events have reduced depths in sector102 relative to37/64; aperture dilution or baseline changes possible, no intrinsic depth variation claimed.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

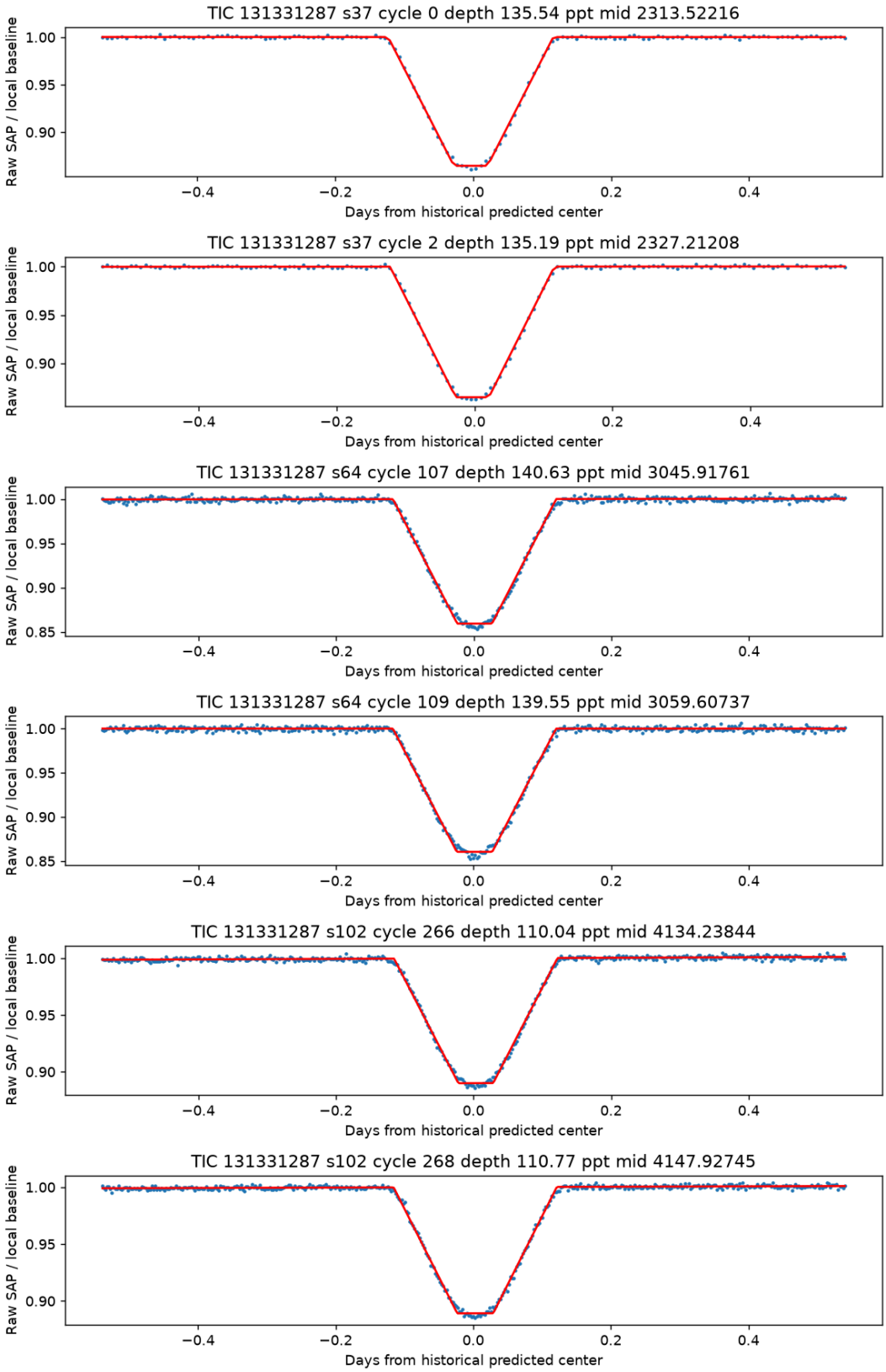
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
37	0	2313.5221586	0.003472	135.542	used in ephemeris
37	2	2327.2120790	0.003472	135.193	used in ephemeris
64	107	3045.9176080	0.001157	140.625	used in ephemeris
64	108	3052.7625806	0.001157	139.738	used in ephemeris
64	109	3059.6073706	0.001157	139.546	used in ephemeris
102	266	4134.2384426	0.001157	110.040	used in ephemeris
102	267	4141.0827260	0.001157	110.358	used in ephemeris
102	268	4147.9274541	0.001157	110.766	used in ephemeris

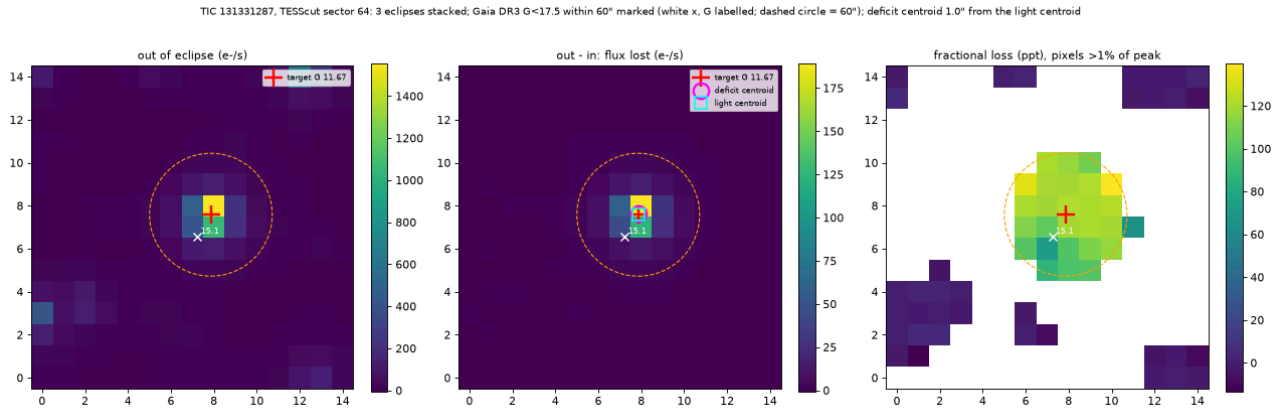
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s64.json, historical/diff_s64.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R339. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R339; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R339; diff_s64.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R340. TIC 48752905

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 48752905: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 52 distinct eclipse events in selected sectors[33, 87]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 25 same-kind midtimes give P=1.7220909305 ± 0.0000011993d, reference eclipse epoch BTJD2202.7901470, kind: Deeper repeating eclipse; even half-period cycles. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S33 deficit peak[8, 8] vs target pixel[7, 8]; deficit-target separation1.683arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.221. Capable neighbours: Gaia 2926491213995591296: 40.26 arcsec, G15.750, required fractional eclipse0.855; Gaia 2926491351434540160: 40.72 arcsec, G13.776, required fractional eclipse0.155; Gaia 2926491042196899200: 47.18 arcsec, G15.412, required fractional eclipse0.632
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths25.065/24.019/21.698ppt, with 28 stacked events. QLP selected-kind fitted median37.566ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3292649124405313971 2, P=1.7220692438 ± 0.0000166135d, e=0.015301, omega=141.9089deg. Closest edge-on conjunction offset0.026703d (0.015506phase), full-SB1-covariance linear propagated Gaia uncertainty0.018956d at the measured epoch. Near-Gaia-reference projected eclipse

offset0.003867d (photometric period projection allowance0.001263d). The approximate central duration4.35h vs observed median2.24h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses establish doubled period relative to historical SAP half-period. Deeper even historical cycles alone define epoch and orbital timing; mixed historical cutout stack is not a same-kind depth measurement.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

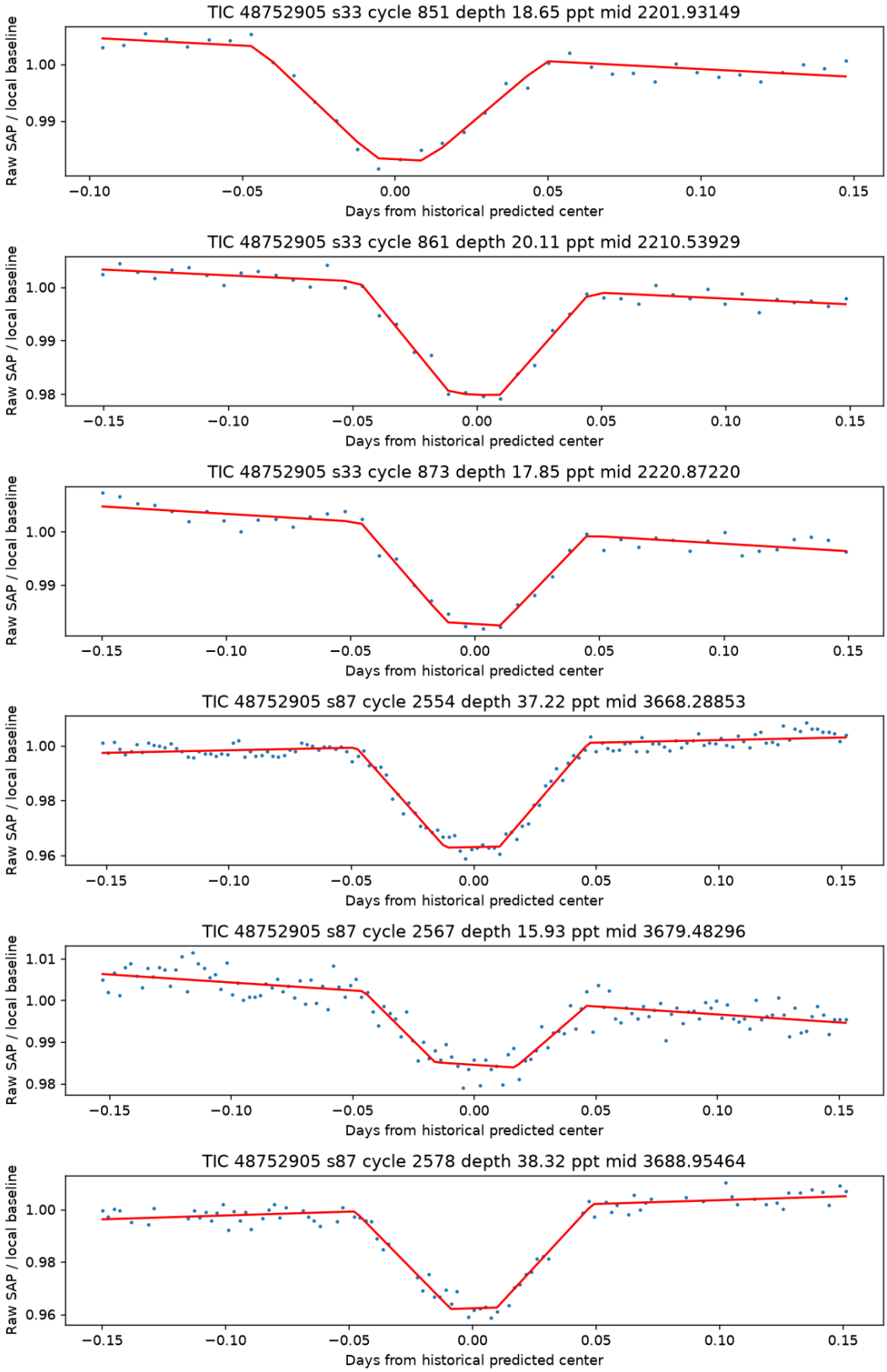
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	851	2201.9314905	0.003472	18.650	measured; excluded from primary fit (secondary)
33	852	2202.7893974	0.003472	38.298	used in ephemeris
33	853	2203.6511250	0.003472	18.214	measured; excluded from primary fit (secondary)
33	854	2204.5118227	0.003472	41.100	used in ephemeris
33	855	2205.3726943	0.003472	20.597	measured; excluded from primary fit (secondary)
33	856	2206.2344529	0.003472	36.151	used in ephemeris
33	857	2207.0949678	0.003472	20.597	measured; excluded from primary fit (secondary)
33	858	2207.9549409	0.003472	36.463	used in ephemeris
33	859	2208.8172361	0.003472	18.513	measured; excluded from primary fit (secondary)
33	860	2209.6795282	0.003472	36.596	used in ephemeris
33	861	2210.5392888	0.003472	20.109	measured; excluded from primary fit (secondary)
33	862	2211.4000198	0.003472	36.405	used in ephemeris
33	863	2212.2629854	0.003472	19.326	measured; excluded from primary fit (secondary)
33	864	2213.1232831	0.003472	37.366	used in ephemeris
33	867	2215.7066632	0.003472	18.369	measured; excluded from primary fit (secondary)
33	868	2216.5670950	0.003472	36.338	used in ephemeris
33	869	2217.4281851	0.003472	17.666	measured; excluded from primary fit (secondary)
33	870	2218.2895162	0.003472	37.164	used in ephemeris
33	871	2219.1522256	0.003472	19.003	measured; excluded from primary fit (secondary)
33	872	2220.0111150	0.003472	35.985	used in ephemeris
33	873	2220.8722004	0.003472	17.851	measured; excluded from primary fit (secondary)
33	875	2222.5969952	0.003472	19.897	measured; excluded from primary fit (secondary)
33	876	2223.4557585	0.003472	36.141	used in ephemeris
33	877	2224.3158238	0.003472	17.879	measured; excluded from primary fit (secondary)

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	878	2225.1776472	0.003472	37.015	used in ephemeris
33	879	2226.0382563	0.003472	17.382	measured; excluded from primary fit (secondary)
33	880	2226.8993957	0.003472	37.341	used in ephemeris
87	2551	3665.7062363	0.001973	16.809	measured; excluded from primary fit (secondary)
87	2552	3666.5673389	0.001157	38.467	used in ephemeris
87	2553	3667.4268908	0.001375	18.654	measured; excluded from primary fit (secondary)
87	2554	3668.2885280	0.001157	37.217	used in ephemeris
87	2555	3669.1488583	0.001157	18.590	measured; excluded from primary fit (secondary)
87	2556	3670.0111680	0.001157	37.626	used in ephemeris
87	2557	3670.8721219	0.001258	18.570	measured; excluded from primary fit (secondary)
87	2558	3671.7337316	0.001157	38.217	used in ephemeris
87	2559	3672.5964494	0.001352	18.699	measured; excluded from primary fit (secondary)
87	2560	3673.4556201	0.001157	38.481	used in ephemeris
87	2561	3674.3174168	0.001491	17.810	measured; excluded from primary fit (secondary)
87	2562	3675.1784426	0.001157	38.433	used in ephemeris
87	2563	3676.0397348	0.001369	18.937	measured; excluded from primary fit (secondary)
87	2567	3679.4829622	0.002117	15.928	measured; excluded from primary fit (secondary)
87	2568	3680.3436343	0.001157	39.680	used in ephemeris
87	2569	3681.2060437	0.001391	17.966	measured; excluded from primary fit (secondary)
87	2570	3682.0671908	0.001157	39.473	used in ephemeris
87	2571	3682.9268450	0.001523	17.770	measured; excluded from primary fit (secondary)
87	2572	3683.7888215	0.001157	39.372	used in ephemeris
87	2573	3684.6510309	0.001812	16.907	measured; excluded from primary fit (secondary)
87	2574	3685.5104438	0.001157	38.125	used in ephemeris
87	2575	3686.3737709	0.001689	16.623	measured; excluded from primary fit (secondary)
87	2576	3687.2327691	0.001157	37.566	used in ephemeris
87	2577	3688.0944441	0.001906	18.420	measured; excluded from primary fit (secondary)
87	2578	3688.9546417	0.001157	38.316	used in ephemeris

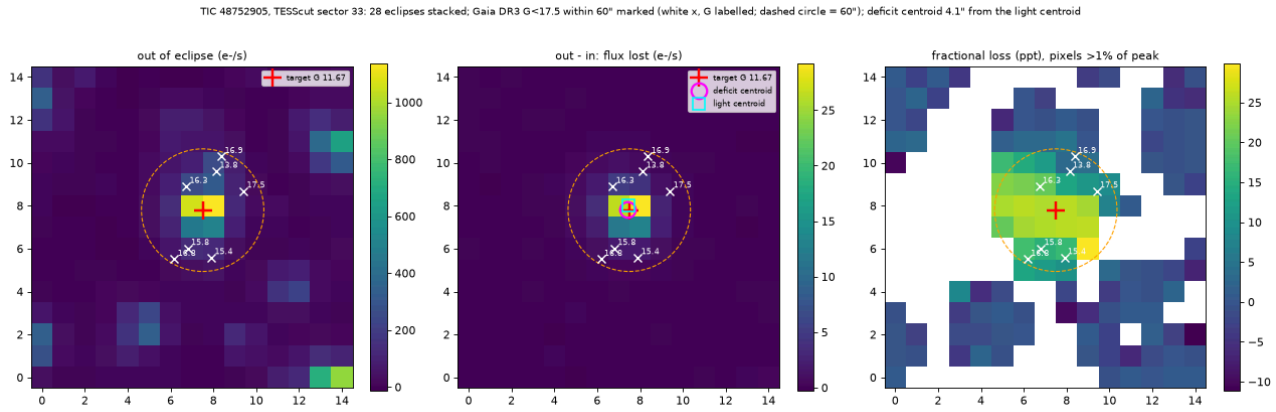
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s33.json, historical/diff_s33.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R340. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R340; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R340; diff_s33.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R341. TIC 337881909

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 337881909: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[13, 93, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=9.6174021260 ± 0.0000203371d, reference eclipse epoch BTJD1661.0486137, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S13 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.252arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.828. Capable neighbours: Gaia 6724638855242197120: 17.77 arcsec, G14.446, required fractional eclipse0.193; Gaia 6724638820882453888: 21.65 arcsec, G16.246, required fractional eclipse0.952; Gaia 6724638889601940224: 21.89 arcsec, G15.135, required fractional eclipse0.351; Gaia 6724638820882457984: 26.79 arcsec, G15.128, required fractional eclipse0.349; Gaia 6724638923961681280: 27.11 arcsec, G15.629, required fractional eclipse0.545; Gaia 6724638820882447232: 47.87 arcsec, G16.037, required fractional eclipse0.788; Gaia 6724638850925800064: 52.52 arcsec, G16.080, required fractional eclipse0.819; Gaia 6724638649083749760: 58.15 arcsec, G15.084, required fractional eclipse0.336
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths13.705/13.040/11.404ppt, with 3 stacked events. QLP selected-kind fitted median13.400ppt. This separately extracted cutout is an alternative reduction of the same TESS

photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR3672463888528215872O, P=9.6159030557 ± 0.0037502071d, e=0.245757, omega=215.8197deg. Closest edge-on conjunction offset0.007154d (0.000744phase), full-SB1-covariance linear propagated Gaia uncertainty0.532172d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.190723d (photometric period projection allowance0.002684d). The approximate central duration9.33h vs observed median8.77h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field: eight capable neighbors, nearest17.8arcsec; deficit0.06pixel from target versus0.80pixel from nearest capable source. Target-centered core favors target, but highPSF fractional residual0.83 limits precise localization. No calibrated errors or resolved binary parameters claimed.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

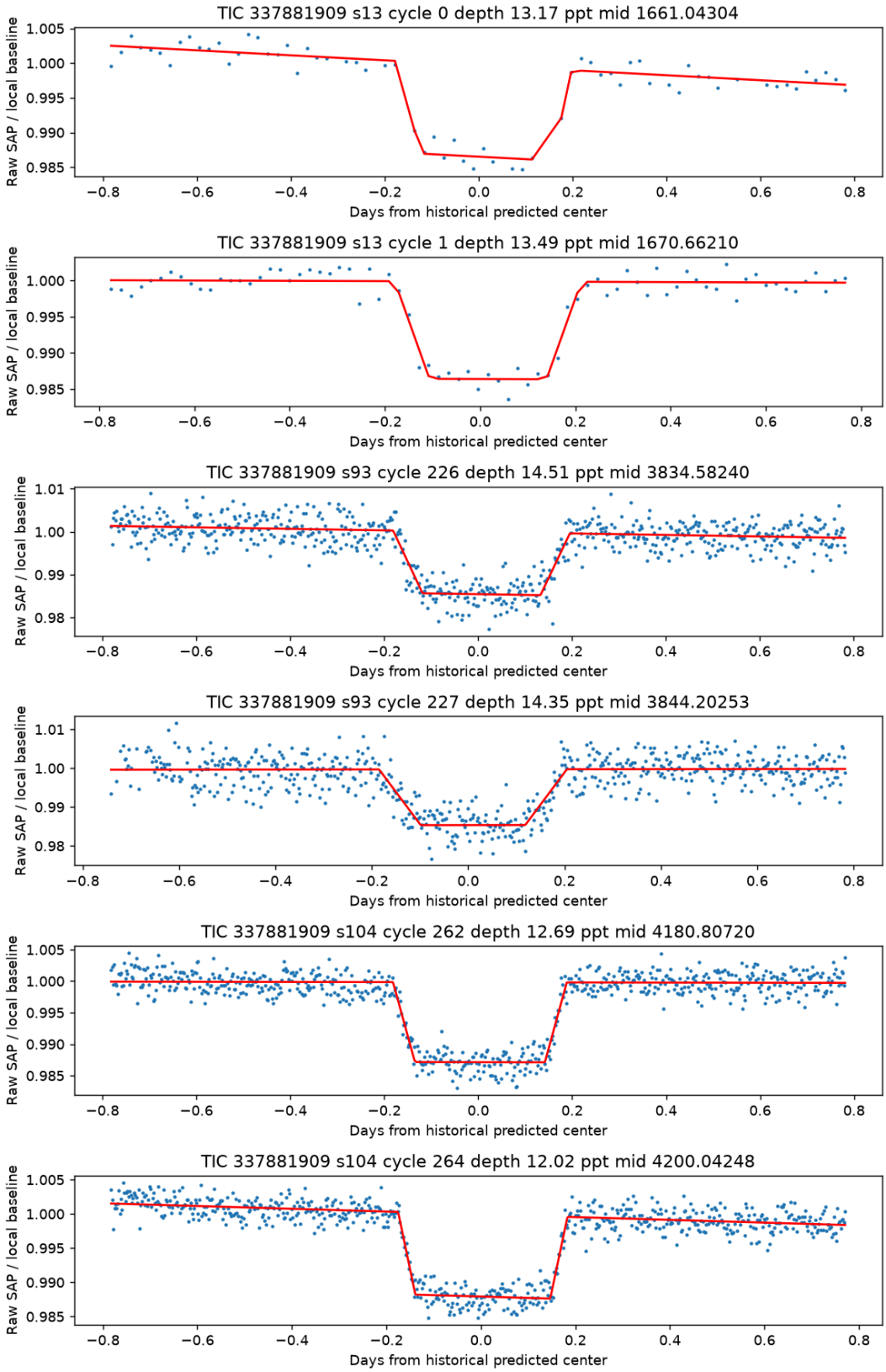
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
13	0	1661.0430448	0.010416	13.173	used in ephemeris
13	1	1670.6620979	0.010416	13.492	used in ephemeris
13	2	1680.2756262	0.010416	13.400	used in ephemeris
93	226	3834.5824036	0.001608	14.514	used in ephemeris
93	227	3844.2025253	0.002441	14.350	used in ephemeris
93	228	3853.8174251	0.002353	13.545	used in ephemeris
104	262	4180.8071992	0.001157	12.689	used in ephemeris
104	263	4190.4246764	0.001823	11.171	used in ephemeris
104	264	4200.0424811	0.001157	12.016	used in ephemeris

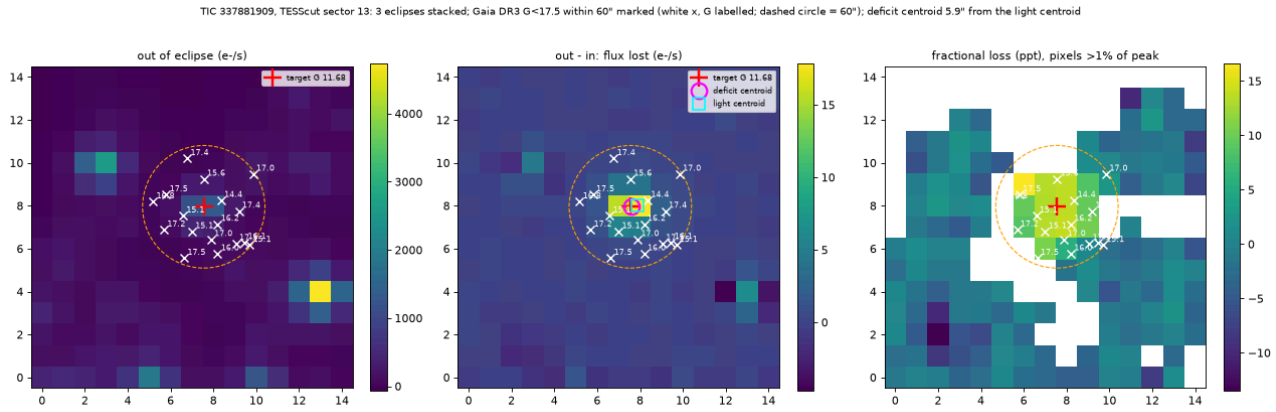
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s13.json, historical/diff_s13.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R341. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R341; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R341; diff_s13.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R342. TIC 274578846

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 274578846: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 5 distinct eclipse events in selected sectors[23, 53, 84]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=12.9678602618 ± 0.0000360930d, reference eclipse epoch BTJD1931.6480036, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S23 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation3.183arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.412. Capable neighbours: Gaia 1413834278144873728: 28.85 arcsec, G12.494, required fractional eclipse0.138
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths37.663/31.396/28.856ppt, with 2 stacked events. QLP selected-kind fitted median45.643ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3141383431250461235 2, P=12.9670803122 ± 0.0021571855d, e=0.248242, omega=61.1714deg. Closest edge-on conjunction offset0.089513d (0.006903phase), full-SB1-covariance linear propagated Gaia uncertainty0.291273d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.003301d (photometric period projection allowance0.004295d). The approximate central

duration6.01h vs observed median3.94h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Same-kind repeat within sector53 and84 secures long-baseline cycle count. Modest aperture/sector depth differences retained.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

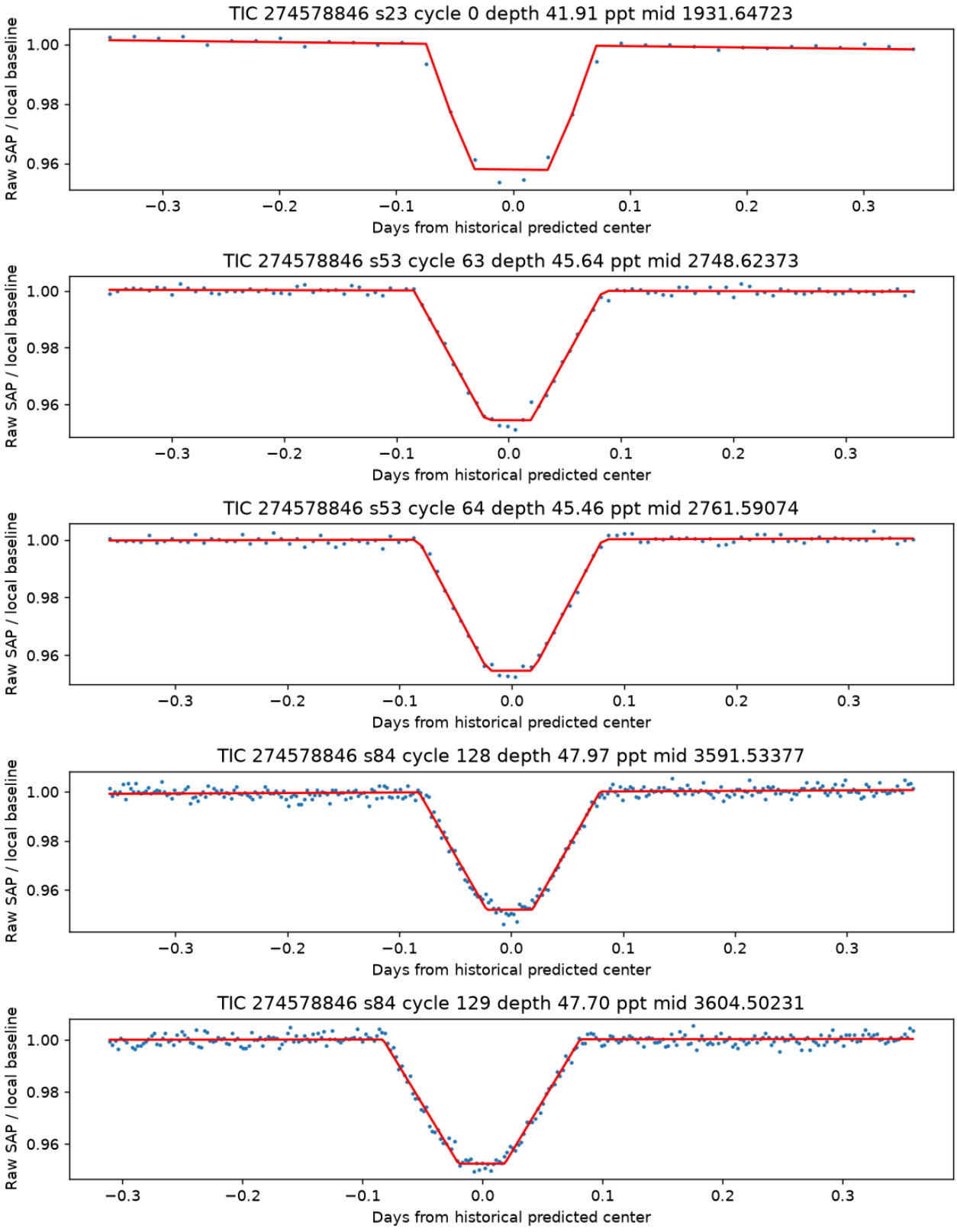
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
23	0	1931.6472336	0.010417	41.906	used in ephemeris
53	63	2748.6237263	0.003472	45.643	used in ephemeris
53	64	2761.5907438	0.003472	45.459	used in ephemeris
84	128	3591.5337707	0.001157	47.966	used in ephemeris
84	129	3604.5023100	0.001157	47.698	used in ephemeris

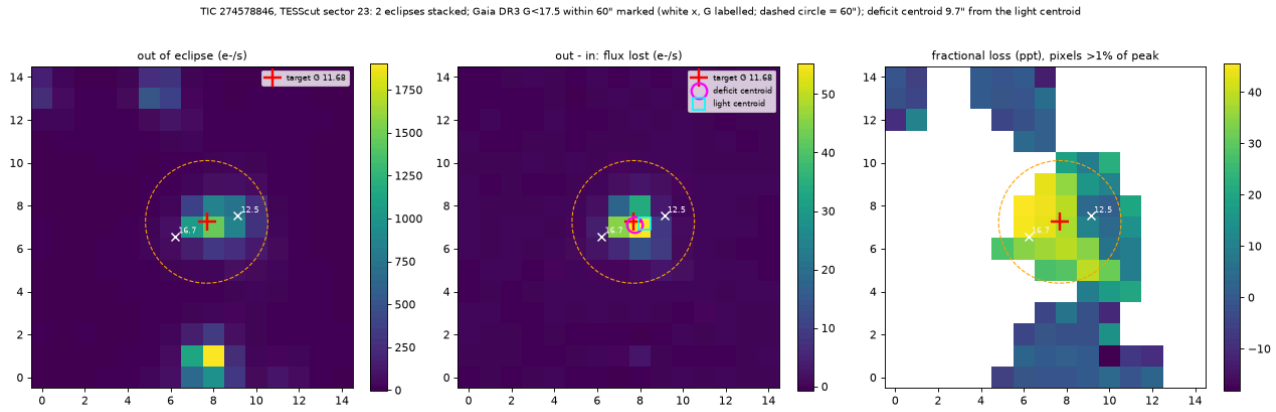
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s23.json, historical/diff_s23.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R342. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R342; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R342; diff_s23.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R343. TIC 14291901

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 14291901: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 11 distinct eclipse events in selected sectors[42, 44, 71]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=5.9988293272 ± 0.0000112890d, reference eclipse epoch BTJD2448.4020518, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S43 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation0.978arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.047. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths40.282/39.535/39.267ppt, with 3 stacked events. QLP selected-kind fitted median43.443ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3502219158807322 88, P=5.9990825867 ± 0.0004188713d, e=0.004308, omega=257.5851deg. Closest edge-on conjunction offset-0.108901d (-0.018153phase), full-SB1-covariance linear propagated Gaia uncertainty0.155500d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.022033d (photometric period projection allowance0.003872d). The approximate central

duration4.74h vs observed median2.89h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses at multiple successive cycles; Gaia conjunction offset retained with propagated orbit covariance and radius/model limitations.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

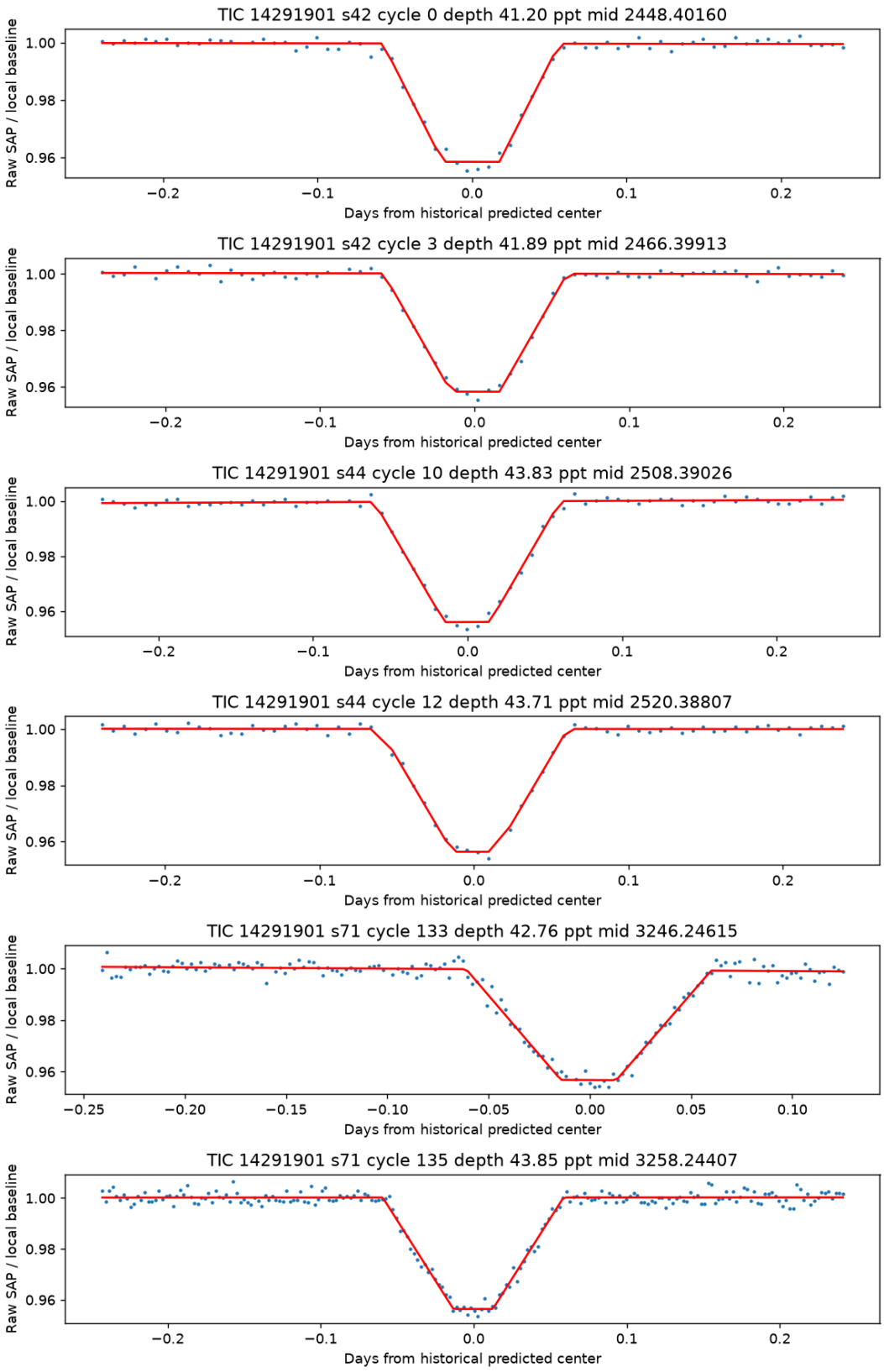
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
42	0	2448.4016012	0.003473	41.196	used in ephemeris
42	1	2454.4008384	0.003473	41.540	used in ephemeris
42	3	2466.3991292	0.003473	41.887	used in ephemeris
44	9	2502.3911272	0.003472	44.937	used in ephemeris
44	10	2508.3902582	0.003472	43.827	used in ephemeris
44	11	2514.3895396	0.003472	44.107	used in ephemeris
44	12	2520.3880726	0.003472	43.706	used in ephemeris
71	132	3240.2473673	0.001157	42.743	used in ephemeris
71	133	3246.2461485	0.001157	42.758	used in ephemeris
71	134	3252.2454786	0.001157	43.443	used in ephemeris
71	135	3258.2440673	0.001157	43.849	used in ephemeris

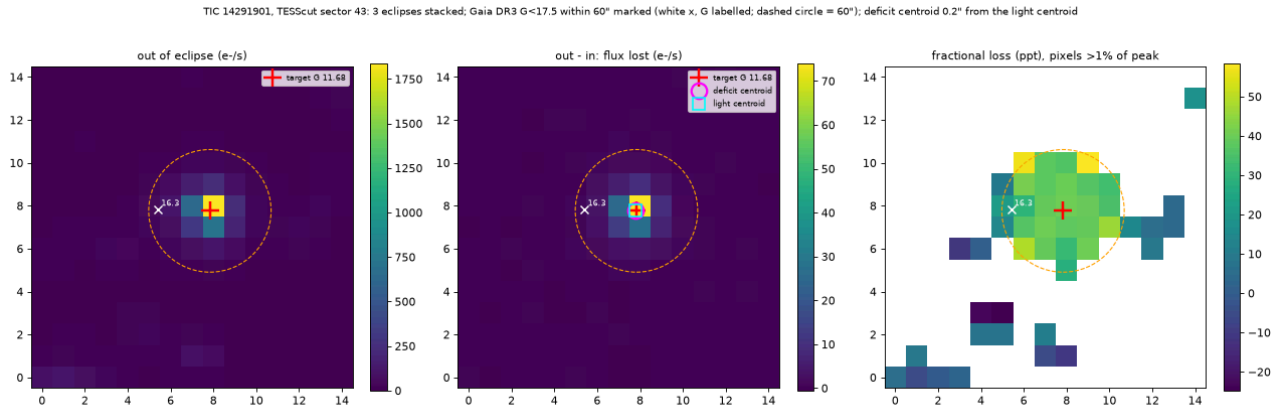
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s43.json, historical/diff_s43.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R343. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R343; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R343; diff_s43.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R344. TIC 180179209

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 180179209: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[11, 65, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=6.3122615616 ± 0.0000079776d, reference eclipse epoch BTJD1601.6364849, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S64 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.874arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.815. Capable neighbours: Gaia 6054997992860786816: 16.65 arcsec, G14.888, required fractional eclipse0.454; Gaia 6054997997204150784: 28.39 arcsec, G14.393, required fractional eclipse0.296; Gaia 6054997962844403840: 34.00 arcsec, G15.410, required fractional eclipse0.722; Gaia 6054997928484667904: 34.34 arcsec, G14.207, required fractional eclipse0.253; Gaia 6054998164658275584: 35.63 arcsec, G12.723, required fractional eclipse0.081; Gaia 6054997924140107776: 37.07 arcsec, G14.060, required fractional eclipse0.224; Gaia 6054997928484660992: 40.14 arcsec, G14.685, required fractional eclipse0.381; Gaia 6054997928484659200: 41.13 arcsec, G14.017, required fractional eclipse0.216; Gaia 6054997928484664704: 42.76 arcsec, G15.405, required fractional eclipse0.718; Gaia 6054998233379003392: 54.73 arcsec, G15.640, required fractional eclipse0.887; Gaia 6054998164659490560: 59.57 arcsec, G15.708, required fractional eclipse0.942

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths23.099/18.880/11.766ppt, with 4 stacked events. QLP selected-kind fitted median24.572ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3605499799285958412 8, P=6.3121869210 ± 0.0008826224d, e=0.294949, omega=289.1618deg. Closest edge-on conjunction offset0.000128d (0.000020phase), full-SB1-covariance linear propagated Gaia uncertainty0.168798d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.014203d (photometric period projection allowance0.001532d). The approximate central duration7.03h vs observed median5.45h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Crowded field: eleven capable neighbors, nearest16.6arcsec; deficit0.23pixel from target versus1.00pixel from nearest capable source. PSF residual0.82 limits localization precision; target-centered core supports attribution at TESS resolution. Final partial/singular sector101 fit excluded from timing.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

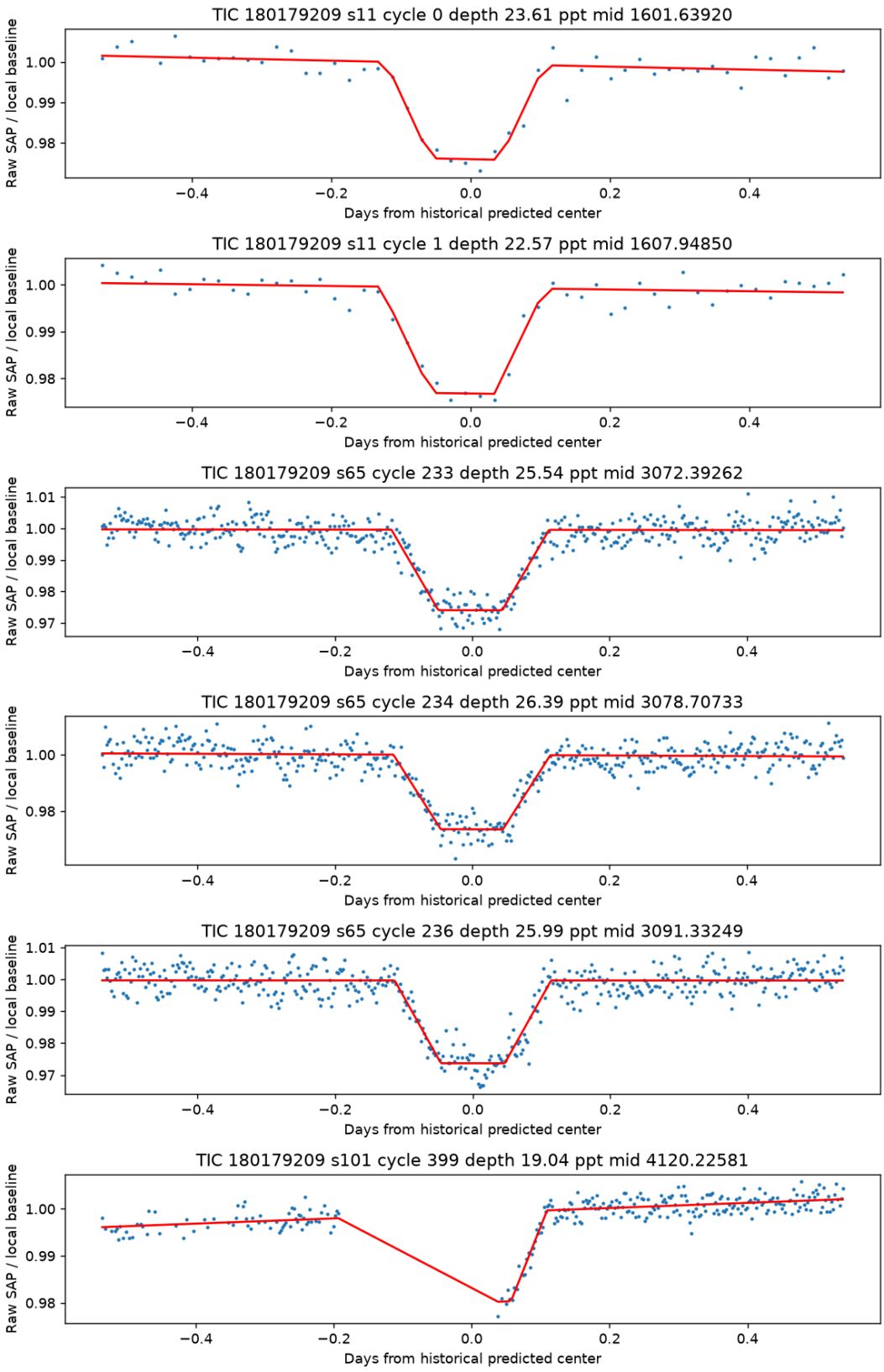
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
11	0	1601.6392042	0.010417	23.608	used in ephemeris
11	1	1607.9484964	0.010417	22.568	used in ephemeris
11	3	1620.5780545	0.010417	23.295	used in ephemeris
65	233	3072.3926234	0.001227	25.536	used in ephemeris
65	234	3078.7073291	0.001640	26.387	used in ephemeris
65	235	3085.0162053	0.001157	25.551	used in ephemeris
65	236	3091.3324864	0.001453	25.994	used in ephemeris
101	397	4107.6044410	0.001157	20.905	used in ephemeris
101	399	4120.2258125	83546.29 5765	19.042	Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.

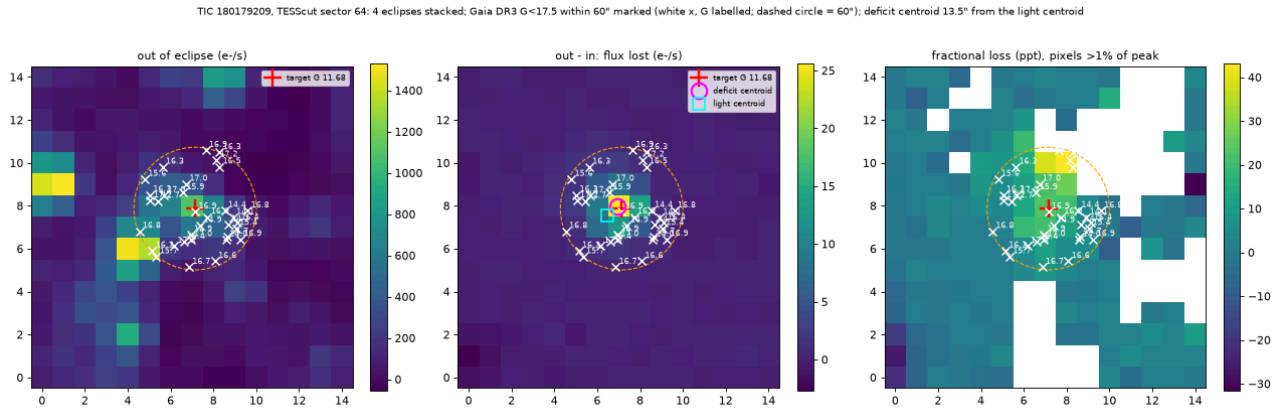
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s64.json, historical/diff_s64.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R344. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R344; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R344; diff_s64.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R345. TIC 351335839

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 351335839: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[13, 101, 104]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=14.8399513286 ± 0.0000448985d, reference eclipse epoch BTJD1655.6300072, kind: Deeper repeating eclipse; even half-period cycles. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S13 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.874arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.992. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths284.455/267.314/185.501ppt, with 4 stacked events. QLP selected-kind fitted median380.798ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3644411058808416268 8, P=14.8359991883 ± 0.0022444509d, e=0.037964, omega=230.7166deg. Closest edge-on conjunction offset0.190357d (0.012831phase), full-SB1-covariance linear propagated Gaia uncertainty0.212663d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.145575d (photometric period projection allowance0.003816d). The approximate central duration7.04h vs observed median5.25h permits noncentral geometry; this radius/minimum-mass

estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipse depths establish doubled orbital period; deeper even historical cycles define ephemeris. Initial duration bound broadened to capture deepV morphology. Poor PSF residual0.99 limits detailed image modeling, but target-centered deficit and no capable listed neighbors support source. Mixed historical cutout stack cannot be equated to primary depth.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

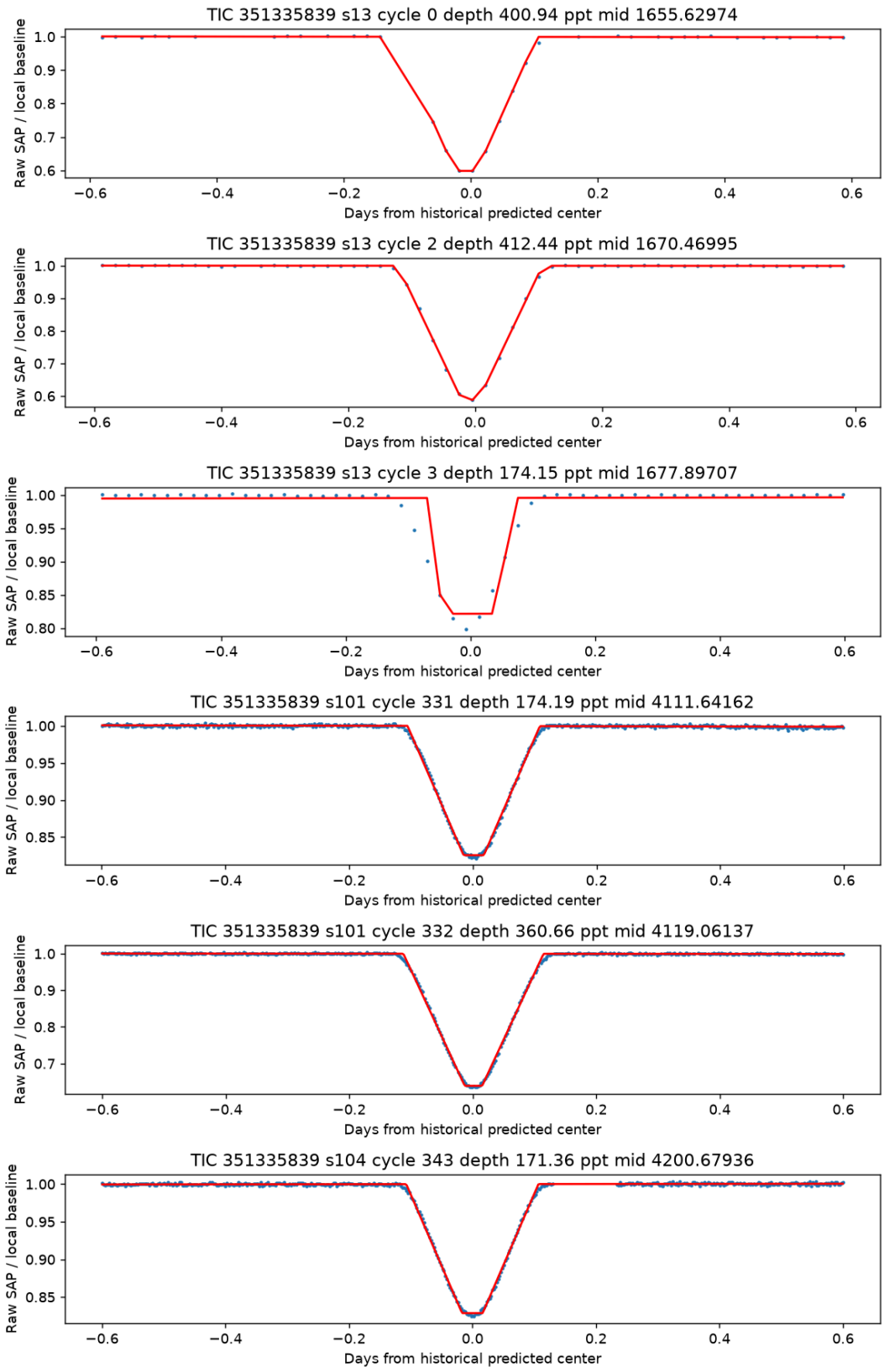
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
13	0	1655.6297408	0.010417	400.937	used in ephemeris
13	2	1670.4699505	0.010417	412.438	used in ephemeris
13	3	1677.8970689	0.010417	174.146	measured; excluded from primary fit (secondary)
101	330	4104.2225387	0.001157	357.044	used in ephemeris
101	331	4111.6416208	0.001157	174.195	measured; excluded from primary fit (secondary)
101	332	4119.0613689	0.001157	360.659	used in ephemeris
104	341	4185.8404540	0.001157	166.111	measured; excluded from primary fit (secondary)
104	343	4200.6793647	0.001157	171.355	measured; excluded from primary fit (secondary)

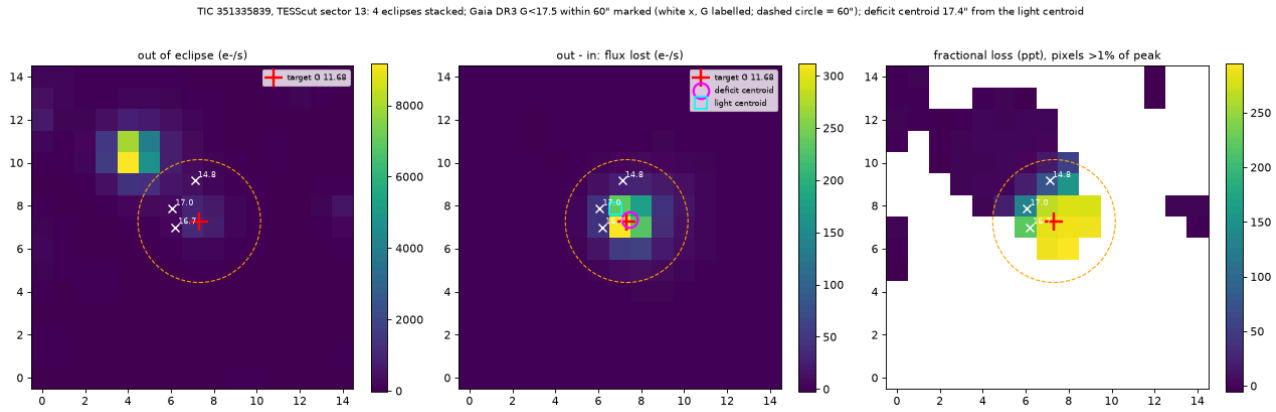
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s13.json, historical/diff_s13.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R345. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R345; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R345; diff_s13.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R346. TIC 89553209

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 89553209: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 6 distinct eclipse events in selected sectors[14, 54, 81]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.8658638287 ± 0.0000359636d, reference eclipse epoch BTJD1695.6867422, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S14 deficit peak[8, 7] vs target pixel[7, 8]; deficit-target separation0.970arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.476. Capable neighbours: Gaia 2024239325242611072: 53.46 arcsec, G14.018, required fractional eclipse0.461
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths10.283/12.097/12.528ppt, with 2 stacked events. QLP selected-kind fitted median51.272ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32024239325242612736, P=11.8678248503 ± 0.0035823200d, e=0.375346, omega=266.7319deg. Closest edge-on conjunction offset-0.231422d (-0.019500phase), full-SB1-covariance linear propagated Gaia uncertainty0.480204d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.015709d (photometric period projection allowance0.003956d). The approximate central

duration12.04h vs observed median11.52h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Sector14 cycle-1 lacks ingress coverage and is excluded from timing. Remaining five measured events include successive sector54/81 pairs and a full sector14 event. Historical prediction drift is fitted freely; partial raw event is not a precise timing. Broad eclipse duration and radius-based central estimate agree approximately.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

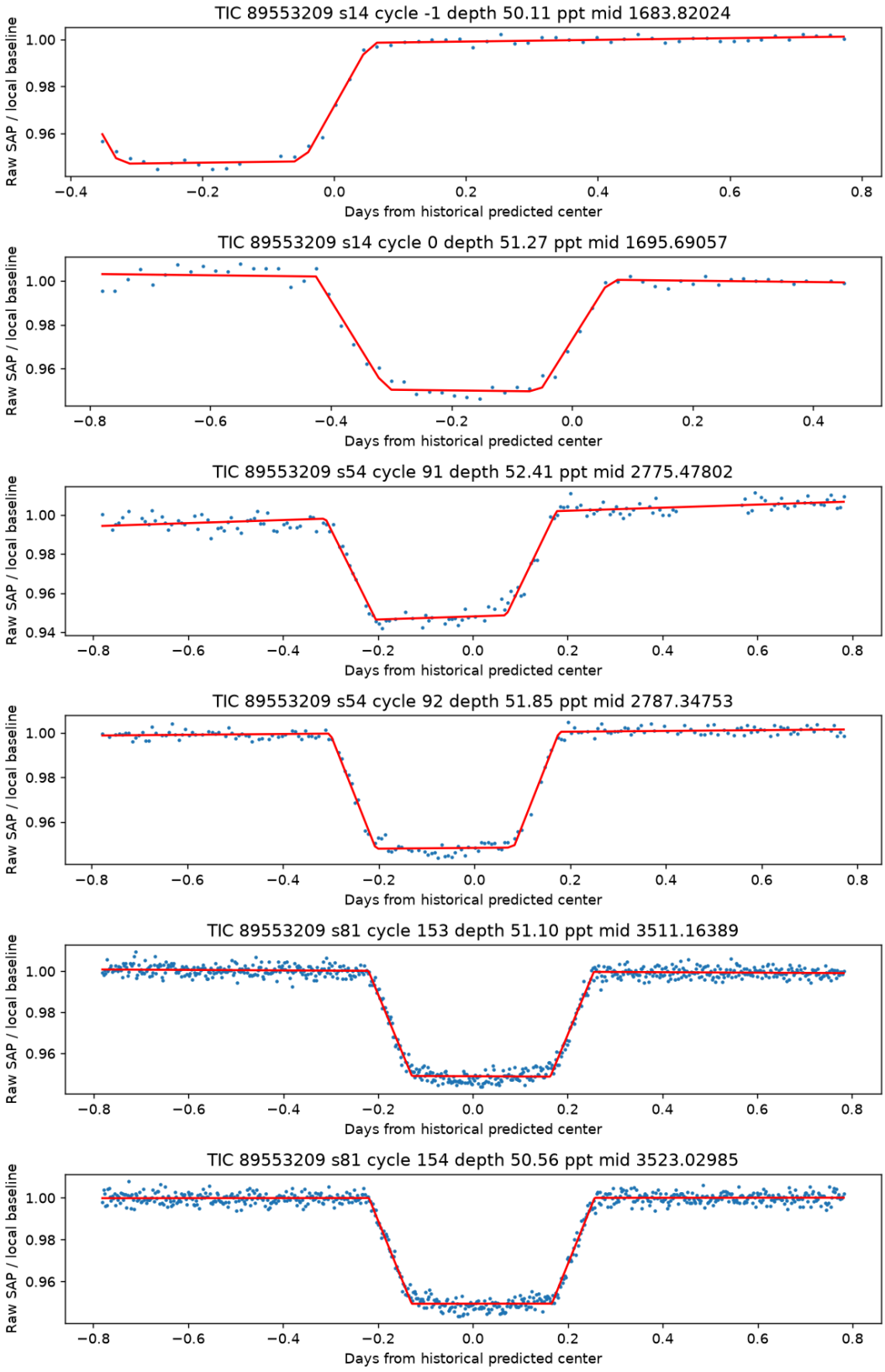
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	-1	1683.8202443	0.010417	50.113	Partially covered eclipse lacks ingress baseline; retained raw morphology but excluded timing.
14	0	1695.6905685	0.010417	51.272	used in ephemeris
54	91	2775.4780234	0.003472	52.415	used in ephemeris
54	92	2787.3475254	0.003472	51.847	used in ephemeris
81	153	3511.1638934	0.001157	51.095	used in ephemeris
81	154	3523.0298522	0.001157	50.557	used in ephemeris

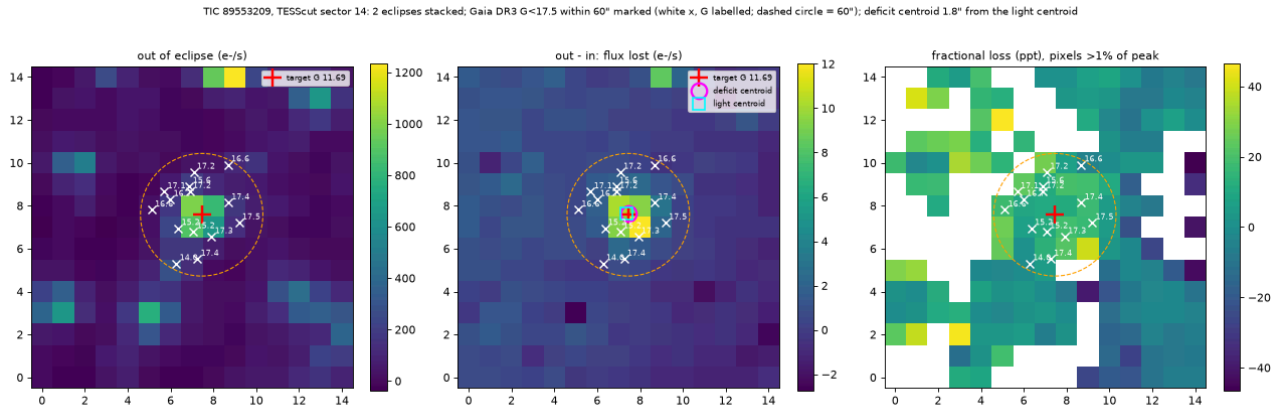
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s14.json, historical/diff_s14.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R346. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R346; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R346; diff_s14.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R347. TIC 60555220

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 60555220: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[7, 35, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=8.4552418596 ±0.0000133662d, reference eclipse epoch BTJD1498.9281820, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S7 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation2.023arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.712. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths225.863/215.449/147.434ppt, with 3 stacked events. QLP selected-kind fitted median258.330ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3551210285665213 2736, P=8.4559112267±0.0006208254d, e=0.016619, omega=326.8573deg. Closest edge-on conjunction offset-0.021730d (-0.002570phase), full-SB1-covariance linear propagated Gaia uncertainty0.089100d at the measured epoch. Near-Gaia-reference projected eclipse offset0.065957d (photometric period projection allowance0.001751d). The approximate central

duration5.52h vs observed median4.28h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

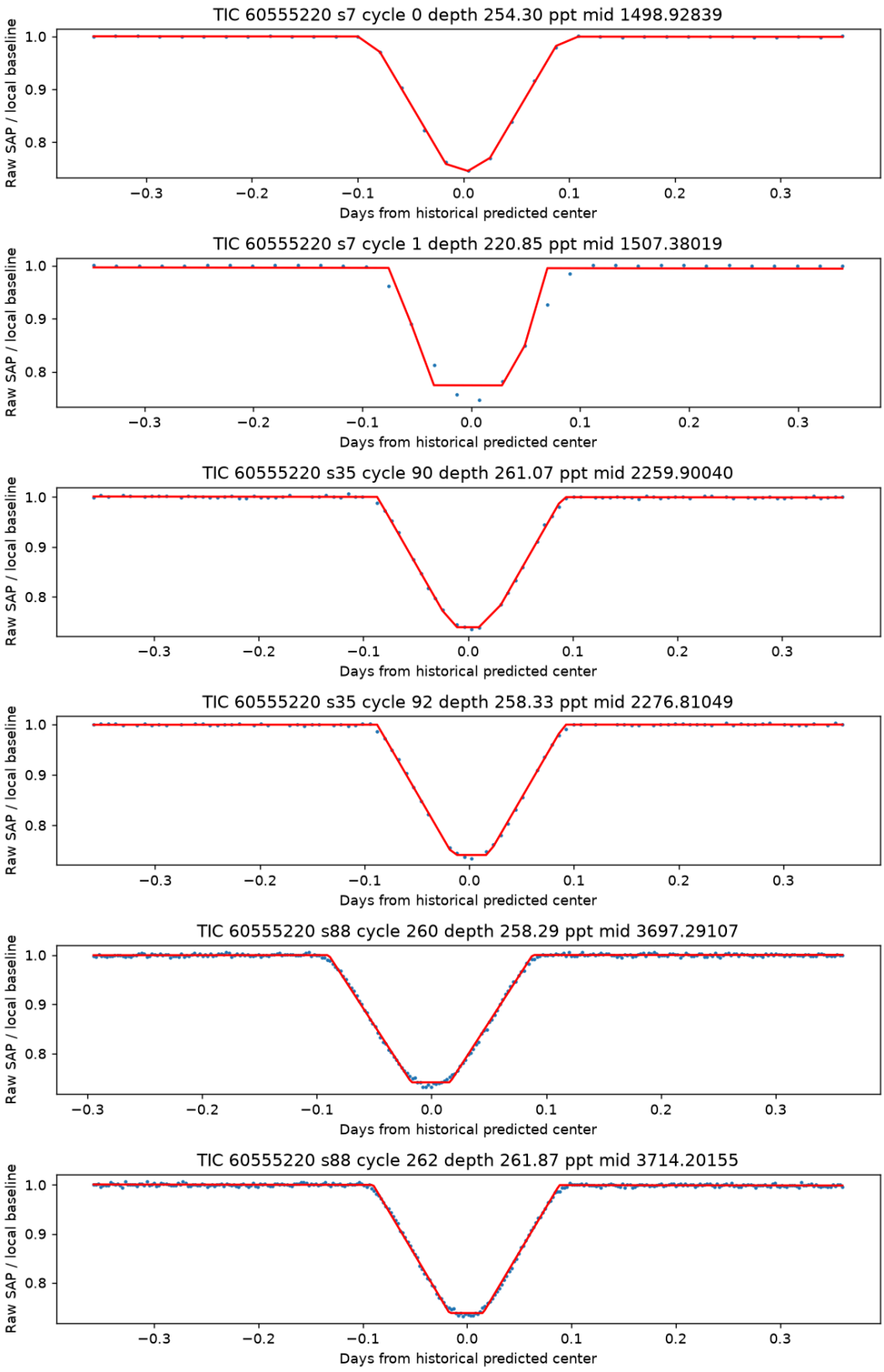
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1498.9283940	0.010417	254.299	included timing center
7	1	1507.3801901	0.010417	220.847	included timing center
35	90	2259.9004015	0.003472	261.070	included timing center
35	92	2276.8104913	0.003472	258.330	included timing center
88	260	3697.2910750	0.001157	258.288	included timing center
88	261	3705.7462821	0.001157	259.446	included timing center
88	262	3714.2015457	0.001157	261.868	included timing center

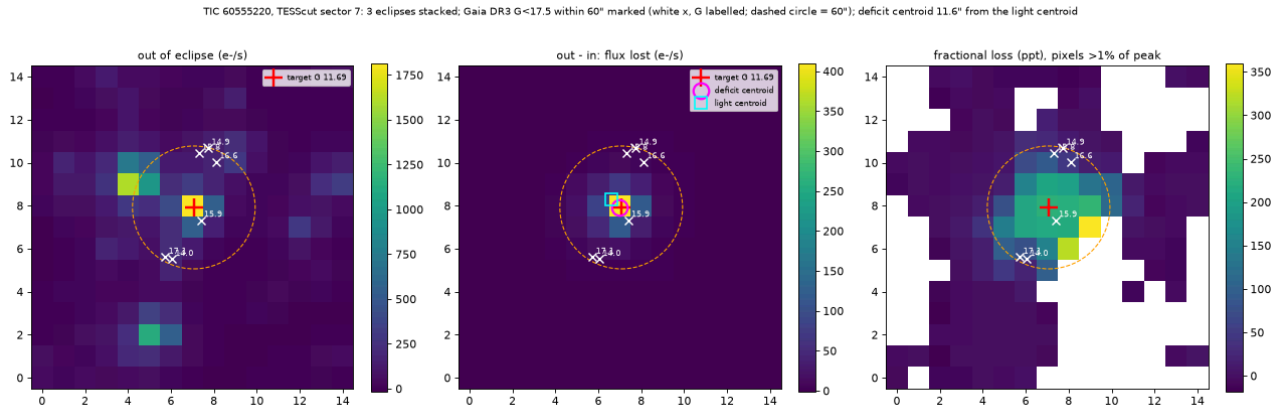
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s07.json, historical/diff_s07.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R347. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R347; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R347; diff_s07.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R348. TIC 142932503

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 142932503: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 3 distinct eclipse events in selected sectors[14, 47, 75]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=18.6886141507 ±0.0000723224d, reference eclipse epoch BTJD1699.2030487, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json; for142932503 the three separated centers, integer-count scan and clean submultiple tests are in helper/alias142.py and helper audit. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S14 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.808arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.433. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths154.268/153.320/151.562ppt, with 1 stacked events. QLP selected-kind fitted median170.639ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3106167066809147648O, P=18.6878640883±0.0038143874d, e=0.507144, omega=148.4350deg. Closest edge-on conjunction offset0.027089d (0.001450phase), full-SB1-covariance linear propagated Gaia uncertainty0.325945d

at the measured epoch. Near-Gaia-reference projected eclipse offset-0.025415d (photometric period projection allowance0.005063d). The approximate central duration5.60h vs observed median4.19h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Three separated measured eclipses at cycles0,48,89. Independent integer-count scan1-25d finds unique near18.69d fit; clean sector47 nulls reject identical deep events at divisors2,3,4. This is bounded alias evidence, not exhaustive all-period exclusion.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

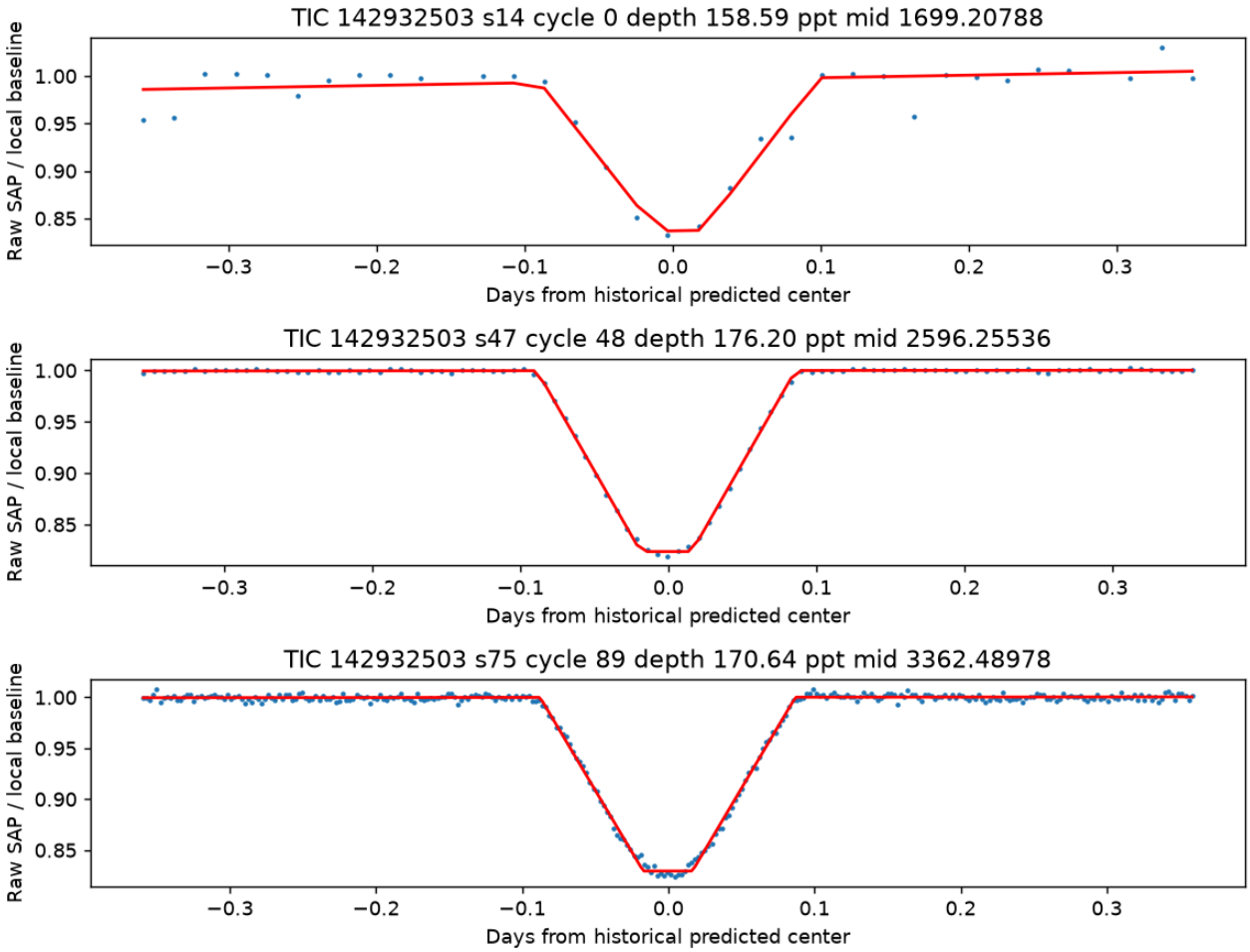
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	0	1699.2078766	0.010416	158.588	included timing center
47	48	2596.2553634	0.003472	176.201	included timing center
75	89	3362.4897779	0.001157	170.639	included timing center

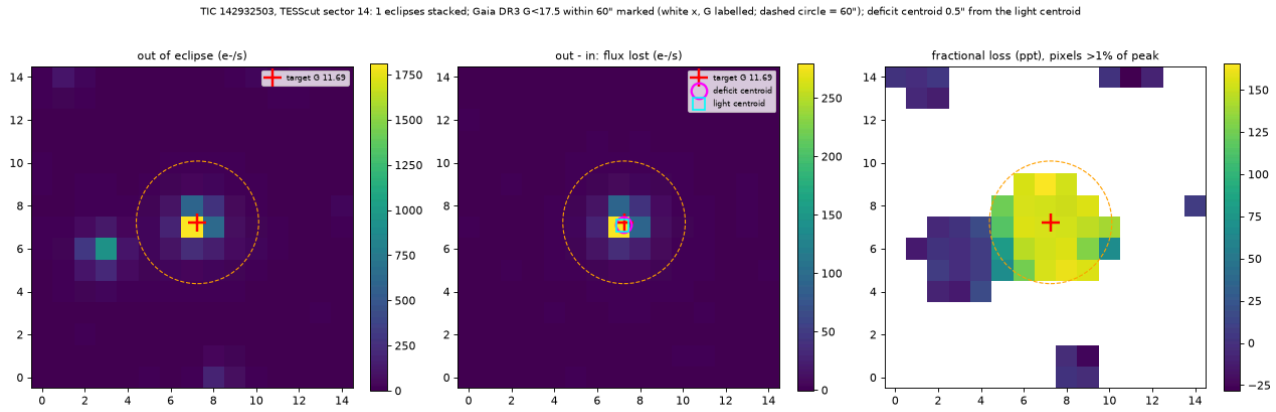
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s14.json, historical/diff_s14.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R348. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R348; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R348; diff_s14.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R349. TIC 256260982

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 256260982: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 16 distinct eclipse events in selected sectors[26, 53, 80]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 16 same-kind midtimes give P=3.2856228182 ±0.0000056568d, reference eclipse epoch BTJD2031.8784385, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S40 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.022arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.197. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths145.050/142.465/141.179ppt, with 5 stacked events. QLP selected-kind fitted median173.195ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3452540572608998 4184, P=3.2853698535±0.0001679014d, e=0.050016, omega=131.5130deg. Closest edge-on conjunction offset0.076348d (0.023239phase), full-SB1-covariance linear propagated Gaia uncertainty0.114725d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.050134d (photometric period projection allowance0.002828d). The approximate central duration4.50h vs observed median4.07h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

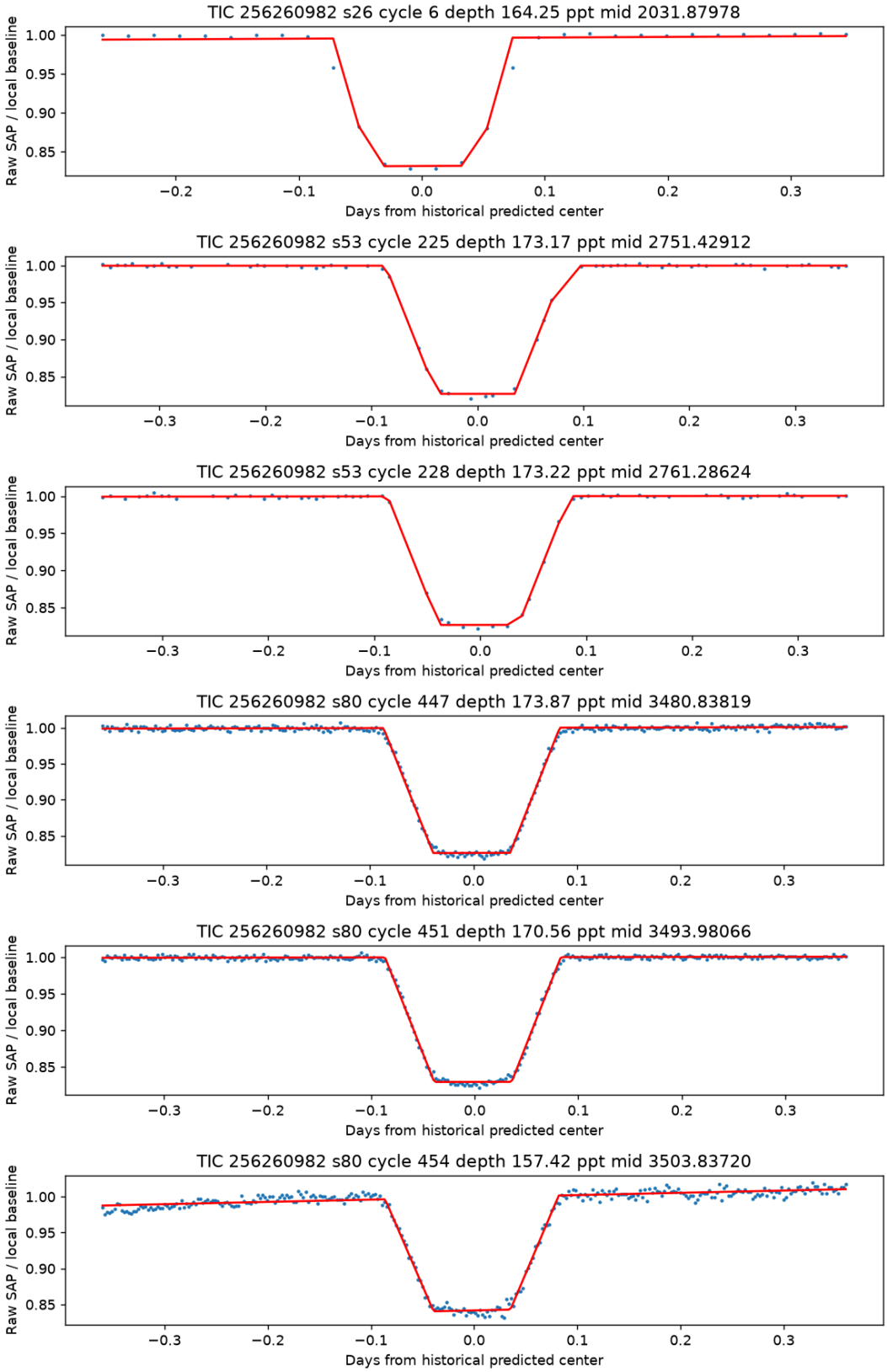
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
26	6	2031.8797815	0.010417	164.255	included timing center
53	223	2744.8581590	0.003472	173.594	included timing center
53	224	2748.1451904	0.003472	174.773	included timing center
53	225	2751.4291247	0.003472	173.172	included timing center
53	226	2754.7158517	0.003472	173.711	included timing center
53	227	2758.0007849	0.003472	173.170	included timing center
53	228	2761.2862405	0.003472	173.218	included timing center
53	229	2764.5725357	0.003472	174.577	included timing center
53	230	2767.8578561	0.003472	174.542	included timing center
80	447	3480.8381873	0.001157	173.873	included timing center
80	448	3484.1238251	0.001157	173.308	included timing center
80	450	3490.6955204	0.001157	156.268	included timing center
80	451	3493.9806597	0.001157	170.559	included timing center
80	452	3497.2661390	0.001157	171.073	included timing center
80	453	3500.5514003	0.001157	169.957	included timing center
80	454	3503.8372006	0.001157	157.425	included timing center

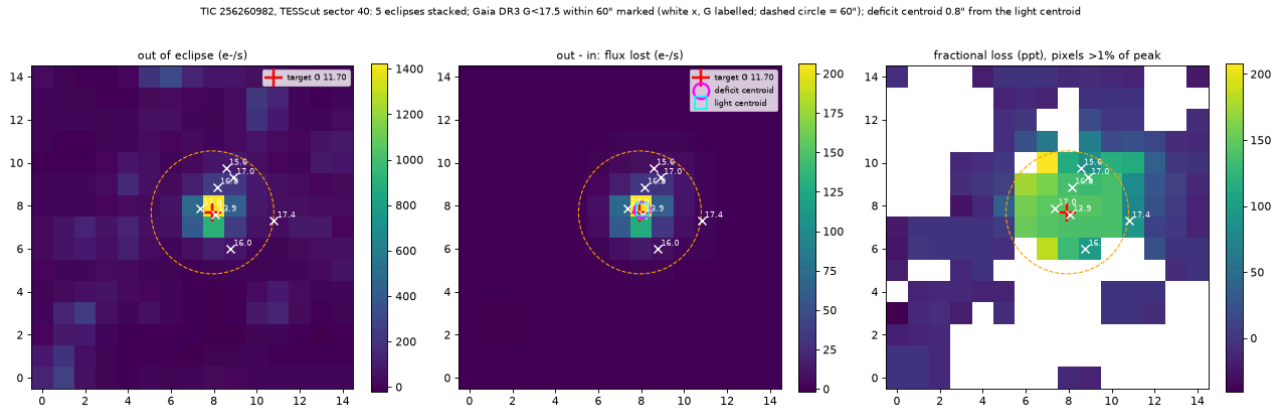
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s40.json, historical/diff_s40.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R349. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R349; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R349; diff_s40.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R350. TIC 11176109

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 11176109: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 13 distinct eclipse events in selected sectors[7, 61, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 13 same-kind midtimes give P=5.0191531715 ±0.0000048262d, reference eclipse epoch BTJD1501.3234240, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S61 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.987arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.565. Capable neighbours: Gaia 5713502188219216384: 16.97 arcsec, G13.638, required fractional eclipse0.762; Gaia 5713502089440751232: 35.79 arcsec, G12.303, required fractional eclipse0.300; Gaia 5713501917642055808: 59.41 arcsec, G13.219, required fractional eclipse0.553
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths100.668/77.592/55.934ppt, with 4 stacked events. QLP selected-kind fitted median110.415ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3571350212380031168, P=5.0189270961±0.0003204633d, e=0.059916, omega=184.1156deg. Closest edge-on conjunction offset-0.004499d (-0.000896phase), full-SB1-covariance linear propagated Gaia

uncertainty0.078599d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.054687d (photometric period projection allowance0.001071d). The approximate central duration4.36h vs observed median3.88h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated eclipses. Target0.142pixel from deficit versus nearest capable neighbor0.722pixel, which would require76percent intrinsic loss. Target favored by concentrated deficit core, without calibrated localization uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

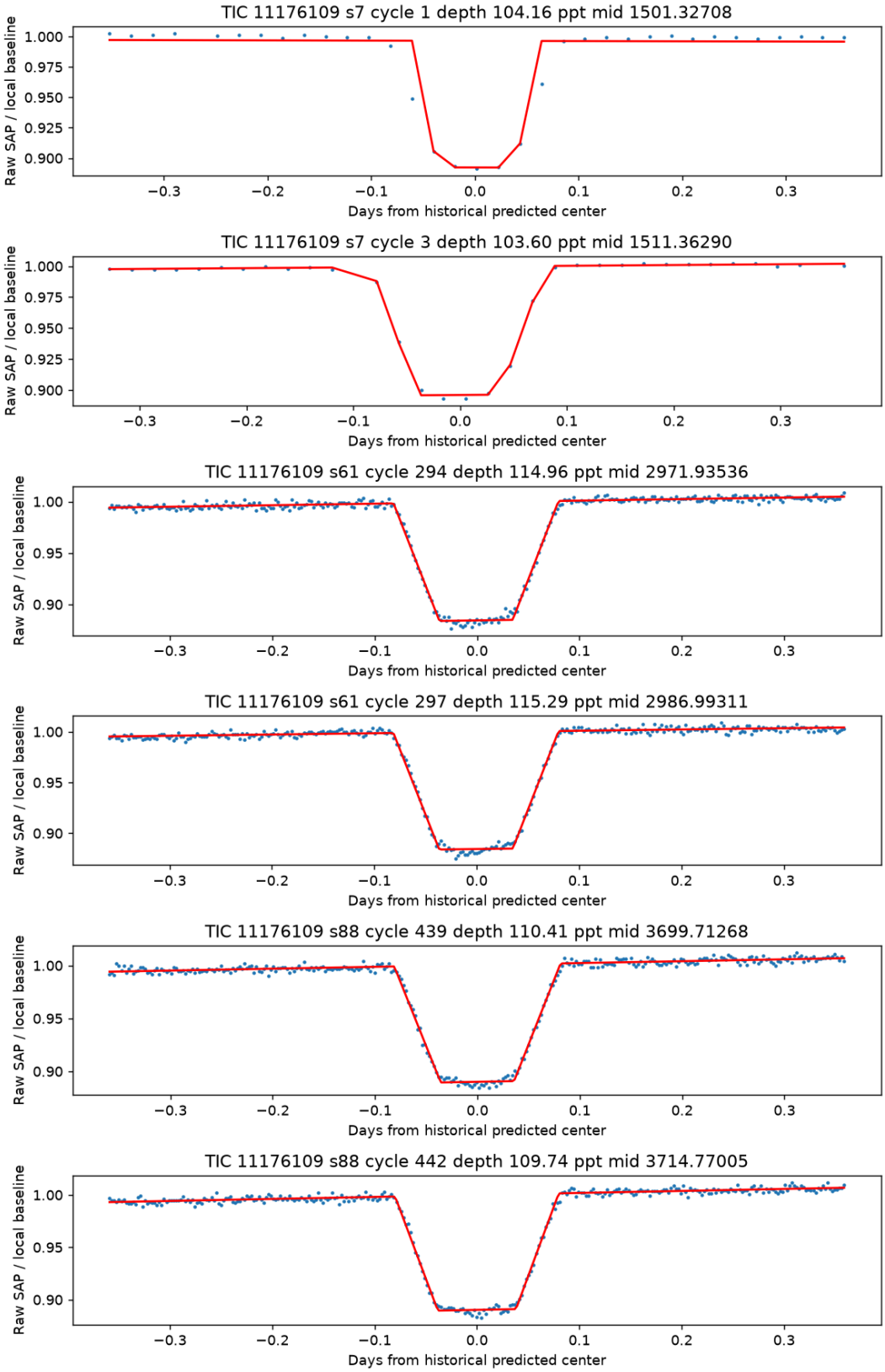
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	1	1501.3270834	0.010417	104.159	included timing center
7	2	1506.3445659	0.010417	106.386	included timing center
7	3	1511.3629019	0.010417	103.604	included timing center
61	293	2966.9158740	0.001157	116.171	included timing center
61	294	2971.9353564	0.001157	114.964	included timing center
61	295	2976.9542299	0.001157	114.557	included timing center
61	296	2981.9734669	0.001157	117.137	included timing center
61	297	2986.9931061	0.001157	115.291	included timing center
88	438	3694.6934857	0.001157	110.545	included timing center
88	439	3699.7126828	0.001157	110.415	included timing center
88	440	3704.7314124	0.001157	110.272	included timing center
88	441	3709.7508639	0.001157	109.855	included timing center
88	442	3714.7700513	0.001157	109.738	included timing center

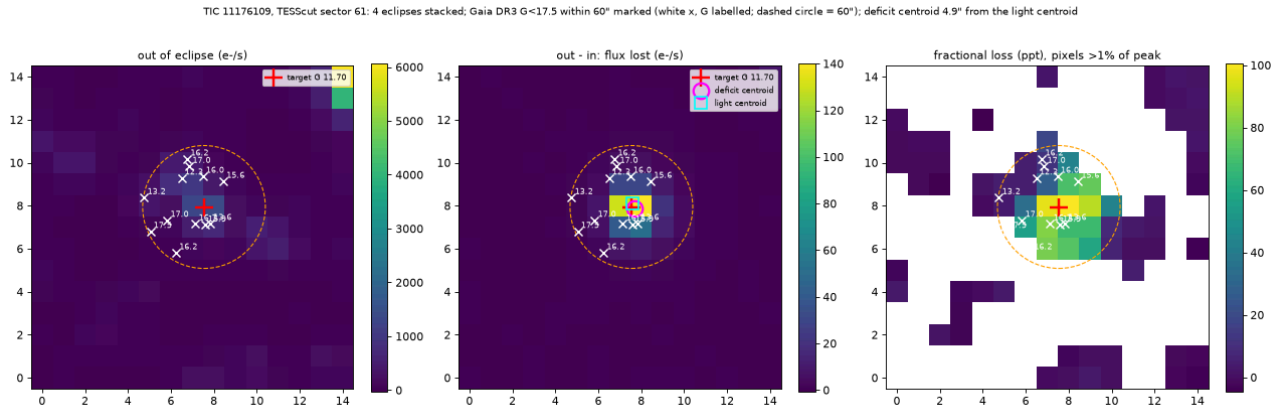
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s61.json, historical/diff_s61.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R350. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R350; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R350; diff_s61.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R351. TIC 279823568

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 279823568: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[21, 47]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=4.4197074796 ±0.0000283837d, reference eclipse epoch BTJD1873.6000620, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S21 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation5.456arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.069. Capable neighbours: Gaia 1012826405920802560: 45.32 arcsec, G16.425, required fractional eclipse0.927
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths12.160/11.633/11.566ppt, with 6 stacked events. QLP selected-kind fitted median11.191ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3101282620405770060 8, P=4.4200216973±0.0006874443d, e=0.100927, omega=16.0548deg. Closest edge-on conjunction offset0.027673d (0.006261phase), full-SB1-covariance linear propagated Gaia uncertainty0.264185d at the measured epoch. Near-Gaia-reference projected eclipse offset0.133250d (photometric period projection allowance0.009537d). The approximate central duration5.27h vs observed median5.65h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

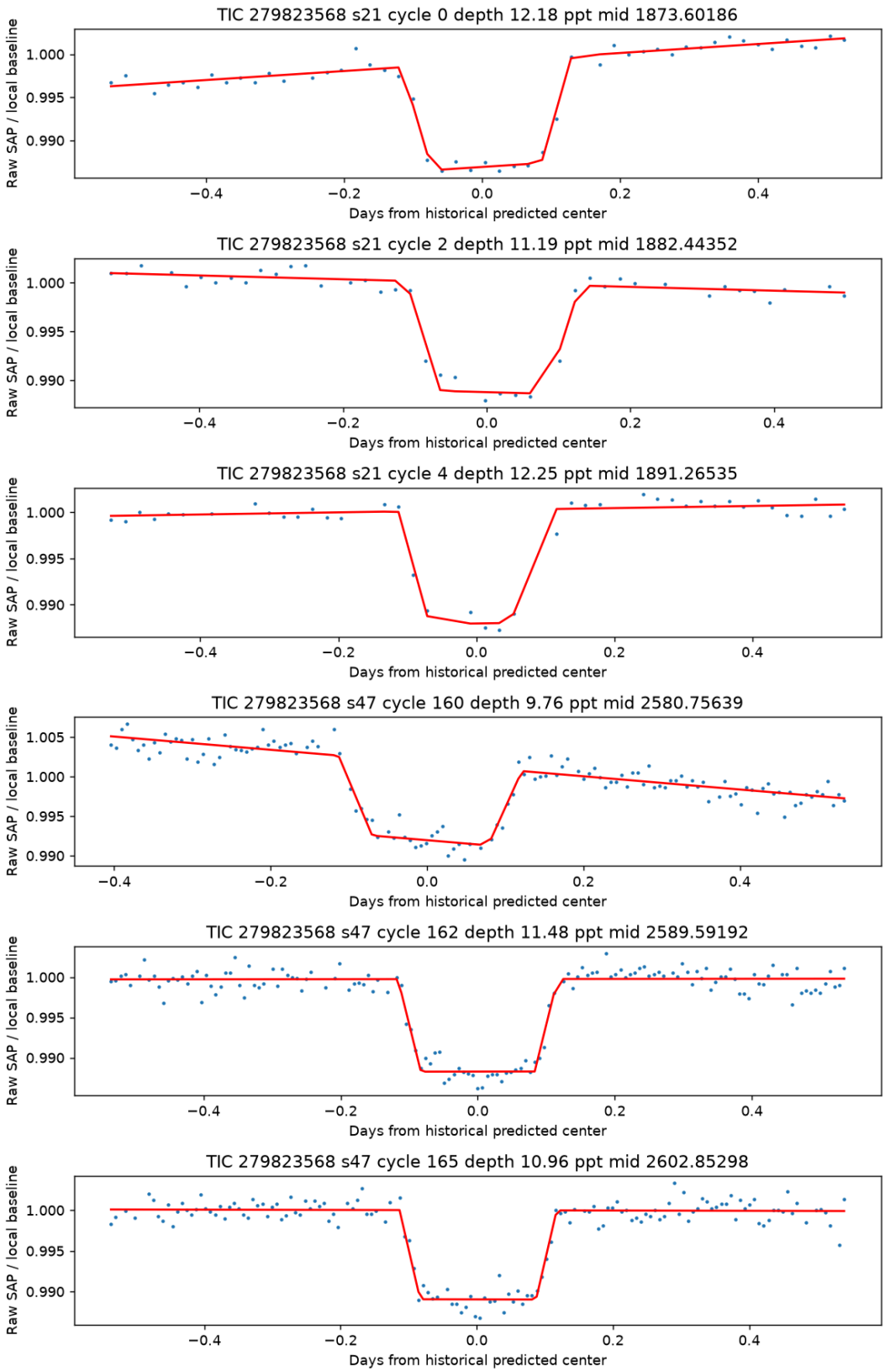
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
21	0	1873.6018611	0.010417	12.177	included timing center
21	1	1878.0207227	0.010417	10.993	included timing center
21	2	1882.4435239	0.010417	11.191	included timing center
21	3	1886.8656536	0.010417	11.658	included timing center
21	4	1891.2653482	0.010417	12.250	included timing center
21	5	1895.6986739	0.010417	11.354	included timing center
47	160	2580.7563940	0.004243	9.759	included timing center
47	161	2585.1713439	0.003472	10.566	included timing center
47	162	2589.5919178	0.003472	11.485	included timing center
47	164	2598.4312080	0.003472	10.915	included timing center
47	165	2602.8529769	0.003472	10.956	included timing center

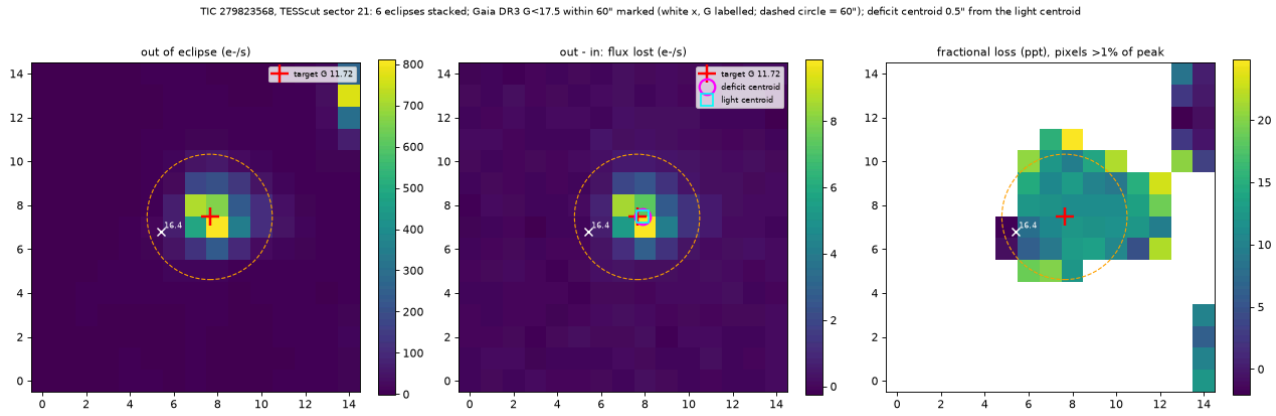
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s21.json, historical/diff_s21.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R351. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R351; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R351; diff_s21.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R352. TIC 179857938

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 179857938: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[11, 65, 101]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=8.8906205223 ±0.0000097587d, reference eclipse epoch BTJD1618.7786820, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S38 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation2.766arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.367. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths129.005/130.652/126.417ppt, with 3 stacked events. QLP selected-kind fitted median117.807ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3605811212337115238 4, P=8.8903150445±0.0017019648d, e=0.365126, omega=46.0190deg. Closest edge-on conjunction offset0.053718d (0.006042phase), full-SB1-covariance linear propagated Gaia uncertainty0.269890d at the measured epoch. Near-Gaia-reference projected eclipse offset0.011562d (photometric period projection allowance0.001347d). The approximate central

duration5.69h vs observed median3.84h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

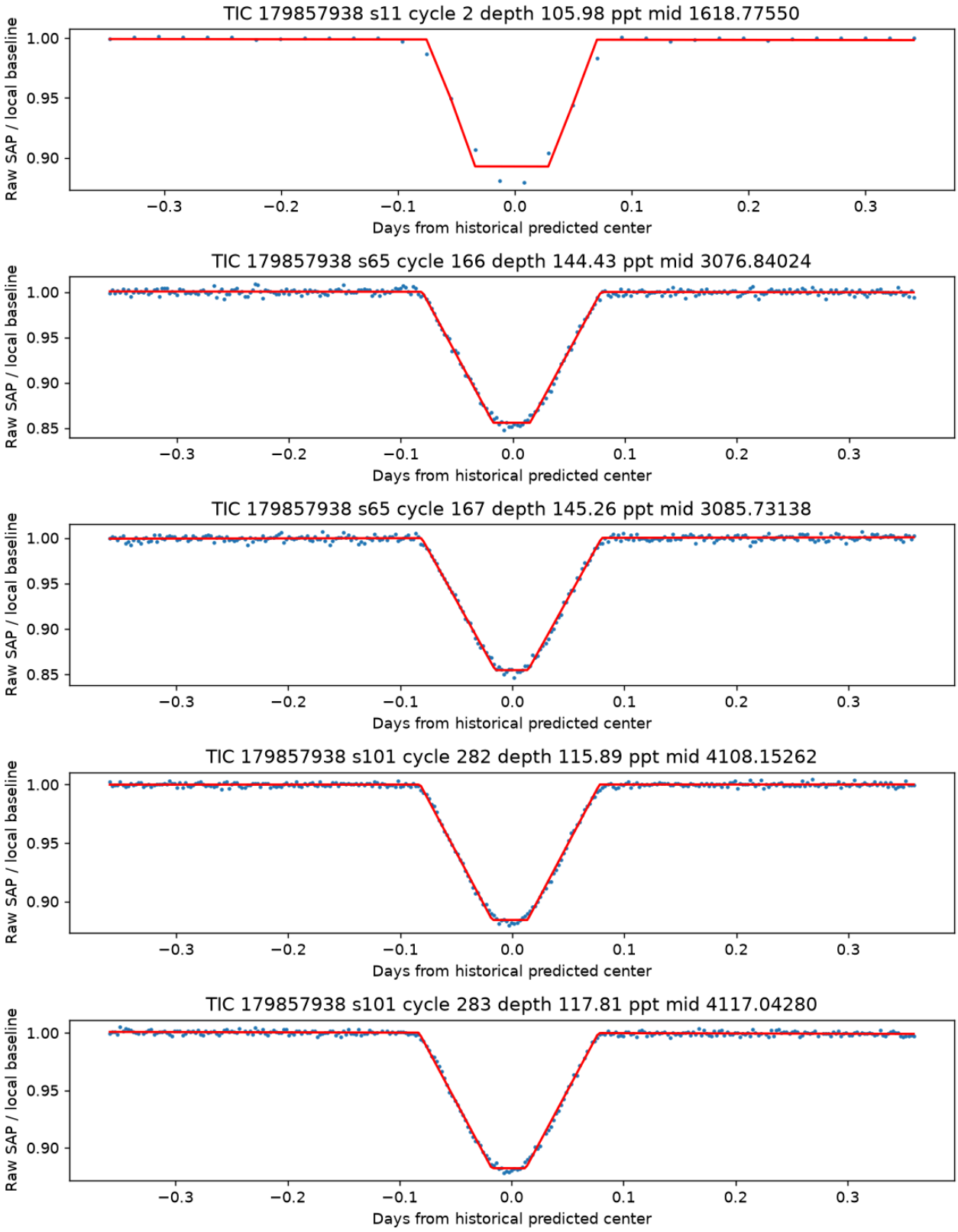
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
11	2	1618.7754970	0.010417	105.976	included timing center
65	166	3076.8402354	0.001157	144.429	included timing center
65	167	3085.7313758	0.001157	145.261	included timing center
101	282	4108.1526208	0.001157	115.891	included timing center
101	283	4117.0428000	0.001157	117.807	included timing center

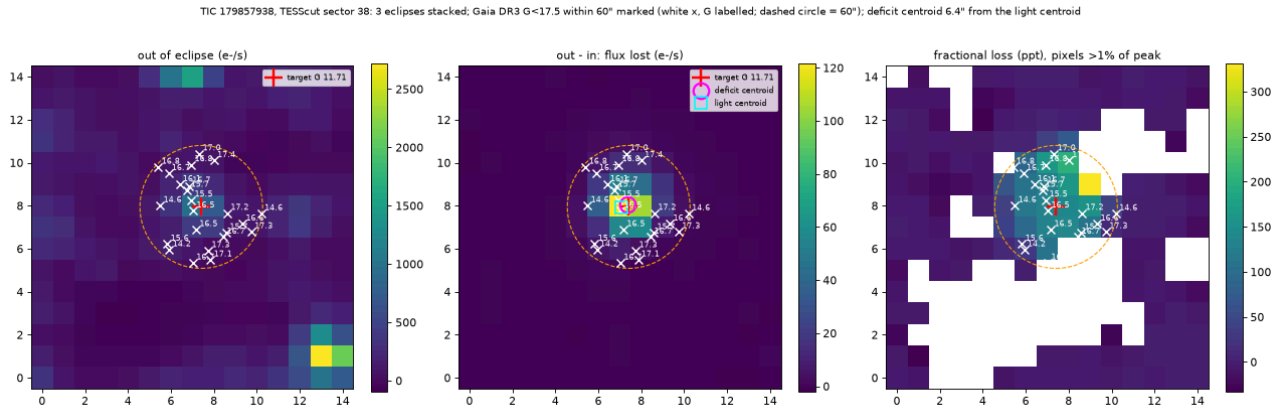
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R352. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R352; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R352; diff_s38.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R353. TIC 220617848

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 220617848: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 5 distinct eclipse events in selected sectors[32, 33]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=9.9217119096 ±0.0009025586d, reference eclipse epoch BTJD2177.2176816, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S33 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.858arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.355. Capable neighbours: Gaia 4802940366659639808: 46.36 arcsec, G15.176, required fractional eclipse0.849
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths33.374/31.962/30.168ppt, with 3 stacked events. QLP selected-kind fitted median36.320ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3480293998440533593 6, P=9.9222605876±0.0022063493d, e=0.263062, omega=113.6466deg. Closest edge-on conjunction offset-0.117564d (-0.011849phase), full-SB1-covariance linear propagated Gaia uncertainty0.384956d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.018802d (photometric period projection allowance0.162461d). The approximate central

duration3.56h vs observed median2.43h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

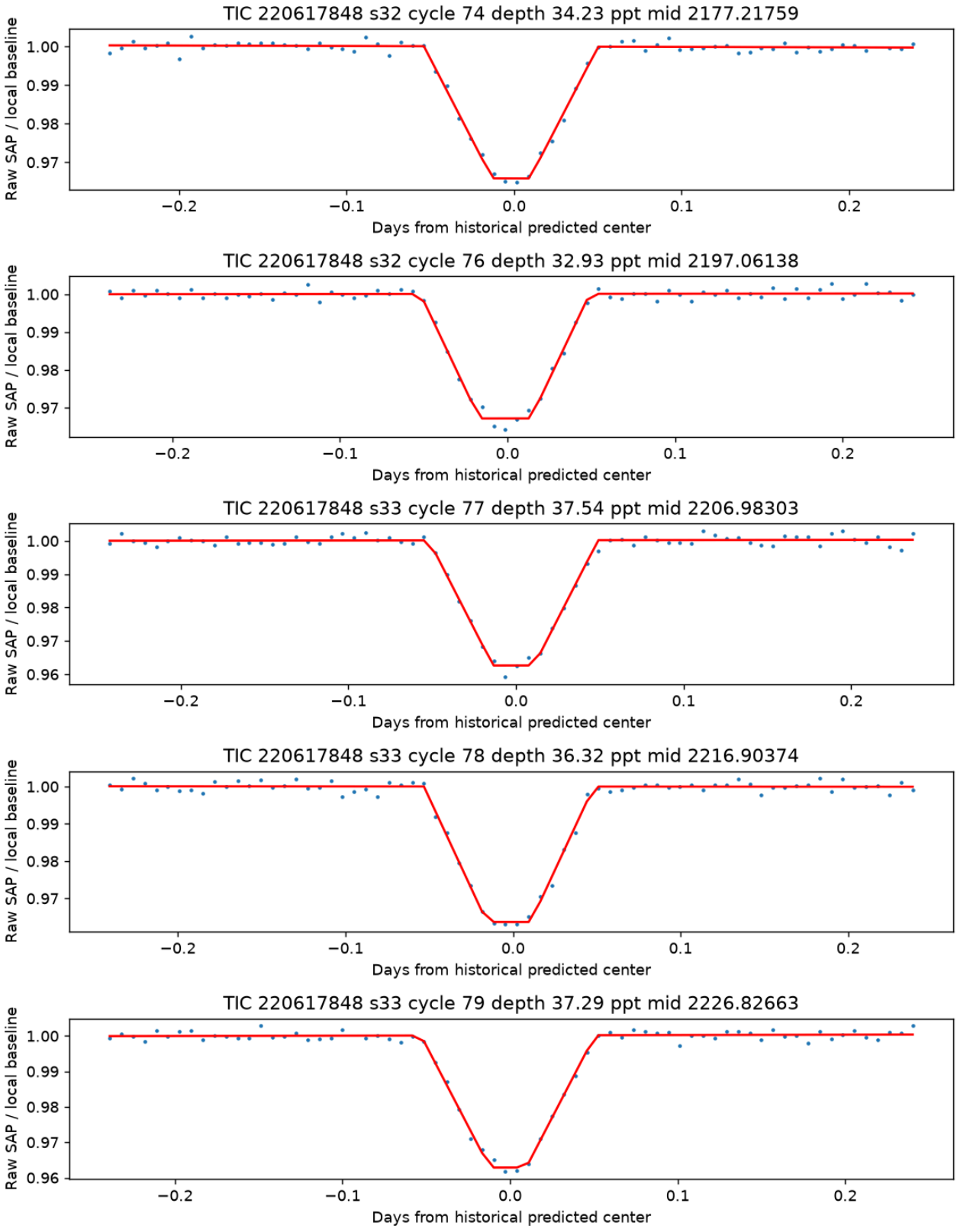
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
32	74	2177.2175890	0.003472	34.230	included timing center
32	76	2197.0613830	0.003472	32.929	included timing center
33	77	2206.9830260	0.003472	37.543	included timing center
33	78	2216.9037423	0.003472	36.320	included timing center
33	79	2226.8266345	0.003472	37.286	included timing center

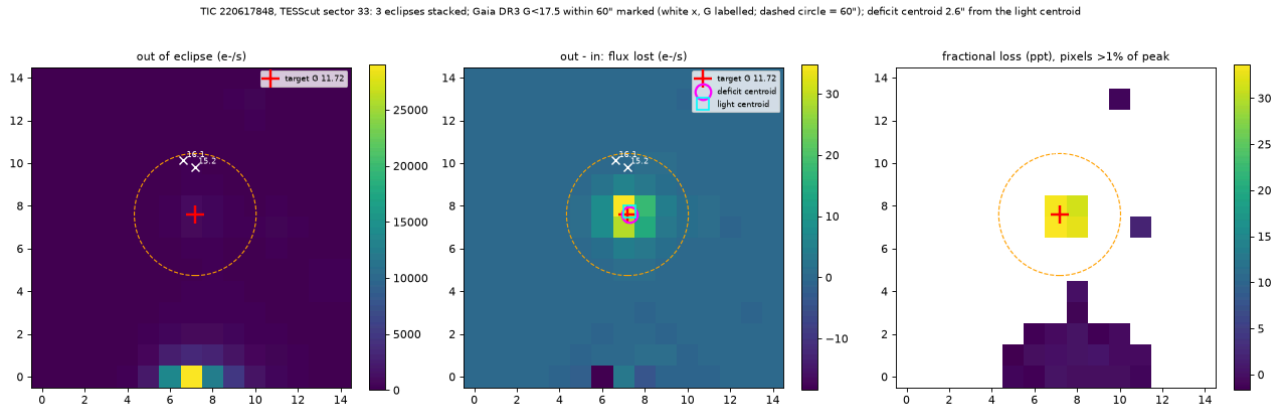
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s33.json, historical/diff_s33.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R353. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R353; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R353; diff_s33.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R354. TIC 254723054

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 254723054: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 7 distinct eclipse events in selected sectors[12, 65, 103]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=13.8861420099 ±0.0000226421d, reference eclipse epoch BTJD1636.3604759, kind: Deeper repeating eclipse; even half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S65 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.094arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.384. Capable neighbours: Gaia 5987885181823038720: 19.91 arcsec, G14.701, required fractional eclipse0.629
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths24.441/23.551/24.559ppt, with 3 stacked events. QLP selected-kind fitted median40.985ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3598788514313164928O, P=13.8892341157±0.0021777940d, e=0.042019, omega=326.0764deg. Closest edge-on conjunction offset-0.077217d (-0.005560phase), full-SB1-covariance linear propagated Gaia uncertainty0.194146d at the measured epoch. Near-Gaia-reference projected eclipse offset0.201072d (photometric period projection allowance0.002038d). The approximate central duration13.11h vs

observed median12.03h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Stable alternating depths establish doubled orbital period. Four deeper even historical half-period events alone define ephemeris; historical cutout stack mixes both kinds. Nearest capable neighbor is0.883pixel from deficit versus target0.100pixel; target favored at image resolution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

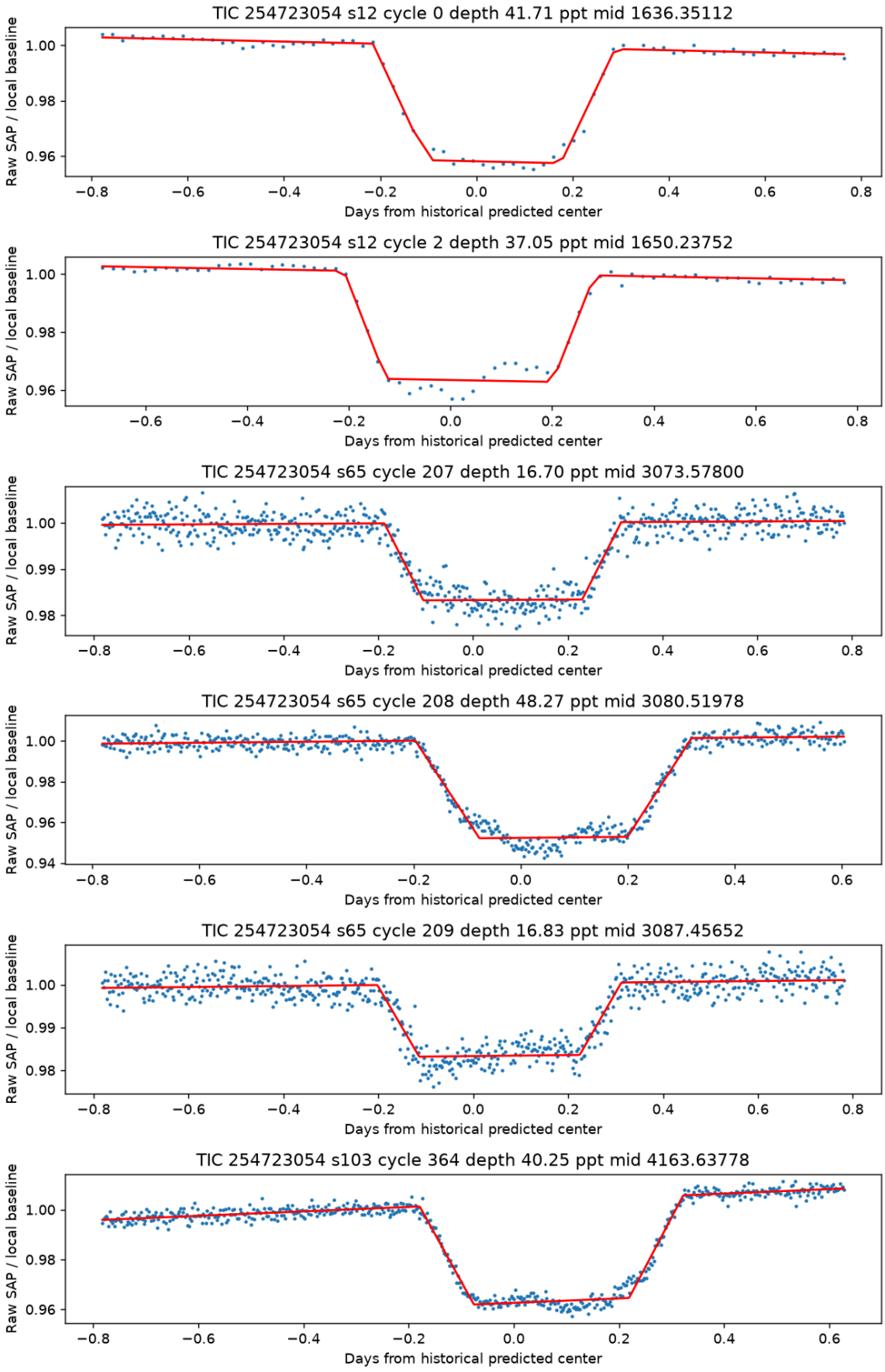
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
12	0	1636.3511235	0.010417	41.715	included timing center
12	2	1650.2375152	0.010417	37.047	included timing center
65	207	3073.5780045	0.001468	16.704	usable center; excluded opposite eclipse kind
65	208	3080.5197751	0.001157	48.266	included timing center
65	209	3087.4565207	0.001633	16.834	usable center; excluded opposite eclipse kind
103	363	4156.6930015	0.002959	14.142	usable center; excluded opposite eclipse kind
103	364	4163.6377848	0.001542	40.255	included timing center

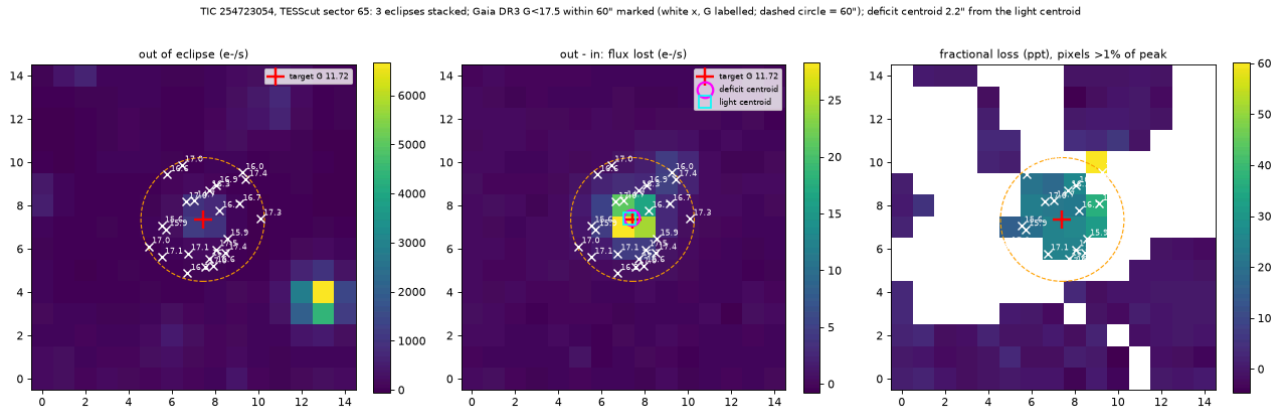
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s65.json, historical/diff_s65.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R354. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R354; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R354; diff_s65.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R355. TIC 257483992

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 257483992: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[3, 30, 97]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=8.7868298591 ±0.0000112166d, reference eclipse epoch BTJD1392.1739129, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.034arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.084. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths21.166/21.235/20.570ppt, with 4 stacked events. QLP selected-kind fitted median18.882ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3514769553608211635 2, P=8.7875714993±0.0016554215d, e=0.153182, omega=203.5610deg. Closest edge-on conjunction offset-0.112963d (-0.012855phase), full-SB1-covariance linear propagated Gaia uncertainty0.222418d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.028416d (photometric period projection allowance0.001279d). The approximate central

duration7.97h vs observed median6.42h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

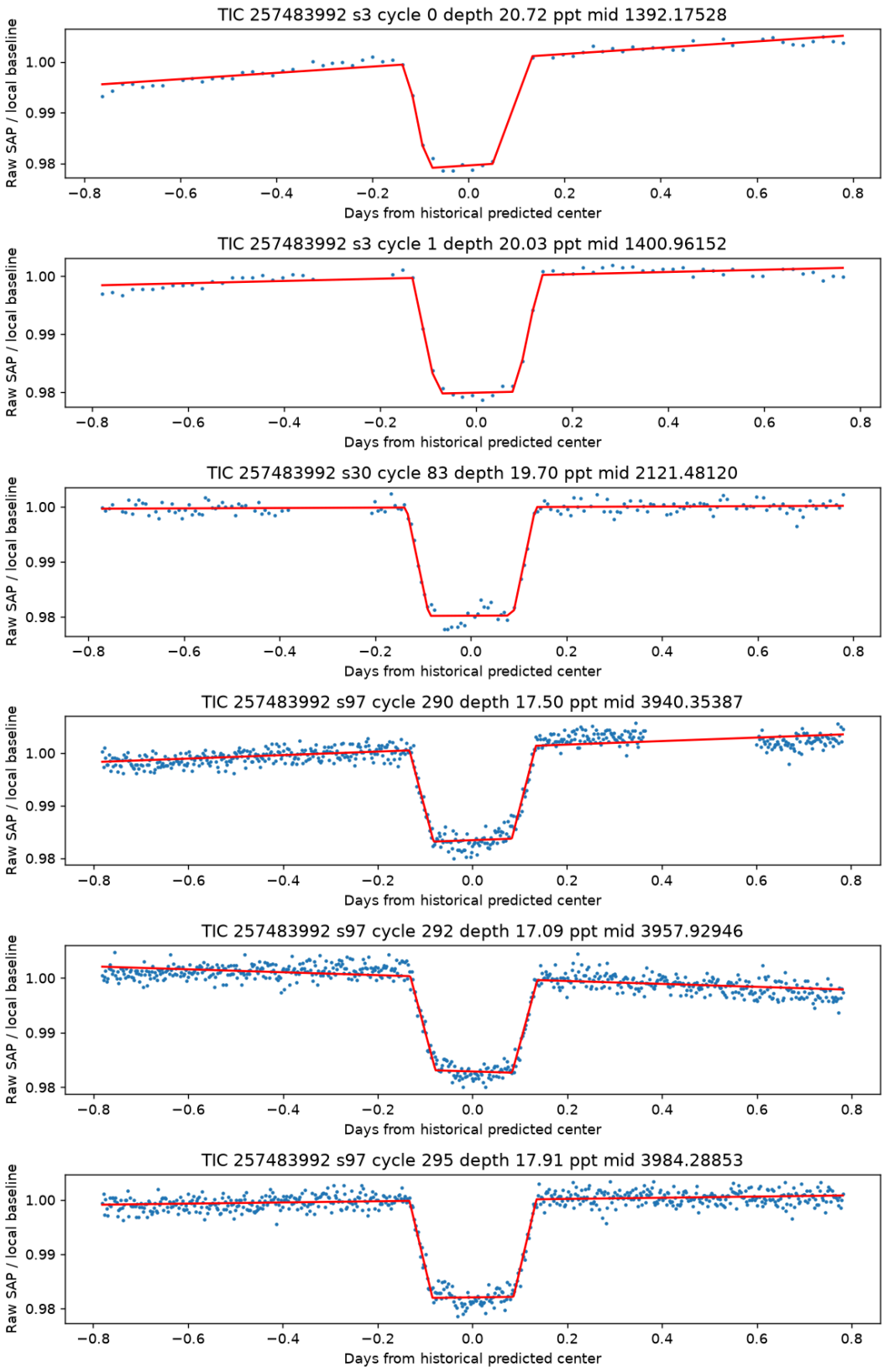
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
3	0	1392.1752764	0.014321	20.715	included timing center
3	1	1400.9615218	0.010417	20.032	included timing center
30	83	2121.4811951	0.003472	19.700	included timing center
30	85	2139.0538306	0.003472	21.420	included timing center
97	290	3940.3538705	0.001157	17.503	included timing center
97	292	3957.9294558	0.001157	17.092	included timing center
97	293	3966.7147318	0.001157	18.065	included timing center
97	295	3984.2885342	0.001157	17.907	included timing center

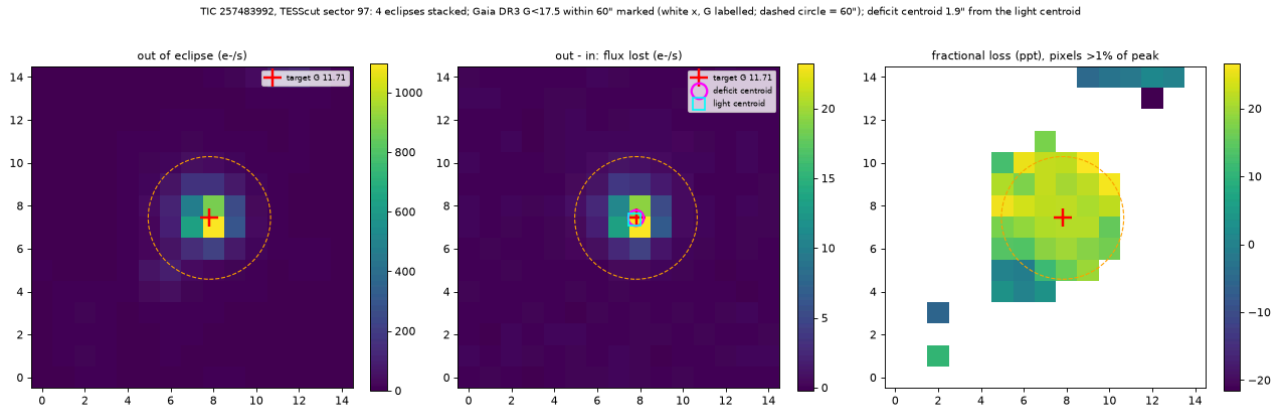
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R355. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R355; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R355; diff_s97.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R356. TIC 388052602

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Updated spectroscopy to test Gaia ephemeris tension

Archived source: sanitized report text.

TIC 388052602: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 21 distinct eclipse events in selected sectors[46, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 10 same-kind midtimes give P=4.5139686736 ±0.0000064790d, reference eclipse epoch BTJD2552.6908935, kind: Deeper repeating eclipse; odd half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S46 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.269arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.546. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths64.003/63.122/58.832ppt, with 11 stacked events. QLP selected-kind fitted median97.691ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3314714969430761408O, P=4.5134701102±0.0001857166d, e=0.019114, omega=5.4854deg. Closest edge-on conjunction offset0.307071d (0.068034phase), full-SB1-covariance linear propagated Gaia uncertainty0.101429d at the measured epoch. Near-Gaia-reference projected eclipse offset0.068259d (photometric period projection allowance0.003103d). The approximate central duration5.61h vs observed median3.48h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Stable alternating depths establish doubled orbital period. Ten deeper odd historical half-period events alone define ephemeris; historical cutout stack mixes both kinds. Gaia conjunction offset0.307d (0.068phase) exceeds the linear propagated0.101d error, but near-reference projected offset is0.06826d with independently propagated Gaia uncertainty0.02888d (about2.36 descriptive sigma) and the measured period differs from Gaia by0.000499d. Independent helper geometry388.py finds about3.03 descriptive sigma at the observed epoch. These are linear model uncertainties, not calibrated total coverage. Period accumulation explains much of the distant-epoch offset; low-eccentricity omega/periastron are strongly correlated. This association tension is retained without a sigma veto; updated spectroscopy would test it. Mass function0.126solar and approximate central duration5.61h versus observed3.48h admit noncentral binary geometry.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

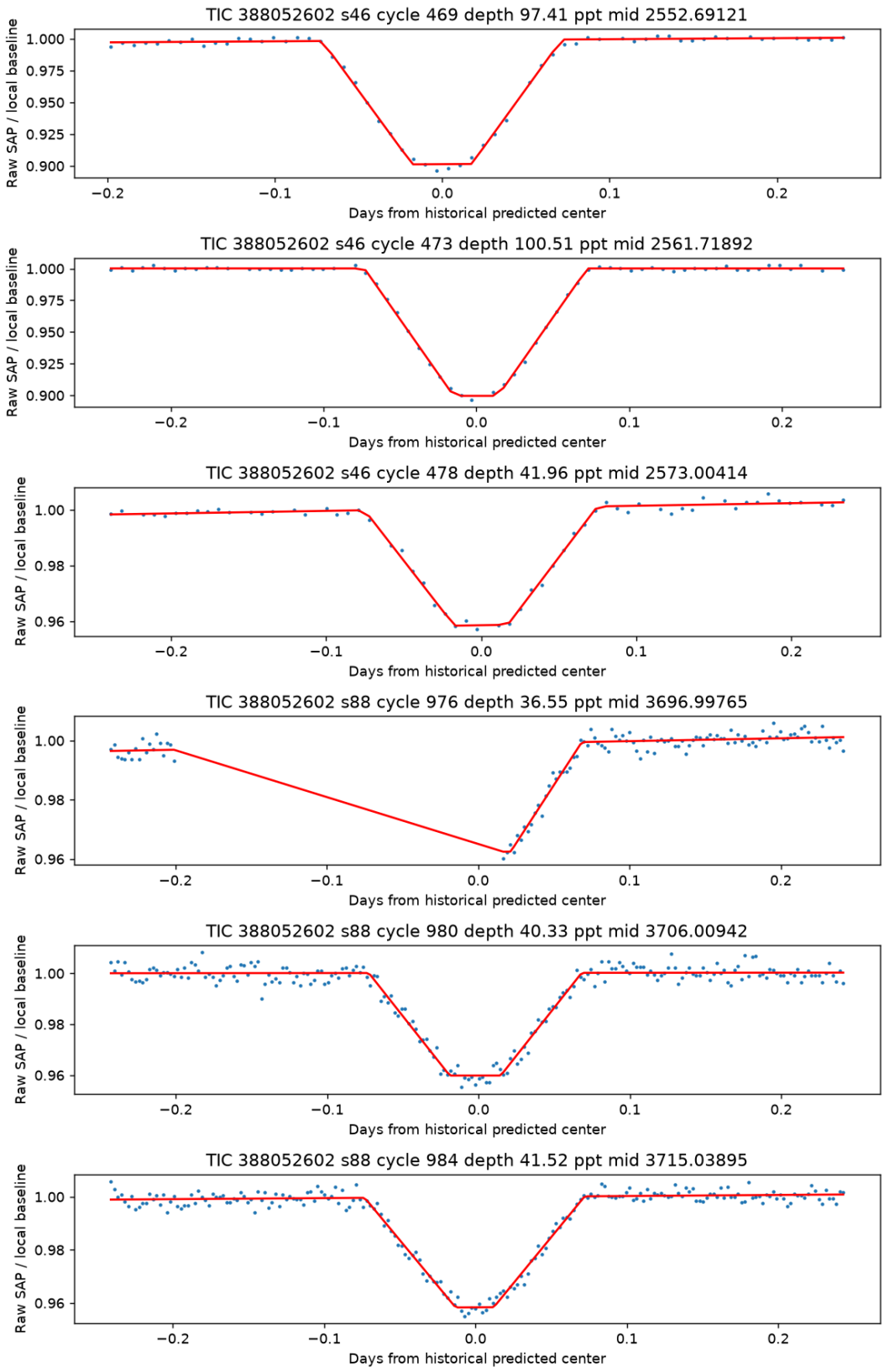
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
46	469	2552.6912073	0.003472	97.409	included timing center
46	470	2554.9481087	0.003472	42.644	usable center; excluded opposite eclipse kind
46	471	2557.2048891	0.003472	101.314	included timing center
46	472	2559.4614116	0.003472	41.894	usable center; excluded opposite eclipse kind
46	473	2561.7189195	0.003472	100.507	included timing center
46	474	2563.9757522	0.003472	42.439	usable center; excluded opposite eclipse kind
46	476	2568.4907964	0.003472	43.866	usable center; excluded opposite eclipse kind
46	477	2570.7464032	0.003472	100.765	included timing center
46	478	2573.0041426	0.003472	41.961	usable center; excluded opposite eclipse kind
46	479	2575.2606474	0.003472	98.805	included timing center
46	480	2577.5170125	0.003472	44.175	usable center; excluded opposite eclipse kind
88	975	3694.7252793	0.001157	97.972	included timing center
88	976	3696.9976478	211733.137317	36.554	excluded singular timing fit
88	977	3699.2388694	0.001157	93.808	included timing center
88	978	3701.4963538	0.001157	42.496	usable center; excluded opposite eclipse kind
88	979	3703.7527191	0.001157	94.557	included timing center
88	980	3706.0094214	0.001157	40.326	usable center; excluded opposite eclipse kind
88	981	3708.2667280	0.001157	95.139	included timing center
88	982	3710.5243085	0.001157	40.072	usable center; excluded opposite eclipse kind
88	983	3712.7809332	0.001157	94.660	included timing center

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
88	984	3715.0389528	0.001157	41.521	usable center; excluded opposite eclipse kind

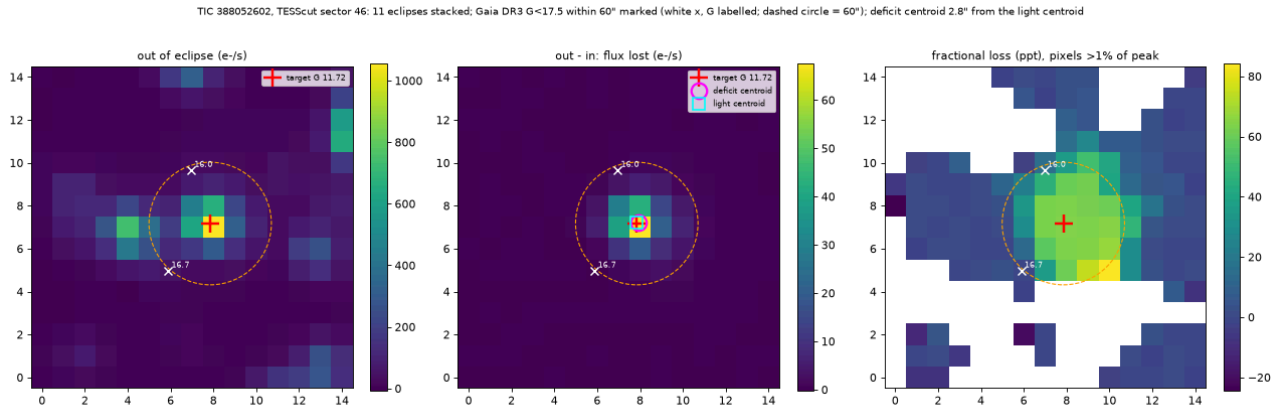
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s46.json, historical/diff_s46.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R356. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R356; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R356; diff_s46.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R357. TIC 157850336

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 157850336: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[14, 81]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=14.9033560025 ± 0.0000664269d, reference eclipse epoch BTJD1683.7854476, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S14 deficit peak[8, 1] vs target pixel[7, 7]; deficit-target separation1.384arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.981. Capable neighbours: Gaia 2098520337930119552: 51.05 arcsec, G15.628, required fractional eclipse0.622
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths17.846/16.645/13.145ppt, with 2 stacked events. QLP selected-kind fitted median17.411ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32098520303570374016, P=14.9046473566 ± 0.0047076703d, e=0.280777, omega=94.9536deg. Closest edge-on conjunction offset-0.095974d (-0.006439phase), full-SB1-covariance linear propagated Gaia uncertainty0.356126d at the measured epoch. Near-Gaia-reference projected eclipse offset0.016374d (photometric period projection allowance0.005779d). The approximate central duration6.79h vs

observed median6.90h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses and successive raw cycles secure recurrence; target deficit0.066pixel versus capable neighbor2.53pixels. Aperture dilution retained.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

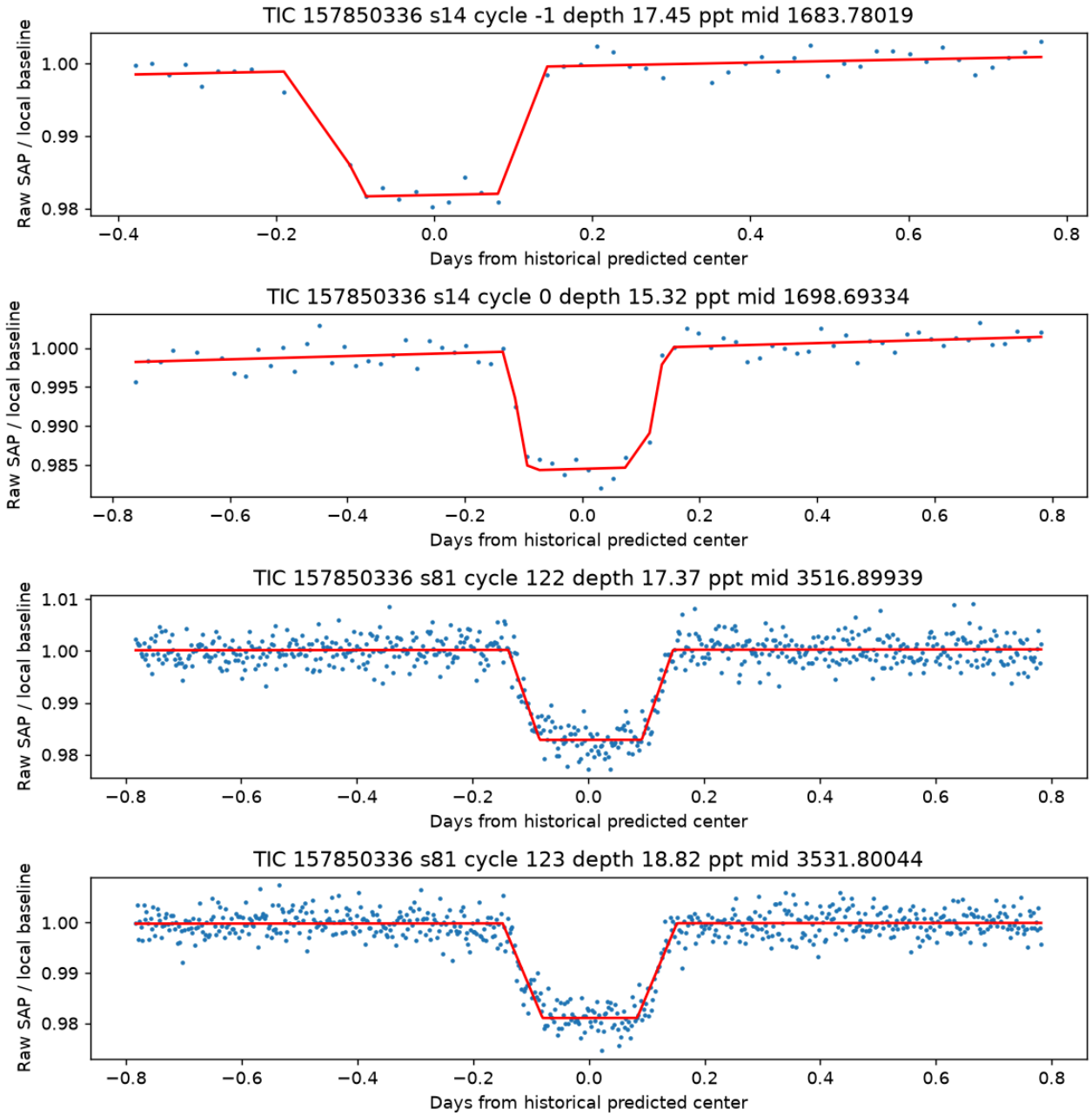
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	-1	1683.7801863	0.010417	17.450	used in ephemeris
14	0	1698.6933446	0.010417	15.321	used in ephemeris
81	122	3516.8993945	0.001157	17.372	used in ephemeris
81	123	3531.8004421	0.001157	18.824	used in ephemeris

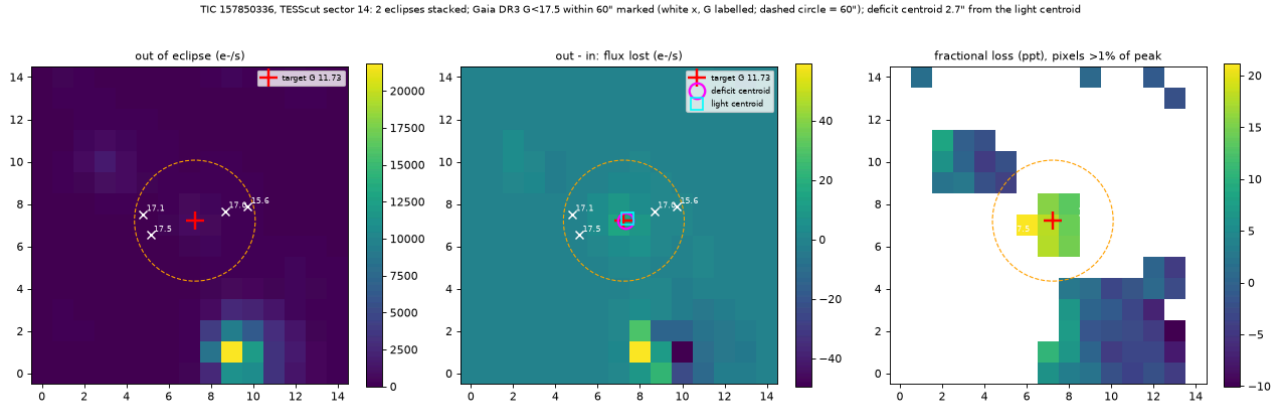
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s14.json, historical/diff_s14.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R357. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R357; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R357; diff_s14.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R358. TIC 233462817

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 233462817: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[49, 77]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=10.1615169478 ± 0.0000405203d, reference eclipse epoch BTJD2644.8490471, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S49 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation11.160arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.554. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths3.174/2.700/2.949ppt, with 2 stacked events. QLP selected-kind fitted median3.817ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3151269249146107520, P=10.1598188069 ± 0.0010289917d, e=0.023474, omega=64.8980deg. Closest edge-on conjunction offset0.416191d (0.040964phase), full-SB1-covariance linear propagated Gaia uncertainty0.245151d at the measured epoch. Near-Gaia-reference projected eclipse offset0.039204d (photometric period projection allowance0.008996d). The approximate central duration5.89h vs

observed median1.49h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Shallow repeated raw eclipses in sectors49/77; no capable listed neighbor. Deficit centroid0.53pixel offset and noisy difference image limit precise localization. Gaia conjunction offset0.041phase is interpreted with full covariance (0.245d), short observed duration allowing noncentral geometry; no arbitrary sigma veto.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

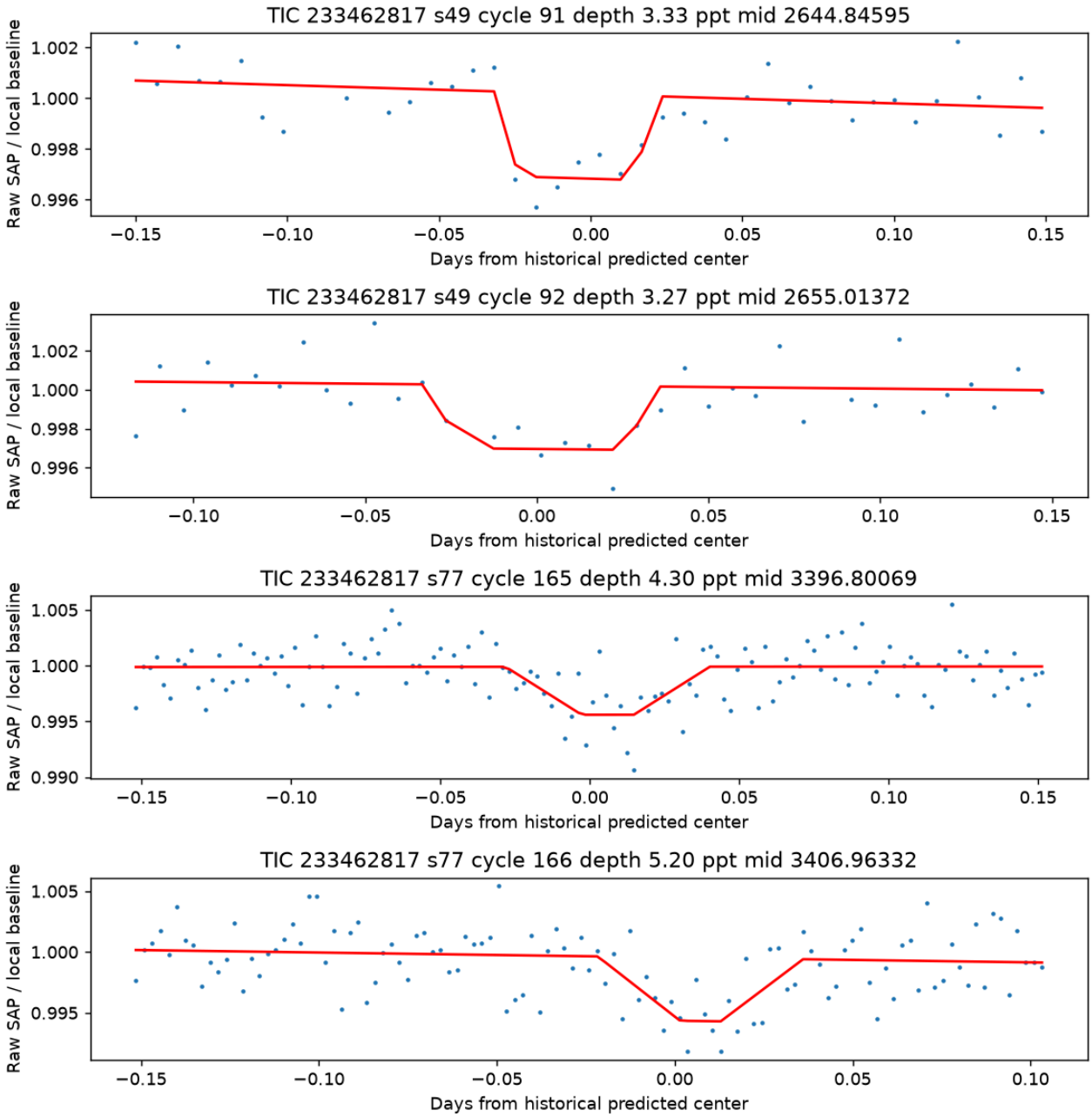
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
49	91	2644.8459518	0.003472	3.330	used in ephemeris
49	92	2655.0137168	0.003472	3.268	used in ephemeris
77	165	3396.8006857	0.002535	4.305	used in ephemeris
77	166	3406.9633220	0.002353	5.197	used in ephemeris

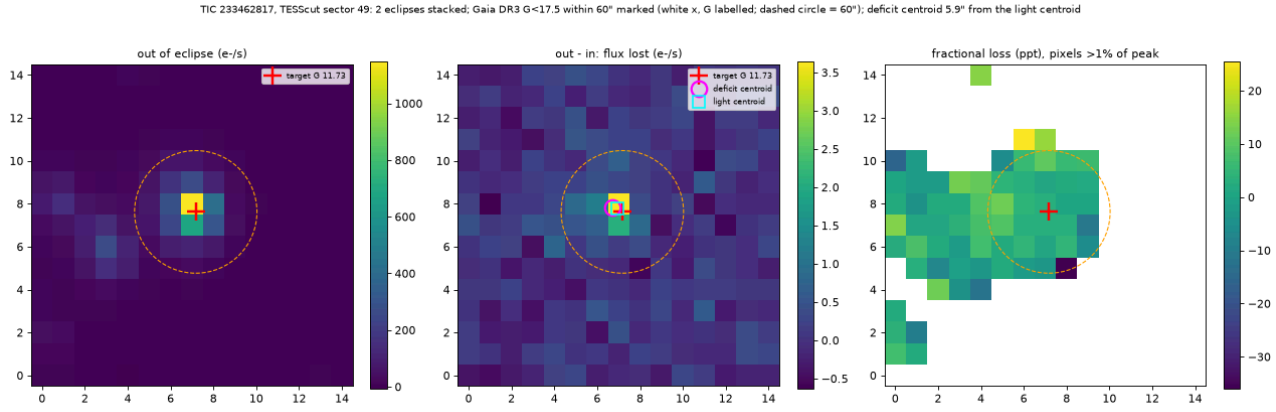
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s49.json, historical/diff_s49.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R358. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R358; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R358; diff_s49.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R359. TIC 48870240

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 48870240: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 6 distinct eclipse events in selected sectors[91]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=2.7256354375 ± 0.0001582455d, reference eclipse epoch BTJD3777.6561442, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S91 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.173arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.595. Capable neighbours: Gaia 6248866704865905536: 29.16 arcsec, G11.866, required fractional eclipse0.079
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths32.789/20.566/17.718ppt, with 7 stacked events. QLP selected-kind fitted median37.000ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3624886632260882124 8, P=2.7257434900 ± 0.0002657921d, e=0.047898, omega=356.3095deg. Closest edge-on conjunction offset-0.069254d (-0.025408phase), full-SB1-covariance linear propagated Gaia uncertainty0.334151d at the measured epoch. Near-Gaia-reference projected eclipse offset0.065055d (photometric period projection allowance0.196699d). The approximate central duration4.18h vs

observed median3.87h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Six measured raw events in sector91 include four successive cycles5-8 and the cycle0/1 pair, establishing recurrence. Capable neighbor1.39pixels from deficit versus target0.056pixel; target core favors attribution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

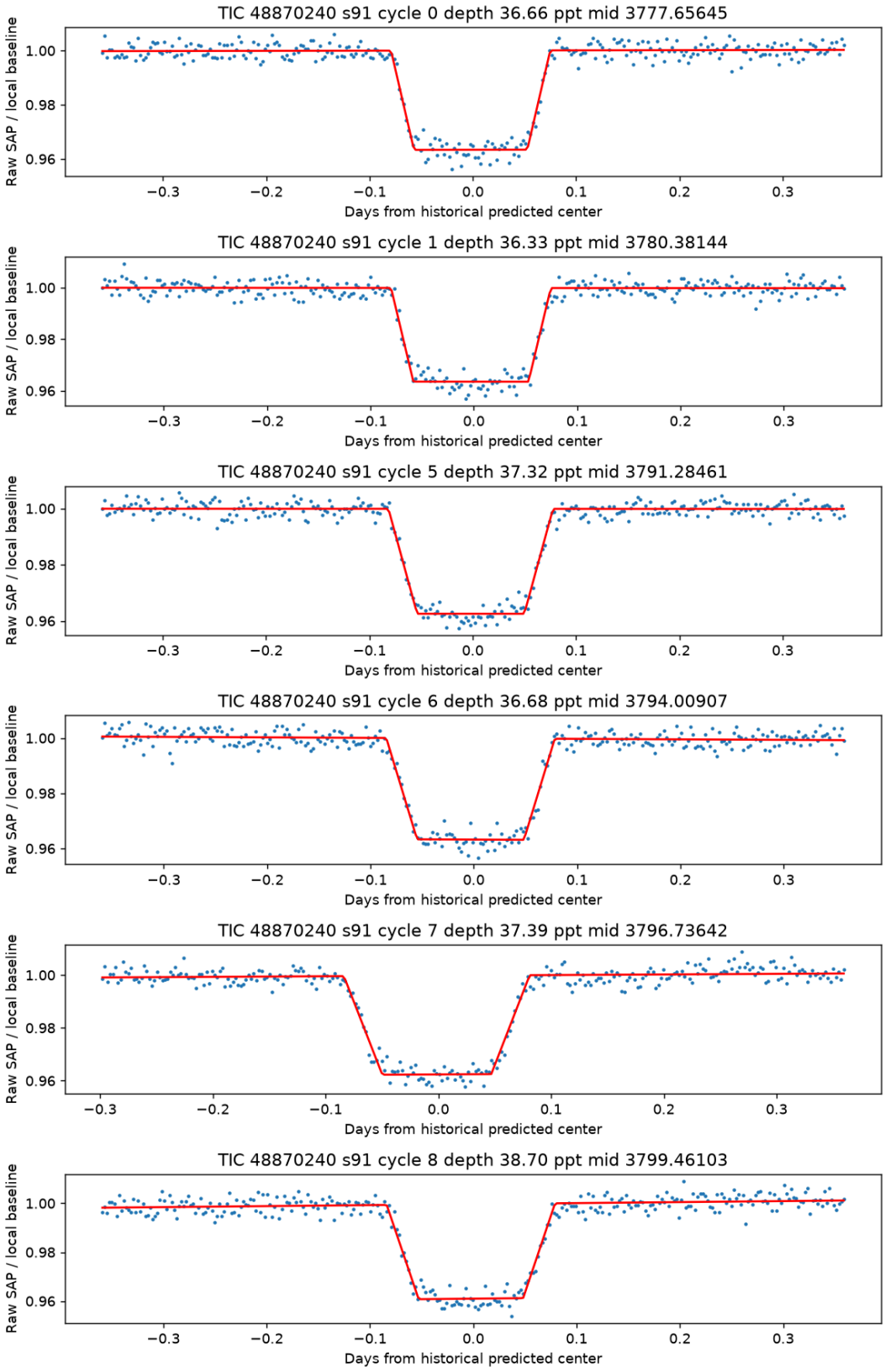
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
91	0	3777.6564539	0.001157	36.662	used in ephemeris
91	1	3780.3814365	0.001157	36.331	used in ephemeris
91	5	3791.2846138	0.001157	37.322	used in ephemeris
91	6	3794.0090677	0.001157	36.677	used in ephemeris
91	7	3796.7364181	0.001157	37.388	used in ephemeris
91	8	3799.4610321	0.001157	38.699	used in ephemeris

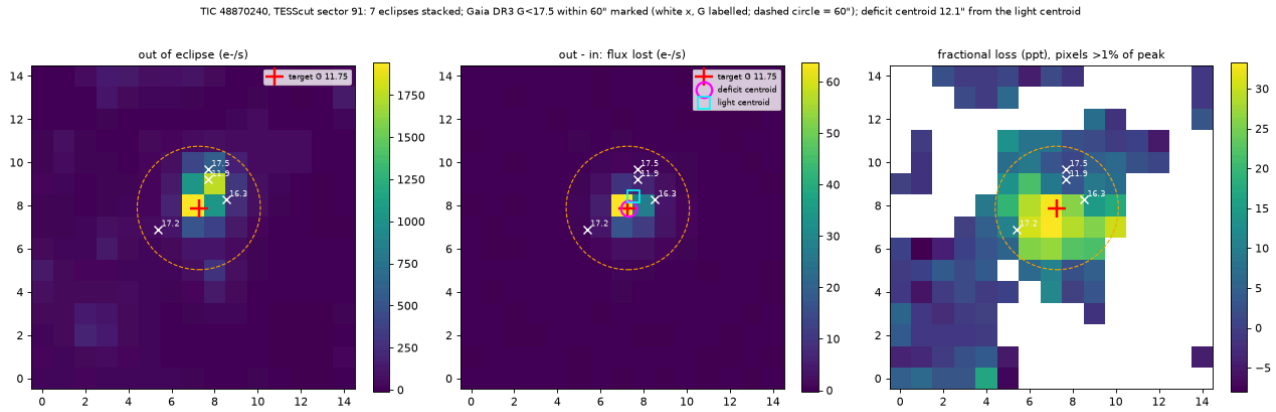
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s91.json, historical/diff_s91.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R359. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R359; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R359; diff_s91.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R360. TIC 178199444

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 178199444: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 3 distinct eclipse events in selected sectors[31]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=6.5485022540 ± 0.0016073438d, reference eclipse epoch BTJD2149.2179543, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S31 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.857arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.964. Capable neighbours: Gaia 5087572522289602944: 59.59 arcsec, G14.889, required fractional eclipse0.700
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths36.052/32.012/21.556ppt, with 4 stacked events. QLP selected-kind fitted median40.179ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3508757262536881689 6, P=6.5478351785 ± 0.0003587164d, e=0.006130, omega=71.8319deg. Closest edge-on conjunction offset-0.031815d (-0.004859phase), full-SB1-covariance linear propagated Gaia uncertainty0.111501d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.211258d (photometric period projection allowance0.432375d). The approximate central duration4.44h vs

observed median2.31h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Three raw events in sector31 include consecutive cycles110/111 and a two-cycle separation108/110. Single selected sector supplies repeated cycles; this is not a single-event case.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

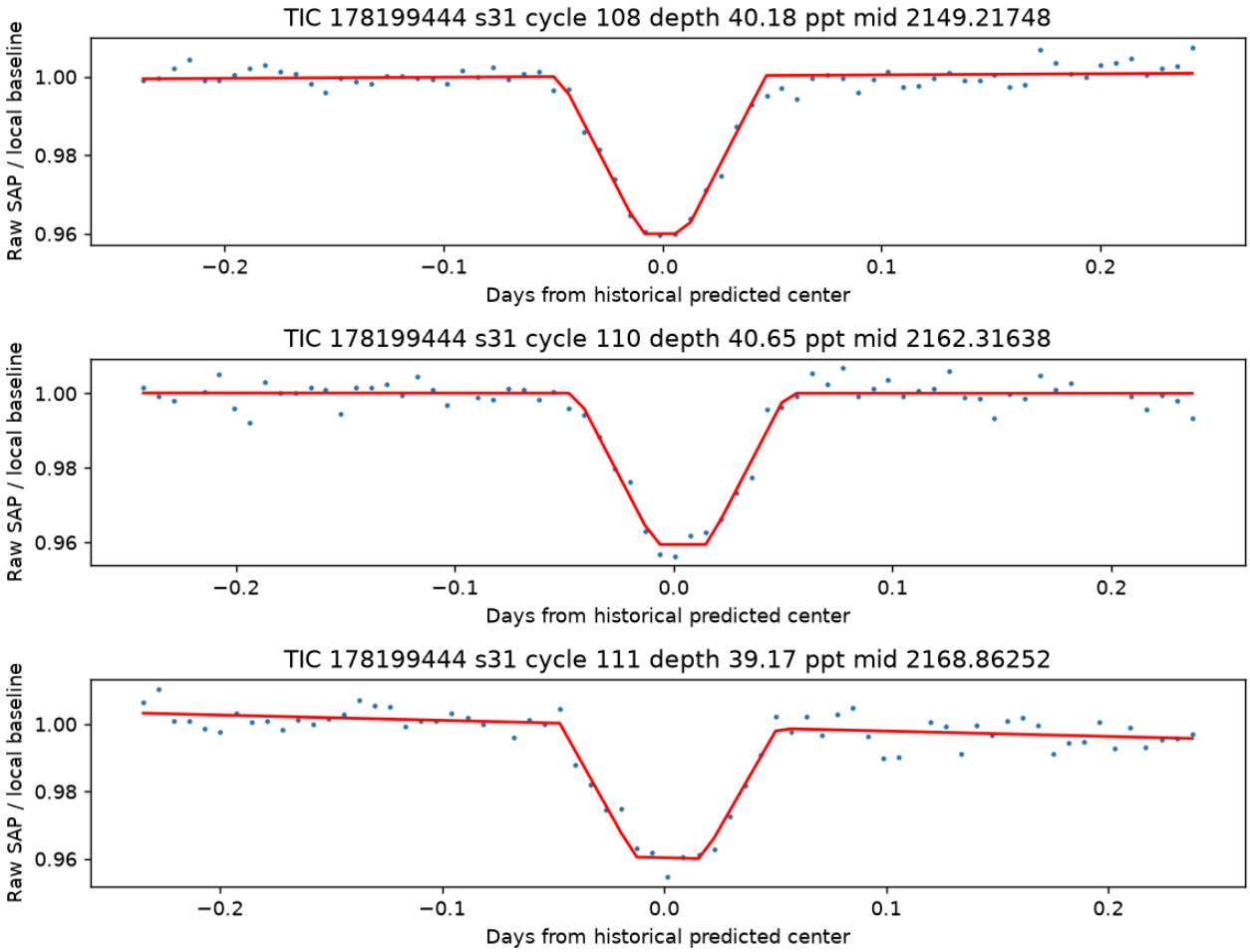
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
31	108	2149.2174819	0.003472	40.179	used in ephemeris
31	110	2162.3163759	0.003472	40.647	used in ephemeris
31	111	2168.8625163	0.003472	39.168	used in ephemeris

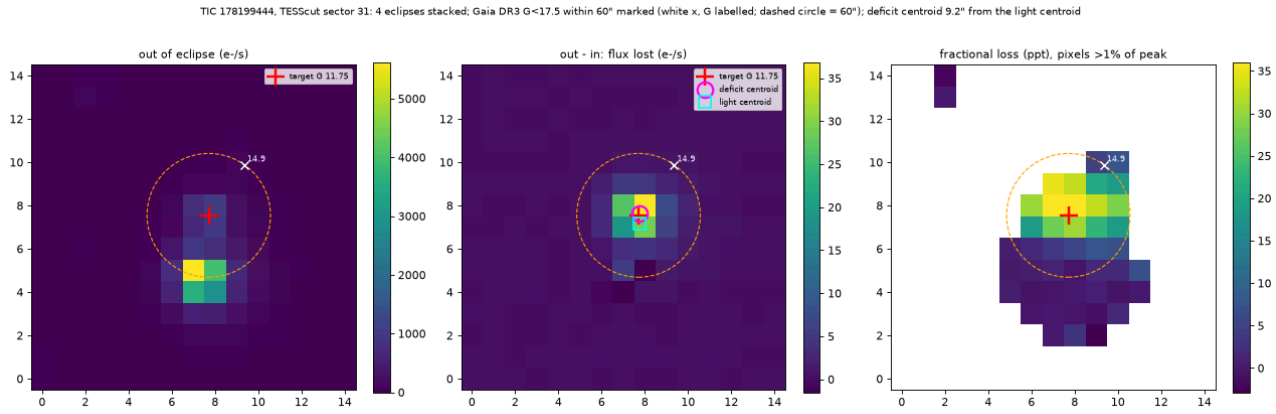
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s31.json, historical/diff_s31.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R360. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R360; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R360; diff_s31.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R361. TIC 375058760

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 375058760: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[1, 36, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=7.1293749205 ± 0.0000079011d, reference eclipse epoch BTJD1329.2625675, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.295arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.055. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths392.706/389.647/379.041ppt, with 6 stacked events. QLP selected-kind fitted median393.515ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3547872223768323046 4, P=7.1284859943 ± 0.0003946301d, e=0.021598, omega=6.5168deg. Closest edge-on conjunction offset0.175983d (0.024687phase), full-SB1-covariance linear propagated Gaia uncertainty0.072295d at the measured epoch. Near-Gaia-reference projected eclipse offset0.058645d (photometric period projection allowance0.001043d). The approximate central duration5.36h vs observed median4.29h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated V eclipses show sector-dependent morphology and fitted depths; raw baseline/shape systematics remain and fitted depths are diluted. No detailed binary characterization claimed.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

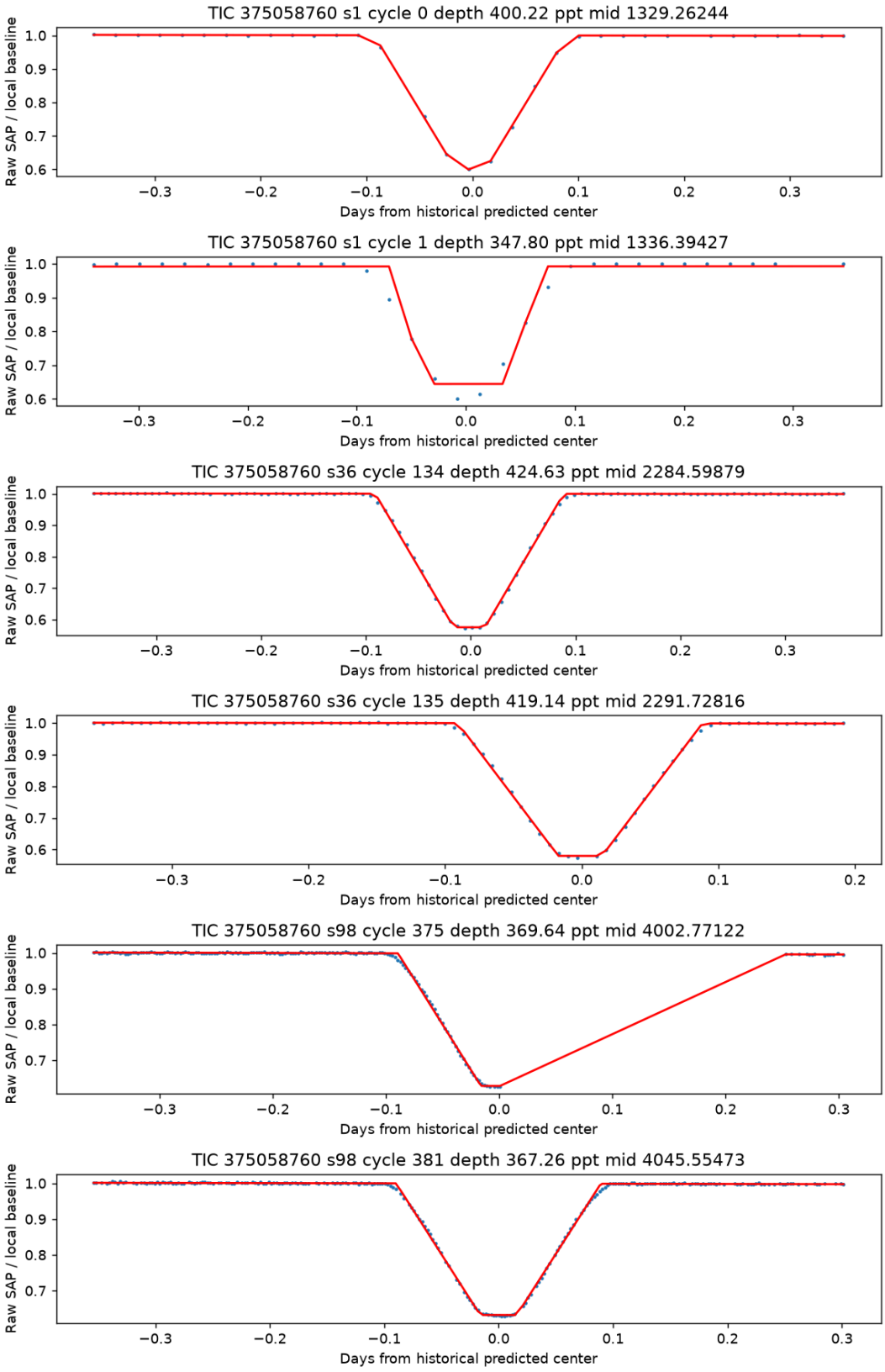
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1329.2624358	0.010417	400.217	used in ephemeris
1	1	1336.3942674	0.010417	347.800	used in ephemeris
1	3	1350.6505991	0.010417	386.813	used in ephemeris
36	134	2284.5987869	0.003472	424.631	used in ephemeris
36	135	2291.7281639	0.003472	419.137	used in ephemeris
36	136	2298.8572551	0.003472	417.119	used in ephemeris
98	375	4002.7712239	42631.591806	369.645	Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
98	379	4031.2953609	0.001157	372.982	used in ephemeris
98	381	4045.5547256	0.001157	367.255	used in ephemeris

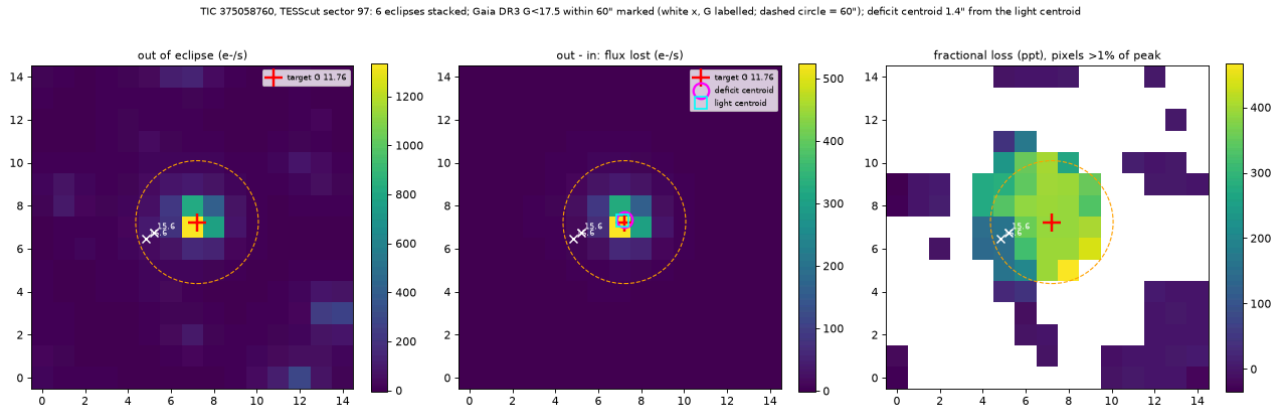
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R361. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R361; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R361; diff_s97.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R362. TIC 167923097

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 167923097: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 5 distinct eclipse events in selected sectors[9, 62, 100]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=15.9435356330 ± 0.0000166784d, reference eclipse epoch BTJD1560.4521364, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S35 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation9.287arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.537. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths226.324/220.069/212.394ppt, with 2 stacked events. QLP selected-kind fitted median241.112ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3546644935145490521 6, P=15.9431608845 ± 0.0084437670d, e=0.133116, omega=261.7914deg. Closest edge-on conjunction offset-0.439652d (-0.027576phase), full-SB1-covariance linear propagated Gaia uncertainty0.577385d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.467008d (photometric period projection allowance0.001218d). The approximate central

duration6.93h vs observed median5.11h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated V eclipses in sectors9/62/100; aperture differences and0.44pixel deficit offset retained; no capable listed neighbor.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

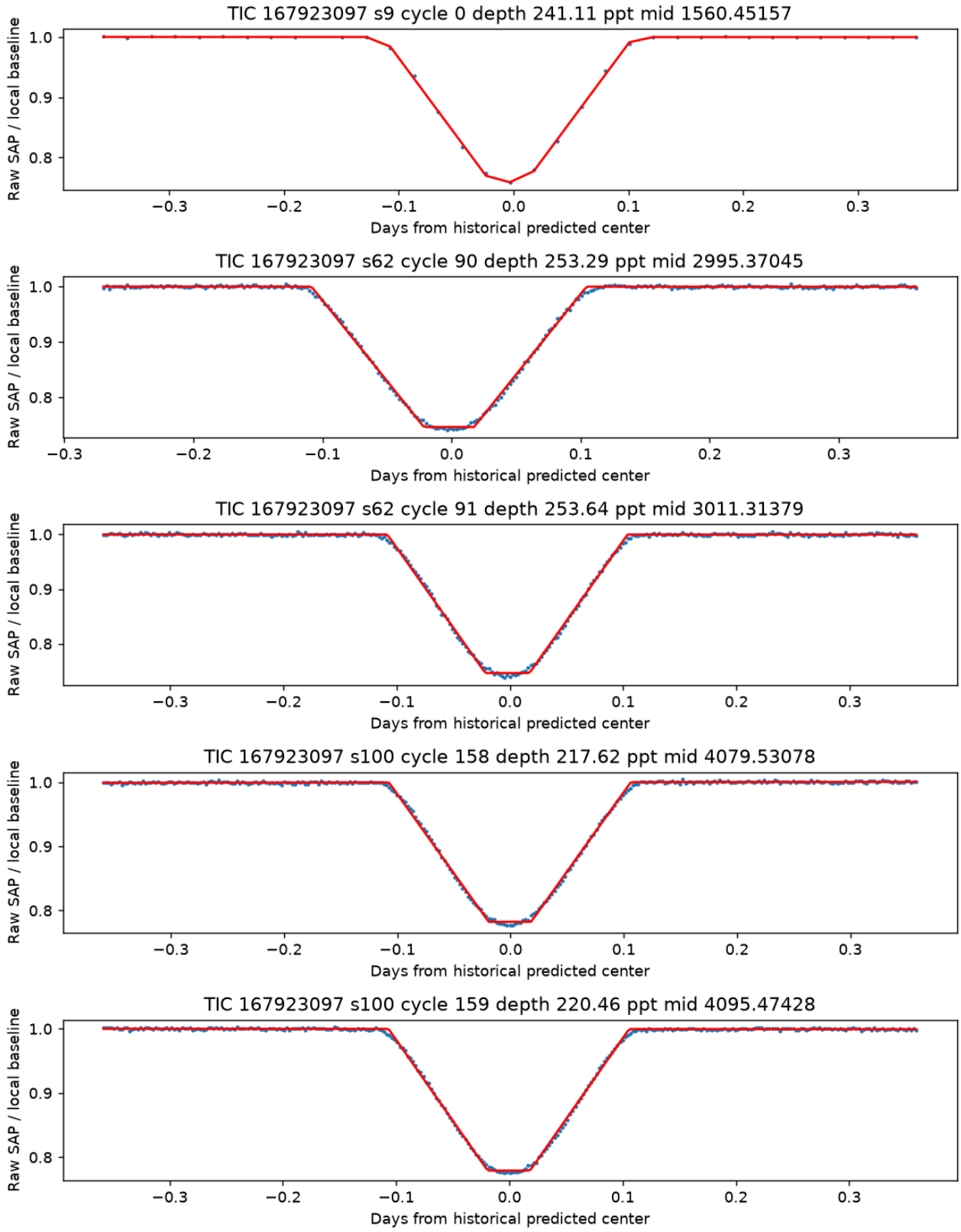
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
9	0	1560.4515699	0.010417	241.112	used in ephemeris
62	90	2995.3704457	0.001157	253.285	used in ephemeris
62	91	3011.3137913	0.001157	253.640	used in ephemeris
100	158	4079.5307793	0.001157	217.623	used in ephemeris
100	159	4095.4742815	0.001157	220.461	used in ephemeris

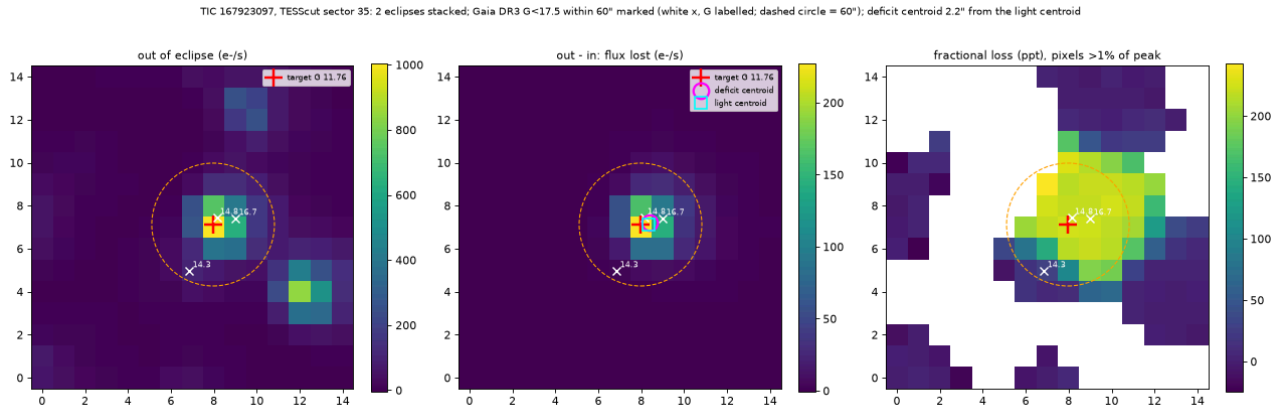
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s35.json, historical/diff_s35.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R362. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R362; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R362; diff_s35.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R363. TIC 406297889

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 406297889: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 23 distinct eclipse events in selected sectors[14, 55, 82]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 23 same-kind midtimes give P=2.8227799299 ± 0.0000039347d, reference eclipse epoch BTJD1685.8956770, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation2.266arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.408. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths87.296/81.926/78.810ppt, with 10 stacked events. QLP selected-kind fitted median95.040ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3186066906917655616O, P=2.8228180213 ± 0.0000945775d, e=0.034549, omega=126.9402deg. Closest edge-on conjunction offset-0.025598d (-0.009068phase), full-SB1-covariance linear propagated Gaia uncertainty0.046441d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.008114d (photometric period projection allowance0.001806d). The approximate central

duration4.60h vs observed median4.16h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated raw events across sectors14/55/82 with stable isolated morphology; no capable listed neighbor.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

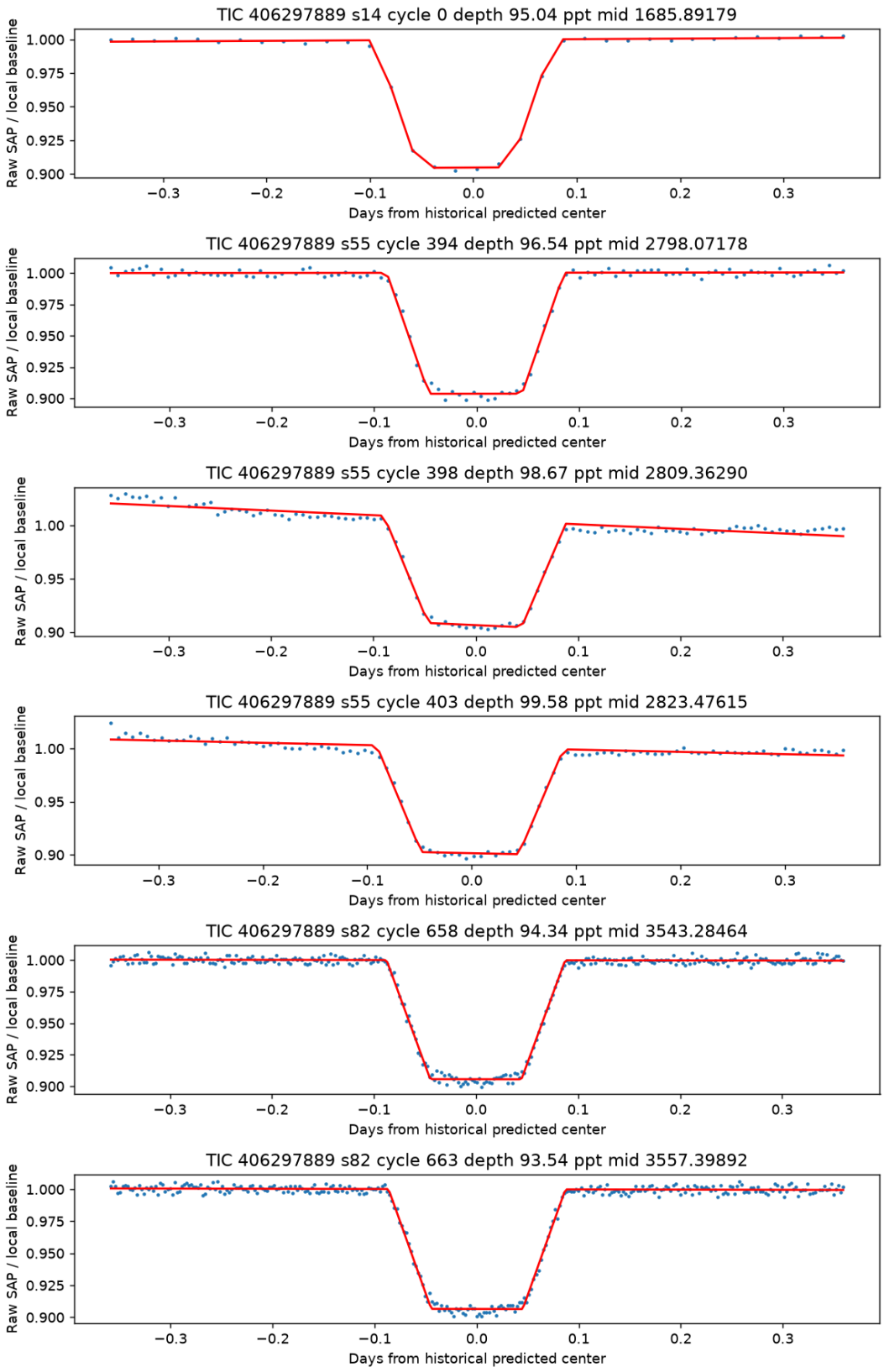
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	0	1685.8917905	0.010417	95.040	used in ephemeris
14	1	1688.7135021	0.010417	96.145	used in ephemeris
14	5	1700.0051986	0.010417	93.668	used in ephemeris
14	6	1702.8279558	0.010417	95.062	used in ephemeris
55	394	2798.0717778	0.003472	96.535	used in ephemeris
55	395	2800.8942891	0.003472	96.278	used in ephemeris
55	396	2803.7169761	0.003472	95.099	used in ephemeris
55	397	2806.5398949	0.003472	97.329	used in ephemeris
55	398	2809.3628982	0.003472	98.674	used in ephemeris
55	399	2812.1855604	0.003472	96.196	used in ephemeris
55	400	2815.0081985	0.003472	96.954	used in ephemeris
55	401	2817.8306117	0.003472	95.109	used in ephemeris
55	402	2820.6534237	0.003472	92.253	used in ephemeris
55	403	2823.4761498	0.003472	99.577	used in ephemeris
82	655	3534.8161824	0.001157	93.909	used in ephemeris
82	656	3537.6390350	0.001157	93.036	used in ephemeris
82	657	3540.4625039	0.001157	93.587	used in ephemeris
82	658	3543.2846434	0.001157	94.343	used in ephemeris
82	659	3546.1075581	0.001157	92.950	used in ephemeris
82	660	3548.9305805	0.001157	94.080	used in ephemeris
82	661	3551.7530043	0.001157	94.672	used in ephemeris
82	662	3554.5760892	0.001157	93.599	used in ephemeris
82	663	3557.3989236	0.001157	93.538	used in ephemeris

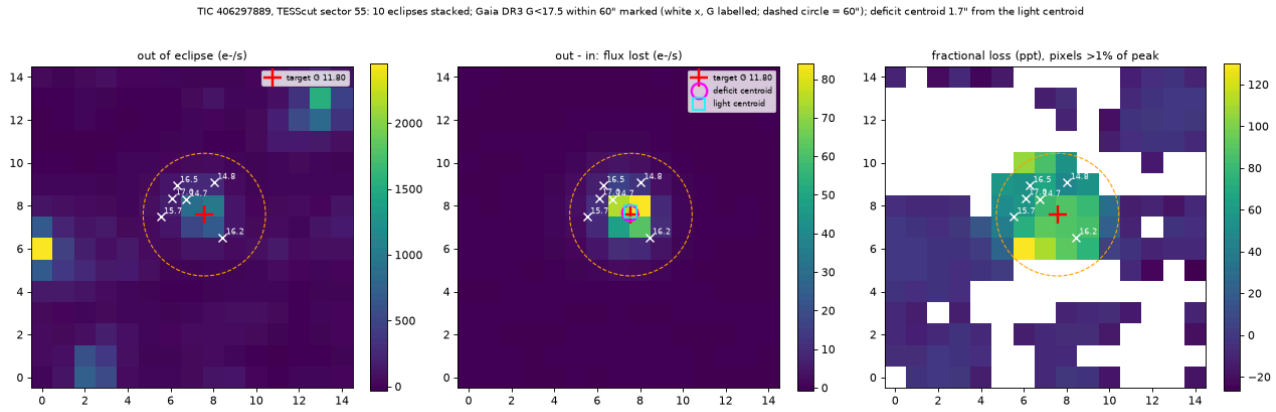
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R363. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R363; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R363; diff_s55.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R364. TIC 160258534

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 160258534: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[15, 56, 83]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=7.7049863920 ± 0.0000097839d, reference eclipse epoch BTJD1730.5945848, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 7] vs target pixel[7, 7]; deficit-target separation1.654arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.519. Capable neighbours: Gaia 1951244157007311360: 49.79 arcsec, G13.296, required fractional eclipse0.491; Gaia 1951244569324172288: 53.83 arcsec, G13.622, required fractional eclipse0.628
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths89.365/86.649/69.479ppt, with 3 stacked events. QLP selected-kind fitted median96.211ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3195124453496447872O, P=7.7037472107 ± 0.0008325053d, e=0.045926, omega=218.2302deg. Closest edge-on conjunction offset0.215587d (0.027985phase), full-SB1-covariance linear propagated Gaia uncertainty0.179708d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.000030d (photometric period projection allowance0.001702d). The approximate central duration4.74h vs observed median3.64h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated eclipses across sectors15/56/83; capable neighbors2.51/2.73pixels from deficit versus target0.079pixel. Pixel core supports target at TESS resolution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

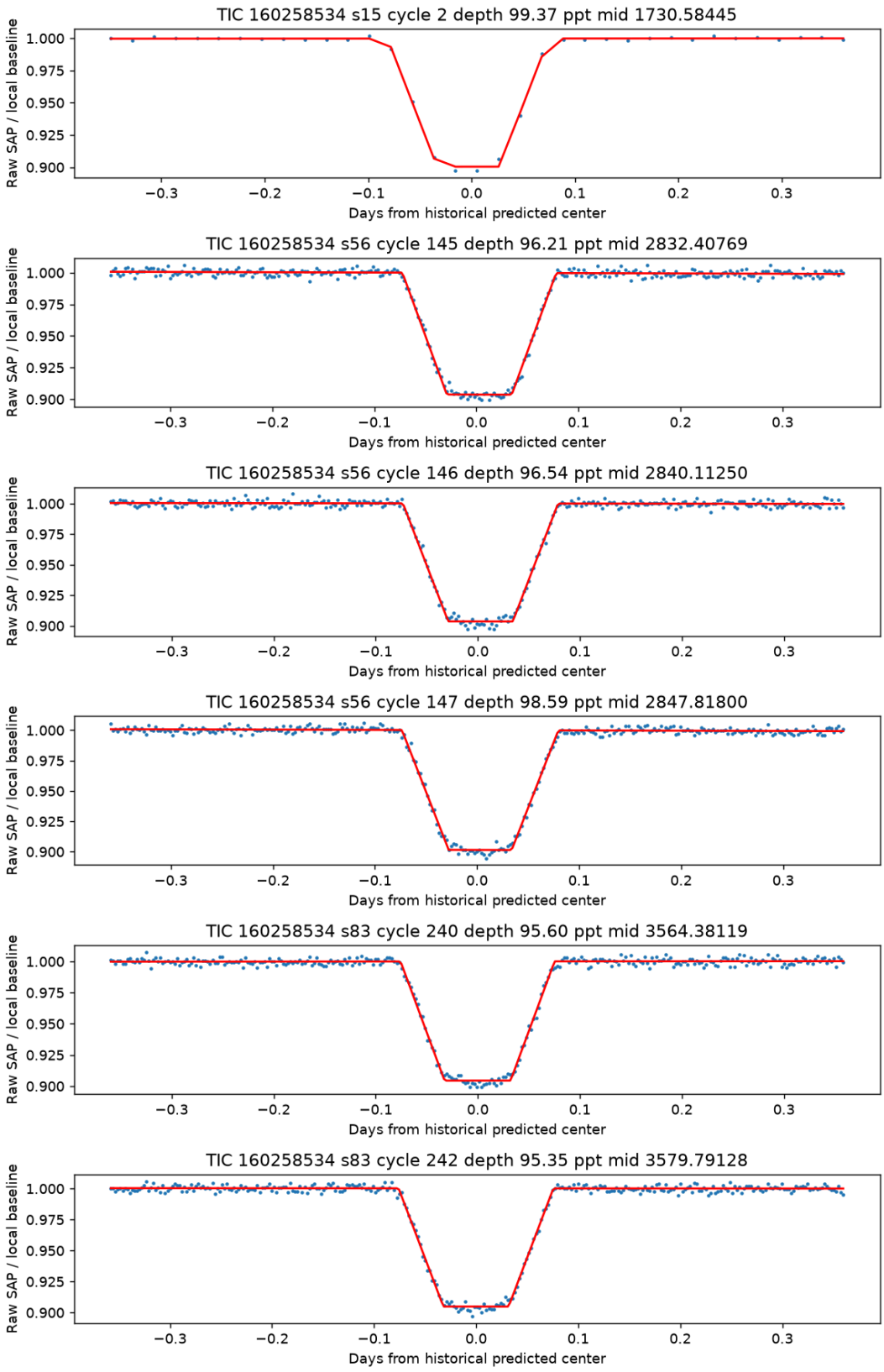
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	2	1730.5844469	0.010417	99.373	used in ephemeris
56	145	2832.4076921	0.001157	96.211	used in ephemeris
56	146	2840.1125037	0.001157	96.537	used in ephemeris
56	147	2847.8179997	0.001157	98.588	used in ephemeris
83	240	3564.3811879	0.001157	95.605	used in ephemeris
83	241	3572.0863352	0.001157	96.052	used in ephemeris
83	242	3579.7912800	0.001157	95.350	used in ephemeris

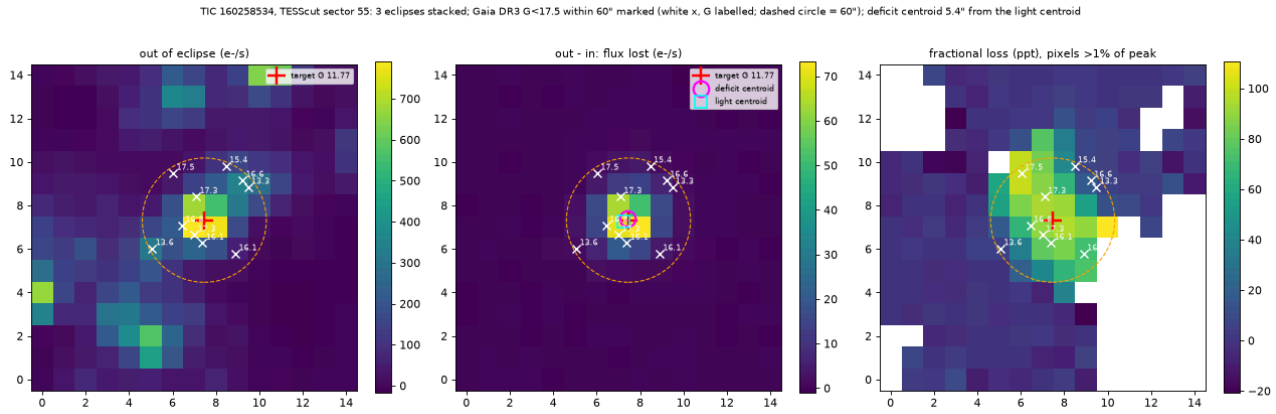
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R364. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R364; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R364; diff_s55.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R365. TIC 16809274

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 16809274: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[15, 82, 83]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=14.7258719722 ± 0.0000898281d, reference eclipse epoch BTJD1714.5124553, kind: Shallower repeating eclipse, even historical half-period cycles; not asserted to be deeper primary. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S56 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.360arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.958. Capable neighbours: Gaia 1897254734463908352: 24.02 arcsec, G13.330, required fractional eclipse0.305
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths55.552/50.247/40.055ppt, with 4 stacked events. QLP selected-kind fitted median58.352ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3189725473016727052 8, P=14.7287368632 ± 0.0029848195d, e=0.026585, omega=301.3580deg. Closest edge-on conjunction offset-0.254052d (-0.017249phase), full-SB1-covariance linear propagated Gaia uncertainty0.301075d at the measured epoch. Near-Gaia-reference projected eclipse offset0.003788d (photometric period projection allowance0.008085d). The approximate central duration15.06h vs observed median14.83h permits noncentral geometry; this radius/minimum-

mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Persistent unequal alternating eclipses establish doubled orbital period relative to historical SAP half-period7.363d. Even historical cycles are the shallower event; only those five measured events define the ephemeris, explicitly not called deeper primary. Odd deeper events remain in raw evidence. Capable neighbor1.046pixels from deficit versus target0.160pixel; target core favors attribution. Historical cutout stack mixes eclipse kinds.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

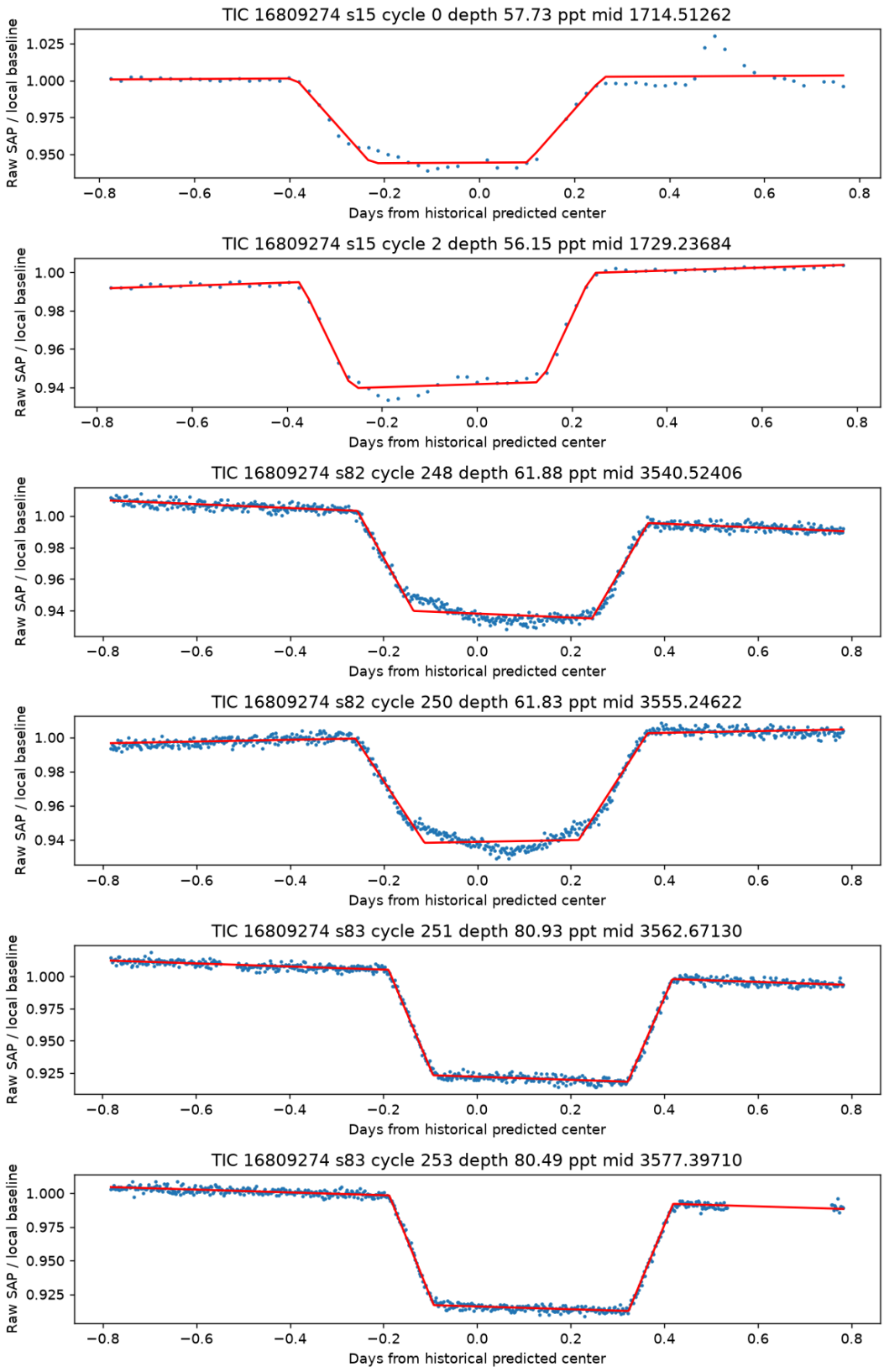
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1714.5126228	0.010417	57.733	used in ephemeris
15	2	1729.2368375	0.010417	56.147	used in ephemeris
82	248	3540.5240558	0.002026	61.879	used in ephemeris
82	249	3547.9459064	0.001157	86.665	measured; excluded from selected shallower-eclipse fit (deeper event)
82	250	3555.2462222	0.001157	61.825	used in ephemeris
83	251	3562.6712971	0.001514	80.932	measured; excluded from selected shallower-eclipse fit (deeper event)
83	252	3569.9665978	0.002939	58.352	used in ephemeris
83	253	3577.3971027	0.001157	80.492	measured; excluded from selected shallower-eclipse fit (deeper event)

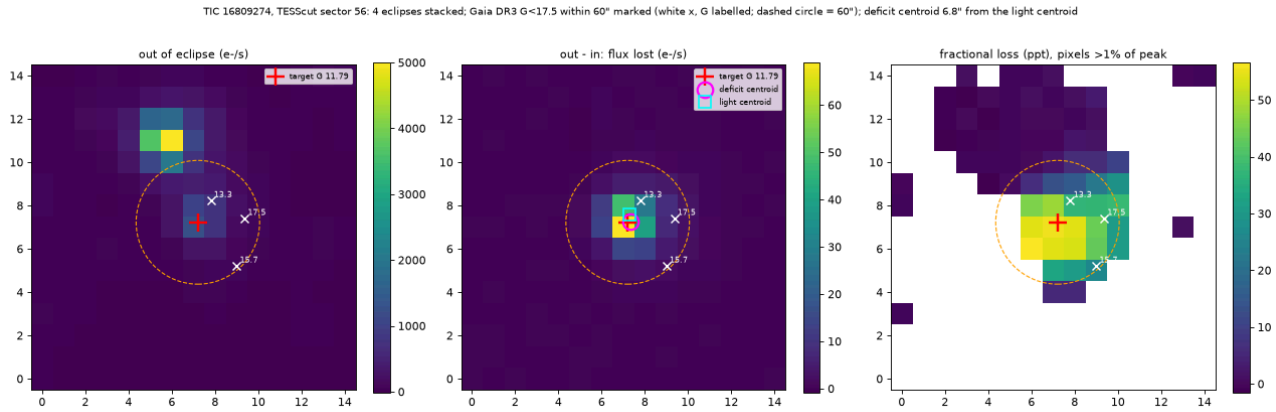
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s56.json, historical/diff_s56.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R365. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R365; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R365; diff_s56.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R366. TIC 279065540

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 279065540: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[58, 85]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=5.2403431586 ± 0.0000058932d, reference eclipse epoch BTJD2883.0673999, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S58 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation5.250arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.844. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths213.830/201.663/167.136ppt, with 6 stacked events. QLP selected-kind fitted median216.202ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3430965367901091456, P=5.2403509794 ± 0.0002055046d, e=0.240755, omega=135.7897deg. Closest edge-on conjunction offset0.000412d (0.000079phase), full-SB1-covariance linear propagated Gaia uncertainty0.104181d at the measured epoch. Near-Gaia-reference projected eclipse offset0.004135d (photometric period projection allowance0.002805d). The approximate central duration5.07h vs

observed median3.31h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated isolated eclipses across sectors58/85; no capable listed neighbor. Aperture dilution retained.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

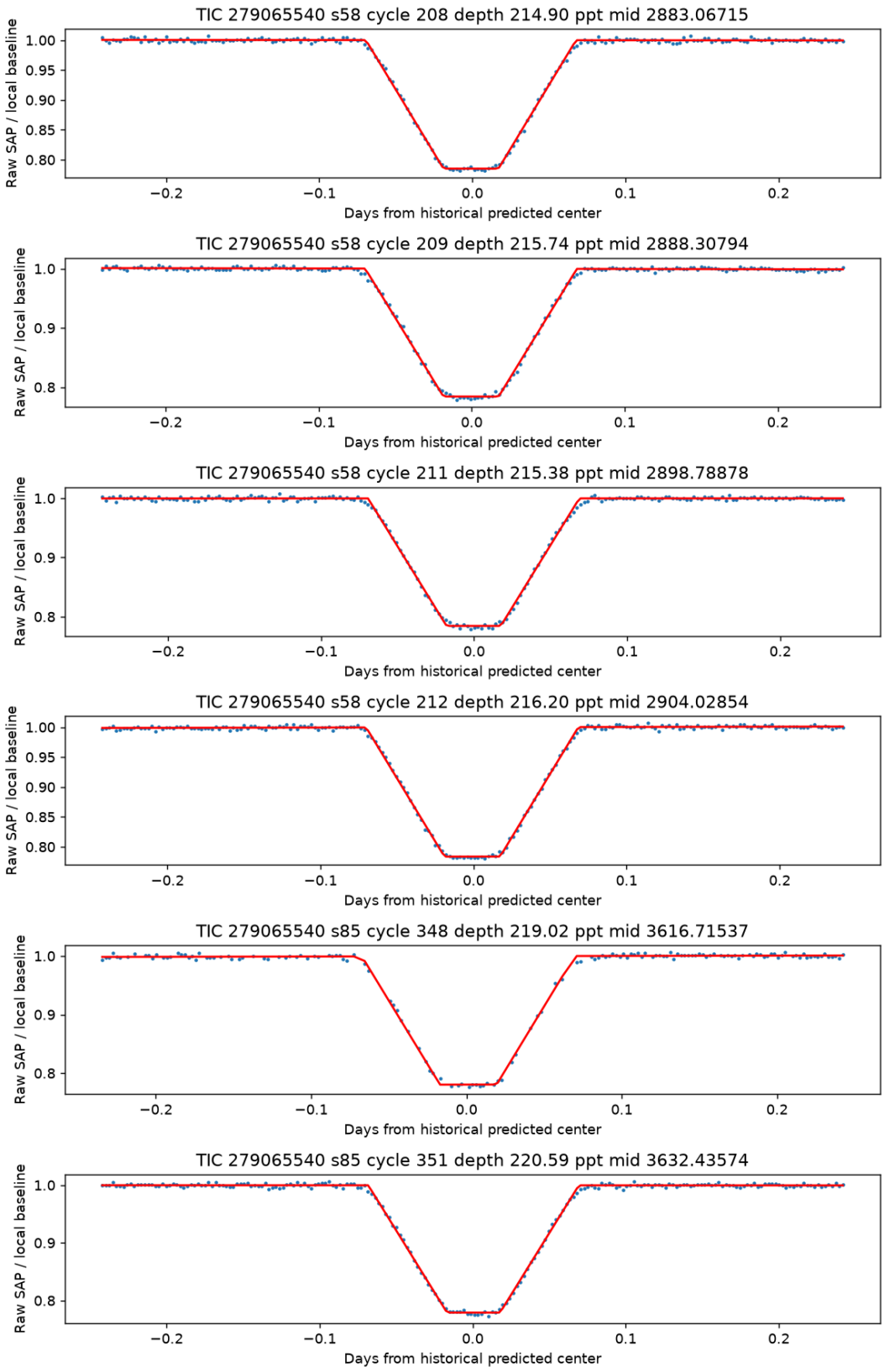
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
58	208	2883.0671535	0.001157	214.898	used in ephemeris
58	209	2888.3079418	0.001157	215.743	used in ephemeris
58	210	2893.5480362	0.001157	215.478	used in ephemeris
58	211	2898.7887838	0.001157	215.383	used in ephemeris
58	212	2904.0285397	0.001157	216.202	used in ephemeris
58	213	2909.2690821	0.001157	218.223	used in ephemeris
85	348	3616.7153682	0.001157	219.016	used in ephemeris
85	349	3621.9565974	0.001157	220.389	used in ephemeris
85	351	3632.4357426	0.001157	220.593	used in ephemeris

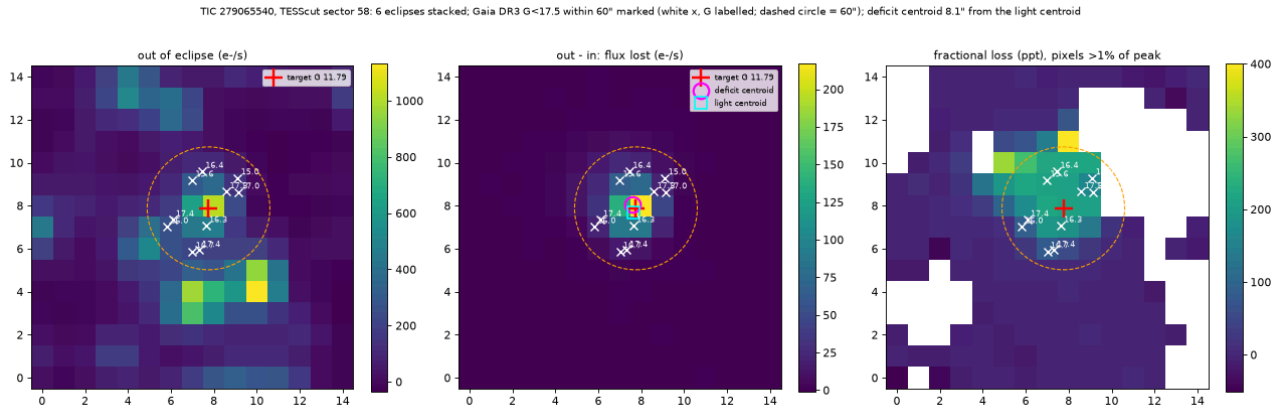
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s58.json, historical/diff_s58.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R366. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R366; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R366; diff_s58.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R367. TIC 439808137

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 439808137: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 10 distinct eclipse events in selected sectors[42, 82, 92]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 10 same-kind midtimes give P=6.4350026636 ±0.0000108672d, reference eclipse epoch BTJD2448.1647207, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S82 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation3.089arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.286. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths102.553/100.621/93.986ppt, with 4 stacked predicted windows. QLP selected-kind fitted median105.158ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3267624023720201254 4, P=6.4354242696±0.0006605117d, e=0.023885, omega=293.9222deg. Closest edge-on conjunction offset-0.099754d (-0.015501phase), full-SB1-covariance linear propagated Gaia uncertainty0.226322d at the measured epoch. Near-Gaia-reference projected eclipse offset0.035159d (photometric period projection allowance0.003477d). The approximate central

duration5.54h vs observed median4.53h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

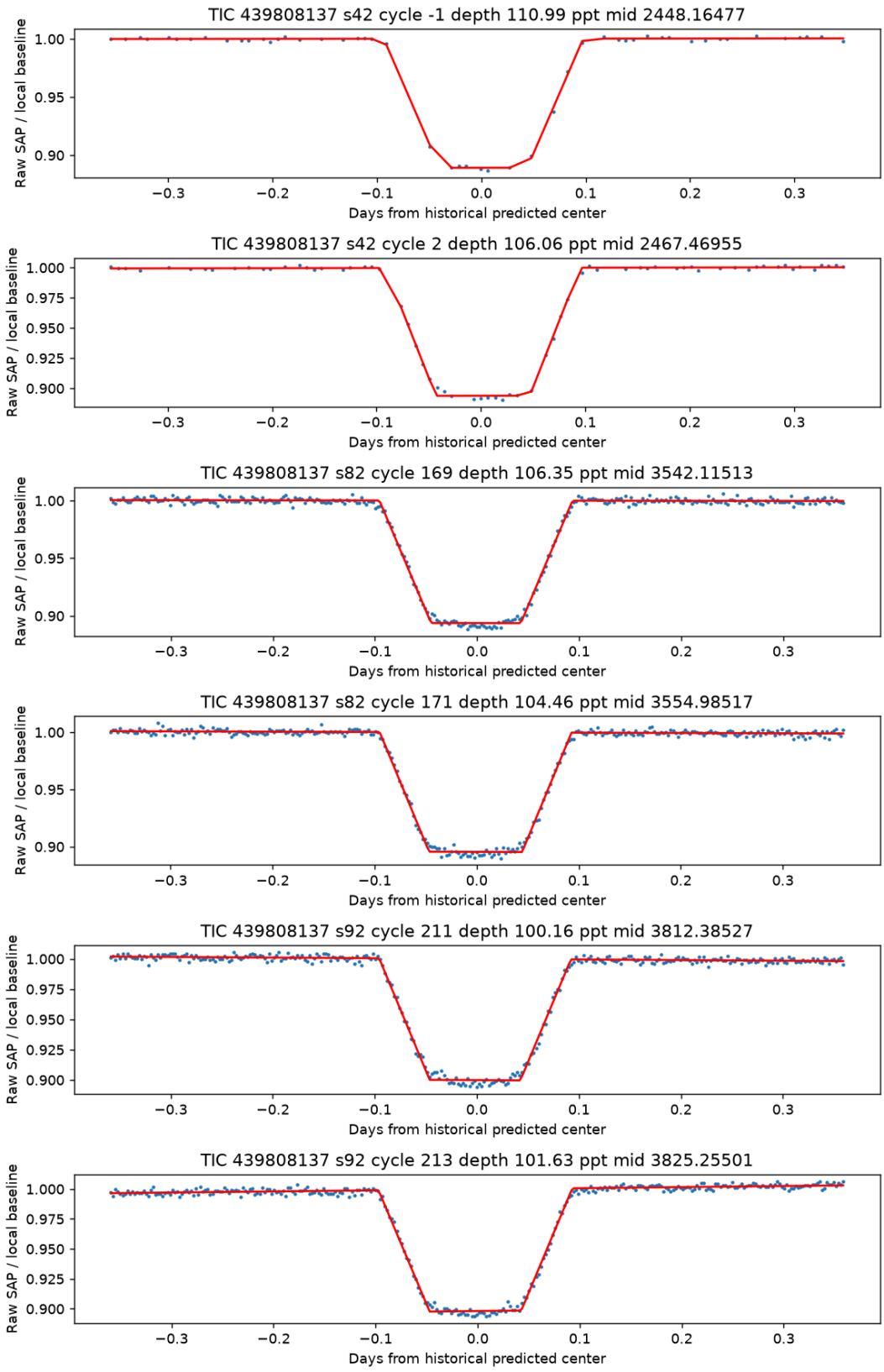
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
42	-1	2448.1647697	0.003472	110.994	included timing center
42	2	2467.4695528	0.003472	106.059	included timing center
82	168	3535.6802741	0.001157	106.751	included timing center
82	169	3542.1151315	0.001157	106.351	included timing center
82	170	3548.5501821	0.001157	105.859	included timing center
82	171	3554.9851673	0.001157	104.457	included timing center
92	210	3805.9503635	0.001158	99.183	included timing center
92	211	3812.3852699	0.001158	100.160	included timing center
92	212	3818.8204620	0.001158	101.345	included timing center
92	213	3825.2550097	0.001158	101.630	included timing center

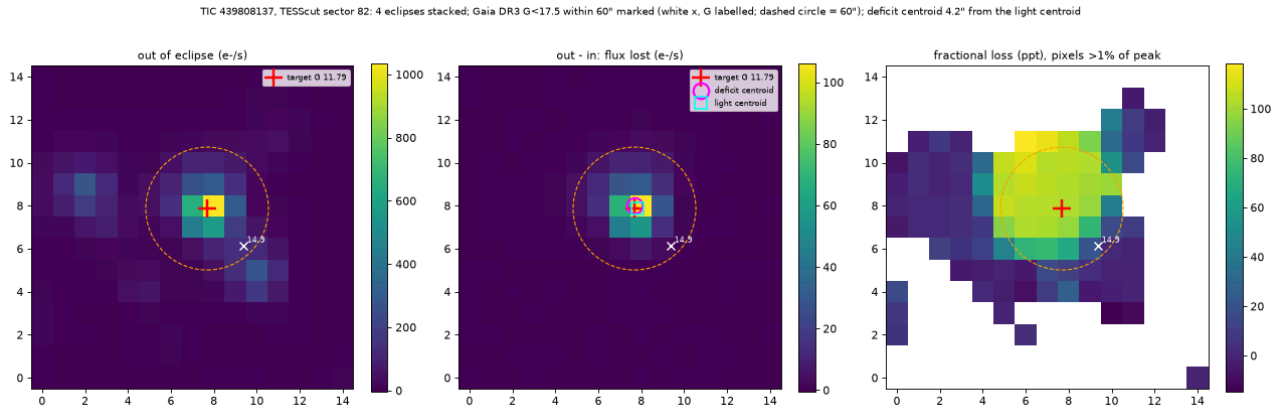
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s82.json, historical/diff_s82.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R367. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R367; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R367; diff_s82.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R368. TIC 382631802

physical_verdict: PLAUSIBLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Joint variability/eclipse model and radii check

Archived source: sanitized report text.

TIC 382631802: PLAUSIBLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: recurrence detected; eclipse interpretation insecure.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 28 distinct eclipse events in selected sectors[1, 36, 97]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 28 same-kind midtimes give P=2.5954869623 ±0.0000430249d, reference eclipse epoch BTJD1328.5030929, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Freshly extracted S97 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation4.628arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.460. Capable neighbours: Gaia 5290921322334100480: 48.54 arcsec, G16.282, required fractional eclipse0.564; Gaia 5290921287974360448: 57.75 arcsec, G16.856, required fractional eclipse0.951
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths5.739/5.663/5.407ppt, with 19 stacked predicted windows. QLP selected-kind fitted median8.947ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: unresolved after completed check.** DR35290921356693837568, P=2.5956559888±0.0001178502d, e=0.268983, omega=78.2783deg. Closest edge-on conjunction offset-0.055532d (-0.021394phase), full-SB1-covariance linear propagated Gaia uncertainty0.064915d at the measured epoch. Near-Gaia-reference projected eclipse

offset0.005655d (photometric period projection allowance0.015575d). The approximate central duration3.21h vs observed median7.58h exceeds the simple model and remains unresolved; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: PLAUSIBLE: recurring V-shaped dips ride structured variability. No stable sharp contacts or flat-bottomed occultation signature is resolved across representative windows; free trapezoids alone do not distinguish eclipses from pulsation or tidal minima. Eclipse interpretation itself is insecure. Free trapezoid duration median7.58h exceeds approximate central3.21h using TIC radius1.64solar and main-sequence minimum-mass companion. Difference centroid favors target0.220pixel versus capable neighbours2.38/2.67pixel; PSF fractional residual0.46 limits precision. Peak is measured within the central target5x5 box, excluding invalid peripheral cutout pixels; peripheral invalid pixels do not decide attribution. Model durations vary1.58-14.05h, indicating baseline/shape contamination. Timing reduced chi-square about126 requires scatter inflation. Geometry and eclipse identity are unresolved after completed checks; alternative variability and revised radii require testing.

Next check: Fit pulsation/background and eclipse jointly; test stellar radii and obtain resolved photometry.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

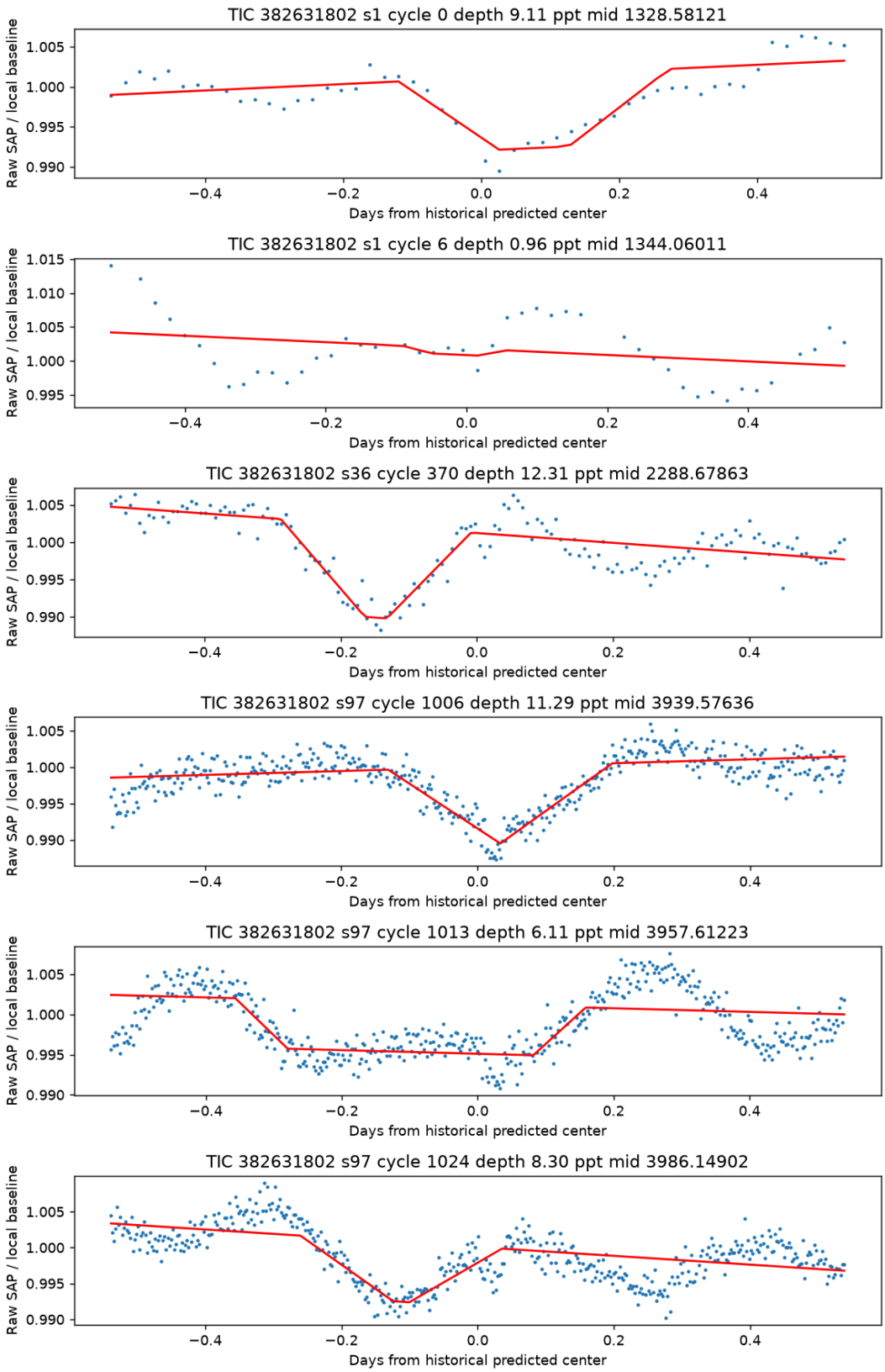
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1328.5812055	0.010417	9.111	included timing center
1	1	1331.0120150	0.019902	8.926	included timing center
1	2	1333.6020632	0.052052	28.663	included timing center
1	3	1336.3461388	0.011229	16.578	included timing center
1	5	1341.4693760	0.026132	13.075	included timing center
1	6	1344.0601145	0.135325	0.960	included timing center
1	7	1346.7001347	0.010793	11.626	included timing center
1	9	1351.8802014	0.022025	12.263	included timing center
36	368	2283.6109148	0.010744	3.464	included timing center
36	369	2286.2843727	0.008316	13.363	included timing center
36	370	2288.6786321	0.007128	12.305	included timing center
36	371	2291.4025187	0.009587	7.672	included timing center
36	373	2296.7469743	0.015301	7.709	included timing center
36	374	2299.1663617	0.005423	7.725	included timing center

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
36	375	2301.9361800	0.006910	13.205	included timing center
36	376	2304.2976439	0.018354	13.907	included timing center
97	1006	3939.5763559	0.002833	11.292	included timing center
97	1007	3942.1418373	0.004789	5.472	included timing center
97	1008	3944.7198978	0.002105	12.902	included timing center
97	1011	3952.4717370	0.010255	5.919	included timing center
97	1012	3955.1845224	0.007913	7.162	included timing center
97	1013	3957.6122346	0.007956	6.112	included timing center
97	1016	3965.5937512	0.003300	7.595	included timing center
97	1017	3968.0753467	0.006071	8.186	included timing center
97	1018	3970.6765769	0.009017	2.254	included timing center
97	1019	3973.3286723	0.003551	12.547	included timing center
97	1023	3983.7445661	0.003163	8.968	included timing center
97	1024	3986.1490173	0.005594	8.300	included timing center

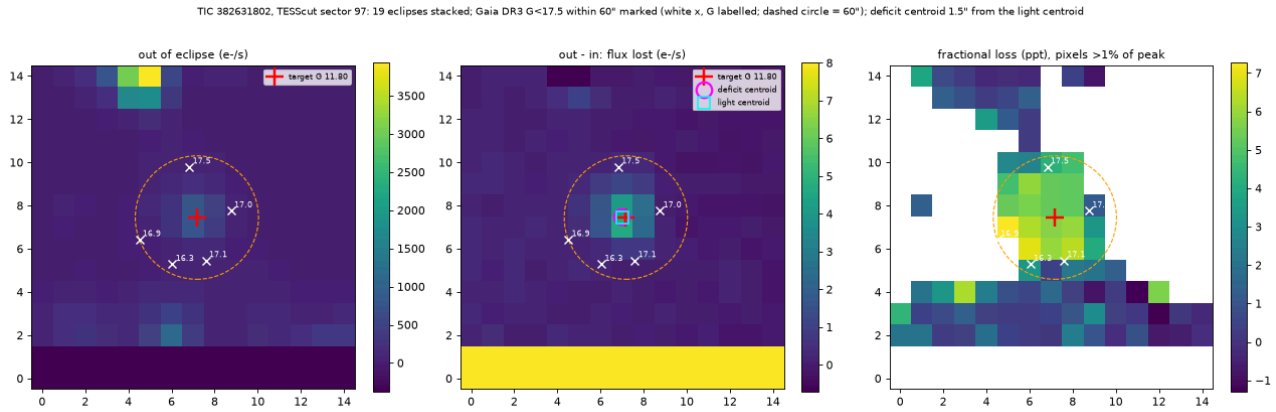
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R368. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R368; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R368; diff_s97.png. New sector-97 difference-image stack from 19 predicted-guided windows. The deficit favours the target, while structured variability and broad-model duration tension leave the eclipse interpretation insecure; this plausible signal is excluded.

R369. TIC 239572145

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 239572145: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 17 distinct eclipse events in selected sectors[15, 55, 82]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=6.3129271986 ±0.0000162239d, reference eclipse epoch BTJD1714.9544368, kind: Deeper eclipse; even historical half-period cycles. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.430arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.792. Capable neighbours: Gaia 1852594221418982144: 28.33 arcsec, G13.286, required fractional eclipse0.278
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths98.238/86.360/78.576ppt, with 8 stacked predicted windows. QLP selected-kind fitted median150.972ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3185259425577871948 8, P=6.3125426224±0.0003139626d, e=0.044112, omega=17.4791deg. Closest edge-on conjunction offset0.190656d (0.030203phase), full-SB1-covariance linear propagated Gaia uncertainty0.071723d at the measured epoch. Near-Gaia-reference projected eclipse offset0.109895d (photometric period projection allowance0.003407d). The approximate central duration6.77h vs observed median5.69h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Stable alternating depths near150/55ppt establish doubled orbital period; six deeper even historical half-period events define ephemeris. Historical pixel stack mixes both eclipse kinds; concentrated target deficit supports localization. Closest Gaia conjunction offset+0.1907d (0.0302phase) versus linear covariance0.0717d is retained as descriptive association tension without arbitrary sigma veto.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

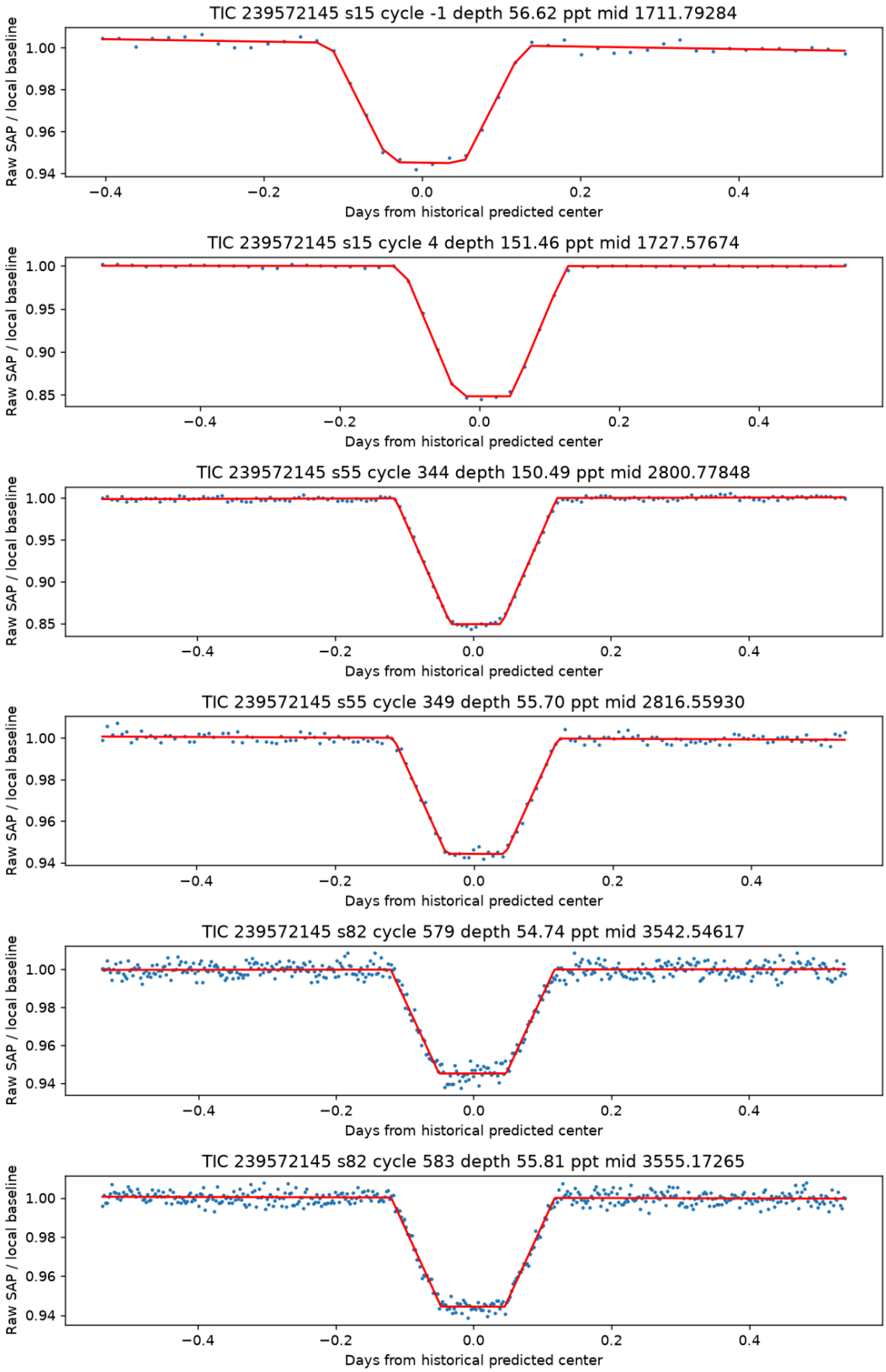
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	-1	1711.7928386	0.010417	56.617	usable center; excluded opposite eclipse kind
15	0	1714.9504919	0.010417	152.615	included timing center
15	1	1718.1060346	0.010417	57.330	usable center; excluded opposite eclipse kind
15	4	1727.5767403	0.010417	151.458	included timing center
15	5	1730.7318327	0.010417	58.018	usable center; excluded opposite eclipse kind
55	343	2797.6227382	0.003472	55.933	usable center; excluded opposite eclipse kind
55	344	2800.7784811	0.003472	150.485	included timing center
55	345	2803.9352346	0.003472	54.691	usable center; excluded opposite eclipse kind
55	348	2813.4047013	0.003472	149.223	included timing center
55	349	2816.5592973	0.003472	55.697	usable center; excluded opposite eclipse kind
55	350	2819.7172580	0.003472	150.059	included timing center
82	577	3536.2338340	0.001157	54.913	usable center; excluded opposite eclipse kind
82	579	3542.5461712	0.001157	54.741	usable center; excluded opposite eclipse kind
82	580	3545.7118361	121226.206053	147.482	excluded singular timing fit
82	581	3548.8590316	0.001157	54.329	usable center; excluded opposite eclipse kind
82	582	3552.0161153	0.001157	151.477	included timing center
82	583	3555.1726501	0.001157	55.813	usable center; excluded opposite eclipse kind

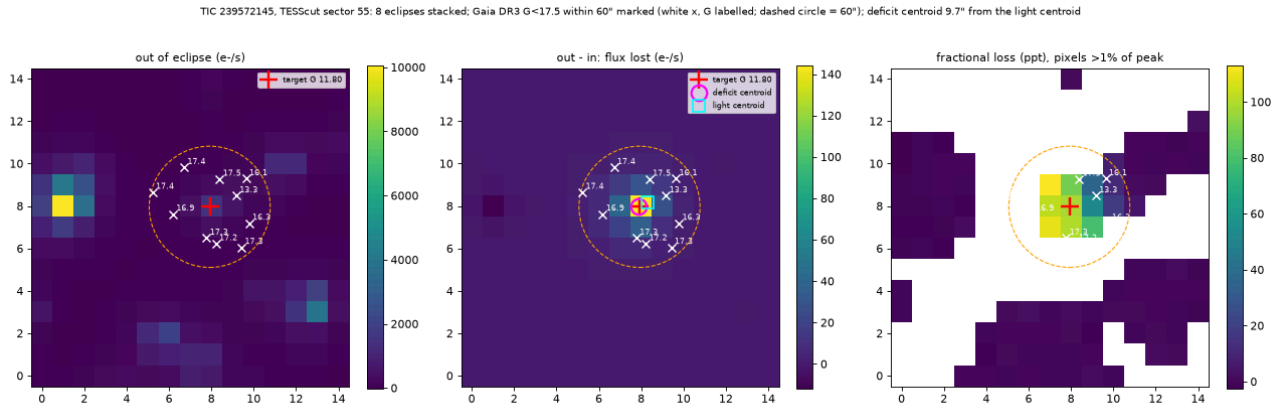
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R369. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R369; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R369; diff_s55.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R370. TIC 90285860

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 90285860: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 3 distinct eclipse events in selected sectors[41, 74]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 3 same-kind midtimes give P=18.5945632879 ±0.0000559615d, reference eclipse epoch BTJD2422.5645534, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S41 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.713arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.356. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths162.253/156.481/137.458ppt, with 2 stacked predicted windows. QLP selected-kind fitted median178.980ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3202580461748610099 2, P=18.5977982181±0.0019708206d, e=0.436172, omega=250.9028deg. Closest edge-on conjunction offset-0.451898d (-0.024298phase), full-SB1-covariance linear propagated Gaia uncertainty0.202050d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.099291d (photometric period projection allowance0.006100d). The approximate central

duration10.47h vs observed median7.88h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Three measured eclipses include within-sector S41 recurrence, testing long-baseline cycle count. Closest Gaia conjunction offset-0.4519d (-0.0243phase) versus propagated0.2020d is retained as descriptive association tension, without arbitrary sigma veto; updated spectroscopy would test it.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

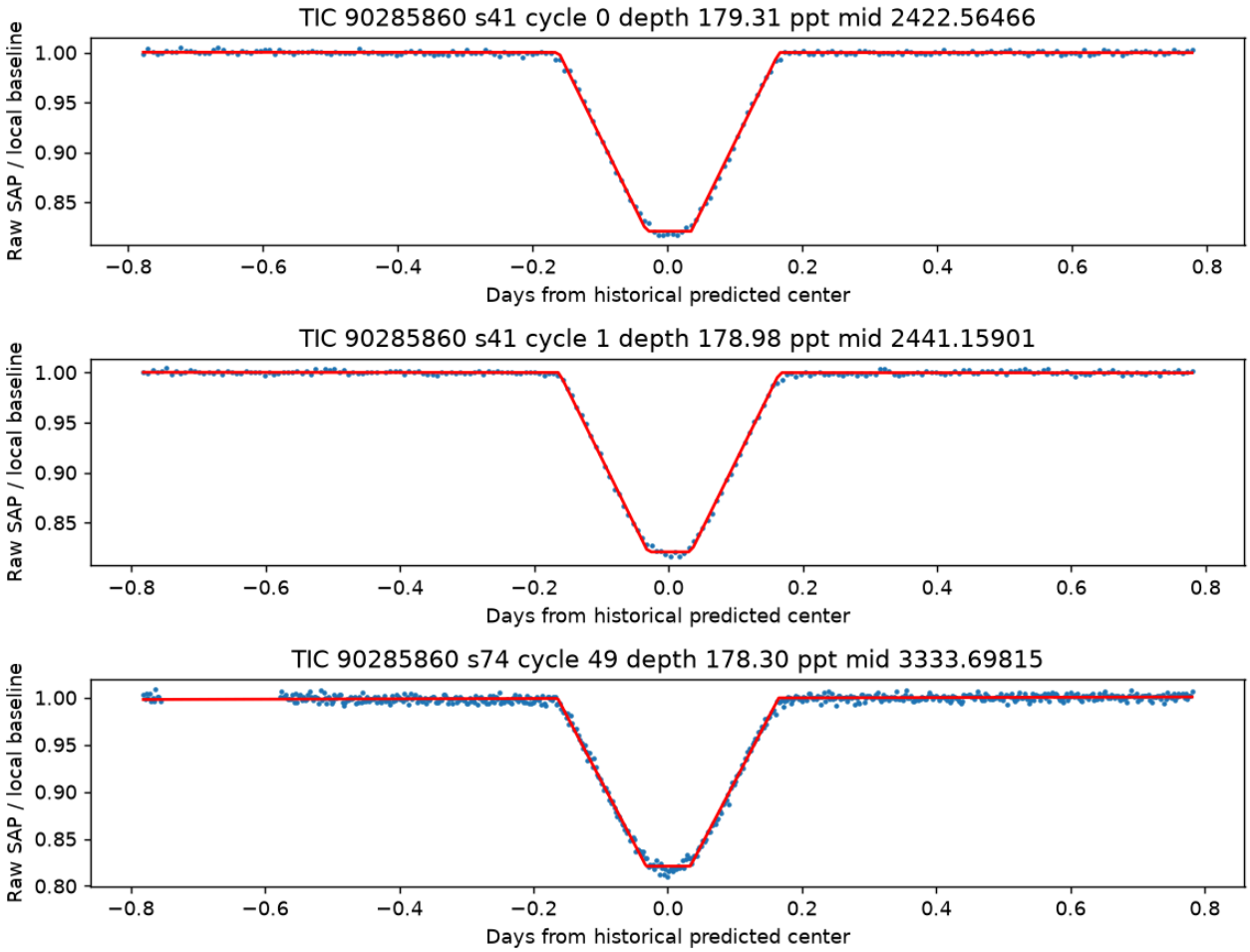
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
41	0	2422.5646616	0.003472	179.306	included timing center
41	1	2441.1590062	0.003472	178.980	included timing center
74	49	3333.6981548	0.001157	178.305	included timing center

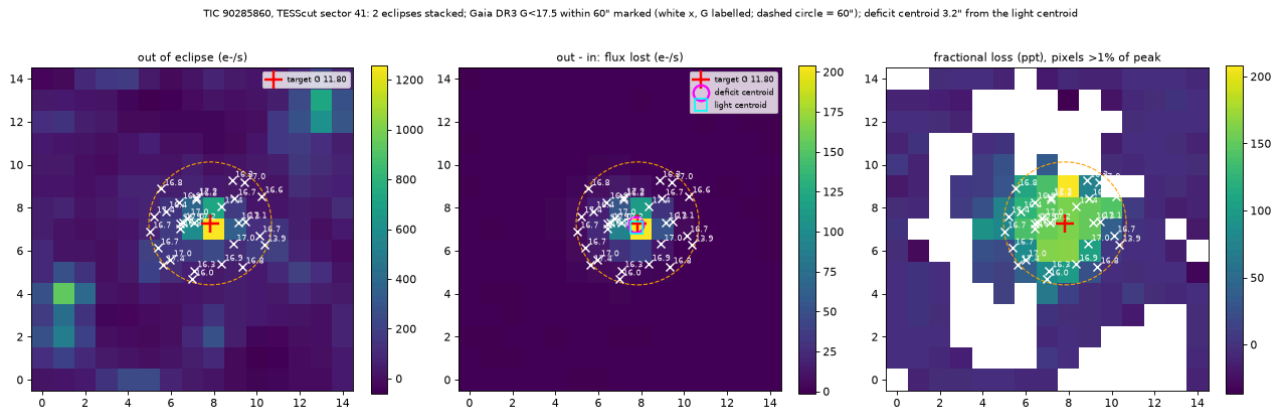
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s41.json, historical/diff_s41.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R370. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R370; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R370; diff_s41.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R371. TIC 74717164

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 74717164: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 12 distinct eclipse events in selected sectors[8, 36, 89]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 12 same-kind midtimes give P=4.7665054846 ±0.0000054999d, reference eclipse epoch BTJD1520.9571959, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S62 deficit peak[7, 8] vs target pixel[7, 7]; deficit-target separation4.426arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.268. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths145.095/137.072/121.642ppt, with 6 stacked predicted windows. QLP selected-kind fitted median161.796ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3542726880006204979 2, P=4.7669038760±0.0002652066d, e=0.011747, omega=19.5123deg. Closest edge-on conjunction offset-0.096457d (-0.020235phase), full-SB1-covariance linear propagated Gaia uncertainty0.084036d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.002038d (photometric period projection allowance0.001303d). The approximate central

duration6.18h vs observed median5.64h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

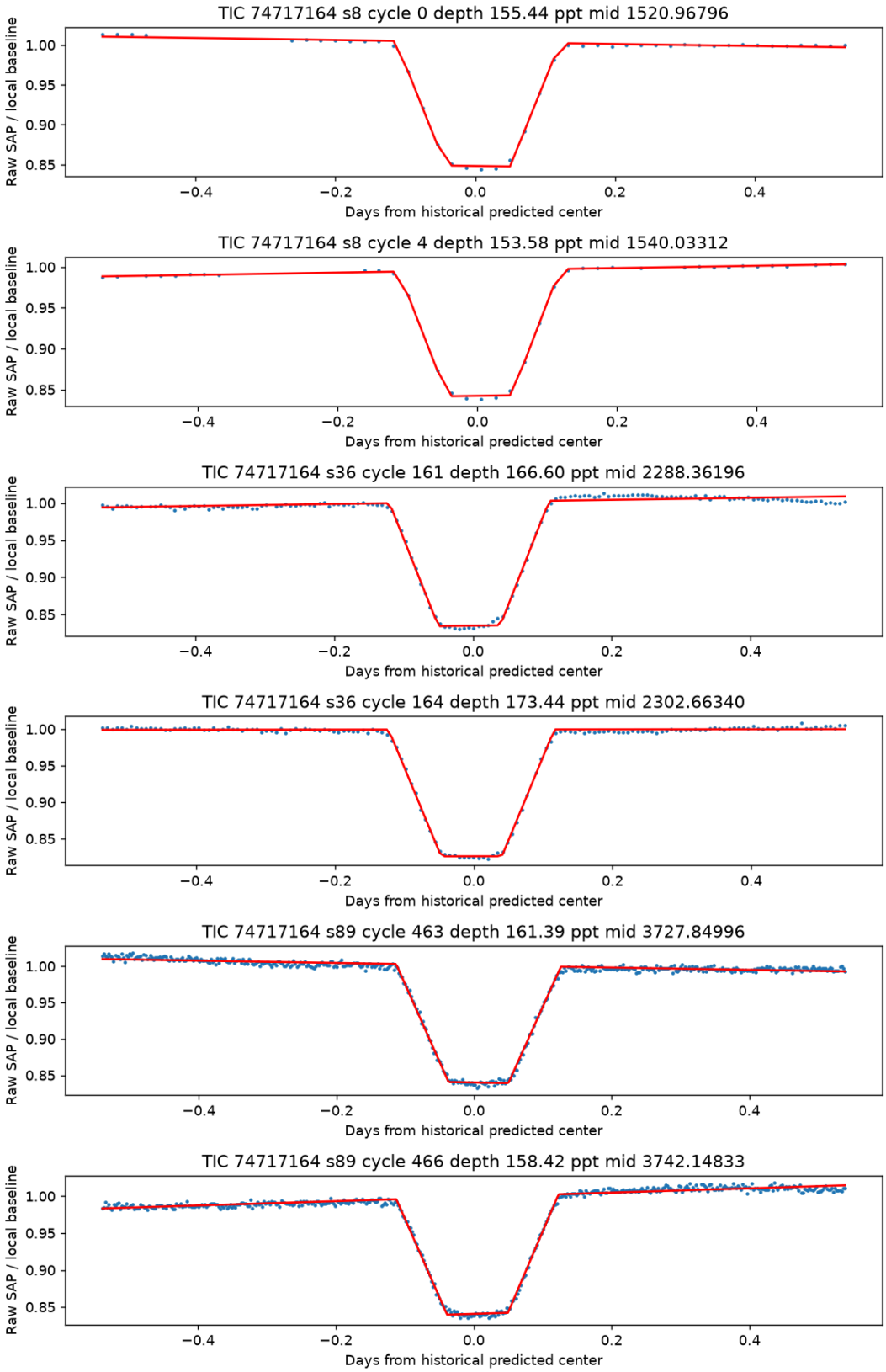
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
8	0	1520.9679571	0.010417	155.437	included timing center
8	1	1525.7335383	0.010417	155.225	included timing center
8	4	1540.0331209	0.010417	153.583	included timing center
36	160	2283.5977269	0.003472	168.836	included timing center
36	161	2288.3619604	0.003472	166.596	included timing center
36	163	2297.8960431	0.003472	167.187	included timing center
36	164	2302.6634024	0.003472	173.439	included timing center
89	462	3723.0825840	0.001157	158.638	included timing center
89	463	3727.8499626	0.001157	161.394	included timing center
89	464	3732.6150904	0.001157	162.197	included timing center
89	465	3737.3829364	0.001157	163.950	included timing center
89	466	3742.1483320	0.001157	158.422	included timing center

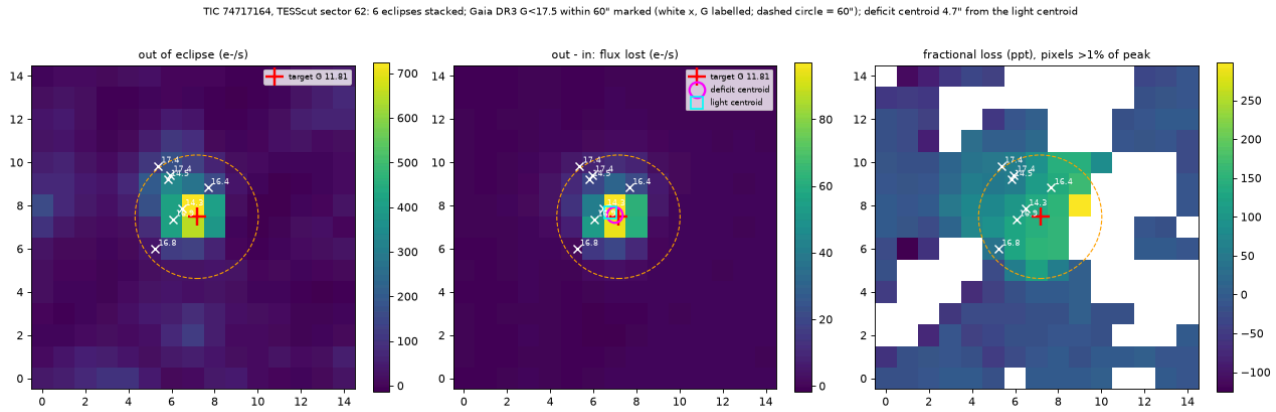
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s62.json, historical/diff_s62.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R371. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R371; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R371; diff_s62.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R372. TIC 134193638

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 134193638: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 15 distinct eclipse events in selected sectors[6, 34, 88]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 15 same-kind midtimes give P=4.0538858989 ±0.0000171546d, reference eclipse epoch BTJD1476.1660599, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S87 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.286arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.191. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths30.037/30.552/28.579ppt, with 5 stacked predicted windows. QLP selected-kind fitted median31.643ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3555934513038192742 4, P=4.0539744870±0.0004631847d, e=0.084801, omega=205.8049deg. Closest edge-on conjunction offset-0.087266d (-0.021526phase), full-SB1-covariance linear propagated Gaia uncertainty0.124551d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.063525d (photometric period projection allowance0.004597d). The approximate central

duration9.97h vs observed median5.67h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated sharp raw eclipses coexist with strong oscillatory variability, especially sectors34/88. Free trapezoids can absorb background oscillation; timing reduced chi-square about9 is scatter inflated and uncalibrated shape/baseline systematics remain.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

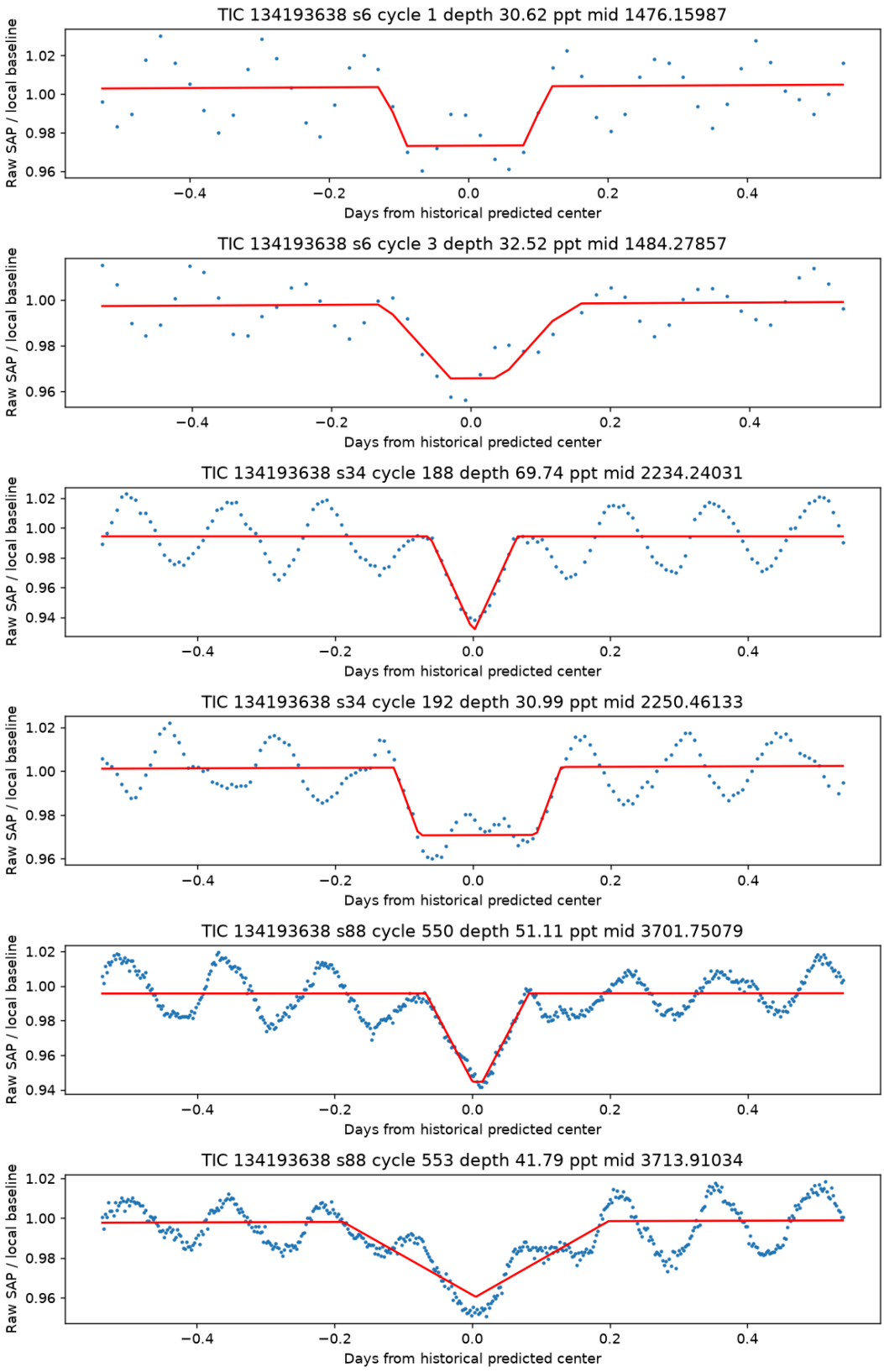
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
6	1	1476.1598733	0.016060	30.617	included timing center
6	2	1480.2272868	0.010417	39.105	included timing center
6	3	1484.2785717	0.010709	32.524	included timing center
6	4	1488.3313258	0.010417	29.815	included timing center
34	187	2230.1775933	0.005450	31.891	included timing center
34	188	2234.2403065	0.005115	69.739	included timing center
34	189	2238.3028897	0.004174	31.095	included timing center
34	191	2246.3996140	0.004258	40.017	included timing center
34	192	2250.4613313	0.004077	30.993	included timing center
88	548	3693.6609777	0.002936	30.594	included timing center
88	549	3697.6724459	0.003413	30.702	included timing center
88	550	3701.7507853	0.002321	51.107	included timing center
88	551	3705.8086631	0.002312	30.201	included timing center
88	552	3709.8482450	0.002514	31.643	included timing center
88	553	3713.9103364	0.004836	41.788	included timing center

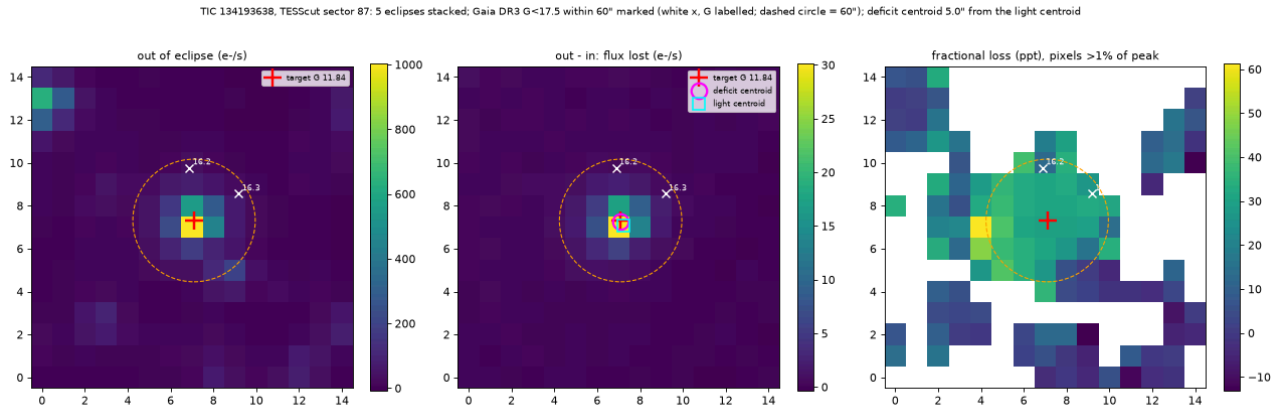
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s87.json, historical/diff_s87.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R372. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R372; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R372; diff_s87.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R373. TIC 34588199

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 34588199: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 11 distinct eclipse events in selected sectors[6, 33, 87]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=5.6182034108 ±0.0000064767d, reference eclipse epoch BTJD1470.7824555, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S6 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation0.933arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.203. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths54.615/54.342/52.630ppt, with 4 stacked predicted windows. QLP selected-kind fitted median52.670ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3294492744700969356 8, P=5.6184294916±0.0004145044d, e=0.061222, omega=210.8797deg. Closest edge-on conjunction offset-0.086507d (-0.015397phase), full-SB1-covariance linear propagated Gaia uncertainty0.104403d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.042873d (photometric period projection allowance0.001250d). The approximate central

duration4.38h vs observed median4.07h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

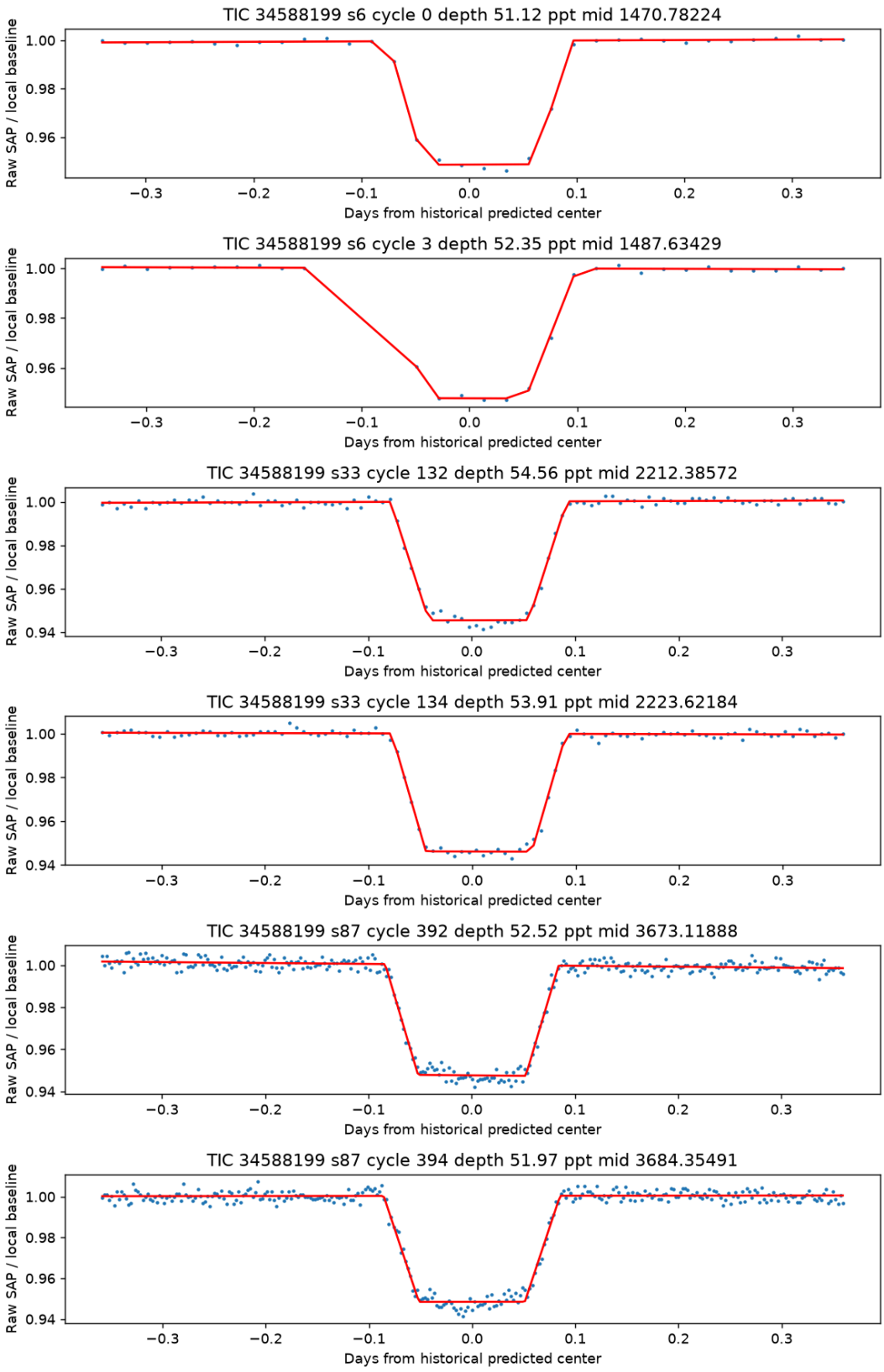
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
6	0	1470.7822351	0.010417	51.124	included timing center
6	2	1482.0183493	0.010417	52.775	included timing center
6	3	1487.6342930	0.010417	52.352	included timing center
33	131	2206.7671282	0.003472	52.667	included timing center
33	132	2212.3857154	0.003472	54.557	included timing center
33	133	2218.0035617	0.003472	53.653	included timing center
33	134	2223.6218359	0.003472	53.915	included timing center
87	391	3667.4993936	0.001157	52.720	included timing center
87	392	3673.1188778	0.001157	52.519	included timing center
87	393	3678.7359728	0.001157	52.670	included timing center
87	394	3684.3549079	0.001157	51.972	included timing center

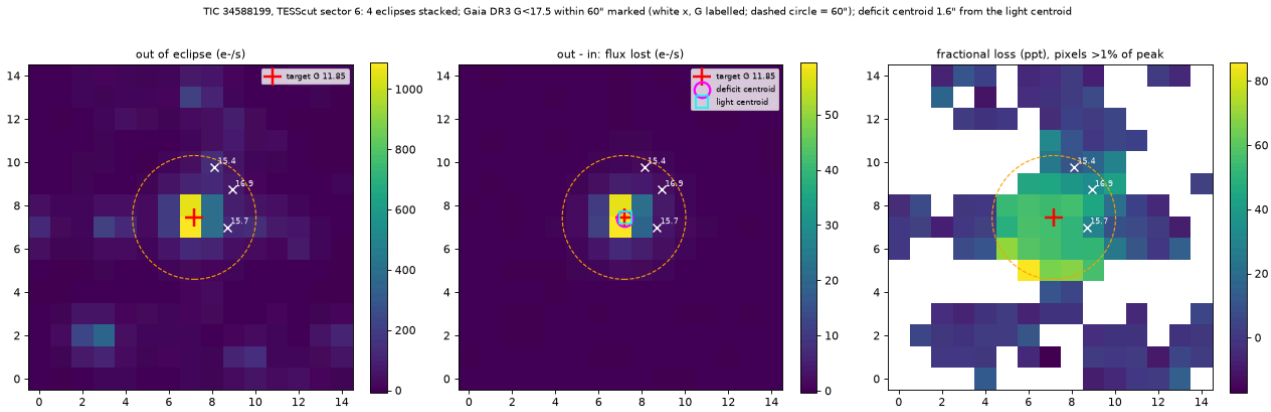
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s06.json, historical/diff_s06.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R373. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R373; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R373; diff_s06.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R374. TIC 146463944

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 146463944: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 9 distinct eclipse events in selected sectors[5, 32, 98]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=6.9896922117 ±0.0000074444d, reference eclipse epoch BTJD1444.7562858, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S98 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.021arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.054. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths274.775/272.253/263.827ppt, with 5 stacked predicted windows. QLP selected-kind fitted median246.336ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3297508464577622489 6, P=6.9895742942±0.0002845635d, e=0.112660, omega=95.4246deg. Closest edge-on conjunction offset0.031213d (0.004466phase), full-SB1-covariance linear propagated Gaia uncertainty0.056178d at the measured epoch. Near-Gaia-reference projected eclipse offset0.013407d (photometric period projection allowance0.001124d). The approximate central duration5.62h vs observed median4.39h

permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

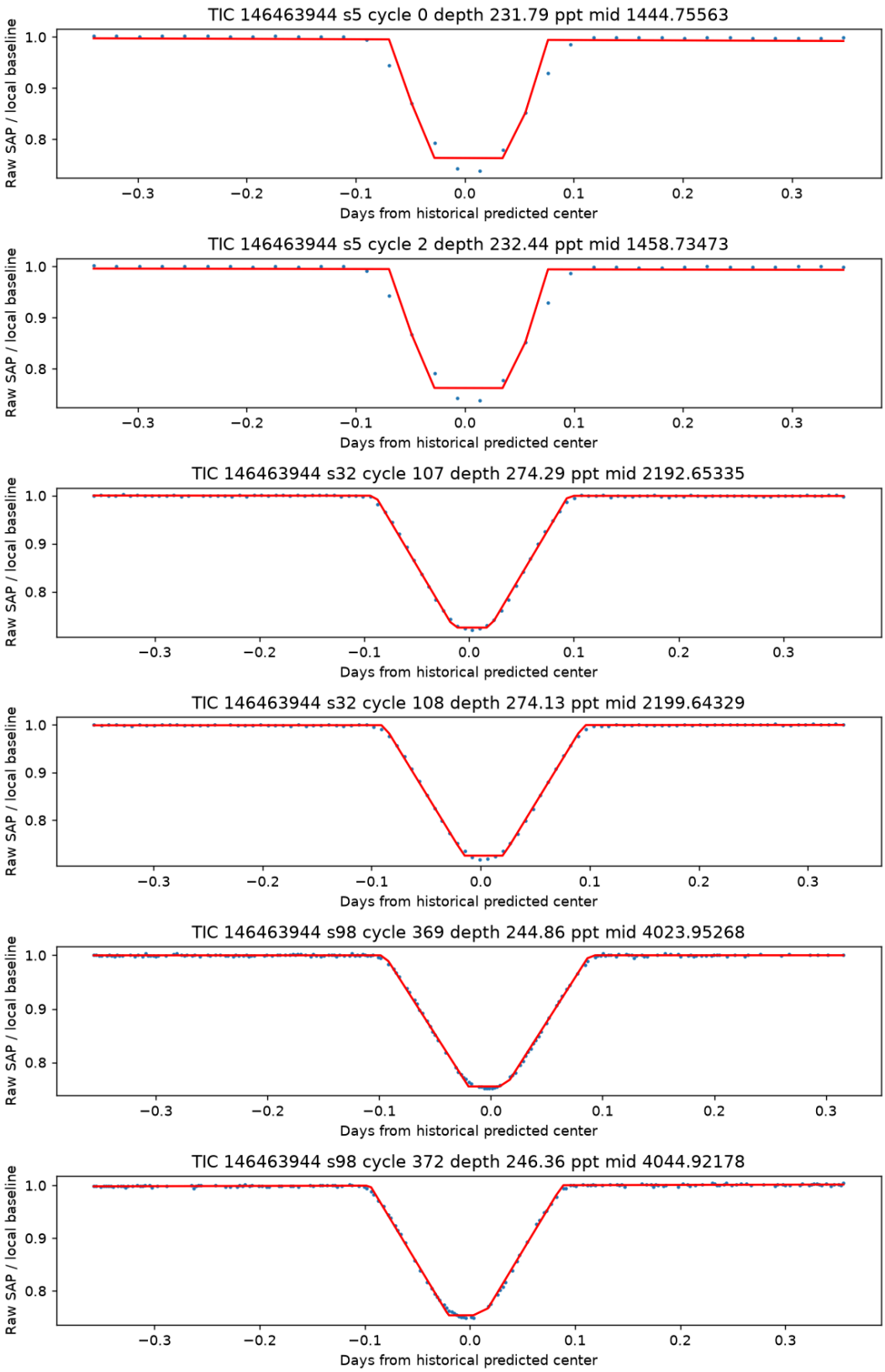
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
5	0	1444.7556265	0.010417	231.789	included timing center
5	2	1458.7347272	0.010417	232.438	included timing center
32	105	2178.6739255	0.003472	273.968	included timing center
32	107	2192.6533520	0.003472	274.285	included timing center
32	108	2199.6432936	0.003472	274.127	included timing center
98	365	3995.9942097	0.001157	241.859	included timing center
98	369	4023.9526829	0.001157	244.863	included timing center
98	370	4030.9421693	0.001157	246.336	included timing center
98	372	4044.9217827	0.001157	246.361	included timing center

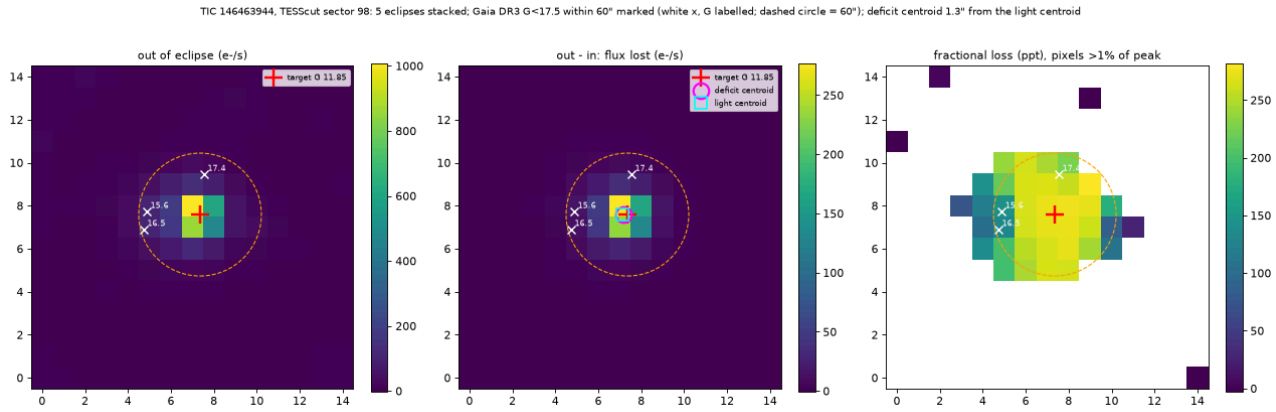
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s98.json, historical/diff_s98.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R374. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R374; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R374; diff_s98.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R375. TIC 113810700

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 113810700: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 22 distinct eclipse events in selected sectors[42, 44, 71]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 22 same-kind midtimes give P=2.7284265924 ±0.0000036417d, reference eclipse epoch BTJD2449.5539561, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S70 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.046arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.433. Capable neighbours: Gaia 60689300937937920: 57.10 arcsec, G15.397, required fractional eclipse0.924
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths33.426/33.144/31.944ppt, with 8 stacked predicted windows. QLP selected-kind fitted median34.363ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR360689193562928512, P=2.7284185865±0.0001165221d, e=0.057487, omega=53.9806deg. Closest edge-on conjunction offset0.028502d (0.010446phase), full-SB1-covariance linear propagated Gaia uncertainty0.094242d at the measured epoch. Near-Gaia-reference projected eclipse offset0.022458d (photometric period projection allowance0.002749d). The approximate central

duration4.70h vs observed median4.23h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

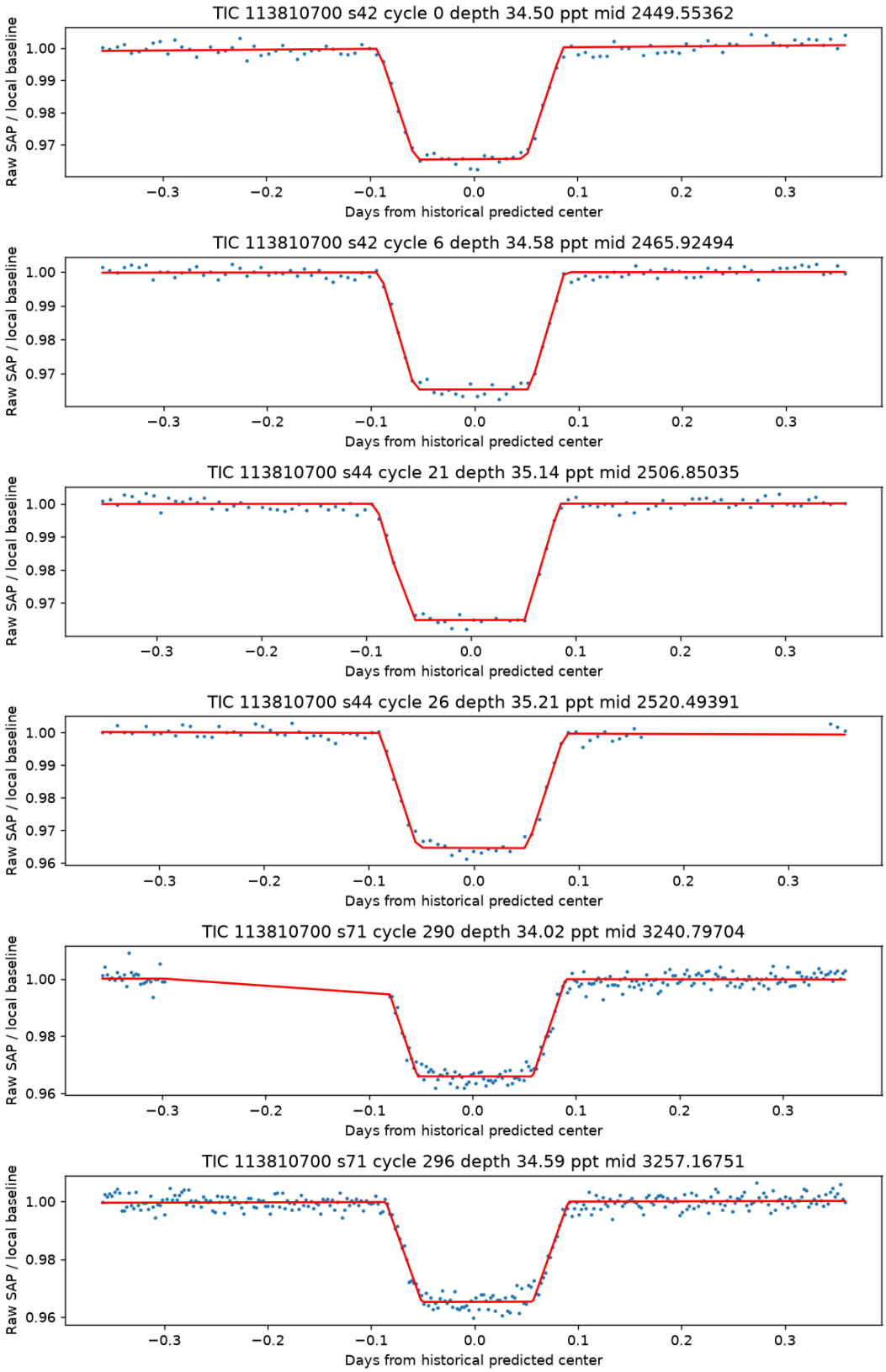
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
42	0	2449.5536236	0.003473	34.496	included timing center
42	1	2452.2824437	0.003473	34.302	included timing center
42	2	2455.0107777	0.003473	34.087	included timing center
42	5	2463.1974482	0.003473	35.917	included timing center
42	6	2465.9249403	0.003473	34.578	included timing center
42	7	2468.6527539	0.003473	34.284	included timing center
42	8	2471.3816855	0.003473	34.125	included timing center
44	19	2501.3941823	0.003472	34.279	included timing center
44	21	2506.8503510	0.003472	35.140	included timing center
44	22	2509.5784516	0.003472	34.790	included timing center
44	24	2515.0357068	0.003472	35.584	included timing center
44	25	2517.7642280	0.003472	34.788	included timing center
44	26	2520.4939123	0.003472	35.210	included timing center
44	27	2523.2212149	0.003472	34.318	included timing center
71	288	3235.3398853	0.001157	34.354	included timing center
71	289	3238.0697129	0.001157	34.204	included timing center
71	290	3240.7970380	0.001157	34.023	included timing center
71	291	3243.5265239	0.001157	34.373	included timing center
71	293	3248.9838239	0.001157	33.364	included timing center
71	294	3251.7112147	0.001157	34.349	included timing center

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
71	295	3254.4404550	0.001157	34.376	included timing center
71	296	3257.1675145	0.001157	34.595	included timing center

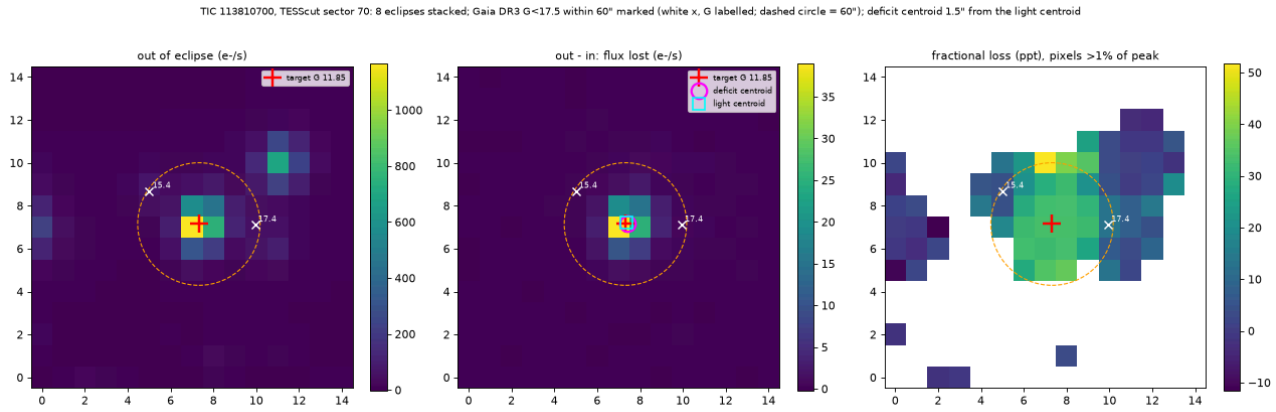
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s70.json, historical/diff_s70.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R375. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R375; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R375; diff_s70.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R376. TIC 120029059

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 120029059: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX show 8 distinct eclipse events in selected sectors[25, 78, 79]. The raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=3.9188796724 ±0.0000270796d, reference eclipse epoch BTJD1992.7941562, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. Local recurrence where covered and alternating depths are recorded in alias-audit.json. No exhaustive blind search claimed. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S79 deficit peak[7, 8] vs target pixel[8, 8]; deficit-target separation5.909arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.423. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths104.749/101.439/94.107ppt, with 4 stacked predicted windows. QLP selected-kind fitted median109.639ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3446573739471280627 2, P=3.9191167014±0.0001278734d, e=0.168077, omega=296.8544deg. Closest edge-on conjunction offset-0.078894d (-0.020131phase), full-SB1-covariance linear propagated Gaia uncertainty0.061003d at the measured epoch. Near-Gaia-reference projected eclipse offset0.018051d (photometric period projection allowance0.011076d). The approximate central duration4.63h vs

observed median3.53h permits noncentral geometry; this radius/minimum-mass estimate is not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Representative raw eclipses recur. Depths remain diluted and sector dependent; timing errors omit uncalibrated extraction and shape systematics.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

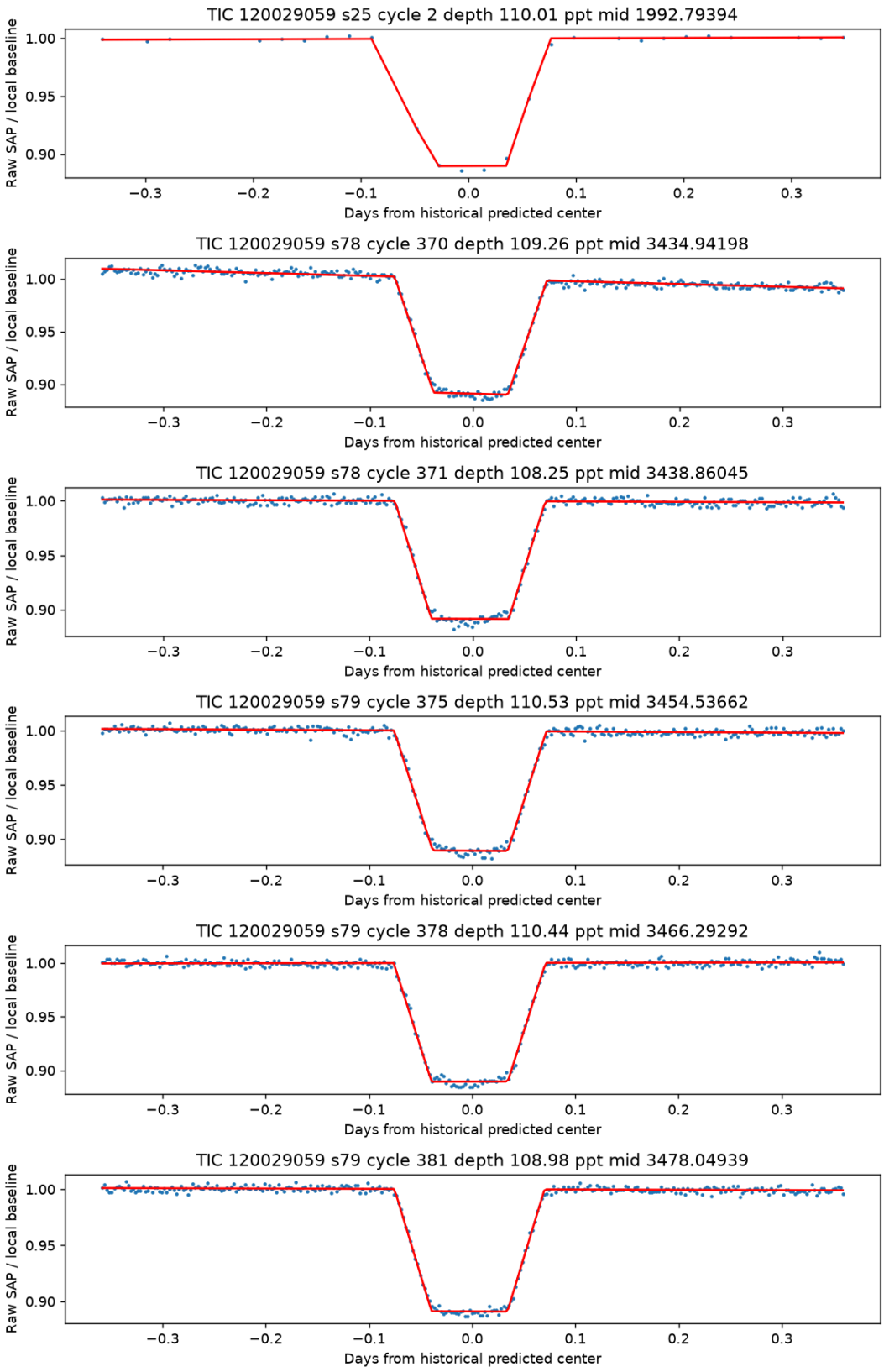
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
25	2	1992.7939376	0.010417	110.015	included timing center
78	370	3434.9419823	0.001157	109.263	included timing center
78	371	3438.8604468	0.001157	108.254	included timing center
78	372	3442.7797493	0.001157	109.075	included timing center
79	375	3454.5366244	0.001157	110.526	included timing center
79	378	3466.2929153	0.001157	110.442	included timing center
79	379	3470.2116903	0.001157	111.699	included timing center
79	381	3478.0493918	0.001157	108.978	included timing center

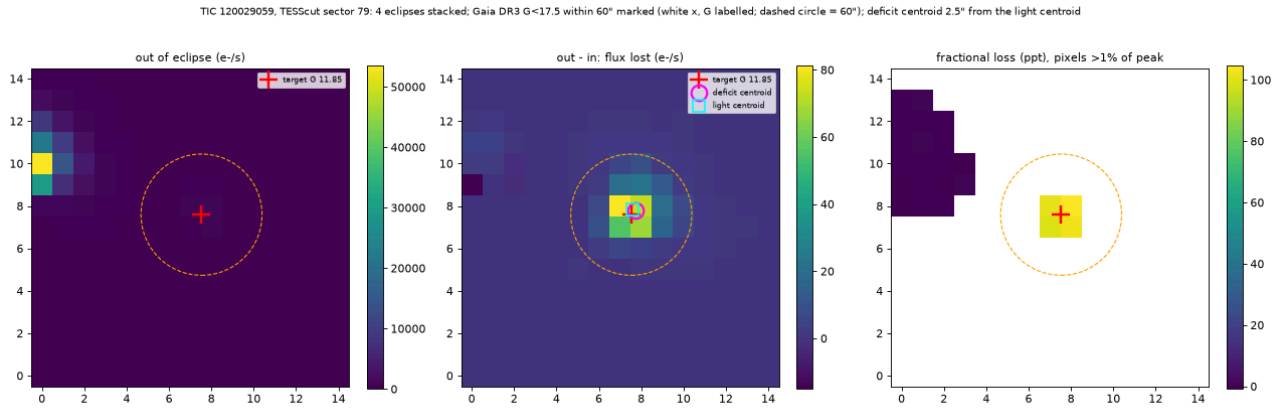
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s79.json, historical/diff_s79.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R376. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R376; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R376; diff_s79.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R377. TIC 66787531

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 66787531: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 6 distinct eclipse events in selected sectors[76, 84]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 6 same-kind midtimes give P=8.4698323955 +/-0.0000408947d, reference eclipse epoch BTJD3372.5340791, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S57 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation4.885arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.438. Capable neighbours: Gaia 1985288988373538176: 29.23 arcsec, G16.005, required fractional eclipse0.649; Gaia 1985288610416416640: 50.84 arcsec, G15.933, required fractional eclipse0.608; Gaia 1985291565353912576: 51.60 arcsec, G15.706, required fractional eclipse0.496
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths14.138/14.120/13.460ppt, with 3 stacked events. QLP selected-kind fitted median16.163ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR31985288988373538432, P=8.4702687962+/-0.0005995472d, e=0.283327, omega=345.7269deg. Closest edge-on conjunction offset-0.194463d (-0.022958phase), full-SB1-covariance linear propagated Gaia uncertainty0.221120d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.040850d (photometric period projection allowance0.014395d). The approximate central duration6.26h vs observed median2.26h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses; deficit0.233pixel from target versus nearest capable neighbor1.675pixels. Aperture dilution limits intrinsic depth.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

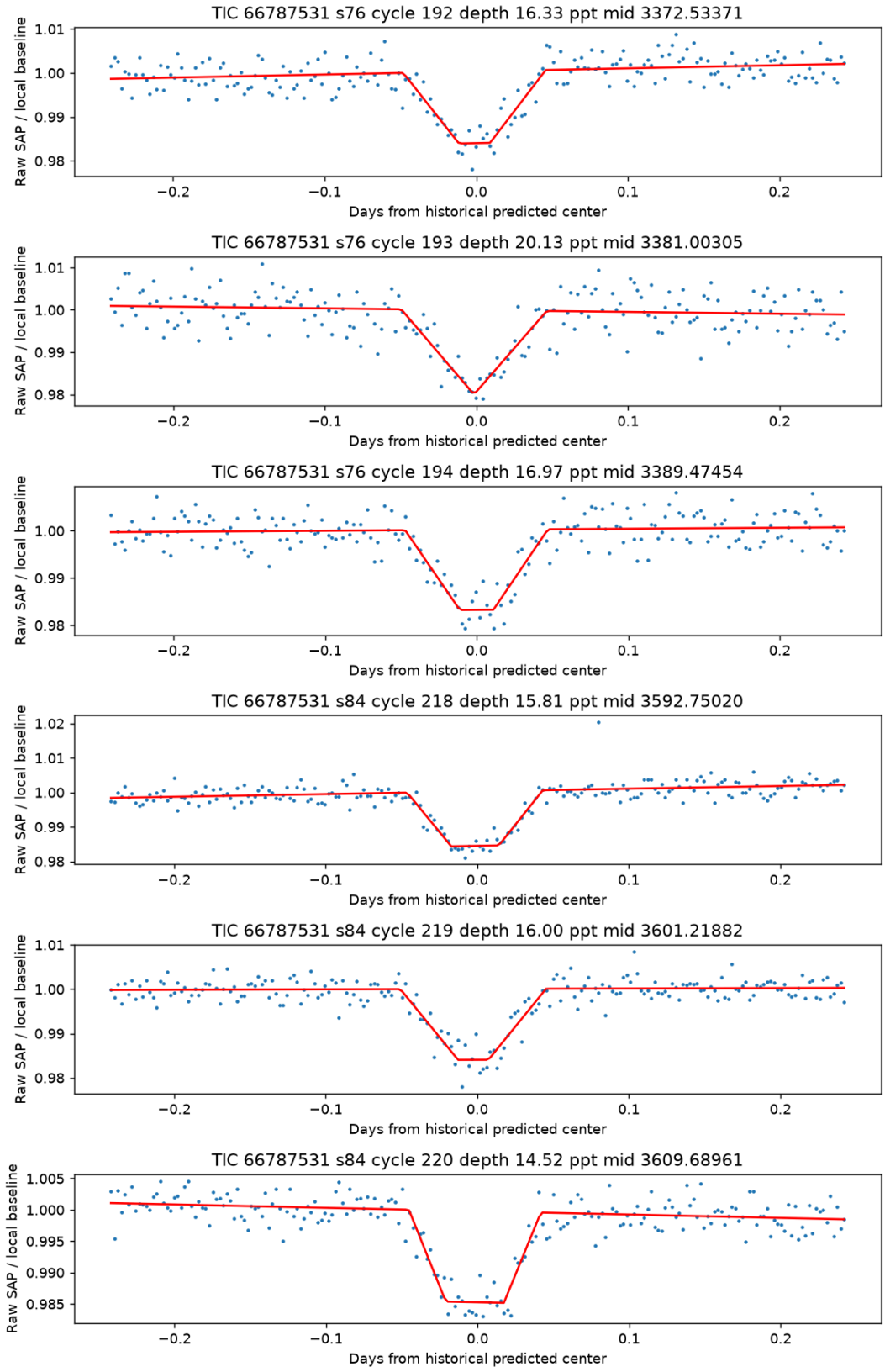
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
76	192	3372.5337087	0.001432	16.326	used in ephemeris
76	193	3381.0030530	0.001689	20.131	used in ephemeris
76	194	3389.4745364	0.001258	16.966	used in ephemeris
84	218	3592.7502013	0.001157	15.806	used in ephemeris
84	219	3601.2188247	0.001157	15.999	used in ephemeris
84	220	3609.6896101	0.001157	14.520	used in ephemeris

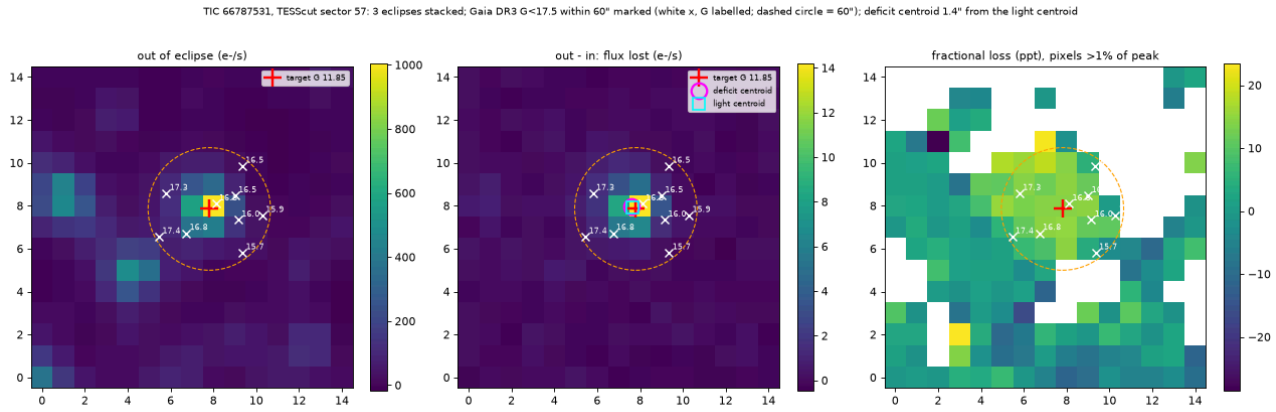
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s57.json, historical/diff_s57.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R377. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R377; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R377; diff_s57.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R378. TIC 382603504

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 382603504: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 40 distinct eclipse events in selected sectors[30, 97]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 17 same-kind midtimes give P=2.8391421780 +/-0.0000023540d, reference eclipse epoch BTJD2118.3611320, kind: Deeper eclipse; even historical half-period cycles. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S97 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.862arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.283. Capable neighbours: Gaia 4701453381830346240: 25.74 arcsec, G16.574, required fractional eclipse0.169; Gaia 4701453381830346368: 26.90 arcsec, G15.507, required fractional eclipse0.064; Gaia 4701453205735853312: 57.51 arcsec, G17.719, required fractional eclipse0.480
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths2.222/2.431/2.231ppt, with 36 stacked events. QLP selected-kind fitted median3.568ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR34701453381830347008, P=2.8391342975+/-0.0000612090d, e=0.037138, omega=190.6897deg. Closest edge-on conjunction offset-0.008284d (-0.002918phase), full-SB1-covariance linear propagated Gaia uncertainty0.045358d at the measured epoch. Near-Gaia-reference projected eclipse offset-

0.013083d (photometric period projection allowance0.001434d). The approximate central duration5.86h vs observed median1.48h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Persistent unequal alternating eclipses: even historical half-period cycles are deeper; only those17 measured events define the orbital ephemeris. Half-period is rejected by unequal parity depths in both sectors. Target deficit0.089pixel versus capable neighbor1.360pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

Measured events (errors are the conservative fit inputs, not calibrated coverage):

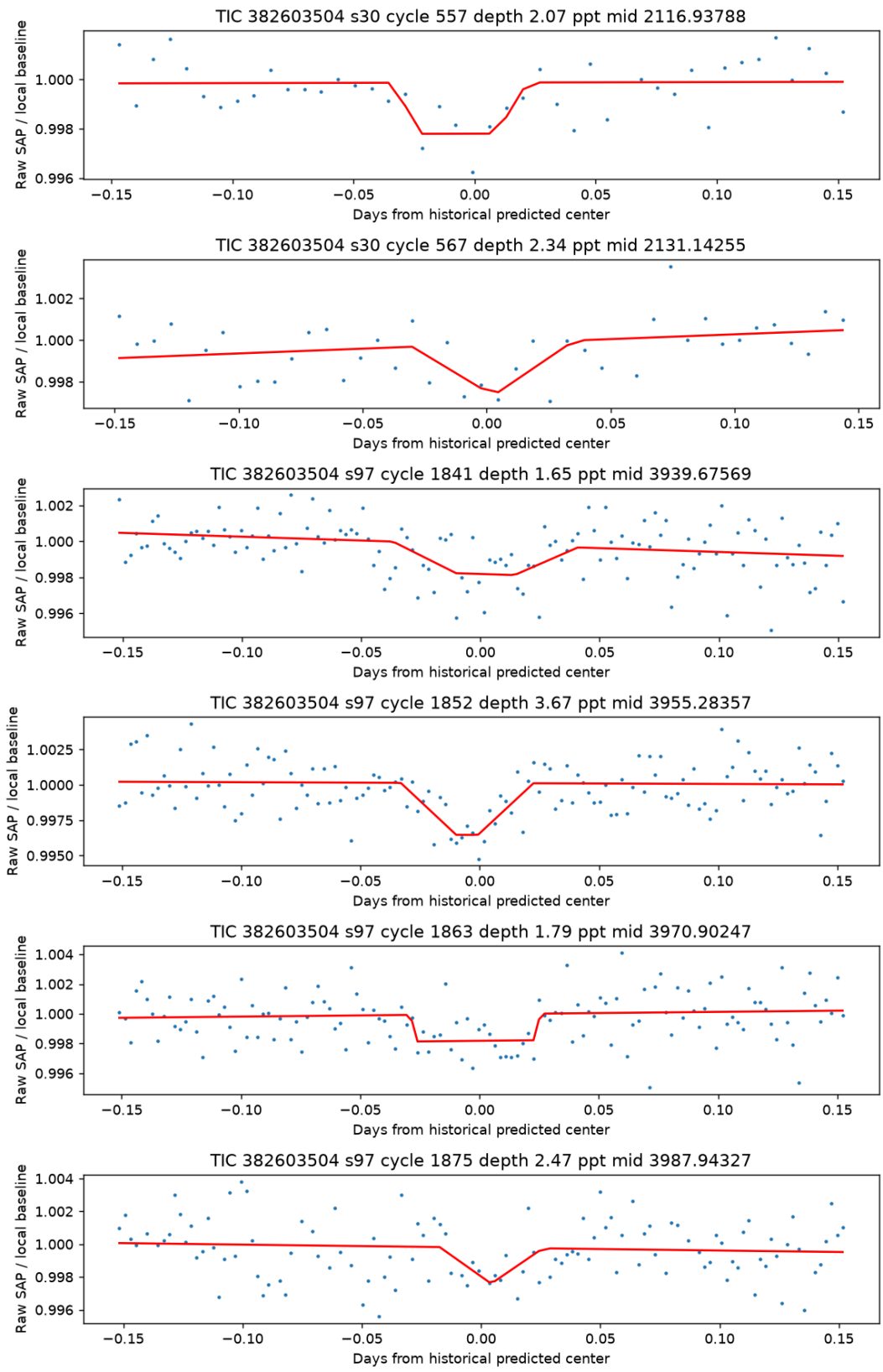
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
30	557	2116.9378844	0.003758	2.072	measured; excluded from primary fit (secondary event)
30	558	2118.3597039	0.003472	4.113	used in ephemeris
30	559	2119.7816907	0.004166	1.532	measured; excluded from primary fit (secondary event)
30	560	2121.2007854	0.003472	4.068	used in ephemeris
30	562	2124.0383936	0.003472	4.503	used in ephemeris
30	563	2125.4597016	0.006833	1.717	measured; excluded from primary fit (secondary event)
30	564	2126.8761905	0.003472	3.570	used in ephemeris
30	567	2131.1425483	0.005153	2.342	measured; excluded from primary fit (secondary event)
30	568	2132.5576719	0.003472	3.353	used in ephemeris
30	569	2133.9644958	0.004286	1.806	measured; excluded from primary fit (secondary event)
30	570	2135.3999099	0.003472	3.358	used in ephemeris
30	572	2138.2347111	0.003472	4.889	used in ephemeris
30	573	2139.6588831	0.005267	1.546	measured; excluded from primary fit (secondary event)
30	574	2141.0778246	0.003472	4.348	Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
97	1840	3938.2487732	0.003216	3.871	used in ephemeris
97	1841	3939.6756944	0.003363	1.648	measured; excluded from primary fit (secondary event)
97	1842	3941.0895928	0.001790	3.568	used in ephemeris
97	1843	3942.5043977	0.004681	2.193	measured; excluded from primary fit (secondary event)
97	1844	3943.9321240	0.002265	2.729	used in ephemeris
97	1845	3945.3436402	0.007403	0.645	measured; excluded from primary fit (secondary event)
97	1849	3951.0188889	0.006166	1.966	measured; excluded from primary fit (secondary event)
97	1850	3952.4482325	0.002438	2.956	used in ephemeris
97	1851	3953.8646719	0.004702	2.938	measured; excluded from primary fit (secondary event)
97	1852	3955.2835655	0.002459	3.667	used in ephemeris
97	1853	3956.7028101	0.003161	2.344	measured; excluded from primary fit (secondary event)
97	1854	3958.1239220	0.002032	3.463	used in ephemeris
97	1855	3959.5446403	0.003948	2.026	measured; excluded from primary fit (secondary event)
97	1859	3965.2198753	0.001157	0.985	Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
97	1860	3966.6448092	0.002995	3.349	used in ephemeris
97	1861	3968.0605634	0.002046	2.121	measured; excluded from primary fit (secondary event)

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
97	1862	3969.4843390	0.002114	2.838	used in ephemeris
97	1863	3970.9024741	0.001759	1.788	measured; excluded from primary fit (secondary event)
97	1864	3972.3225758	0.003392	3.452	used in ephemeris
97	1865	3973.7357642	0.002947	2.138	measured; excluded from primary fit (secondary event)
97	1869	3979.4232988	0.007569	0.968	measured; excluded from primary fit (secondary event)
97	1871	3982.2492902	0.010649	0.965	measured; excluded from primary fit (secondary event)
97	1872	3983.6787296	0.001157	3.062	Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
97	1873	3985.0974385	0.003611	3.079	measured; excluded from primary fit (secondary event)
97	1874	3986.5136329	0.002738	4.700	used in ephemeris
97	1875	3987.9432722	0.004322	2.474	measured; excluded from primary fit (secondary event)

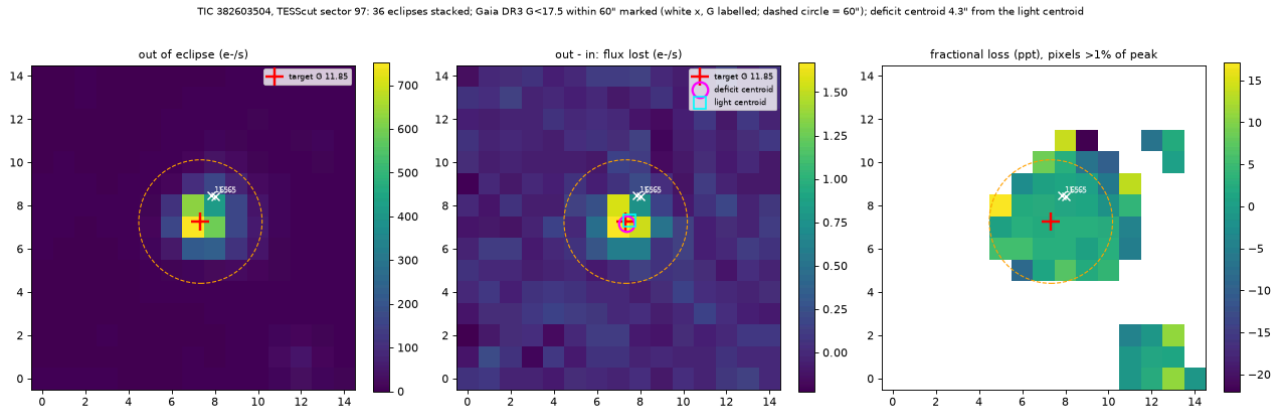
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R378. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R378; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R378; diff_s97.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R379. TIC 305651410

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 305651410: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[15, 56, 83]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=9.0437491336 +/-0.0000163681d, reference eclipse epoch BTJD1713.5371983, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S55 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation1.245arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.552. Capable neighbours: Gaia 1970480765776344576: 11.46 arcsec, G15.095, required fractional eclipse0.273; Gaia 1970480731416602240: 22.41 arcsec, G16.071, required fractional eclipse0.650; Gaia 1970480731416603904: 27.53 arcsec, G14.779, required fractional eclipse0.207; Gaia 1970480765776343040: 31.00 arcsec, G16.117, required fractional eclipse0.678; Gaia 1970480697056864384: 31.24 arcsec, G15.472, required fractional eclipse0.380; Gaia 1970480765776350464: 31.53 arcsec, G13.473, required fractional eclipse0.072; Gaia 1970480765776345088: 40.75 arcsec, G16.360, required fractional eclipse0.845; Gaia 1970480692753954688: 45.53 arcsec, G15.647, required fractional eclipse0.445; Gaia 1970480765776357760: 50.87 arcsec, G16.188, required fractional eclipse0.723; Gaia 1970480800136095616: 54.22 arcsec, G15.066, required fractional eclipse0.266; Gaia 1970480834495821184: 54.26 arcsec, G16.336, required fractional eclipse0.827; Gaia

1970480795828502144: 57.10 arcsec, G15.860, required fractional eclipse0.538; Gaia 1970492551166619008: 59.04 arcsec, G16.503, required fractional eclipse0.962

4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths12.607/11.644/9.927ppt, with 3 stacked events. QLP selected-kind fitted median13.481ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3197048073141660403 2, P=9.0416084657+/-0.0035950535d, e=0.190327, omega=178.9507deg. Closest edge-on conjunction offset-0.145031d (-0.016040phase), full-SB1-covariance linear propagated Gaia uncertainty0.461576d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.457568d (photometric period projection allowance0.002390d). The approximate central duration6.12h vs observed median4.52h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated flat-bottom eclipses; crowded field includes capable neighbor0.550pixel from deficit versus target0.059pixel. Full difference core favors target but subpixel crowding limits precision.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

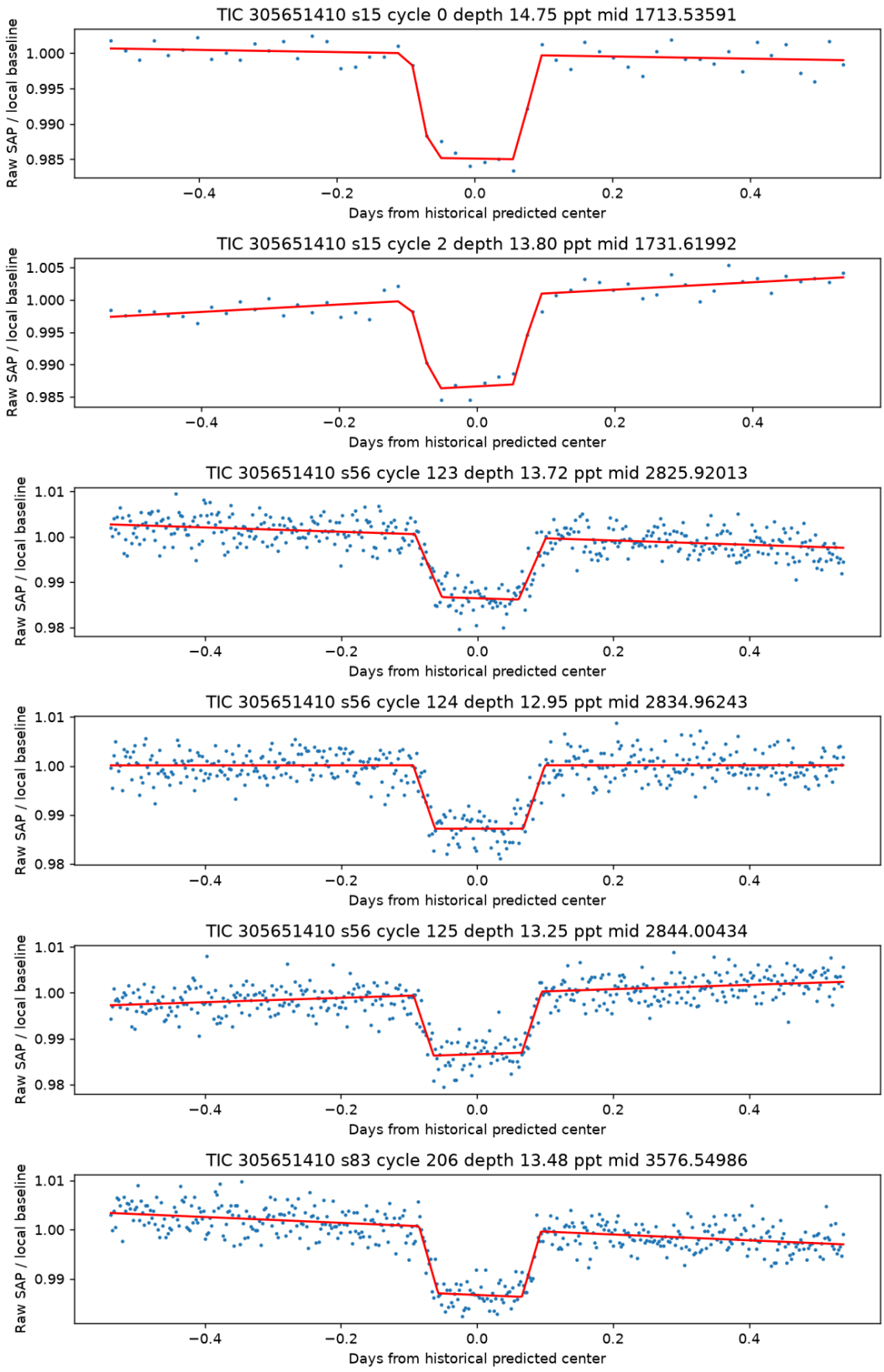
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
15	0	1713.5359134	0.010417	14.752	used in ephemeris
15	2	1731.6199227	0.010417	13.799	used in ephemeris
56	123	2825.9201271	0.001621	13.719	used in ephemeris
56	124	2834.9624255	0.001328	12.950	used in ephemeris
56	125	2844.0043409	0.001419	13.250	used in ephemeris
83	205	3567.5052335	0.001635	13.256	used in ephemeris
83	206	3576.5498593	0.001601	13.481	used in ephemeris

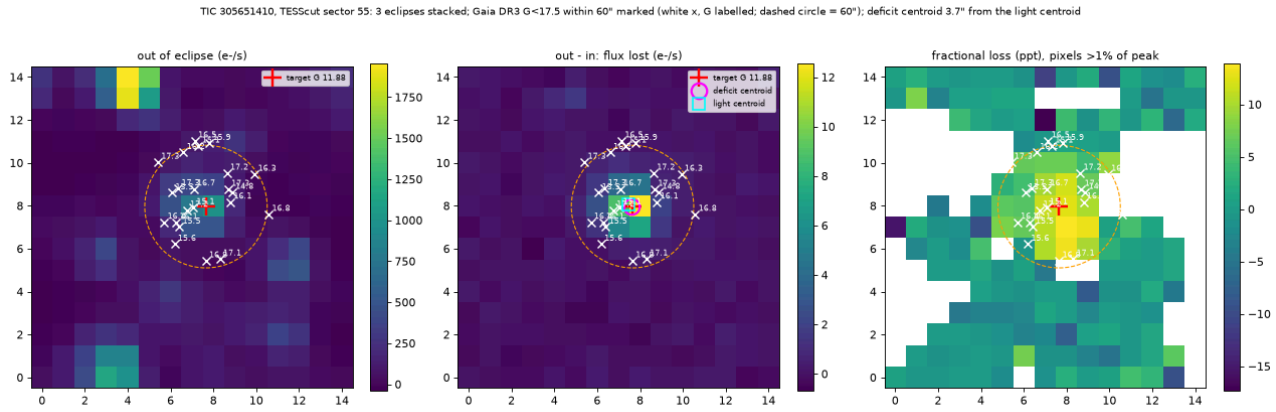
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s55.json, historical/diff_s55.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R379. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R379; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R379; diff_s55.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R380. TIC 343340931

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry / multi-source PRF localization

Archived source: sanitized report text.

TIC 343340931: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[12, 93, 103]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=6.1520562408 +/-0.0000181837d, reference eclipse epoch BTJD1630.7654348, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: unresolved after check.** Historical S13 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation3.996arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.153. Capable neighbours: Gaia 5801411754927216640: 2.88 arcsec, G14.241, required fractional eclipse0.205
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths21.316/20.867/20.807ppt, with 4 stacked events. QLP selected-kind fitted median20.390ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3580141175492721702 4, P=6.1527422755+/-0.0006366165d, e=0.140362, omega=312.4429deg. Closest edge-on conjunction offset-0.155060d (-0.025202phase), full-SB1-covariance linear propagated Gaia uncertainty0.137697d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.016481d (photometric period projection allowance0.003673d). The approximate central duration4.79h vs observed median9.82h is a radius/minimum-mass illustration, not a detailed

binary fit. TIC radius/mass are unavailable for this target; the solar fallback is not a physical duration constraint. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Attribution remains unresolved: capable neighbor only2.879arcsec from target, required intrinsic eclipse0.205; deficit distances0.190pixel to target and0.282pixel to neighbor are insufficient to distinguish these sources reliably. TIC mass/radius absent; approximate solar-fallback duration4.79h cannot constrain observed9.82h. Gaia association remains possible; source assignment needs resolved photometry.

Next check: Resolved eclipse photometry or multi-source PRF localization of this signal.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

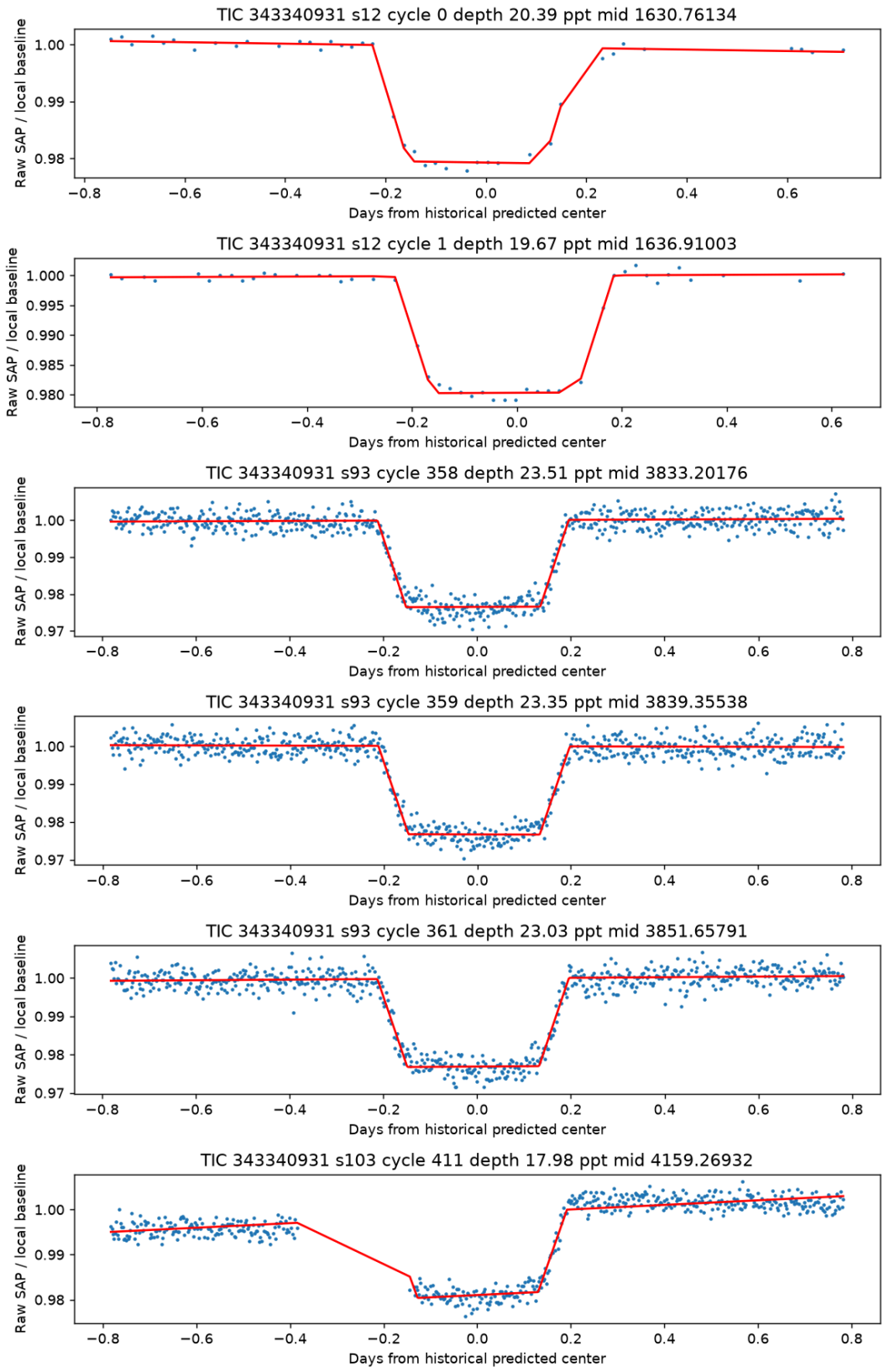
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
12	0	1630.7613410	0.010417	20.390	used in ephemeris
12	1	1636.9100316	0.010417	19.672	used in ephemeris
12	3	1649.2165764	0.010417	20.254	used in ephemeris
93	358	3833.2017649	0.001157	23.512	used in ephemeris
93	359	3839.3553771	0.001157	23.355	used in ephemeris
93	360	3845.5052080	0.001157	22.784	used in ephemeris
93	361	3851.6579128	0.001157	23.029	used in ephemeris
103	410	4153.1062442	0.001157	18.409	used in ephemeris
103	411	4159.2693171	0.003821	17.983	used in ephemeris

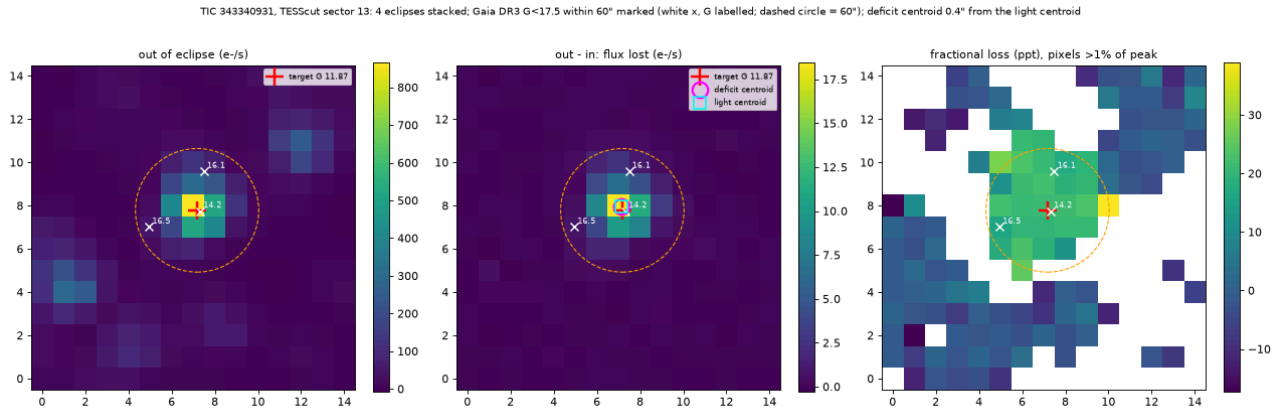
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s13.json, historical/diff_s13.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R380. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R380; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R380; diff_s13.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R381. TIC 421951253

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 421951253: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 11 distinct eclipse events in selected sectors[1, 68, 95]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=5.1102857278 +/-0.0000053139d, reference eclipse epoch BTJD1327.5694426, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S28 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.504arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.101. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths43.861/43.412/43.001ppt, with 4 stacked events. QLP selected-kind fitted median42.001ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3681996379463091878 4, P=5.1105848877+/-0.0003627777d, e=0.081883, omega=85.1163deg. Closest edge-on conjunction offset-0.052807d (-0.010333phase), full-SB1-covariance linear propagated Gaia uncertainty0.073267d at the measured epoch. Near-Gaia-reference projected eclipse offset0.002239d (photometric period projection allowance0.000978d). The approximate central duration5.61h vs observed median4.43h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated isolated eclipses across widely separated sectors; no capable listed neighbor. Diluted fitted depths vary by sector aperture.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

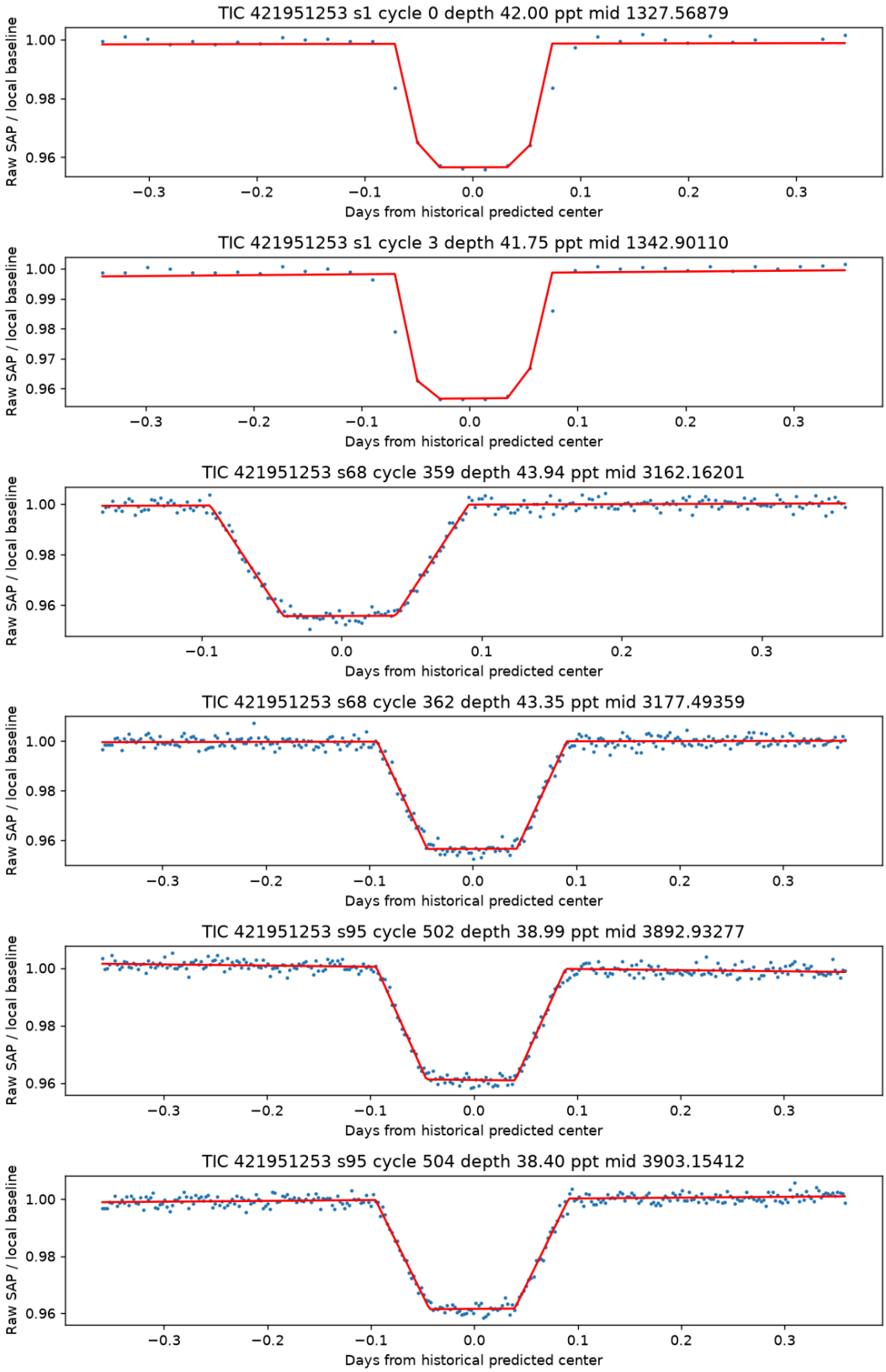
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1327.5687926	0.010417	42.001	used in ephemeris
1	1	1332.6798393	0.010417	43.810	used in ephemeris
1	3	1342.9011012	0.010417	41.749	used in ephemeris
68	358	3157.0511831	0.001157	43.389	used in ephemeris
68	359	3162.1620065	0.001157	43.939	used in ephemeris
68	361	3172.3824542	0.001157	43.371	used in ephemeris
68	362	3177.4935875	0.001157	43.353	used in ephemeris
95	501	3887.8224600	0.001157	39.253	used in ephemeris
95	502	3892.9327738	0.001157	38.989	used in ephemeris
95	503	3898.0427180	0.001157	37.906	used in ephemeris
95	504	3903.1541154	0.001157	38.400	used in ephemeris

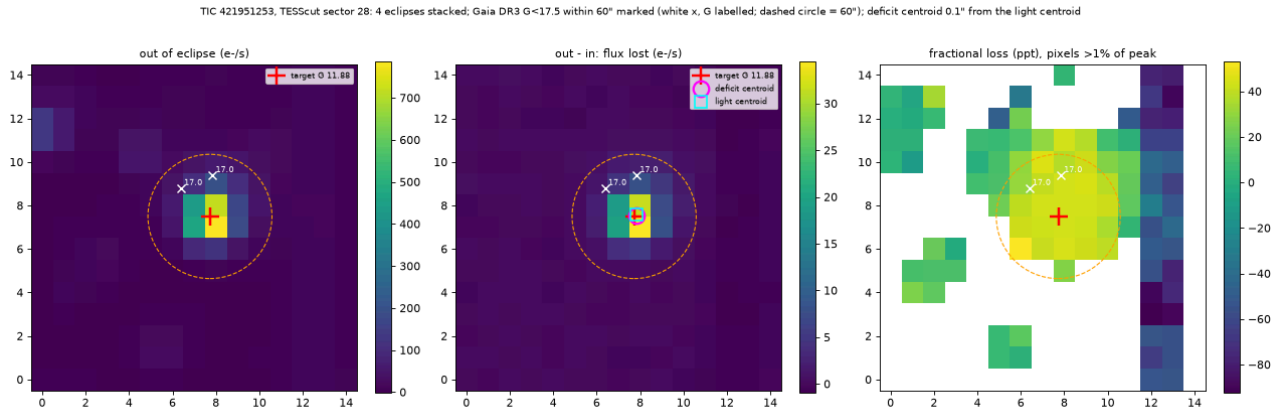
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s28.json, historical/diff_s28.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R381. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R381; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R381; diff_s28.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R382. TIC 128825695

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 128825695: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 21 distinct eclipse events in selected sectors[56, 83]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 11 same-kind midtimes give P=4.8028045149 +/-0.0000048724d, reference eclipse epoch BTJD2826.0721681, kind: Deeper eclipse; even historical half-period cycles. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S56 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation1.753arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.408. Capable neighbours: Gaia 1904568750234102400: 51.23 arcsec, G12.586, required fractional eclipse0.043
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths13.142/12.399/10.344ppt, with 12 stacked events. QLP selected-kind fitted median17.106ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3190456871587436416O, P=4.8031801465+/-0.0002357494d, e=0.067203, omega=233.4166deg. Closest edge-on conjunction offset-0.165546d (-0.034466phase), full-SB1-covariance linear propagated Gaia uncertainty0.124919d at the measured epoch. Near-Gaia-reference projected eclipse offset0.024899d (photometric period projection allowance0.002470d). The approximate central duration6.52h vs observed median3.05h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating eclipses establish doubled orbital period relative to historical SAP half branch. Even11 deeper events define ephemeris; odd secondary timings excluded. Target deficit0.083pixel versus capable neighbor2.531pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

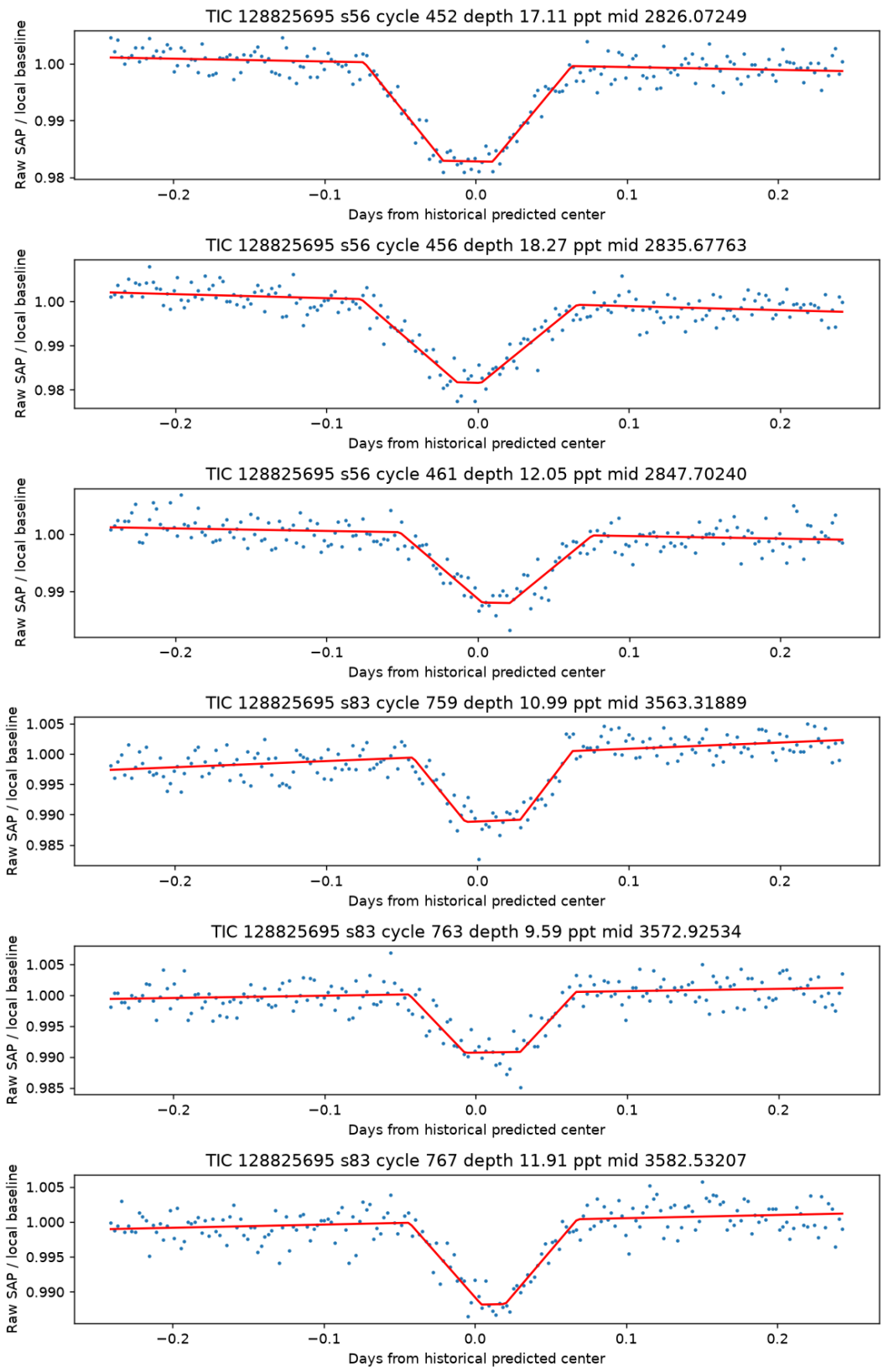
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
56	452	2826.0724916	0.001157	17.106	used in ephemeris
56	453	2828.4942892	0.001352	10.922	measured; excluded from primary fit (secondary event)
56	454	2830.8743699	0.001157	16.895	used in ephemeris
56	455	2833.2930554	0.001578	12.377	measured; excluded from primary fit (secondary event)
56	456	2835.6776301	0.001249	18.272	used in ephemeris
56	458	2840.4807618	0.001157	15.987	used in ephemeris
56	459	2842.9021433	0.001198	11.381	measured; excluded from primary fit (secondary event)
56	460	2845.2829200	0.001157	17.889	used in ephemeris
56	461	2847.7023979	0.001479	12.048	measured; excluded from primary fit (secondary event)
56	462	2850.0877416	0.001722	18.482	used in ephemeris
56	463	2852.5049818	0.001716	7.258	measured; excluded from primary fit (secondary event)
83	758	3560.9028777	0.001157	16.620	used in ephemeris
83	759	3563.3188879	0.001538	10.988	measured; excluded from primary fit (secondary event)
83	760	3565.7020372	0.001157	17.010	used in ephemeris
83	761	3568.1223736	0.001157	9.544	measured; excluded from primary fit (secondary event)
83	762	3570.5065417	0.001157	15.349	used in ephemeris
83	763	3572.9253431	0.001345	9.593	measured; excluded from primary fit (secondary event)
83	764	3575.3101004	0.001724	19.003	used in ephemeris
83	765	3577.7288782	0.001240	11.536	measured; excluded from primary fit (secondary event)
83	766	3580.1130087	0.001157	19.051	used in ephemeris
83	767	3582.5320671	0.001262	11.906	measured; excluded from primary fit (secondary event)

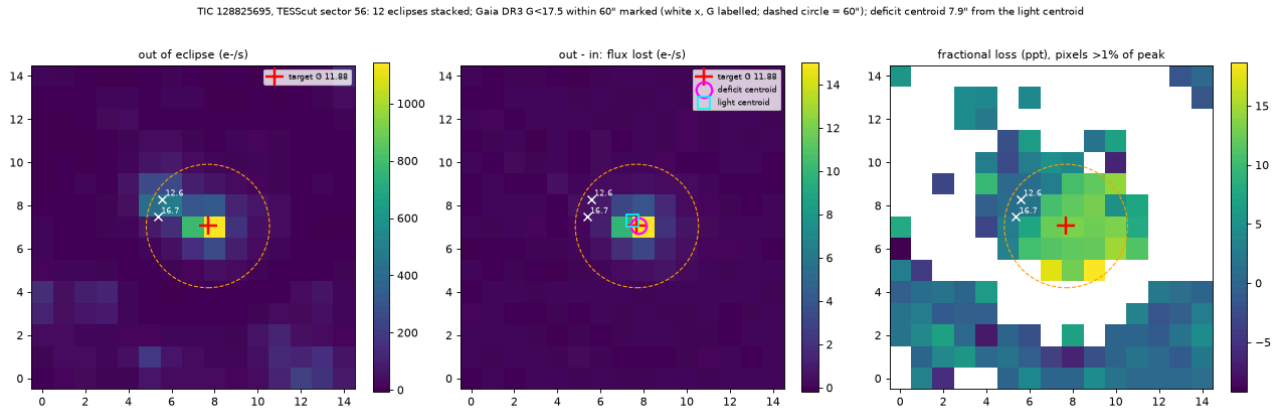
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s56.json, historical/diff_s56.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R382. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R382; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R382; diff_s56.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R383. TIC 283635479

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 283635479: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[17, 57, 84]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=12.4540146131 +/-0.0000236056d, reference eclipse epoch BTJD1768.3738400, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S17 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation4.322arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.653. Capable neighbours: Gaia 1996693191697286144: 38.43 arcsec, G11.906, required fractional eclipse0.234
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths118.799/99.677/70.809ppt, with 2 stacked events. QLP selected-kind fitted median125.079ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3199669319169728371 2, P=12.4547731122+/-0.0018532069d, e=0.172986, omega=141.3381deg. Closest edge-on conjunction offset-0.206500d (-0.016580phase), full-SB1-covariance linear propagated Gaia uncertainty0.254473d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.122307d (photometric period projection allowance0.002620d). The approximate central duration5.52h vs observed median3.96h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four repeated deep eclipses across sectors17/57/84; sector84 consecutive cycles146/147 test integer recurrence. Target deficit0.206pixel versus capable neighbor1.714pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

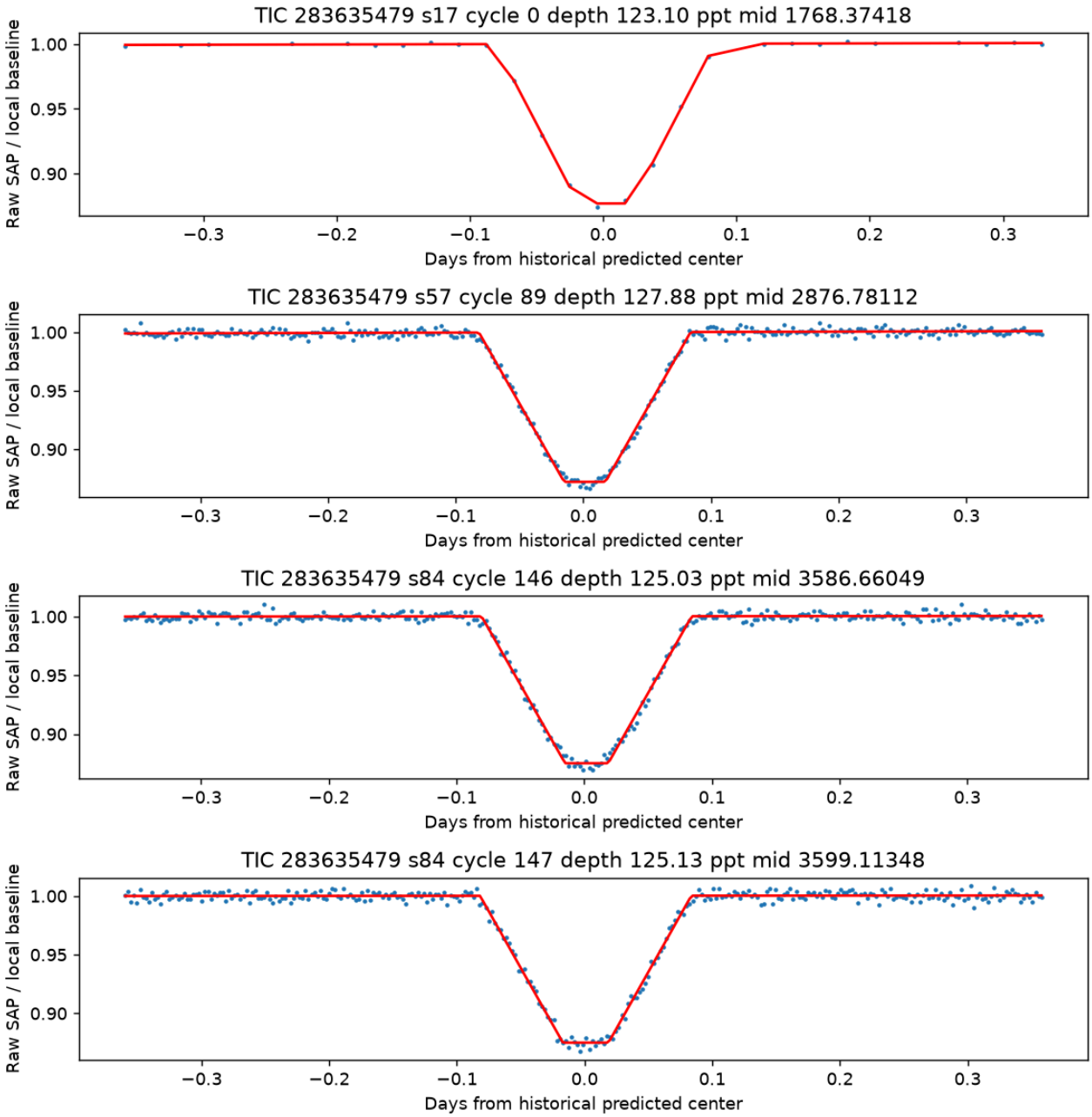
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
17	0	1768.3741759	0.010417	123.098	used in ephemeris
57	89	2876.7811212	0.001157	127.880	used in ephemeris
84	146	3586.6604921	0.001157	125.032	used in ephemeris
84	147	3599.1134849	0.001157	125.126	used in ephemeris

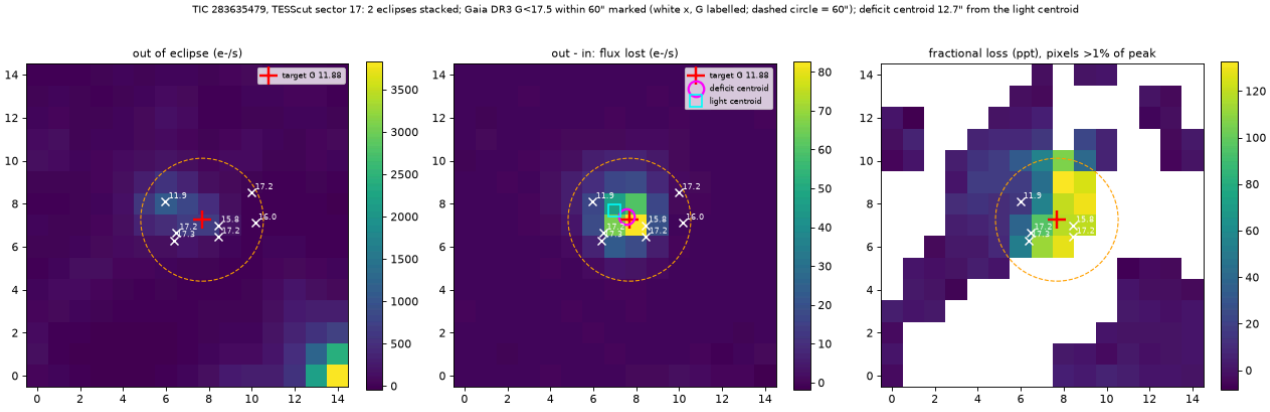
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s17.json, historical/diff_s17.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R383. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R383; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R383; diff_s17.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R384. TIC 9262715

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 9262715: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[20, 47, 60]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=7.3625774772 +/-0.0000348536d, reference eclipse epoch BTJD1846.2999364, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S47 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.380arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.643. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths100.072/97.050/85.495ppt, with 4 stacked events. QLP selected-kind fitted median109.006ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3923996517914746624, P=7.3632161676+/-0.0009799956d, e=0.069654, omega=157.7983deg. Closest edge-on conjunction offset-0.175911d (-0.023890phase), full-SB1-covariance linear propagated Gaia uncertainty0.198272d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.049450d (photometric period projection allowance0.006901d). The approximate central duration5.42h vs observed median3.68h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Seven deep repeated eclipses across sectors20/47/60, including sector47 consecutive cycles100-103. No capable listed neighbor.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

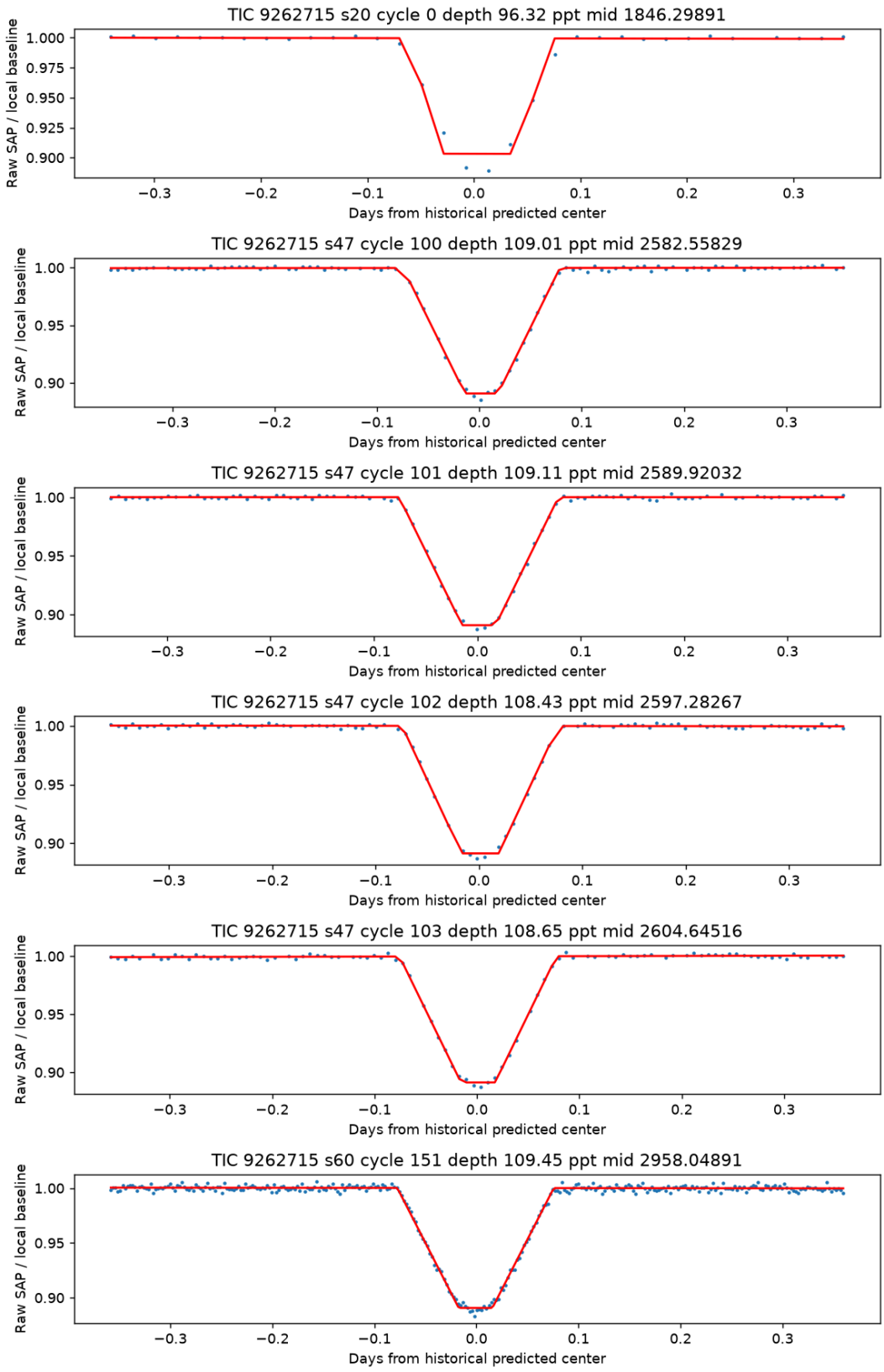
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
20	0	1846.2989112	0.010417	96.324	used in ephemeris
47	100	2582.5582853	0.003472	109.006	used in ephemeris
47	101	2589.9203232	0.003472	109.112	used in ephemeris
47	102	2597.2826705	0.003472	108.430	used in ephemeris
47	103	2604.6451631	0.003472	108.646	used in ephemeris
60	149	2943.3241963	0.001157	110.151	used in ephemeris
60	151	2958.0489055	0.001157	109.449	used in ephemeris

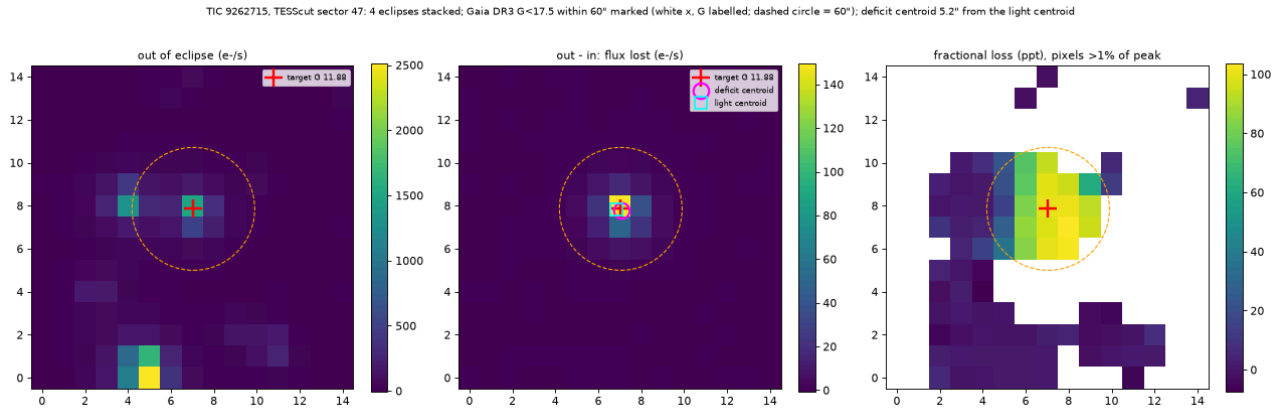
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s47.json, historical/diff_s47.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R384. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R384; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R384; diff_s47.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R385. TIC 394620087

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 394620087: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 5 distinct eclipse events in selected sectors[54, 81]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=6.8947331077 +/-0.0000236096d, reference eclipse epoch BTJD2771.8036128, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S40 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.578arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.547. Capable neighbours: Gaia 2045989451938846336: 30.87 arcsec, G13.589, required fractional eclipse0.594
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths101.759/96.507/82.787ppt, with 2 stacked events. QLP selected-kind fitted median104.974ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32045989413267218816, P=6.8950679495+/-0.0005730378d, e=0.015067, omega=339.7470deg. Closest edge-on conjunction offset-0.094934d (-0.013768phase), full-SB1-covariance linear propagated Gaia uncertainty0.216787d at the measured epoch. Near-Gaia-reference projected eclipse offset0.020921d (photometric period projection allowance0.008169d). The approximate central duration7.07h vs observed median5.28h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Five deep repeated eclipses across sectors54/81, including consecutive cycles266/267. Target deficit0.123pixel versus capable neighbor1.436pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

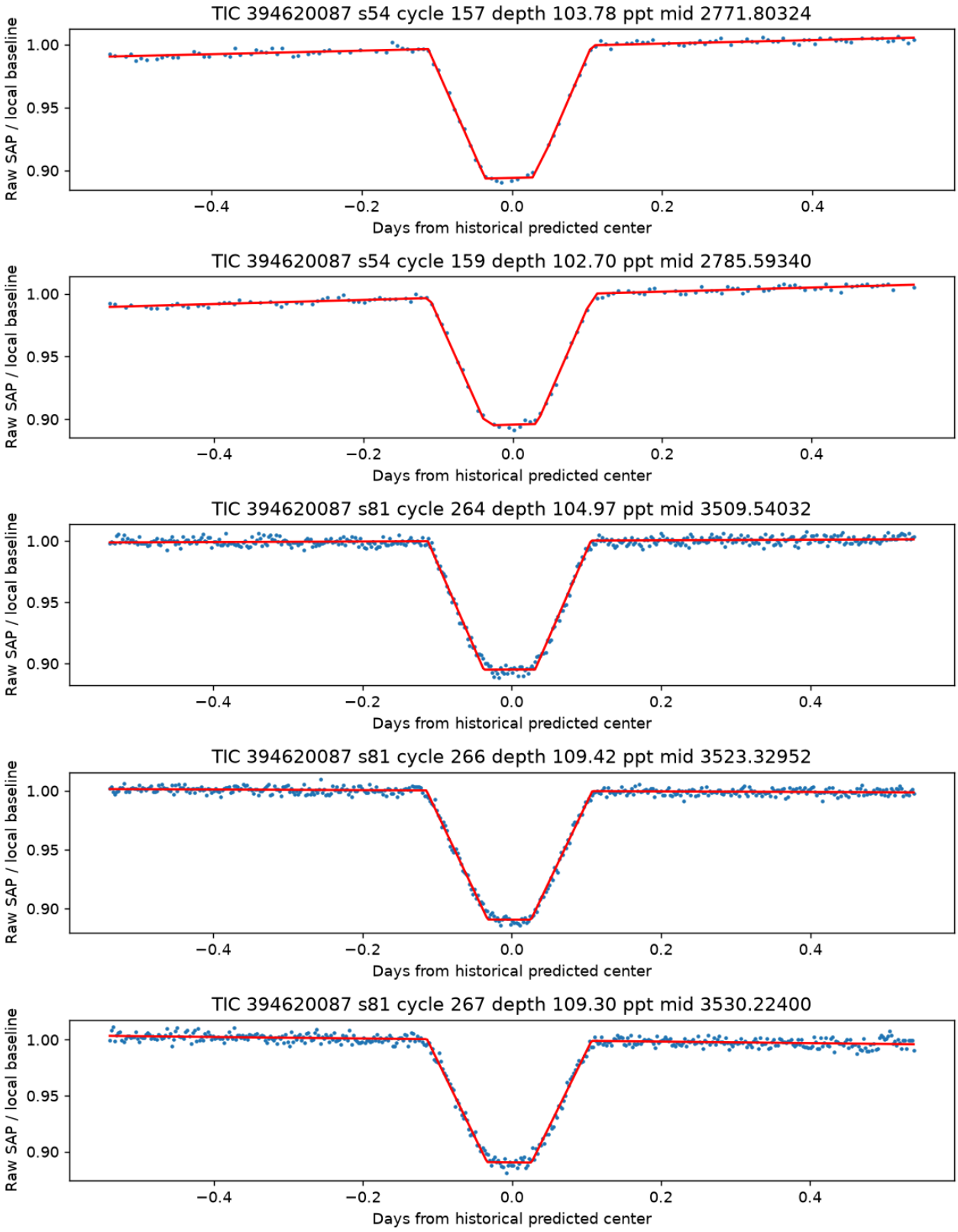
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
54	157	2771.8032357	0.003472	103.783	used in ephemeris
54	159	2785.5933979	0.003472	102.695	used in ephemeris
81	264	3509.5403160	0.001157	104.974	used in ephemeris
81	266	3523.3295217	0.001157	109.417	used in ephemeris
81	267	3530.2240002	0.001157	109.299	used in ephemeris

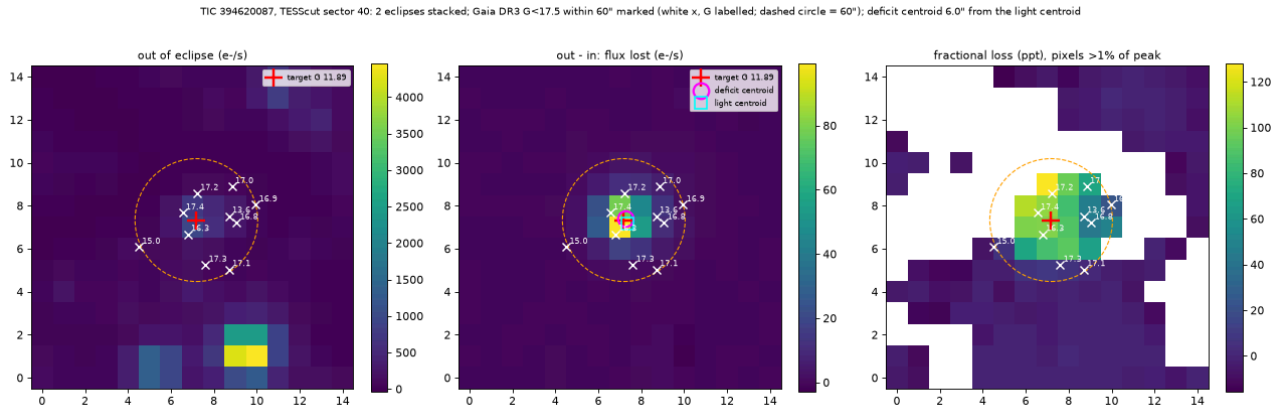
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s40.json, historical/diff_s40.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R385. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R385; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R385; diff_s40.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R386. TIC 443900773

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 443900773: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[7, 46, 88]. The representative raw-event plot and numeric raw-windows.csv were inspected. Broadly repeated morphology is retained even when individual shallow fits are noisy. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=9.3257982069 +/-0.0000153302d, reference eclipse epoch BTJD1496.6547499, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S7 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.805arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.082. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths351.148/350.810/345.926ppt, with 3 stacked events. QLP selected-kind fitted median379.302ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3649664385352561664, P=9.3257311444+/-0.0008100741d, e=0.014226, omega=280.9488deg. Closest edge-on conjunction offset0.022938d (0.002460phase), full-SB1-covariance linear propagated Gaia uncertainty0.121549d at the measured epoch. Near-Gaia-reference projected eclipse offset0.014958d (photometric period projection allowance0.001824d). The approximate central duration5.30h vs observed median4.31h

is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep repeated flat-bottom eclipses across sectors7/46/88; no capable listed neighbor. Fitted depths differ by aperture and sector; no detailed binary characterization.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

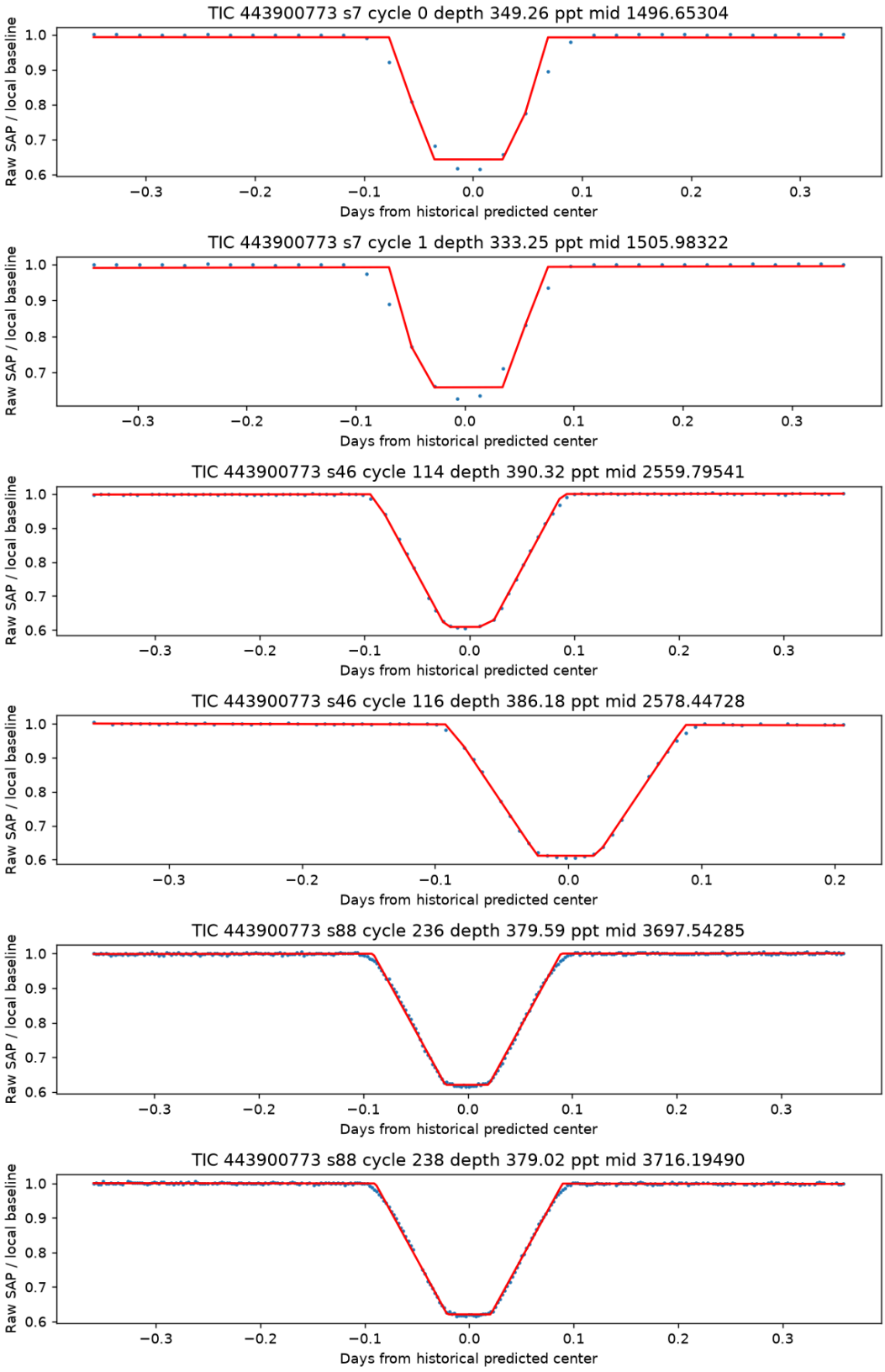
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
7	0	1496.6530390	0.010417	349.261	used in ephemeris
7	1	1505.9832182	0.010417	333.251	used in ephemeris
46	114	2559.7954116	0.003472	390.319	used in ephemeris
46	115	2569.1217730	0.003472	387.013	used in ephemeris
46	116	2578.4472772	0.003472	386.179	used in ephemeris
88	236	3697.5428520	0.001157	379.586	used in ephemeris
88	237	3706.8690316	0.001157	378.635	used in ephemeris
88	238	3716.1948982	0.001157	379.018	used in ephemeris

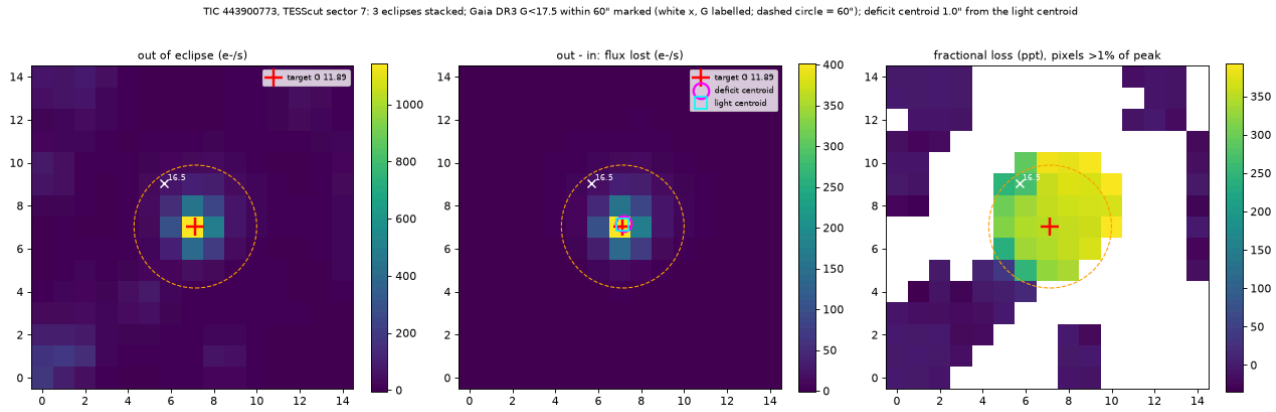
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s07.json, historical/diff_s07.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are diffstack.py and c3_sapdepth.py in the source sb1-scale tree. No historical evidence overwritten.

Stable review key: R386. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R386; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R386; diff_s07.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R387. TIC 438188407

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 438188407: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 5 fitted event windows in sectors [6, 45, 87] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 5 same-kind measured centers give P = 12.9836788156 +/- 0.0000267918 d and reference epoch BTJD 1479.9810460. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 33 deficit-target distance 0.173 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 3331368224678745984: separation 49.058 arcsec, required intrinsic fractional eclipse 0.6145, deficit distance 2.329 pixel; Gaia 3331368057179759104: separation 49.704 arcsec, required intrinsic fractional eclipse 0.4503, deficit distance 2.248 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 227.353/172.916/101.983 ppt from 2 stacked events; current selected-kind QLP median 243.437 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 3331368263337545984, P = 12.9857302396 +/- 0.0030179827 d, e = 0.079374, omega = 18.5457 deg. Closest edge-on conjunction is offset -0.270572 d (-0.020836 phase), with full SB1 covariance linear propagated

uncertainty 0.318309 d at measured epoch. Projection near Gaia reference gives offset -0.098252 d; photometric period projection allowance 0.002251 d excludes covariance and model systematics. Approximate central duration 11.75 h versus fitted median 10.12 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Deep repeated flat-bottom eclipses in three widely separated sectors. Capable neighbors are more than 2.24 pixels from the deficit, versus target 0.173 pixel. Gaia period drift is acknowledged through propagated covariance; no arbitrary formal-error veto is used.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

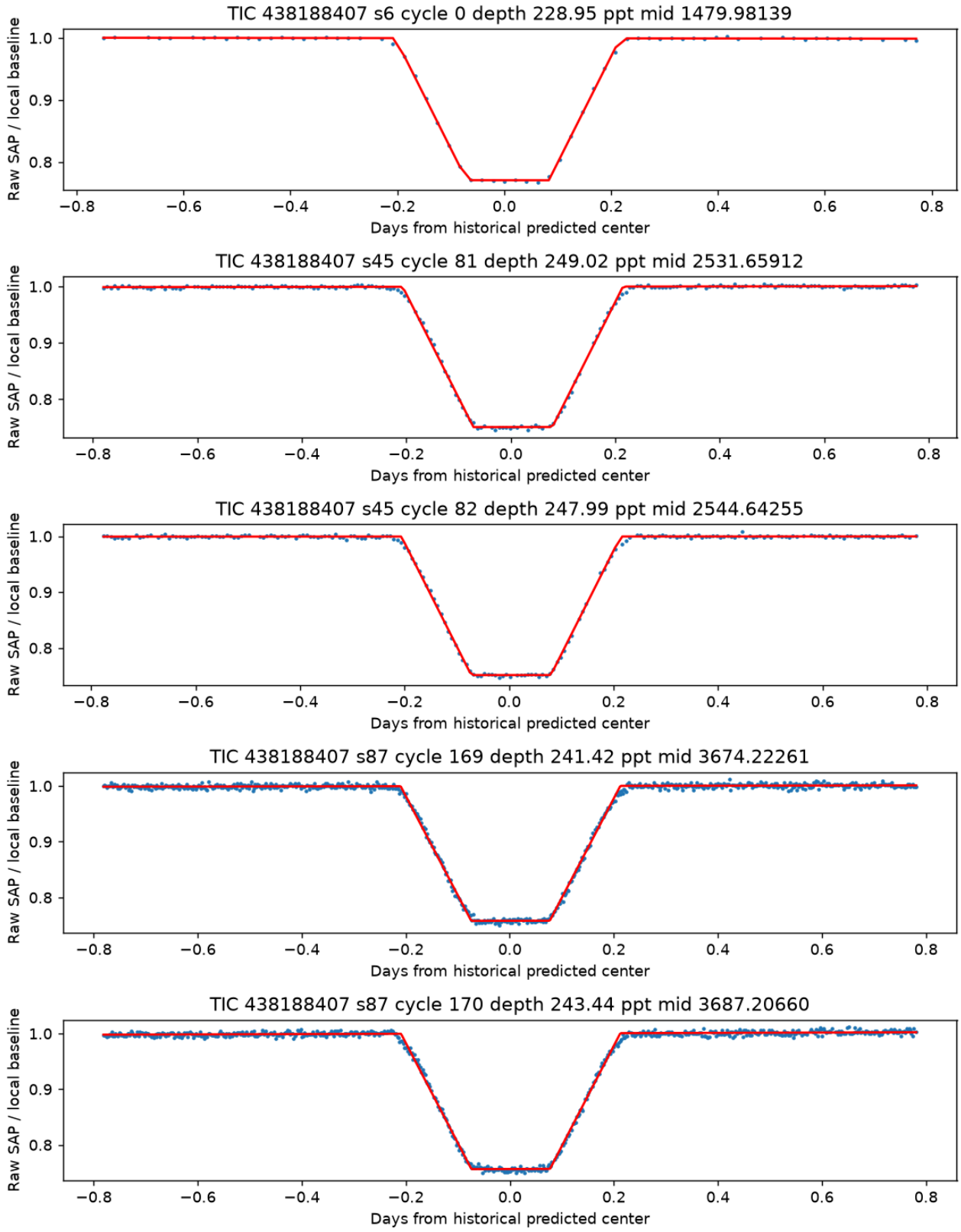
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
6	0	1479.9813931	0.0104167	228.946	used
45	81	2531.6591248	0.0034724	249.018	used
45	82	2544.6425545	0.0034724	247.986	used
87	169	3674.2226098	0.0011574	241.423	used
87	170	3687.2066030	0.0011574	243.437	used

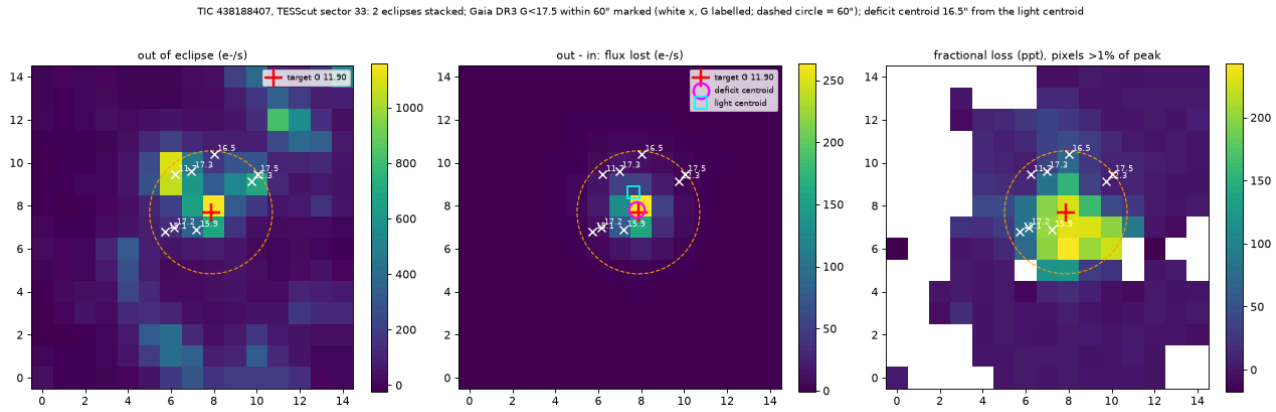
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R387. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R387; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R387; diff_s33.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R388. TIC 62337815

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 62337815: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 4 fitted event windows in sectors [76, 83] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 4 same-kind measured centers give P = 11.5008928530 +/- 0.0000784016 d and reference epoch BTJD 3378.3037354. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 55 deficit-target distance 0.249 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 2166153703341315968: separation 12.253 arcsec, required intrinsic fractional eclipse 0.2344, deficit distance 0.833 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 16.077/16.162/15.507 ppt from 2 stacked events; current selected-kind QLP median 20.247 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 2166153699041263360, P = 11.5003902549 +/- 0.0013799967 d, e = 0.048491, omega = 318.6830 deg. Closest edge-on conjunction is offset 0.067766 d (0.005892 phase), with full SB1 covariance linear propagated uncertainty 0.357581 d at measured epoch. Projection near Gaia reference gives offset -0.062910 d;

photometric period projection allowance 0.020384 d excludes covariance and model systematics. Approximate central duration 7.79 h versus fitted median 3.44 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Four isolated flat-bottom eclipses in sectors 76/83 with consecutive cycles in each. The 12.25 arcsec capable neighbor is 0.833 pixel from the deficit versus target 0.249 pixel; full image favors target, with residual subpixel crowding limitation.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

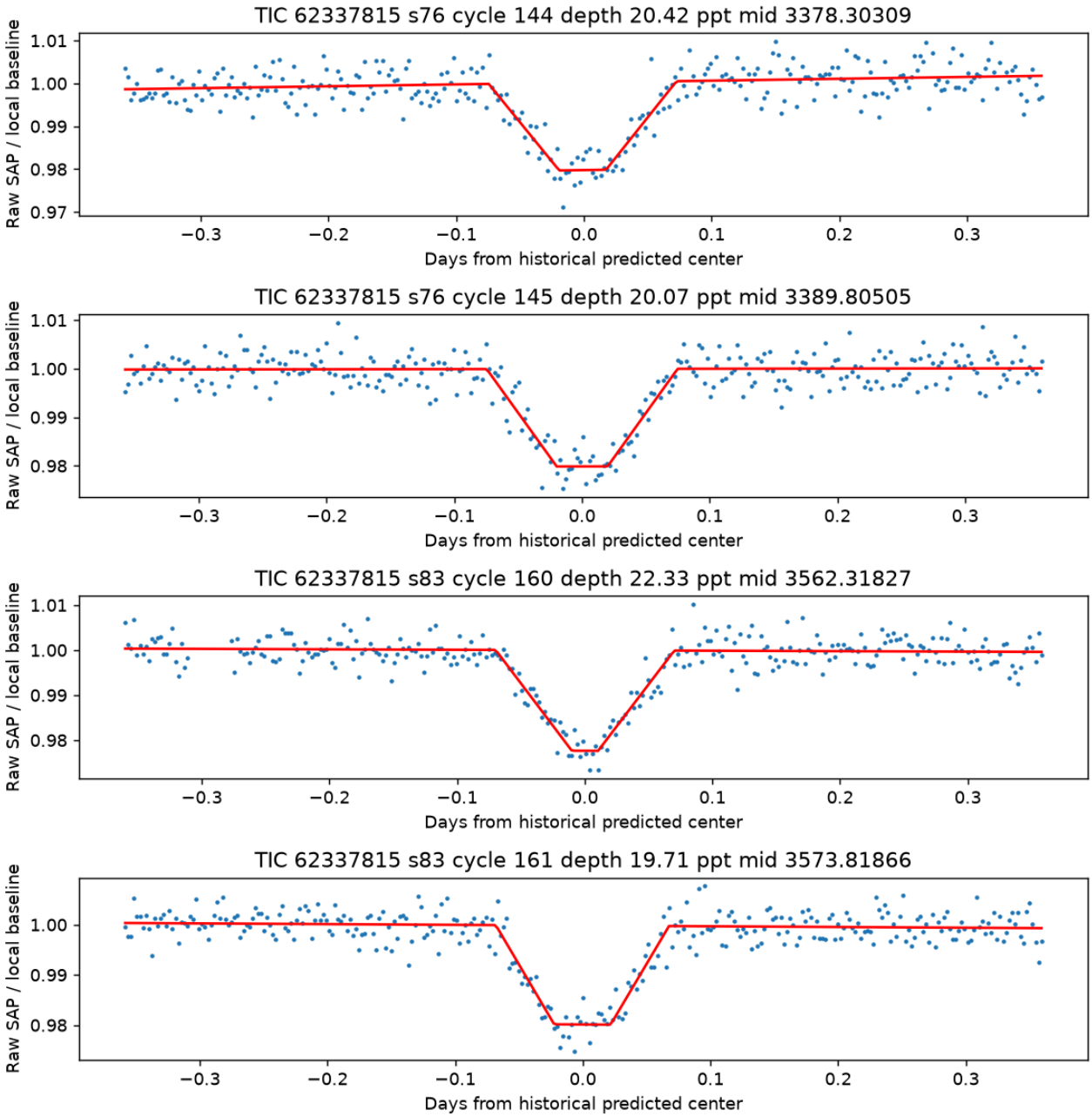
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
76	144	3378.3030917	0.0015034	20.425	used
76	145	3389.8050547	0.0012106	20.069	used
83	160	3562.3182704	0.0011574	22.328	used
83	161	3573.8186563	0.0011574	19.705	used

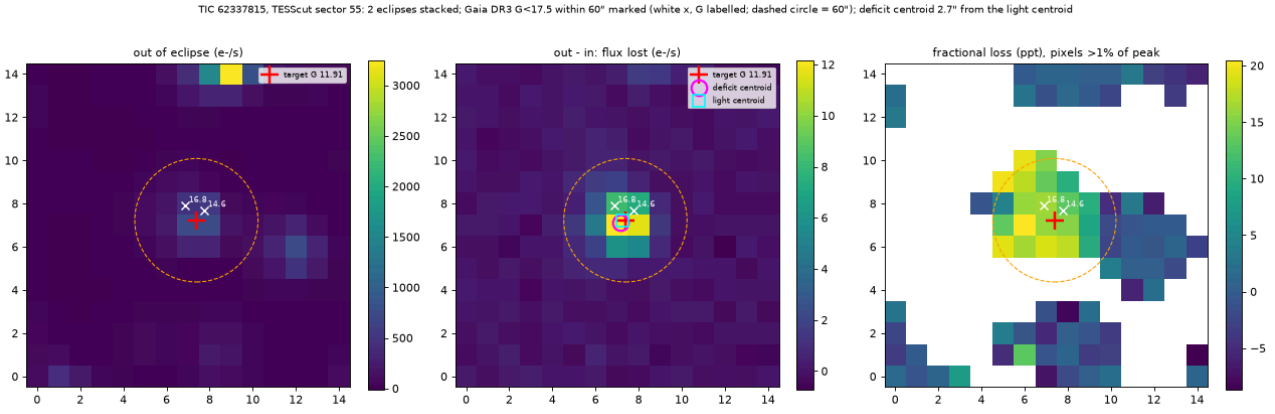
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s55.json, historical/diff_s55.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R388. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R388; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R388; diff_s55.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R389. TIC 399285183

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 399285183: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 7 fitted event windows in sectors [54, 81] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 7 same-kind measured centers give P = 3.8060033593 +/- 0.0000370371 d and reference epoch BTJD 2771.5481179. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 81 deficit-target distance 0.142 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 4242940643455267968: separation 26.034 arcsec, required intrinsic fractional eclipse 0.3477, deficit distance 1.135 pixel; Gaia 4242937654158029312: separation 38.743 arcsec, required intrinsic fractional eclipse 0.2899, deficit distance 1.942 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 12.498/11.543/10.458 ppt from 6 stacked events; current selected-kind QLP median 13.833 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 4242940643455270144, P = 3.8059045398 +/- 0.0002783071 d, e = 0.147943, omega = 124.5770 deg. Closest edge-on conjunction is offset 0.034084 d (0.008956 phase), with full SB1 covariance linear propagated

uncertainty 0.161791 d at measured epoch. Projection near Gaia reference gives offset -0.027777 d; photometric period projection allowance 0.023185 d excludes covariance and model systematics. Approximate central duration 5.16 h versus fitted median 3.05 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Seven repeated localized shallow eclipses, with clean flat surrounding baseline in sector 81 and consecutive measured cycles. Capable neighbors are at least 1.135 pixels from the deficit versus target 0.142 pixel.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

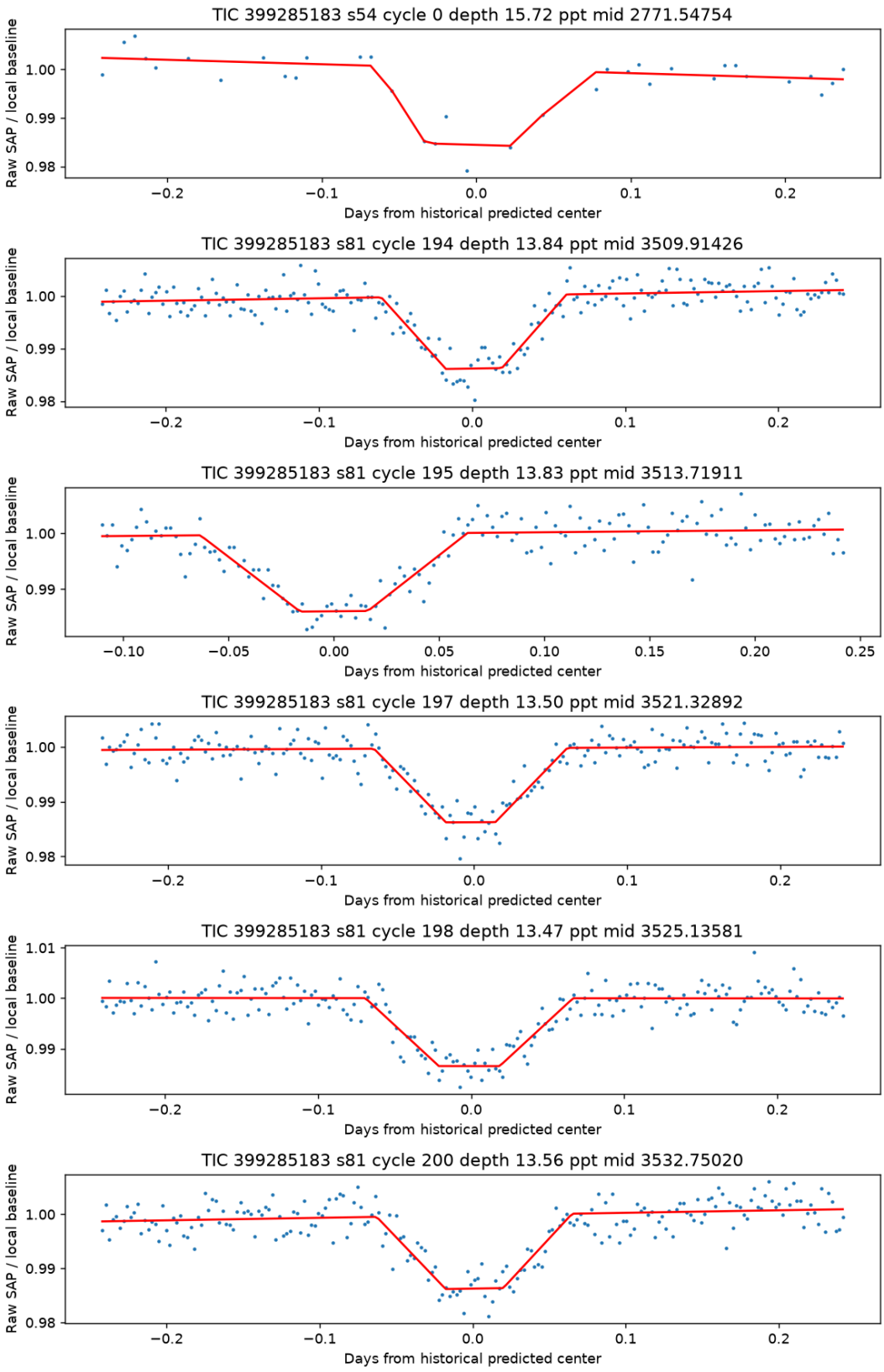
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
54	0	2771.5475403	0.0069444	15.721	used
81	194	3509.9142646	0.0011574	13.839	used
81	195	3513.7191080	0.0014885	13.833	used
81	197	3521.3289178	0.0011574	13.503	used
81	198	3525.1358128	0.0014356	13.468	used
81	199	3528.9429709	0.0018431	14.612	used
81	200	3532.7502021	0.0016008	13.555	used

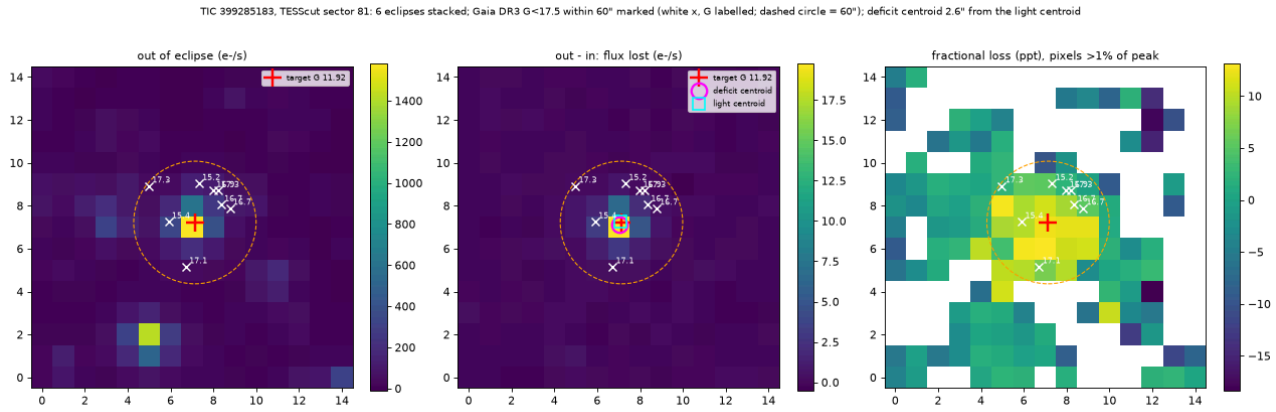
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s81.json, historical/diff_s81.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R389. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R389; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R389; diff_s81.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R390. TIC 399868192

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 399868192: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 7 fitted event windows in sectors [5, 71, 98] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 6 same-kind measured centers give P = 6.4834644669 +/- 0.0000125529 d and reference epoch BTJD 1457.0655796. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 98 deficit-target distance 0.057 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 3291551648758099840: separation 37.753 arcsec, required intrinsic fractional eclipse 0.1130, deficit distance 1.860 pixel; Gaia 3291551614399361152: separation 42.464 arcsec, required intrinsic fractional eclipse 0.0417, deficit distance 2.031 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 29.732/24.057/11.964 ppt from 6 stacked events; current selected-kind QLP median 29.230 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 3291551648758101248, P = 6.4836906195 +/- 0.0007693937 d, e = 0.248840, omega = 343.5160 deg. Closest edge-on conjunction is offset 0.204507 d (0.031542 phase), with full SB1 covariance linear propagated uncertainty 0.176224 d at

measured epoch. Projection near Gaia reference gives offset 0.241823 d; photometric period projection allowance 0.002071 d excludes covariance and model systematics. Approximate central duration 5.40 h versus fitted median 4.88 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Repeated flat-bottom eclipses across sectors 5/71/98. Sector 98 cycle 398 lacks ingress and has singular center covariance, so it is excluded from timing. Other consecutive events anchor the recurrence. Capable neighbors are at least 1.86 pixels from deficit versus target 0.057 pixel.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

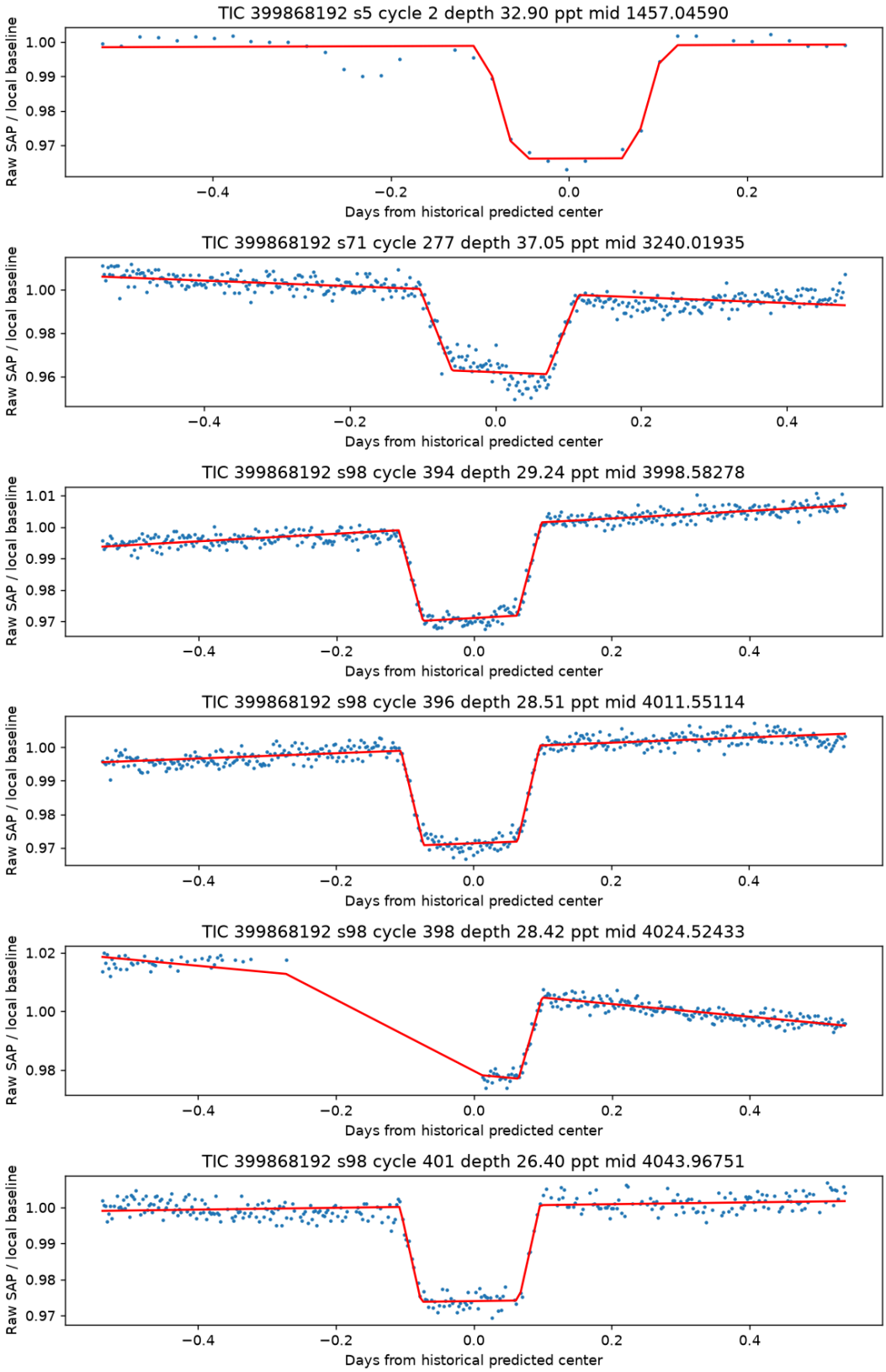
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
5	2	1457.0459022	0.0104168	32.896	used
71	277	3240.0193501	0.0013255	37.051	used
98	394	3998.5827791	0.0014039	29.239	used
98	396	4011.5511436	0.0011574	28.515	used
98	398	4024.5243308	304938.5784592	28.421	Partially covered eclipse lacks ingress baseline; retained raw morphology but excluded timing.
98	399	4031.0005506	0.0020469	29.221	used
98	401	4043.9675139	0.0011574	26.397	used

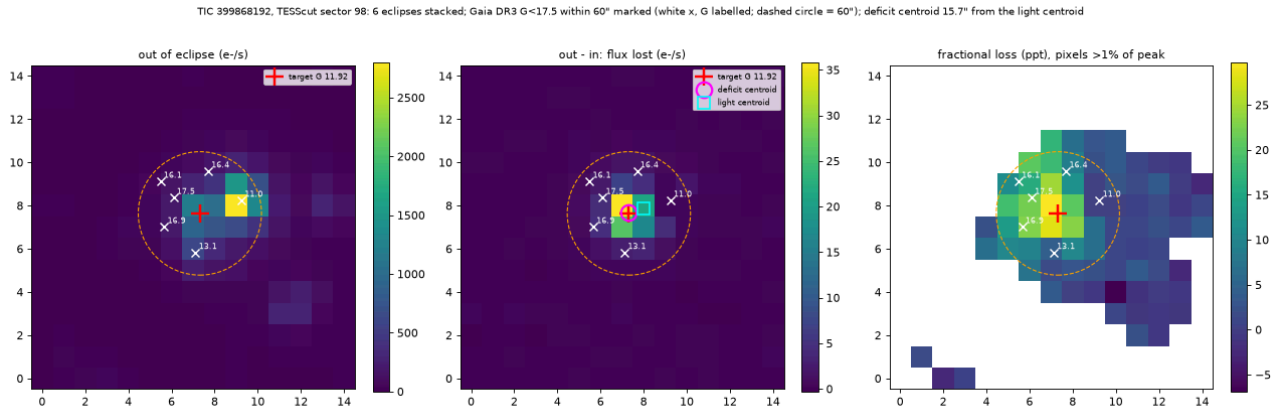
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s98.json, historical/diff_s98.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R390. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R390; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R390; diff_s98.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R391. TIC 457003443

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 457003443: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 13 fitted event windows in sectors [43, 45, 72] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 13 same-kind measured centers give P = 4.6443443385 +/- 0.0000089036 d and reference epoch BTJD 2479.0598944. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 43 deficit-target distance 0.218 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 3379491619541073280: separation 29.470 arcsec, required intrinsic fractional eclipse 0.2631, deficit distance 1.708 pixel; Gaia 3379493092713030656: separation 43.785 arcsec, required intrinsic fractional eclipse 0.1467, deficit distance 2.360 pixel; Gaia 3379491580884499072: separation 45.533 arcsec, required intrinsic fractional eclipse 0.3374, deficit distance 2.052 pixel; Gaia 3379491516461860608: separation 49.600 arcsec, required intrinsic fractional eclipse 0.2856, deficit distance 2.277 pixel; Gaia 3379493028290345728: separation 51.857 arcsec, required intrinsic fractional eclipse 0.0408, deficit distance 2.845 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 8.270/8.372/7.000 ppt from 6 stacked events; current selected-kind QLP median 9.585 ppt. These are reductions of the same TESS photons, not

independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.

5. **Gaia orbital geometry: pass with limitations.** DR3 3379491615243515392, P = 4.6449635222 +/- 0.0019033067 d, e = 0.097546, omega = 281.9464 deg. Closest edge-on conjunction is offset -0.132312 d (-0.028485 phase), with full SB1 covariance linear propagated uncertainty 0.925255 d at measured epoch. Projection near Gaia reference gives offset 0.146321 d; photometric period projection allowance 0.004007 d excludes covariance and model systematics. Approximate central duration 7.92 h versus fitted median 4.14 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Repeated isolated flat-bottom shallow eclipses across sectors 43/45/72. Full deficit image favors target at 0.218 pixel; capable neighbors are at least 1.708 pixels away. Aperture dilution and baseline variability limit depth interpretation.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

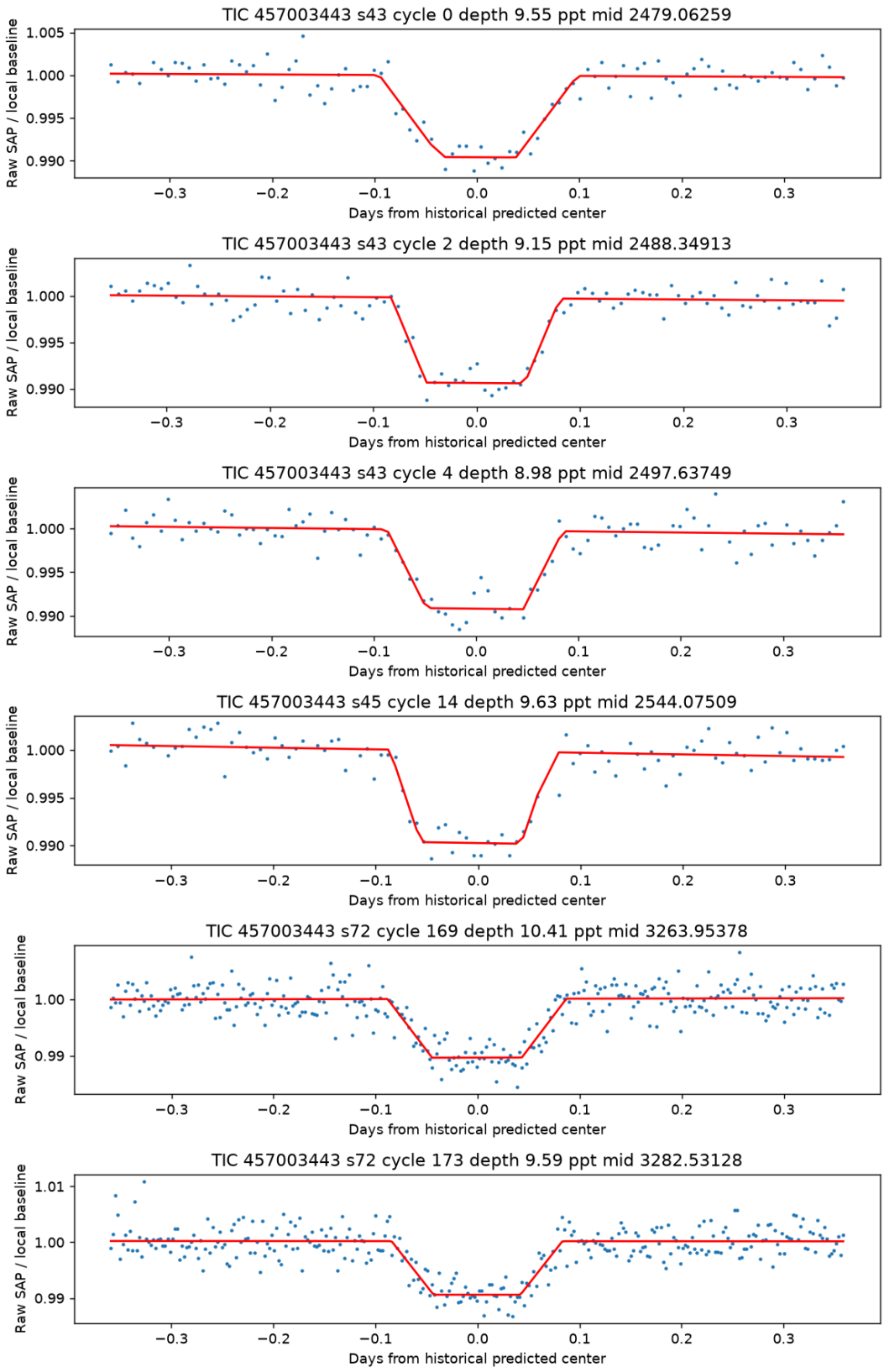
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
43	0	2479.0625871	0.0034726	9.552	used
43	1	2483.7041251	0.0034726	8.606	used
43	2	2488.3491318	0.0034726	9.154	used
43	3	2492.9908462	0.0034726	9.107	used
43	4	2497.6374866	0.0034726	8.975	used
45	11	2530.1479548	0.0034725	9.448	used
45	12	2534.7972437	0.0034725	9.810	used
45	14	2544.0750911	0.0034725	9.629	used
45	15	2548.7237772	0.0034725	9.585	used
72	169	3263.9537807	0.0017904	10.407	used
72	170	3268.5973875	0.0018398	9.784	used
72	172	3277.8889397	0.0019366	9.990	used
72	173	3282.5312831	0.0016369	9.589	used

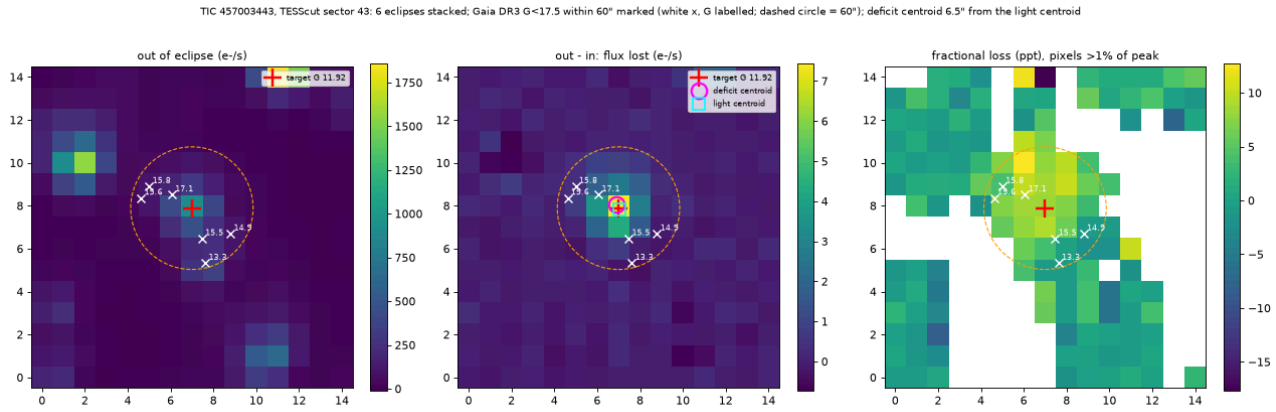
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s43.json, historical/diff_s43.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R391. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R391; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R391; diff_s43.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R392. TIC 76020303

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 76020303: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 35 fitted event windows in sectors [43, 45, 72] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 17 same-kind measured centers give P = 3.7052792516 +/- 0.0000054871 d and reference epoch BTJD 2475.6703770. Epoch kind: Even historical half-period cycles; indistinguishable-depth primary/secondary identity ambiguous. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 43 deficit-target distance 0.057 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 3428849314988561024: separation 43.711 arcsec, required intrinsic fractional eclipse 0.9305, deficit distance 2.120 pixel; Gaia 3428849040110655872: separation 50.558 arcsec, required intrinsic fractional eclipse 0.4995, deficit distance 2.414 pixel; Gaia 3428849074470410880: separation 52.996 arcsec, required intrinsic fractional eclipse 0.3558, deficit distance 2.448 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 63.661/57.191/40.992 ppt from 12 stacked events; current selected-kind QLP median 68.620 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 3428849104531235072, P = 3.7053469413 +/- 0.0001501581 d, e = 0.070651, omega = 327.6200 deg. Closest edge-on conjunction is offset

-0.106060 d (-0.028623 phase), with full SB1 covariance linear propagated uncertainty 0.092846 d at measured epoch. Projection near Gaia reference gives offset -0.067950 d; photometric period projection allowance 0.003089 d excludes covariance and model systematics. Approximate central duration 6.80 h versus fitted median 4.33 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Near-equal alternating eclipses recur every 1.85264 d. Same-parity even events give the reported orbital period 3.705279 d, consistent with Gaia 3.705347 d. Photometry alone retains the half-period interpretation because parity depths are indistinguishable; Gaia supplies the orbital interpretation, not an unannounced alias replacement. Both conjunctions are plausible for the paired eclipses. The target is 0.057 pixel from deficit versus capable neighbors beyond 2.12 pixels.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

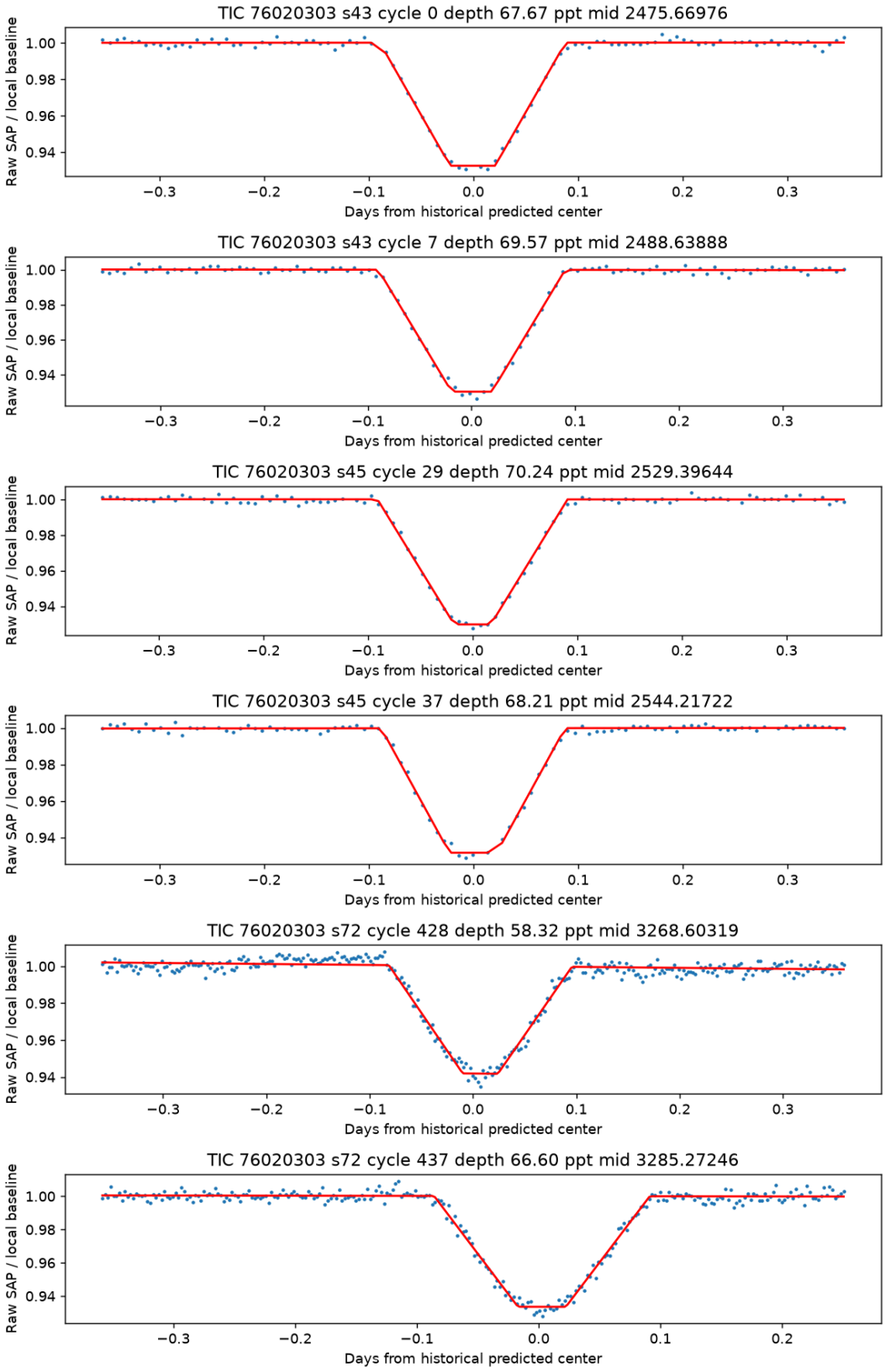
Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
43	0	2475.6697589	0.0034726	67.665	used
43	1	2477.5236281	0.0034726	69.322	other eclipse parity; excluded from same-kind ephemeris
43	2	2479.3754730	0.0034726	68.964	used
43	3	2481.2282474	0.0034726	70.635	other eclipse parity; excluded from same-kind ephemeris
43	4	2483.0810813	0.0034726	67.395	used
43	5	2484.9336102	0.0034726	68.310	other eclipse parity; excluded from same-kind ephemeris
43	7	2488.6388780	0.0034726	69.566	other eclipse parity; excluded from same-kind ephemeris
43	8	2490.4903457	0.0034726	68.699	used
43	9	2492.3435930	0.0034726	68.698	other eclipse parity; excluded from same-kind ephemeris
43	10	2494.1962290	0.0034726	67.655	used
43	11	2496.0490936	0.0034726	72.071	other eclipse parity; excluded from same-kind ephemeris
43	12	2497.9021391	0.0034726	68.620	used
45	28	2527.5436132	0.0034724	69.909	used
45	29	2529.3964439	0.0034724	70.240	other eclipse parity; excluded from same-kind ephemeris
45	30	2531.2496872	0.0034724	69.897	used
45	31	2533.1019885	0.0034724	69.582	other eclipse parity; excluded from same-kind ephemeris
45	32	2534.9542965	0.0034724	69.811	used
45	33	2536.8078710	0.0034724	68.673	other eclipse parity; excluded from same-kind ephemeris
45	35	2540.5125340	0.0034724	69.639	other eclipse parity; excluded from same-kind ephemeris
45	36	2542.3653463	0.0034724	69.139	used
45	37	2544.2172234	0.0034724	68.207	other eclipse parity; excluded from same-kind ephemeris
45	38	2546.0698243	0.0034724	68.563	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
45	39	2547.9224540	0.0034724	70.204	other eclipse parity; excluded from same-kind ephemeris
45	40	2549.7802128	0.0034724	58.158	used
72	425	3263.0412498	0.0011575	68.936	other eclipse parity; excluded from same-kind ephemeris
72	426	3264.8940083	0.0011575	69.238	used
72	427	3266.7469286	0.0011575	68.077	other eclipse parity; excluded from same-kind ephemeris
72	428	3268.6031886	0.0011575	58.321	used
72	429	3270.4522278	0.0011575	70.801	other eclipse parity; excluded from same-kind ephemeris
72	430	3272.3048371	0.0011575	69.120	used
72	433	3277.8616510	0.0011575	68.759	other eclipse parity; excluded from same-kind ephemeris
72	434	3279.7153501	0.0011575	68.571	used
72	435	3281.5672412	0.0011575	67.446	other eclipse parity; excluded from same-kind ephemeris
72	436	3283.4202577	0.0011575	67.873	used
72	437	3285.2724572	0.0011575	66.604	other eclipse parity; excluded from same-kind ephemeris

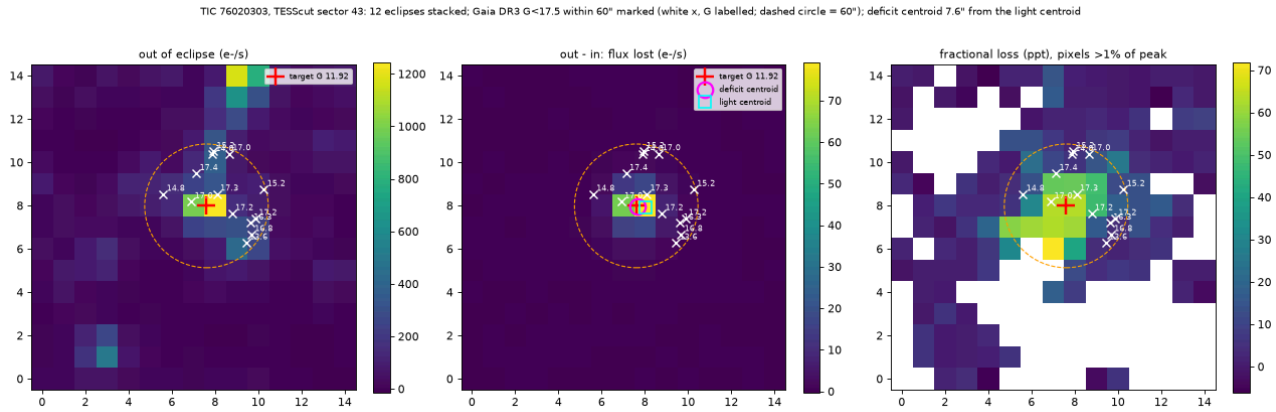
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s43.json, historical/diff_s43.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R392. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R392; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R392; diff_s43.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R393. TIC 251798466

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 251798466: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 4 fitted event windows in sectors [54, 81] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 3 same-kind measured centers give P = 9.0605146129 +/- 0.0008183987 d and reference epoch BTJD 3508.7680130. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 81 deficit-target distance 0.063 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 4234971348825403520: separation 57.192 arcsec, required intrinsic fractional eclipse 0.4300, deficit distance 2.789 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 175.432/175.338/146.016 ppt from 2 stacked events; current selected-kind QLP median 197.169 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 4234971451904627968, P = 9.0594815327 +/- 0.0012896050 d, e = 0.080359, omega = 152.2532 deg. Closest edge-on conjunction is offset 0.360772 d (0.039823 phase), with full SB1 covariance linear propagated uncertainty 0.487312 d at measured epoch. Projection near Gaia reference gives offset 0.005392 d; photometric period

projection allowance 0.281529 d excludes covariance and model systematics. Approximate central duration 7.29 h versus fitted median 5.13 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Three complete deep eclipses in sector 81, consecutive cycles 81-83, measure the period. Earlier sector 54 cycle 0 is a visibly partial raw repeat and supports recurrence but contributes no precision timing. Only the short complete-event baseline sets the stated formal error. Target deficit 0.063 pixel versus capable neighbor 2.789 pixels.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

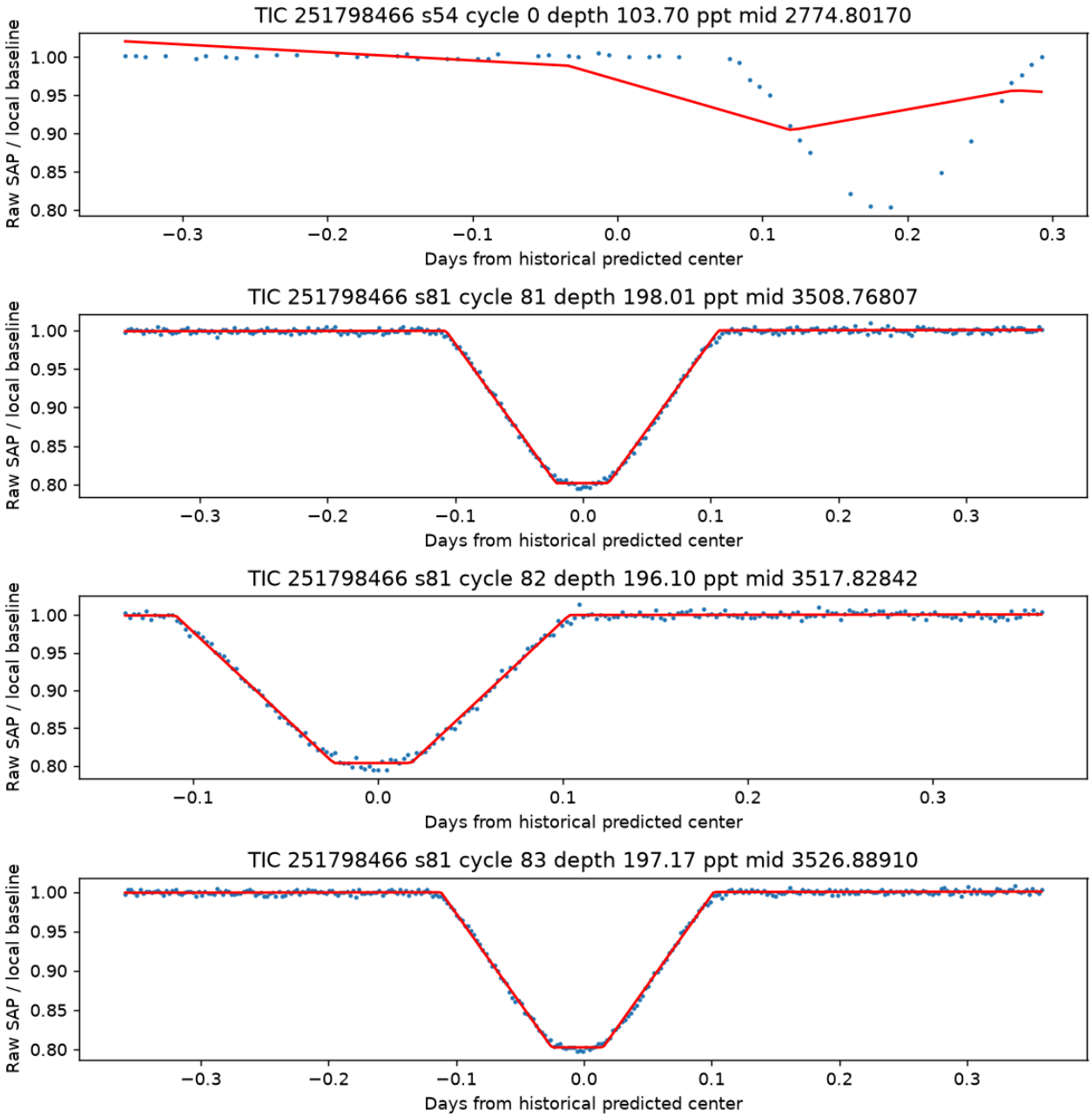
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
54	0	2774.8017000	0.0069444	103.705	Partially covered eclipse lacks ingress baseline; retained raw morphology but excluded timing.
81	81	3508.7680670	0.0011574	198.014	used
81	82	3517.8284196	0.0011574	196.105	used
81	83	3526.8890962	0.0011574	197.169	used

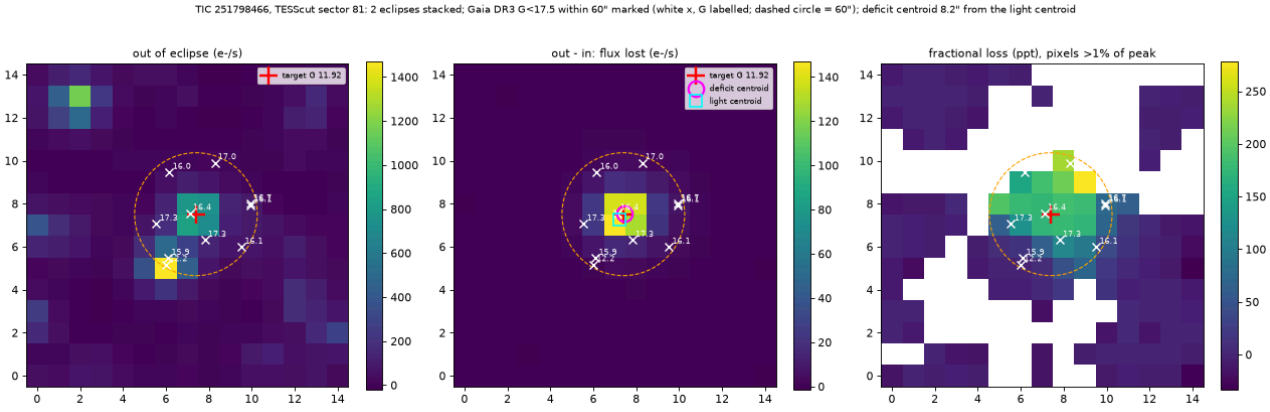
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s81.json, historical/diff_s81.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R393. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R393; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R393; diff_s81.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R394. TIC 363264601

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or calibrated multi-source PRF localization

Archived source: sanitized report text.

TIC 363264601: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 9 fitted event windows in sectors [100, 103] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 9 same-kind measured centers give P = 4.6066819596 +/- 0.0002625463 d and reference epoch BTJD 4076.6896945. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: unresolved after completed check.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 66 deficit-target distance 0.093 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 5808791951859897216: separation 5.595 arcsec, required intrinsic fractional eclipse 0.0185, deficit distance 0.350 pixel; Gaia 5808792054939115264: separation 21.193 arcsec, required intrinsic fractional eclipse 0.4070, deficit distance 0.918 pixel; Gaia 5808791638319283328: separation 36.431 arcsec, required intrinsic fractional eclipse 0.8786, deficit distance 1.817 pixel; Gaia 5808791986219638784: separation 36.982 arcsec, required intrinsic fractional eclipse 0.6397, deficit distance 1.699 pixel; Gaia 5808792054939117184: separation 41.095 arcsec, required intrinsic fractional eclipse 0.0749, deficit distance 1.871 pixel; Gaia 5808792054939117568: separation 42.632 arcsec, required intrinsic fractional eclipse 0.6112, deficit distance 1.937 pixel; Gaia 5808791986219637120: separation 43.123 arcsec, required intrinsic fractional eclipse 0.0652, deficit distance 2.033 pixel; Gaia 5808791986219640960: separation 45.848 arcsec, required intrinsic fractional eclipse 0.1185, deficit distance 2.092 pixel; Gaia 5808792020579377024: separation 48.371 arcsec, required intrinsic fractional eclipse 0.0762, deficit distance 2.293 pixel; Gaia 5808791986219634432: separation 51.140 arcsec, required intrinsic fractional eclipse 0.5430, deficit distance 2.455 pixel Required

intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.

4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 3.366/3.092/3.043 ppt from 6 stacked events; current selected-kind QLP median 4.209 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 5808791947555627648, P = 4.6068777227 +/- 0.0001919322 d, e = 0.090713, omega = 286.1038 deg. Closest edge-on conjunction is offset -0.248803 d (-0.054007 phase), with full SB1 covariance linear propagated uncertainty 0.160371 d at measured epoch. Projection near Gaia reference gives offset -0.091997 d; photometric period projection allowance 0.210300 d excludes covariance and model systematics. Approximate central duration 5.28 h versus fitted median 2.12 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Repeated localized 4-5 ppt depressions have flat surrounding baseline in sectors 100/103 and support genuine eclipses. Attribution remains unresolved after a completed check: a 5.595 arcsec neighbor can supply the signal with 1.85 percent intrinsic loss. Target/neighbor deficit distances 0.093/0.350 pixel favor the target but do not reliably distinguish these close sources without a calibrated localization. This is PROBABLE because the eclipses repeat and attribution is unresolved, not because their morphology is uncertain.

Next check: Resolved eclipse photometry or calibrated multi-source PRF localization.

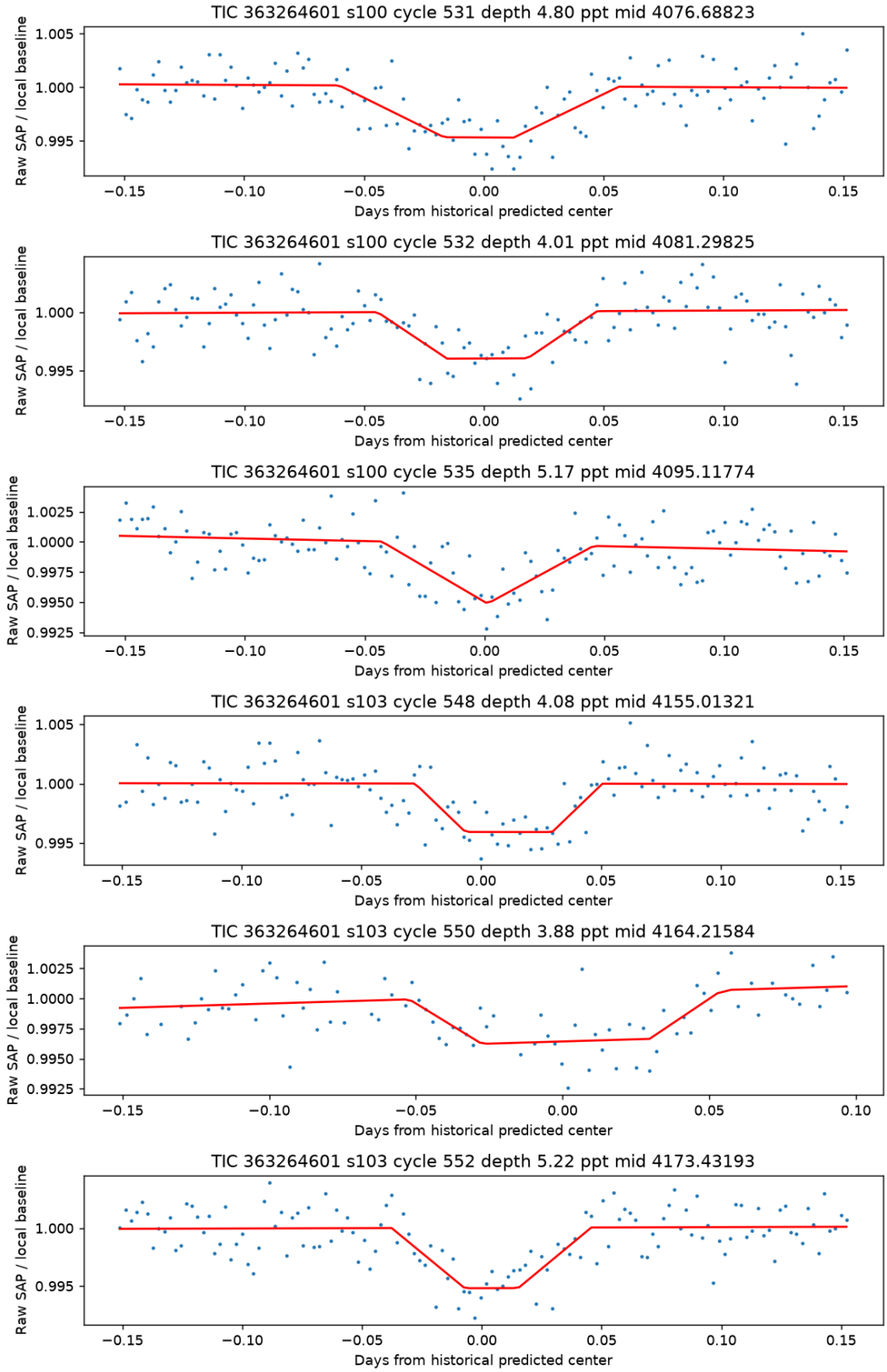
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
100	531	4076.6882345	0.0025348	4.796	used
100	532	4081.2982518	0.0027791	4.013	used
100	534	4090.5078394	0.0026478	4.451	used
100	535	4095.1177446	0.0029427	5.171	used
103	548	4155.0132103	0.0020838	4.079	used
103	549	4159.6001259	0.0021269	4.209	used
103	550	4164.2158434	0.0030199	3.876	used
103	551	4168.8212895	0.0029483	3.544	used
103	552	4173.4319275	0.0024537	5.218	used

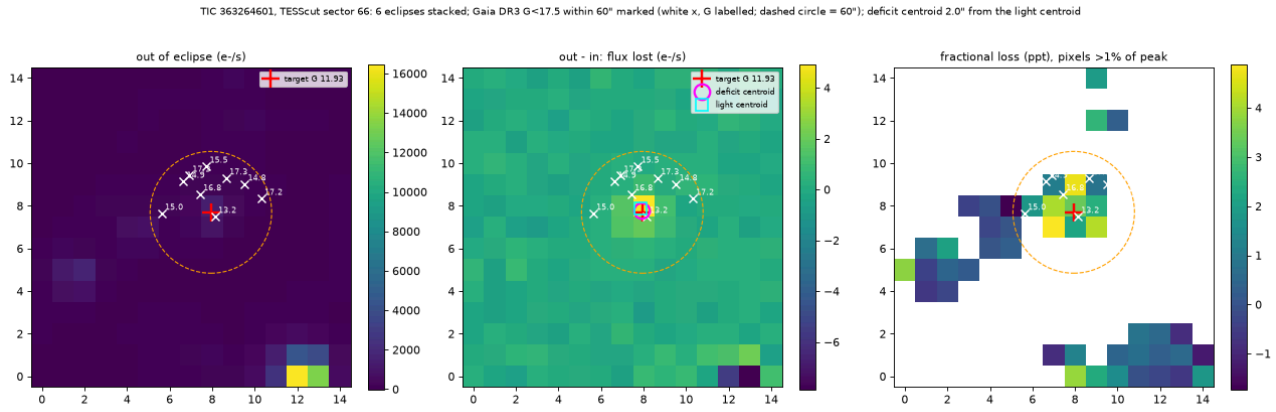
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s66.json, historical/diff_s66.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R394. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R394; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R394; diff_s66.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R395. TIC 399089869

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 399089869: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 5 fitted event windows in sectors [23, 46, 91] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 5 same-kind measured centers give P = 8.8537610758 +/- 0.0000176369 d and reference epoch BTJD 1937.8424876. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 46 deficit-target distance 0.221 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: No capable listed source within the saved 60 arcsec cone. Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 106.328/108.693/103.991 ppt from 2 stacked events; current selected-kind QLP median 119.002 ppt. These are reductions of the same TESS photons, not independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.
5. **Gaia orbital geometry: pass with limitations.** DR3 3902848326747296000, P = 8.8536629216 +/- 0.0012623065 d, e = 0.154456, omega = 84.8249 deg. Closest edge-on conjunction is offset 0.064106 d (0.007241 phase), with full SB1 covariance linear propagated uncertainty 0.229109 d at measured epoch. Projection near Gaia reference gives offset 0.046929 d; photometric period projection allowance 0.003086 d excludes covariance and model systematics.

Approximate central duration 5.54 h versus fitted median 3.68 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Deep repeated isolated eclipses across sectors 23/46/91; sector 46 consecutive cycles 70/71 test recurrence. No capable neighbor in the saved 60 arcsec cone. Aperture-dependent dilution limits intrinsic depth.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

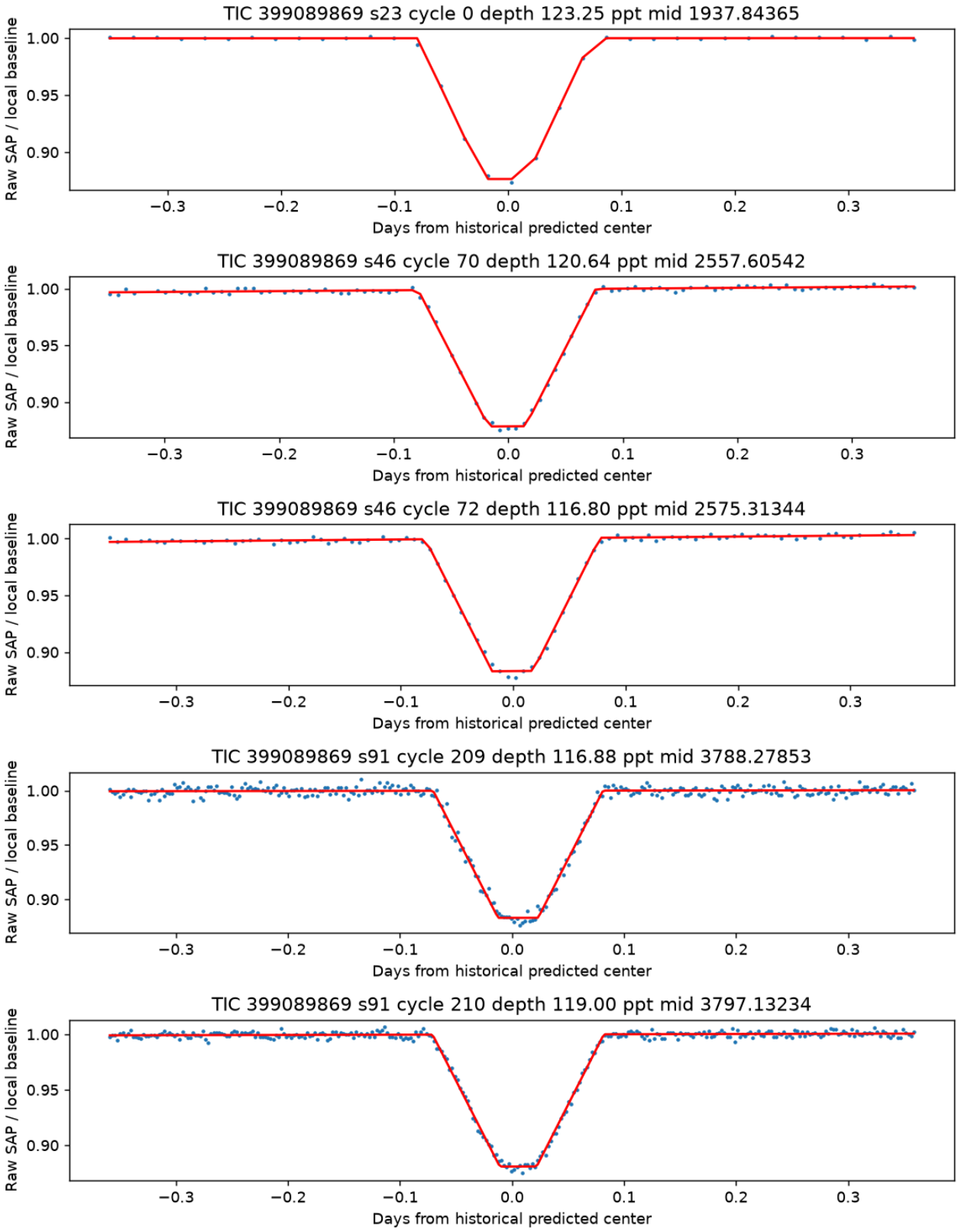
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
23	0	1937.8436528	0.0104166	123.249	used
46	70	2557.6054165	0.0034726	120.642	used
46	72	2575.3134408	0.0034726	116.801	used
91	209	3788.2785328	0.0011573	116.881	used
91	210	3797.1323400	0.0011573	119.002	used

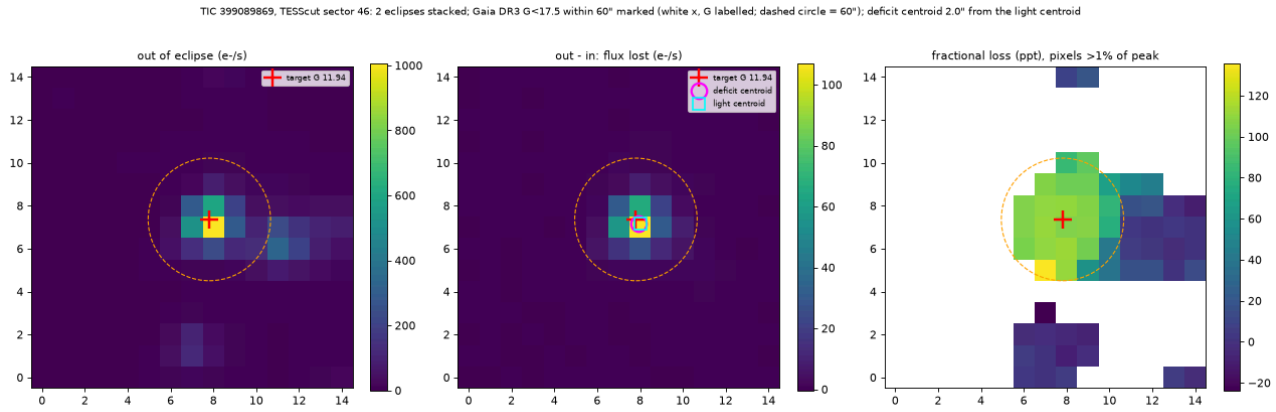
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s46.json, historical/diff_s46.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R395. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R395; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R395; diff_s46.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R396. TIC 329559935

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry or spectroscopy for precision characterization

Archived source: sanitized report text.

TIC 329559935: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. Current section 2.7 five-check standard applied; historical screening is not itself admission evidence.

1. **Raw flux: pass.** 12 fitted event windows in sectors [80] were examined in raw-events.png and numeric raw-windows.csv. Repeated localized ingress/egress depressions and surrounding baseline were visually inspected; trapezoid fits alone are not the eclipse interpretation. QLP raw SAP_FLUX with quality zero is used for these measurements. Coverage requirements are in review_measure.py; uncovered/flagged windows are not clean nulls. At most three spread available sectors were initially selected; no mandatory all-event or all-sector analysis is implied.
2. **Measured timing: pass.** 5 same-kind measured centers give P = 2.6037352194 +/- 0.0001586838 d and reference epoch BTJD 3480.4072501. Epoch kind: Odd historical half-period cycles; deeper eclipses. Predicted historical centers only seed windows and cycle counts, not measured timings. Within-sector recurrence/parity tests are retained in alias-audit.json; this is not an exhaustive blind alias scan. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Partial/singular fits remain documented and excluded; the epoch may denote a selected repeating eclipse rather than an independently established deeper primary.
3. **Source attribution: pass at TESS resolution.** Original cutout WCS/Gaia identity audited in pixel-audit.json. Saved sector 80 deficit-target distance 0.081 pixel; full deficit image examined against every flux-capable listed source. Capable neighbors: Gaia 4253888274641830912: separation 26.962 arcsec, required intrinsic fractional eclipse 0.1384, deficit distance 1.362 pixel; Gaia 4253888996196371968: separation 47.822 arcsec, required intrinsic fractional eclipse 0.9953, deficit distance 2.243 pixel; Gaia 4253888240282092416: separation 51.643 arcsec, required intrinsic fractional eclipse 0.7951, deficit distance 2.503 pixel; Gaia 4253888377721051648: separation 57.036 arcsec, required intrinsic fractional eclipse 0.5436, deficit distance 2.733 pixel; Gaia 4253888201559031424: separation 57.314 arcsec, required intrinsic fractional eclipse 0.5383, deficit distance 2.821 pixel; Gaia 4253888240282111232: separation 59.694 arcsec, required intrinsic fractional eclipse 0.4583, deficit distance 2.845 pixel Required intrinsic fractional eclipse and observed blended pixel fractional loss have different dilution and are not directly compared to exclude neighbors. The approximate G-band budget and saved 60 arcsec cone limit this check.
4. **Alternative reduction: pass.** Separately extracted background-subtracted target-centered TESScut 1x1/3x3/5x5 apertures show depths 54.387/43.686/32.724 ppt from 11 stacked events; current selected-kind QLP median 87.969 ppt. These are reductions of the same TESS photons, not

independent observing occasions. Dilution, aperture differences and mixed primary/secondary stacks can change depth; no claim of equal intrinsic depths is made.

5. **Gaia orbital geometry: pass with limitations.** DR3 4253888270275983360, P = 2.6038620064 +/- 0.0002119814 d, e = 0.081014, omega = 116.3438 deg. Closest edge-on conjunction is offset -0.048017 d (-0.018441 phase), with full SB1 covariance linear propagated uncertainty 0.202816 d at measured epoch. Projection near Gaia reference gives offset 0.102480 d; photometric period projection allowance 0.188358 d excludes covariance and model systematics. Approximate central duration 5.84 h versus fitted median 3.70 h is an illustration using TIC radius/mass and a minimum-mass main-sequence companion, not a fitted geometry. No arbitrary sigma veto; gaia-geometry.json preserves all assumptions and both conjunctions.

Deciding interpretation and limitations: Only sector 80 supplied usable selected QLP data. Unequal alternating depths (odd median about 87.97 ppt; even about 41.28 ppt) establish the doubled orbital period, with five measured odd deeper eclipses spanning cycles -1 to 15. Half-period recurrence mixes two unequal eclipse types. Measured 2.603735 d agrees with Gaia 2.603862 d within the descriptive timing uncertainty; queue 2.60568 d is a historical search value, not adopted. Target deficit 0.081 pixel versus capable neighbors beyond 1.362 pixels.

Next check: Optional resolved photometry or spectroscopy for precision characterization.

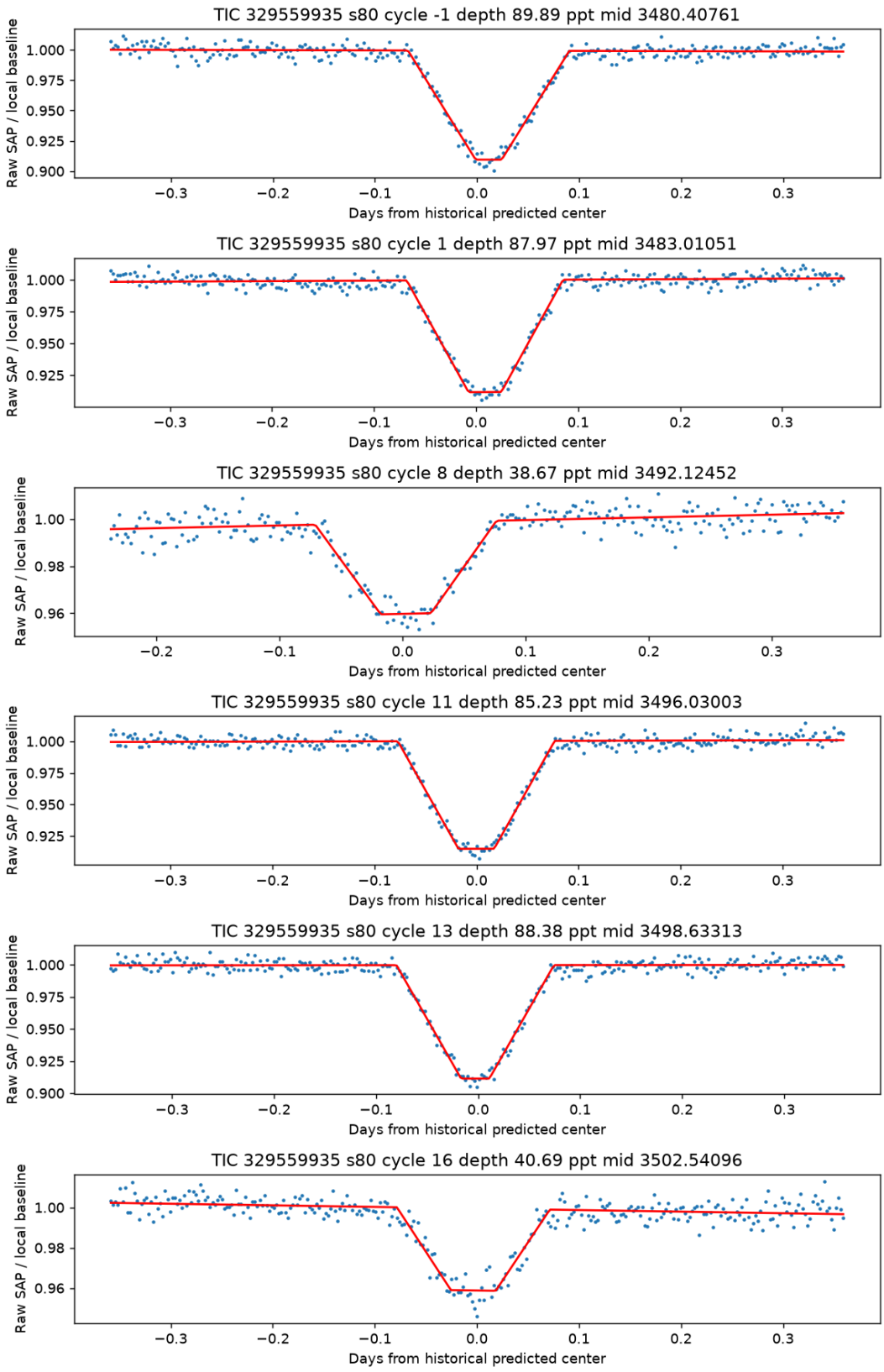
Catalogue novelty: see central ../../catalogues/coverage-matrix.csv; this report does not establish exhaustive literature priority. Current observed repeats remove a single-cycle availability question; the short-baseline cases explicitly state their evidence.

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing input error d	Depth ppt	Timing status
80	-1	3480.4076100	0.0011574	89.887	used
80	0	3481.7081686	0.0011574	40.781	other eclipse parity; excluded from same-kind ephemeris
80	1	3483.0105084	0.0011574	87.969	used
80	2	3484.3132274	0.0011574	44.308	other eclipse parity; excluded from same-kind ephemeris
80	8	3492.1245218	0.0011574	38.674	other eclipse parity; excluded from same-kind ephemeris
80	10	3494.7283572	0.0011574	49.196	other eclipse parity; excluded from same-kind ephemeris
80	11	3496.0300255	0.0011574	85.232	used
80	12	3497.3318479	0.0011574	41.278	other eclipse parity; excluded from same-kind ephemeris
80	13	3498.6331281	0.0011574	88.381	used
80	14	3499.9358438	0.0011574	49.396	other eclipse parity; excluded from same-kind ephemeris
80	15	3501.2371534	0.0011574	87.212	used
80	16	3502.5409560	0.0011574	40.690	other eclipse parity; excluded from same-kind ephemeris

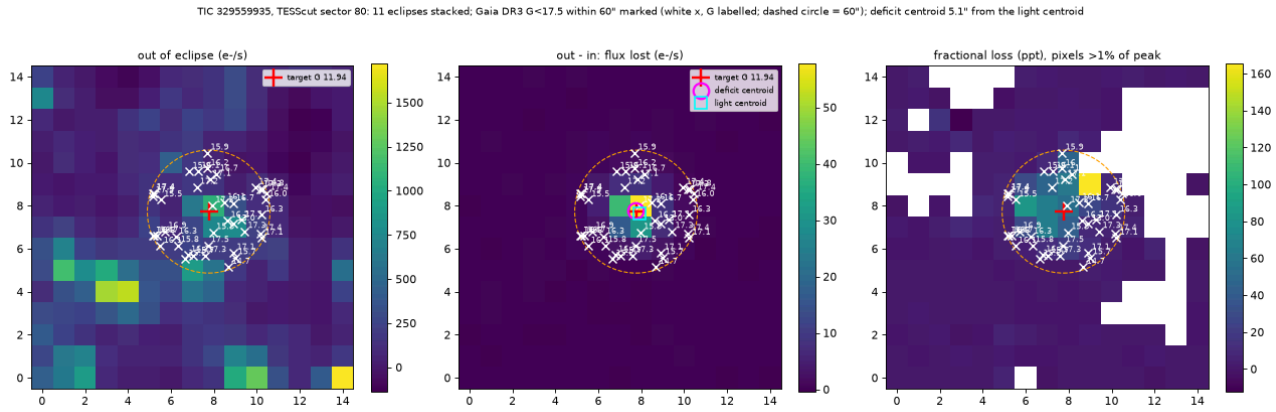
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, historical/diff_s80.json, historical/diff_s80.png, historical/gaia.json; batch ../pixel-audit.json, ../alias-audit.json, ../gaia-geometry.json, ../helper-audit.md and ../SHA256SUMS.json. Raw cutout: [local path omitted] Historical extraction definitions: [local path omitted] and c3_sapdepth.py. Reused source hashes in source-hashes.json; no historical evidence overwritten.

Stable review key: R396. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R396; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R396; diff_s80.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R397. TIC 119193784

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Clean intermediate-cycle eclipse photometry to test orbital aliases

Archived source: sanitized report text.

TIC 119193784: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[50, 91]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: conditional orbital alias.** 4 same-kind midtimes give P=10.4453149260 +/-0.0000244041d, reference eclipse epoch BTJD2669.7487457, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S50 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation4.008arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.570. Capable neighbours: Gaia 3645530720010524672: 41.07 arcsec, G12.275, required fractional eclipse0.268
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths112.056/101.074/78.404ppt, with 2 stacked events. QLP selected-kind fitted median122.868ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3364553078873000128O, P=10.4445071004+/-0.0023469310d, e=0.204078, omega=254.9518deg. Closest edge-on conjunction offset0.062988d (0.006031phase), full-SB1-covariance linear propagated Gaia uncertainty0.556629d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.113118d (photometric period projection allowance0.005320d). The approximate central

duration6.07h vs observed median4.30h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four deep V-shaped/truncated-bottom events across S50/91, all odd historical10.445d cycles. Intermediate cycles0 and106 have zero quality-zero points within0.2d (42 and173 flagged points respectively), so neither is a clean null. Available QLP inventory lists only S50/91. Two20.8906d-separated pairs establish real repetition, but10.4453149d period is conditional on the Gaia-informed cycle count;20.8906d and integer-count aliases are not independently rejected. This missing orbital-alias check retains PROBABLE despite completed bounded inspection. Target deficit0.191pixel versus capable neighbor2.279pixels; dilution limits intrinsic depth.

Next check: Obtain clean intermediate-cycle photometry to distinguish10.445d from20.8906d orbital aliases.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

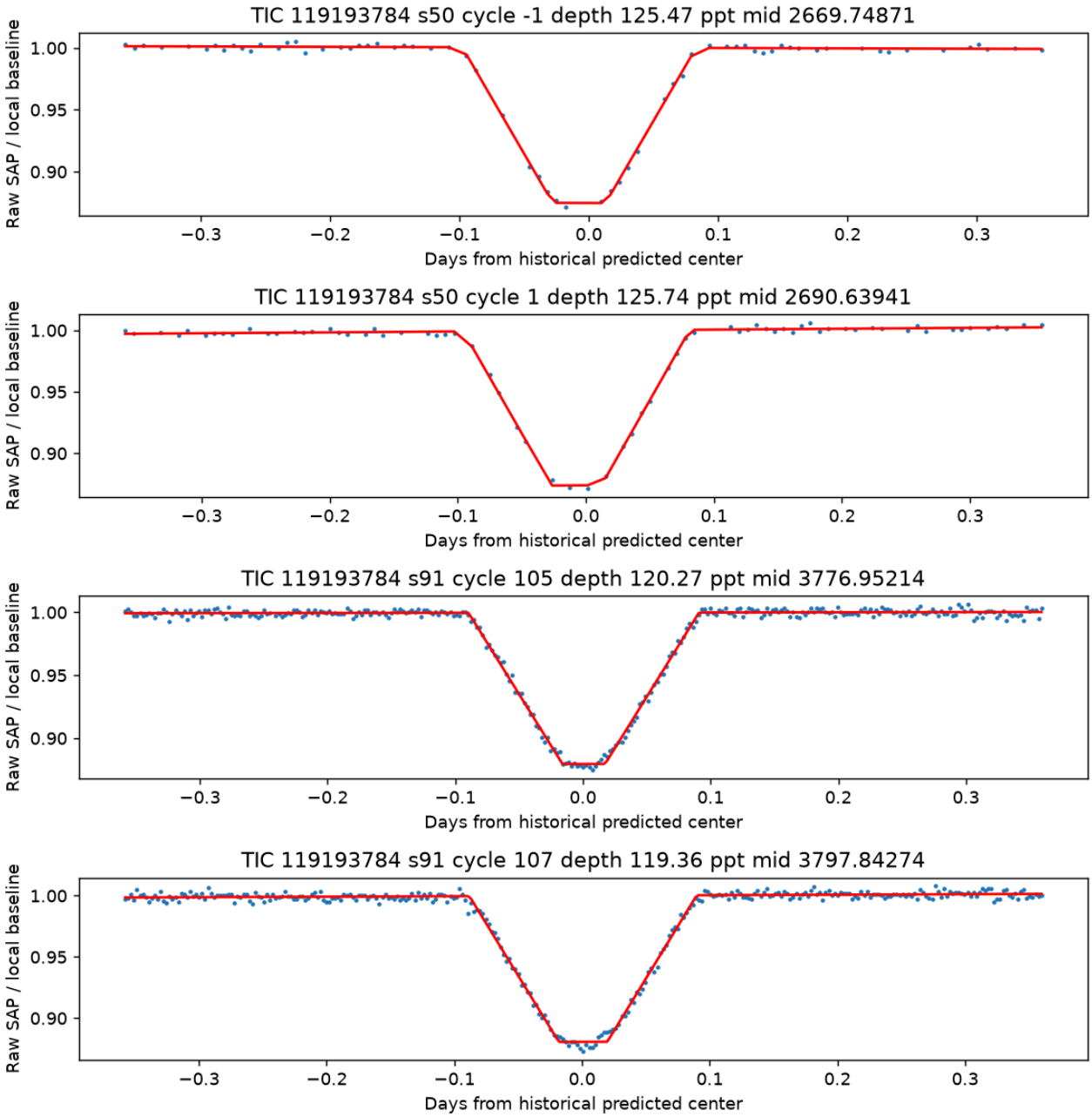
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
50	-1	2669.7487060	0.003472	125.471	used in conditional ephemeris
50	1	2690.6394133	0.003472	125.744	used in conditional ephemeris
91	105	3776.9521434	0.001157	120.265	used in conditional ephemeris
91	107	3797.8427424	0.001157	119.358	used in conditional ephemeris

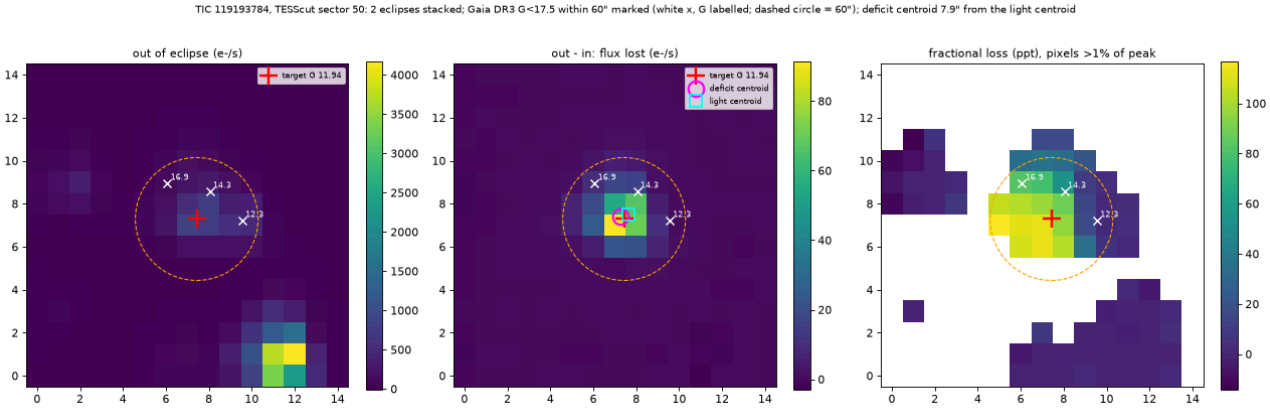
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s50.json, historical/diff_s50.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R397. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R397; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R397; diff_s50.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R398. TIC 300148401

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 300148401: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[6, 33, 98]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=8.7876379642 +/-0.0000126985d, reference eclipse epoch BTJD1469.3015561, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S32 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation1.674arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.504. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths45.737/46.373/45.311ppt, with 3 stacked events. QLP selected-kind fitted median52.895ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR32884583087778756224, P=8.7880603922+/-0.0014557943d, e=0.142219, omega=170.3694deg. Closest edge-on conjunction offset0.026216d (0.002983phase), full-SB1-covariance linear propagated Gaia uncertainty0.174132d at the measured epoch. Near-Gaia-reference projected eclipse offset0.078174d (photometric period projection allowance0.001562d). The approximate central duration6.82h vs observed median5.07h

is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated narrow eclipse cores on curved variable baseline; early S6 is sparse at30min cadence, while S33/98 resolve recurrence. Linear baseline and depth precision are limited by curvature; no capable listed neighbor.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

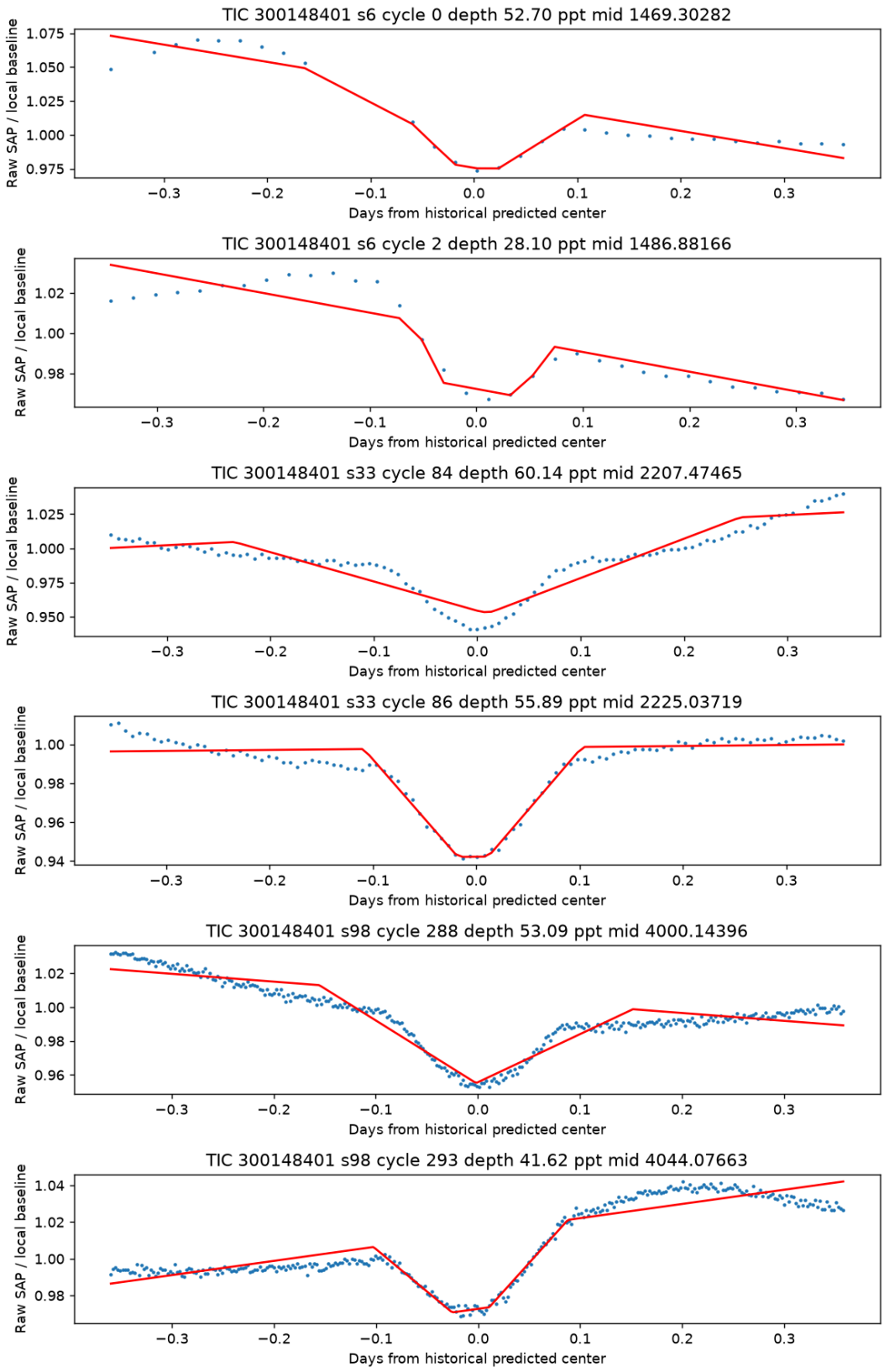
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
6	0	1469.3028231	0.010417	52.704	used in ephemeris
6	2	1486.8816622	0.021990	28.101	used in ephemeris
33	84	2207.4746549	0.005693	60.142	used in ephemeris
33	85	2216.2472697	0.003472	56.352	used in ephemeris
33	86	2225.0371901	0.003472	55.892	used in ephemeris
98	288	4000.1439610	0.003506	53.086	used in ephemeris
98	291	4026.5043511	0.001561	41.323	used in ephemeris
98	293	4044.0766302	0.003304	41.625	used in ephemeris

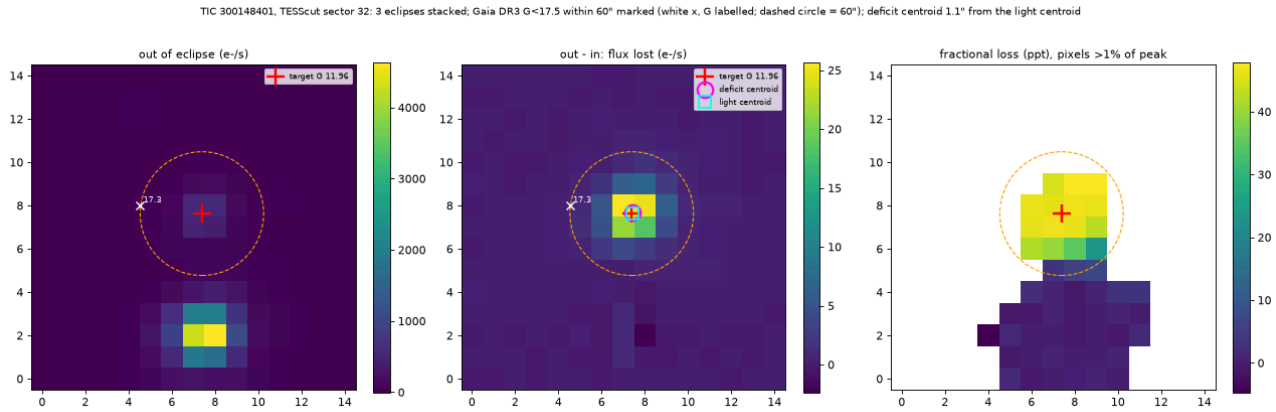
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s32.json, historical/diff_s32.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R398. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R398; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R398; diff_s32.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R399. TIC 167125054

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 167125054: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[1, 36, 98]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 8 same-kind midtimes give P=9.1223775152 +/-0.0000113717d, reference eclipse epoch BTJD1326.8709038, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S1 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation5.019arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.417. Capable neighbours: Gaia 5283286313593528192: 38.02 arcsec, G12.828, required fractional eclipse0.203
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths62.675/55.530/44.226ppt, with 3 stacked events. QLP selected-kind fitted median60.330ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3528328610313326220 8, P=9.1224018521+/-0.0010396464d, e=0.006947, omega=77.3211deg. Closest edge-on conjunction offset-0.079784d (-0.008746phase), full-SB1-covariance linear propagated Gaia uncertainty0.167676d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.077277d (photometric period projection allowance0.001171d). The approximate central duration10.42h vs observed median9.16h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated broad flat-bottom eclipses in S1/36/99; depth near60ppt. Target deficit0.239pixel versus capable neighbor2.036pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

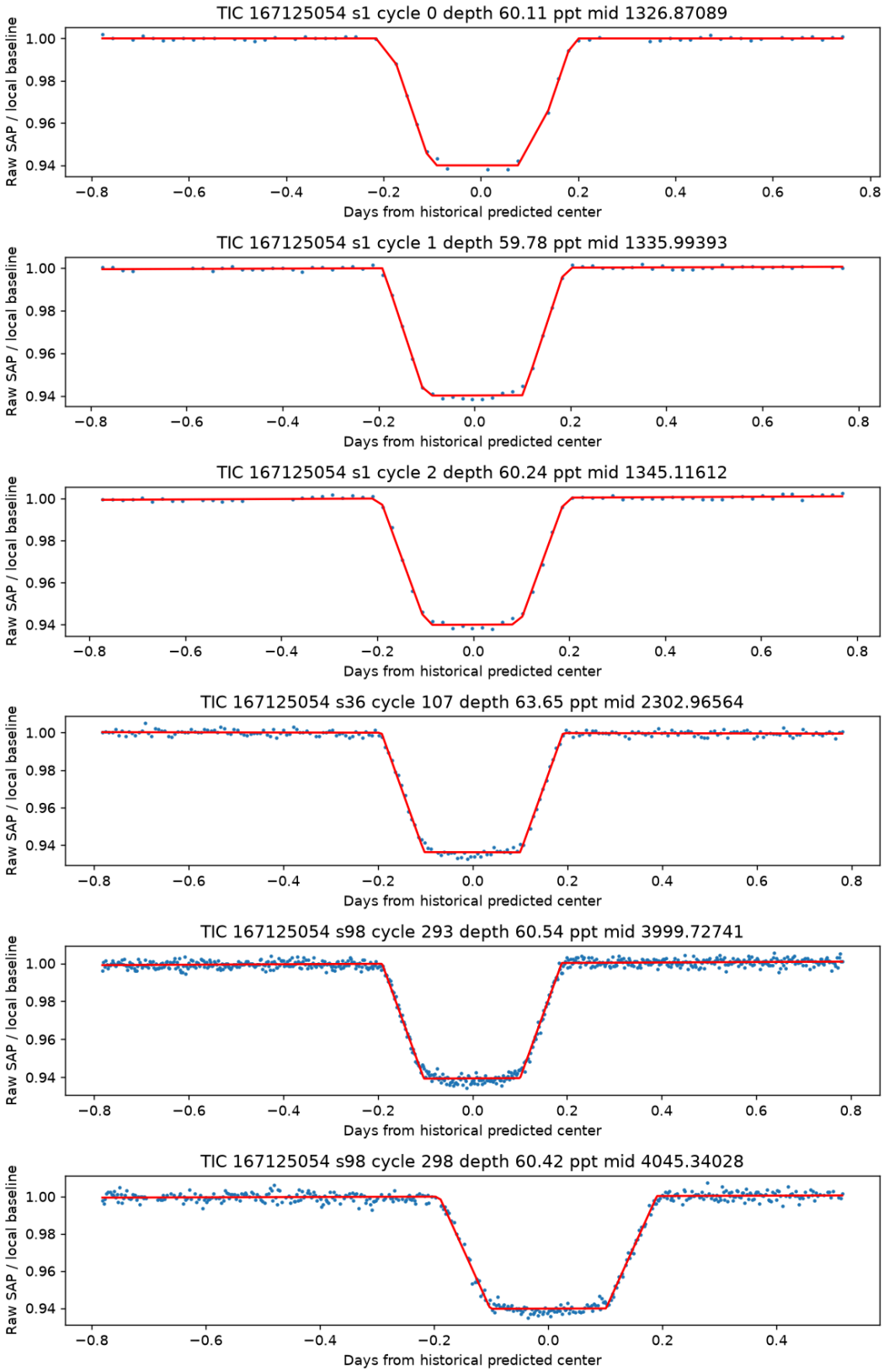
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
1	0	1326.8708926	0.010417	60.114	used in ephemeris
1	1	1335.9939294	0.010417	59.777	used in ephemeris
1	2	1345.1161249	0.010417	60.243	used in ephemeris
36	105	2284.7201107	0.003472	63.700	used in ephemeris
36	107	2302.9656415	0.003472	63.649	used in ephemeris
98	293	3999.7274128	0.001157	60.535	used in ephemeris
98	296	4027.0938736	0.001157	59.743	used in ephemeris
98	298	4045.3402771	0.001157	60.416	used in ephemeris

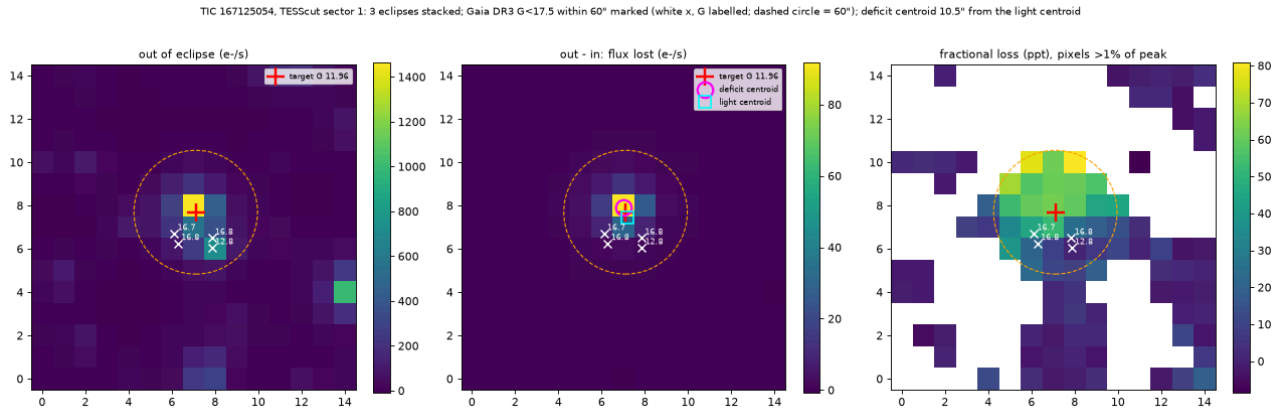
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s01.json, historical/diff_s01.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R399. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R399; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R399; diff_s01.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R400. TIC 60514418

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 60514418: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 5 distinct eclipse events in selected sectors[43, 44, 71]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=11.6952161043 +/-0.0000339692d, reference eclipse epoch BTJD2475.7039155, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S44 deficit peak[8, 8] vs target pixel[8, 8]; deficit-target separation0.747arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.202. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths35.497/34.959/34.188ppt, with 2 stacked events. QLP selected-kind fitted median38.420ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3153384449055145856, P=11.6940432568+/-0.0012138569d, e=0.063286, omega=211.5158deg. Closest edge-on conjunction offset0.160480d (0.013723phase), full-SB1-covariance linear propagated Gaia uncertainty0.190129d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.048287d (photometric period projection allowance0.006047d). The approximate central duration8.77h vs observed median3.72h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated V-shaped eclipses in S43/44/71; consecutive S44 cycles test recurrence. No capable listed neighbor; observed duration3.72h is shorter than illustrative central duration8.77h, compatible with noncentral eclipse.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

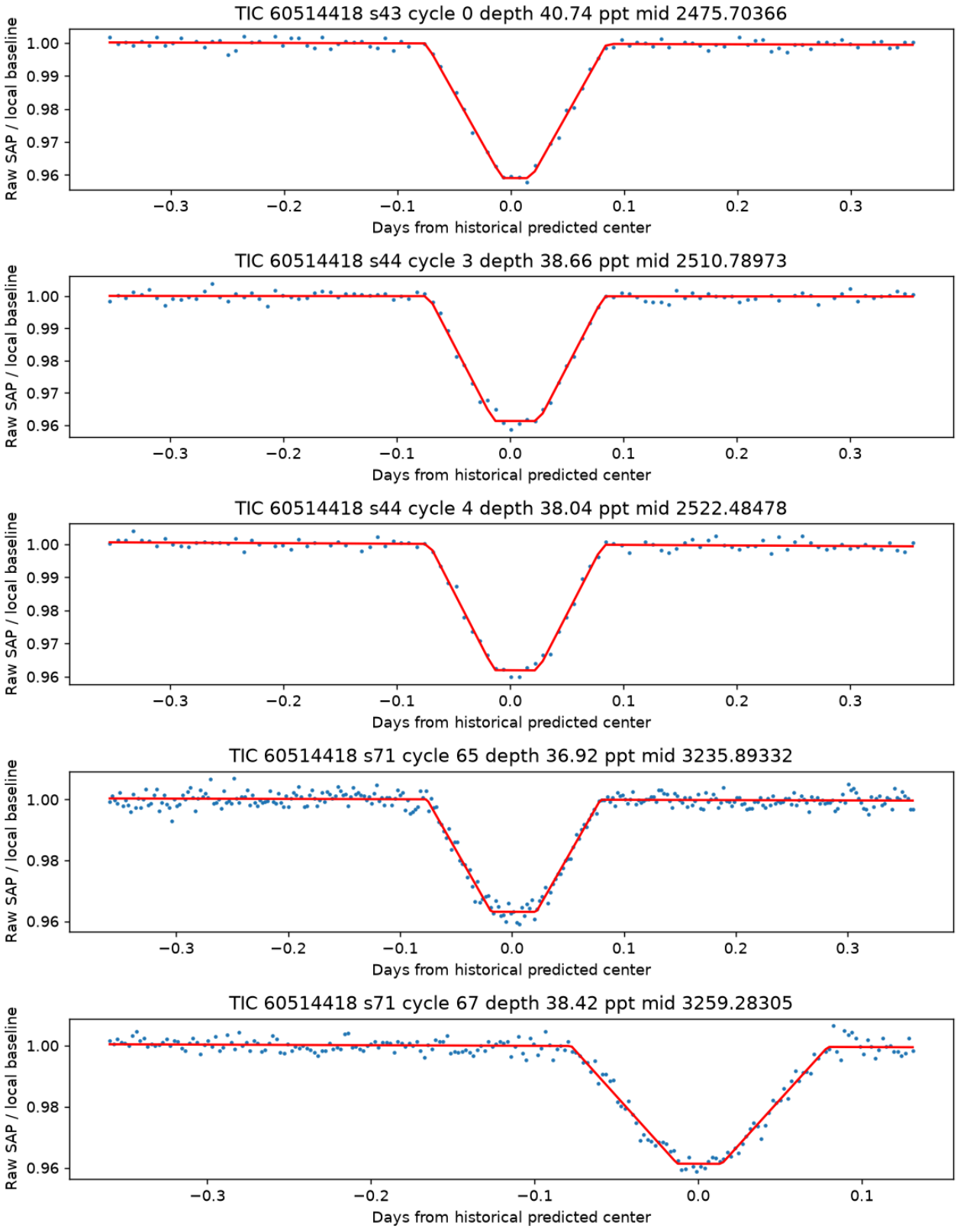
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
43	0	2475.7036610	0.003473	40.742	used in ephemeris
44	3	2510.7897311	0.003472	38.658	used in ephemeris
44	4	2522.4847784	0.003472	38.037	used in ephemeris
71	65	3235.8933198	0.001157	36.920	used in ephemeris
71	67	3259.2830468	0.001157	38.420	used in ephemeris

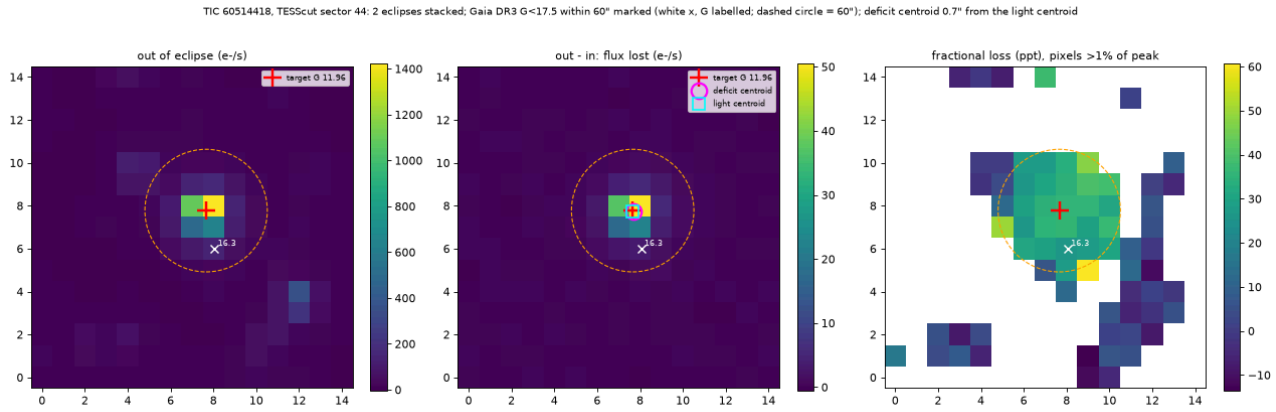
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s44.json, historical/diff_s44.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R400. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R400; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R400; diff_s44.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R401. TIC 257047728

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 257047728: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 2 distinct eclipse events in selected sectors[48]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 2 same-kind midtimes give P=15.2070674831 +/-0.0049104284d, reference eclipse epoch BTJD2612.2132742, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S40 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation0.295arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.154. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths21.672/22.494/22.800ppt, with 2 stacked events. QLP selected-kind fitted median24.411ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3170097228519837644 8, P=15.2066086810+/-0.0035990214d, e=0.279558, omega=176.3644deg. Closest edge-on conjunction offset0.002539d (0.000167phase), full-SB1-covariance linear propagated Gaia uncertainty0.530874d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.064446d (photometric period projection allowance0.716923d). The approximate central duration7.47h vs observed median3.46h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Two fully covered S48 eclipses are consecutive at15.20707d, directly testing recurrence rather than long-baseline integer counting. Historical S40 two-event difference stack supplies separate raw-pixel morphology; no capable listed neighbor. Timing precision is single-sector and not calibrated total uncertainty.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

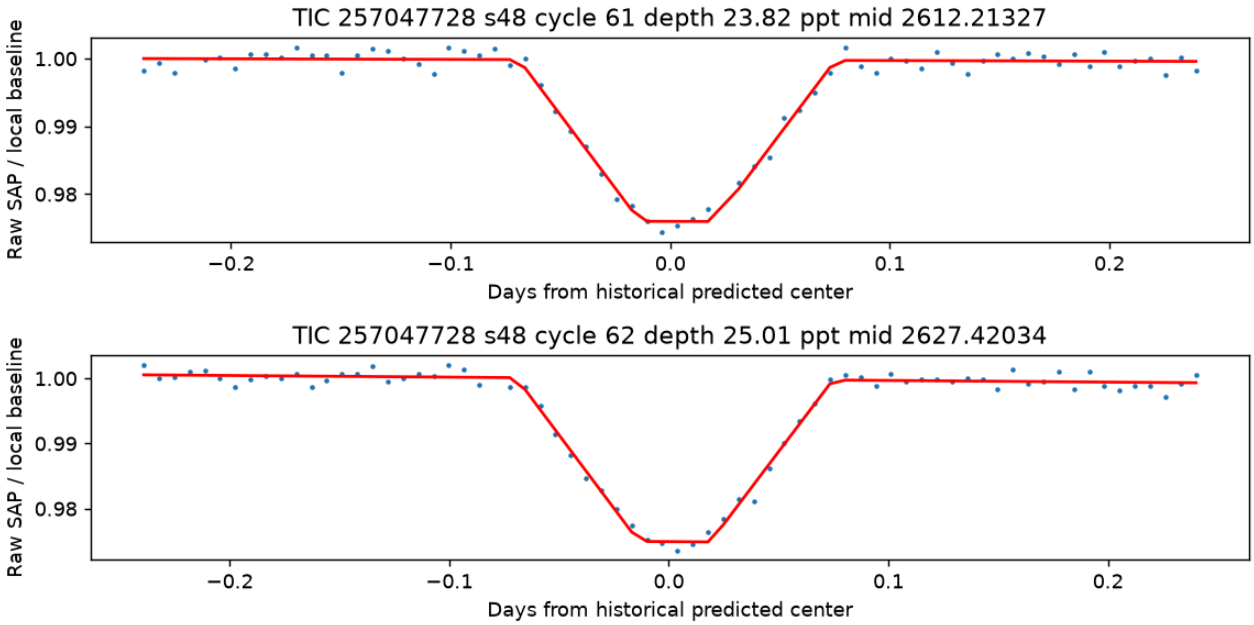
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
48	61	2612.2132742	0.003472	23.816	used in ephemeris
48	62	2627.4203417	0.003472	25.005	used in ephemeris

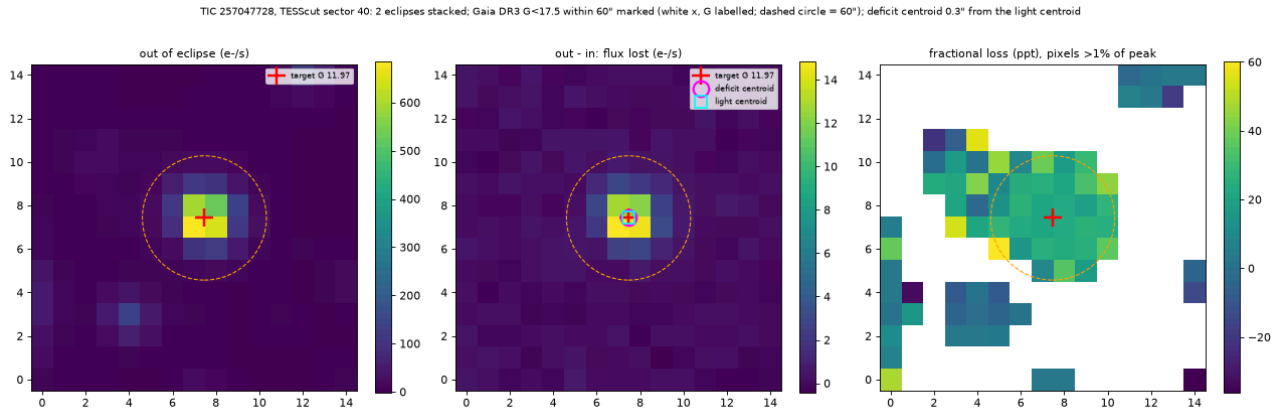
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s40.json, historical/diff_s40.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R401. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R401; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R401; diff_s40.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R402. TIC 168917716

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 168917716: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[14, 74, 82]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 5 same-kind midtimes give P=13.4024383530 +/-0.0000718553d, reference eclipse epoch BTJD1699.2149087, kind: Selected alternating eclipse; even half-period cycles; depth parity nearly equal, orbital identity informed by Gaia. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S41 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation1.800arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.666. Capable neighbours: Gaia 2072042036374420736: 16.82 arcsec, G15.817, required fractional eclipse0.581; Gaia 2072040558905683072: 41.98 arcsec, G16.071, required fractional eclipse0.730; Gaia 2072040352747228672: 44.54 arcsec, G14.587, required fractional eclipse0.198; Gaia 2072042105093927936: 52.14 arcsec, G15.474, required fractional eclipse0.428; Gaia 2072040558905685632: 54.35 arcsec, G15.169, required fractional eclipse0.327; Gaia 2072040348430214784: 56.78 arcsec, G13.458, required fractional eclipse0.081; Gaia 2072040558905691648: 58.77 arcsec, G15.239, required fractional eclipse0.348
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths17.954/18.356/16.103ppt, with 4 stacked events. QLP selected-kind fitted median16.506ppt. This separately extracted cutout is an alternative reduction of the same TESS

photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.

5. **Gaia orbital geometry: pass with stated limitations.** DR32072041830215983360, P=13.4010756298+/-0.0017605164d, e=0.043450, omega=234.1643deg. Closest edge-on conjunction offset0.196176d (0.014639phase), full-SB1-covariance linear propagated Gaia uncertainty0.189543d at the measured epoch. Near-Gaia-reference projected eclipse offset0.062630d (photometric period projection allowance0.007042d). The approximate central duration12.09h vs observed median4.68h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Shallow repeating eclipses across S14/74/82 with nearly equal alternating depths. Photometry supports6.701d repetition; selected even cycles give13.40244d consistent with Gaia orbit. Equal depths alone cannot reject half-period orbital identity; retained doubled branch explicitly uses Gaia and eclipse parity, with primary identity unresolved. Deficit0.086pixel versus nearest capable neighbor0.859pixel.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

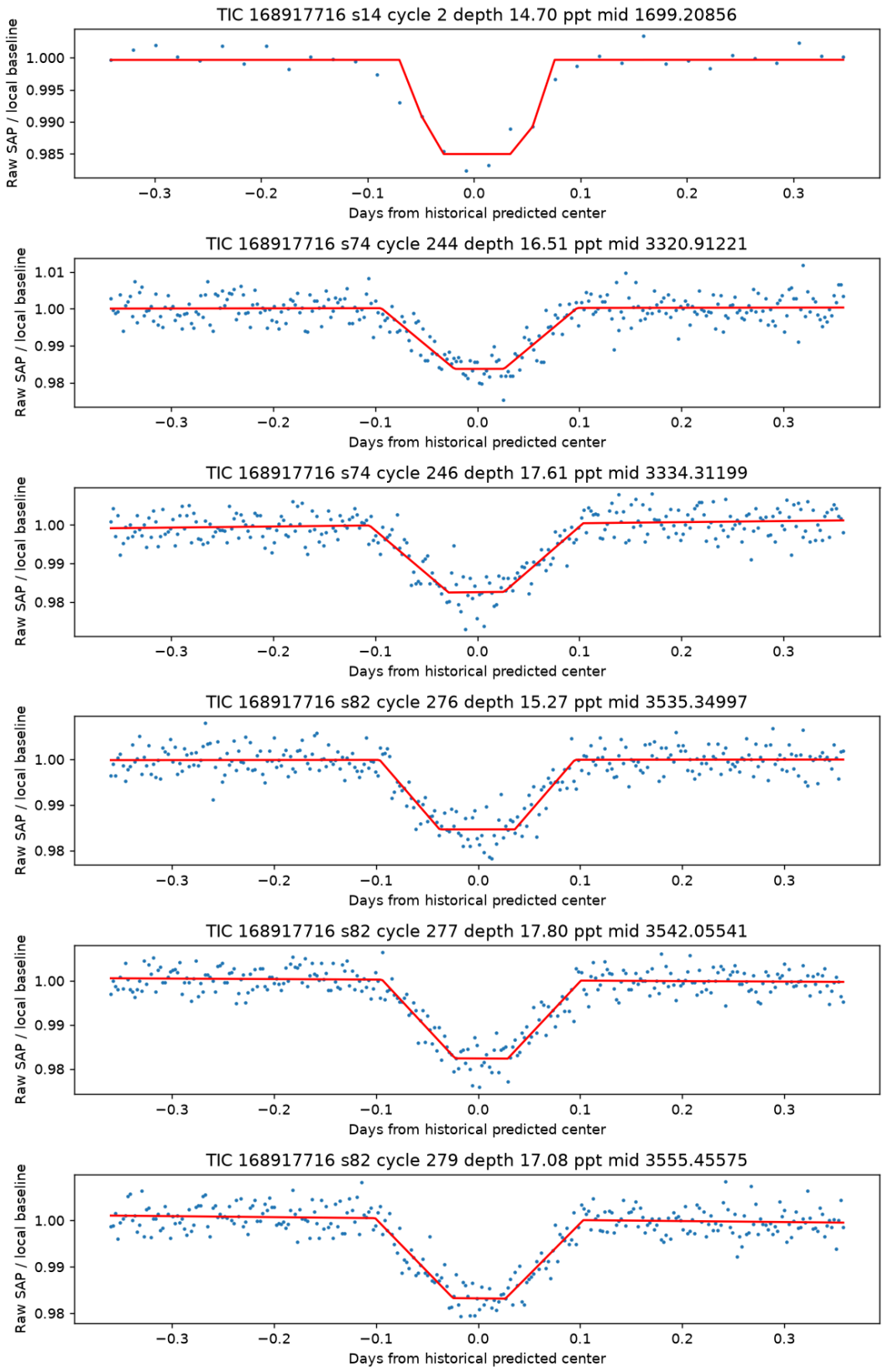
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	2	1699.2085569	0.010417	14.696	used in ephemeris
74	244	3320.9122083	0.002050	16.506	used in ephemeris
74	246	3334.3119945	0.002017	17.610	used in ephemeris
82	276	3535.3499738	0.001446	15.270	used in ephemeris
82	277	3542.0554081	0.001394	17.804	measured; excluded opposite parity (identity unresolved)
82	278	3548.7495208	0.001473	16.810	used in ephemeris
82	279	3555.4557456	0.001531	17.083	measured; excluded opposite parity (identity unresolved)

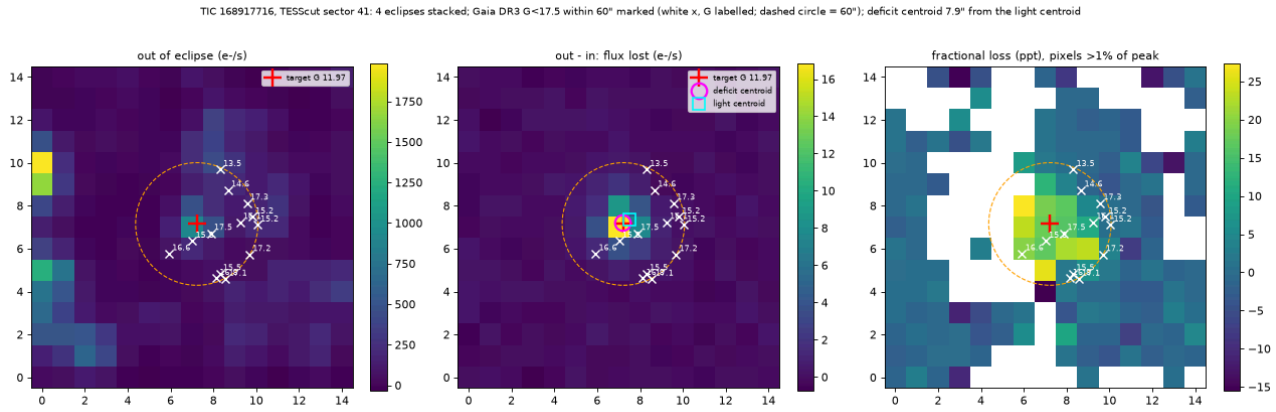
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s41.json, historical/diff_s41.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R402. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R402; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R402; diff_s41.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R403. TIC 156103314

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 156103314: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 9 distinct eclipse events in selected sectors[14, 41, 74]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 9 same-kind midtimes give P=8.1521424200 +/-0.0000185155d, reference eclipse epoch BTJD1691.0673712, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S40 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation3.626arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.140. Capable neighbours: Gaia 1129836918680283008: 36.63 arcsec, G14.967, required fractional eclipse0.510
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths32.550/31.110/29.172ppt, with 4 stacked events. QLP selected-kind fitted median31.470ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR31129836845665117312, P=8.1517909446+/-0.0005995836d, e=0.151241, omega=286.0880deg. Closest edge-on conjunction offset0.030193d (0.003704phase), full-SB1-covariance linear propagated Gaia uncertainty0.085527d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.026043d (photometric period projection allowance0.002962d). The approximate central duration8.41h vs observed median6.18h

is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Repeated flat-bottom eclipses across S14/41/74 with stable31-32ppt depth. Target deficit0.173pixel versus capable neighbor1.864pixels.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

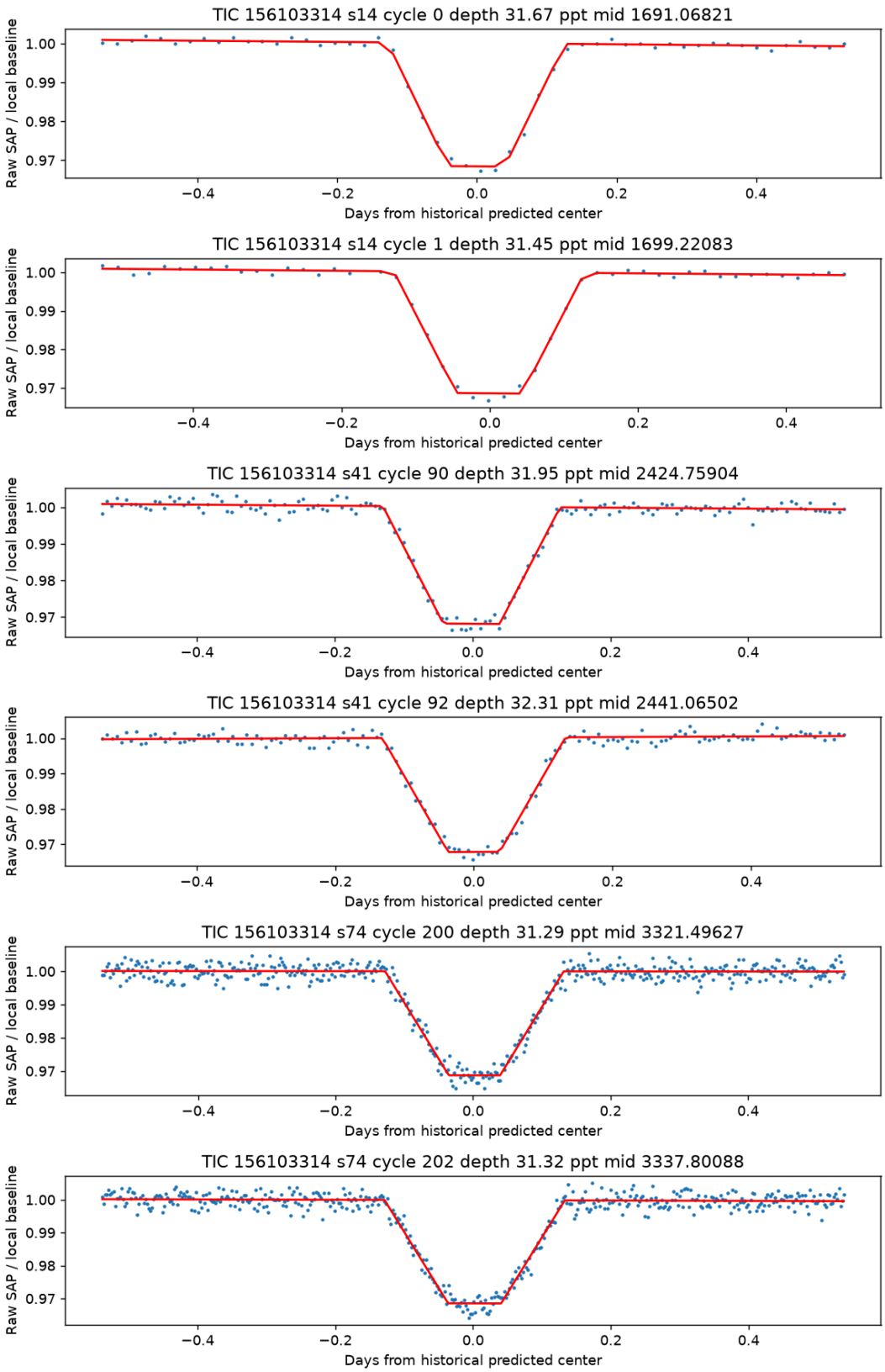
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
14	0	1691.0682147	0.010417	31.668	used in ephemeris
14	1	1699.2208326	0.010417	31.449	used in ephemeris
14	2	1707.3742829	0.010417	31.470	used in ephemeris
41	90	2424.7590385	0.003472	31.947	used in ephemeris
41	92	2441.0650222	0.003472	32.312	used in ephemeris
74	199	3313.3417474	0.001157	32.794	used in ephemeris
74	200	3321.4962720	0.001157	31.290	used in ephemeris
74	201	3329.6488137	0.001157	30.378	used in ephemeris
74	202	3337.8008805	0.001157	31.325	used in ephemeris

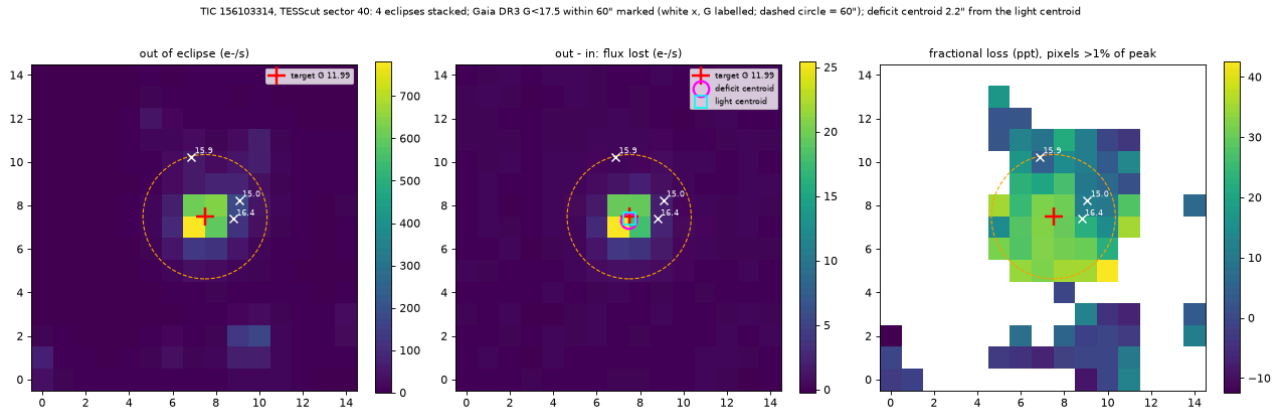
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s40.json, historical/diff_s40.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R403. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R403; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R403; diff_s40.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R404. TIC 140340353

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 140340353: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 13 distinct eclipse events in selected sectors[33, 87]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 13 same-kind midtimes give P=3.3434088582 +/-0.0000031883d, reference eclipse epoch BTJD2203.1906137, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S33 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation0.867arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.174. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths43.911/42.702/41.059ppt, with 8 stacked events. QLP selected-kind fitted median45.287ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3332271761078881088O, P=3.3440197736+/-0.0003574652d, e=0.098550, omega=327.8916deg. Closest edge-on conjunction offset-0.167185d (-0.049995phase), full-SB1-covariance linear propagated Gaia uncertainty0.161169d at the measured epoch. Near-Gaia-reference projected eclipse offset0.164542d (photometric period projection allowance0.001731d). The approximate central duration6.07h vs

observed median3.22h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Thirteen narrow eclipses across S33/87 repeat coherently. No capable listed neighbor. Gaia conjunction offset-0.05phase (-0.167d) compared with full-covariance0.161d uncertainty: association is physically consistent within orbital-epoch uncertainty, without arbitrary phase veto.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

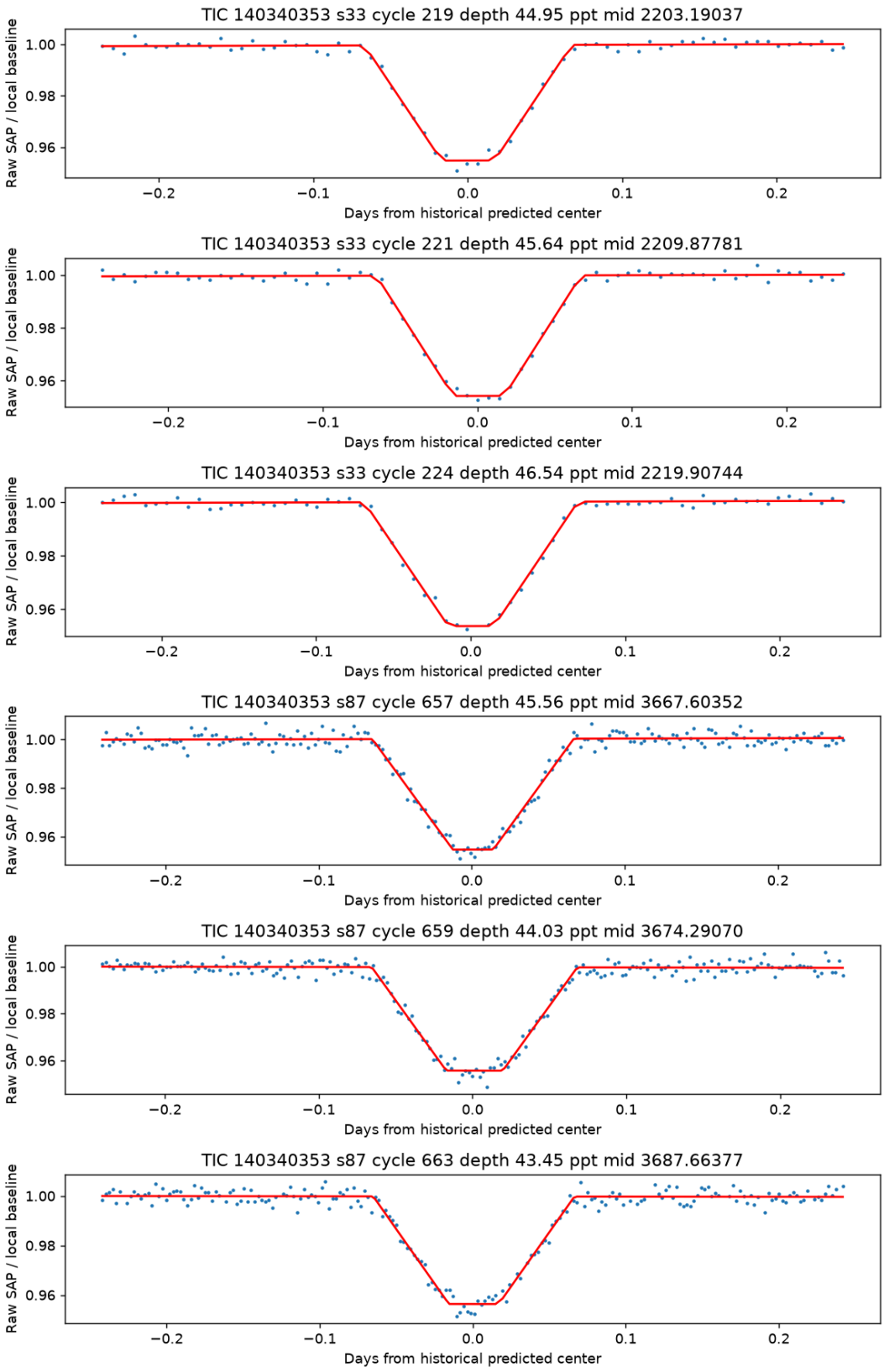
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
33	219	2203.1903722	0.003472	44.952	used in ephemeris
33	220	2206.5339597	0.003472	45.587	used in ephemeris
33	221	2209.8778085	0.003472	45.639	used in ephemeris
33	223	2216.5645577	0.003472	45.287	used in ephemeris
33	224	2219.9074408	0.003472	46.540	used in ephemeris
33	225	2223.2508490	0.003472	45.839	used in ephemeris
33	226	2226.5945579	0.003472	45.128	used in ephemeris
87	657	3667.6035206	0.001157	45.557	used in ephemeris
87	658	3670.9466228	0.001157	45.877	used in ephemeris
87	659	3674.2907028	0.001157	44.033	used in ephemeris
87	661	3680.9777823	0.001157	45.002	used in ephemeris
87	662	3684.3211172	0.001157	45.068	used in ephemeris
87	663	3687.6637723	0.001157	43.446	used in ephemeris

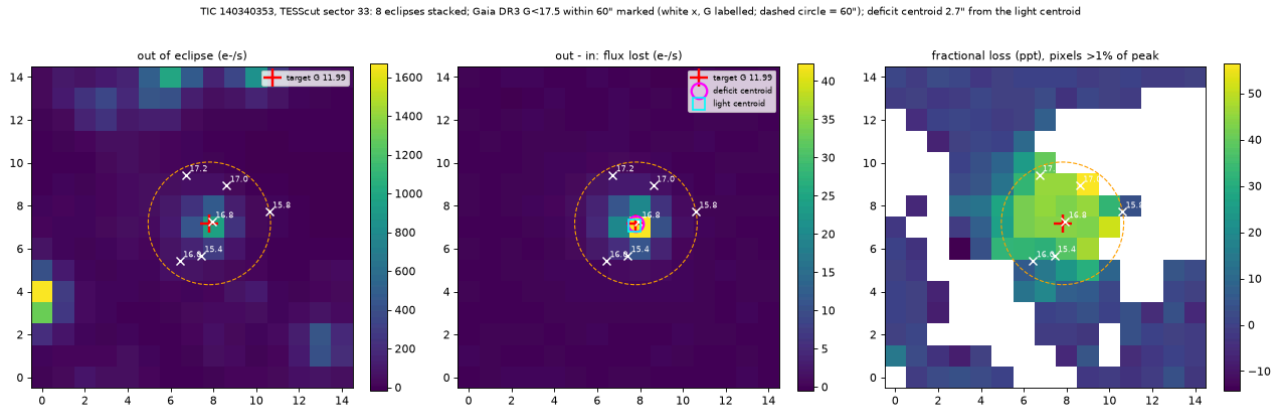
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s33.json, historical/diff_s33.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R404. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R404; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R404; diff_s33.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R405. TIC 143467555

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 143467555: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 22 distinct eclipse events in selected sectors[19, 59, 86]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 13 same-kind midtimes give P=4.2735185639 +/-0.0000041911d, reference eclipse epoch BTJD1820.1507383, kind: Deeper eclipse; odd historical half-period cycles. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S59 deficit peak[7, 8] vs target pixel[7, 8]; deficit-target separation4.348arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.339. Capable neighbours: Gaia 177237602207452544: 30.18 arcsec, G13.098, required fractional eclipse0.536
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths105.058/86.319/73.841ppt, with 11 stacked events. QLP selected-kind fitted median162.430ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3177237602207451008, P=4.2735828118+/-0.0002613483d, e=0.016118, omega=94.3267deg. Closest edge-on conjunction offset-0.015015d (-0.003513phase), full-SB1-covariance linear propagated Gaia uncertainty0.072119d at the measured epoch. Near-Gaia-reference projected eclipse offset0.006508d (photometric period projection allowance0.001404d). The approximate central duration4.96h vs observed median3.70h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Unequal alternating depths persist across S19/59/86: odd deeper events about145/167/162ppt versus even76/82/84ppt. Only13 odd deeper measured events define4.27351856d orbital period; half-period aliases are rejected by persistent parity morphology. Target deficit0.207pixel; mixed-kind historical difference stack still locates the signal.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

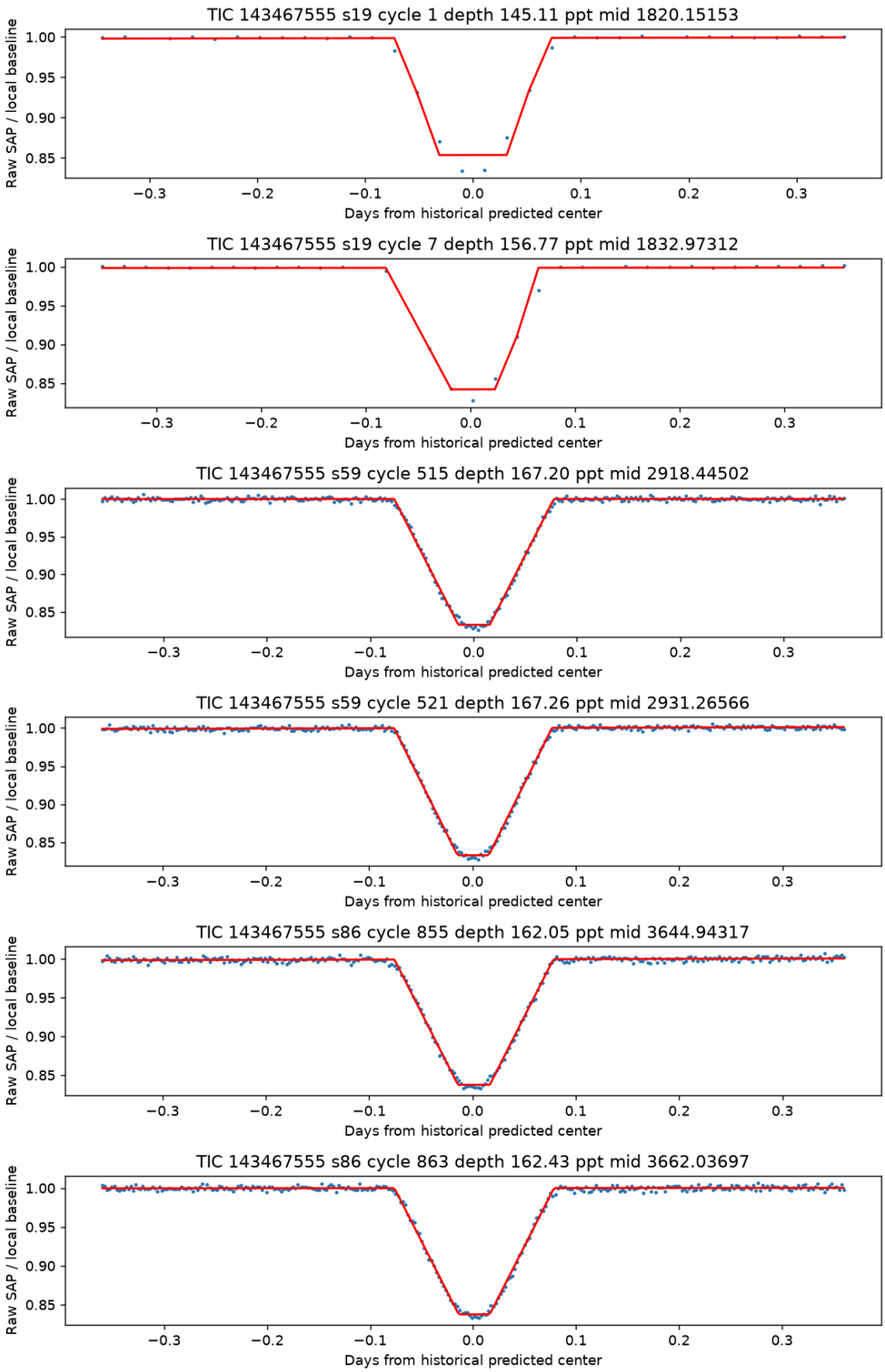
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
19	1	1820.1515257	0.010417	145.111	used in ephemeris
19	3	1824.4234335	0.010417	144.987	used in ephemeris
19	4	1826.5613202	0.010417	81.624	measured; excluded secondary event
19	6	1830.8371340	0.010417	70.159	measured; excluded secondary event
19	7	1832.9731222	0.010417	156.774	used in ephemeris
59	512	2912.0338269	0.001157	84.175	measured; excluded secondary event
59	513	2914.1715086	0.001157	166.425	used in ephemeris
59	514	2916.3075708	0.001157	81.877	measured; excluded secondary event
59	515	2918.4450150	0.001157	167.197	used in ephemeris
59	516	2920.5820861	0.001157	78.611	measured; excluded secondary event
59	519	2926.9921263	0.001157	167.430	used in ephemeris
59	520	2929.1290645	0.001157	82.458	measured; excluded secondary event
59	521	2931.2656587	0.001157	167.260	used in ephemeris
59	522	2933.4021701	0.001157	85.244	measured; excluded secondary event
59	523	2935.5388296	0.001157	168.958	used in ephemeris
86	853	3640.6693219	0.001157	161.944	used in ephemeris
86	855	3644.9431665	0.001157	162.051	used in ephemeris
86	856	3647.0791881	0.001157	82.834	measured; excluded secondary event
86	857	3649.2168370	0.001157	162.340	used in ephemeris
86	859	3653.4906809	0.001157	163.215	used in ephemeris
86	862	3659.9001930	0.001157	84.565	measured; excluded secondary event
86	863	3662.0369654	0.001157	162.430	used in ephemeris

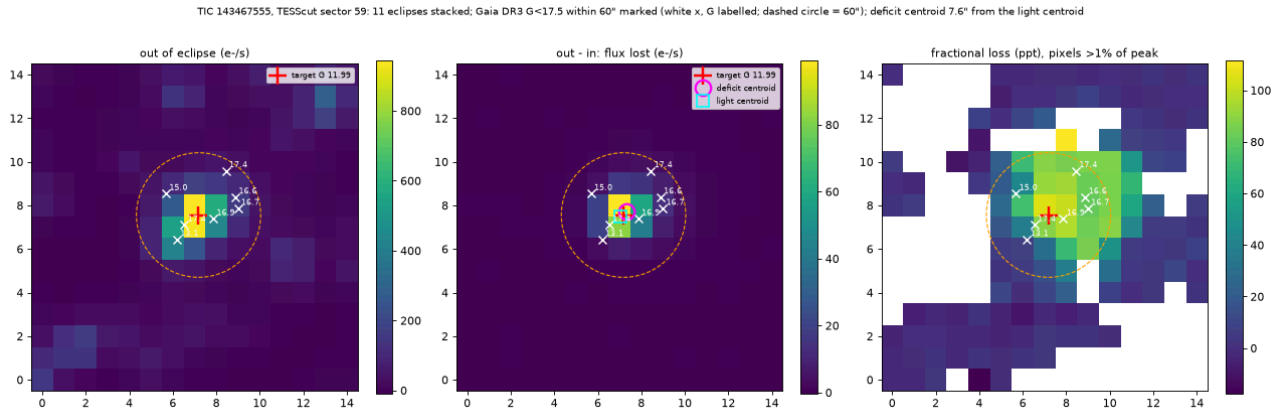
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s59.json, historical/diff_s59.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R405. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R405; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R405; diff_s59.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R406. TIC 57358968

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 57358968: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 7 distinct eclipse events in selected sectors[45, 71, 72]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=7.4961579330 +/-0.0000213829d, reference eclipse epoch BTJD2528.4204021, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S45 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.927arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.637. Capable neighbours: Gaia 3365438142952202496: 58.97 arcsec, G13.277, required fractional eclipse0.350
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths80.702/75.505/59.984ppt, with 3 stacked events. QLP selected-kind fitted median91.699ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3336542638762309657 6, P=7.4952754576+/-0.0007670721d, e=0.474132, omega=152.8161deg. Closest edge-on conjunction offset0.050735d (0.006769phase), full-SB1-covariance linear propagated Gaia uncertainty0.220431d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.200771d (photometric period projection allowance0.006094d). The approximate central duration4.94h vs observed median3.69h

is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Seven deep V-shaped eclipses across S45/71/72 with multiple consecutive local cycles; target-localized difference core and capable-neighbor geometry support attribution.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

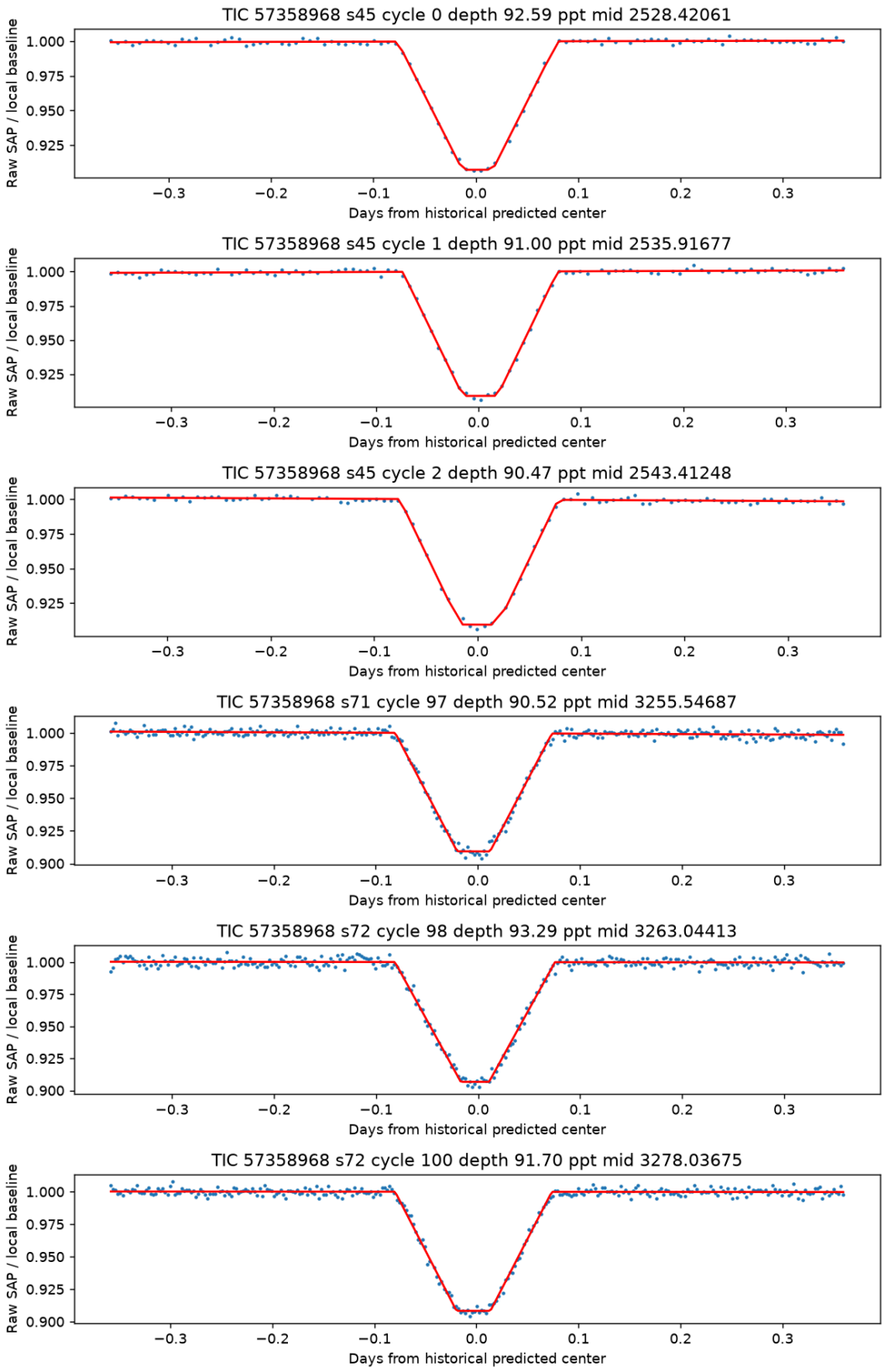
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
45	0	2528.4206133	0.003472	92.594	used in ephemeris
45	1	2535.9167680	0.003472	91.002	used in ephemeris
45	2	2543.4124789	0.003472	90.469	used in ephemeris
71	97	3255.5468668	0.001158	90.521	used in ephemeris
72	98	3263.0441349	0.001157	93.289	used in ephemeris
72	99	3270.5400579	0.001157	93.177	used in ephemeris
72	100	3278.0367542	0.001157	91.699	used in ephemeris

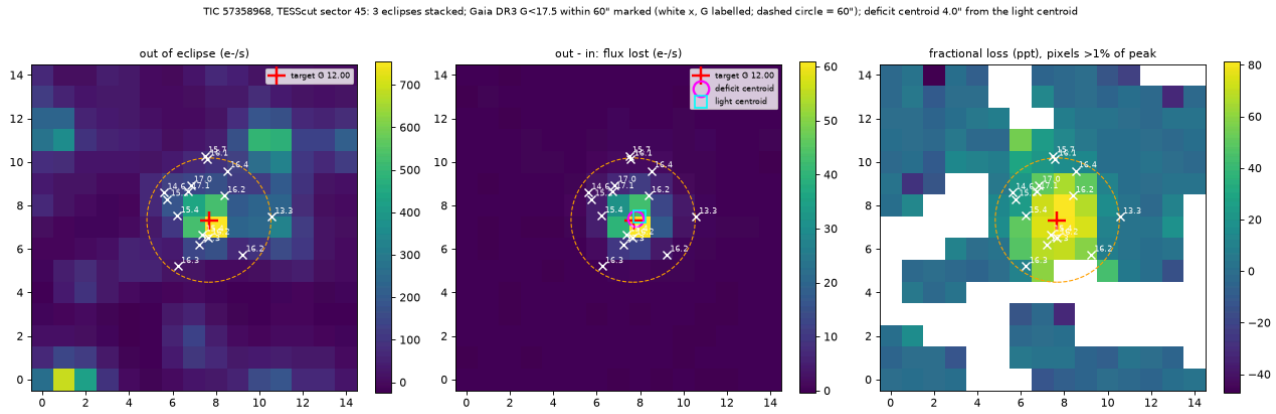
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s45.json, historical/diff_s45.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten.

Stable review key: R406. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R406; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R406; diff_s45.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R407. TIC 348302938

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 348302938: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 14 distinct eclipse events in selected sectors[61, 88]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 14 same-kind midtimes give P=3.6439561479 +/-0.0000165768d, reference eclipse epoch BTJD2966.8369124, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S88 deficit peak[8, 8] vs target pixel[8, 7]; deficit-target separation5.678arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.501. Capable neighbours: Gaia 5724552009498100864: 23.97 arcsec, G15.762, required fractional eclipse0.105; Gaia 5724552211356494720: 25.07 arcsec, G16.345, required fractional eclipse0.178; Gaia 5724552005198026112: 45.97 arcsec, G17.629, required fractional eclipse0.573; Gaia 5724551940778622464: 56.93 arcsec, G17.131, required fractional eclipse0.364
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths2.496/2.706/2.806ppt, with 8 stacked events. QLP selected-kind fitted median4.557ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR35724552009498098432, P=3.6439522672+/-0.0003950777d, e=0.097407, omega=267.6439deg. Closest edge-on conjunction offset-0.047381d (-0.013003phase), full-SB1-covariance linear propagated Gaia

uncertainty0.314984d at the measured epoch. Near-Gaia-reference projected eclipse offset-0.050125d (photometric period projection allowance0.011720d). The approximate central duration4.73h vs observed median2.37h is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Fourteen shallow narrow V-shaped eclipses repeat in S61/88; individual depths scatter at several ppt noise. Deficit0.270pixel from target versus nearest capable neighbor1.107pixels. Consecutive cycles constrain recurrence; parity depths are consistent within noise. Nearly equal primary/secondary events could leave a doubled orbital alias photometrically ambiguous; Gaia supports the retained branch.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

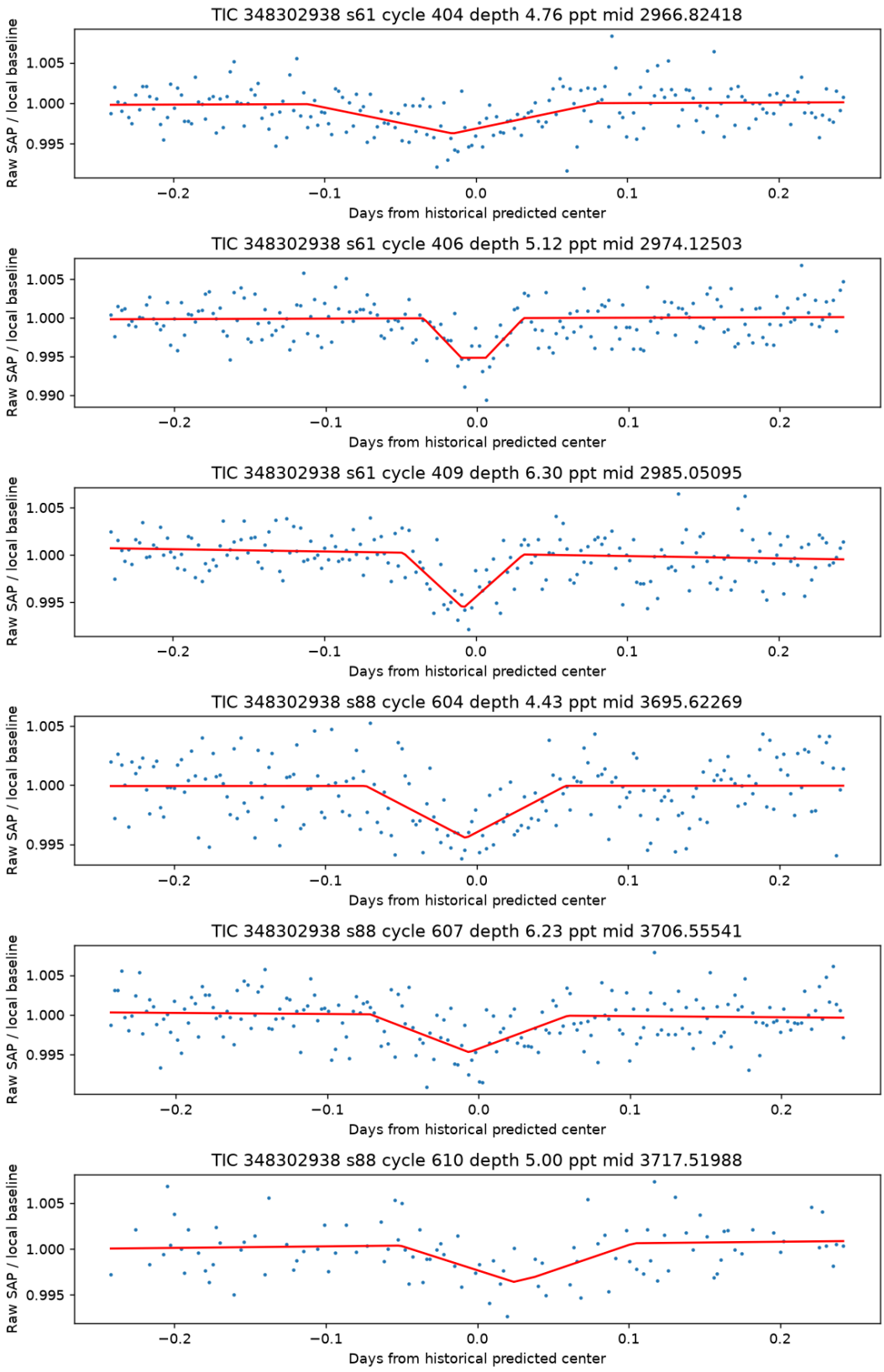
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
61	404	2966.8241799	0.006847	4.757	used in ephemeris
61	405	2970.4831648	0.002995	4.088	used in ephemeris
61	406	2974.1250280	0.002526	5.125	used in ephemeris
61	407	2977.7748846	0.002680	4.731	used in ephemeris
61	408	2981.4141019	0.002771	4.682	used in ephemeris
61	409	2985.0509515	0.002354	6.305	used in ephemeris
88	603	3691.9795432	0.005384	3.563	used in ephemeris
88	604	3695.6226915	0.005366	4.432	used in ephemeris
88	605	3699.2769256	0.005225	4.080	used in ephemeris
88	606	3702.9116340	0.003064	3.874	used in ephemeris
88	607	3706.5554139	0.004973	6.227	used in ephemeris
88	608	3710.2084557	0.003975	4.014	used in ephemeris
88	609	3713.8504577	0.004090	2.649	used in ephemeris
88	610	3717.5198846	0.008513	4.999	used in ephemeris

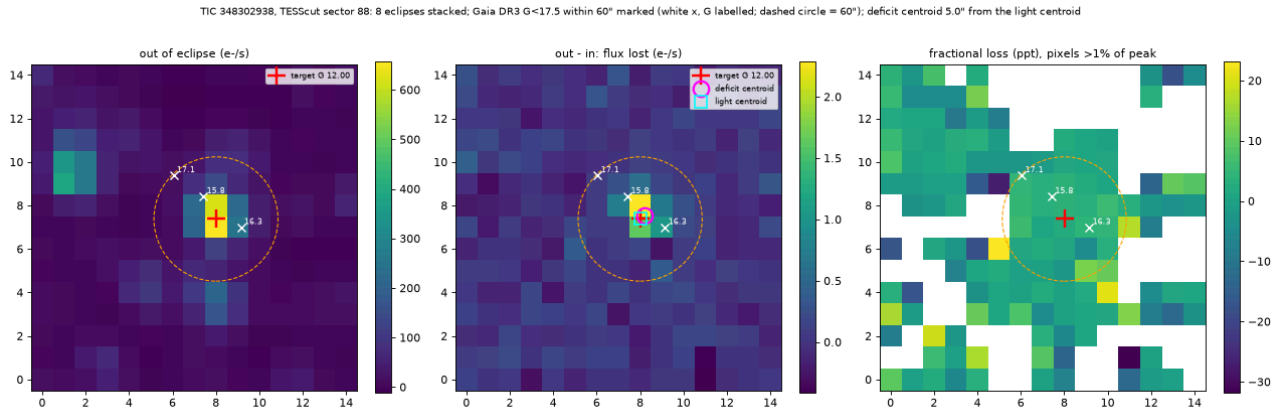
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s88.json, historical/diff_s88.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten. Historical inventory G=12.0059; actual nested Gaia DR3 G=11.996688. All three are inside actual DR3 G11.5-12 despite inventory values above12; this preserves the inventory/DR3 distinction.

Stable review key: R407. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R407; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R407; diff_s88.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R408. TIC 451086560

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 451086560: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 8 distinct eclipse events in selected sectors[10, 64, 100]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 7 same-kind midtimes give P=8.3004291559 +/-0.0000091255d, reference eclipse epoch BTJD1576.7194332, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S37 deficit peak[8, 7] vs target pixel[8, 7]; deficit-target separation2.006arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.985. Capable neighbours: Gaia 5346004174848323200: 32.54 arcsec, G13.548, required fractional eclipse0.780; Gaia 5346004106128831488: 59.25 arcsec, G8.021, required fractional eclipse0.155
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths68.641/16.758/6.125ppt, with 3 stacked events. QLP selected-kind fitted median151.848ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3534600417484830771 2, P=8.2999349364+/-0.0006655746d, e=0.019055, omega=57.1282deg. Closest edge-on conjunction offset0.133686d (0.016107phase), full-SB1-covariance linear propagated Gaia uncertainty0.089273d at the measured epoch. Near-Gaia-reference projected eclipse offset0.063012d (photometric period projection allowance0.001305d). The approximate central duration8.37h vs observed median6.96h

is a radius/minimum-mass illustration, not a detailed binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Deep flat-bottom eclipses repeat in S10/64/100. Partial S64 cycle179 ingress is excluded from timing. Aperture dilution and changing baseline can contribute to depth variation; deficit0.096pixel from target versus nearest capable neighbor1.679pixels. Adjacent S100 cycles directly test8.30d recurrence. A doubled orbital alias with equal eclipse kinds is not ruled out by parity alone; Gaia supports the retained branch.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

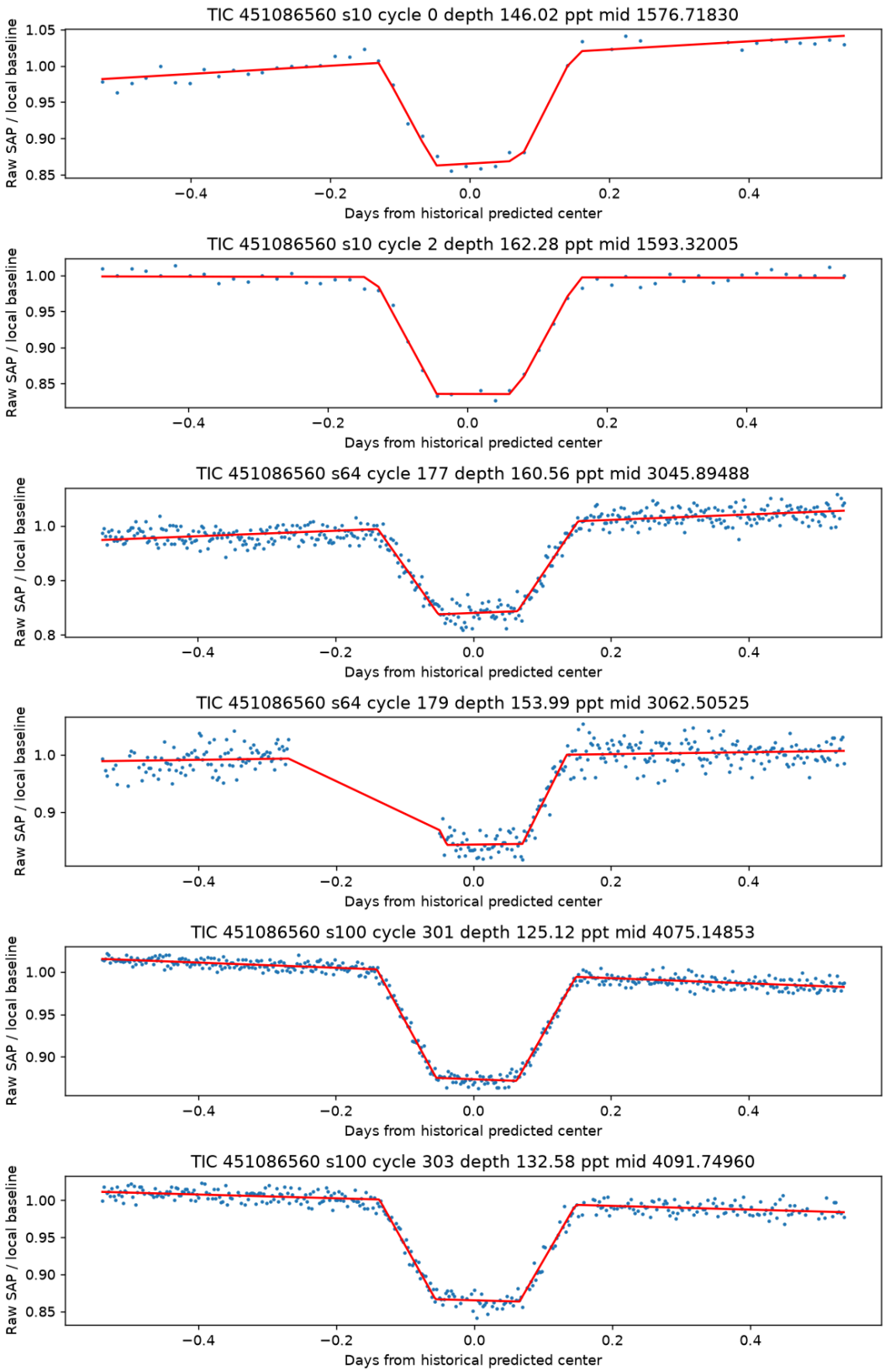
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
10	0	1576.7183031	0.010417	146.024	used in ephemeris
10	2	1593.3200517	0.010417	162.278	used in ephemeris
64	177	3045.8948798	0.001768	160.563	used in ephemeris
64	178	3054.1960875	0.001157	175.333	used in ephemeris
64	179	3062.5052542	0.002057	153.986	Partial ingress coverage across gap; retained raw morphology but excluded timing.
100	301	4075.1485308	0.001388	125.121	used in ephemeris
100	302	4083.4489310	0.001157	151.848	used in ephemeris
100	303	4091.7496016	0.001157	132.578	used in ephemeris

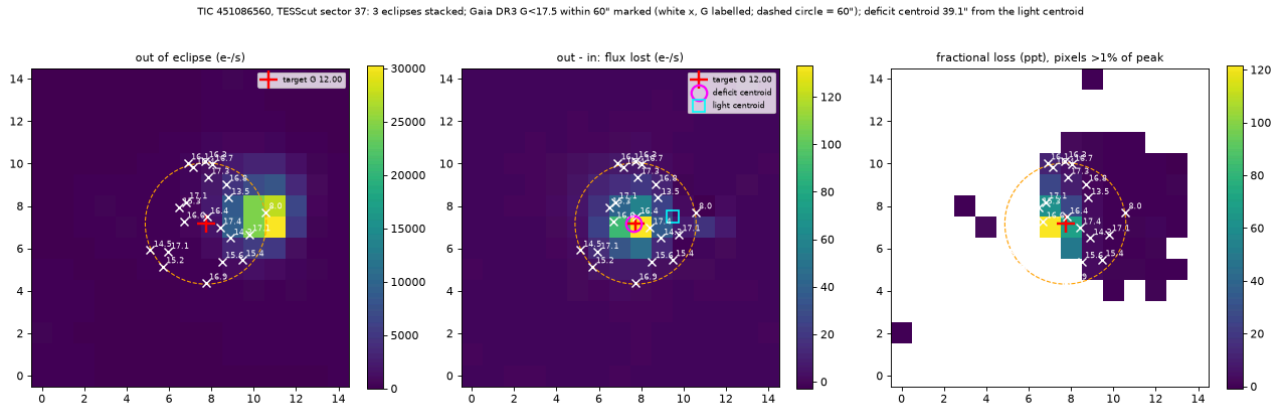
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s37.json, historical/diff_s37.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten. Historical inventory G=12.0086; actual nested Gaia DR3 G=11.996699. All three are inside actual DR3 G11.5-12 despite inventory values above12; this preserves the inventory/DR3 distinction.

Stable review key: R408. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R408; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R408; diff_s37.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

R409. TIC 229308711

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization

Archived source: sanitized report text.

TIC 229308711: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate. This review applies current section2.7 and does not promote the historical screening label automatically.

1. **Raw events: pass.** Free trapezoid fits to quality-zero raw QLP SAP_FLUX contain 4 distinct eclipse events in selected sectors[6, 71]. The representative raw-event plot and numeric raw-windows.csv were inspected. The raw morphology, ingress/egress and local baselines were inspected independently of fitted curves; target-specific interpretation follows below. Local linear baseline, free depth, center, duration and ingress were fitted; QLP detrended flux is not used as the deciding raw check. Representative selection is at most three spread available QLP sectors, with coverage/window requirements recorded by the script. Uncovered or flagged windows were never called nulls.
2. **Eclipse timing: pass.** 4 same-kind midtimes give P=14.6174947319 +/-0.0000862147d, reference eclipse epoch BTJD1471.0933055, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Partial or singular center-covariance fits flagged timing_usable=false are excluded from ephemeris and remain documented in measurements.json. Historical event coordinates in difference-image JSON were predictions and are not used as measured timings. Cycle counts were seeded by historical photometric search period, then fit freely across the selected baseline. This is not a blind all-alias search. Error meaning: Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.
3. **Target attribution: pass at TESS image resolution.** Historical S6 deficit peak[7, 7] vs target pixel[7, 7]; deficit-target separation2.484arcsec. Full saved difference image, original cutout identity and capable neighbours were audited. PSF-scaled image fractional residual0.289. Capable neighbours: None within60arcsec under the approximate G-band flux-budget test.
4. **Alternative reduction: pass.** Background-subtracted target-centered TESScut aperture gives 1x1/3x3/5x5 depths15.273/15.231/13.308ppt, with 2 stacked events. QLP selected-kind fitted median27.974ppt. This separately extracted cutout is an alternative reduction of the same TESS photons, not independent observing occasions. Dilution, aperture differences and, where relevant, mixed primary/secondary stacks prevent a claim of equal intrinsic depths.
5. **Gaia orbital geometry: pass with stated limitations.** DR3332794035974048076 8, P=14.6215831706+/-0.0049952480d, e=0.028473, omega=271.7243deg. Closest edge-on conjunction offset-0.249469d (-0.017062phase), full-SB1-covariance linear propagated Gaia uncertainty0.546625d at the measured epoch. Near-Gaia-reference projected eclipse offset0.053076d (photometric period projection allowance0.006380d). The approximate central duration18.67h vs observed median12.78h is a radius/minimum-mass illustration, not a detailed

binary fit. Full covariance and assumptions are in gaia-geometry.json. No arbitrary sigma veto is used.

Deciding limitations and interpretation: Four long flat-bottom eclipses repeat in consecutive pairs in S6 and71. Curved/sloped out-of-eclipse variability and asymmetric troughs limit simple trapezoid midtimes. The timing fit allows scatter inflation when needed (none applied here), but does not include total shape systematics. Historical custom pixel stack confirms target deficit0.118pixel; no capable listed neighbor. Consecutive pairs test recurrence and long-baseline cycle count. Equal eclipse kinds could conceal a doubled orbital alias; Gaia supports the retained branch.

Next check: Optional resolved photometry or updated spectroscopy for improved binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result applies; see ../../catalogues/coverage-matrix.csv. A separate source in the search cone does not establish a target match. This physical report does not assert exhaustive literature priority.

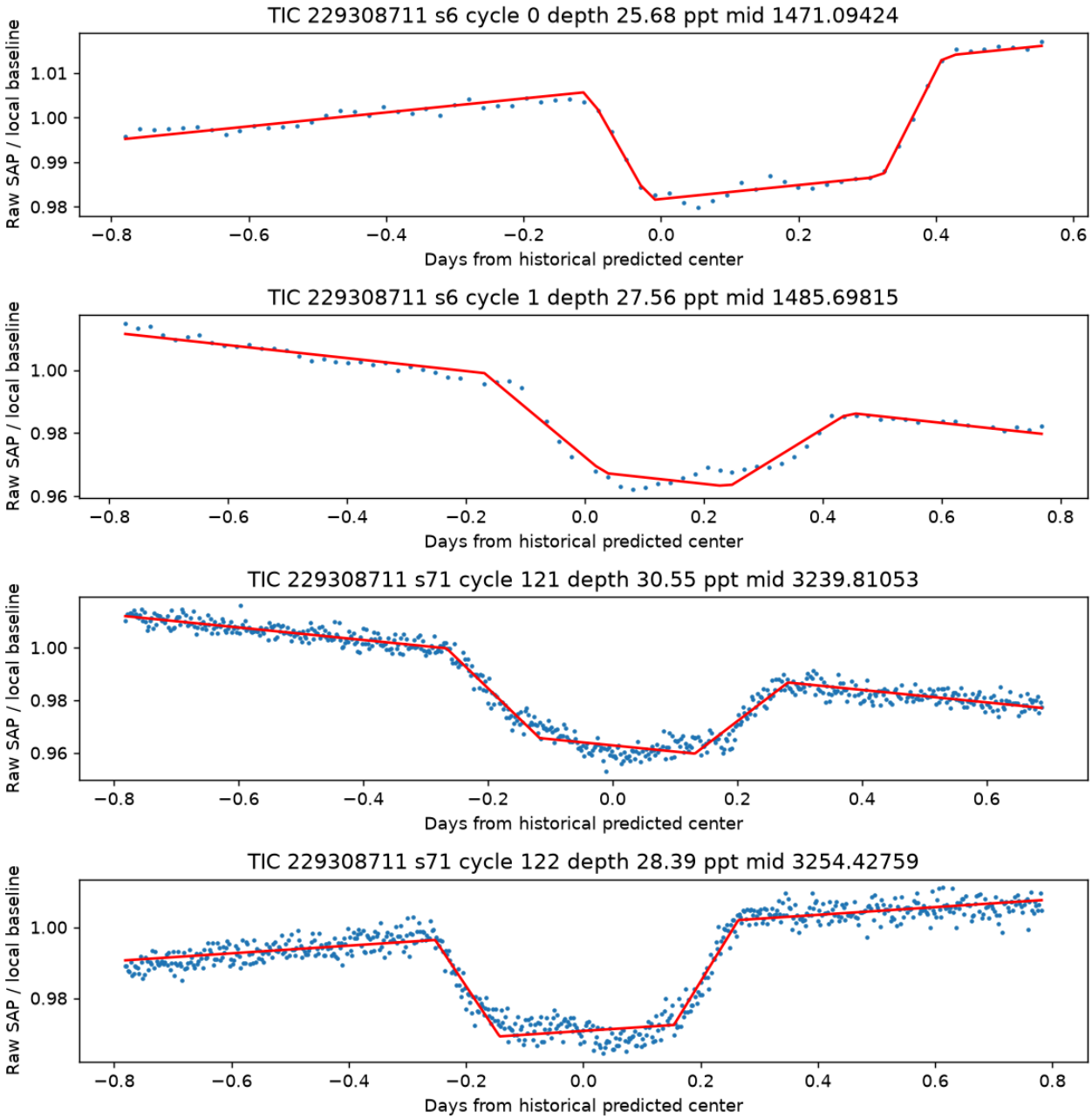
Measured events (errors are the conservative fit inputs, not calibrated coverage):

Sector	Historical photometric cycle	Measured midpoint BTJD	Timing error d	Fitted depth ppt	Timing status
6	0	1471.0942435	0.010417	25.684	used in ephemeris
6	1	1485.6981493	0.037975	27.562	used in ephemeris
71	121	3239.8105347	0.007298	30.553	used in ephemeris
71	122	3254.4275867	0.003354	28.386	used in ephemeris

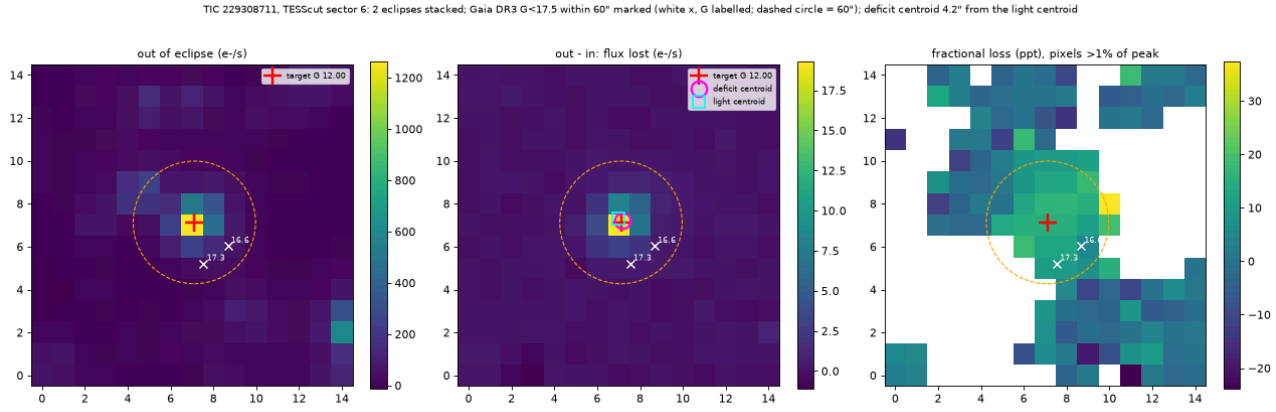
Evidence (original source layout; use the public evidence links below for exported locations): measurements.json, raw-events.png, raw-windows.csv, shared-qlp QLP FITS; historical/diff_s06.json, historical/diff_s06.png and historical/gaia.json; ../gaia-geometry.json; ../helper-audit.md; ../SHA256SUMS.json. Original cutout retained at [local path omitted] Historical pipeline definitions are [local path omitted] and [local path omitted] No historical evidence overwritten. Historical inventory G=12.0132; actual nested Gaia DR3 G=11.995216. All three are inside actual DR3 G11.5-12 despite inventory values above12; this preserves the inventory/DR3 distinction.

Stable review key: R409. Deciding evidence and catalogue coverage. The export manifest maps original hashes to sanitized files. Analysis scripts are archival and require the recorded external data; local paths are omitted.

Direct public evidence: measured events, raw eclipse plots, raw flux windows, Gaia geometry, independent audit, difference image.



R409; raw-events.png. Representative actual raw-flux eclipse windows. The full report retains timing, cadence-quality and alias qualifications.



R409; diff_s06.png. Target-pixel eclipse-deficit evidence. The full report retains capable-neighbour and angular-resolution qualifications.

Spectroscopic-orbit period folds: brightest remaining screened candidates

The next 130 screened candidates in actual Gaia DR3 G order (12.0036-12.3371) have 103 physically confirmed, 24 probable and three plausible outcomes. Two confirmed known recoveries are excluded, leaving 101 confirmed and 24 probable unique additions. The same bounded physical, catalogue and source-attribution criteria apply; the fixed balanced 120-target comparison remains unchanged. Those final 136 screened candidates are reviewed in the following section.

Periods and epochs below retain their per-object eclipse-kind and error meanings. Same-sector QLP and TESScut share photons; distinct sectors provide separate observing occasions on the same instrument. Specific morphology, alias and localization limits remain in each full report.